VU Medical Zone

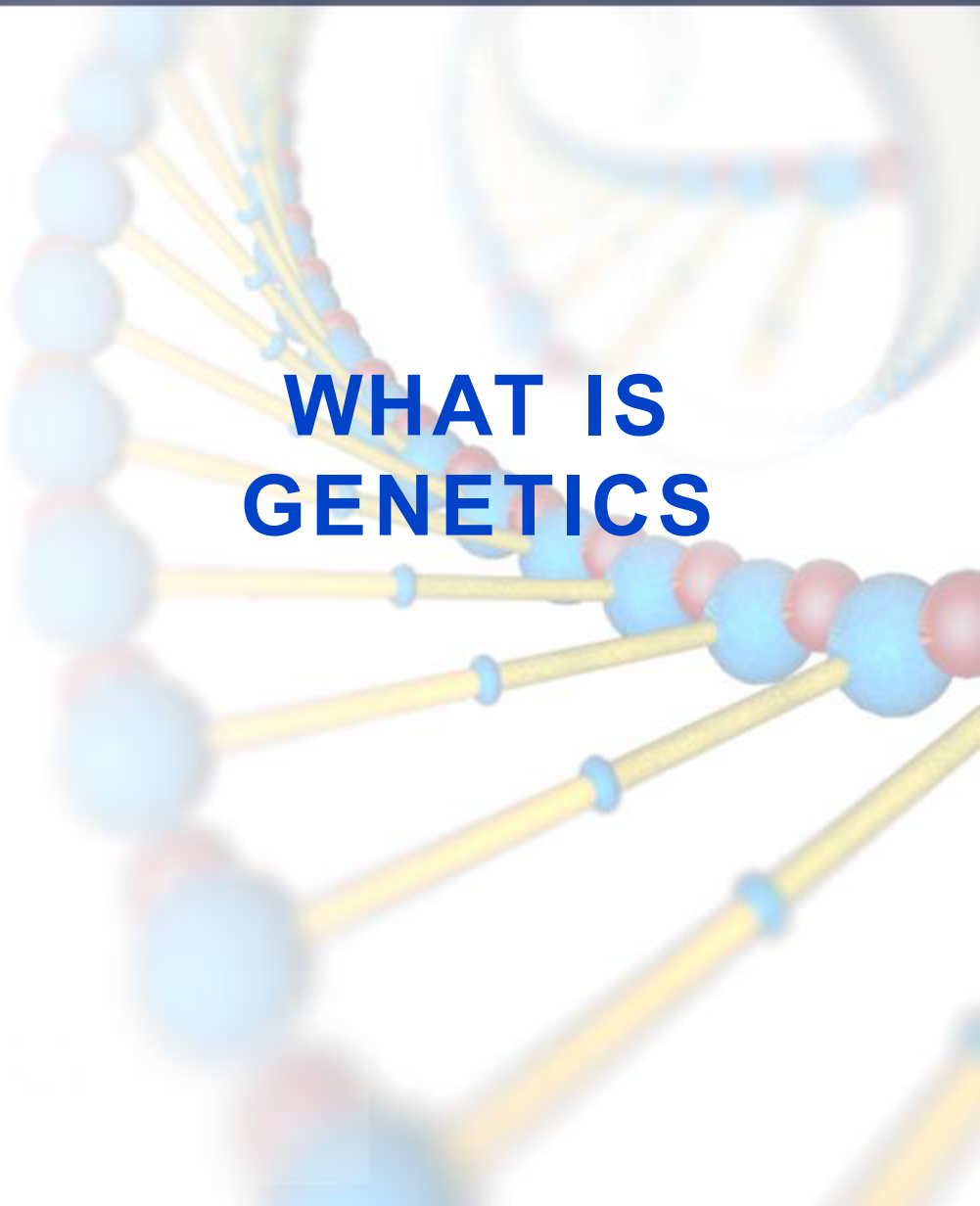
ADMIN:

AMAAN KHAN

BIO 301 MIDTERM PPT

LECTURE NO 1 TO 82

Introduction to Genetics



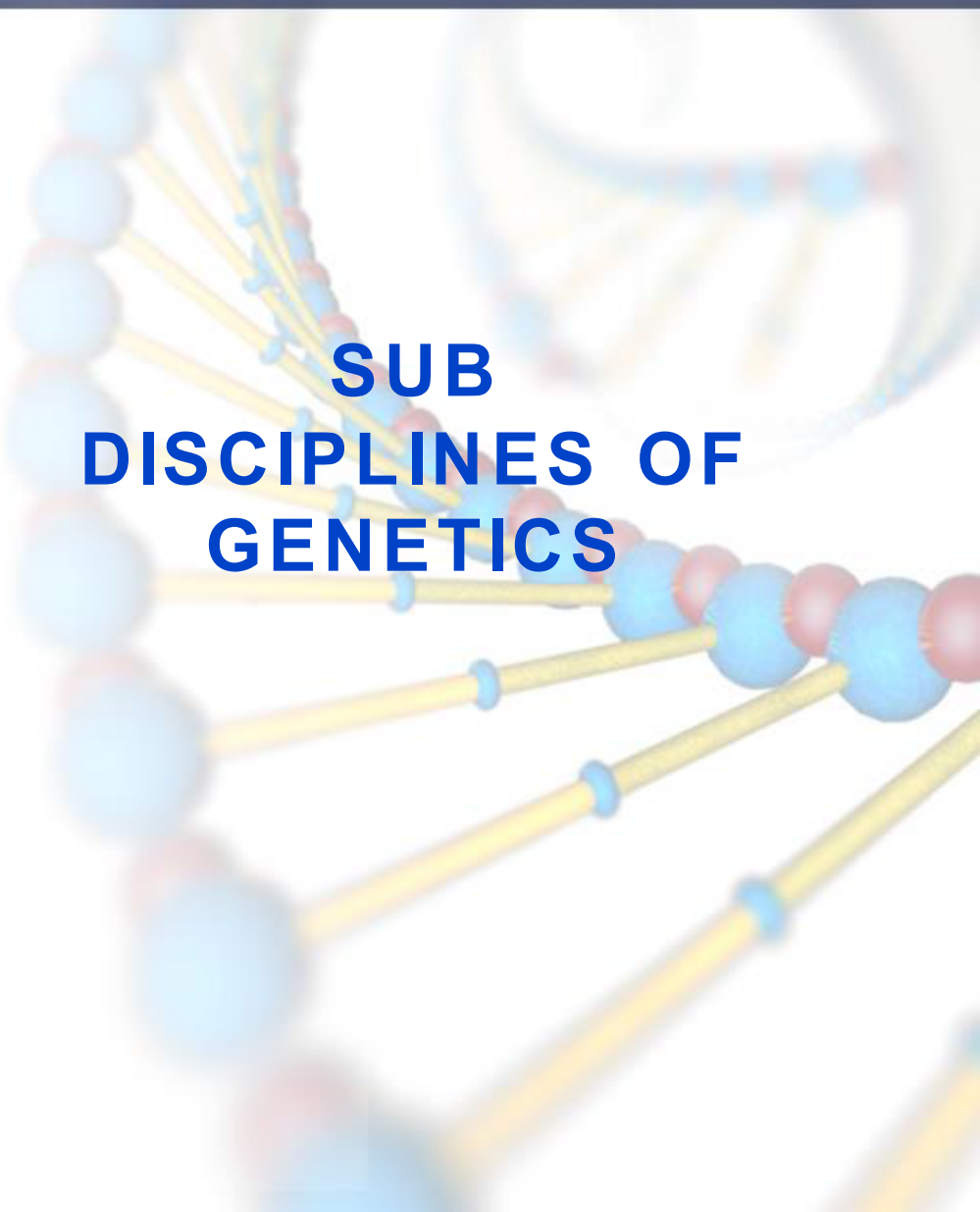
WHAT IS GENETICS

What is Genetics

DEFINITION

- Study of genes, heredity, variation.
- A field of Biology.
- The principles of heredity.
- Mendel unaware chromosomes, gene.

Introduction to Genetics



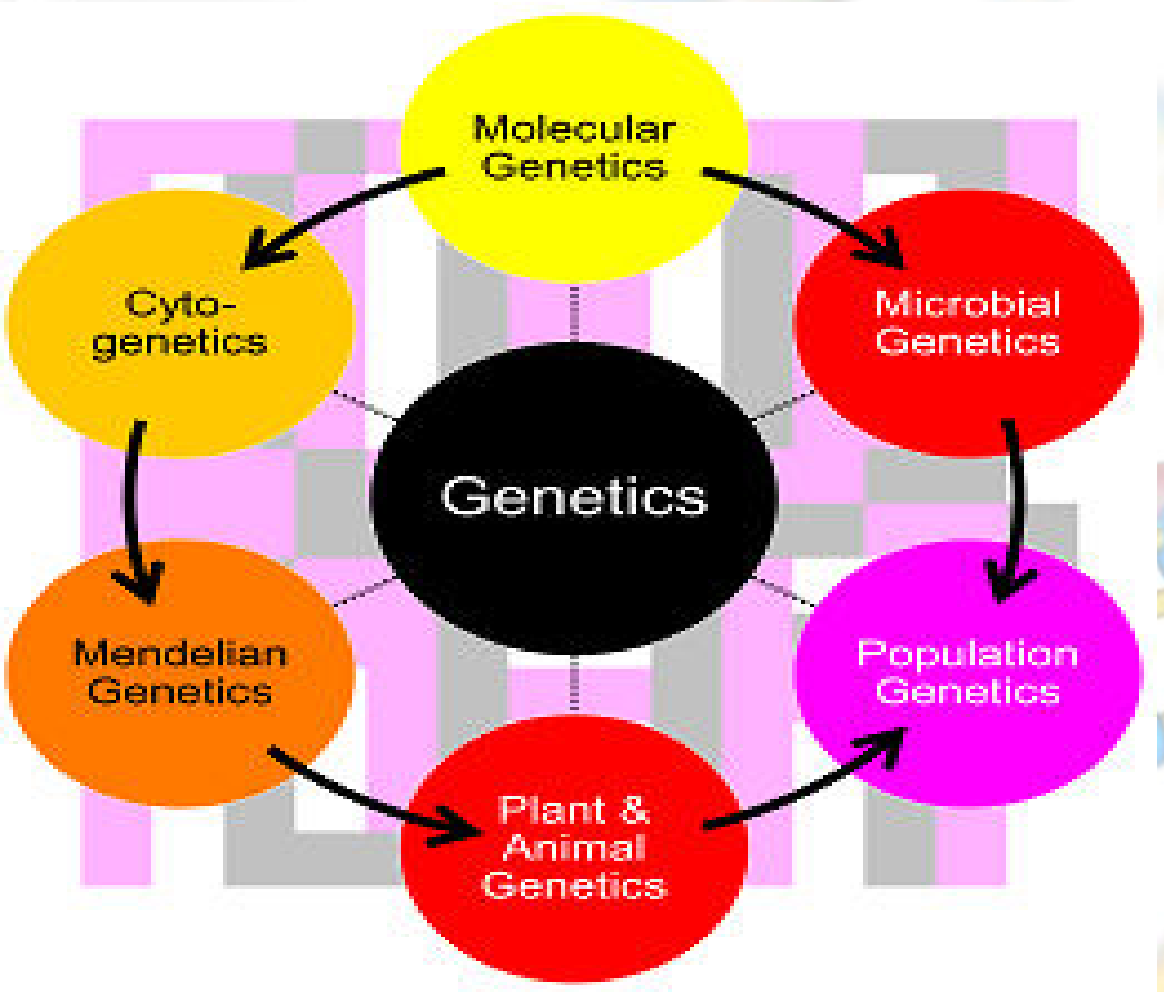
SUB DISCIPLINES OF GENETICS

Sub Disciplines of Genetics

FOUR SUB DISCIPLINES

- Transmission (Classical) Genetics
- Population Genetics
- Quantitative Genetics
- Molecular Genetics

Sub Disciplines of Genetics



Sub Disciplines of Genetics

DEVELOPMENTS IN GENETICS

- Historically, transmission genetics developed first, followed by population genetics, quantitative genetics and finally molecular genetics.

Sub Disciplines of Genetics

TRANSMISSION OR CLASSICAL GENETICS

- Deals with movement of genes and genetic traits from parents to offspring. Mendel s' Laws
- Deals with genetic recombinations.

Sub Disciplines of Genetics

POPULATION GENETICS

- Study traits in a group of population.
- Studies heredity in groups for traits determined by one or a few genes.

Sub Disciplines of Genetics

QUANTITATIVE GENETICS

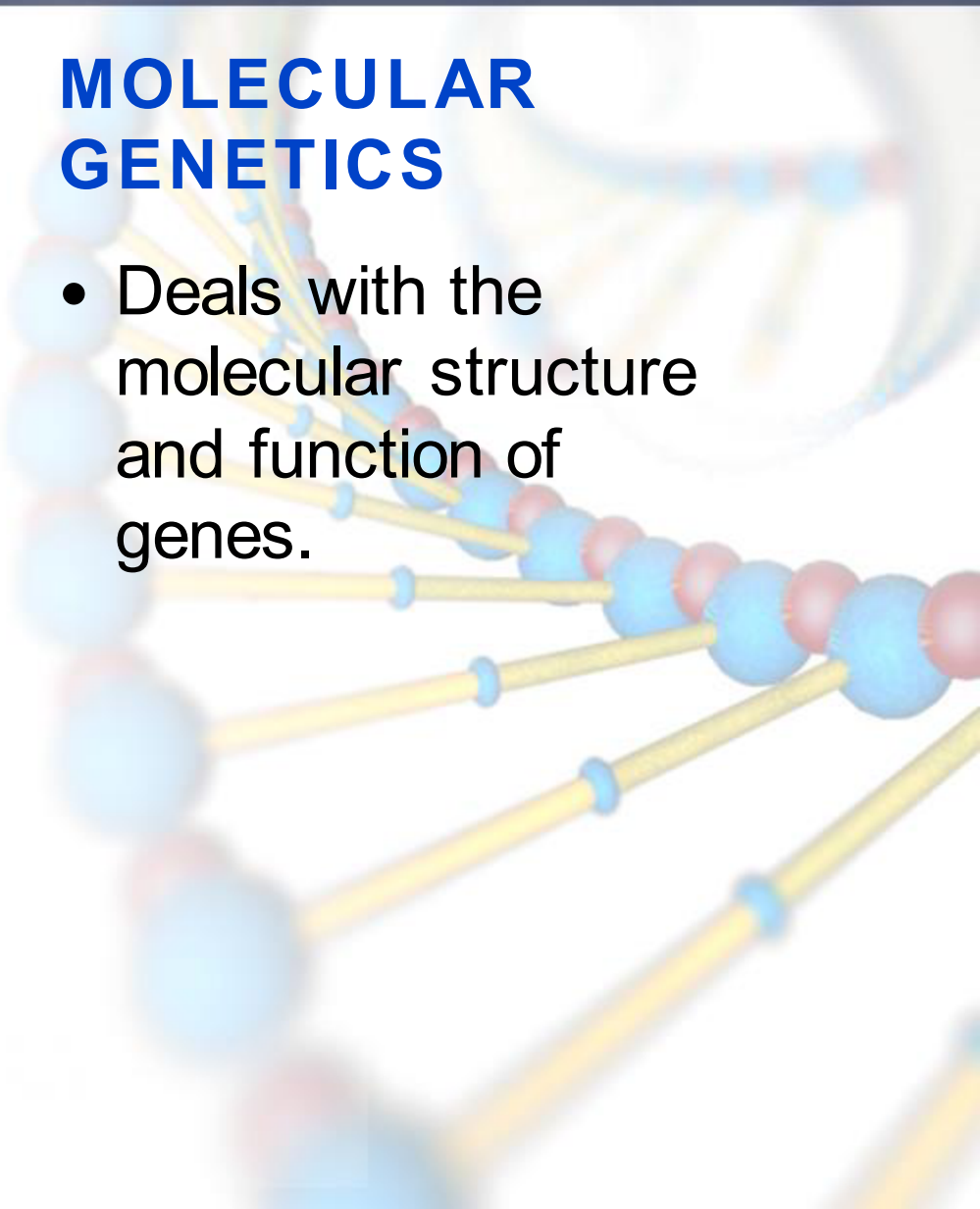
- Studies group hereditary for traits determined by many genes simultaneously.

- Skin color, height, eye color

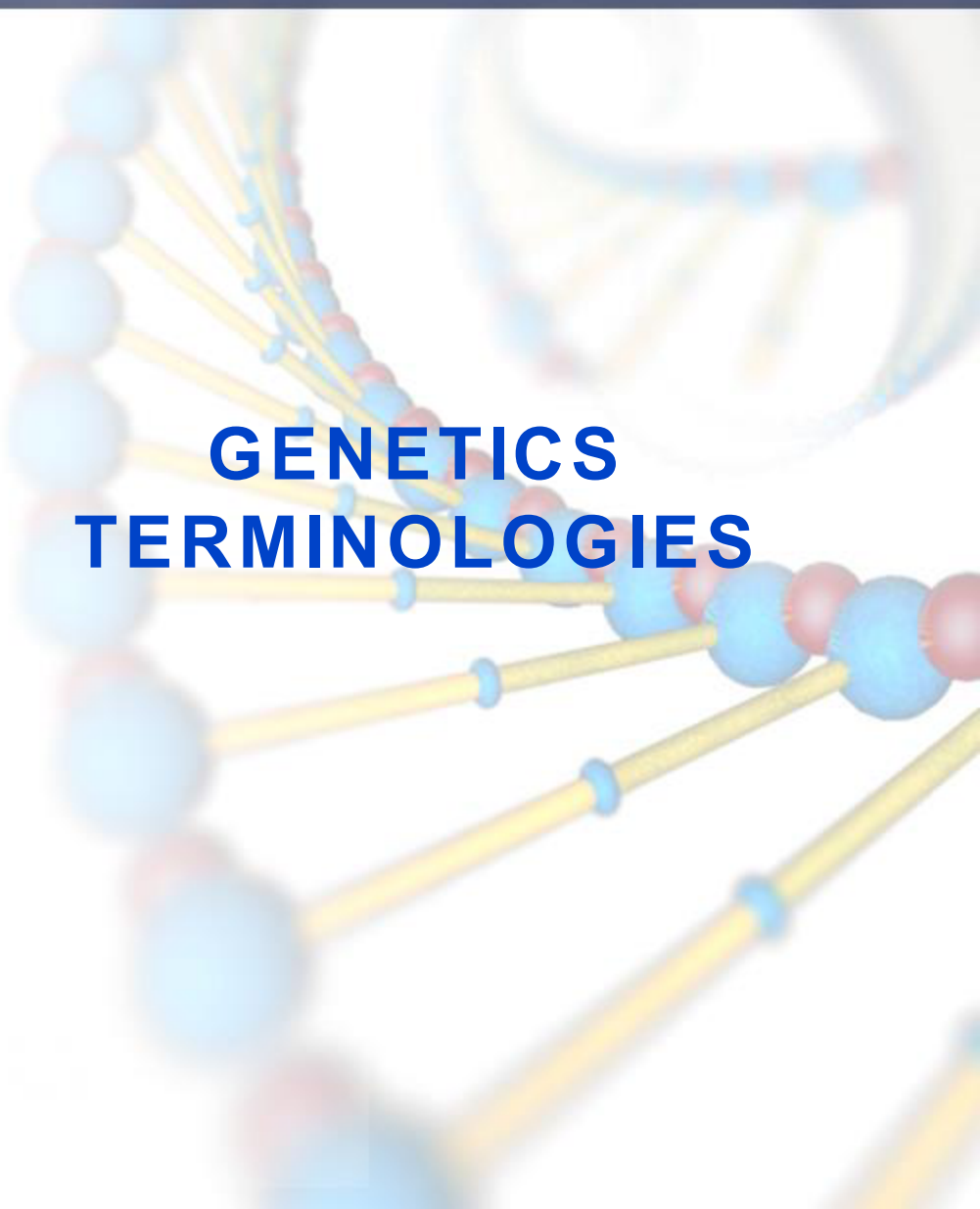
Sub Disciplines of Genetics

MOLECULAR GENETICS

- Deals with the molecular structure and function of genes.



Introduction to Genetics



GENETICS TERMINOLOGIES

Genetics Terminologies

GENETICS TERMINOLOGIES

- Character: heritable feature (skin color, height etc).
- Trait: variant for a character (i.e. brown, black, white etc).
- True-breed: all offspring of same variety.

Genetics Terminologies

DIFFERENT GENERATIONS OF A CROSS

- P generation (parents).
- F1 generation (1st filial generation).
- F2 generation (2nd filial generation).

Genetics Terminologies

PURE CROSS AND HYBRID CROSS

- Pure Cross

True breeding X True breeding

WW X ww

- Hybrid Cross

F1 generation X F1 generation

Ww X Ww

Genetics Terminologies

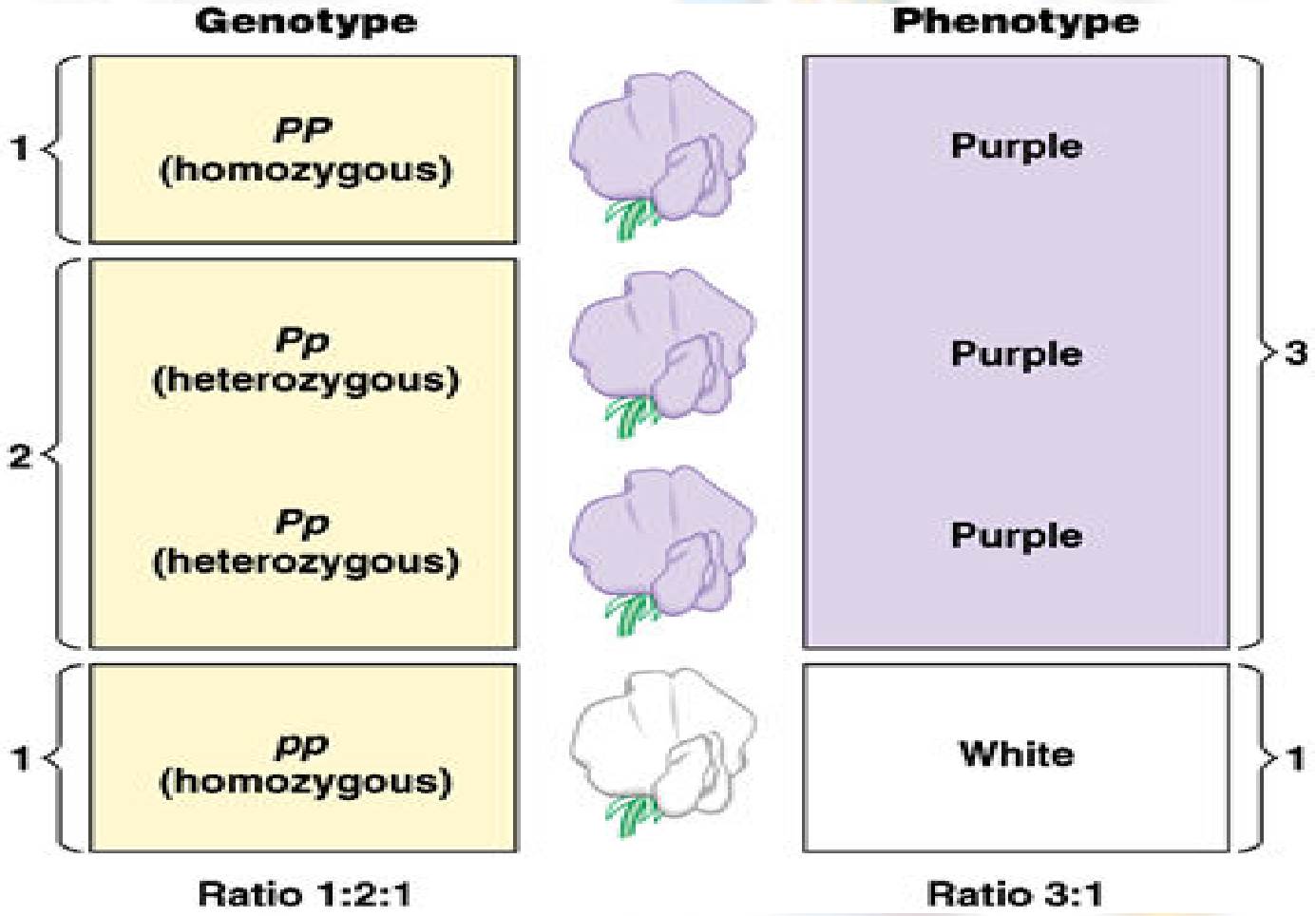
GENOTYPE AND PHENOTYPE

- Genetic make-up of an organism is called Genotype.

- Physical appearance of an organism is called Phenotype.

Genetics Terminologies

GENOTYPE AND PHENOTYPE

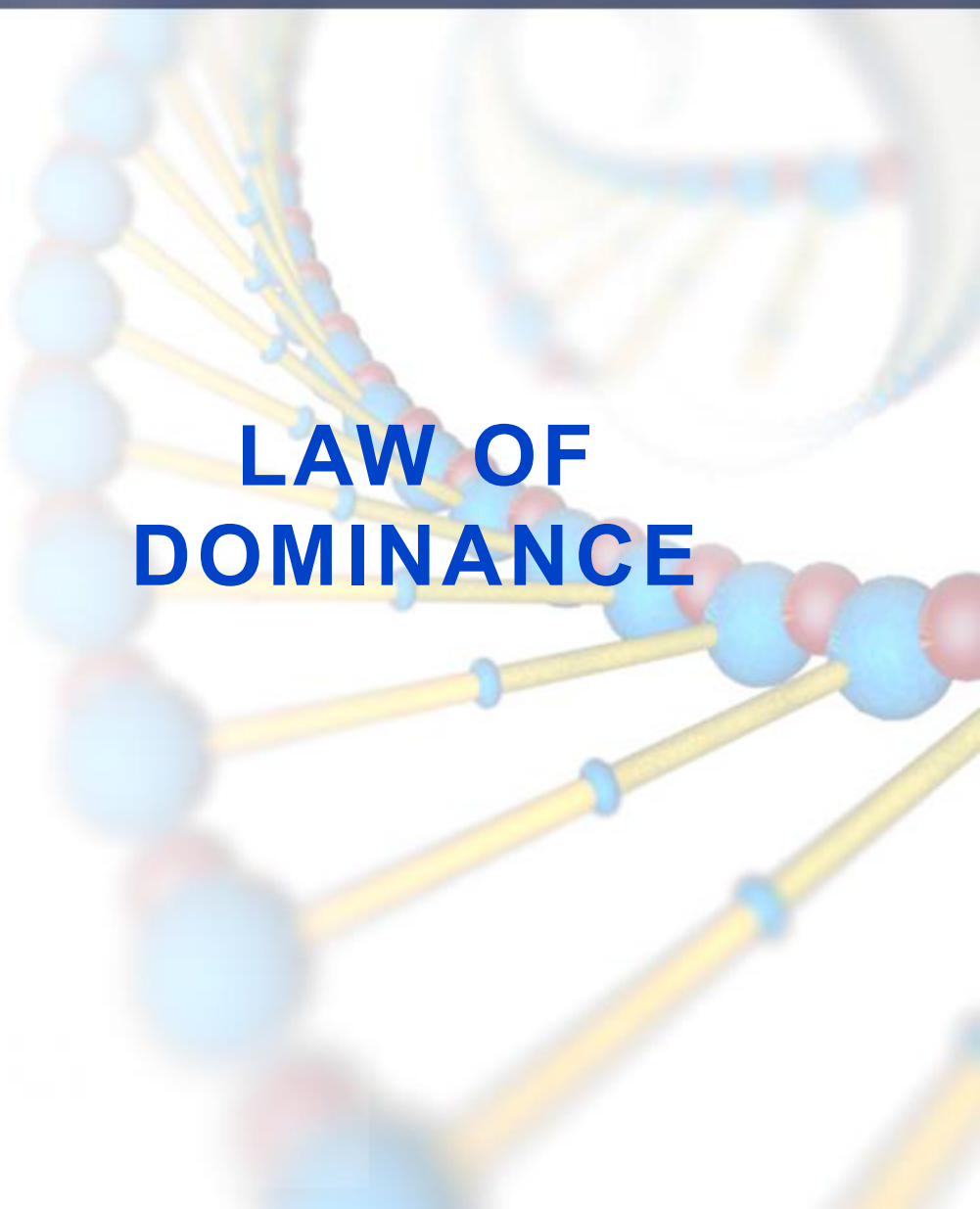


Genetics Terminologies

DOMINANT AND RECESSIVE

- Dominant: when one characteristic expresses itself over the other i.e. round over wrinkled.
- Recessive: The trait that does not show through in the first generation i.e. wrinkled.

Introduction to Genetics



LAW OF DOMINANCE

Law of Dominance

LAW OF DOMINANCE

- In a cross of parents, that are pure for contrasting traits, only one form of the trait will appear in the next generation.
- In the monohybrid cross, one version will be disappeared.

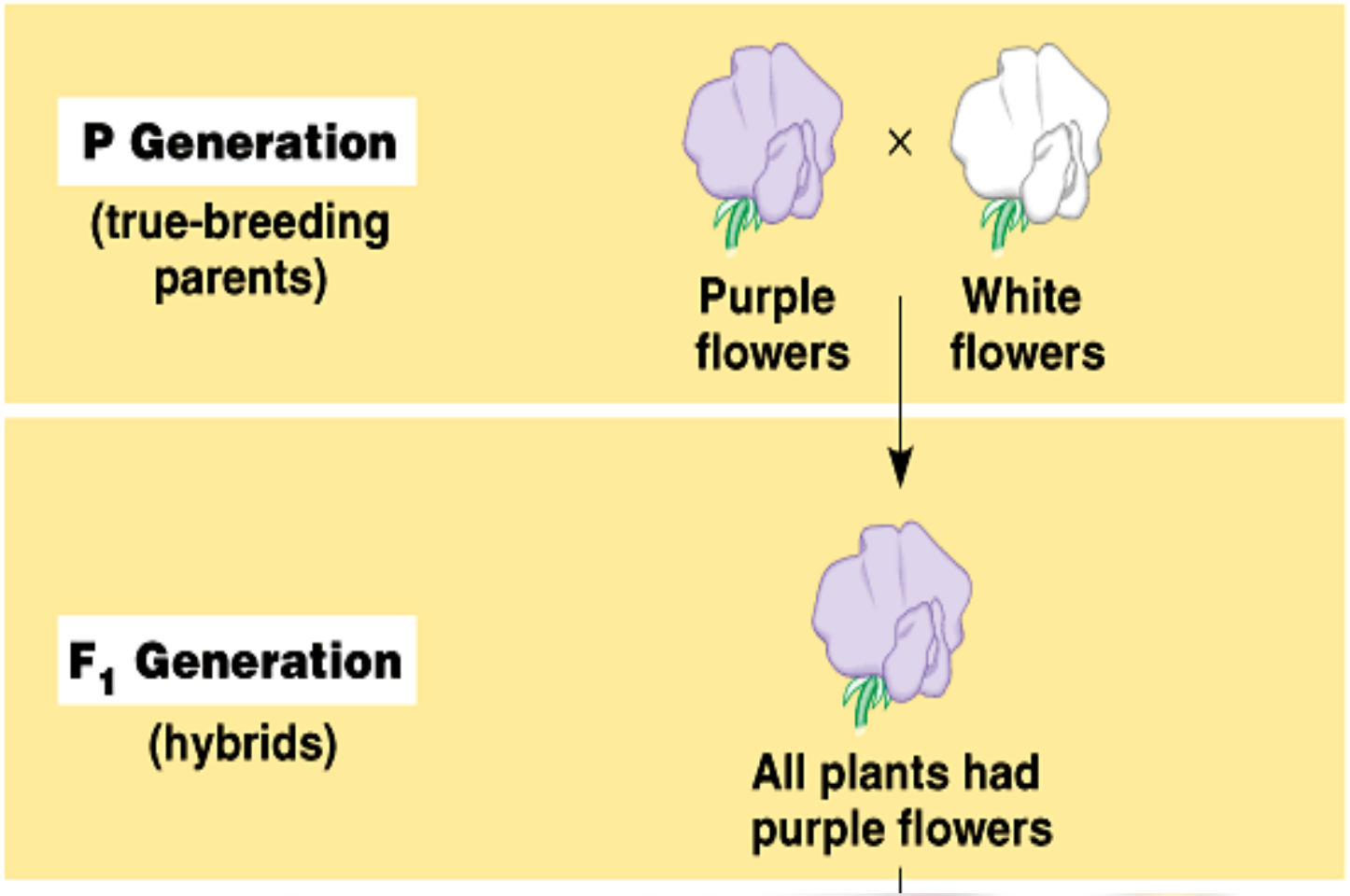
Law of Dominance

EXAMPLE

- Cross of two true-breeding plants.
- First - purple color while second -white color flowers.
- F1 generation, all plants with purple flowers.
- Purple color dominant to white

Law of Dominance

EXAMPLE



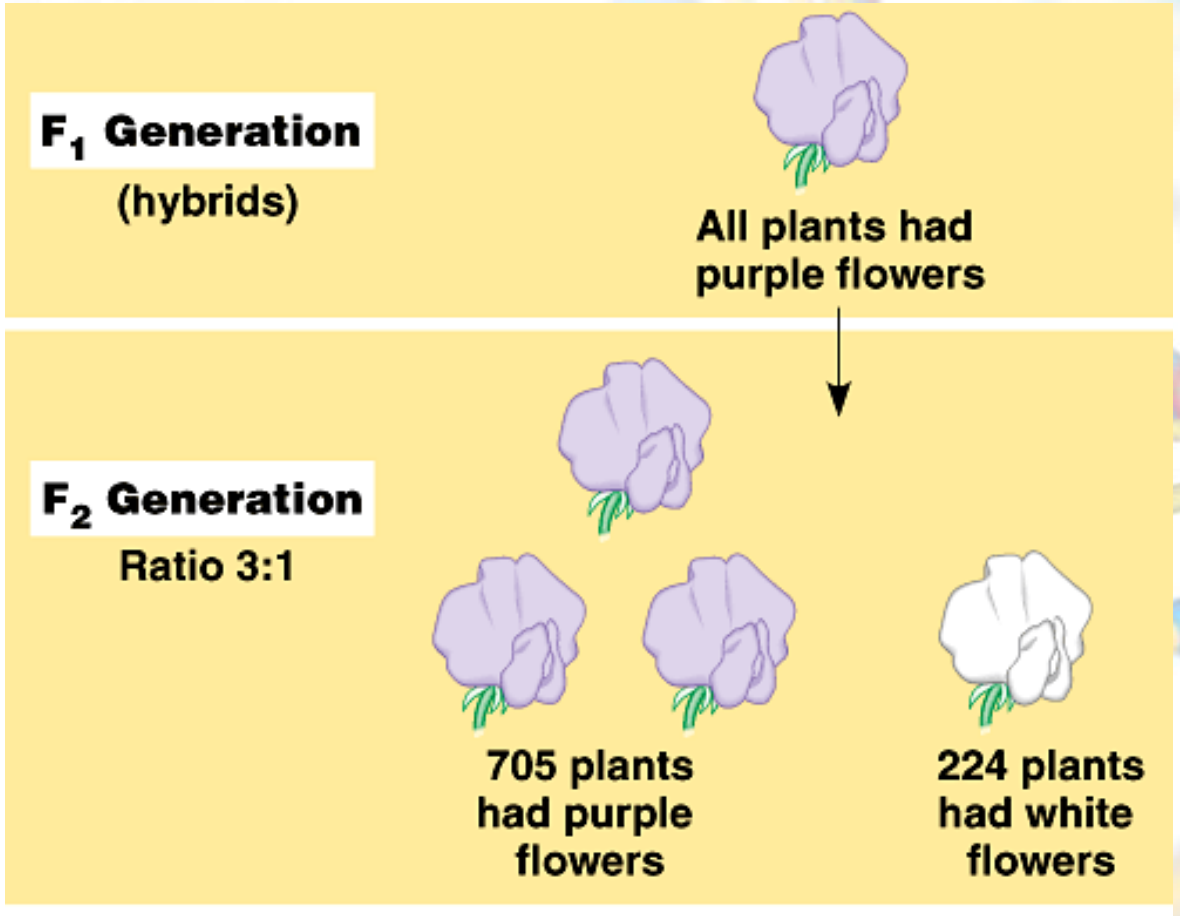
Law of Dominance

LAW OF DOMINANCE AND CURRENT MODEL OF INHERITANCE

- F1 cross, produced F2 plants.
- Lost-trait was re-appeared.
- Formulation of the current model of inheritance.

Law of Dominance

LAW OF DOMINANCE AND F2 GENERATION



Law of Dominance

HYBRIDS WILL SHOW DOMINANT PHENOTYPE

- Example:

  PP = purple pp = white

  Pp = purple

Law of Dominance

LAW OF DOMINANCE

- Every gene has two alleles that can code for a trait.
- One allele is dominant.
- One allele is recessive.

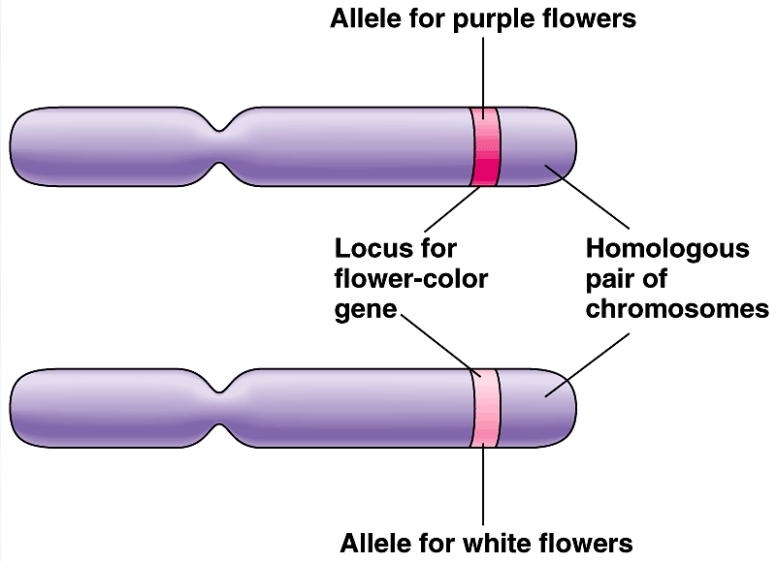
Law of Dominance

MODERN CONCEPT

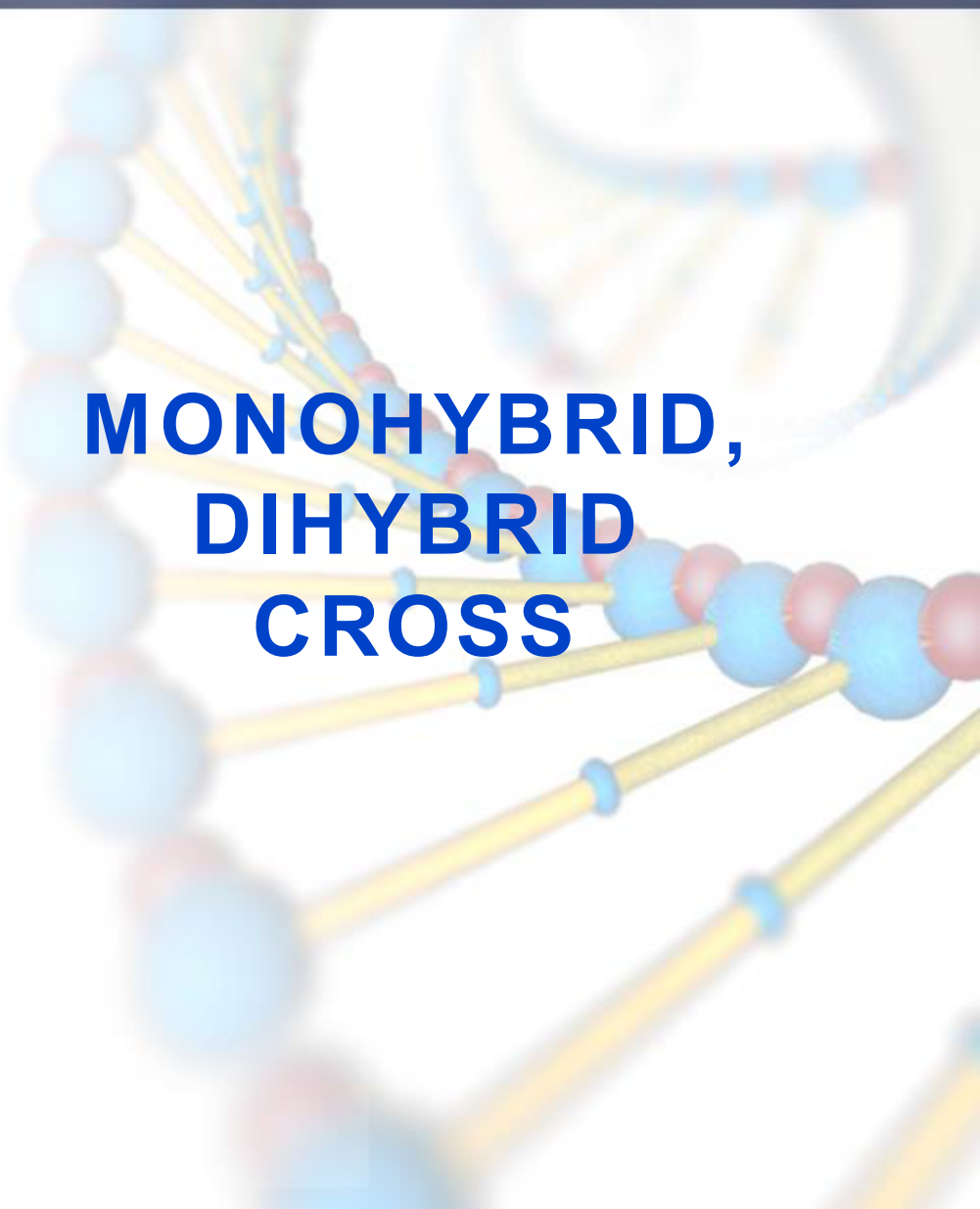
- The gene for a particular inherited character resides at a specific locus (position) on homologous chromosome.
- For each character, an organism inherits two alleles, one from each of the parent.

Law of Dominance

MODERN CONCEPT



Introduction to Genetics



MONOHYBRID, DIHYBRID CROSS

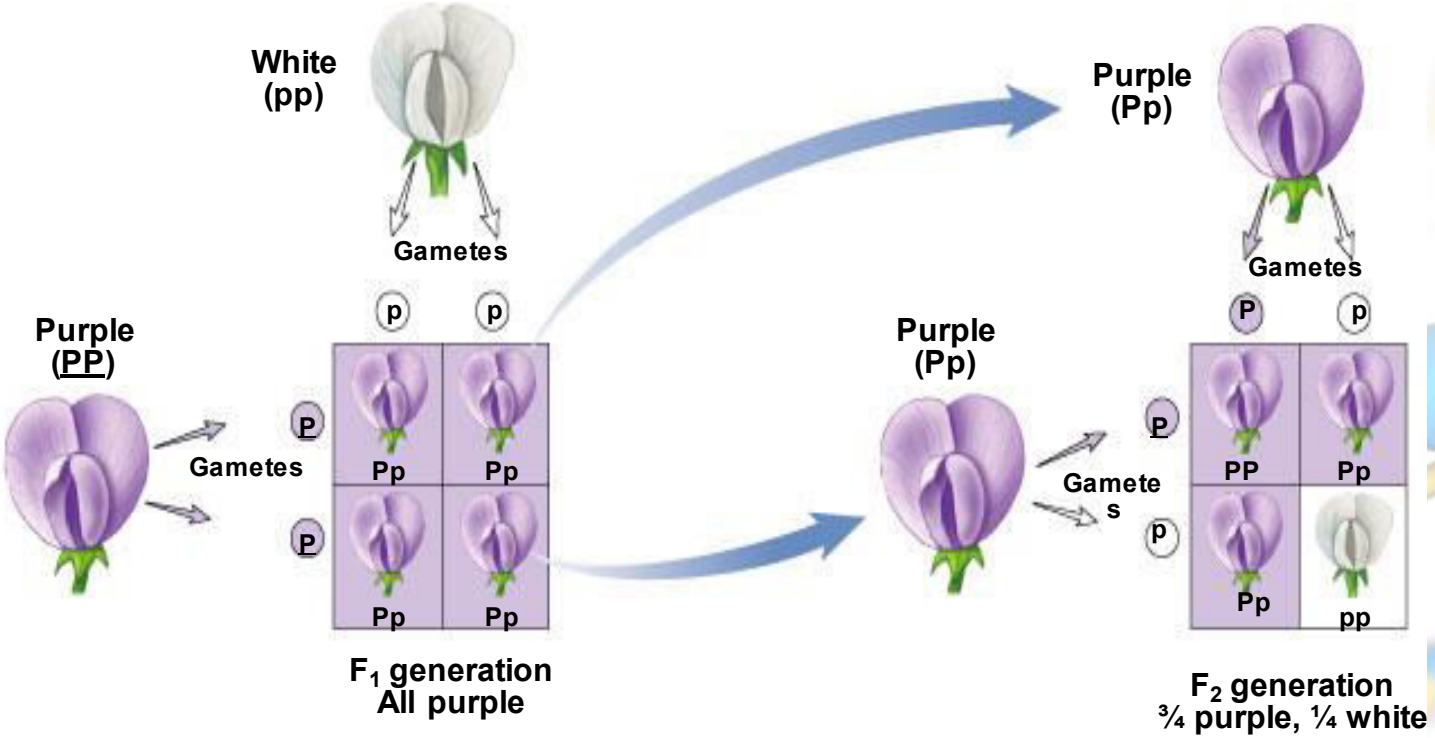
Monohybrid, Dihybrid Cross

MONOHYBRID CROSS

- Cross-fertilization of true-breeding plants which are different in just one character.

Monohybrid, Dihybrid Cross

MONOHYBRID CROSS



Monohybrid, Dihybrid Cross

RESULT OF MONOHYBRID CROSS

- With monohybrid cross, Mendel determined the segregation of alleles at single gene locus.

Monohybrid, Dihybrid Cross

DIHYBRID CROSS HYPOTHESIS

- Dihybrid cross determines that alleles at two different gene loci segregate dependently or independently.
- Segregation is dependent or independent

Monohybrid, Dihybrid Cross

EXPERIMENT

- Seed shape is controlled by one gene while seed color is controlled by a different gene.
- Mendel crossed two pure-breeding plants: one with round/yellow seeds, other with green/wrinkled seeds.

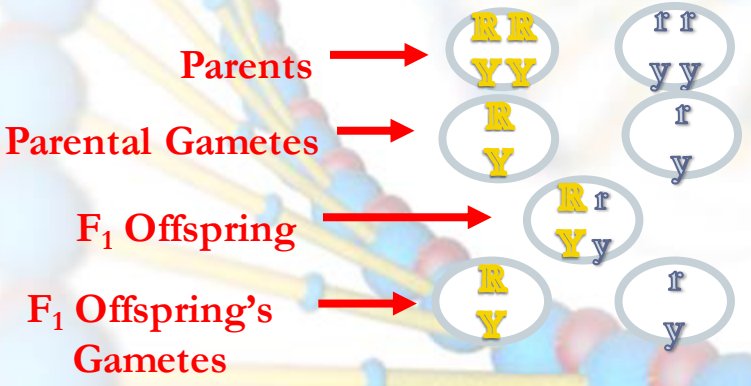
Monohybrid, Dihybrid Cross

DEPENDENT SEGREGATION

- Alleles at the two gene loci segregate together, and are transmitted as a unit.

Monohybrid, Dihybrid Cross

DEPENDENT ASSORTMENT



F2 plants

3 round/yellow :1 wrinkled/green

Monohybrid, Dihybrid Cross

INDEPENDENT ASSORTMENT

- Alleles at the two gene loci segregate independently, and are not transmitted as a unit.

- Plants producing gametes with different allele combinations.

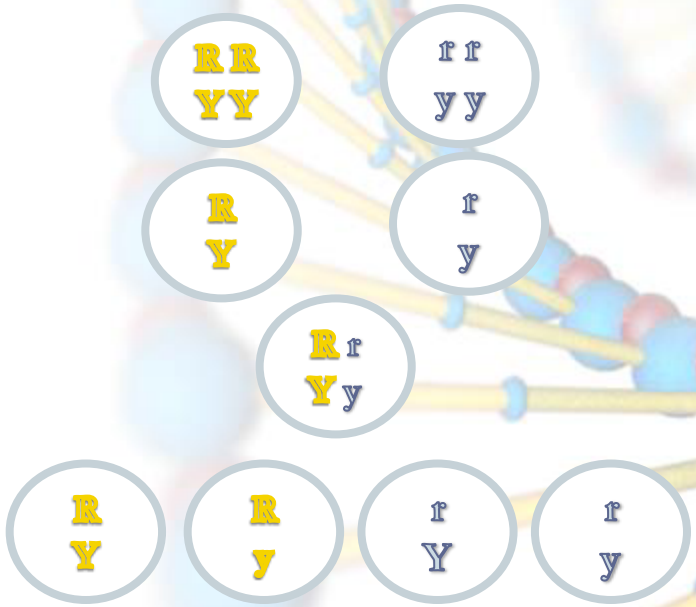
Monohybrid, Dihybrid Cross

Parents

Parental Gametes

F_1 Offspring

F_1 Gametes



Monohybrid, Dihybrid Cross

INDEPENDENT ASSORTMENT

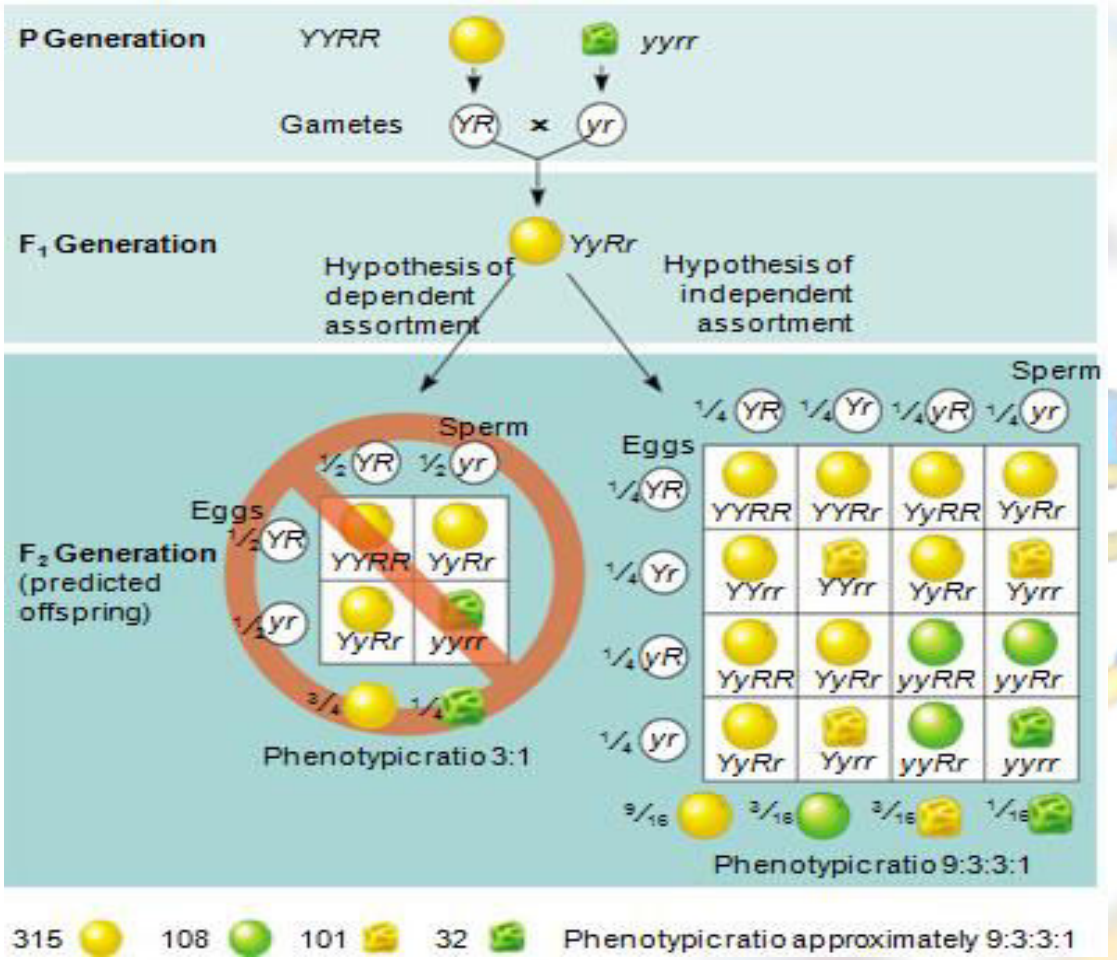
- How are two characters transmitted from parents to offspring?

  As a package?

  Independently?

Monohybrid, Dihybrid Cross

INDEPENDENT ASSORTMENT



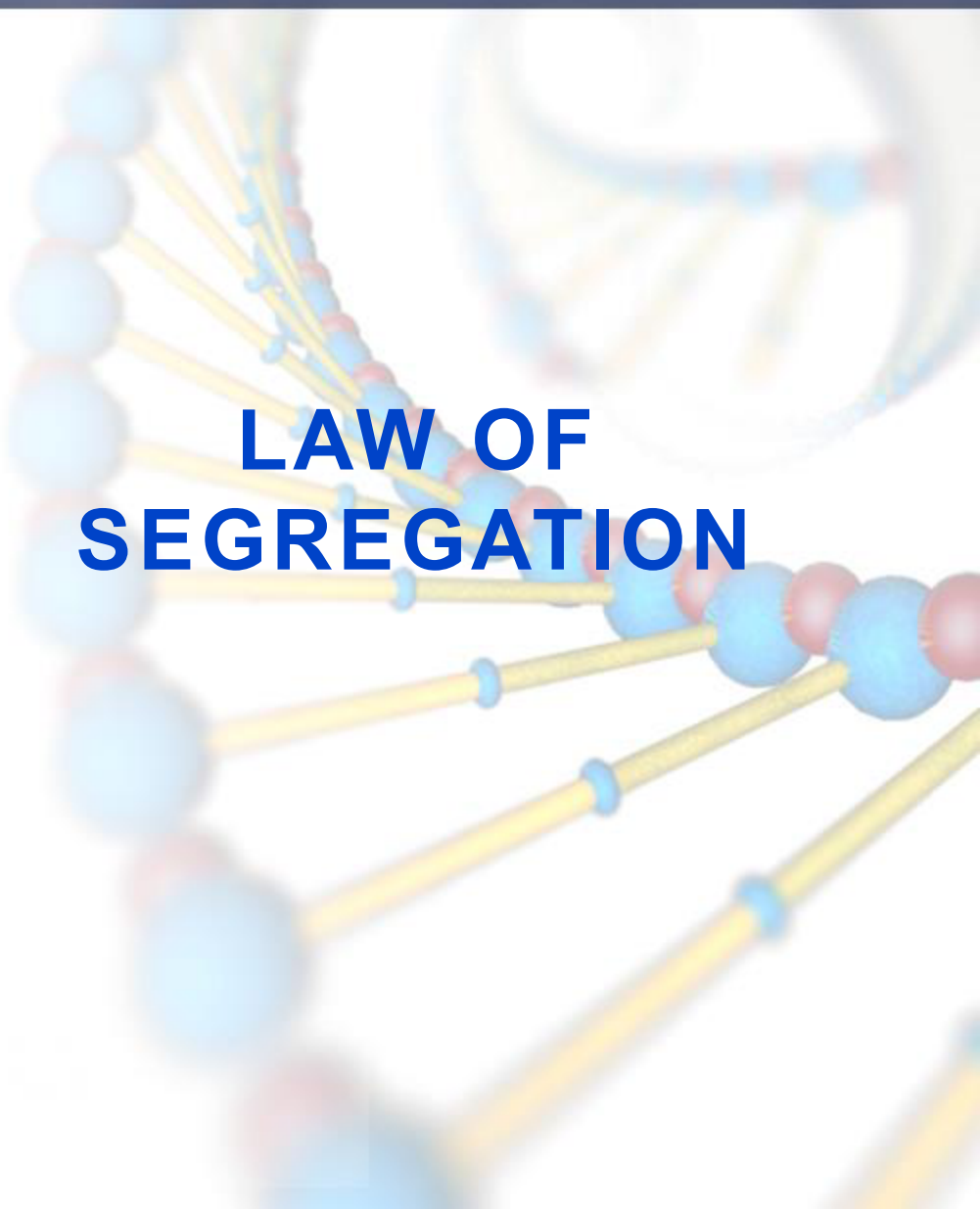
Monohybrid, Dihybrid Cross

MONOHYBRID CROSS AND DIHYBRID CROSS

- Cross fertilization of plants that differ in just one character

- Cross fertilization of plants that differ in two or more characters

Introduction to Genetics



LAW OF SEGREGATION

Law of Segregation

LAW OF SEGREGATION

- During the formation of gametes, the two alleles responsible for a trait separate from each other.
- Alleles are then "recombined" during the process of fertilization.

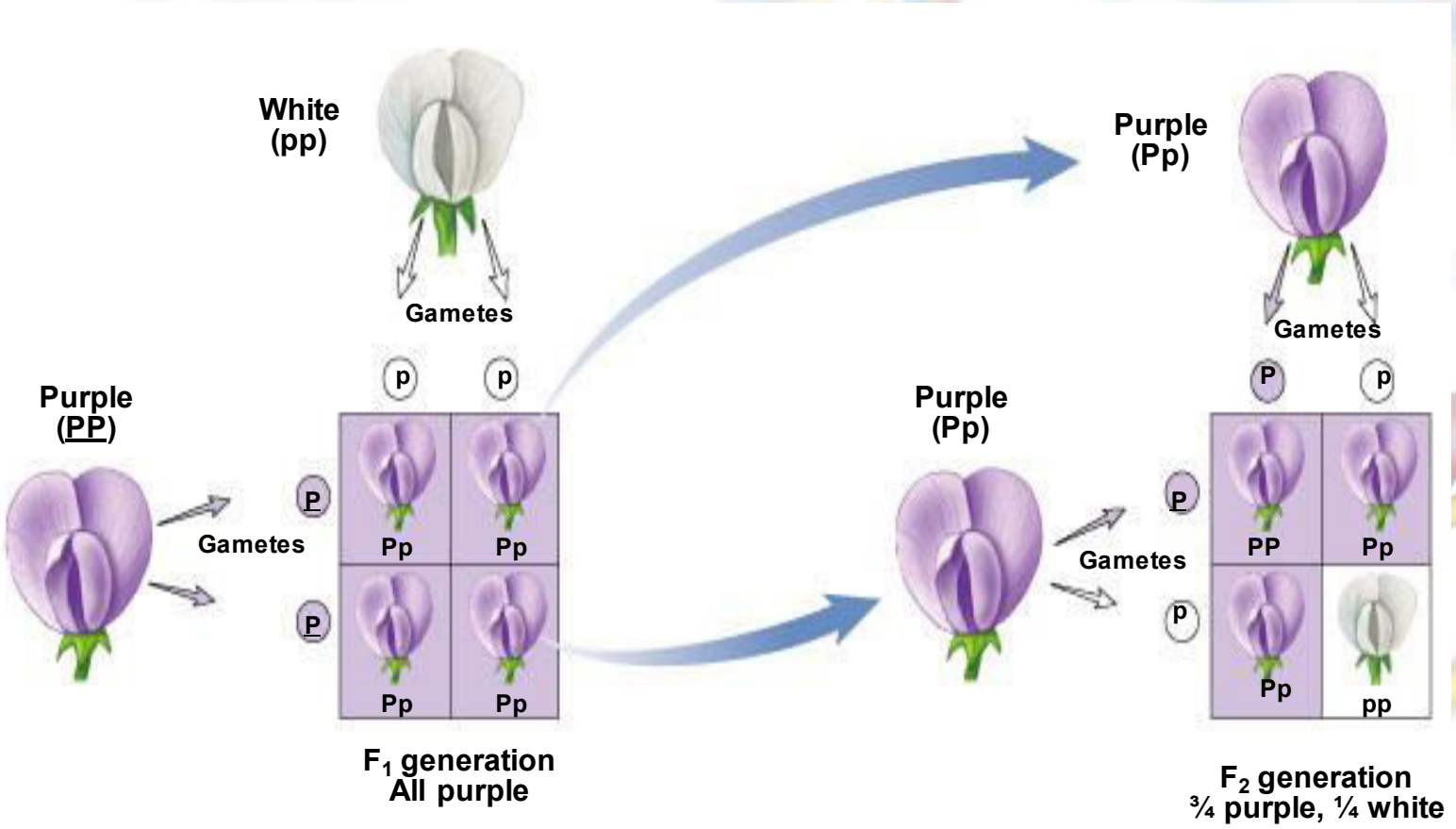
Law of Segregation

EXAMPLE 1

- During a cross, gametes of each parent first segregate.
- Then, gametes combine to give new genotype.
- Resulting phenotype will be according to genotype.

Law of Segregation

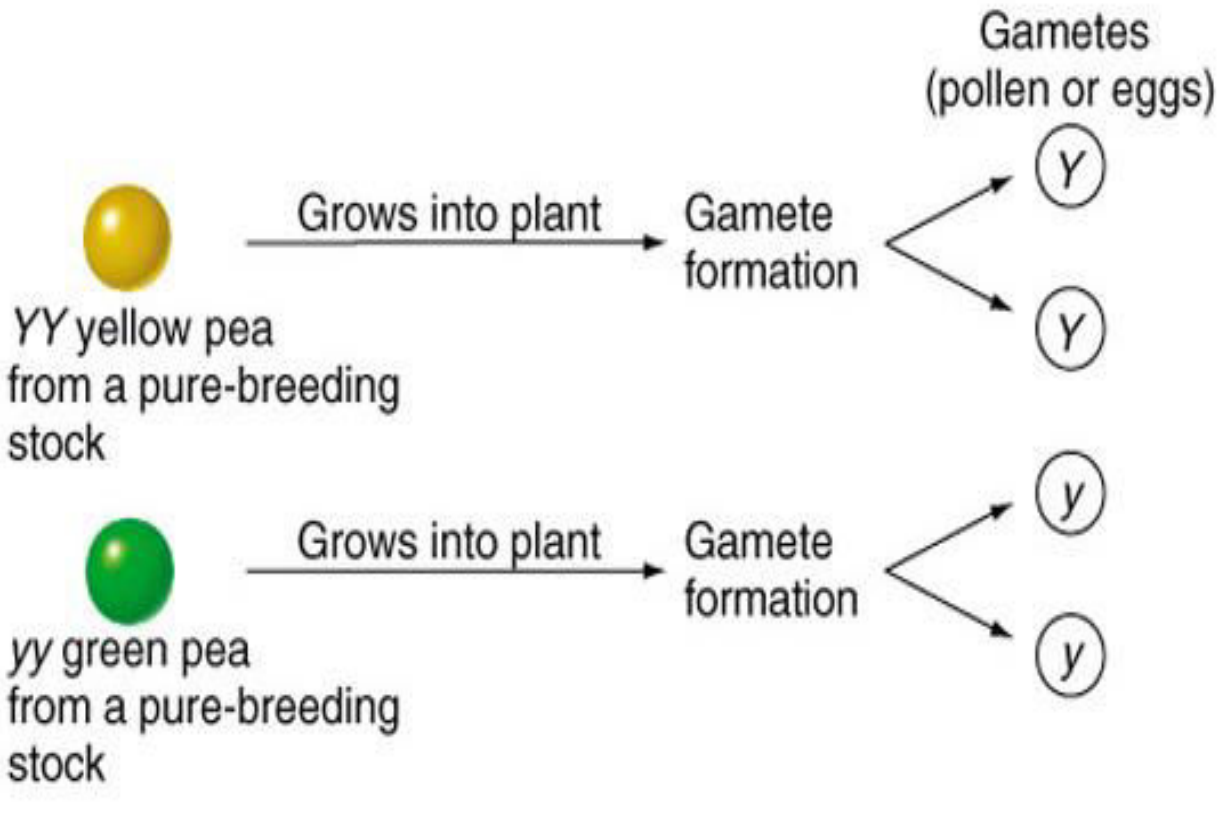
EXAMPLE 1



Law of Segregation

EXAMPLE 2

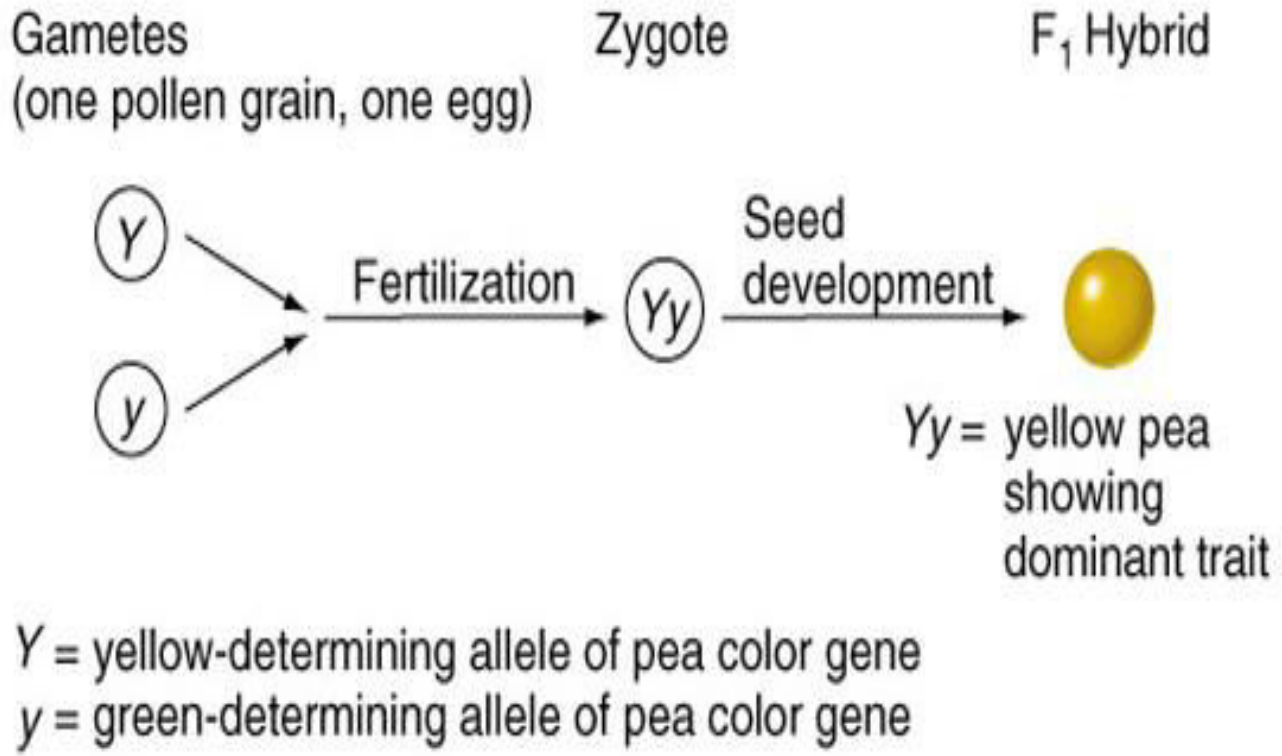
- Gametes formation, where alleles separate.



Law of Segregation

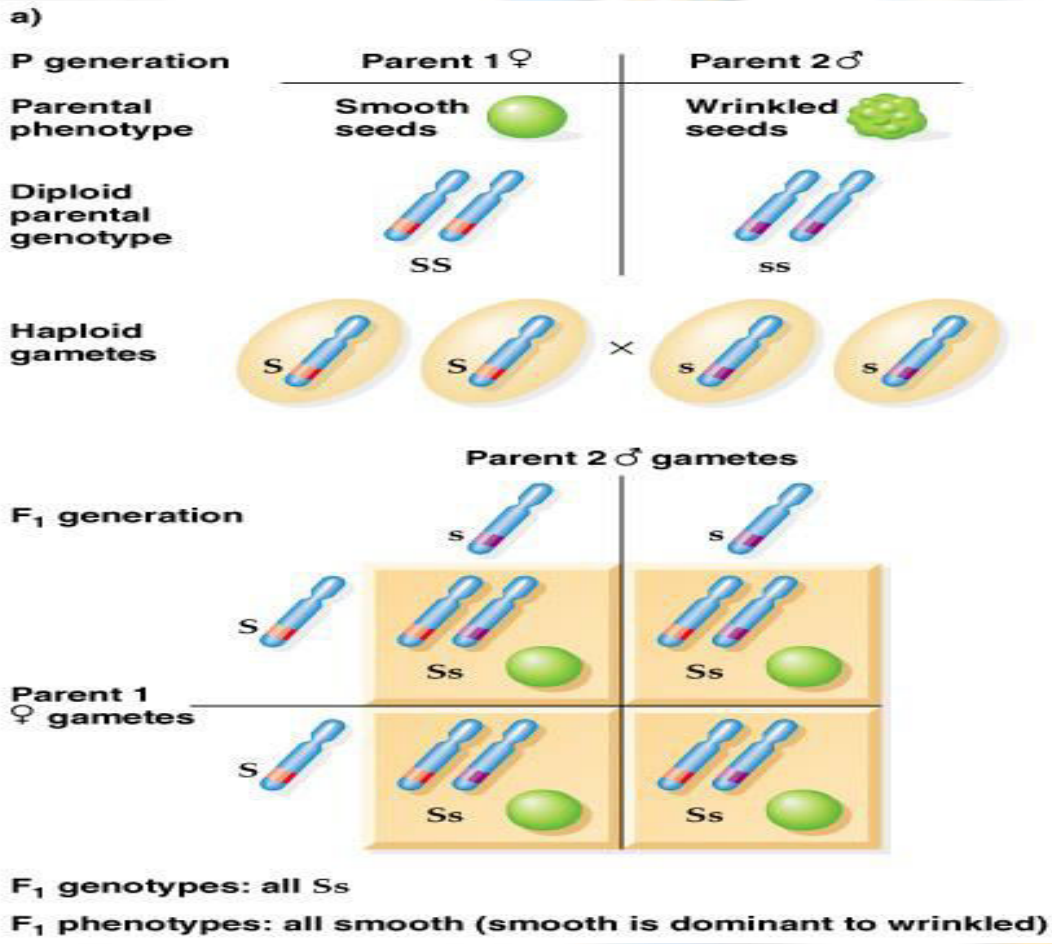
EXAMPLE 2

- During fertilization, gametes re-unite.



Law of Segregation

MODERN CONCEPT



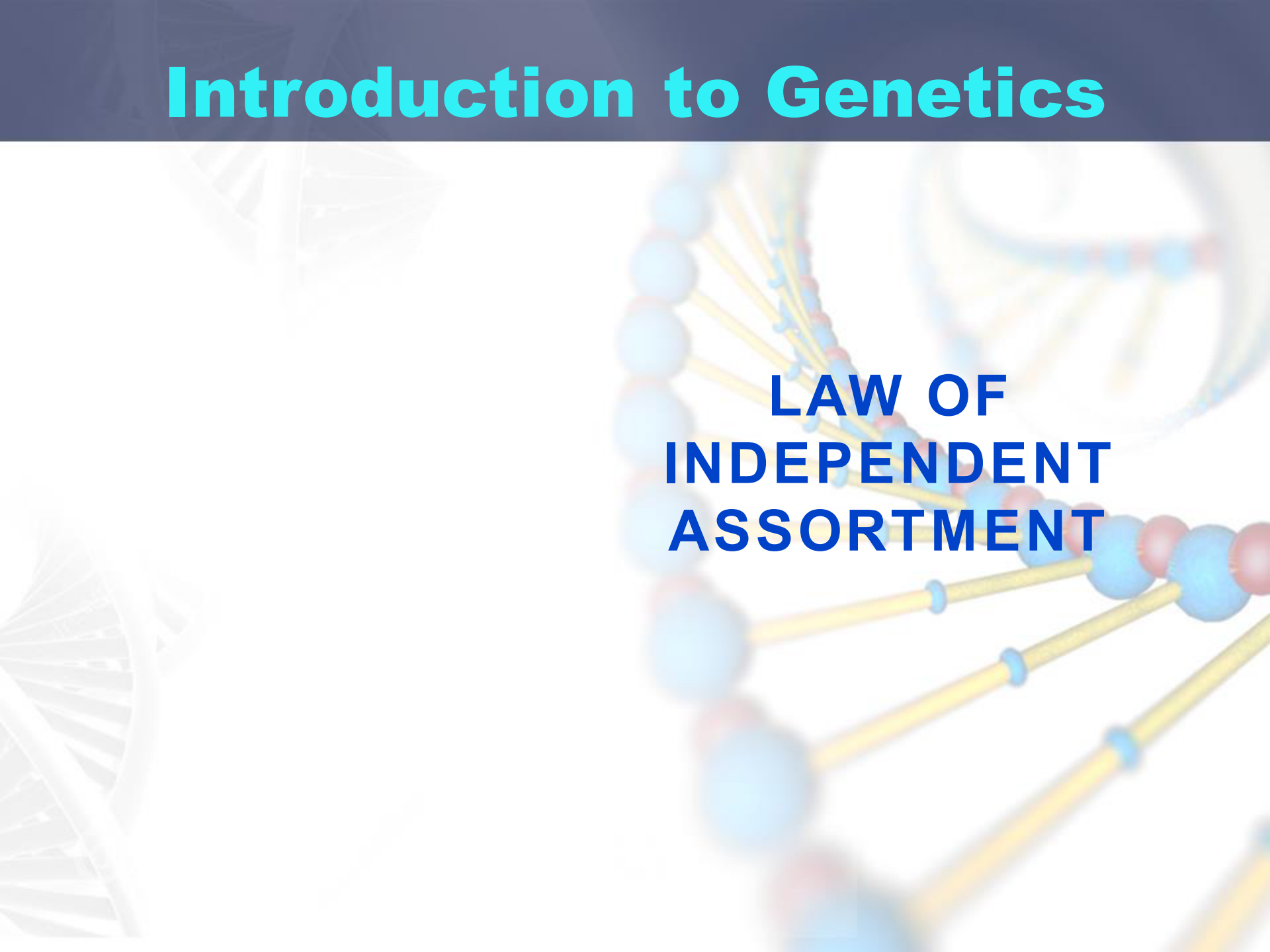
Law of Segregation

LAW OF SEGREGATION STATES

- Every gene has two alleles that can code for a trait.
- Alleles separate during gametogenesis and reunite during fertilization.

Introduction to Genetics

LAW OF INDEPENDENT ASSORTMENT



Law of Independent Assortment

INDEPENDENT ASSORTMENT

- Alleles for different traits are distributed to sex cells independently of one another.
- Traits are transmitted to offspring independently of one another.

Law of Independent Assortment

EXAMPLE

- Dihybrid cross, true-breeding plants for two traits.
- For example, a plant that had green pod color and yellow seed color was cross-pollinated with a plant that had yellow pod color and green seeds.

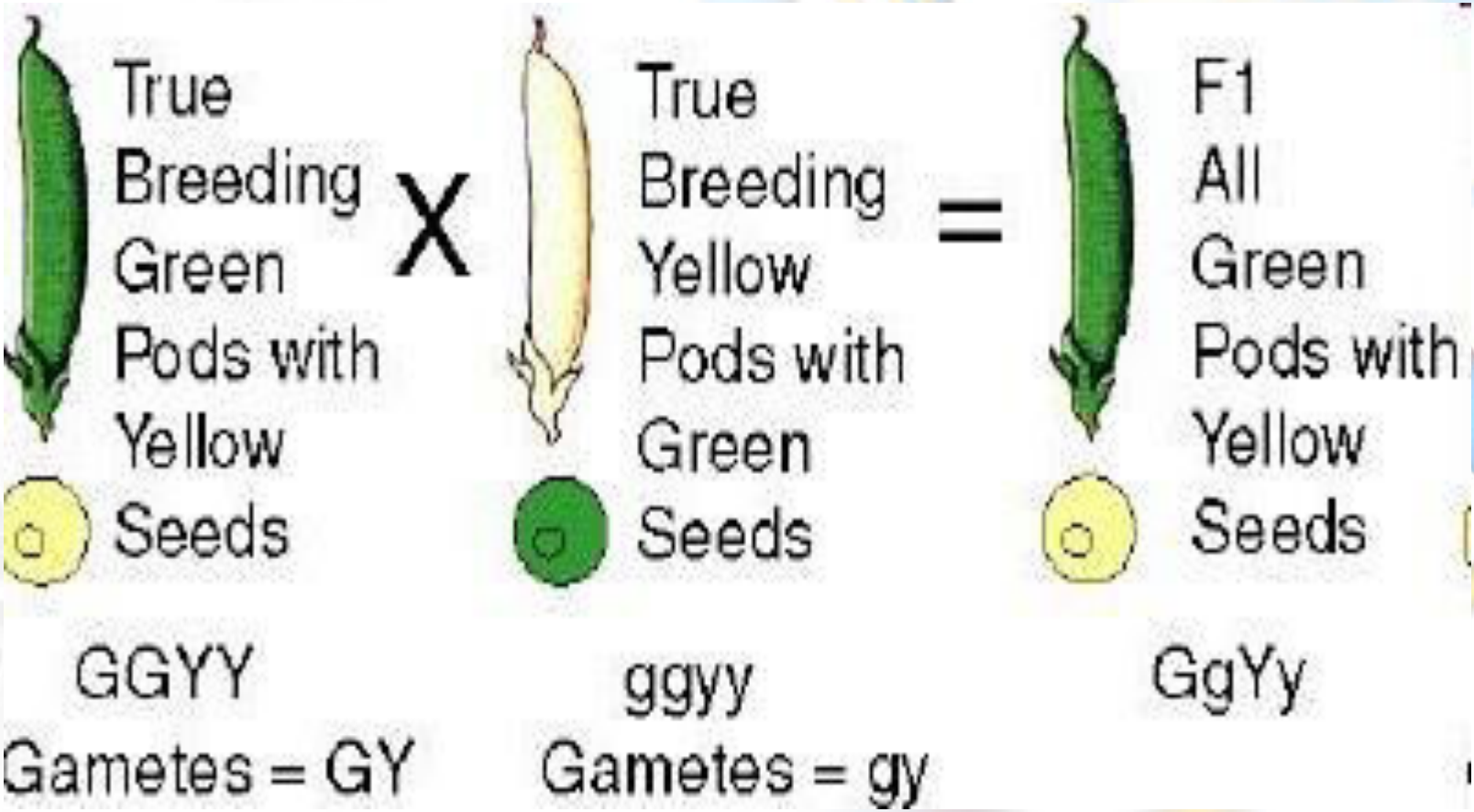
Law of Independent Assortment

EXAMPLE

- The traits for green pod color (GG) and yellow seed color (YY) are dominant.
- Yellow pod color (gg) and green seed color (yy) are recessive.
- F1 plants heterozygous GgYy

Law of Independent Assortment

F1 PLANTS



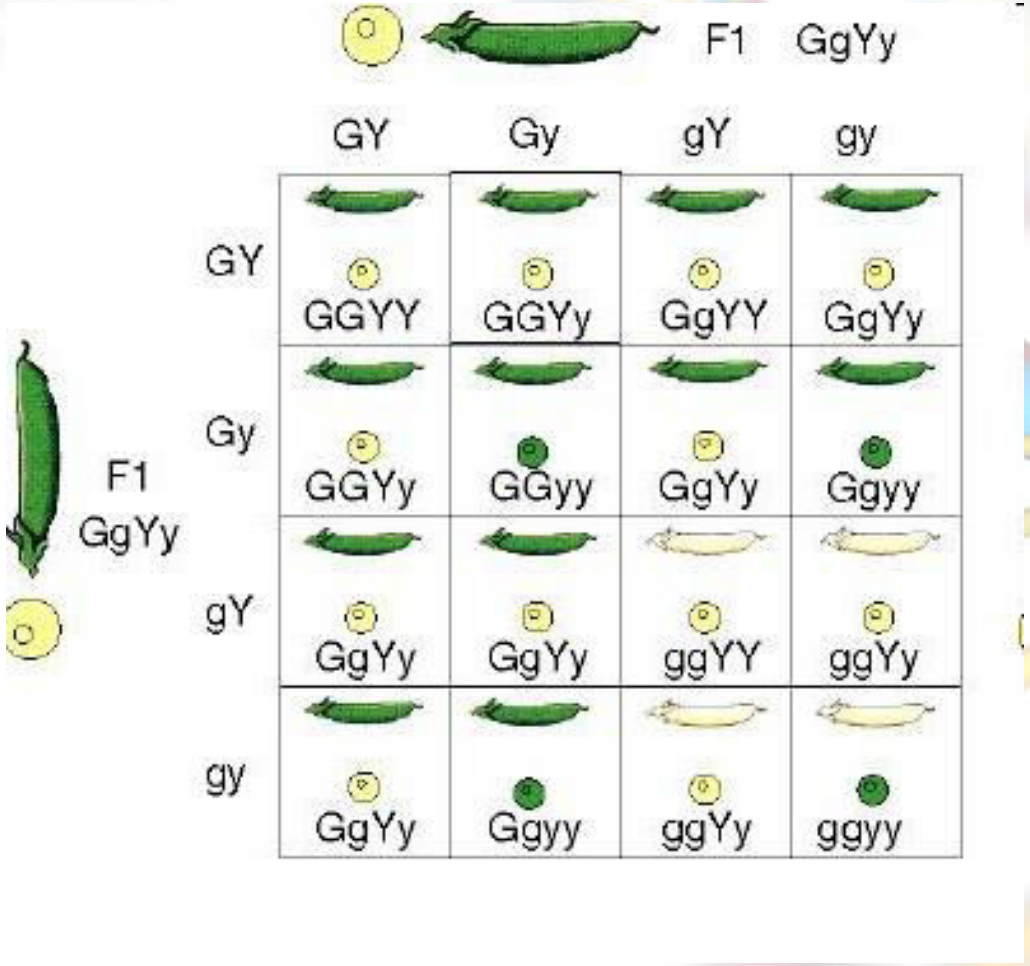
Law of Independent Assortment

F2 PLANTS

- After observing the results of dihybrid cross, Mendel allowed all of the F1 plants to self-pollinate.
- He referred to these offspring as F2 generation.
- Mendel noticed a 9:3:3:1 ratio.

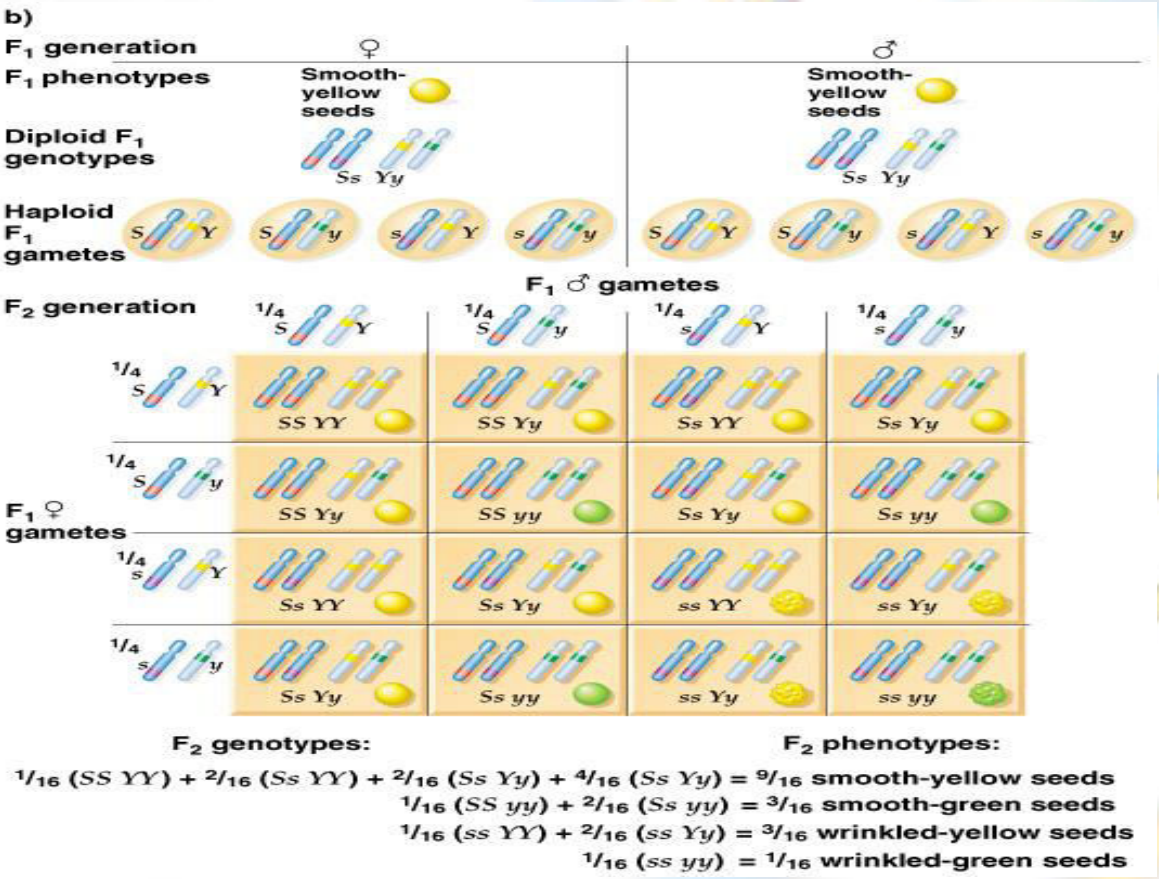
Law of Independent Assortment

GENOTYPES OF F2 PLANTS



Law of Independent Assortment

MODERN CONCEPT

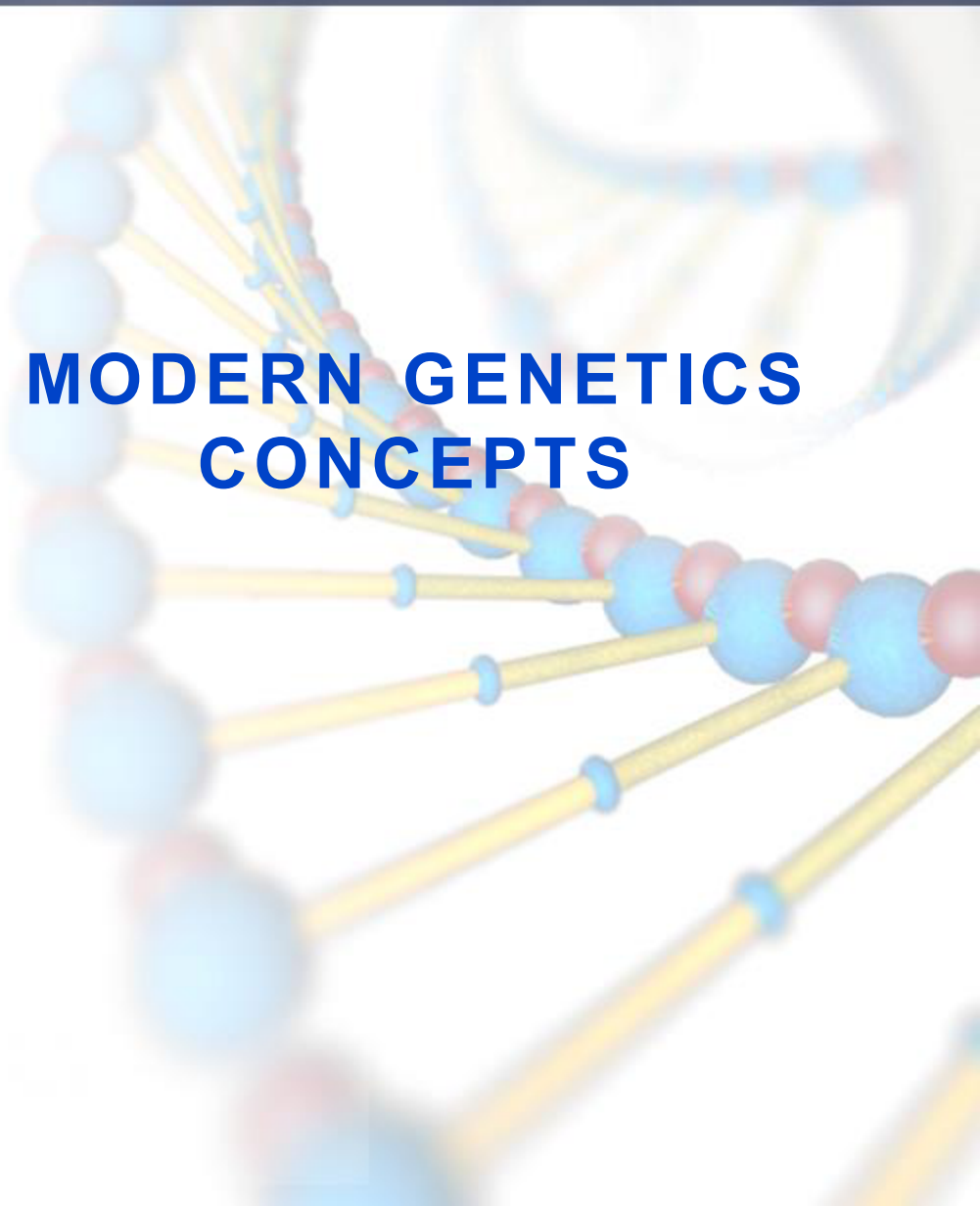


Law of Independent Assortment

CONCLUSION

- The results support the hypothesis of independent assortment.
- The alleles for seed color and seed shape sort into gametes independently of each other.

Introduction to Genetics



MODERN GENETICS CONCEPTS

Modern Genetics Concepts

GENETICS CONCEPTS

- Chromosomes and Gene (s)
- Mendelian Genetics and Non Mendelian Genetics
- Genotypes and Phenotypes
- Mutations and Polymorphism

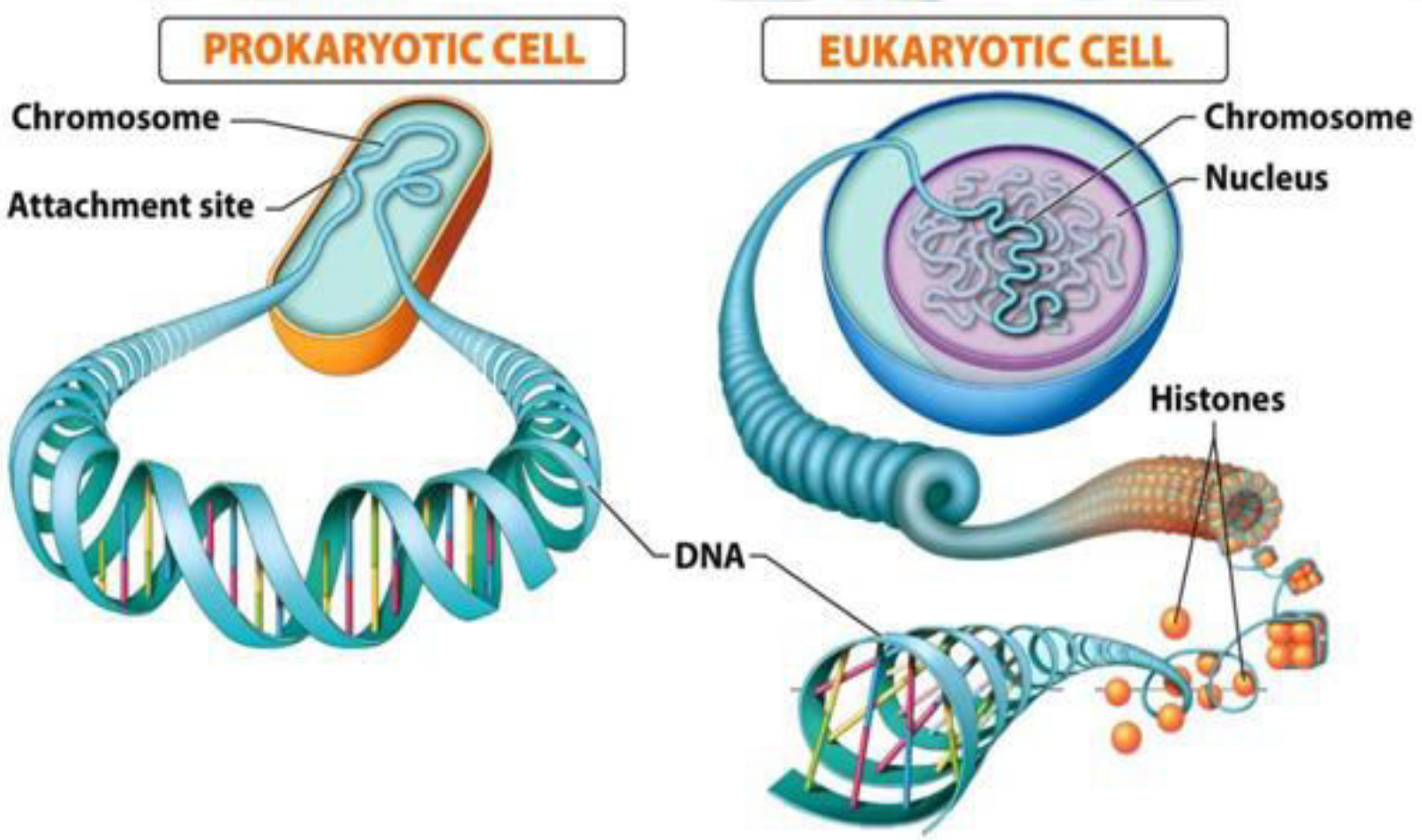
Modern Genetics Concepts

WHAT IS CHROMOSOME

- Genetic material in cells is organized into chromosomes.
- Prokaryotes generally have one circular chromosome.
- Eukaryotes - linear chromosomes in their nuclei.

Modern Genetics Concepts

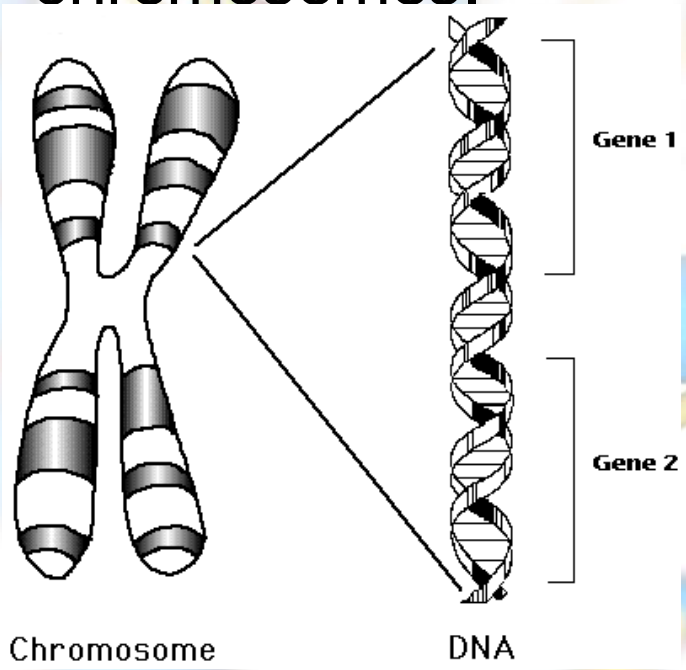
CHROMOSOME



Modern Genetics Concepts

WHAT IS GENE

- A heredity unit on chromosomes.



Modern Genetics Concepts

MENDELIAN GENETICS

- Mendelian genetics - genes that obey Mendel's laws.
- Most genes follow a Mendelian pattern of inheritance, However, there are many that don't.

Modern Genetics Concepts

NON-MENDELIAN GENETICS

- Patterns of inheritance that deviate from a Mendelian pattern.
- Linkage - Non-Mendelian.

  Maternal effect, epigenetic inheritance, Mitochondrial

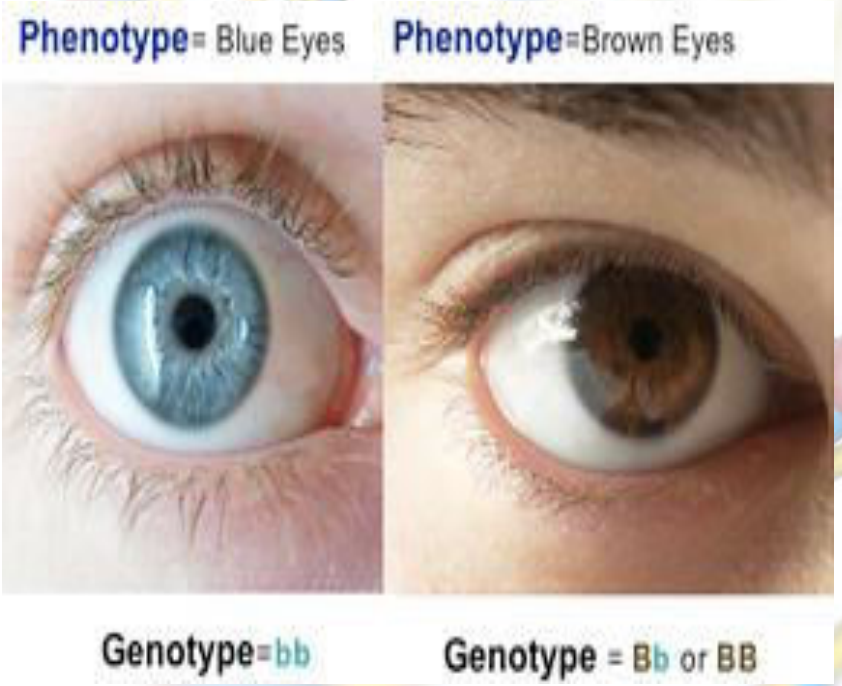
Modern Genetics Concepts

GENOTYPE AND PHENOTYPE

- Genetic make up of an organism.
- Physical appearance of a organism.

Modern Genetics Concepts

Genotype	Phenotype
EE Homozygous dominant	Detached Earlobes 
Ee Heterozygous	Detached Earlobes 
ee Homozygous recessive	Attached Earlobes 



Modern Genetics Concepts

WHAT IS MUTATION

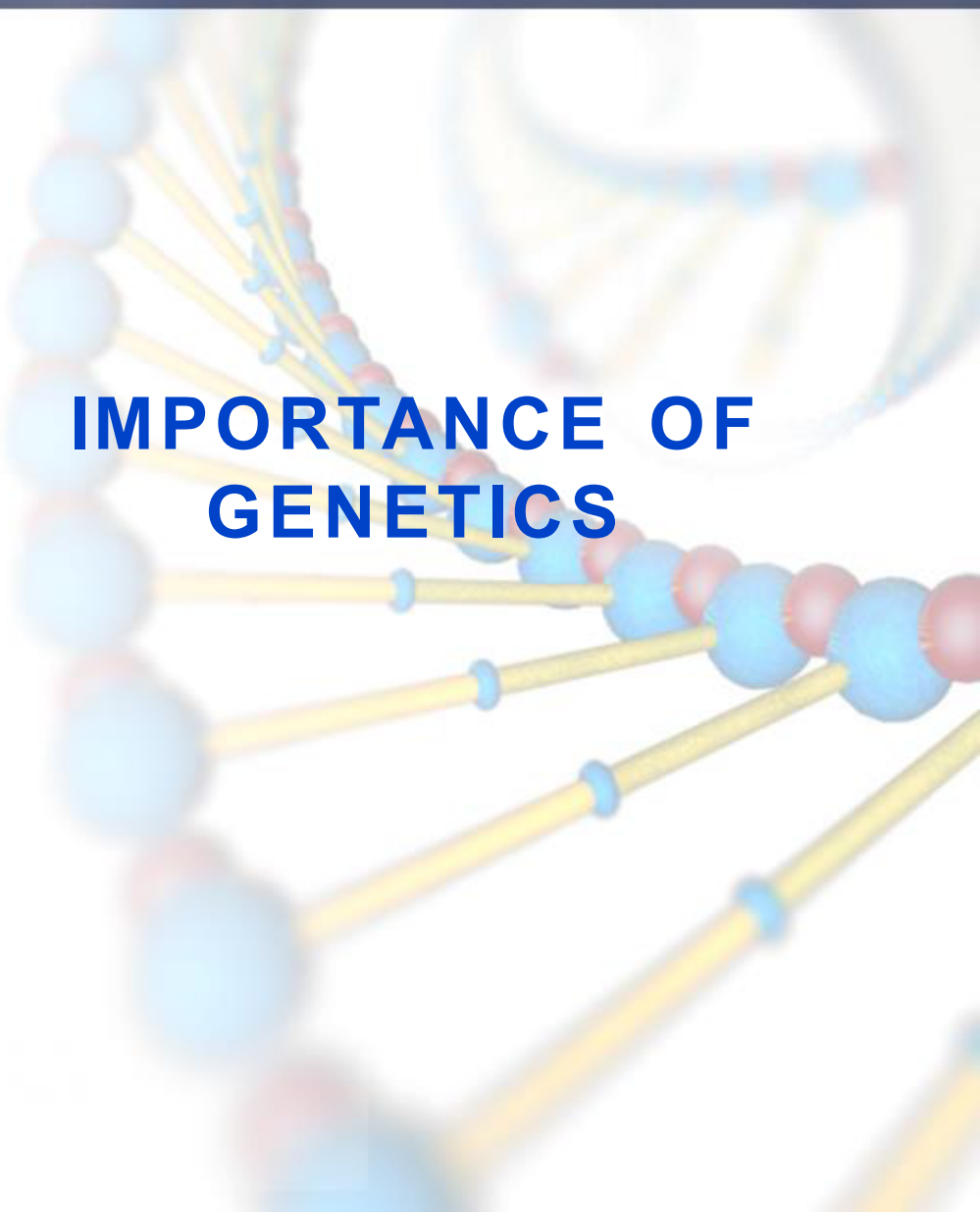
- Indicate "a change" while in other disciplines it is used to indicate "a disease-causing change".
- Mutations may be harmful, beneficial, may have no effect.

Modern Genetics Concepts

POLYMORPHISM

- A DNA polymorphism is 1–2% of a population.
- Polymorphisms can be single base or thousands of bases.
- May or may not have effects.

Introduction to Genetics



IMPORTANCE OF GENETICS

Importance of Genetics

GENETICS IS IMPORTANT TO EVERYONE

- Important to everyone
- All patients
  - Family history
  - Medical history
- Scientists

Importance of Genetics

CHARACTERISTIC OF OUR GENETIC DATA

- Genetic information is
  - Personal
  - Permanent
  - Predictive

Importance of Genetics

GENETICS HELPS US IN MANY WAYS

- Diagnostics and treatments
- Predictive, we can do planning
- Screening (newborn, selected population, carriers)
- Pharmacogenetic (individualized medicine)

Importance of Genetics

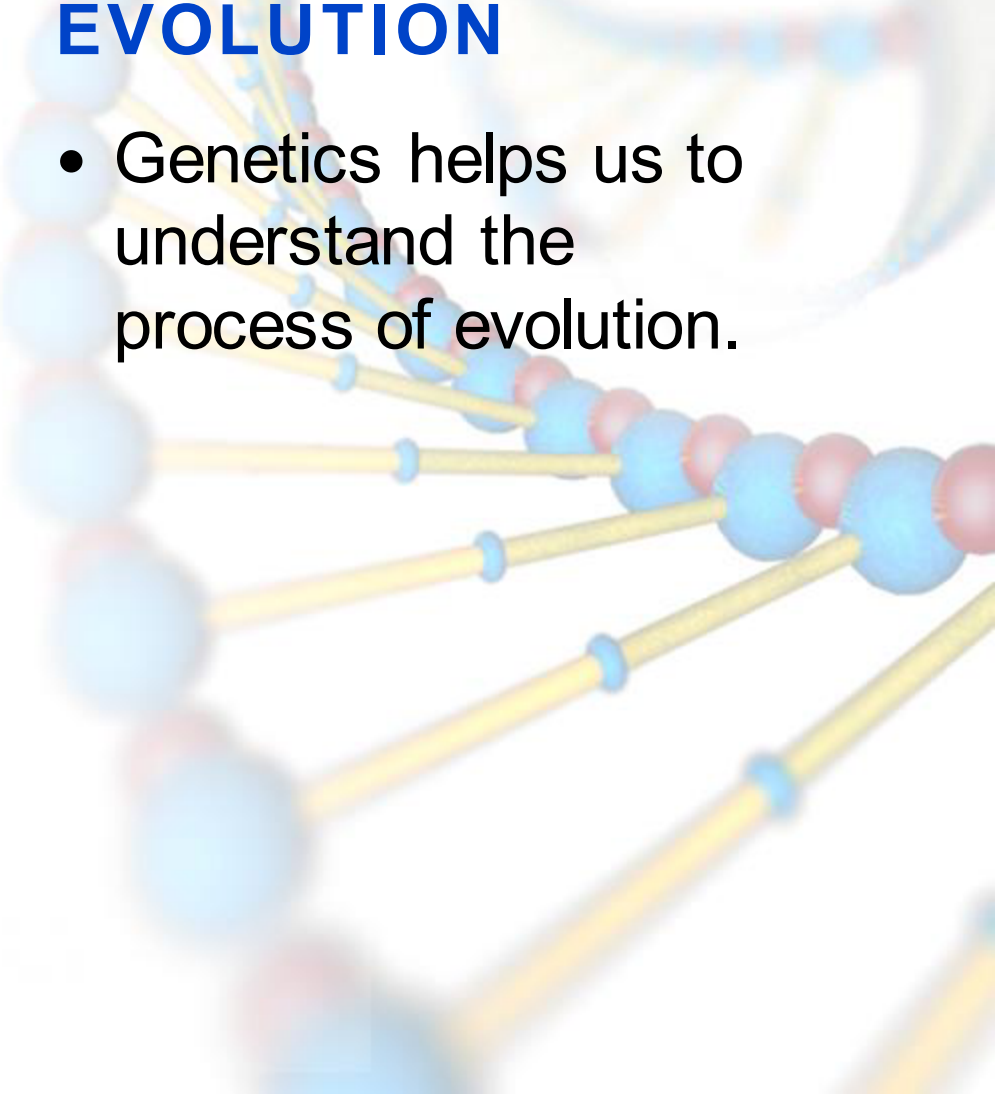
LONG TERM EFFECTS OF GENETICS

- Insurance
- Employment
- Disability
- Medical care
- Lifestyle, Family/children

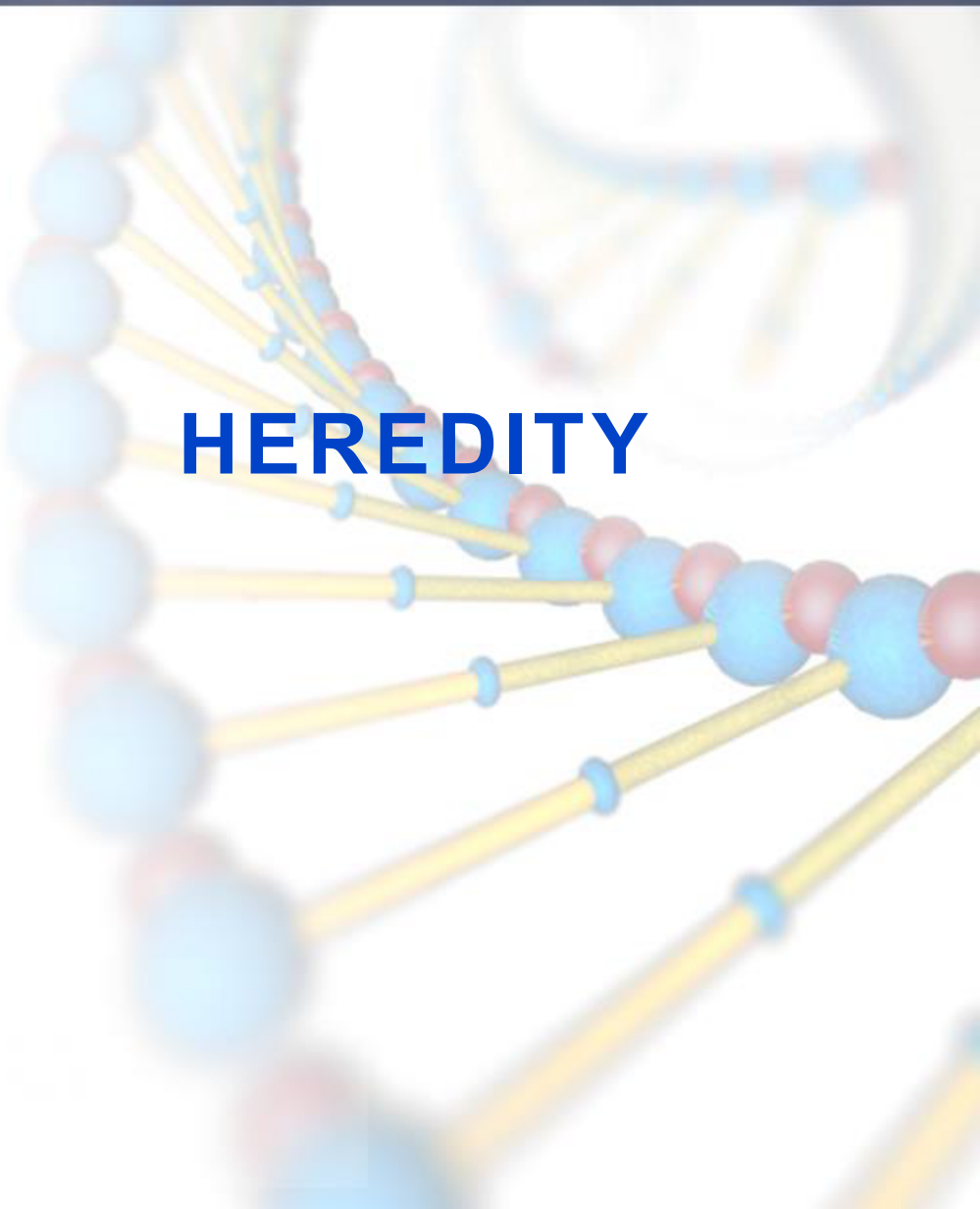
Importance of Genetics

GENETICS AND EVOLUTION

- Genetics helps us to understand the process of evolution.



Heredity and Variation



HEREDITY

Heredity

WHAT IS HEREDITY

- Transmission of traits from parents to offspring.
- Is transmission blending or particulate.

Heredity

IS INHERITANCE BLENDING OR PARTICULATE

- Till 19th century …biologists believed, it was blending.
- Problematic, new genetic variations would quickly be diluted, not passed onward.

Heredity

BLENDING HYPOTHESIS

- According to blending hypothesis, genetic material of the parents mixes during fertilization.
- Blue and yellow paints blend to make green paint.

Heredity

BLENDING HYPOTHESIS DISCREDITED

- Blending inheritance was not appealing and quickly discredited by Gregor Mendel's experiments, which showed that inheritance is particulate.

Heredity

PARTICULATE HYPOTHESIS

- According to particulate hypothesis, parents pass on distinct heritable units.
- Those distinct heritable units are now called as genes.

Heredity

PARTICULATE INHERITANCE

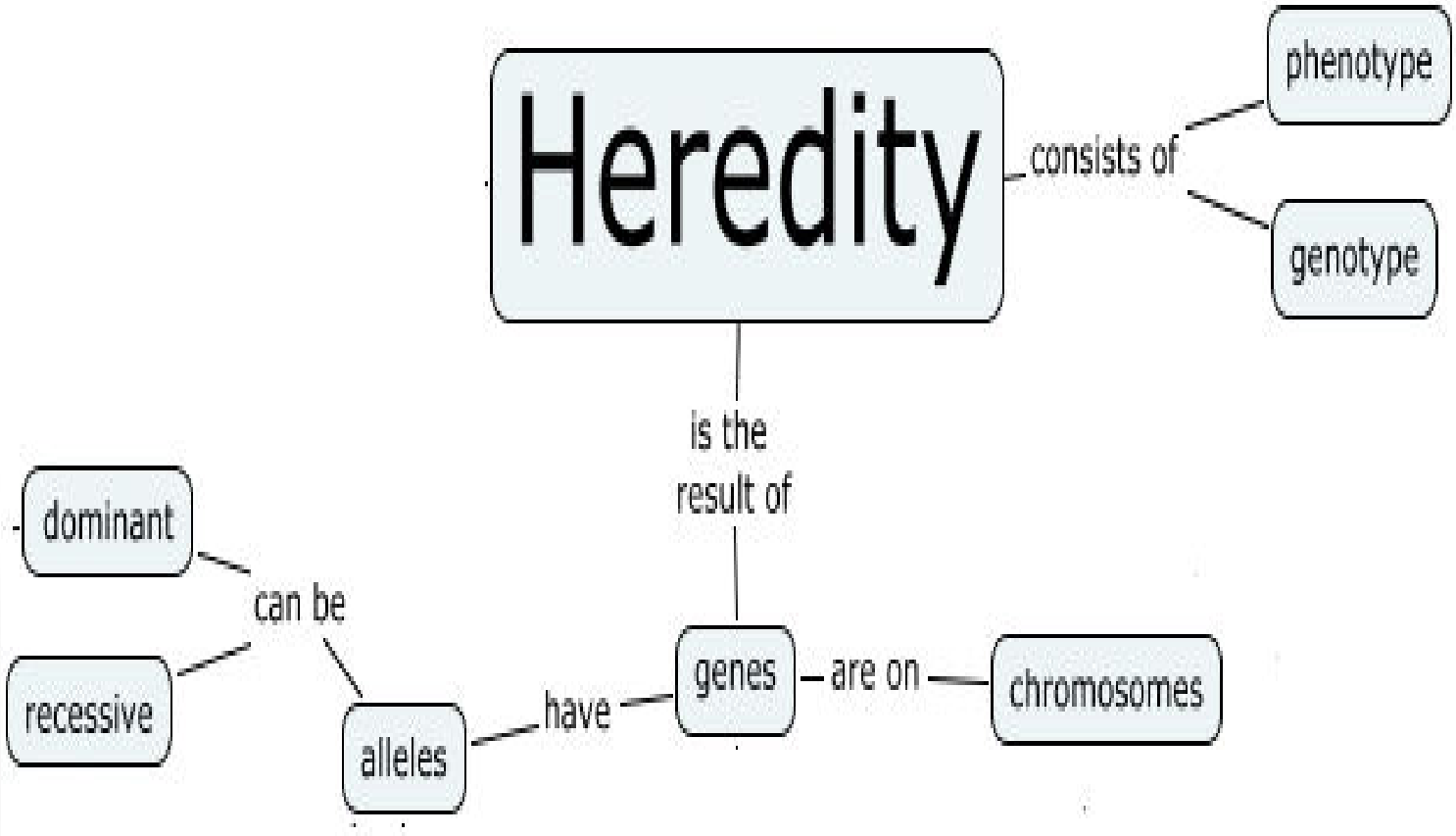
- Mendel showed characters, or what we now call alleles, were inherited unchanged.
- Pattern of inheritance of these characters gave us the first definition of a gene.

Heredity

MENDEL AND PARTICULATE INHERITANCE

- Mendelian inheritance patterns involve genes that obey Mendel's laws. Most genes follow a Mendelian pattern of inheritance, However, there are many that don't.

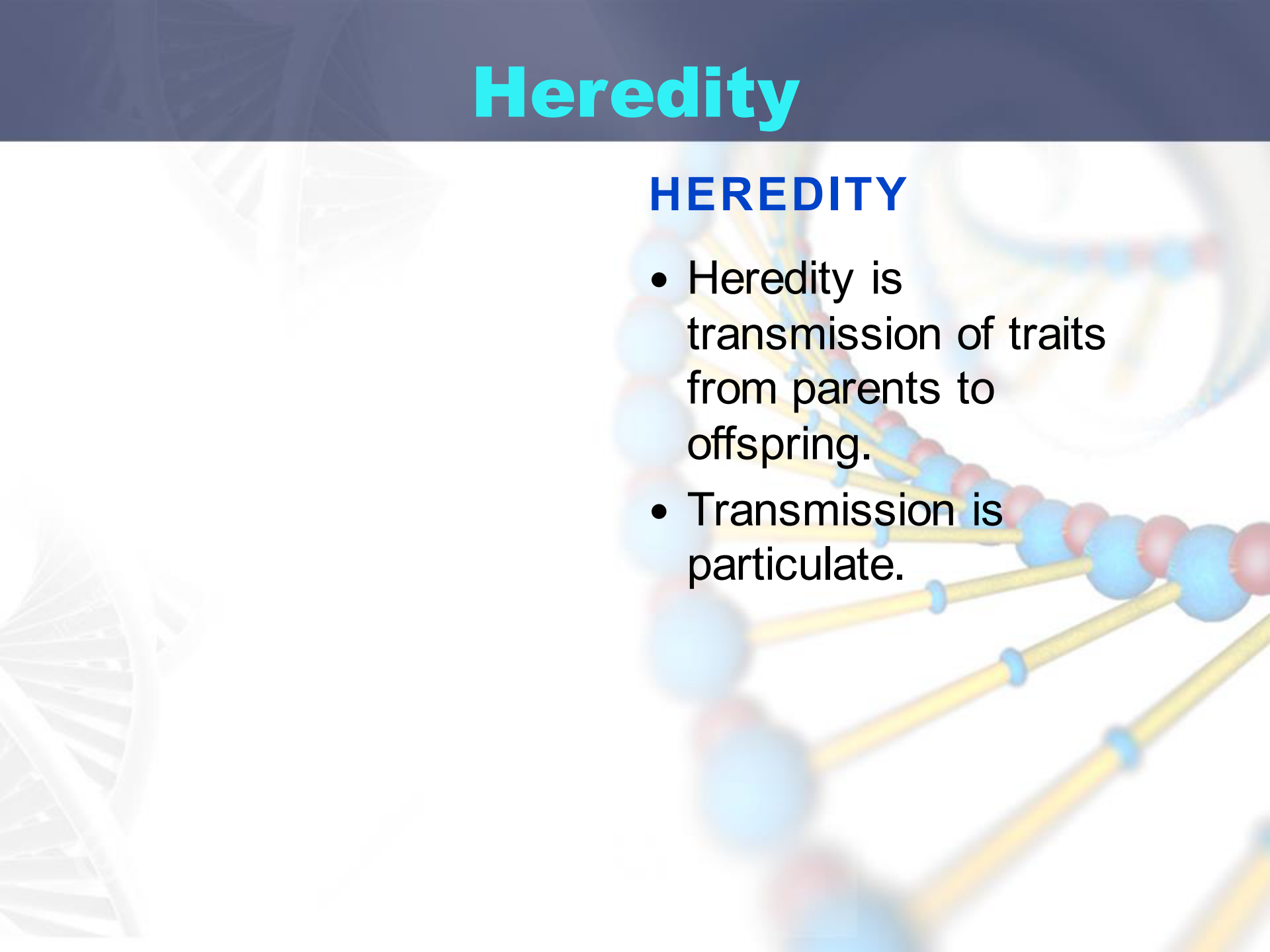
Heredity


Heredity
consists of
phenotype
genotype
is the result of
dominant
can be
recessive
alleles
have
genes
are on
chromosomes

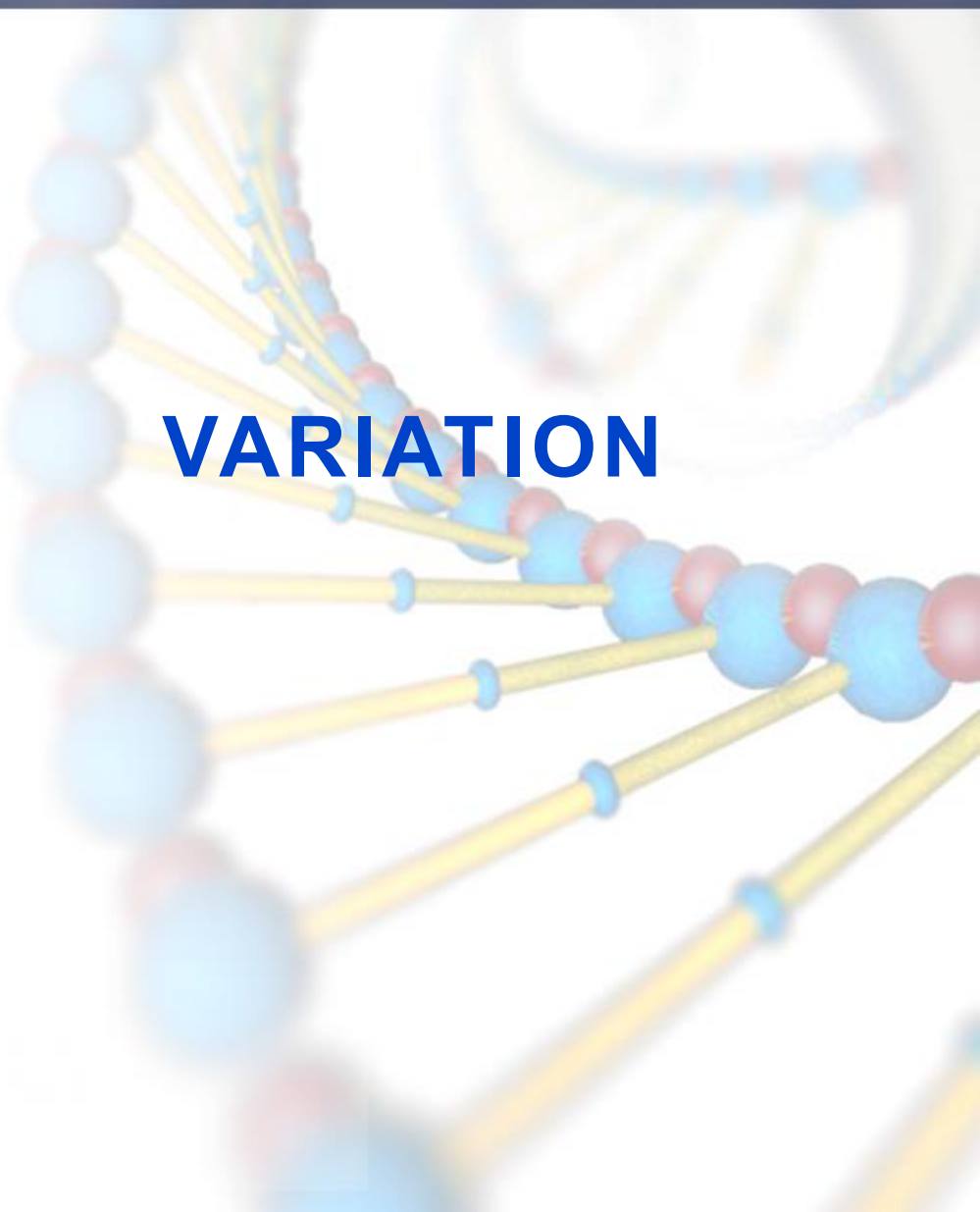
Heredity

HEREDITY

- Heredity is transmission of traits from parents to offspring.
- Transmission is particulate.



Heredity and Variation



VARIATION

Variation

CONCEPT OF VARIATION

- Differences can be seen in individuals of the same species, such as height, weight, color etc.
- Differences due to influence of genotype and environment.

Variation

SNAIL SHELLS

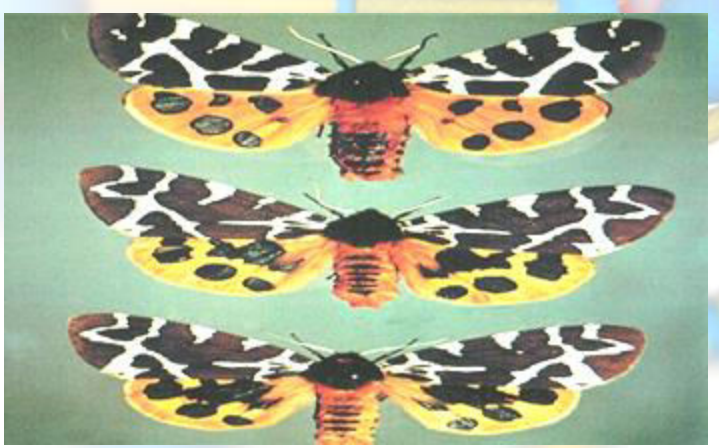
- Their color, presence/absence of bands, number of bands etc



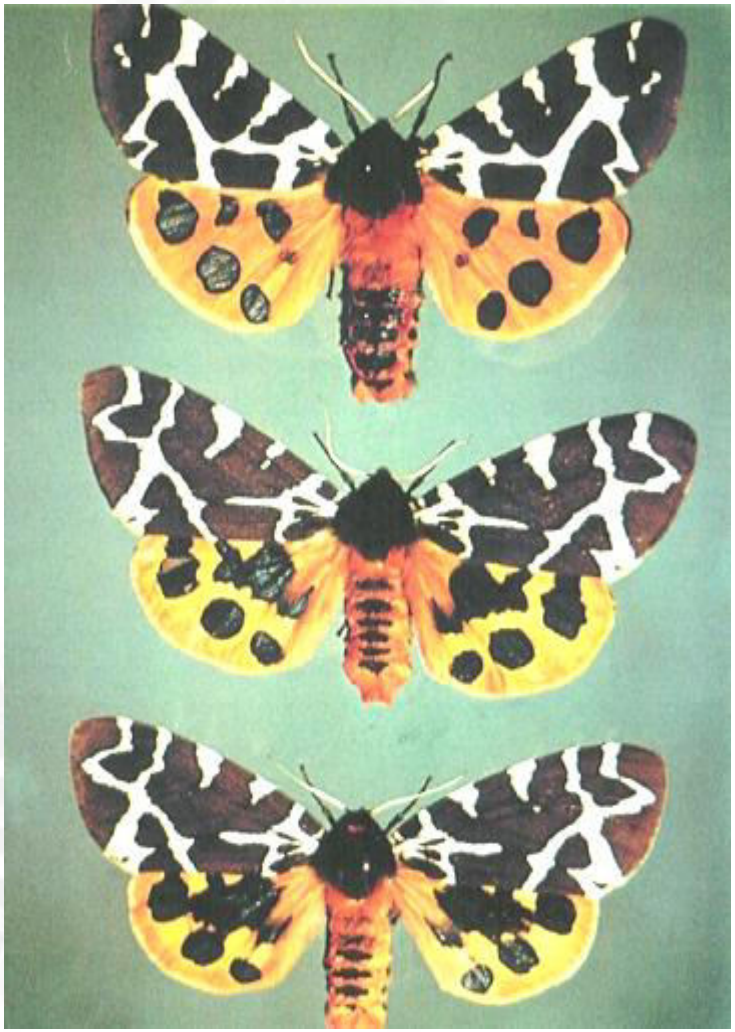
Variation

TIGER MOTHS

- These tiger moths are from the same family. What variations can you see ?



Variation



EXAMPLE - TIGER MOTHS

- Uppermost moth larger;
- Dark bands on abdomen differ in numbers and thickness;
- Patterns of forewings are quite different from each other;
- Number, shape distribution of black spots on hind wings.

Variation

VARIATION IN HUMANS

- You will already be familiar with many of the variations between individuals - humans (Homo sapiens).

Variation



VARIATIONS IN HUMANS

- Variations in skin color
- Variation in hair color
- Variation in hair curliness
- Variation in eye color
- Variation in sex

Variation

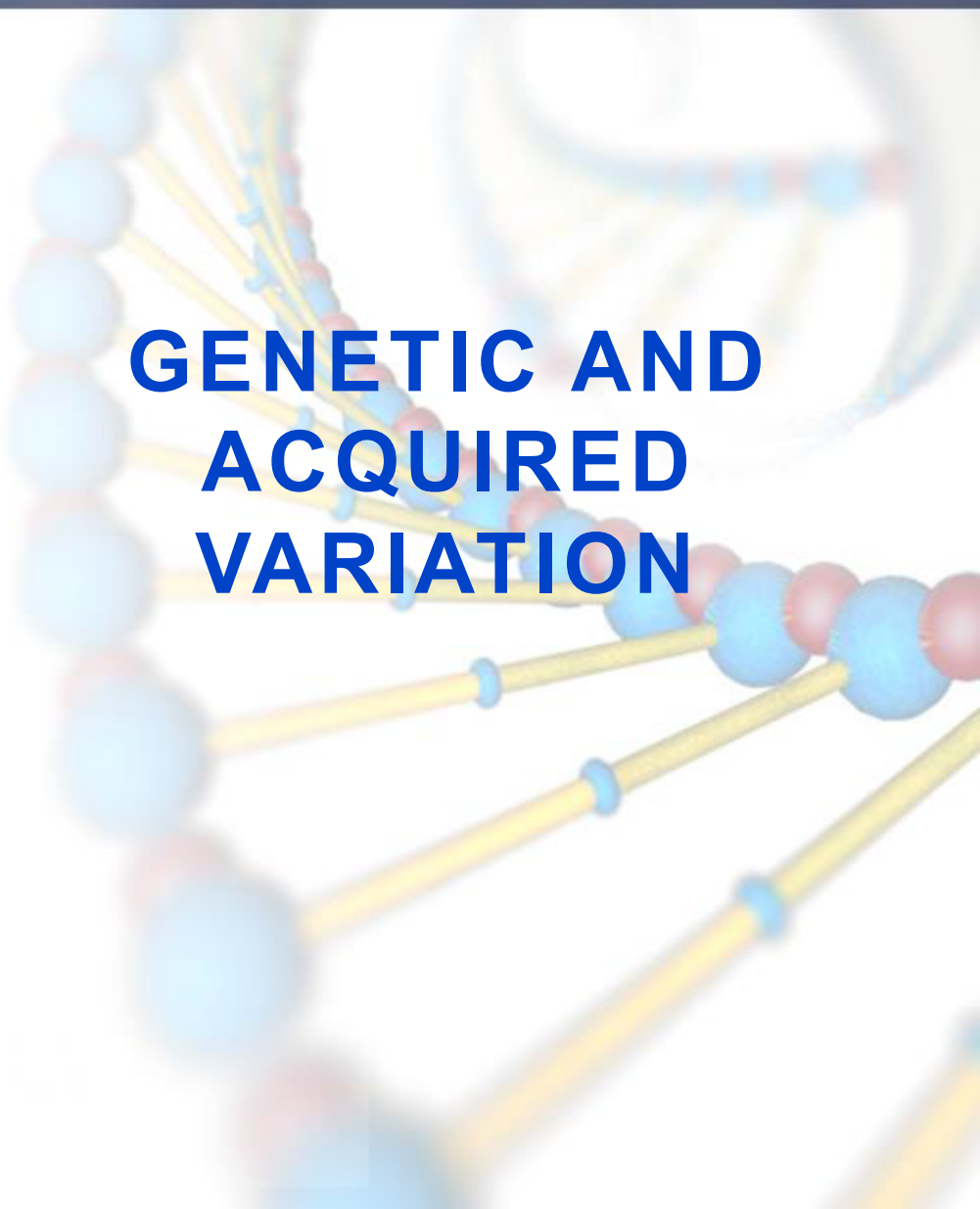
CONCLUSION

- It can be concluded that variation can seen among individuals of same species.

- It can be due to genetic makeup, environment or by combination of both.

Heredity and Variation

GENETIC AND ACQUIRED VARIATION



Genetic & Acquired Variation

INHERITED VARIATIONS

- Inherited variations result from the activity of genes.
- Genetically controlled and cannot be altered.
- Blood groups, finger prints and sex etc.

Genetic & Acquired Variation

ACQUIRED VARIATION

- Acquired variation result from an individual's activities or nutrition or from environment conditions during a lifetime.

Genetic & Acquired Variation

ACQUIRED VARIATION

- Acquired characteristics can not be inherited. Language, obesity, athletic skills etc

Genetic & Acquired Variation

APPLES

- Different position on same tree

North side, upper branches

South side, upper branches



North side, lower branches

South side, lower branches

Genetic & Acquired Variation

ACQUIRED VARIATION

- The differences between the apples are acquired during the growing season.

- The differences will not be inherited.

Genetic & Acquired Variation

WHY THIS VARIATION

- Environmental conditions caused the differences in size of the apples ?

- The upper branches more sunlight than lower branches.

Genetic & Acquired Variation

VARIATION BY GENETIC AND ENVIRONMENT

- Many variations - both genetic and environmental.
- Height depend on what genes you inherit and the amount of food you get.

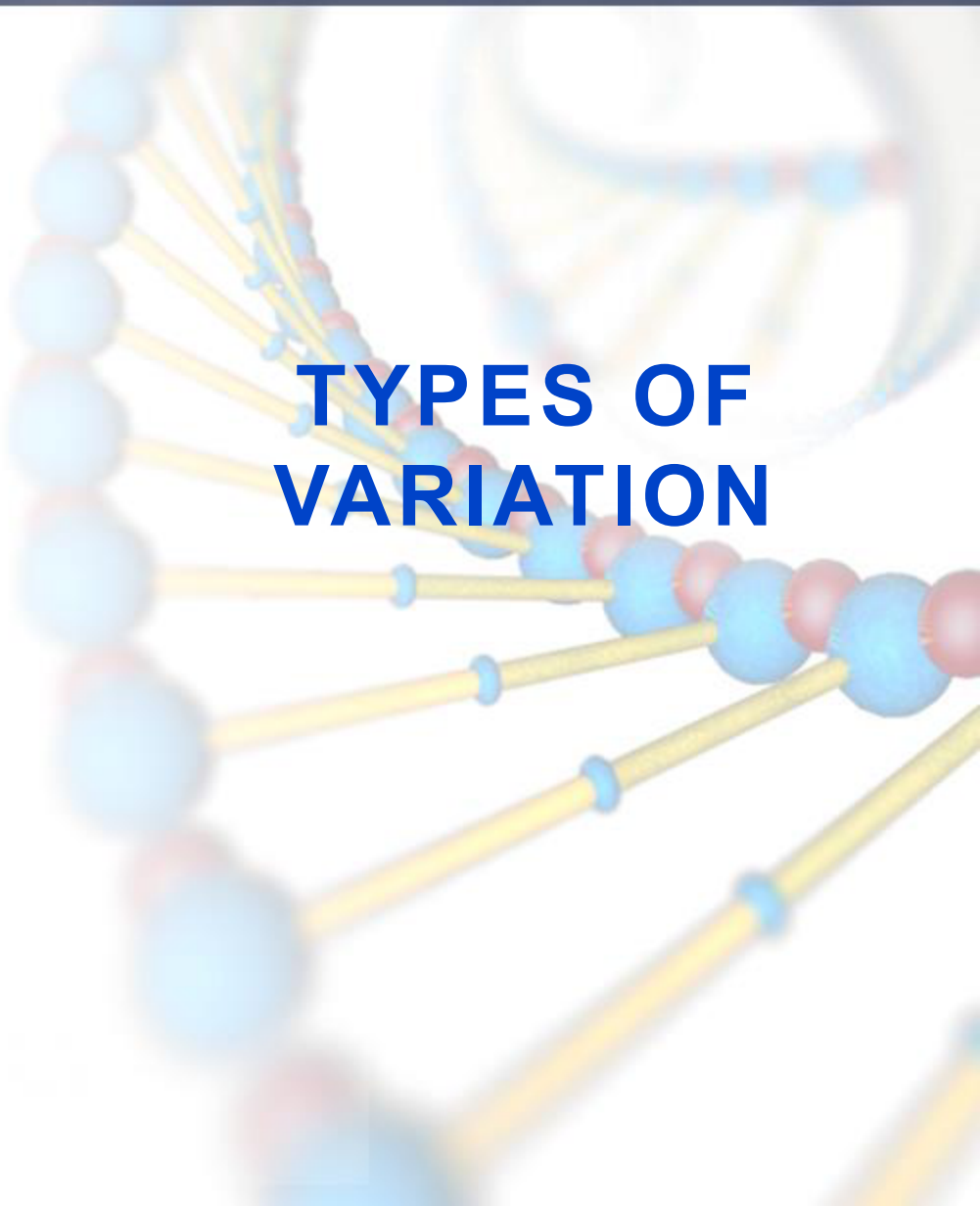
Genetic & Acquired Variation

CONCLUSION

- Variation can be seen among individuals of same species.
- It can be due to genetic makeup, environment or by combination of both.

Heredity and Variation

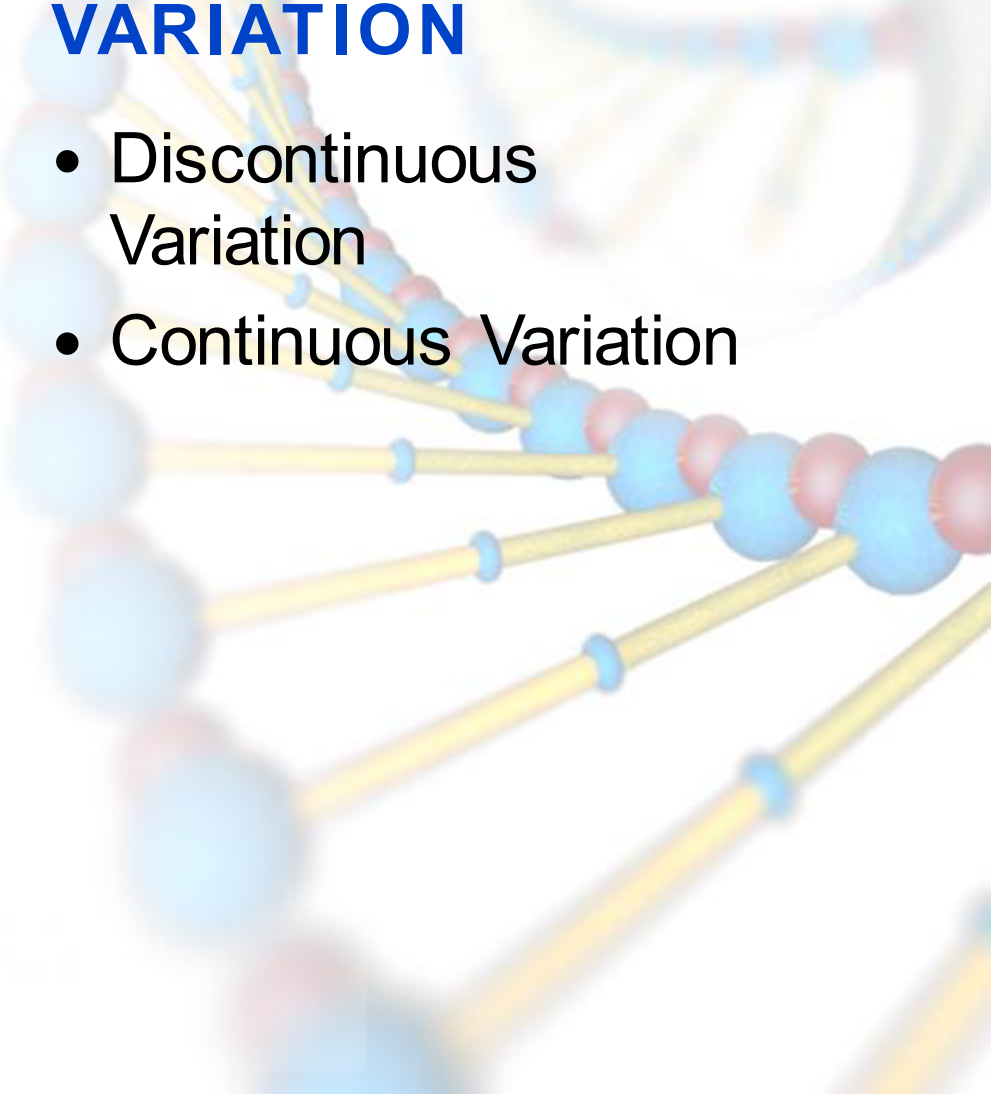
TYPES OF VARIATION



Types of Variation

TWO TYPES OF VARIATION

- Discontinuous Variation
- Continuous Variation



Types of Variation

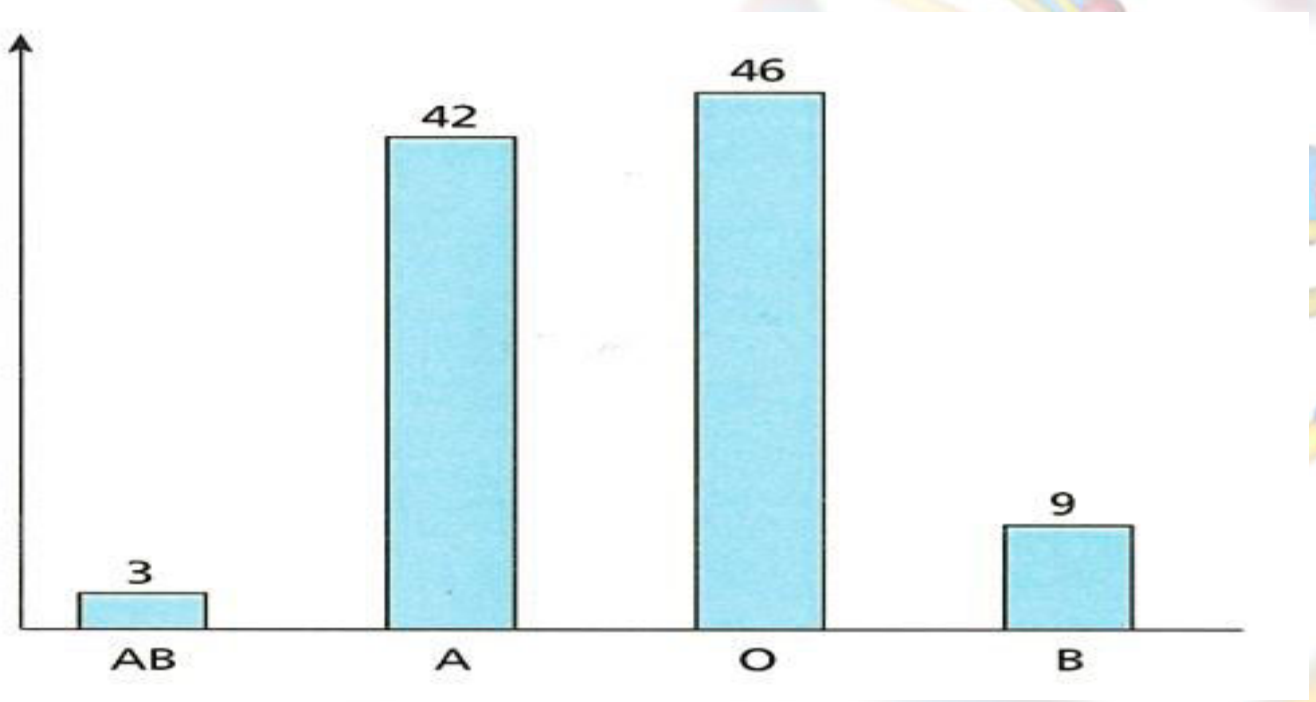
DISCONTINUOUS VARIATION

- Discontinuous variations are entirely genetically controlled.
- They cannot be altered by external conditions. Blood group, color blind, dwarfism are genetically controlled.

Types of Variation

GRAPH FOR DISCONTINUOUS

- Discontinuous variation in blood group. The figures ...not to fit a smooth curve.



Types of Variation

CONTINUOUS VARIATION

- Continuous variation describes the situation in which there are many intermediates between the extremes.
- Shade of hair color between black and blond.

Types of Variation

HAIR COLOR-CONTINUOUS VARIATION



Types of Variation

SEVERAL PAIR OF GENES INVOLVED

- The genome AA BB CC DD -black hair while the genome aa bb cc dd - blond hair.
- Genomes AaBbCcDd or AABbCCdd or aaBBccDd and all the other possible combinations – intermediate.

Types of Variation

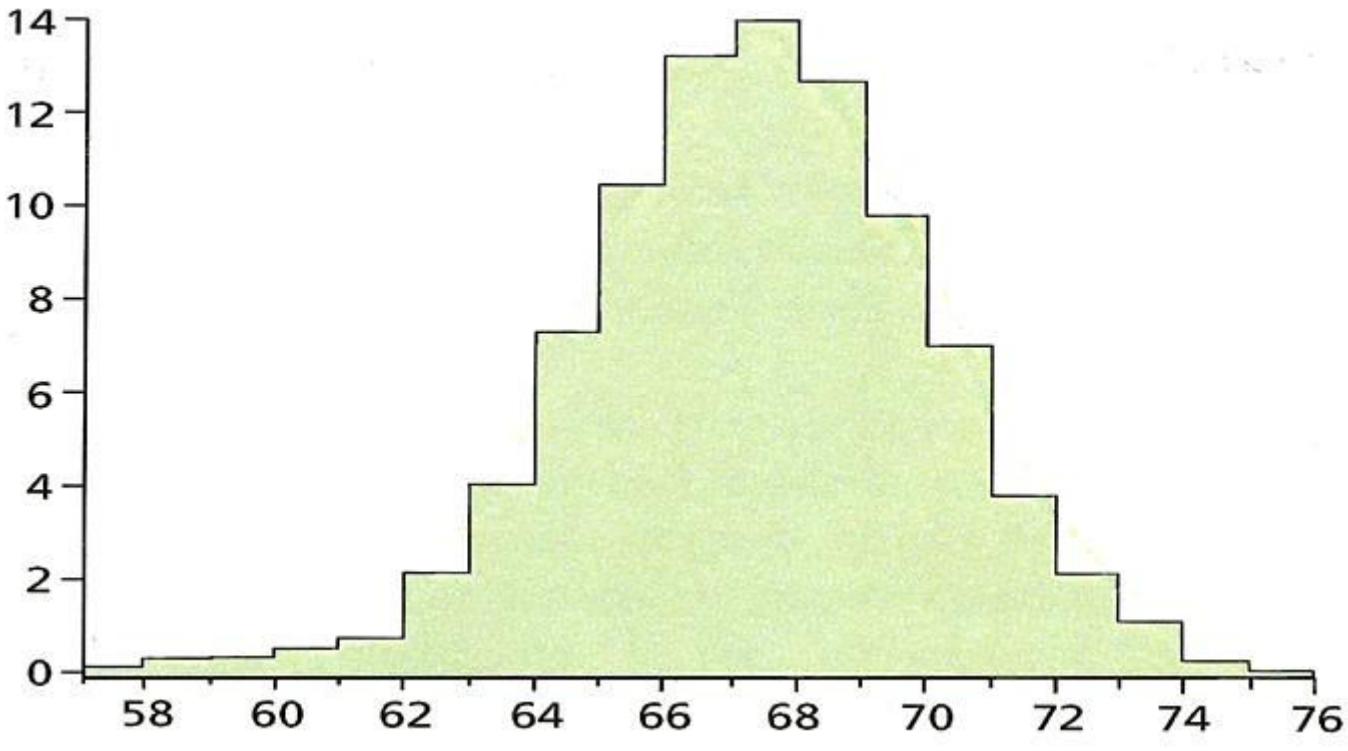
FACTORS - CONTINUOUS VARIATION

- Continuous variation occurs –when controlled by the genes and the environment.
- Height depend on the genes and on the amount of food you eat during your growing.

Types of Variation

GRAPH-CONTINUOUS VARIATION

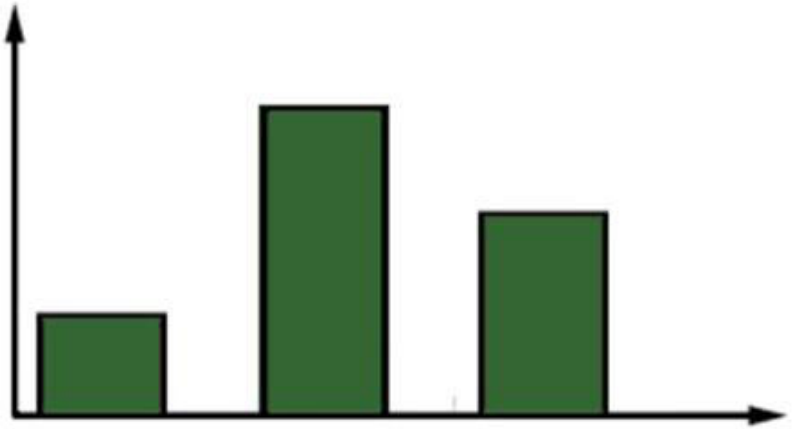
- Smooth curve graph - height.



Types of Variation

COMPARISON





Continuous Variation

- No distinct catagories
- Tends to be quantitative
- Controlled by a lot of genes
- Strongly influenced by the environment

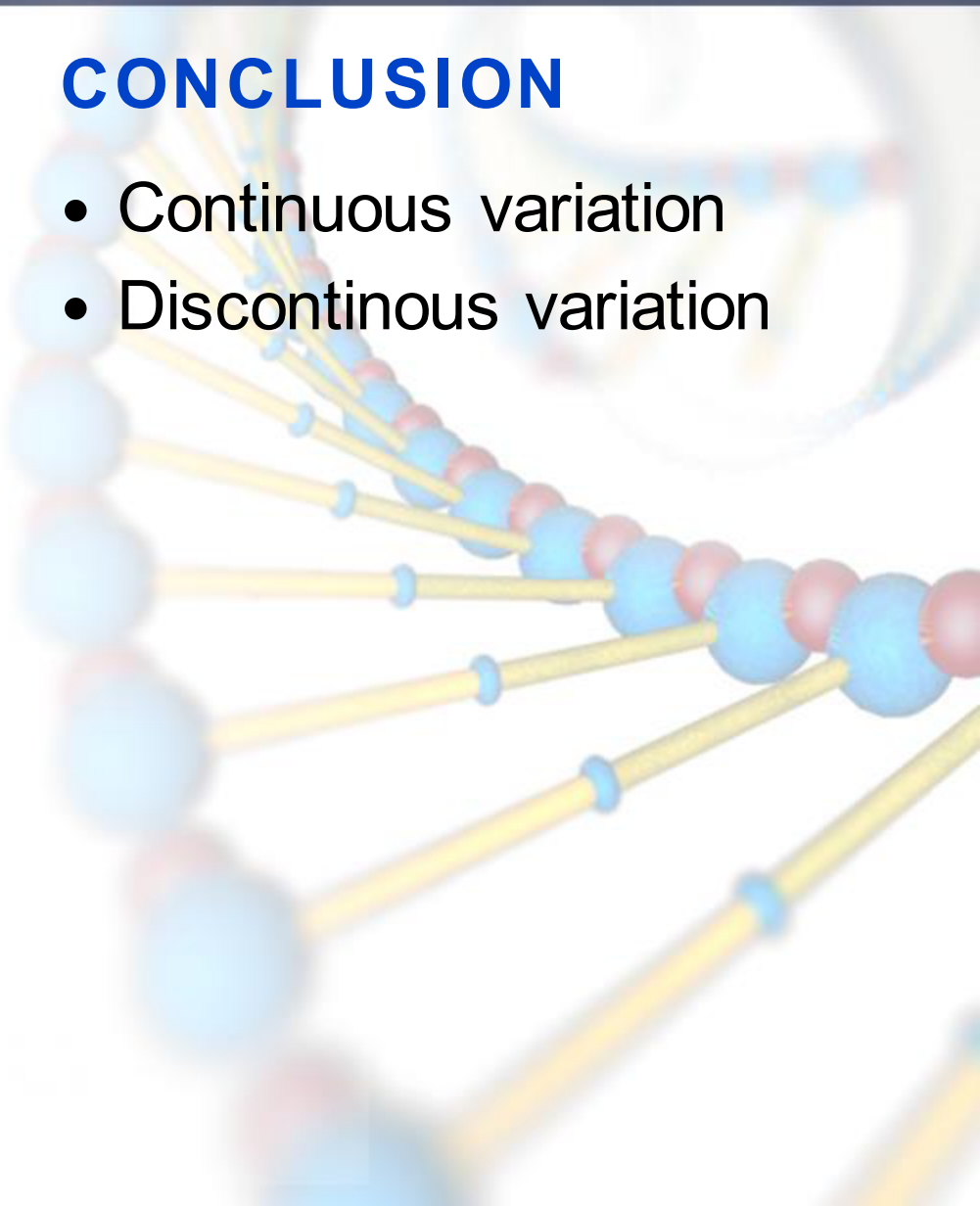
Discontinuous Variation

- Distinct catagories
- Tends to be qualitative
- Controlled by a few genes
- Unaffected by the environment

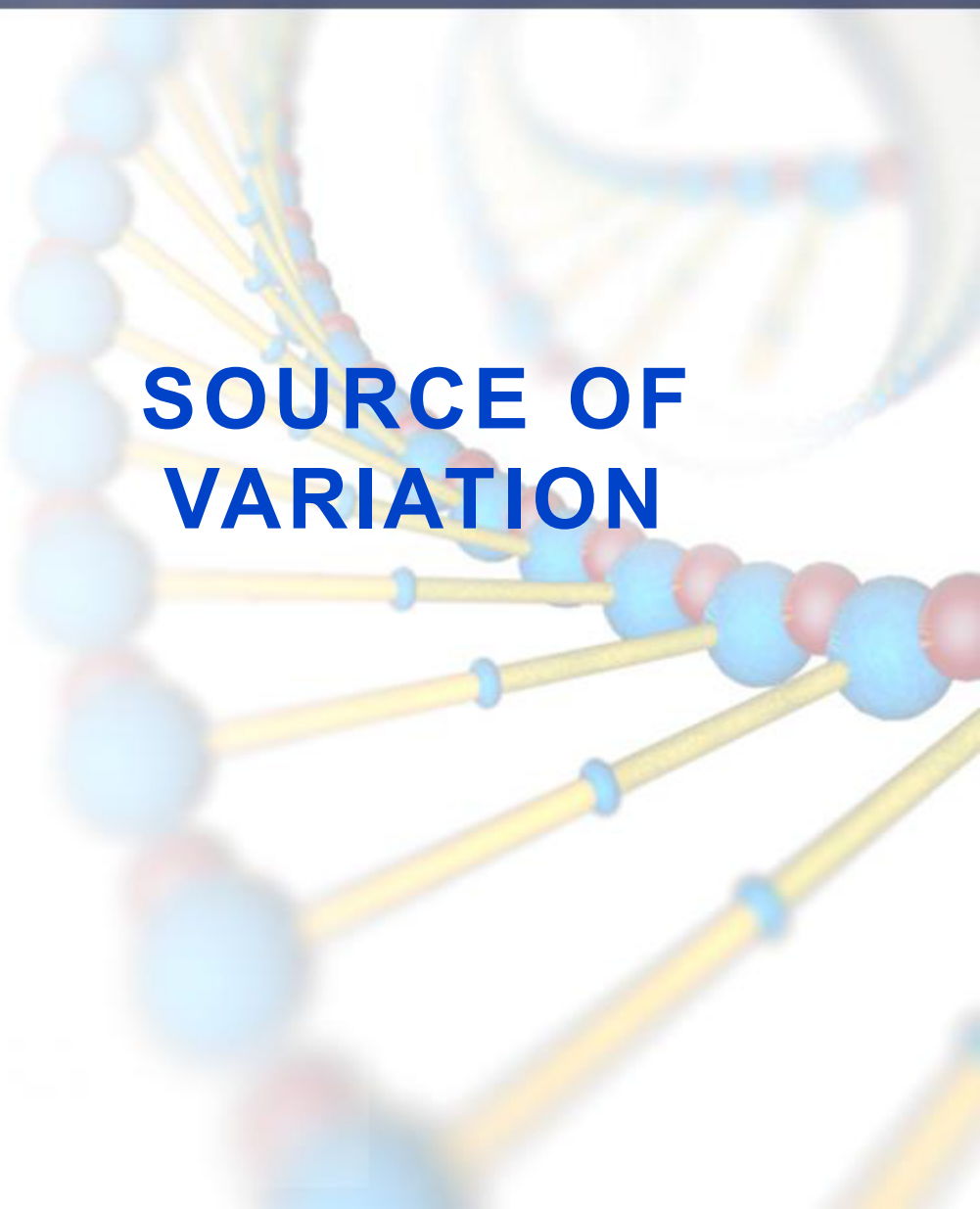
Types of Variation

CONCLUSION

- Continuous variation
- Discontinous variation



Heredity and Variation

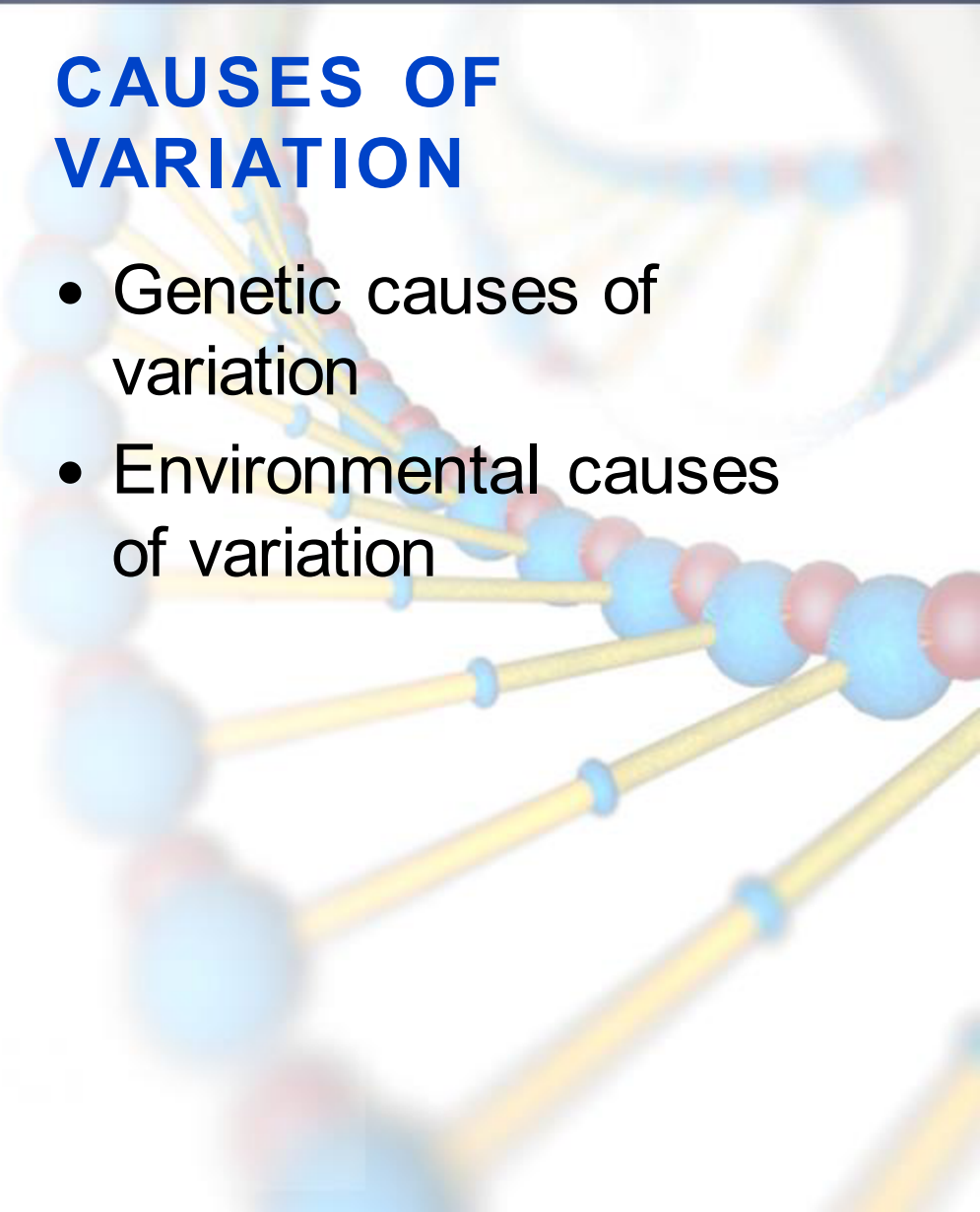


SOURCE OF VARIATION

Source of Variation

CAUSES OF VARIATION

- Genetic causes of variation
- Environmental causes of variation



Source of Variation

GENETIC CAUSES OF VARIATION

- Meiosis
- Sexual reproduction
- Mutations

Source of Variation

ENVIRONMENTAL CAUSES OF VARIATION

- Nutrients
- Drugs
- Temperature
- Physical training

Source of Variation

GENETIC CAUSES OF VARIATION	ENVIRONMENTAL CAUSES
Meiosis – here pairs of chromosomes exchange genes & split from each other producing gametes which are not identical	Nutrients – the food we eat & minerals available to plants
Sexual reproduction – allows for recombination of genetic material from each parent in new ways in the zygote	Drugs – e.g. thalidomide was given to pregnant women in 1960s caused deformities
Mutations – these are random mistakes in the replication of DNA which make the new DNA slightly different to the original	Temperature – affects the rate of enzyme-controlled rxns, e.g. plants grown in warm conditions show faster rates of photosynthesis hence grow faster and better than those in colder environments
	Physical training – when you use muscles you increase their size & power e.g. athletes develop strong muscles as they train for their sport

Source of Variation

CONCLUSION

- Variation from genetic causes are inherited from parents to offspring.
- Inherited variation - raw material for natural selection.
- Variations from environmental causes are not inherited.

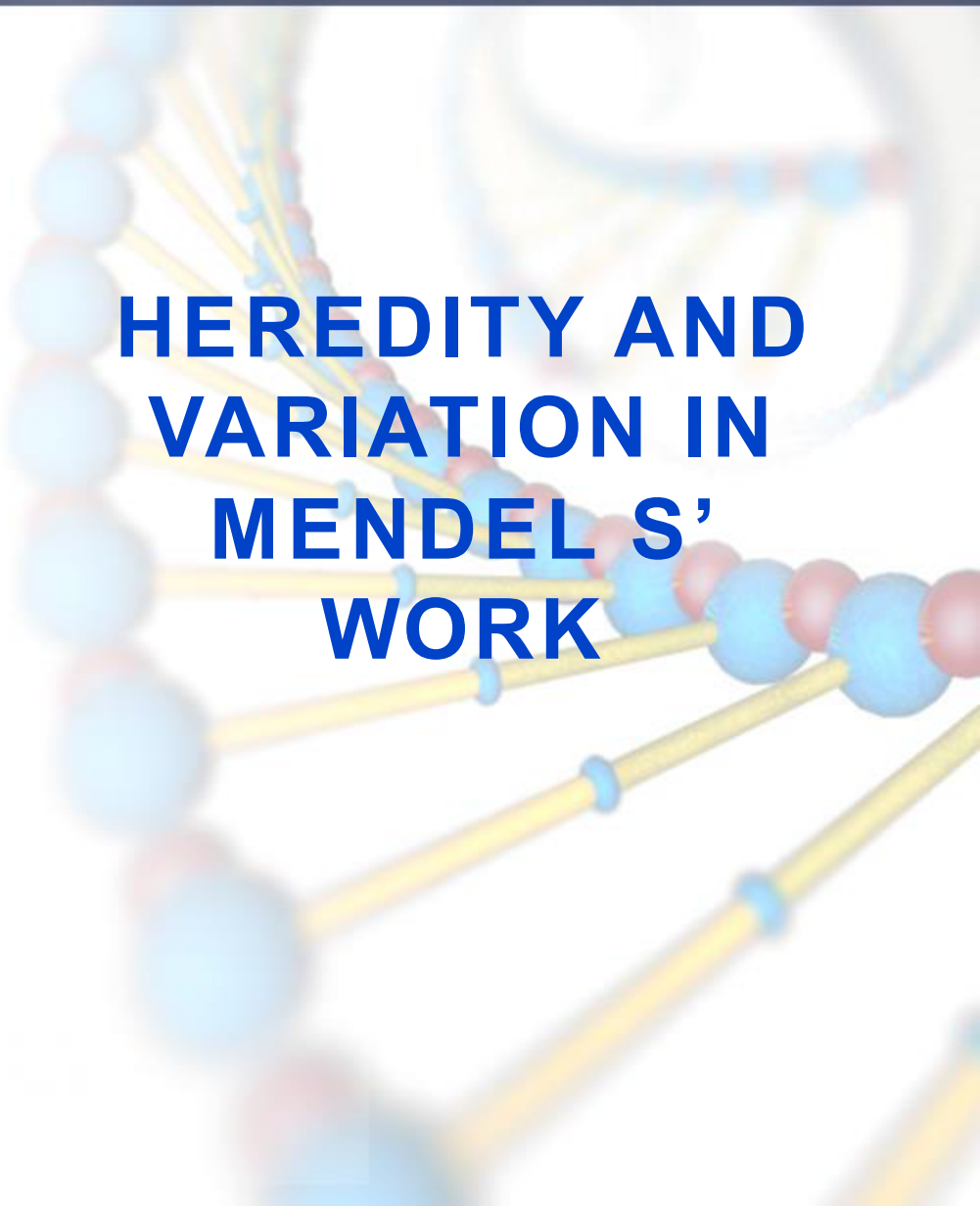
Source of Variation

CONCLUSION

- Variations that result from acquired characteristics are not acted upon by natural selection & thus do not affect evolution.

Heredity and Variation

HEREDITY AND VARIATION IN MENDEL S' WORK



Heredity/Variation-Mendel work

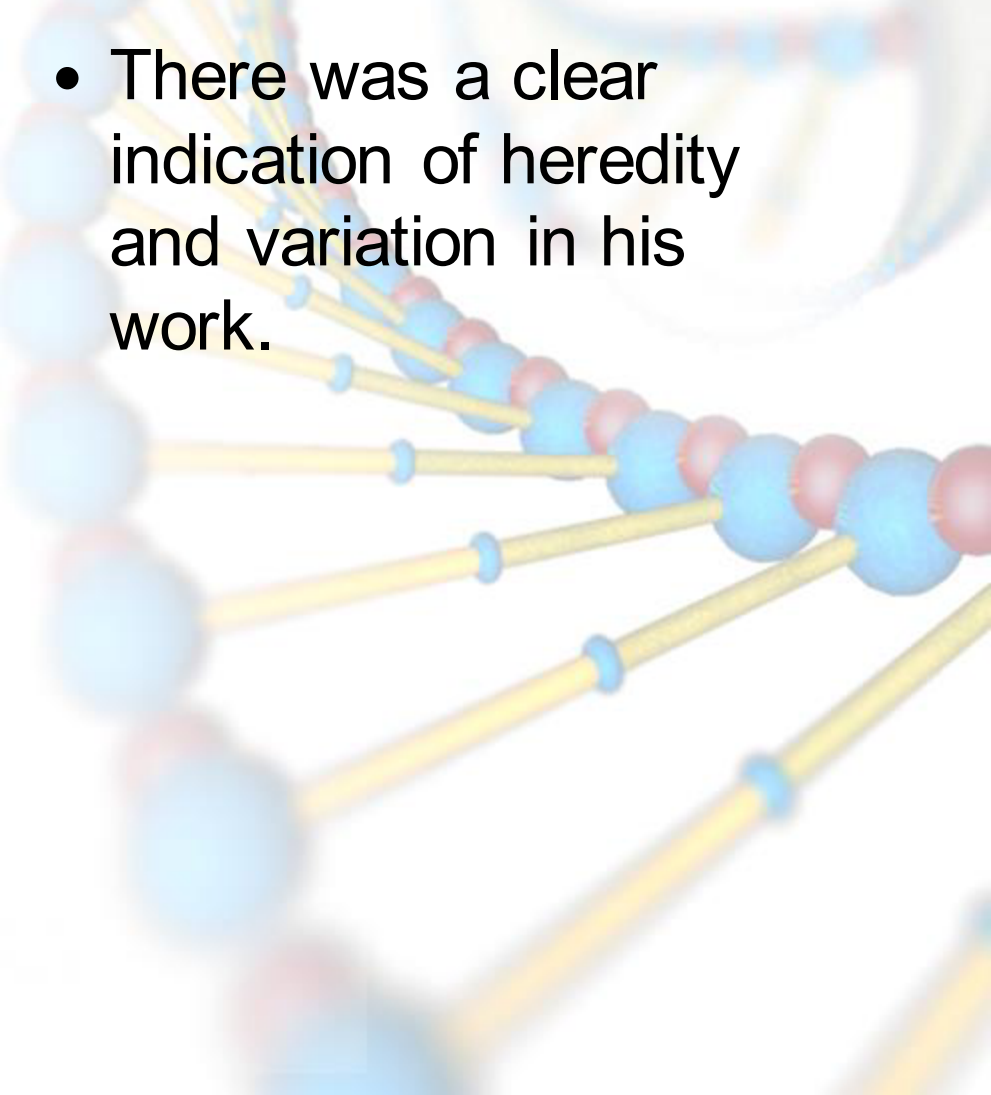
HEREDITY IN MENDEL WORK

- Gregor Mendel – inheritance how characteristics were passed …generation.
- Pure-breeding plants e.g. tall plants produced only tall offspring & purple flowered plants produced only purple.

Heredity/Variation-Mendel work

MENDEL WORK

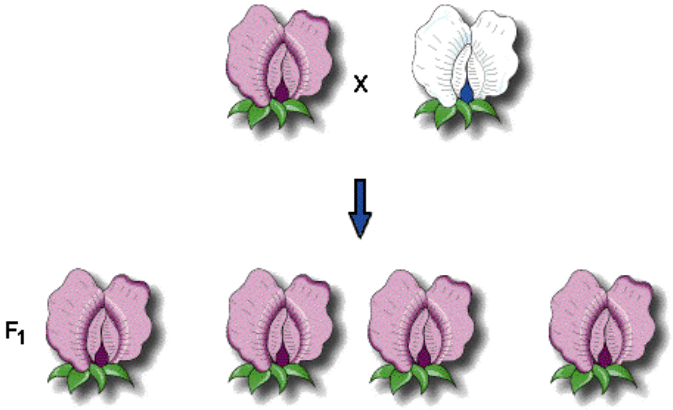
- There was a clear indication of heredity and variation in his work.

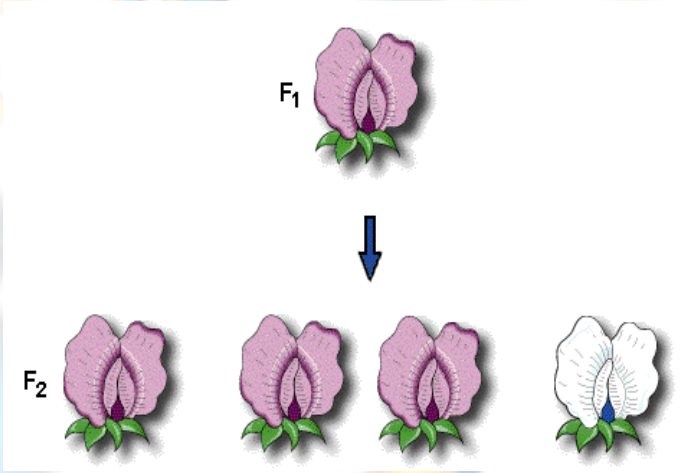


Heredity/Variation-Mendel work

MENDEL WORK

- Both heredity and variation.



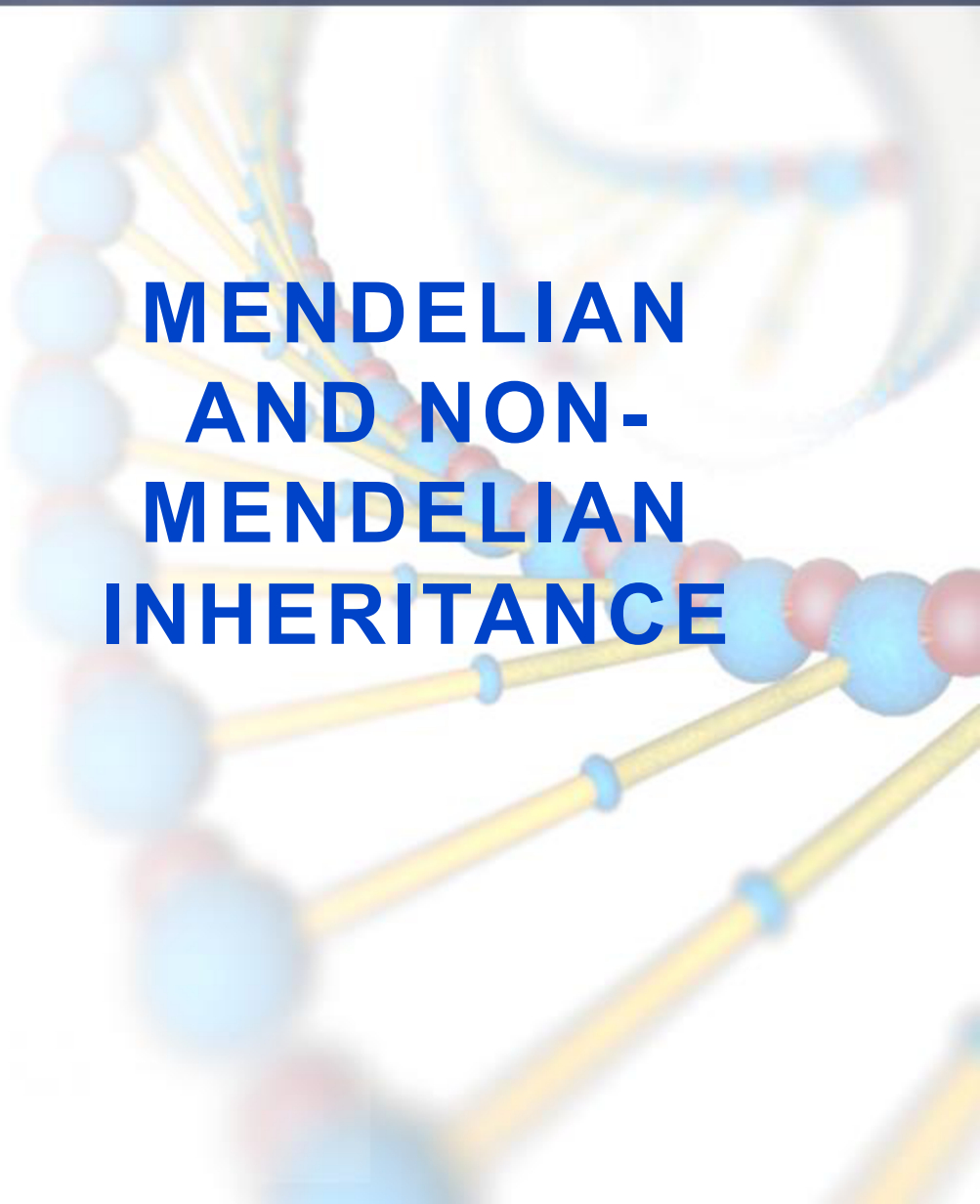


Heredity/Variation-Mendel work

CONCLUSION

- Variation is widespread in nature.
- Variation is inherited according to genetic laws and not solely by chance.

Essentials of Genetics



MENDELIAN AND NON-MENDELIAN INHERITANCE

Mendelian, Non-Mendelian Inheritance

MENDELIAN INHERITANCE

- Gregor Mendel's model of inheritance describes
- Each trait is controlled by a single gene.
- Each gene has two alleles.
- A clear dominant-recessive relationship between alleles.

Mendelian, Non-Mendelian Inheritance

MENDELIAN INHERITANCE

- Mendelian inheritance involve the genes that obey Mendel's laws:
  Law of dominance
  Law of segregation
  Law of independent assortment.

Mendelian, Non-Mendelian Inheritance

MENDELIAN INHERITANCE

- Phenotypes will obey patterns;
- Autosomal dominant
- Autosomal recessive
- X-linked etc.

Mendelian, Non-Mendelian Inheritance

NON-MENDELIAN

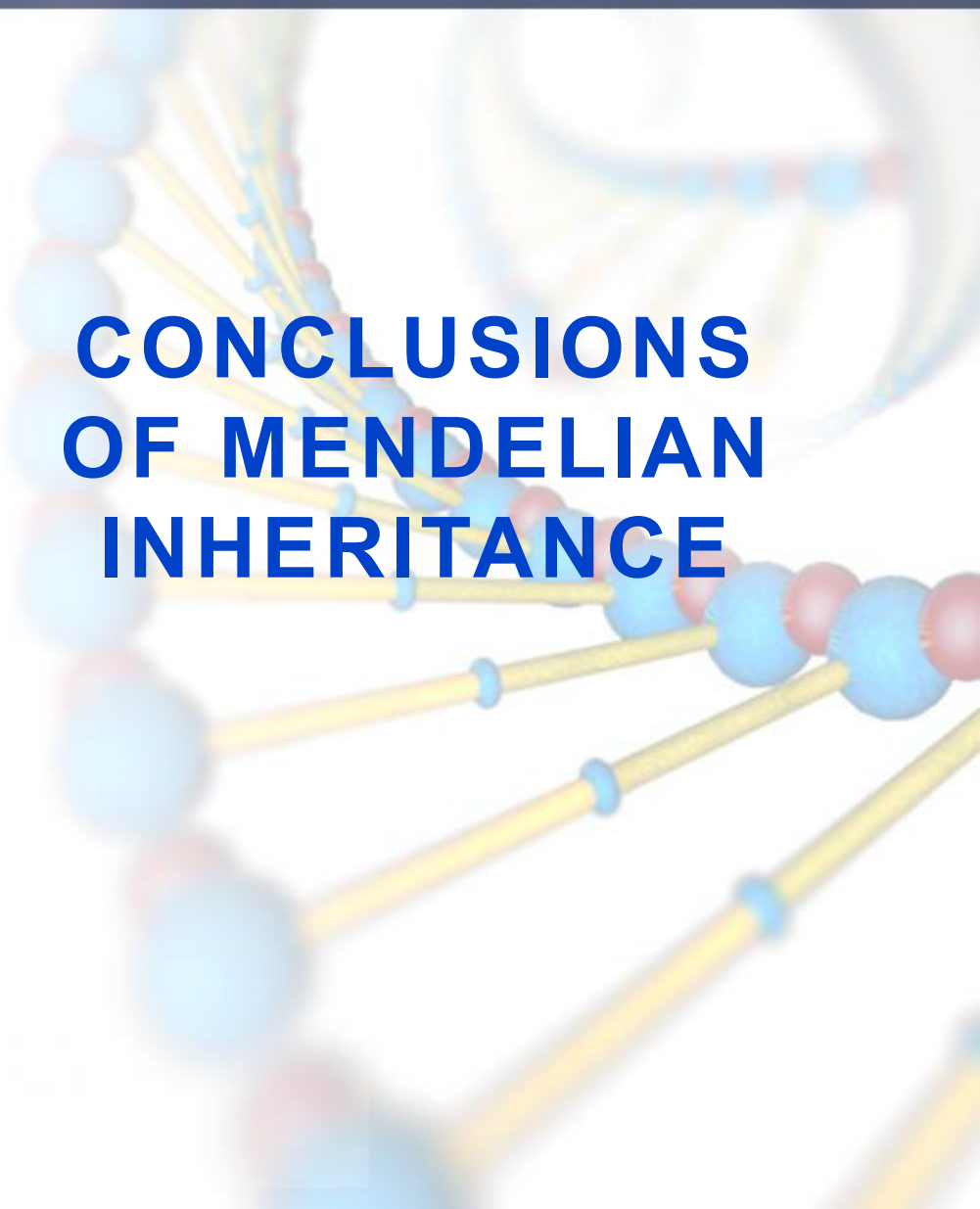
- Pattern of inheritance that deviate from Mendelian pattern of inheritance is called as Non-Mendelian inheritance.
- Those genes that do not follow principles of heredity formulated by Gregor Mendel.

Mendelian, Non-Mendelian Inheritance

EXAMPLES OF NON-MENDELIAN INHERITANCE

- Mitochondrial inheritance
- Genomic (parental) imprinting
- Incomplete dominance
- Co-dominance
- Multiple alleles
- Epigenetics

Essentials of Genetics



CONCLUSIONS OF MENDELIAN INHERITANCE

Mendelian Inheritance

MENDELIAN INHERITANCE

- Mendel s' conclusions were based on three laws:
- Law of Dominance
- Law of Segregation
- Law of Independent Assortment

Mendelian Inheritance

CONCLUSION- LAW OF DOMINANCE

- Cross of true-breeding strains resemble only one of the parents.
- Smooth seeds (S) are completely dominant to wrinkled seeds (s)

Mendelian Inheritance

CONCLUSION - LAW OF SEGREGATION

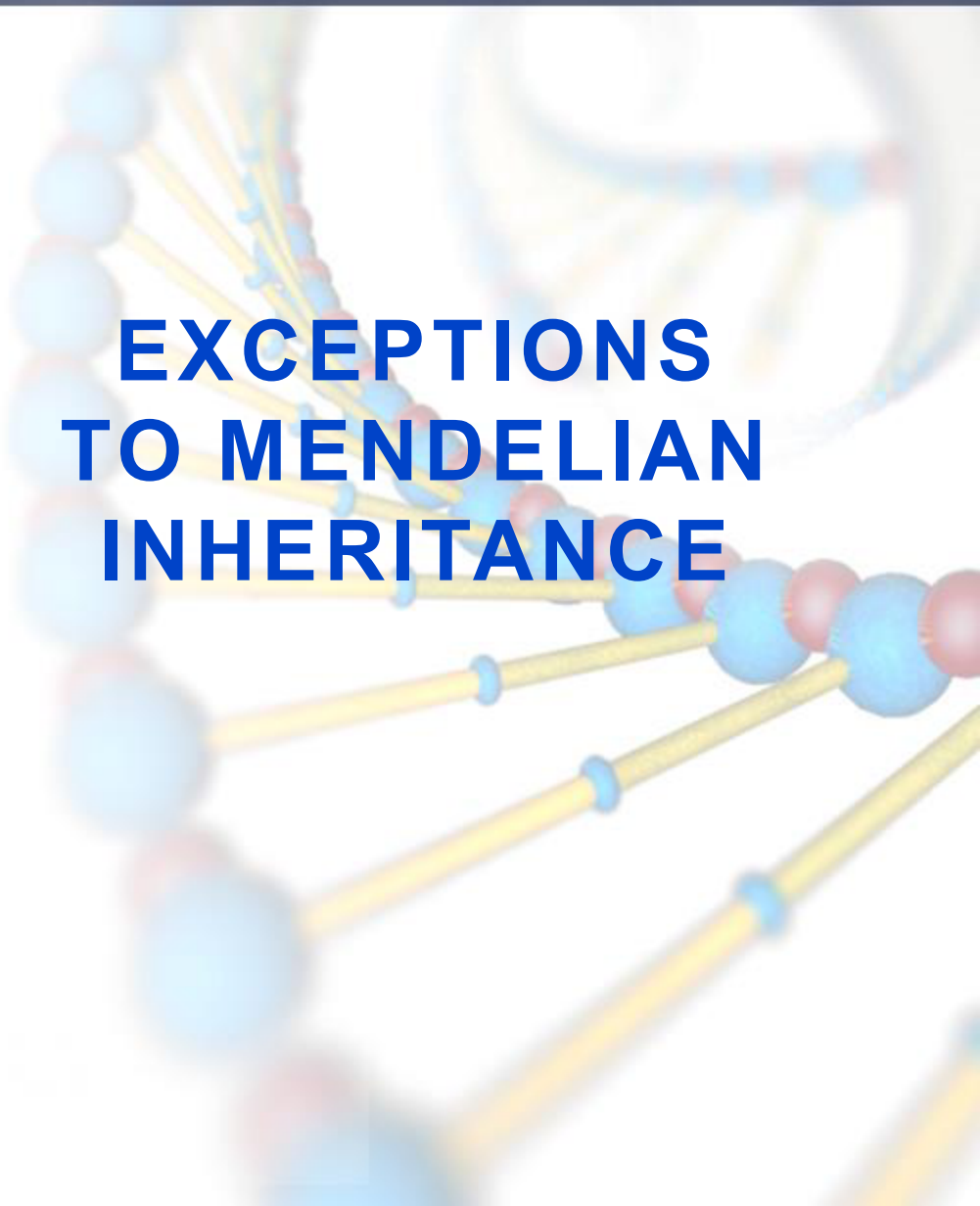
- Two members of a gene pair segregate (separate) from each other during the formation of gametes.

Mendelian Inheritance

CONCLUSION- LAW OF INDEPENDENT ASSORTMENT

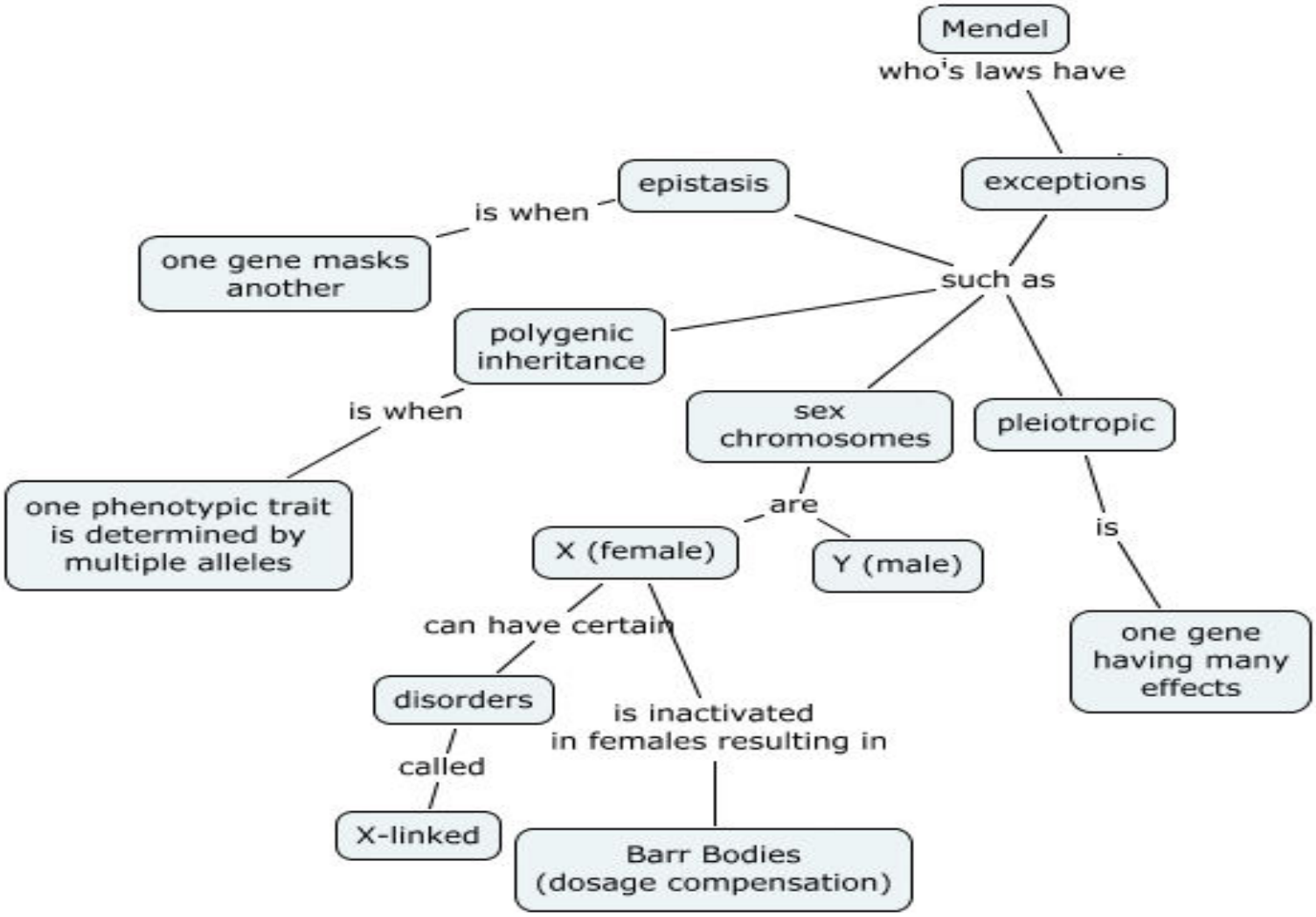
- Alleles for different traits assort independently of one another.
- Genes on different chromosomes independently behave.

Essentials of Genetics



EXCEPTIONS TO MENDELIAN INHERITANCE

Exceptions-Mendelian Inheritance



Exceptions-Mendelian Inheritance

EXCEPTIONS TO MENDELIAN INHERITANCE

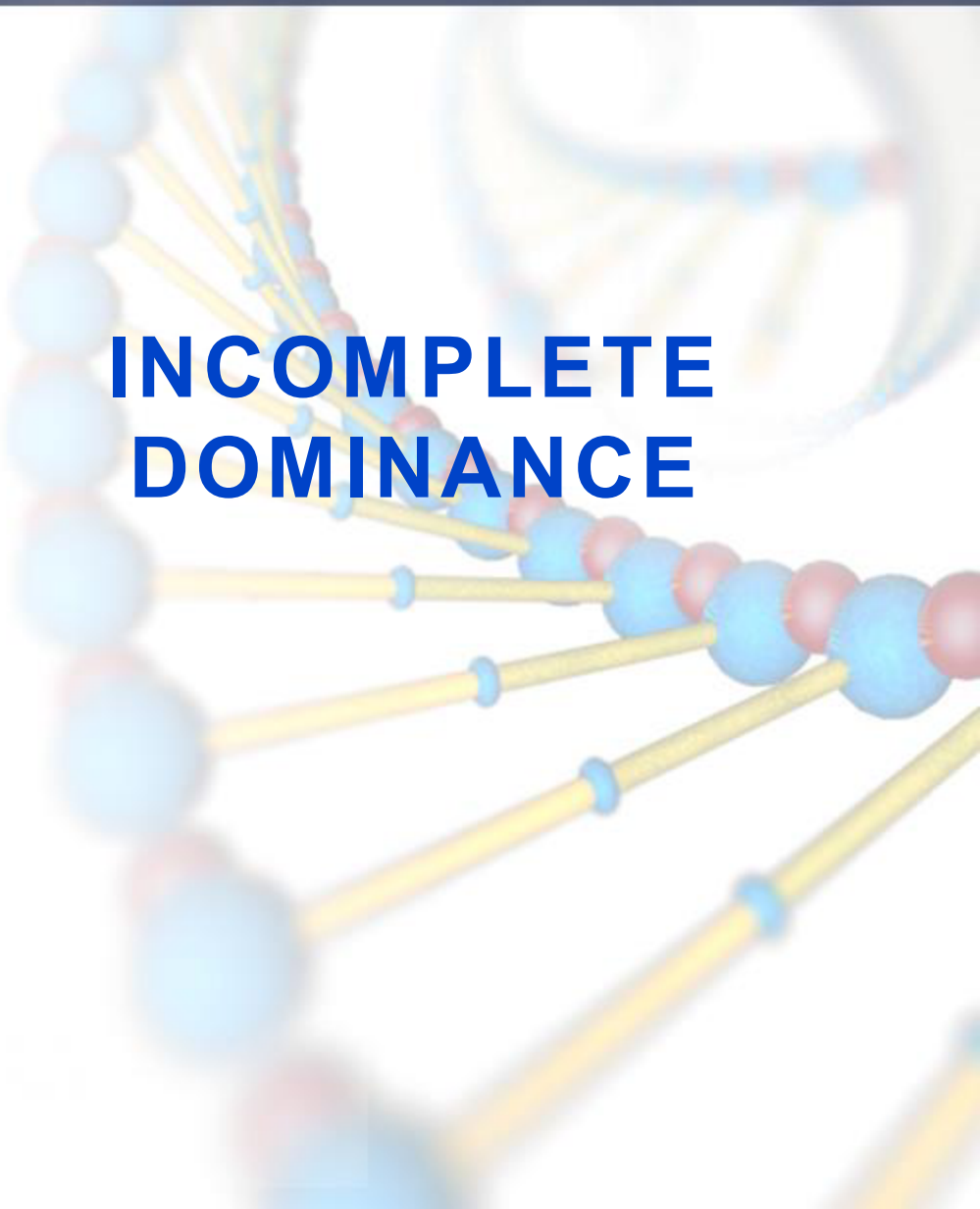
- Incomplete dominance
- Codominance
- Multiple alleles
- Polygenic traits
- Epistasis

Exceptions-Mendelian Inheritance

EXCEPTIONS TO MENDELIAN INHERITANCE

- Pleiotropy
- Environmental effects on gene expression
- Linkage
- Sex linkage

Essentials of Genetics



INCOMPLETE DOMINANCE

Incomplete Dominance

INCOMPLETE DOMINANCE

- Some alleles for a gene are not completely dominant.
- An intermediate phenotype – between two extremes.
- None of the allele is dominant.

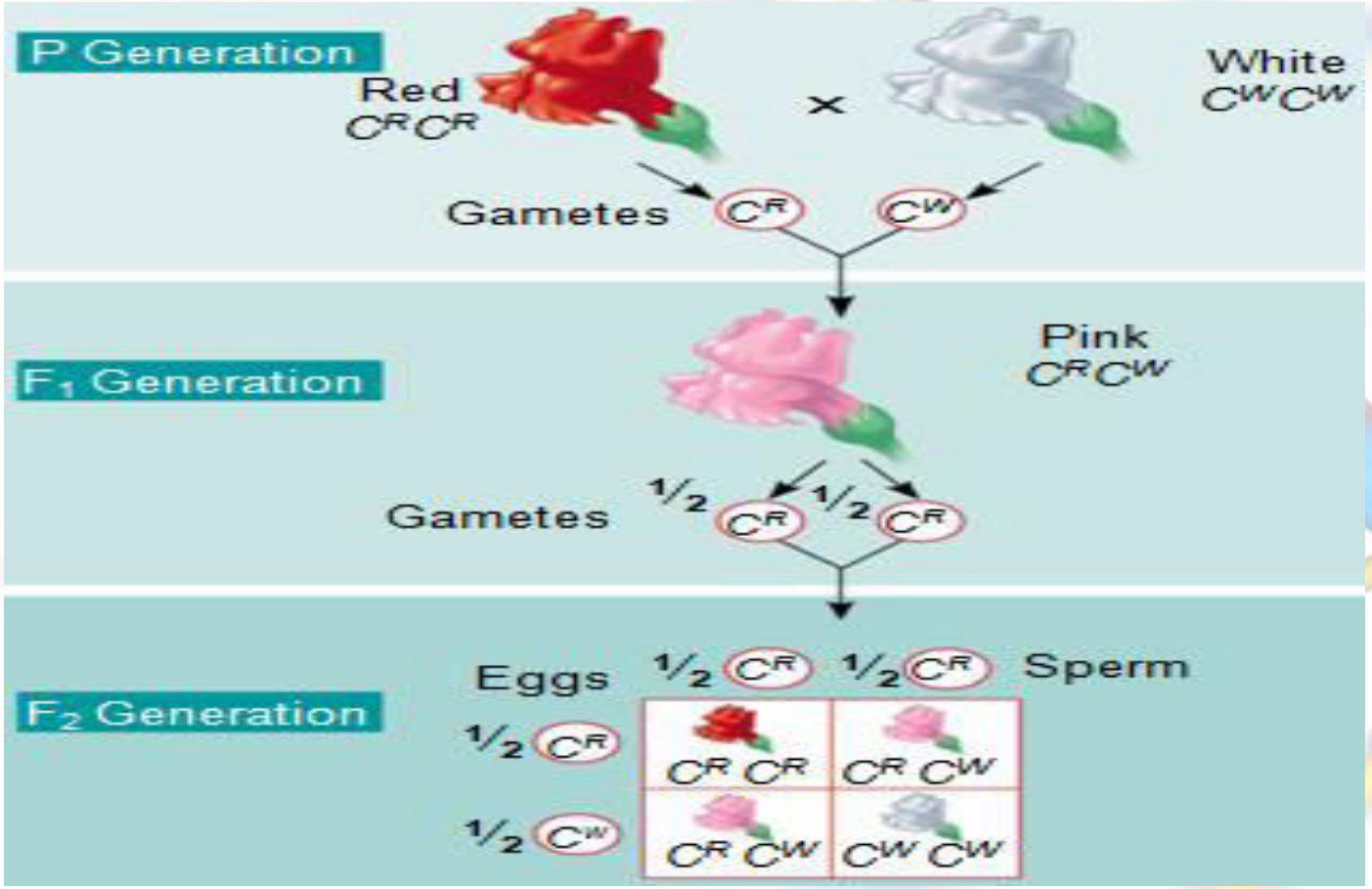
Incomplete Dominance

INCOMPLETE DOMINANCE

- As none of the alleles is dominant. As a result of a cross, a “new” phenotype appears that is a blend of both alleles.

Incomplete Dominance

JAPANESE 4 O' CLOCK FLOWERS



Incomplete Dominance

GENOTYPES OF JAPANESE 4 O' CLOCK PLANT

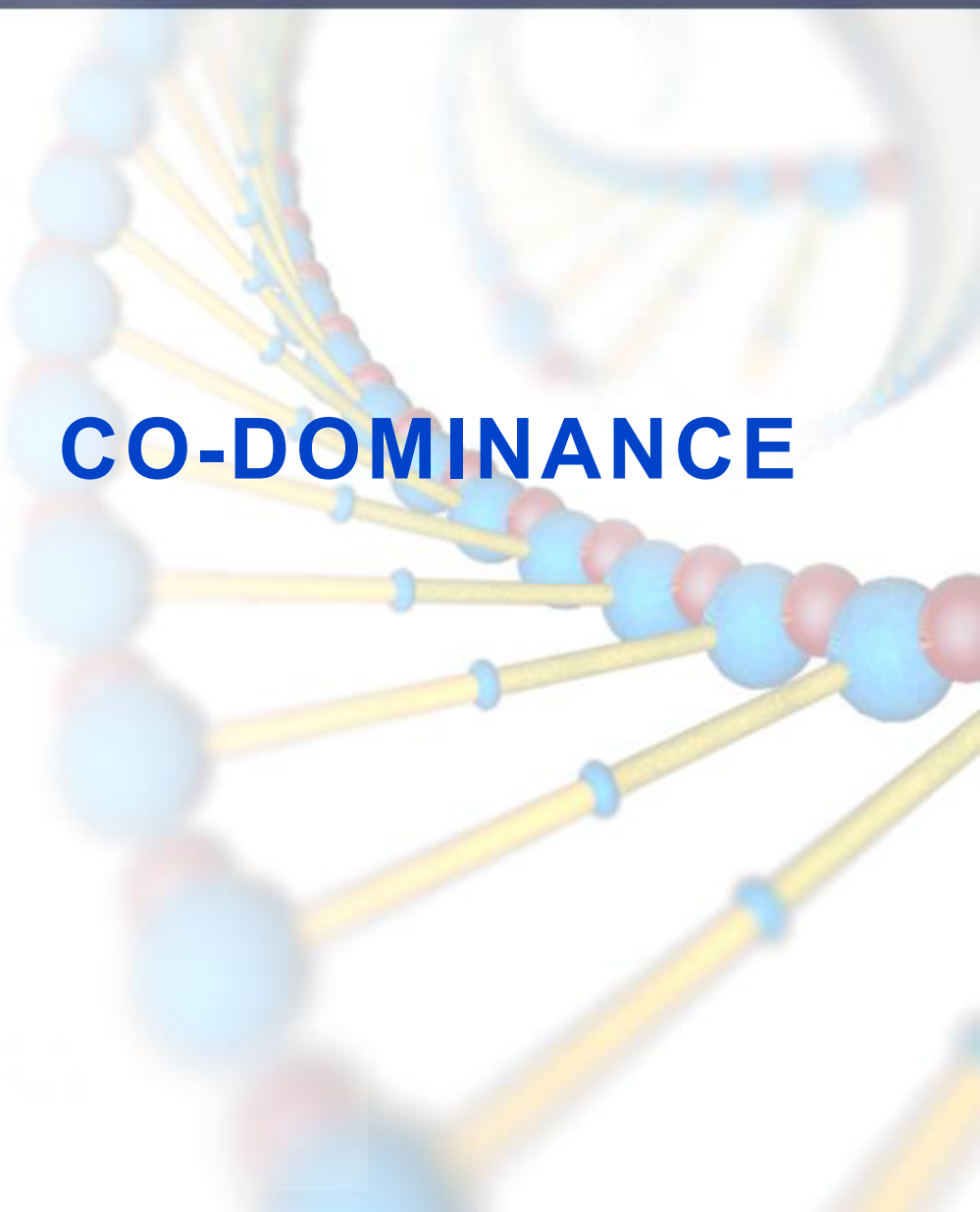
- Red flower plant genotype = **RR**
- White flower plant genotype = **WW**
- Pink flower plant genotype = **RW**

Incomplete Dominance

CONCLUSION

- Incomplete dominance is a situation in which neither allele is dominant.
- New phenotype appears.
- New phenotype is blend of both alleles.

Essentials of Genetics



CO-DOMINANCE

Co-Dominance

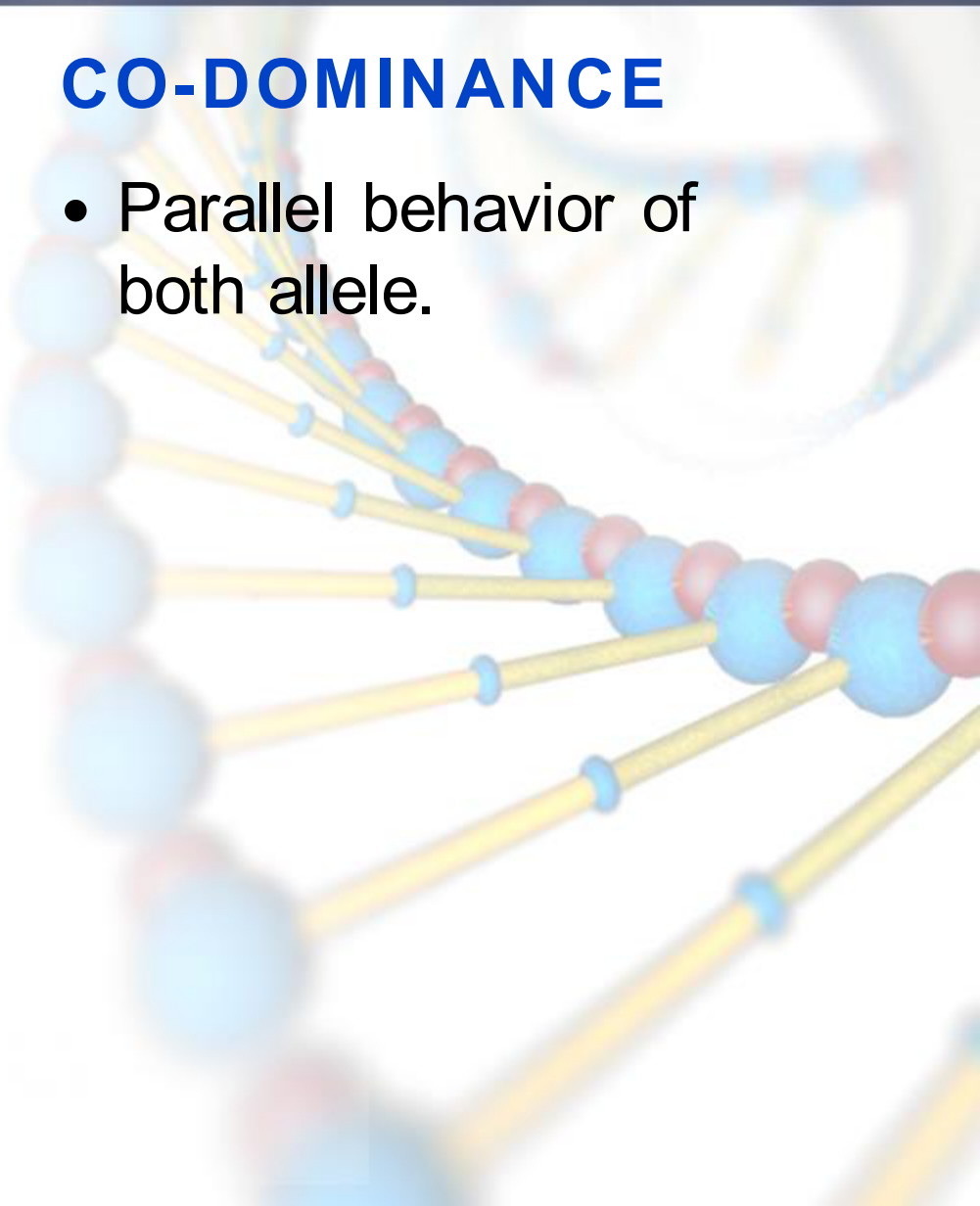
CO-DOMINANCE

- A situation, when both alleles appear in the phenotype.
- Neither allele is dominant.
- Both alleles are expressed in a heterozygous individuals.

Co-Dominance

CO-DOMINANCE

- Parallel behavior of both allele.



Co-Dominance

EXAMPLE 1 BLOOD GROUP

Genotype	Phenotype (Blood Group)	Red Blood Cells
$I^A I^A$ or $I^A i$	A	
$I^B I^B$ or $I^B i$	B	
$I^A I^B$	AB	
ii	O	

Co-Dominance

EXAMPLE 2 ROAN CATTLE INHERITANCE

- Roan color in cattle appears as a mix of red and white colors.

  crcr = **red hairs**

  cwcw = **white hairs**

  crcw = **roan coat**

Co-Dominance

ROAN CATTLE INHERITANCE

- Roan color in cattle appears as a mix of red and white.



Co-Dominance

EXAMPLE 3 APPALOOSA HORSE

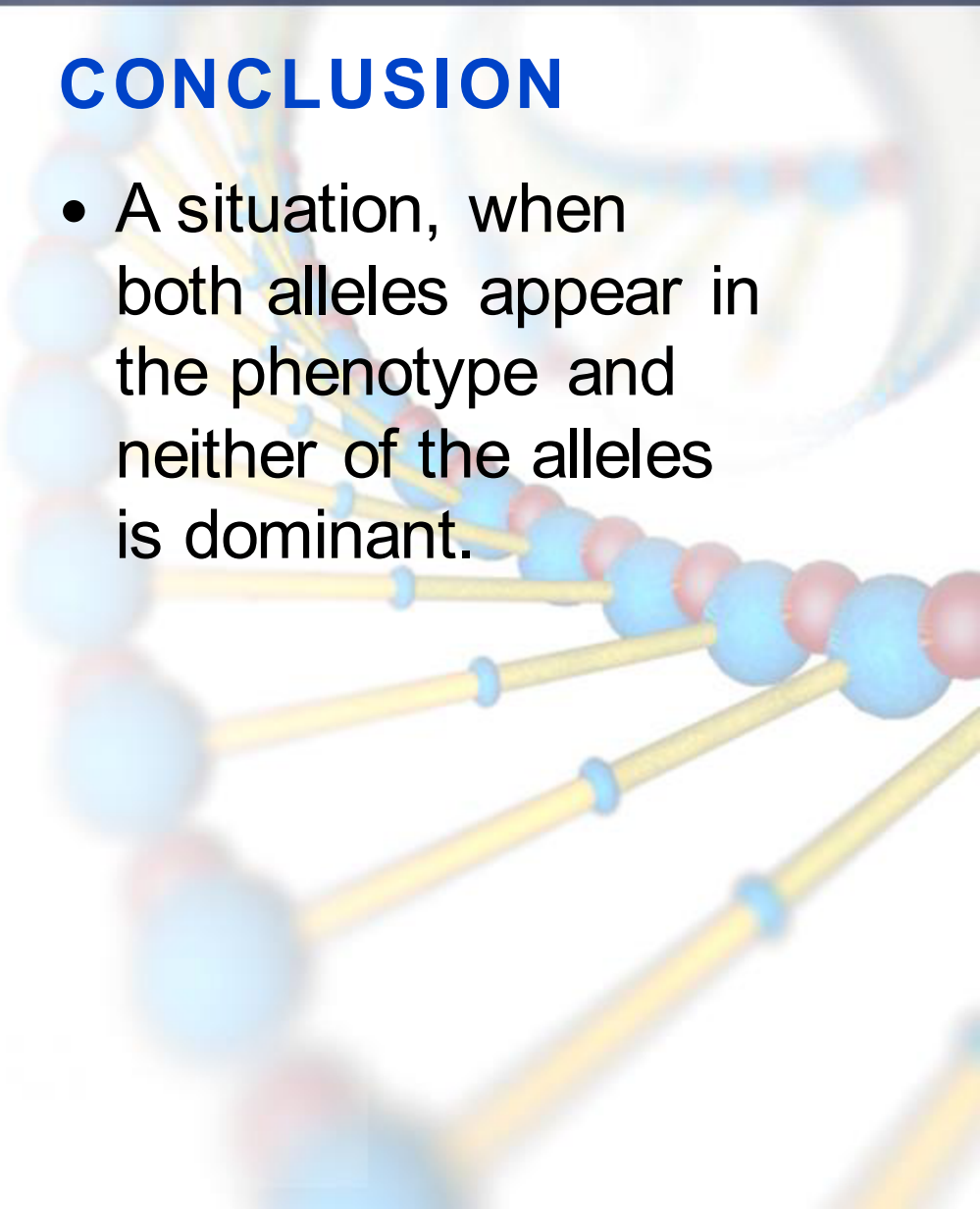
- Gray horse dominant to white horse.
- Heterozygous is appaloosa – a white horse with gray spots.



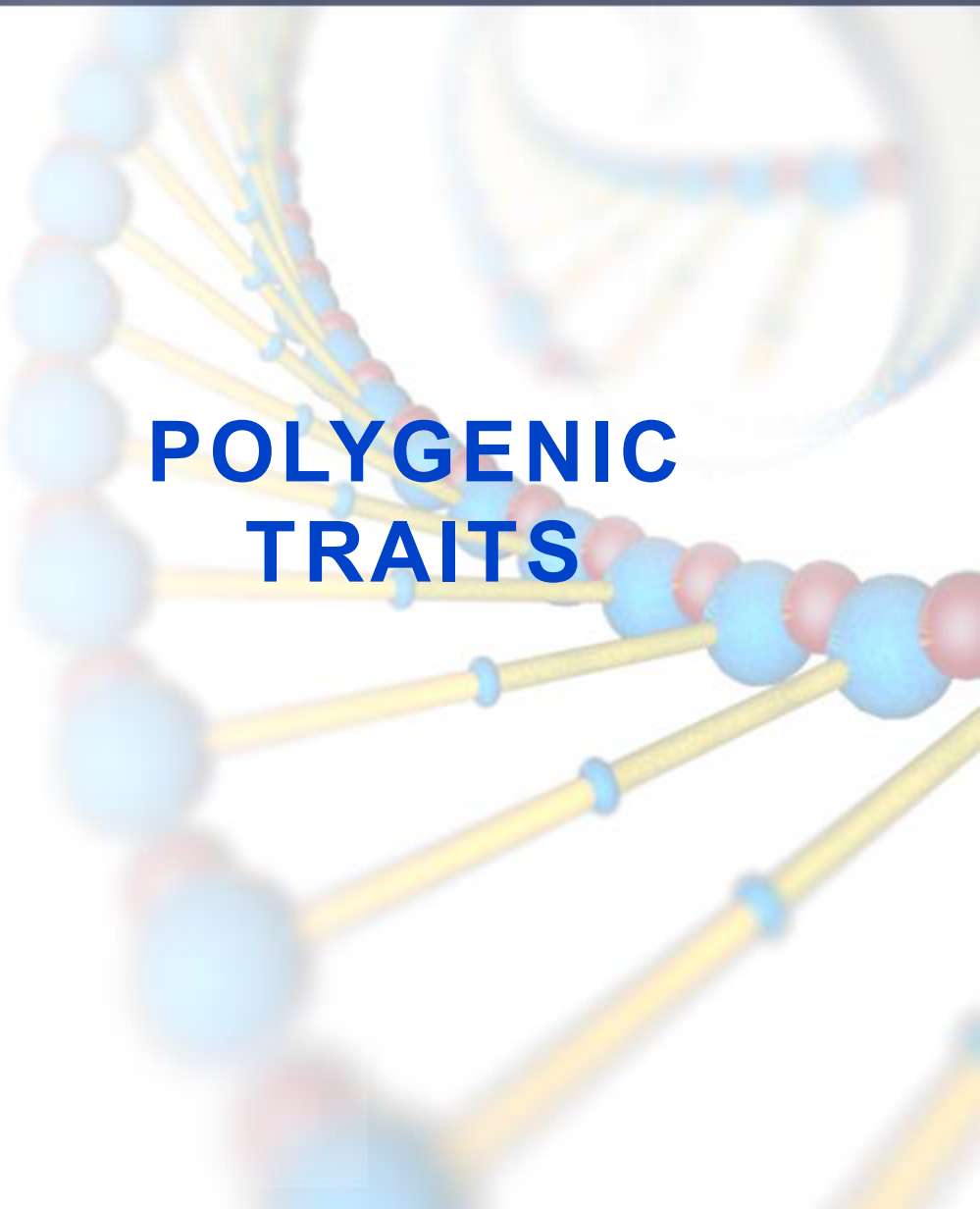
Co-Dominance

CONCLUSION

- A situation, when both alleles appear in the phenotype and neither of the alleles is dominant.



Essentials of Genetics



POLYGENIC TRAITS

Polygenic Traits

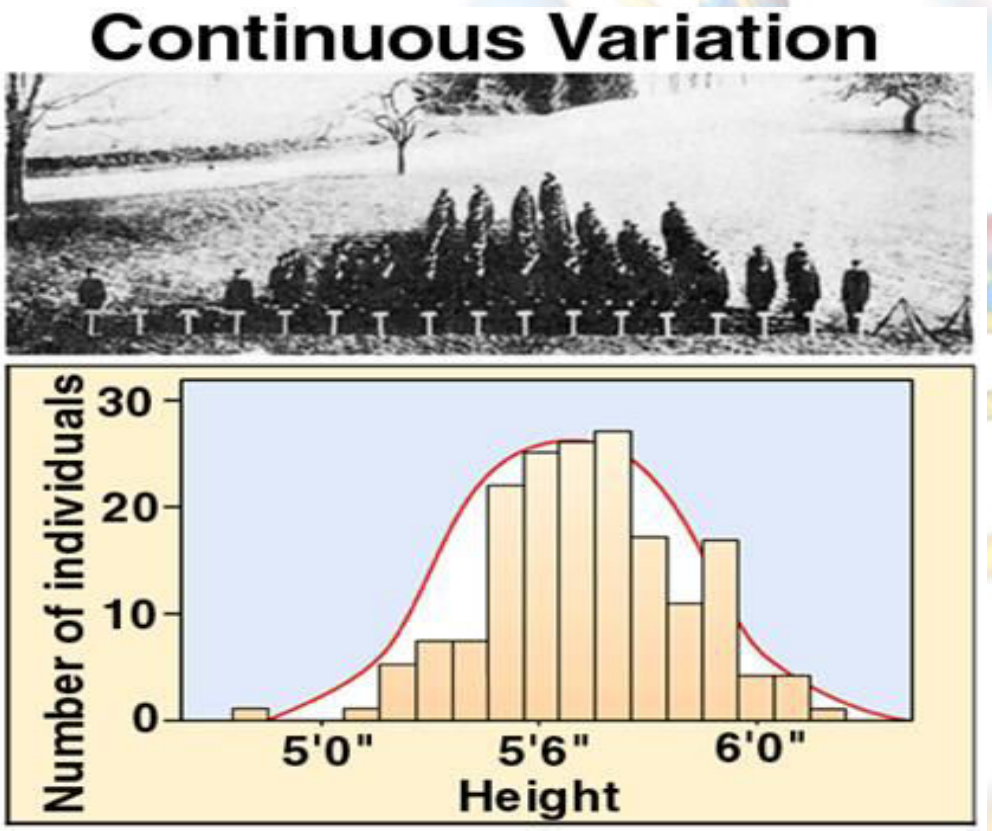
POLYGENIC TRAITS

- Most traits are not controlled by a single gene locus, but by the combined interaction of many gene loci.
- These traits are called polygenic traits.

Polygenic Traits

HUMANS HEIGHT

- Graph for continuous variation



Polygenic Traits

HUMANS EYES COLOR

Controlled by many genes.



Polygenic Traits

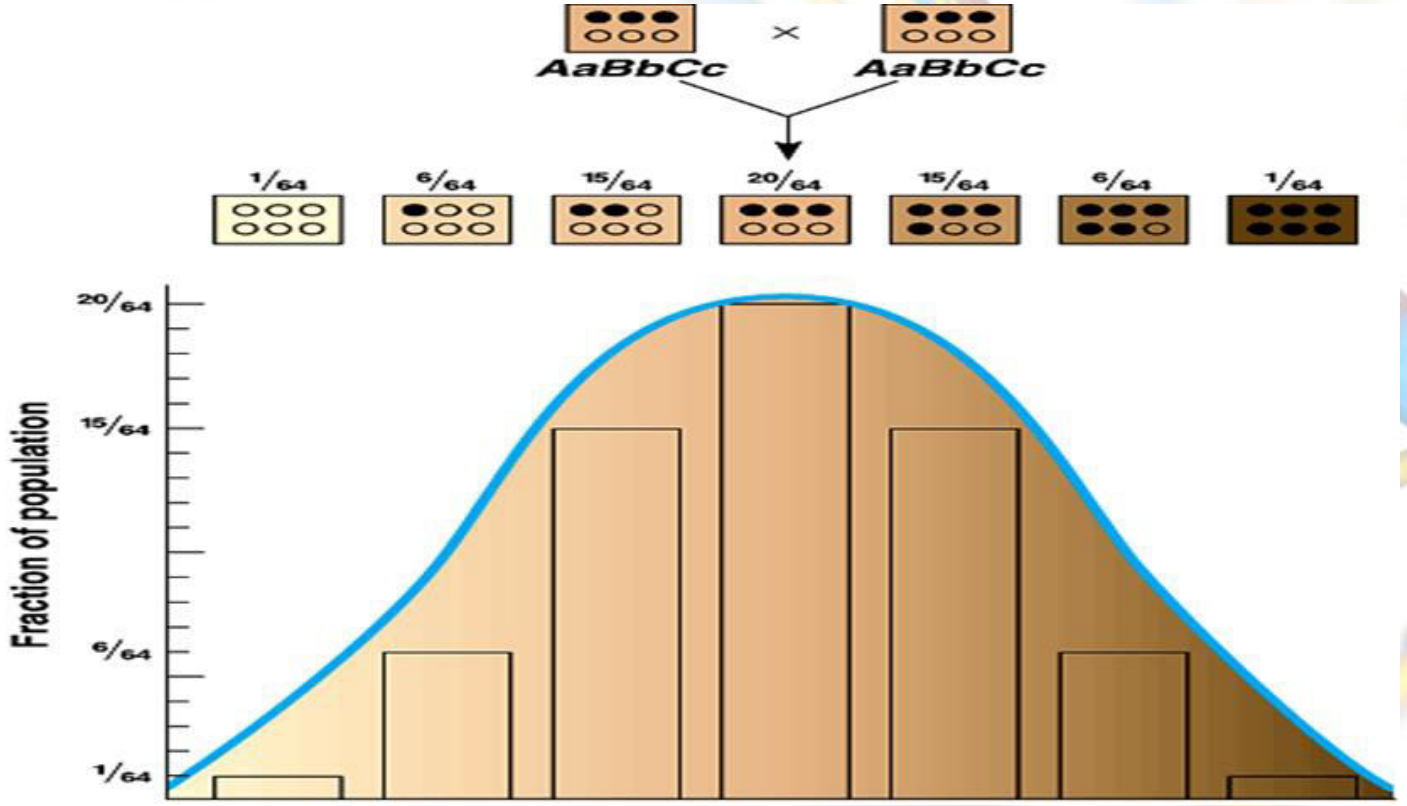
PIGMENTATIONS IN HUMANS

- Qualitative traits usually indicate polygenic inheritance; effect from two/more genes.
- Skin color pigmentation in humans is controlled by at least three separately inherited genes.

Polygenic Traits

PIGMENTATION IN HUMANS

- Controlled by three genes.

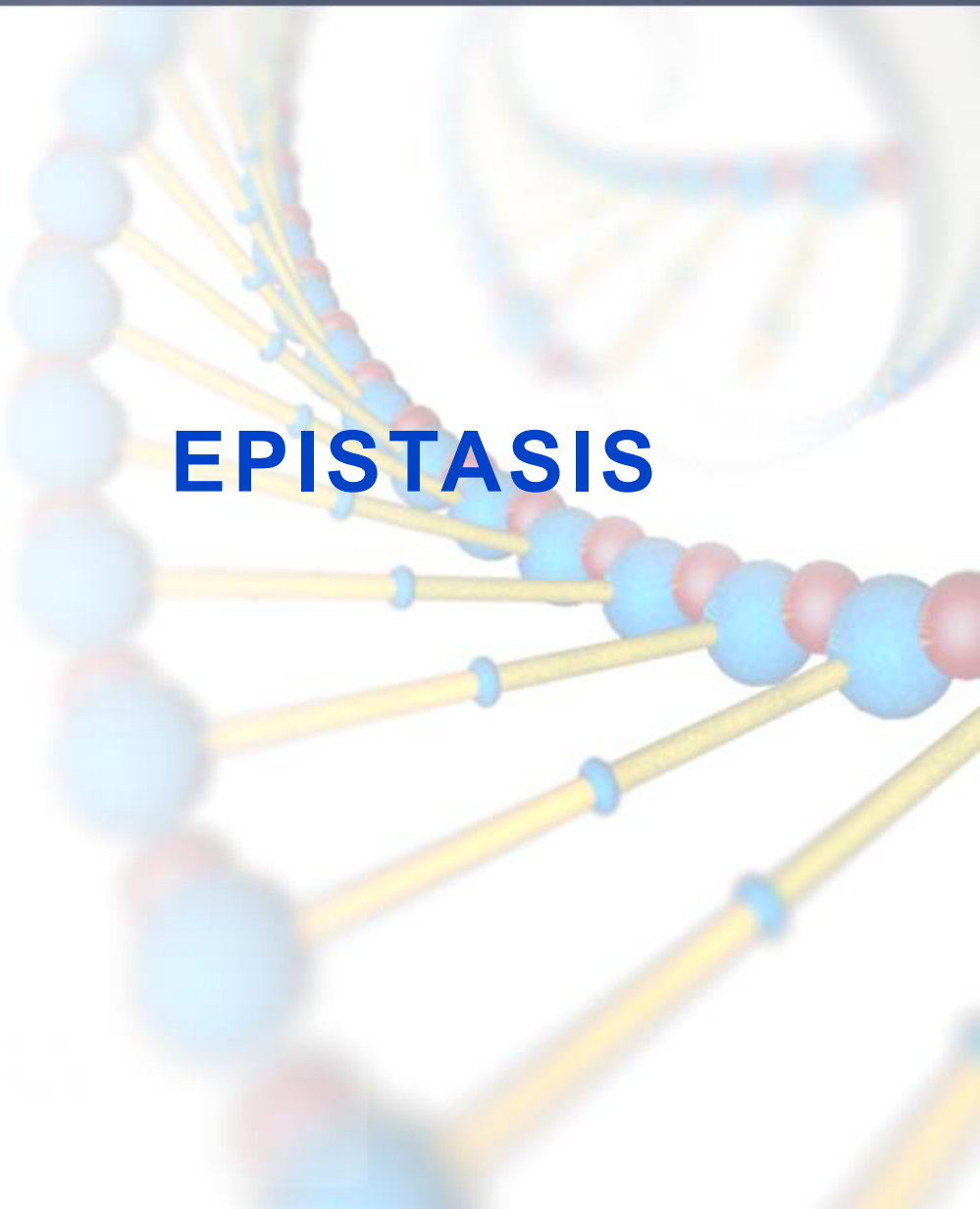


Polygenic Traits

CONCLUSION

- Polygenic traits are those traits which are controlled by additive effect of many genes.
- Qualitative traits are usually polygenic in nature.

Essentials of Genetics



EPISTASIS

Epistasis

EPISTASIS

- Alleles at one gene locus can hide or prevent expression of alleles at second gene locus.
- Labrador retrievers one gene locus affects coat color by controlling pigment eumelanin deposited in the fur.

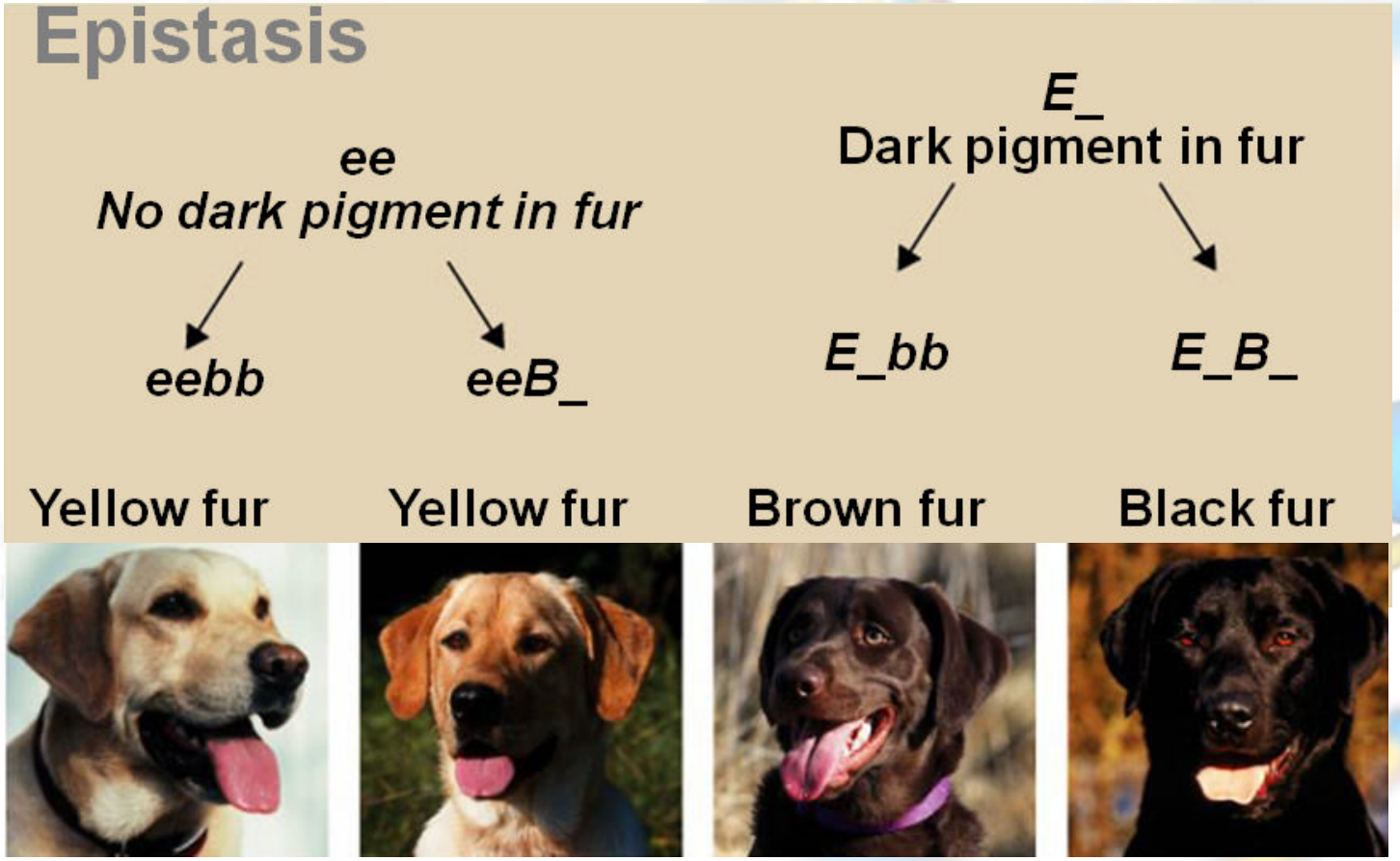
Epistasis

EXAMPLE

- A dominant allele B for black, recessive allele b brown coat, a second gene (e) locus controls eumelanin deposited in fur.
- Dogs homozygous recessive at this locus (ee) will have yellow fur no matter which alleles are at first locus.

Epistasis

EPISTASIS IN DOGS FUR



Epistasis

EPISTASIS IN HORSES FUR

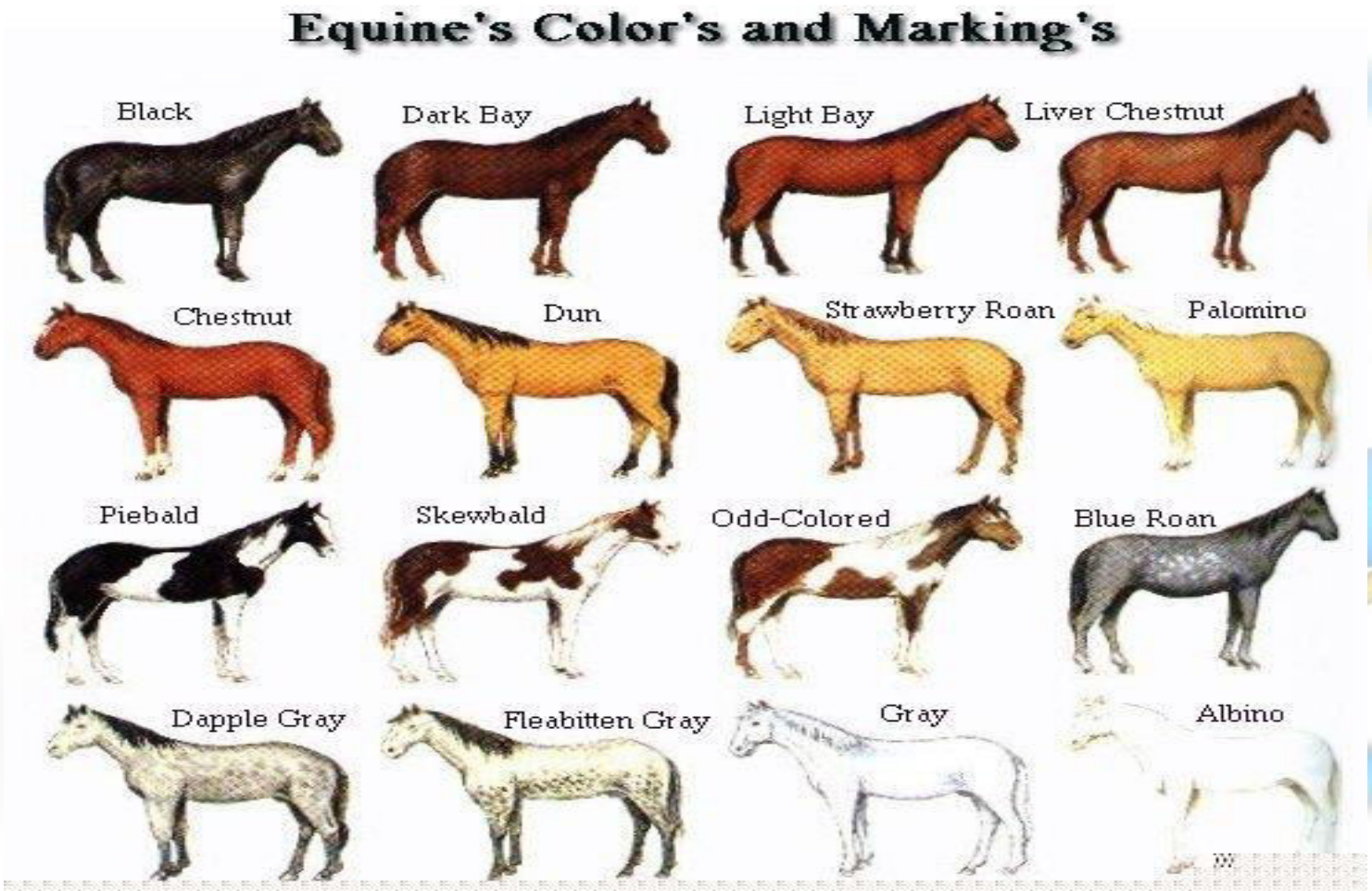
- Horse coloration involves 2 or more gene pairings.
- EE or Ee is for black.
- ee is for red (sorrel).
- Other genes can add effects to base colors.

Epistasis

EXAMPLE

- Bay is AA, EE – black with agouti gene;
- Buckskin is AA, EE, CcrC – bay with cremello gene,
- Dun is AA, EE, Dd – bay with dun gene;
- Palomino is ee, CcrC – sorrel with cremello gene.

Epistasis

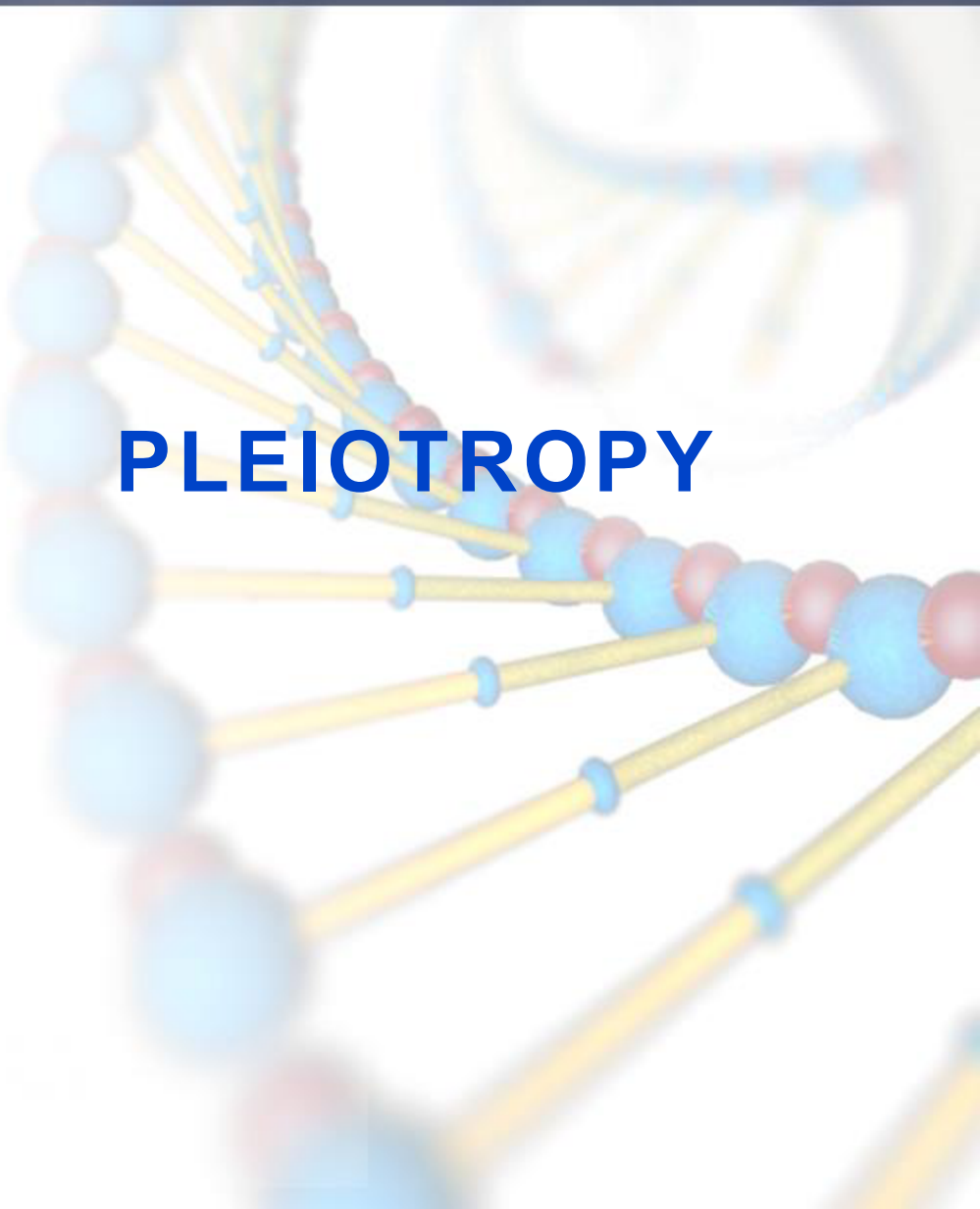

Equine's Color's and Marking's
Black
Dark Bay
Light Bay
Liver Chestnut
Chestnut
Dun
Strawberry Roan
Palomino
Piebald
Skewbald
Odd-Colored
Blue Roan
Dapple Gray
Fleabitten Gray
Gray
Albino

Epistasis

CONCLUSION

- Alleles at one gene locus can hide or prevent expression of alleles at second gene locus is called as epistatsis.

Essentials of Genetics



PLEIOTROPY

Pleiotropy

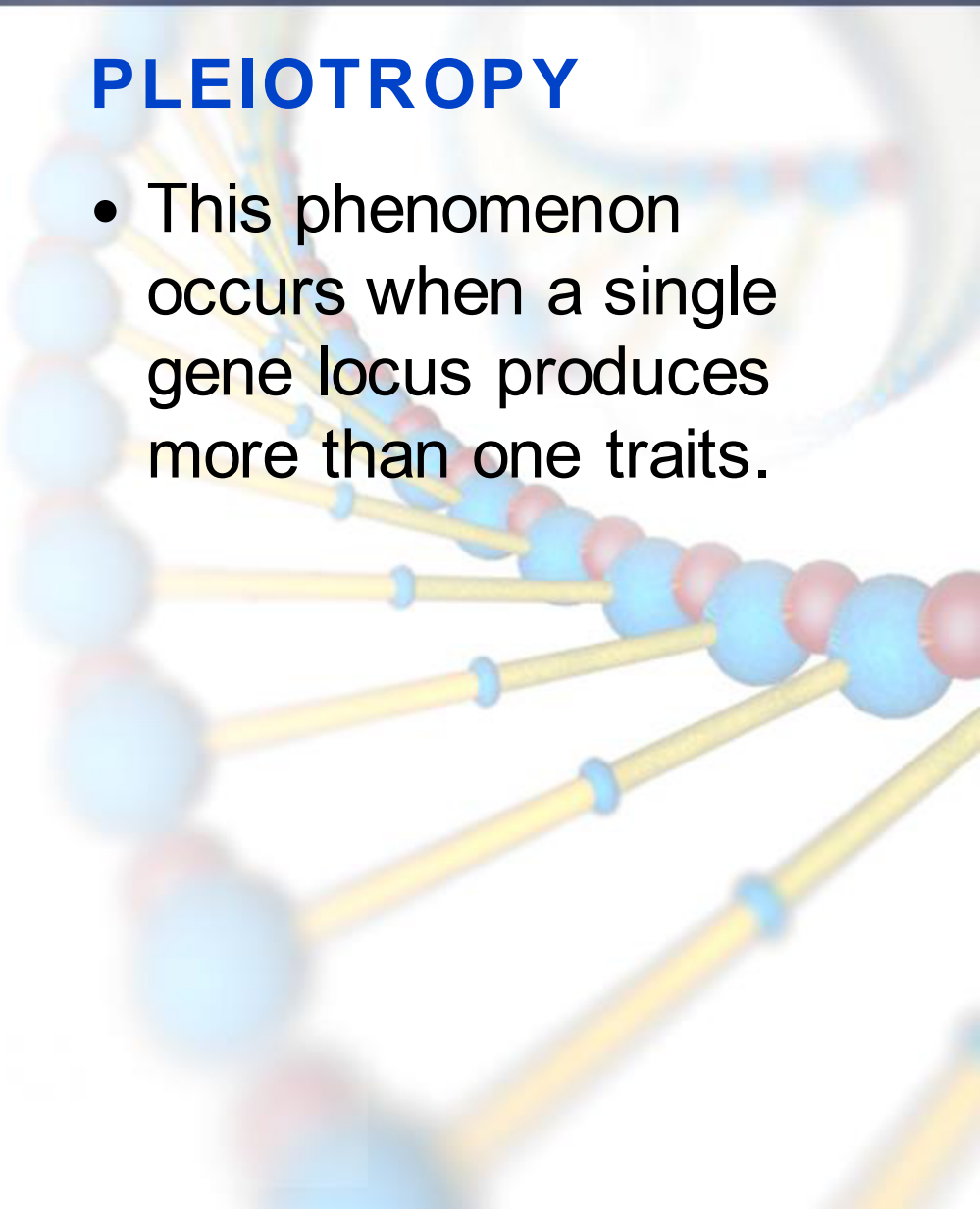
PLEIOTROPY

- The ability of a gene to affect an organism in many ways is called pleiotropy.
- Pleiotropy refers to an allele which has more than one effect on the phenotype.

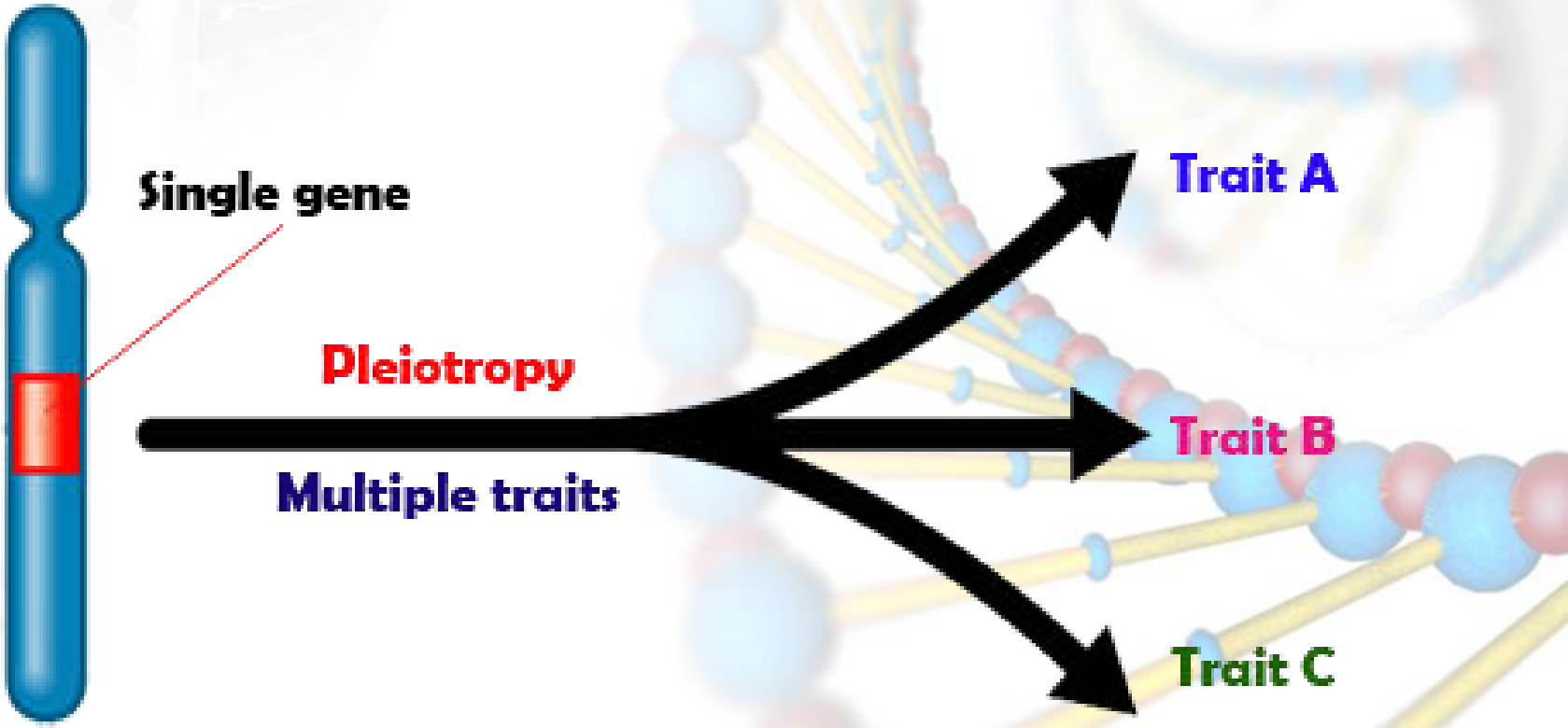
Pleiotropy

PLEIOTROPY

- This phenomenon occurs when a single gene locus produces more than one traits.



Pleiotropy


Single gene
Pleiotropy
Multiple traits
Trait A
Trait B
Trait C

Pleiotropy

PLEIOTROPIC EFFECTS IN HUMANS

- Example:

  Sickle cell anemia

  Cystic fibrosis

Pleiotropy

Two copies of the sickle-cell allele

↓

All hemoglobin is the sickle-cell (abnormal) variety

↓

Abnormal hemoglobin crystallizes when oxygen content of blood is low, causing red blood cells to become sickle-shaped



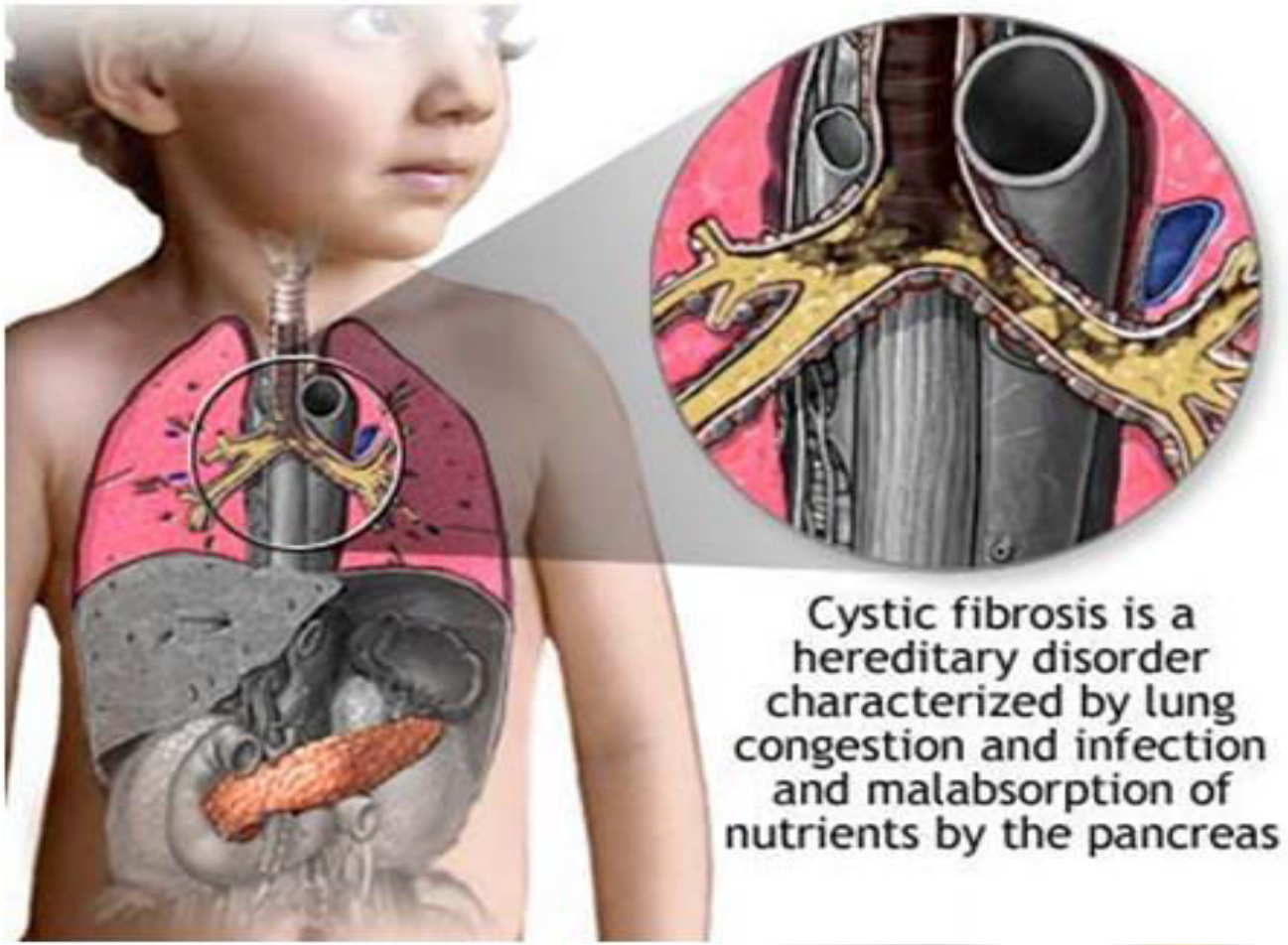
Normal cells



Sickled cells

Pleiotropy

CYSTIC FIBROSIS

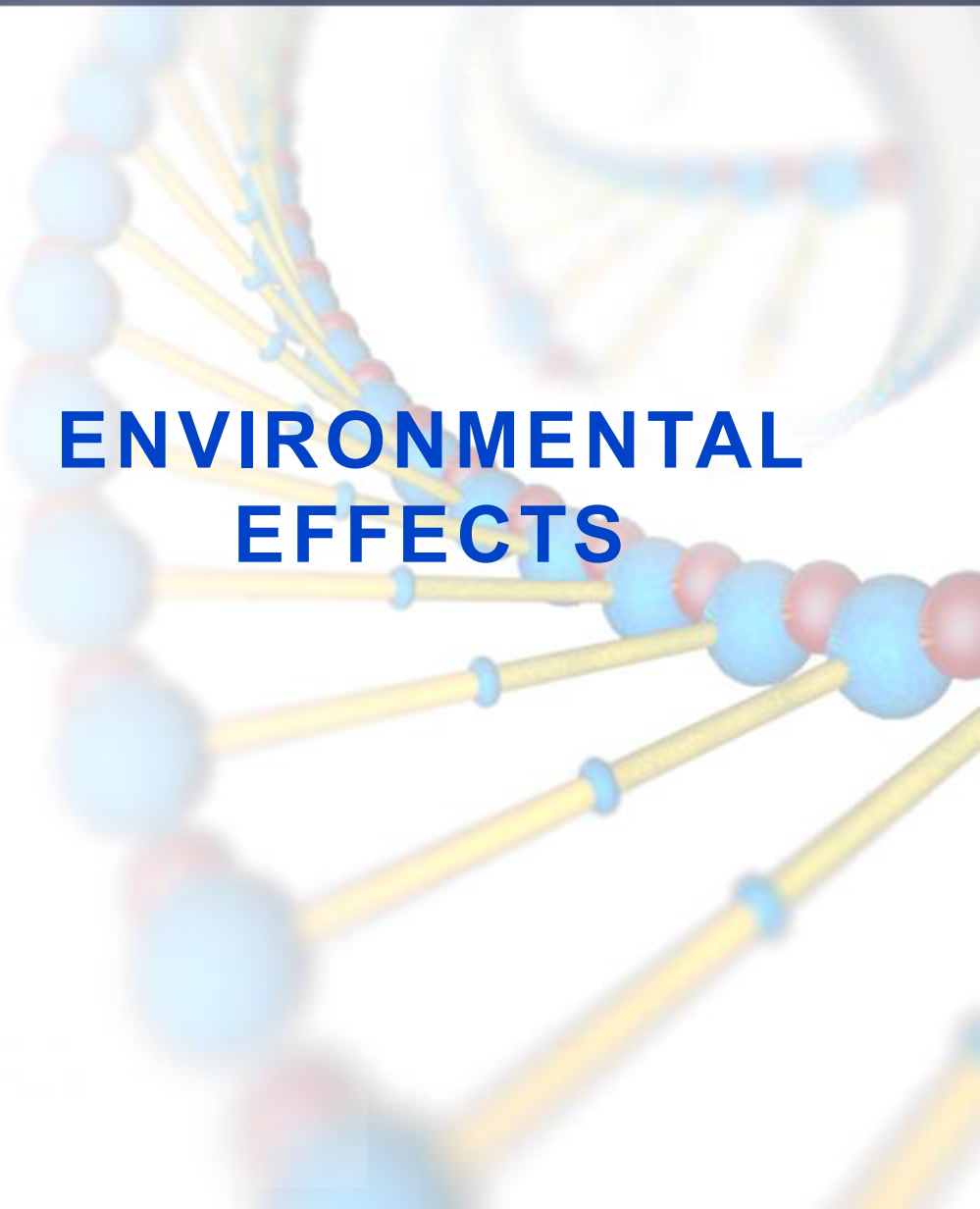


Pleiotropy

HUNTINGTON DISEASE

- One gene – more than one phenotypes
- **Phenotype 1**: progressive dementia at age of 40-50 years
- **Phenotype 2**: death in about 5 years after onset of disease.

Essentials of Genetics



ENVIRONMENTAL EFFECTS

Environmental Effects

ENVIRONMENTAL EFFECTS ON GENE EXPRESSION

- The phenotype of an organism depends not only on which genes it has (genotype), but also on the environment under which it develops.

Environmental Effects

INTERACTION - GENOTYPE AND ENVIRONMENT

- Although scientists agree that phenotype depends on a complex interaction between genotype and environment.
- Debate and controversy about the relative importance of both.

Environmental Effects

ENVIRONMENTAL EFFECTS

- Allele expression may be affected by environmental conditions
- Examples:

  - Coat color in arctic foxes
  - Himalayan rabbits
  - Siamese cats

Environmental Effects

EXPLANATION

- ch allele affected by temp >33 C tyrosinase enzyme inactivates and reduces melanin pigment.



Environmental Effects

Himalayan rabbits



Siamese cats



Environmental Effects

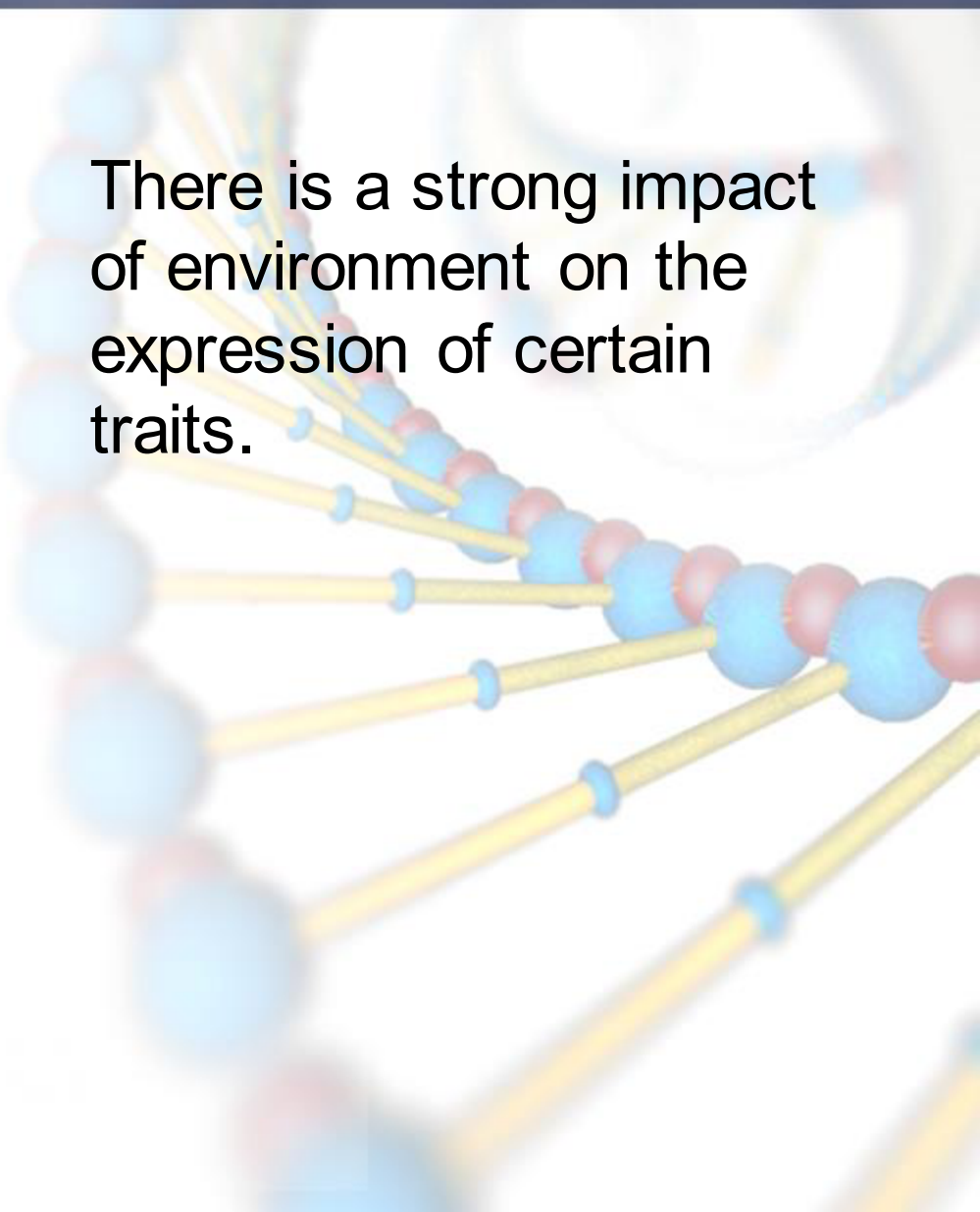
ENVIRONMENTAL EFFECT ON PHENOTYPE

- pH of the soil will change the color of hydrangea flowers.

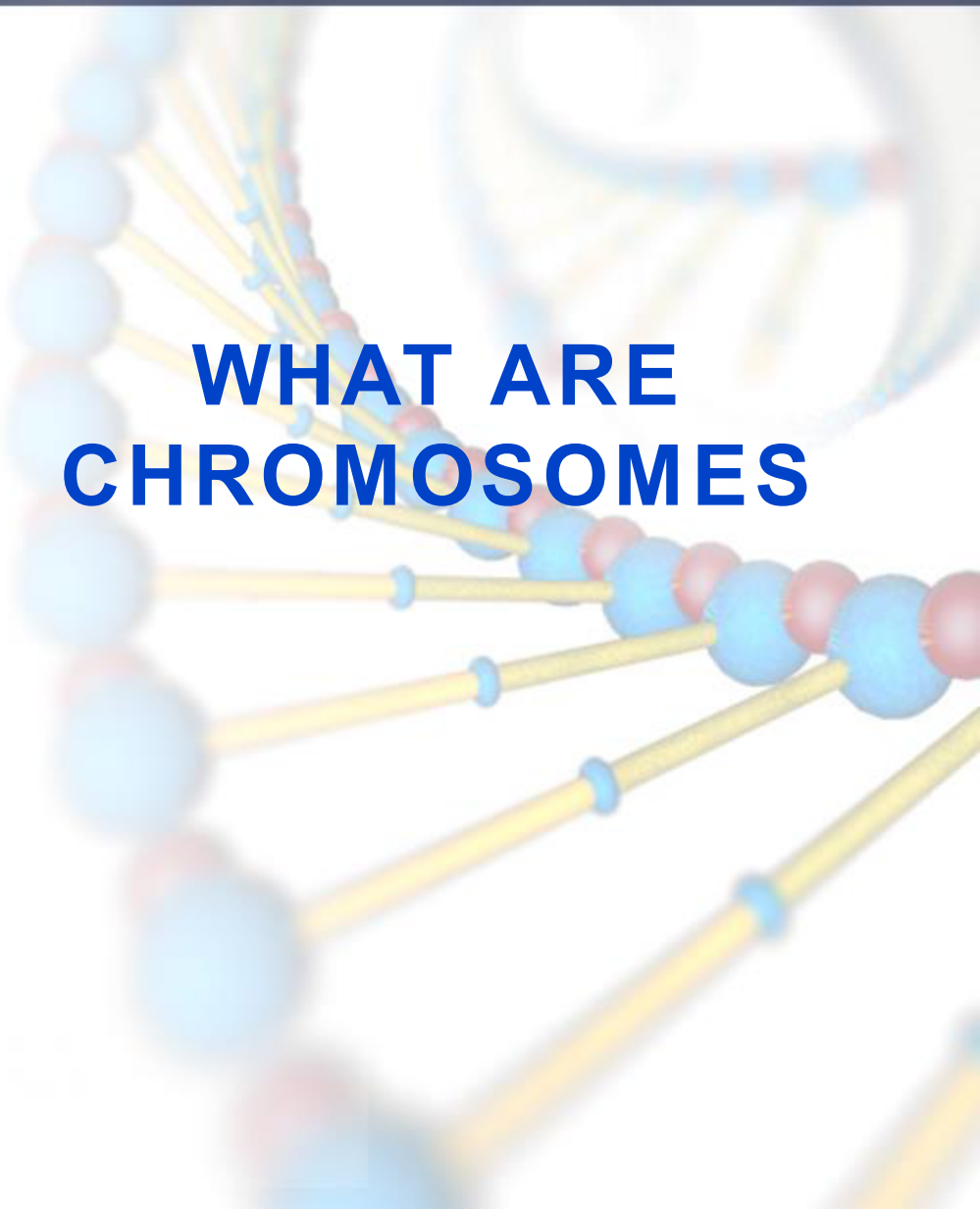


Environmental Effects

There is a strong impact of environment on the expression of certain traits.



Chromosome Structure



WHAT ARE CHROMOSOMES

What are Chromosomes

CHROMOSOMES

- Chromosomes are the rod-shaped, filamentous bodies present in the nucleus.
- They are the carriers of genes.

What are Chromosomes

CHROMOSOMES

- Chromosome not visible in active nucleus due to high water content but can be seen during cell division.

What are Chromosomes

CHROMOSOMES

- Chromosomes were described by Strausberger in 1875.
- Chromosome was first used by Waldeyer in 1888.
- They were given the name chromosome (Chromo = color; Soma = body).

What are Chromosomes

CHROMOSOMES NUMBERS CAN BE COUNTED

- Their numbers can be counted easily only during mitotic metaphase.
- Chromosomes are composed of thin chromatin threads called chromatin fibers.

What are Chromosomes

CHROMATIN FIBERS FOLDING

- Chromatin fibers undergo folding, coiling and supercoiling during prophase so that the chromosomes become progressively thicker and smaller.

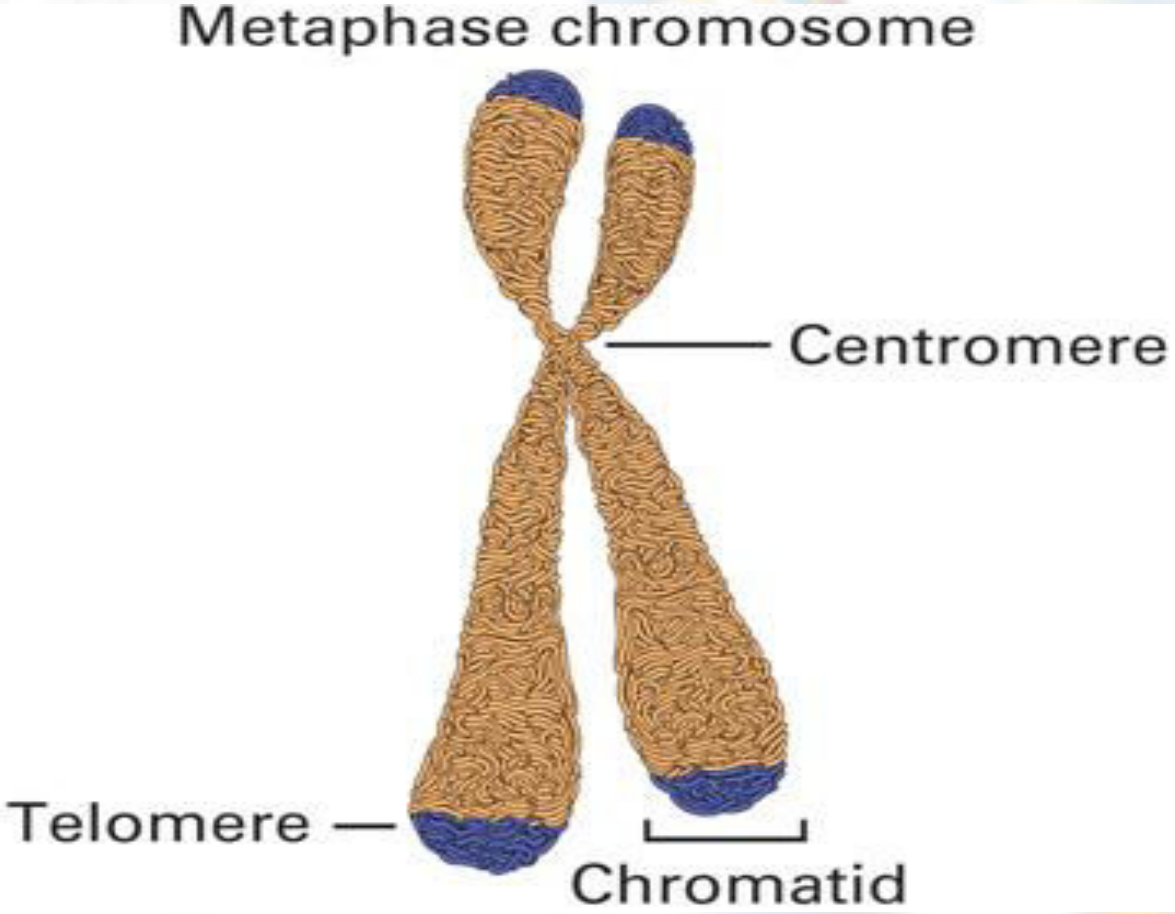
What are Chromosomes

CHROMOSOMES

- At the end of cell division, the fibers uncoil and extend as fine chromatin threads, which are not visible at light microscope.

What are Chromosomes

A TYPICAL CHROMOSOME

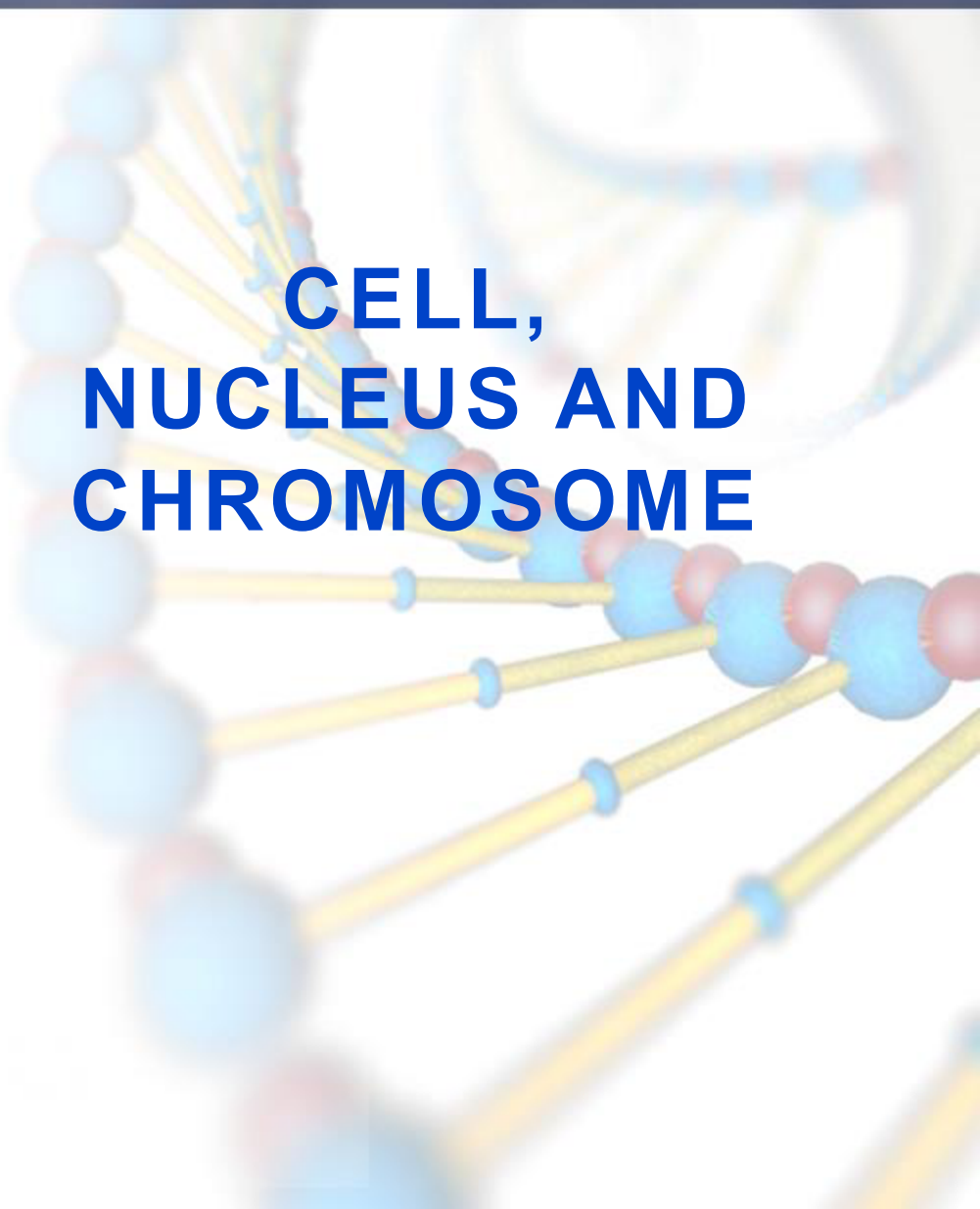


What are Chromosomes

CHROMOSOMES-CONCLUSION

- Chrosomes are rod shaped filamentous bodies present in the prokaryotic/eukaryotic cells.

Chromosome Structure



CELL, NUCLEUS AND CHROMOSOME

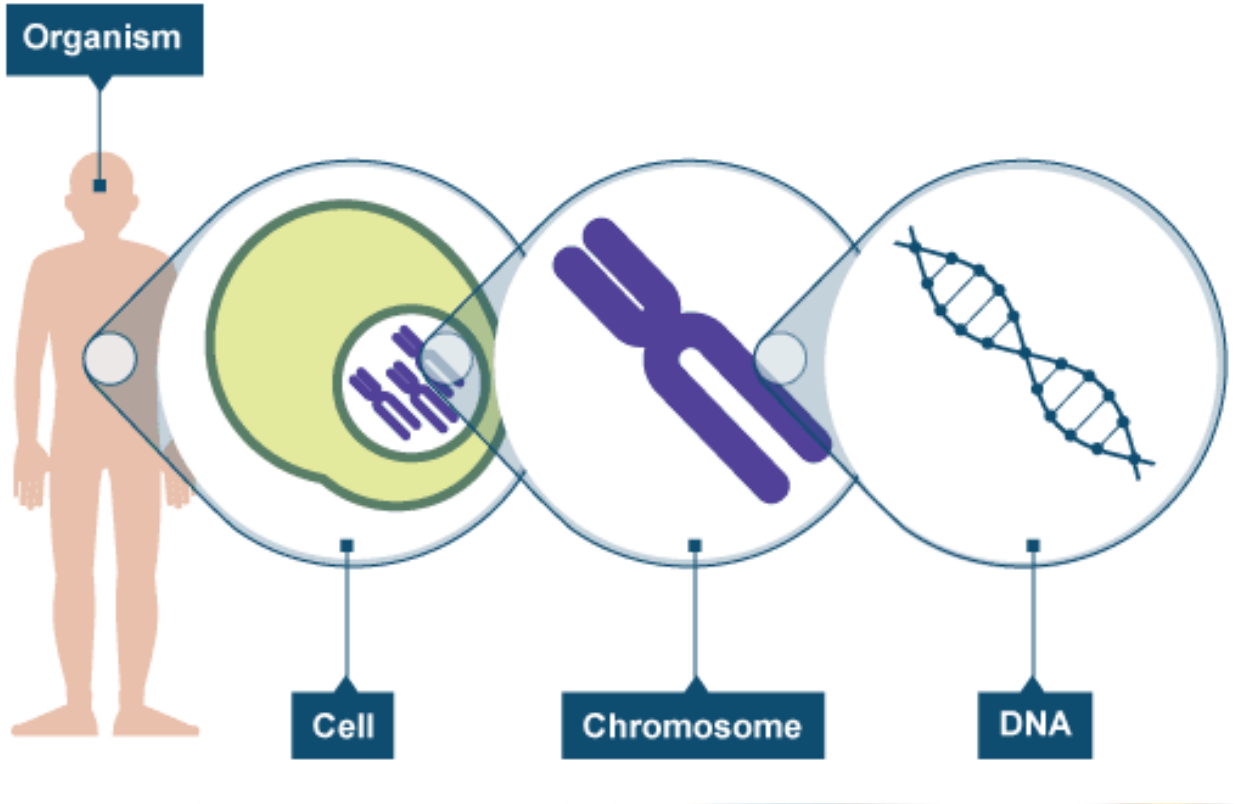
Cell, Nucleus and Chromosome

CELL, NUCLEUS AND CHROMOSOMES

- Living organisms are made up of cells.
- Within cell, there is nucleus or nucleoid region.
- DNA or chromosomes are present in nucleus/nucleoid region.

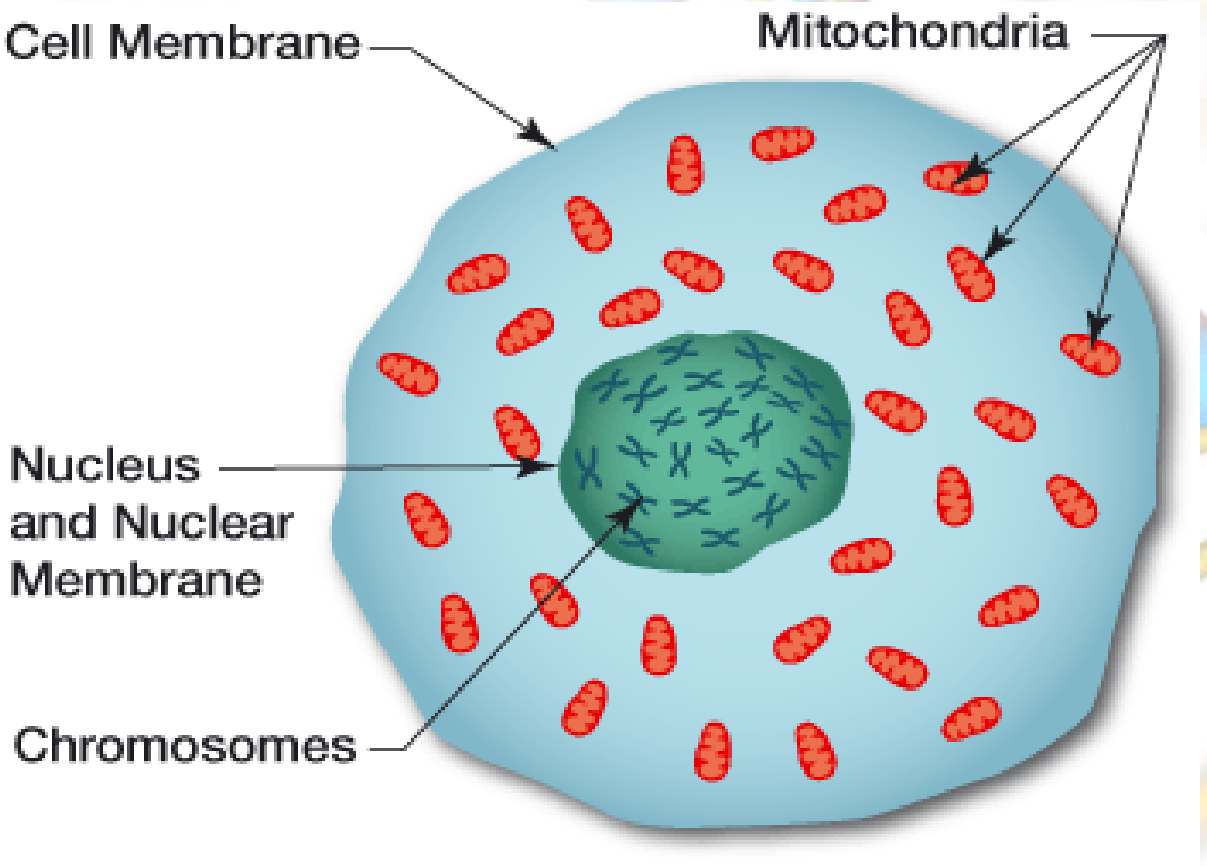
Cell, Nucleus and Chromosome

LIVING ORGANISM MADE UP OF CELLS

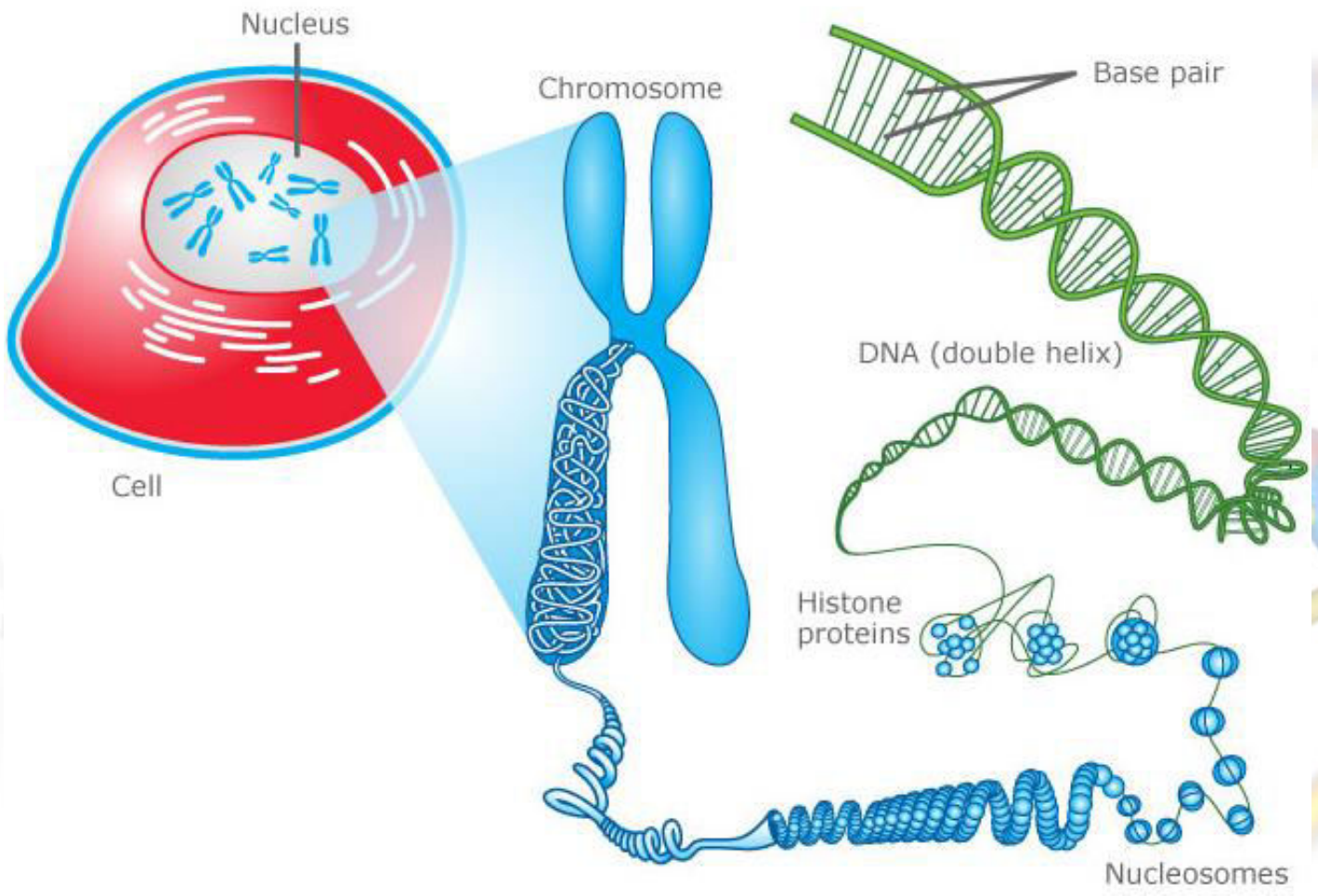


Cell, Nucleus and Chromosome

CHROMOSOMES ARE PRESENT WITHIN NUCLEUS



Cell, Nucleus and Chromosome



Cell, Nucleus and Chromosome

CHROMOSOMES

- Chromosome structure made of DNA and associated proteins.
- Chromosome carries part or all of a cell's genetic information.

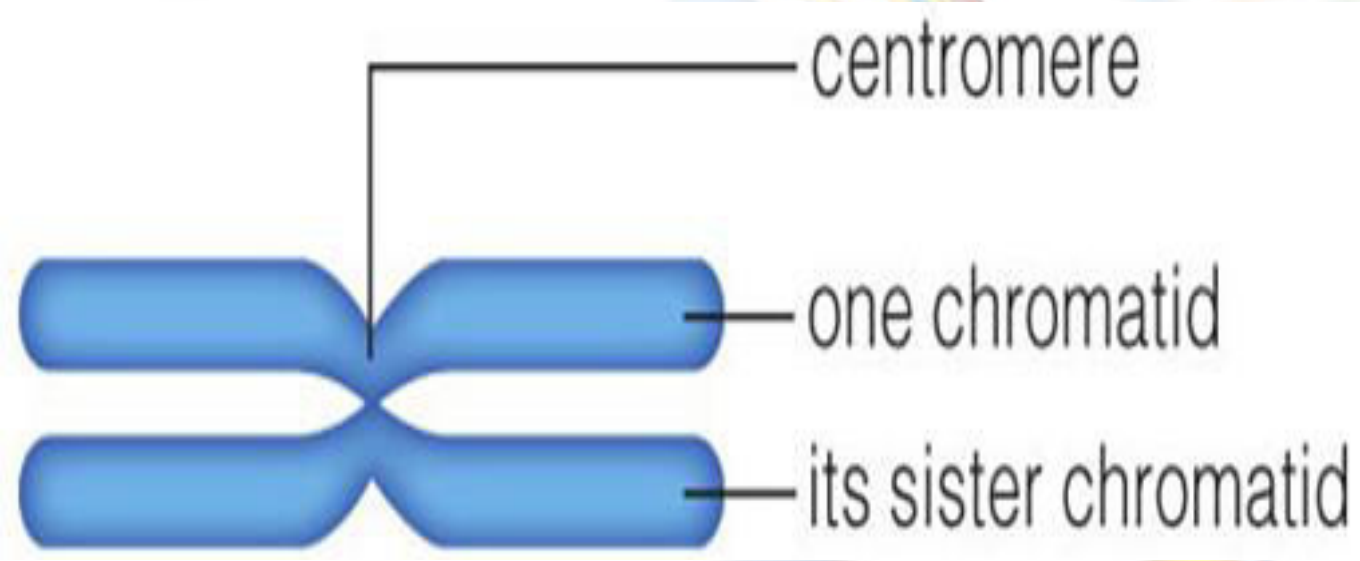
Cell, Nucleus and Chromosome

CHROMOSOME STRUCTURE -

- Sister chromatids
- Centromere is constricted region, where chromatids are attached.

Cell, Nucleus and Chromosome

CHROMOSOME STRUCTURE



Cell, Nucleus and Chromosome

CHROMOSOME CONSTITUTES

- A chromosome constitutes DNA molecule + protein
- Histones and supercoiled DNA

Chromosome Structure

CHROMOSOME NUMBERS IN DIFFERENT SPECIES

Chromosome in Different Species

VARIATION IN NUMBER OF CHROMOSOMES

- Different organisms have different number of chromosomes.

Chromosome in Different Species

VARIATION IN CHROMOSOME NUMBER

- Most organisms have chromosomes numbers between 12 to 50.
- Round worm has only two chromosomes.
- Protozoa having 300 or more.

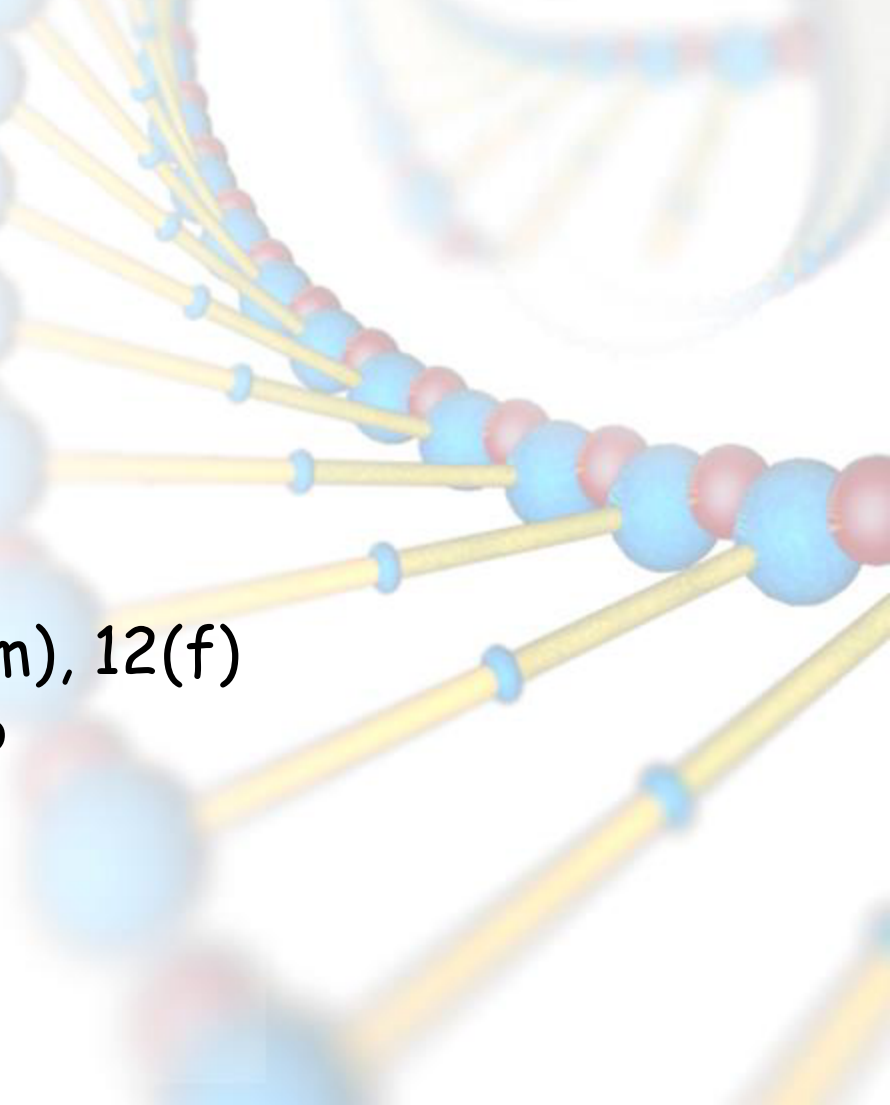
Chromosome in Different Species

CHROMOSOME NUMBER IN DIFFERENT ORGANISMS

- Fungi have 3-8 chromosomes.
- Angiosperms have from 8 – 16 (most common number 12).

Chromosome in Different Species

Organism	No. chromosomes
Human	46
Chimpanzee	48
Dog	78
Horse	64
Chicken	78
Goldfish	94
Fruit fly	8
Mosquito	6
Nematode	11(m), 12(f)
Horsetail	216
Sequoia	22
Round worm	2



Chromosome in Different Species

CHROMOSOME NUMBERS

Organism	No. chromosomes
Onion	16
Mold	16
Carrot	20
Tomato	24
Tobacco	48
Rice	24
Maize	20
Haploppus gracilis	4
Crepis capillaris	6

Chromosome in Different Species

CHROMOSOMES IN HUMANS

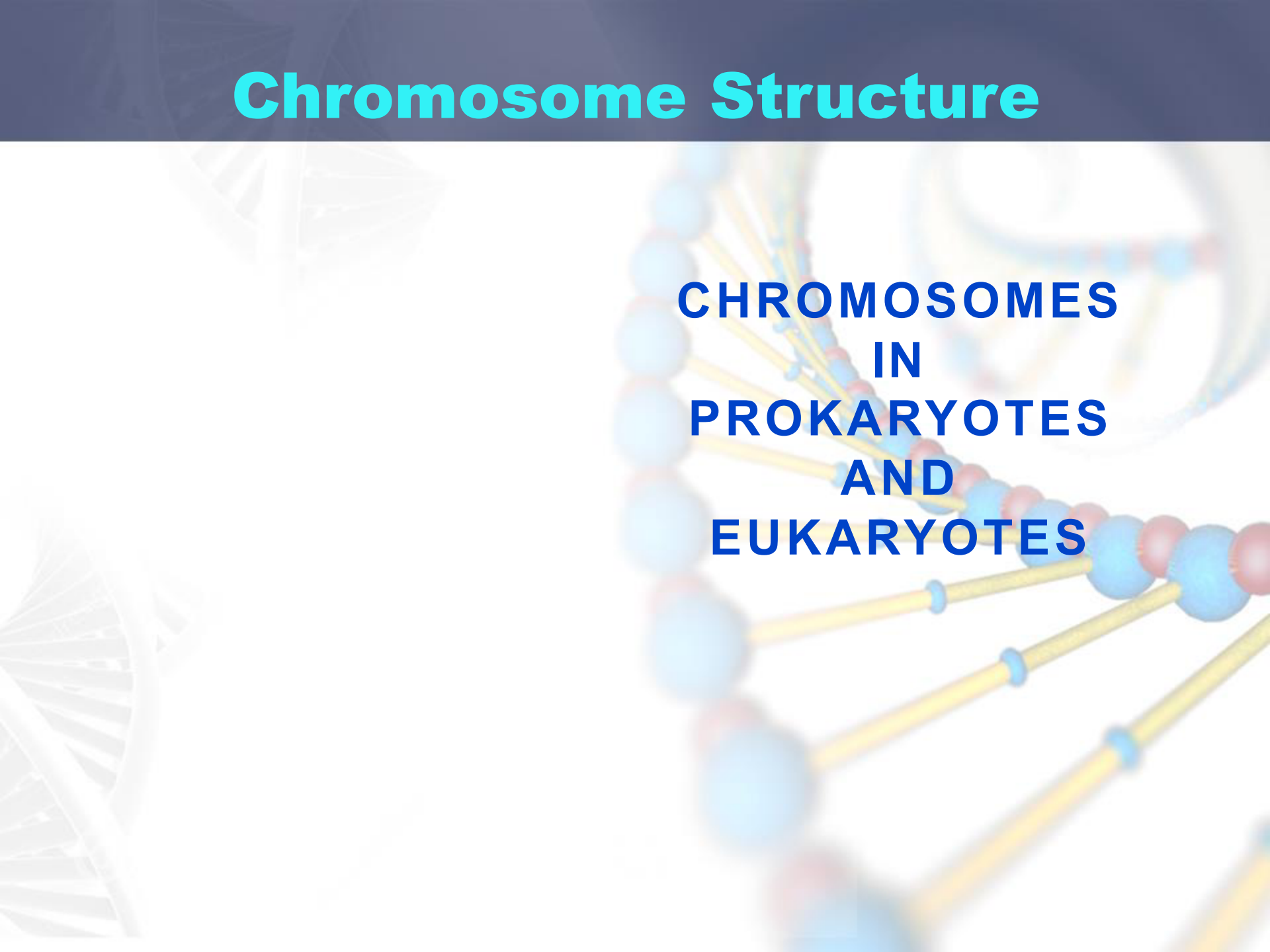
- 46 chromosomes

  23 pairs

  22 pair of autosomes

  one pair of sex chromosomes

Chromosome Structure



CHROMOSOMES IN PROKARYOTES AND EUKARYOTES

Chromosomes-Prokaryotes/Eukaryotes

CHROMOSOMES IN PROKARYOTES AND EUKARYOTES

- Chromosomes are present in both kind of cells.
- Although they are different in both type of cells.

Chromosomes-Prokaryotes/Eukaryotes

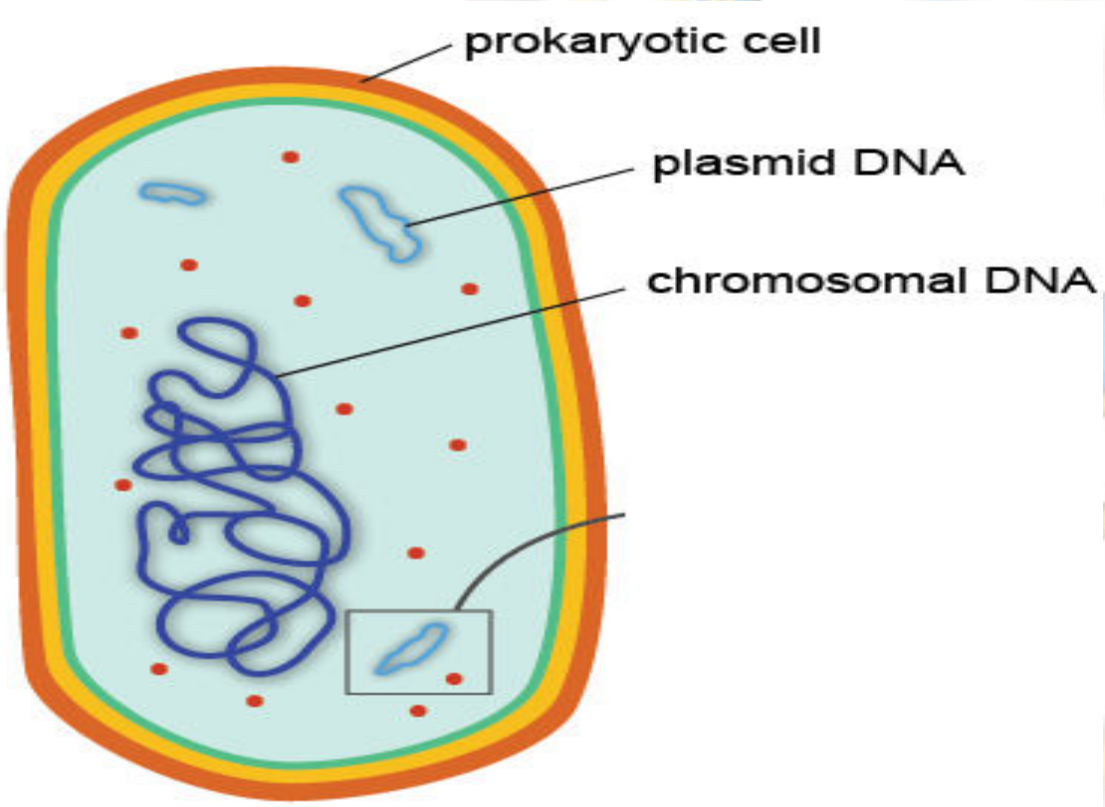
CHROMOSOMES IN PROKARYOTES

- Single chromosome plus plasmids
- Circular chromosome
- Made only of DNA
- Found in cytoplasm

Chromosomes-Prokaryotes/Eukaryotes

CHROMOSOMES IN PROKARYOTES

- Single chromosome, plasmid



Chromosomes-Prokaryotes/Eukaryotes

CHROMOSOMES IN EUKARYOTES

- Many chromosomes.
- Linear chromosomes.
- Made of chromatin, a nucleoprotein – histones.
- Found in a nucleus.

Chromosomes-Prokaryotes/Eukaryotes

CHROMOSOMES IN EUKARYOTES

- Eukaryotic chromosomes are condensed and visible during cell division.
- At the beginning of mitosis can be seen - consist of two threads (sister chromatids) joined by a centromere.

Chromosomes-Prokaryotes/Eukaryotes

CHROMOSOMES IN EUKARYOTES

- The sister chromatids are identical copies.
- During mitosis, the sister chromatids separate and are placed into two daughter nuclei.

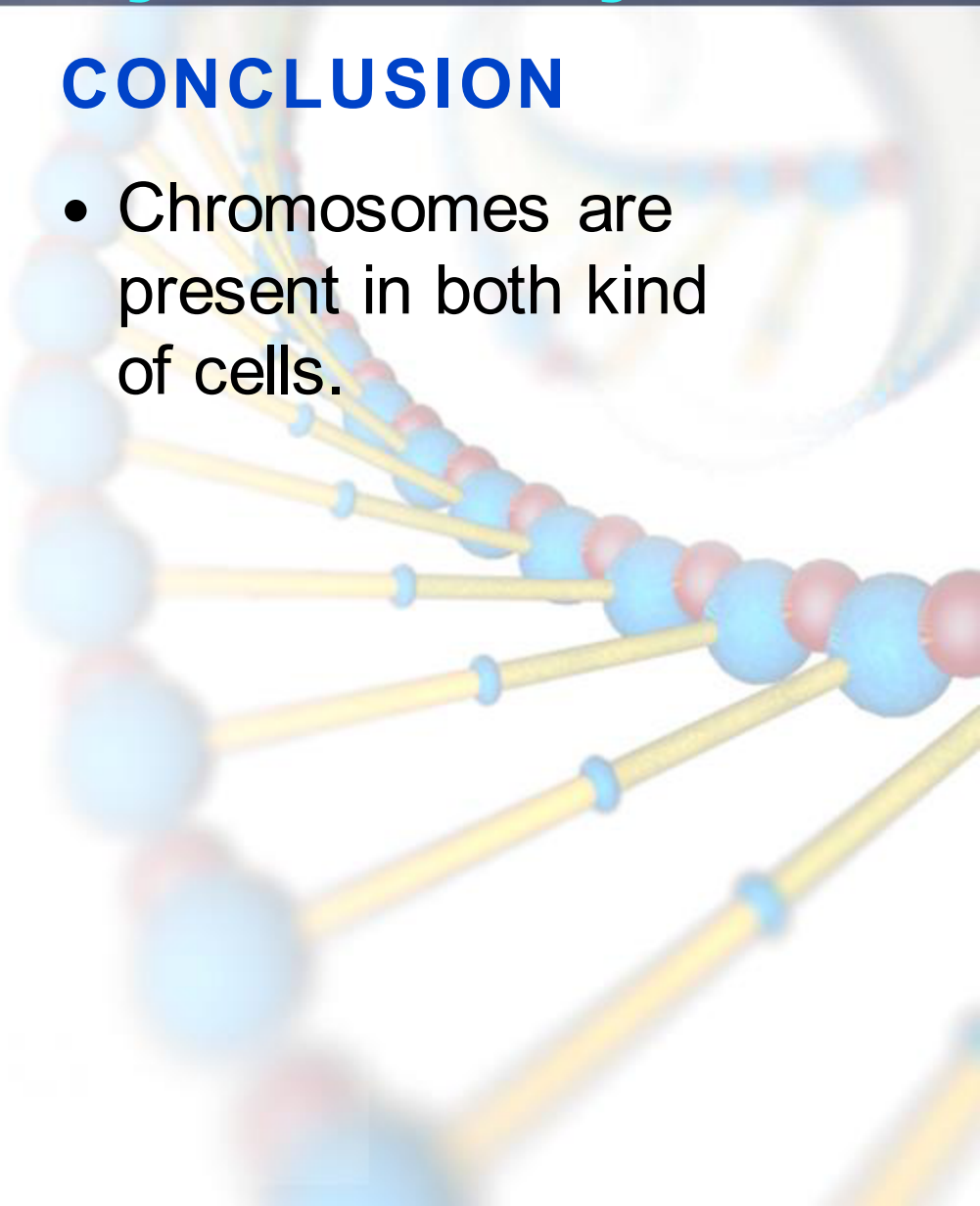
Chromosomes-Prokaryotes/Eukaryotes

PROKARYOTES	EUKARYOTES
single chromosome plus plasmids	many chromosomes
circular chromosome	linear chromosomes
made only of DNA	made of DNA and histones.
found in cytoplasm	found in a nucleus
copies its chromosome and divides immediately afterwards	copies chromosomes, then the cell grows, then goes through mitosis into two groups

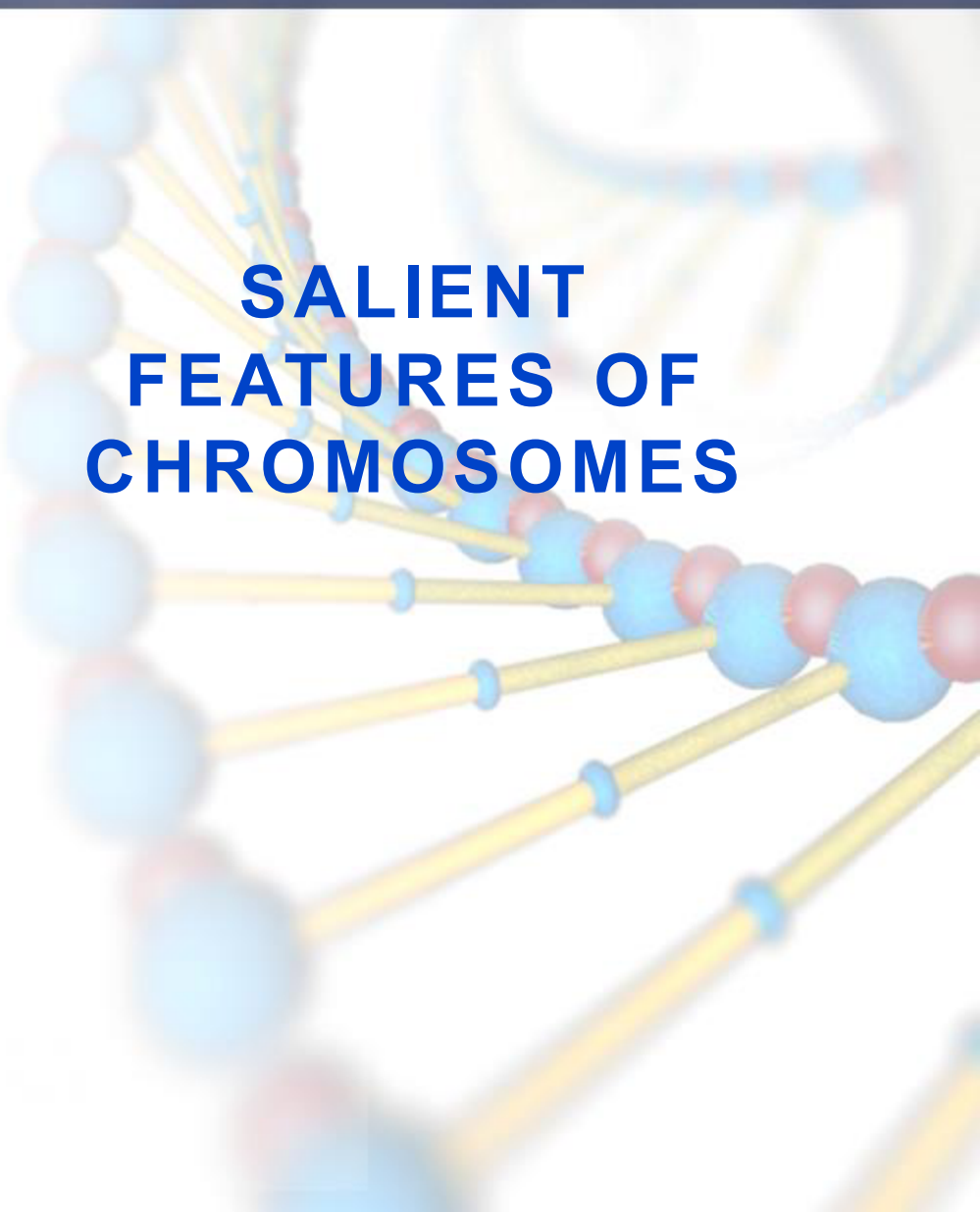
Chromosomes-Prokaryotes/Eukaryotes

CONCLUSION

- Chromosomes are present in both kind of cells.



Chromosome Structure

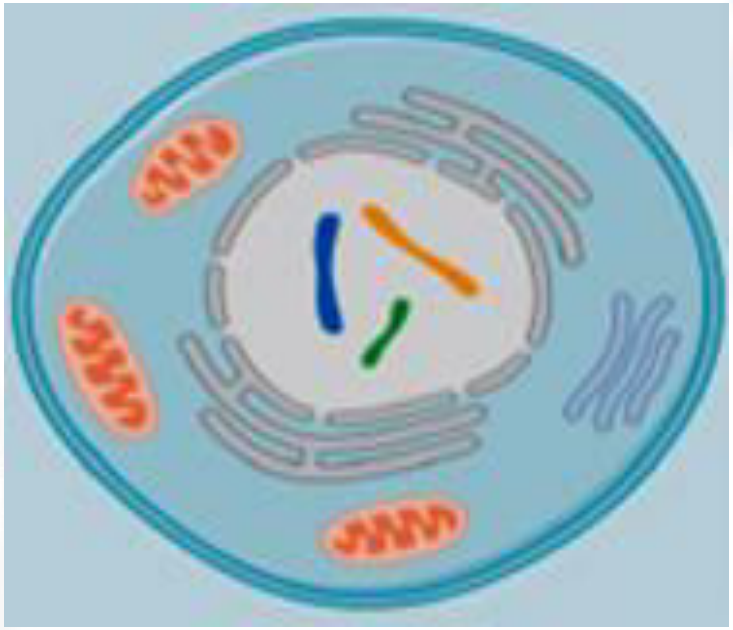


SALIENT FEATURES OF CHROMOSOMES

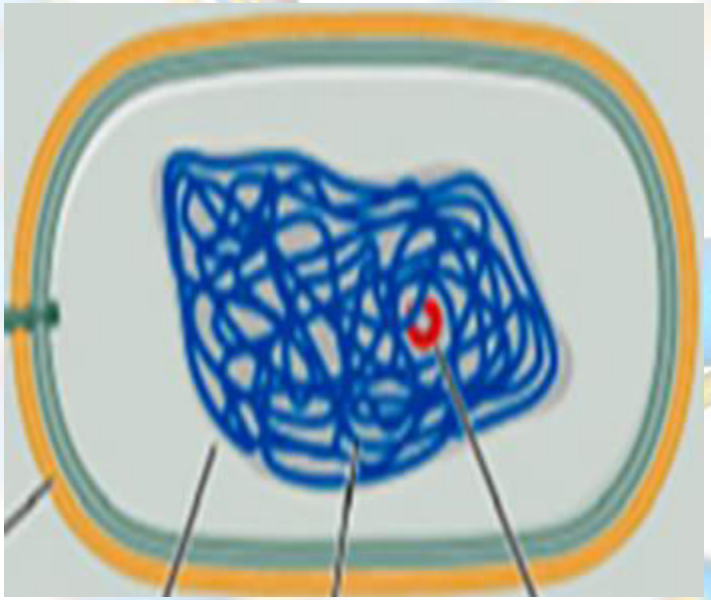
Salient features of Chromosomes

SHAPE OF CHROMOSOMES

- Chromosomes can be circular or linear



Chromosomes in Eukaryotic cell

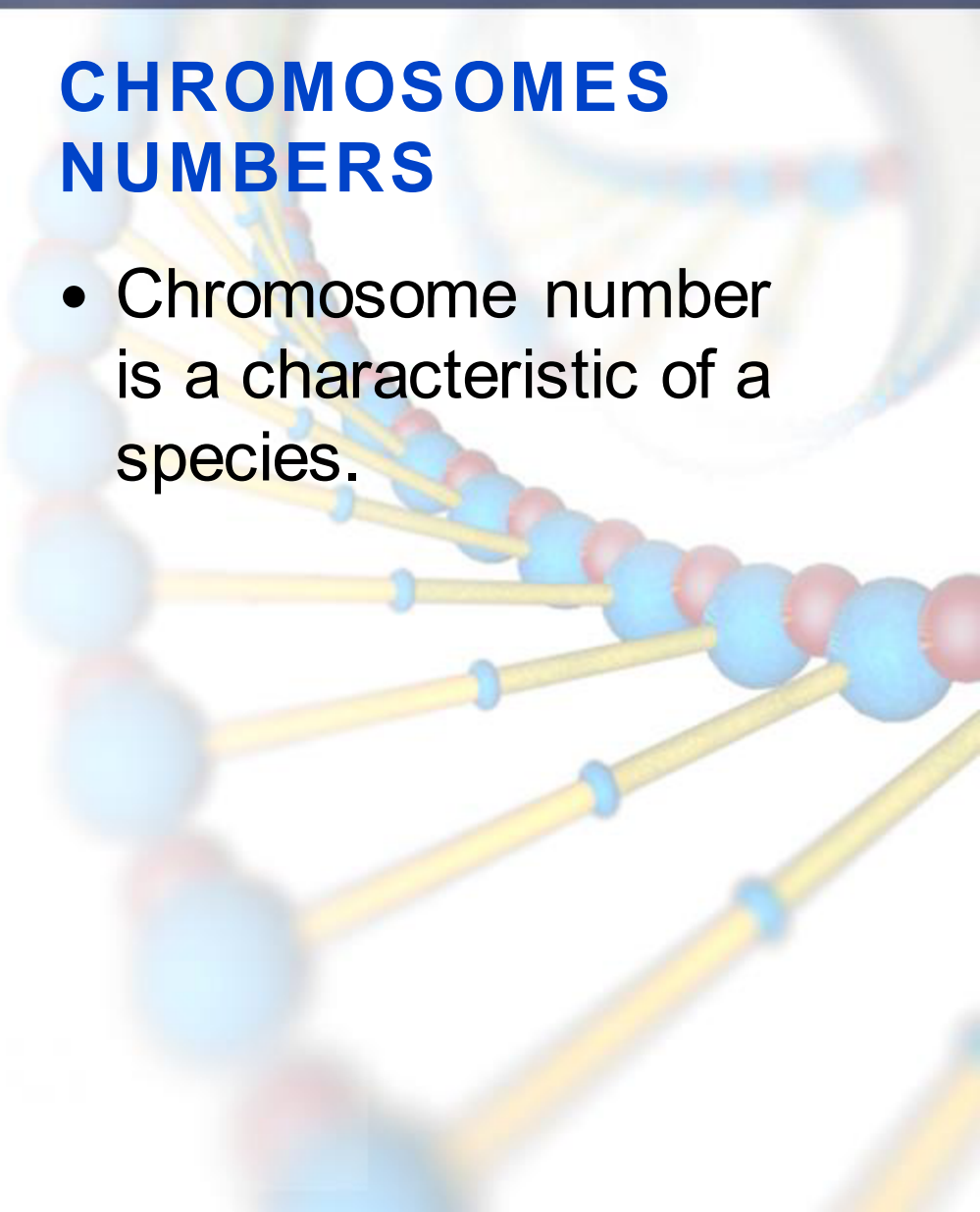


Chromosome in Prokaryotic cell

Salient features of Chromosomes

CHROMOSOMES NUMBERS

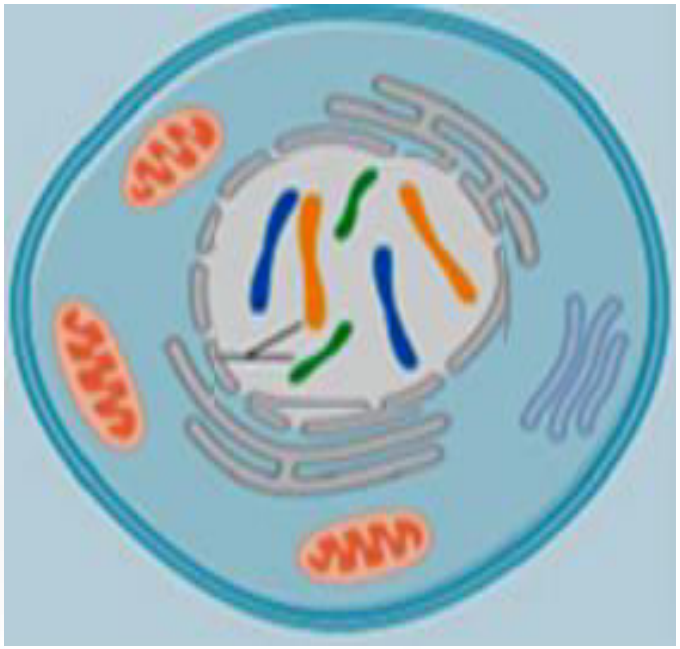
- Chromosome number is a characteristic of a species.



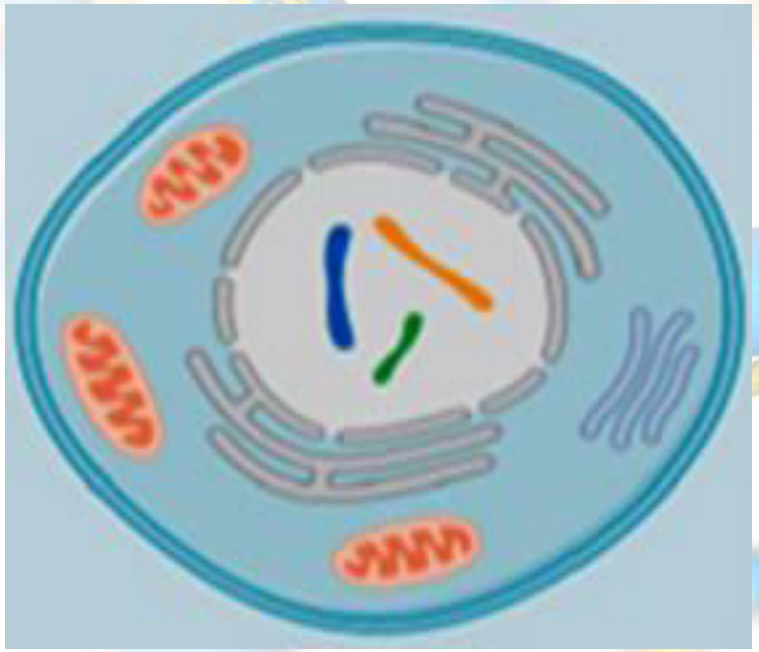
Salient features of Chromosomes

COPY NUMBER OF CHROMOSOMES

- Chromosome can be haploid, diploid or polyploid.



Diploid



Haploid

Salient features of Chromosomes

GENES ON CHROMOSOMES

- Number of genes are different in each of the species.
- Humans 25,000 - 27,000
- Rice 45,000
- Drosophila 13,700
- Yeast 5,800

Salient features of Chromosomes

GENE DENSITY ON CHROMOSOMES

- Prokaryotes have higher gene density than eukaryotes per Mb of DNA.

Salient features of Chromosomes

GENOME AND GENE DENSITY

Species	Genome size (Mb)	Approximate number of genes*	Gene density (genes/Mb)*
PROKARYOTES (bacteria)			
Mycoplasma genitalium	0.58	500	860
Streptococcus pneumonia	2.2	2,300	1,060
Escherichia coli K-12	4.6	4,400	950
Agrobacterium tumefaciens	5.7	5,400	960
Sinorhizobium meliloti	6.7	6,200	930
EUKARYOTES (animals)			
Fungi			
Saccharomyces cerevisiae	12	5,800	480
Schizosaccharomyces pombe	12	4,900	410
Protozoa			
Tetrahymena thermophila	220	> 20,000	> 90
Invertebrates			
Caenorhabditis elegans	97	19,000	200
Drosophila melanogaster	180	13,700	80
Strongylocentrotus purpuratus	845	~22,000	~26
Locusta migratoria	5,000	nd	nd
Vertebrates			
Fugu rubripes	365	> 31,000	> 85
Homo sapiens	2,900	27,000	9.3
Mus musculus	2,500	29,000	12
Plants			
Arabidopsis thaliana	125	25,500	200
Oryza sativa (rice)	430	> 45,000	> 100
Zea mays	2,200	> 45,000	> 20
Fritillaria assyriaca 1(tulip)	120,000	nd	nd

Salient features of Chromosomes

GENOME CHARACTERISTICS

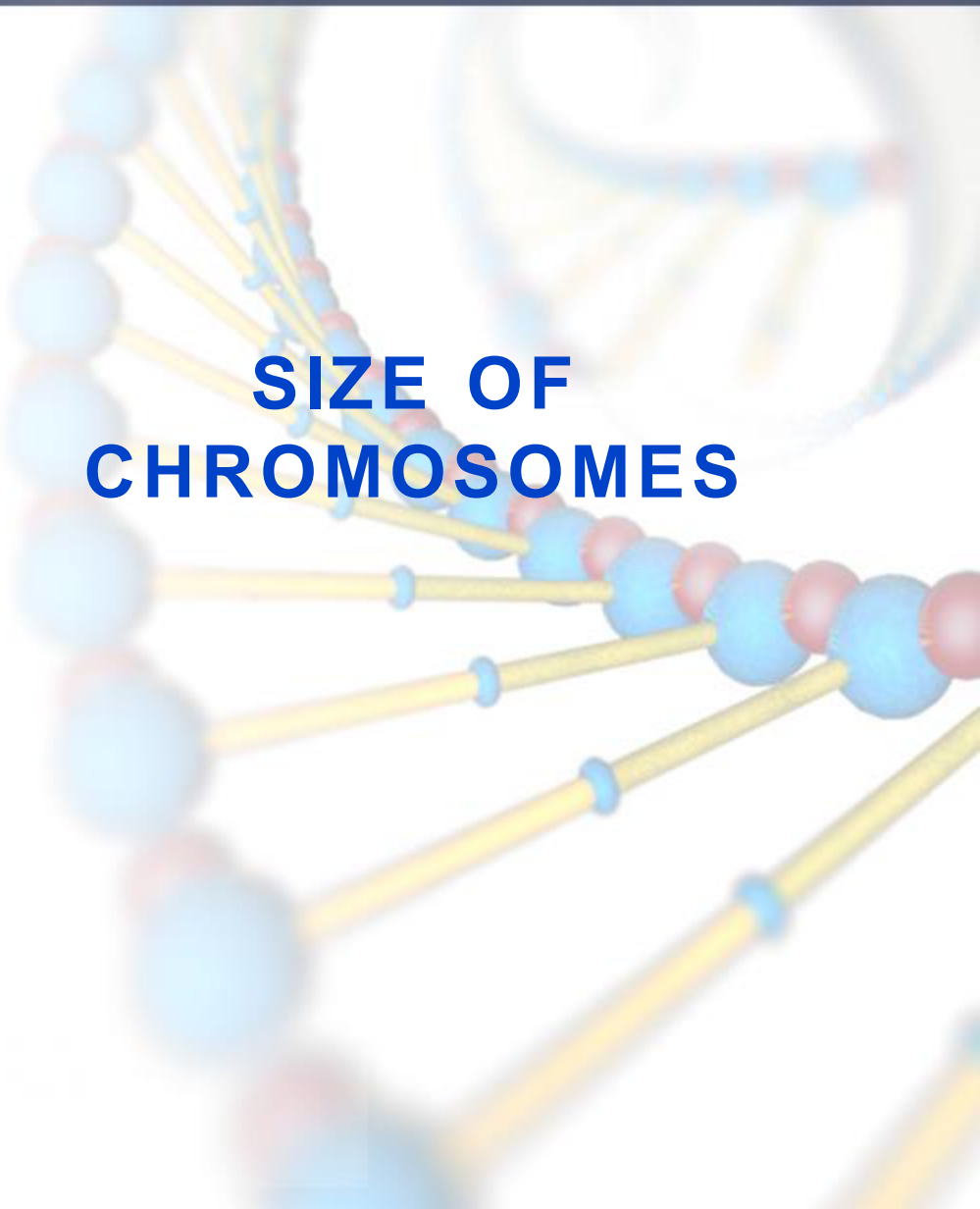
- Prokaryotes have less genome size as compared to eukaryotes.
- Prokaryotes have less number of genes as compared to eukaryotes.

Salient features of Chromosomes

GENOME CHARACTERISTICS

- Prokaryotes have higher gene density as compared to eukaryotes.

Chromosome Structure



SIZE OF CHROMOSOMES

Size of Chromosomes

SIZE OF CHROMOSOMES

- The size of the chromosomes varies between 0.5 μ to 32 μ in length.
- Diameter varies - 0.2 μ and 3.0 μ.
- The longest chromosomes found in Trillium - 32 μ.

Size of Chromosomes

SIZE OF CHROMOSOMES

- The giant chromosomes found in diptera and they may be as long as 300 μ and up to 10 μ in diameter.
- In general, plants have longer chromosomes than animals.

Size of Chromosomes

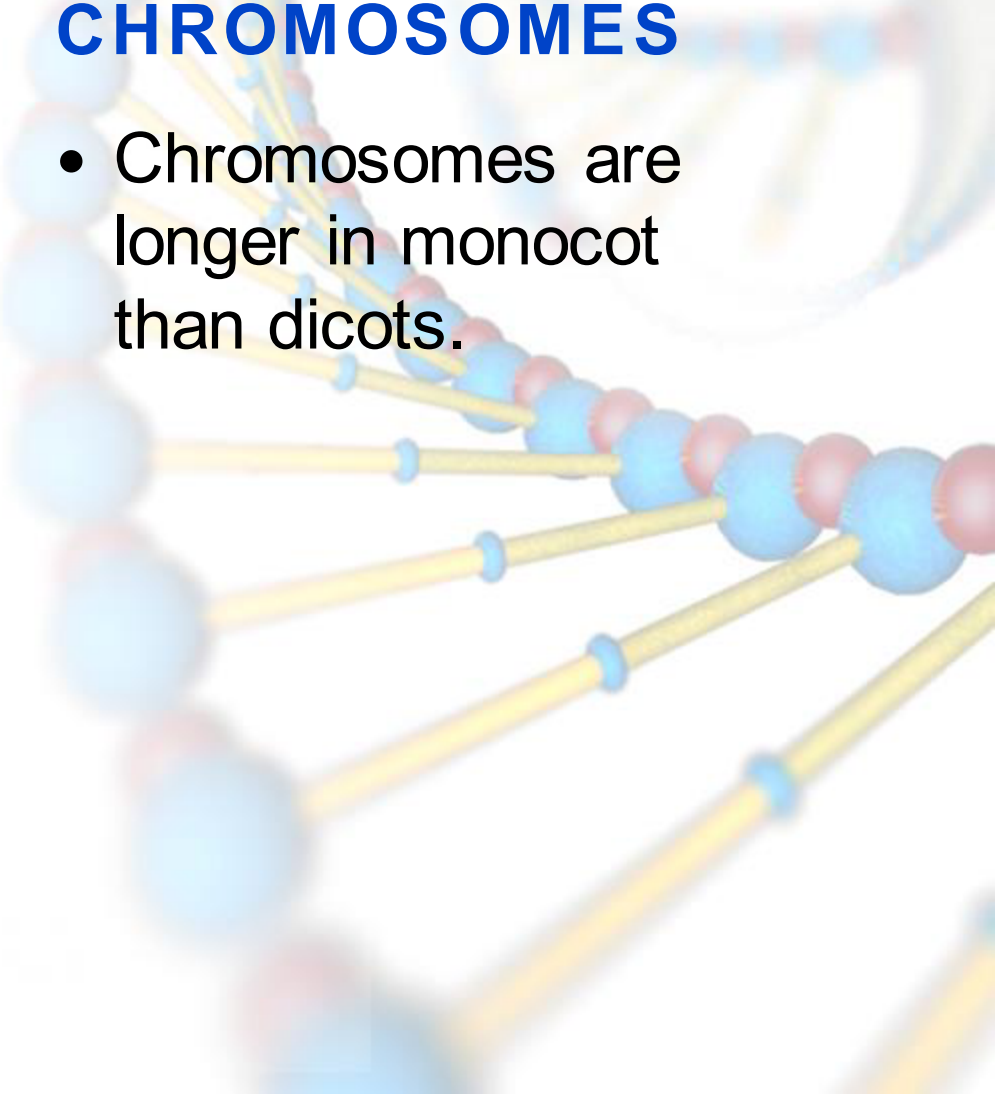
SIZE OF CHROMOSOMES

- Species having less chromosome number usually have long chromosomes.
- Dicots in general, have a higher number of chromosome than monocots.

Size of Chromosomes

SIZE OF CHROMOSOMES

- Chromosomes are longer in monocot than dicots.



Size of Chromosomes

SIZE AT STAGES OF MITOSIS

- Interphase: chromosome are longest & thinnest.
- Prophase: there is a progressive decrease in their length accompanied with an increase in thickness.

Size of Chromosomes

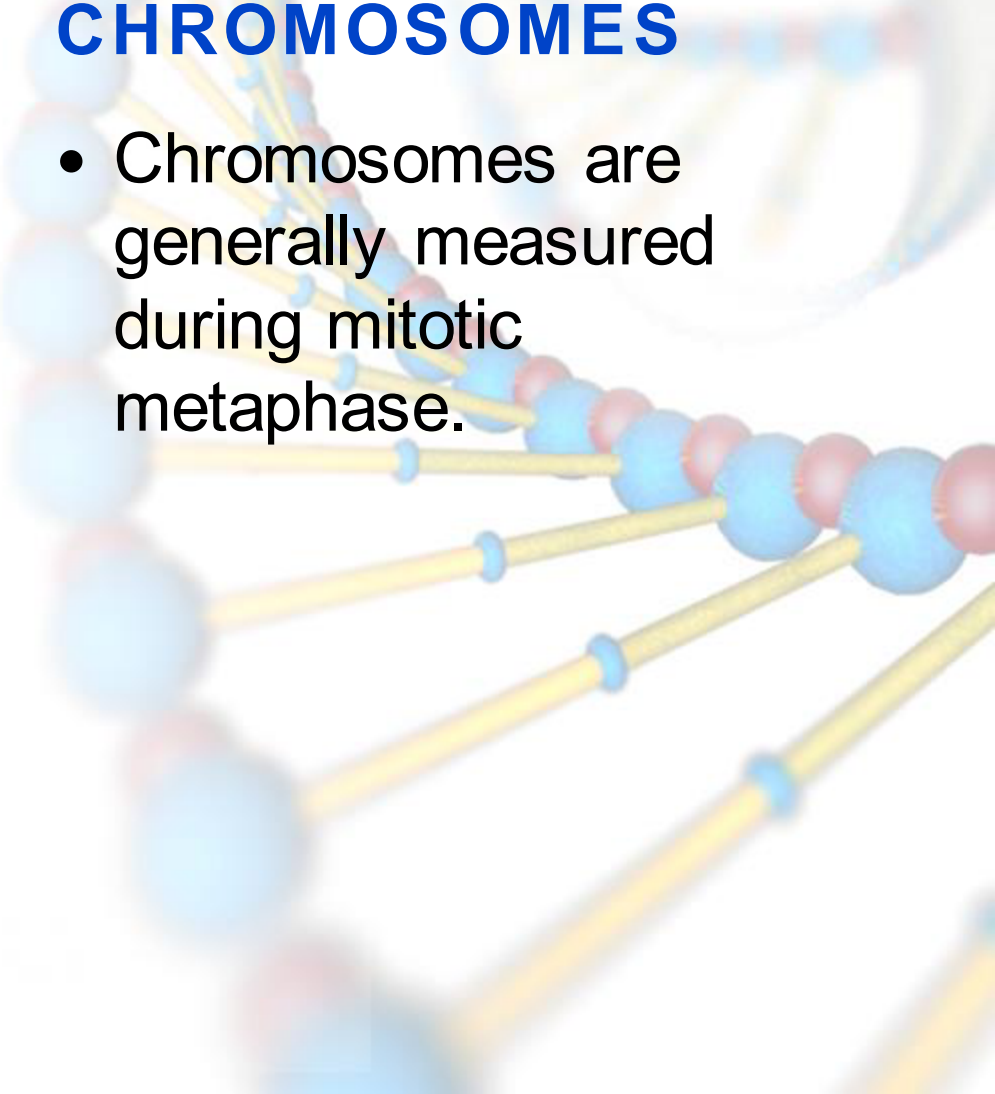
SIZE AT STAGES OF MITOSIS

- Anaphase: chromosomes are smallest.
- Metaphase: chromosomes can be easily observed.
- Chromosomes are studied during metaphase

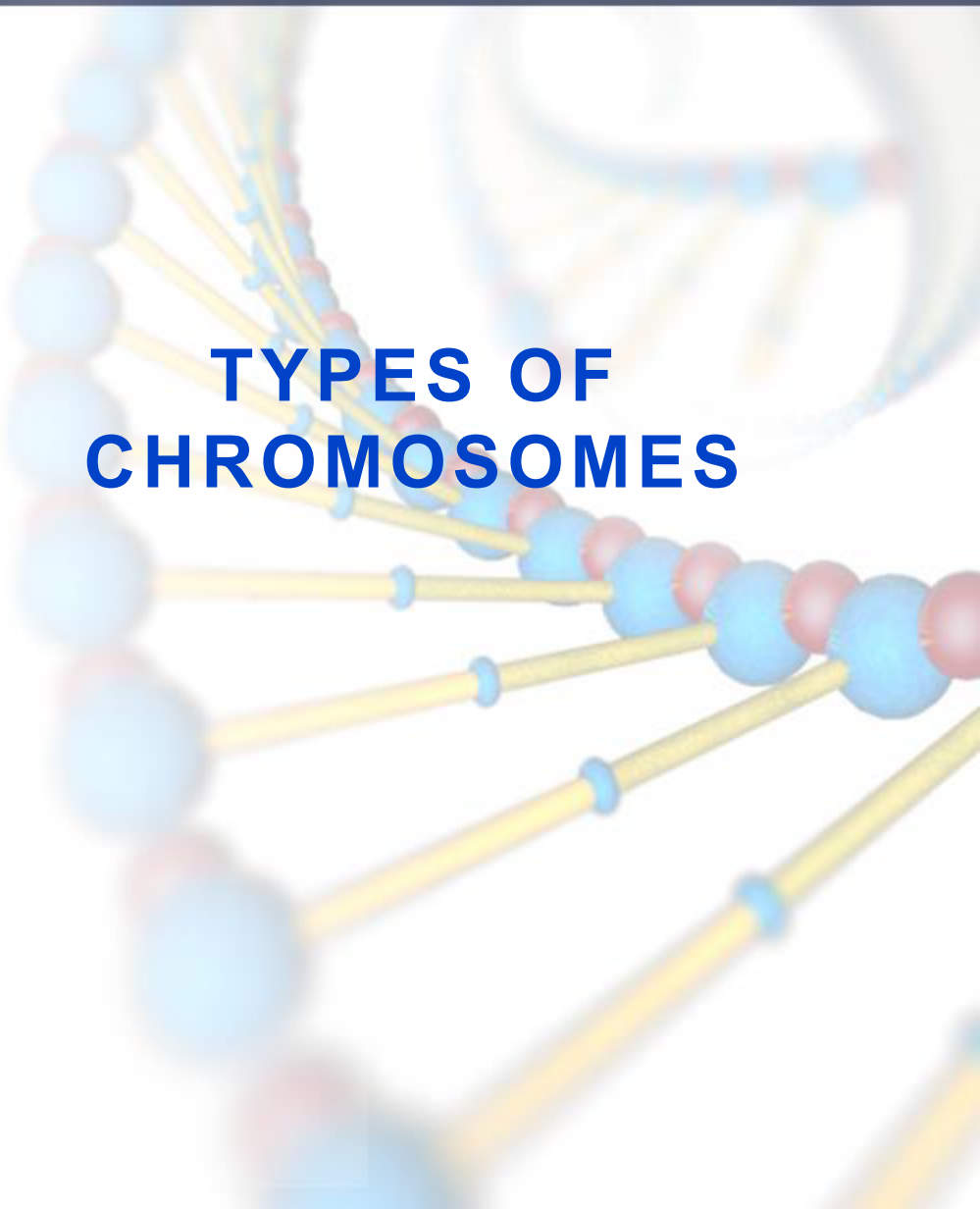
Size of Chromosomes

MEASUREMENT OF CHROMOSOMES

- Chromosomes are generally measured during mitotic metaphase.



Chromosome Structure



TYPES OF CHROMOSOMES

Types of Chromosomes

TYPES OF CHROMOSOMES

- Two types of eukaryotic chromosomes:
  - Autosomes
  - Sex chromosomes

Types of Chromosomes

AUTOSOMES

- Paired chromosomes with the same length, shape, centromere location, and genes.
- Any chromosomes other than sex chromosomes are called as autosomes.

Types of Chromosomes

SEX CHROMOSOMES

- Members of a pair of chromosomes that differ between males and females.
- Bases on this pair of chromosomes, sex is identified.

Types of Chromosomes

TYPE OF CHROMOSOMES BASED ON CENTROMERE POSITION

- Chromosomes may differ in the position of the centromere.

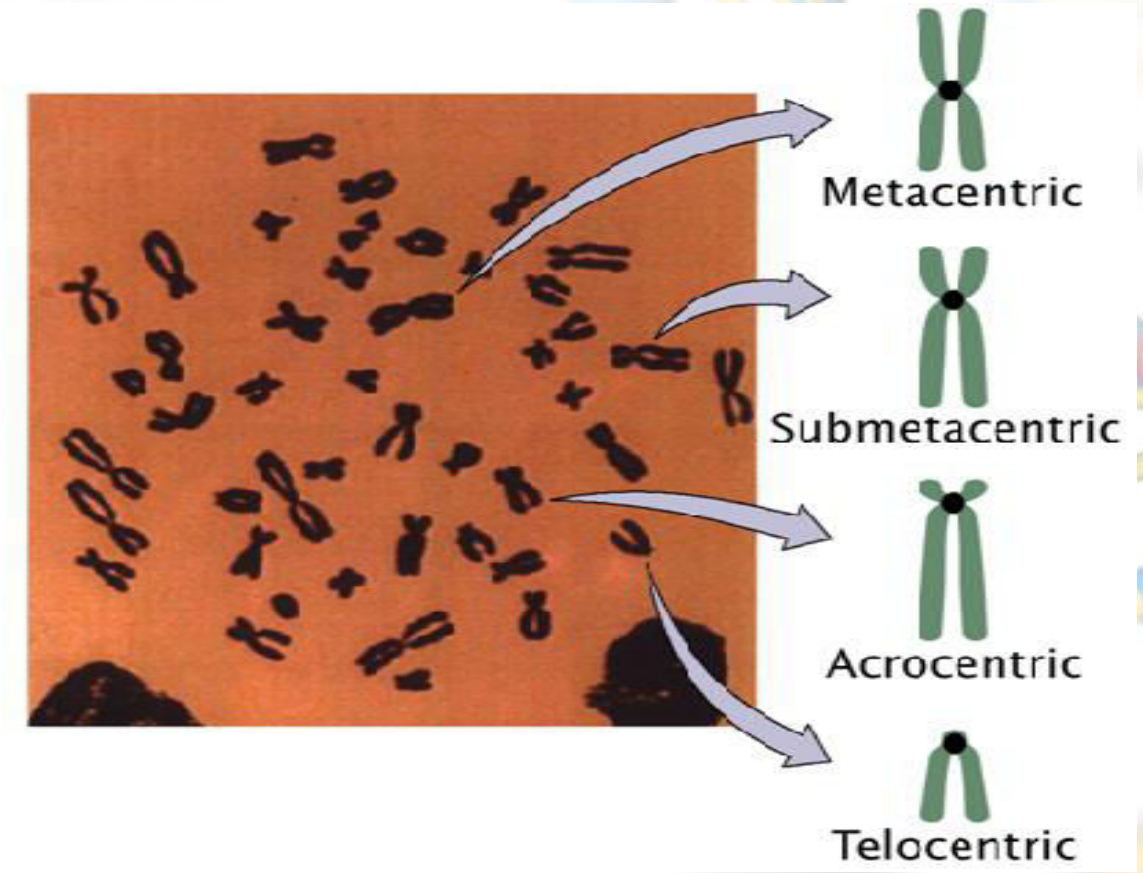
Types of Chromosomes

TYPE OF CHROMOSOMES - CENTROMERE

- Centromere is near the middle, metacentric
- Centromere is toward one end, acrocentric or submetacentric
- Centromere is very near to end, telocentric.

Types of Chromosomes

TYPES OF CHROMOSOMES BASED ON CENTROMERE POSITION



Types of Chromosomes



A - Metacentric chromosome
B - Sub-metacentric chromosome
C - Acrocentric chromosome
D - Telocentric chromosome

Types of Chromosomes

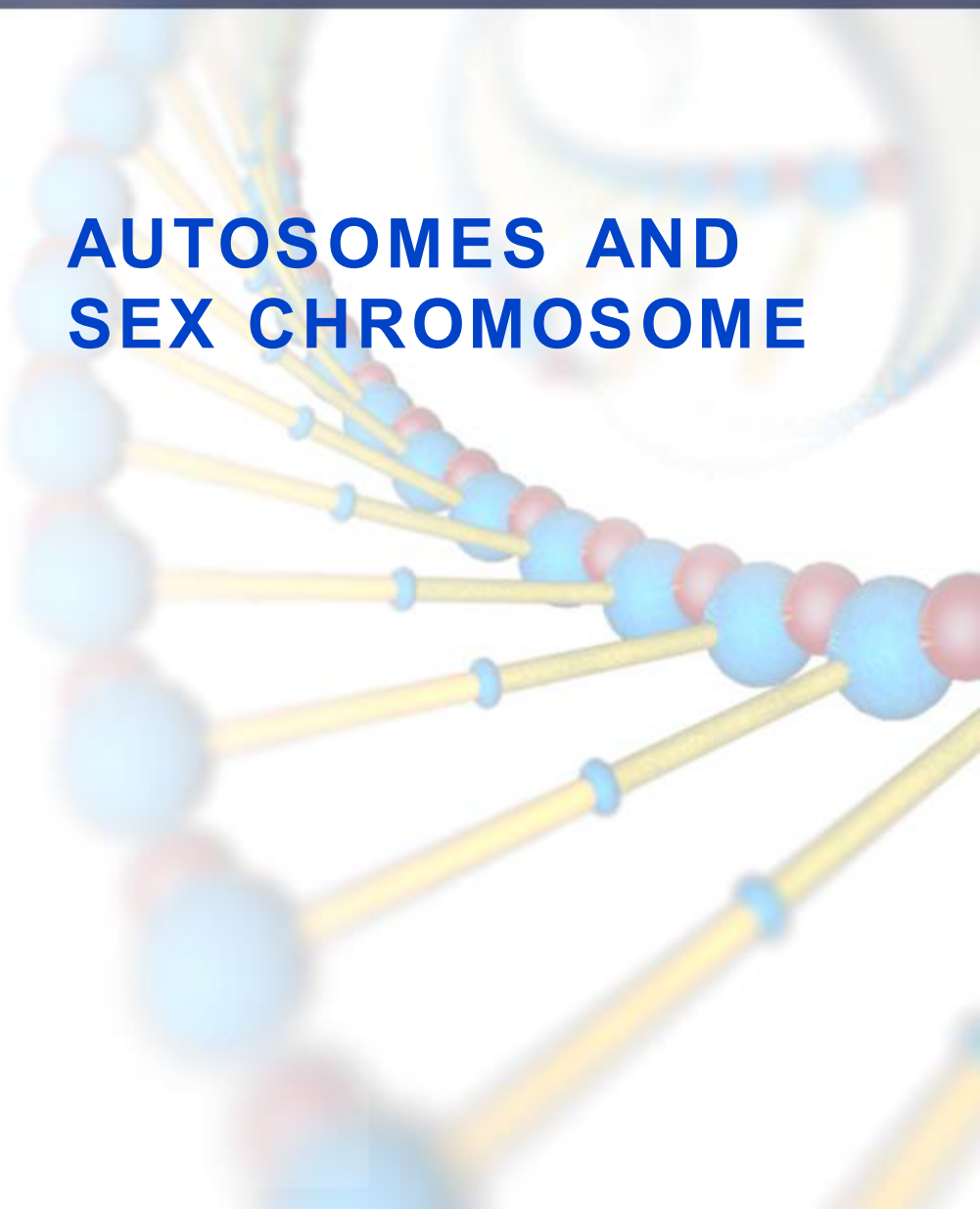
AUTOSOMES AND SEX CHROMOSOME IN DIFFERENT SPECIES

	Autosomal pair			Sex chromosome	
	Diploid (2n)	No. of metacentrics	No. of acrocentric or telocentric	X	Y
Cat	38	16	2	M	M
Dog	78	0	38	M	A
Pig	38	12	6	M	M
Goat	60	0	29	A	M
Sheep	54	3	23	A	M
Cow	60	0	29	M	M
Horse	64	13	18	M	A

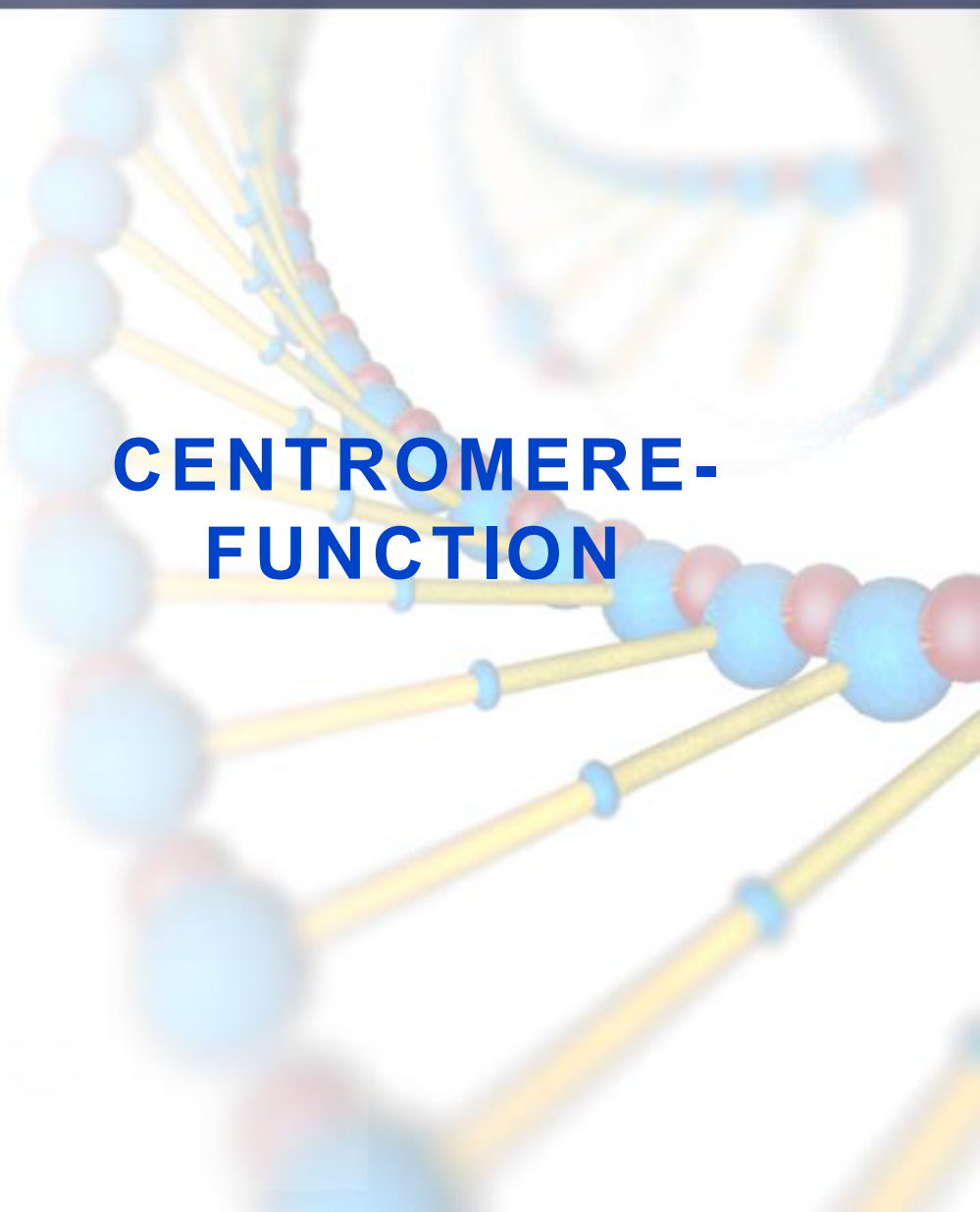
M – Metacentric; A – Acrocentric

Types of Chromosomes

AUTOSOMES AND
SEX CHROMOSOME



Chromosome Structure



CENTROMERE-FUNCTION

Centromere - Function

CENTROMERE

- Centromere and telomeres are two essential features of all eukaryotic chromosomes.
- Each provide a unique function i.e., absolutely necessary for the stability of the chromosomes.

Centromere - Function

CENTROMERE

- Centromere are required for the segregation of the chromosomes during meiosis and mitosis.
- Telomeres provide terminal stability to the chromosome and ensure its survival.

Centromere - Function

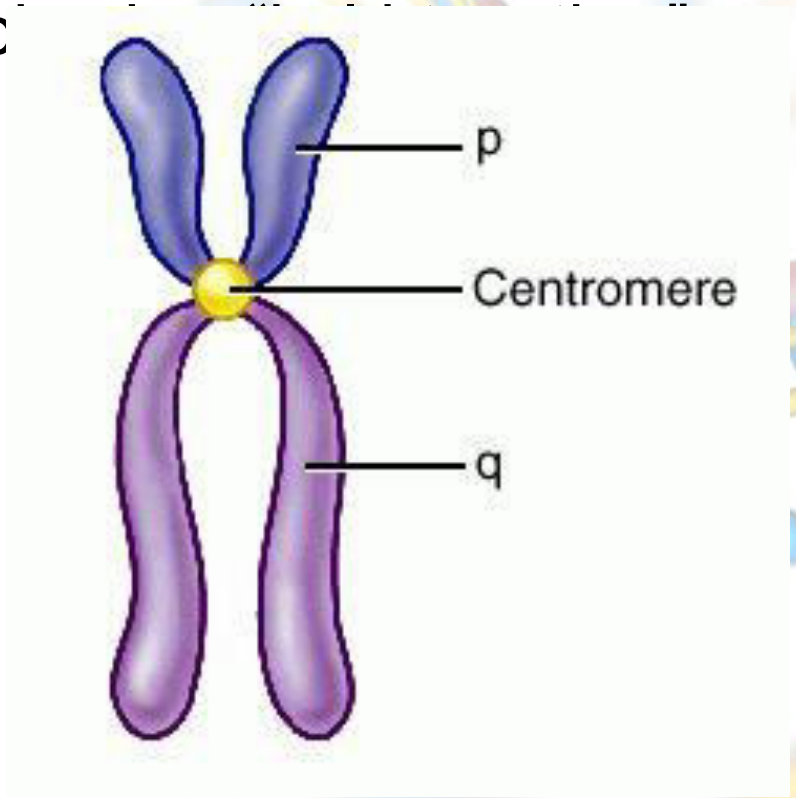
CENTROMERE

- The region where two sister chromatids of a chromosome joined or “held together” during mitotic metaphase is called centromere.
- When chromosomes stained they show a dark-stained region that is the centromere.

Centromere - Function

CENTROMERE

- The region where two sister chromatids of a chromosome jc



Centromere - Function

CENTROMERE

- During mitosis, the centromere that is shared by the sister chromatids must divide so that the chromatids can migrate to opposite poles of the cell.

Centromere - Function

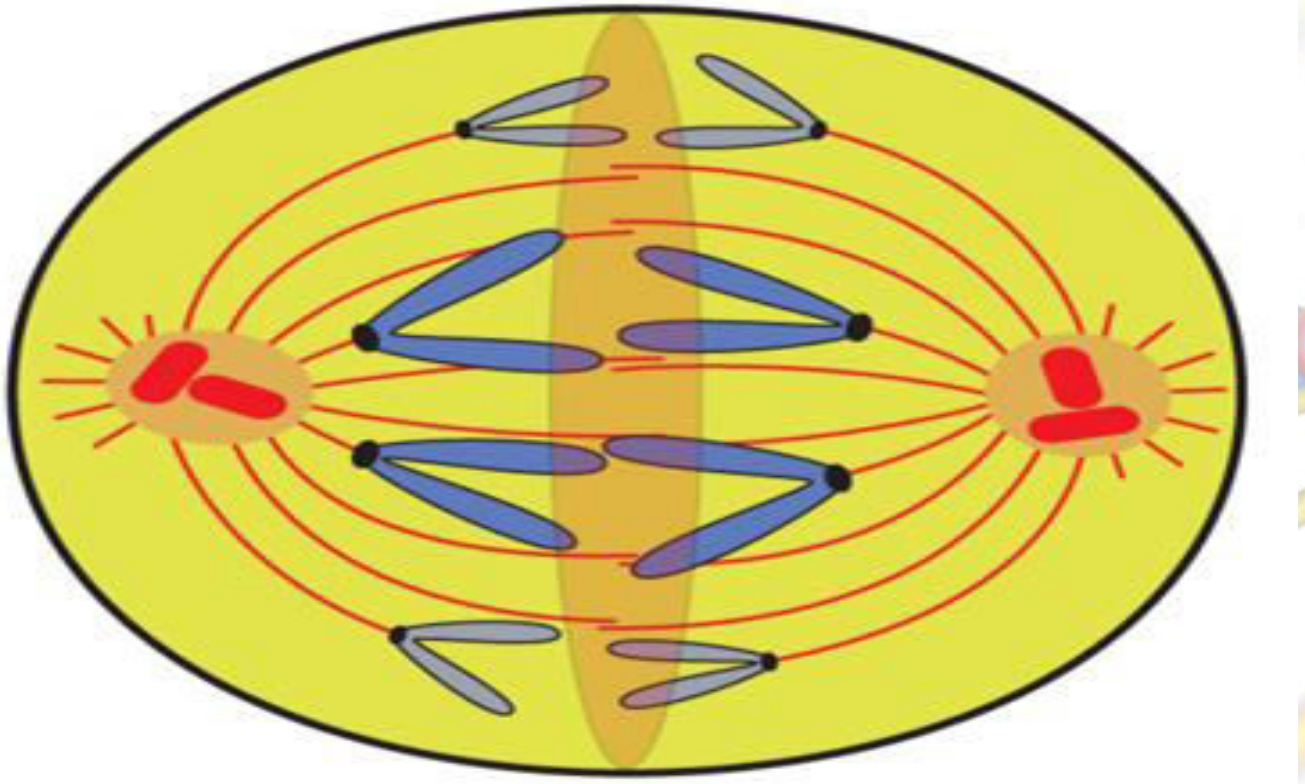
CENTROMERE

- Centromere is an important component of chromosome structure and segregation.
- Centromeres are the first parts of chromosomes to be seen moving towards the opposite poles during anaphase.

Centromere - Function

CENTROMERE

- Centromere moving towards the opposite p

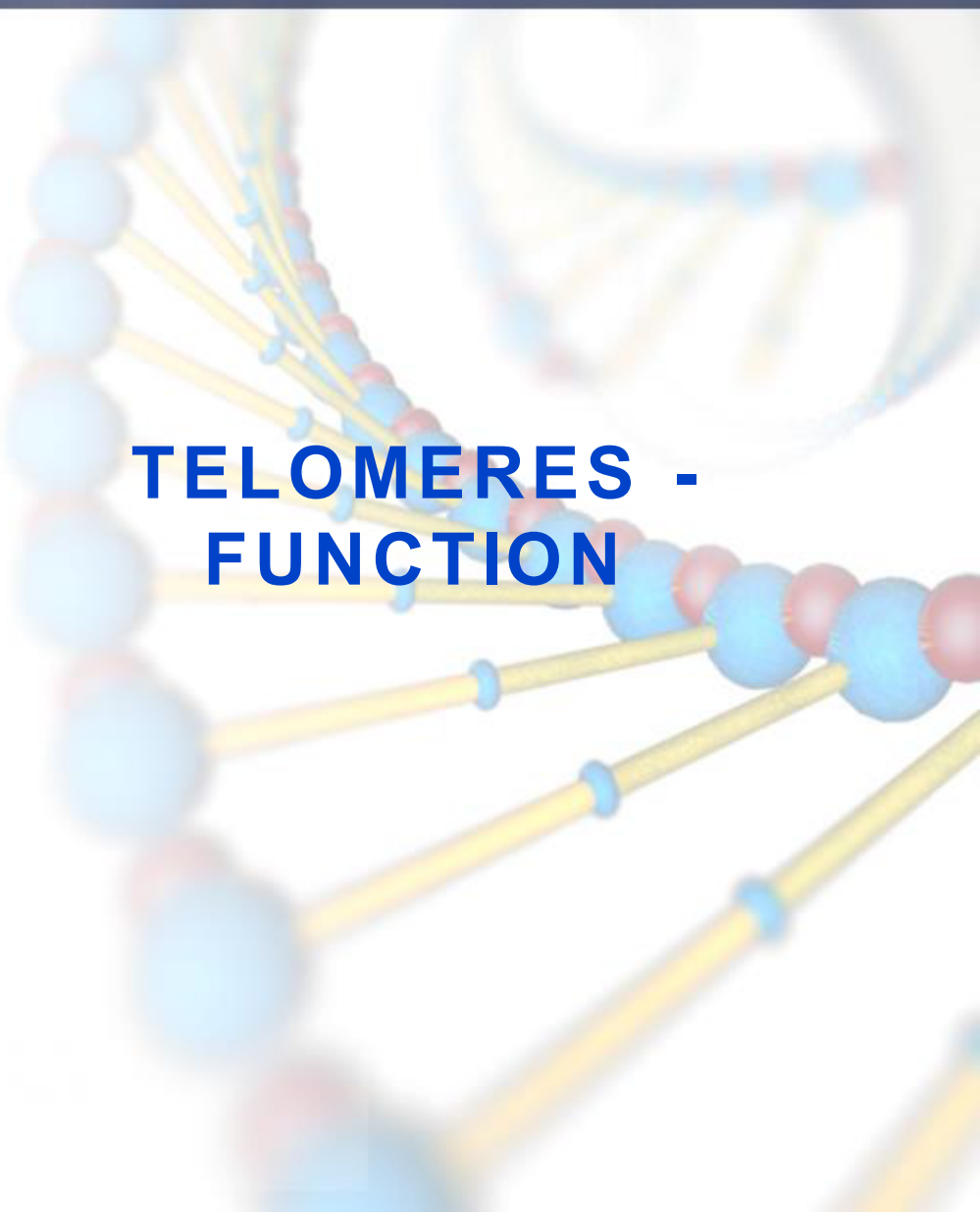


Centromere - Function

CENTROMERE

- The remaining regions of chromosomes lag behind and appear as if they were being pulled by the centromere.

Chromosome Structure



TELOMERES - FUNCTION

Telomeres - Function

WHAT ARE TELOMERES

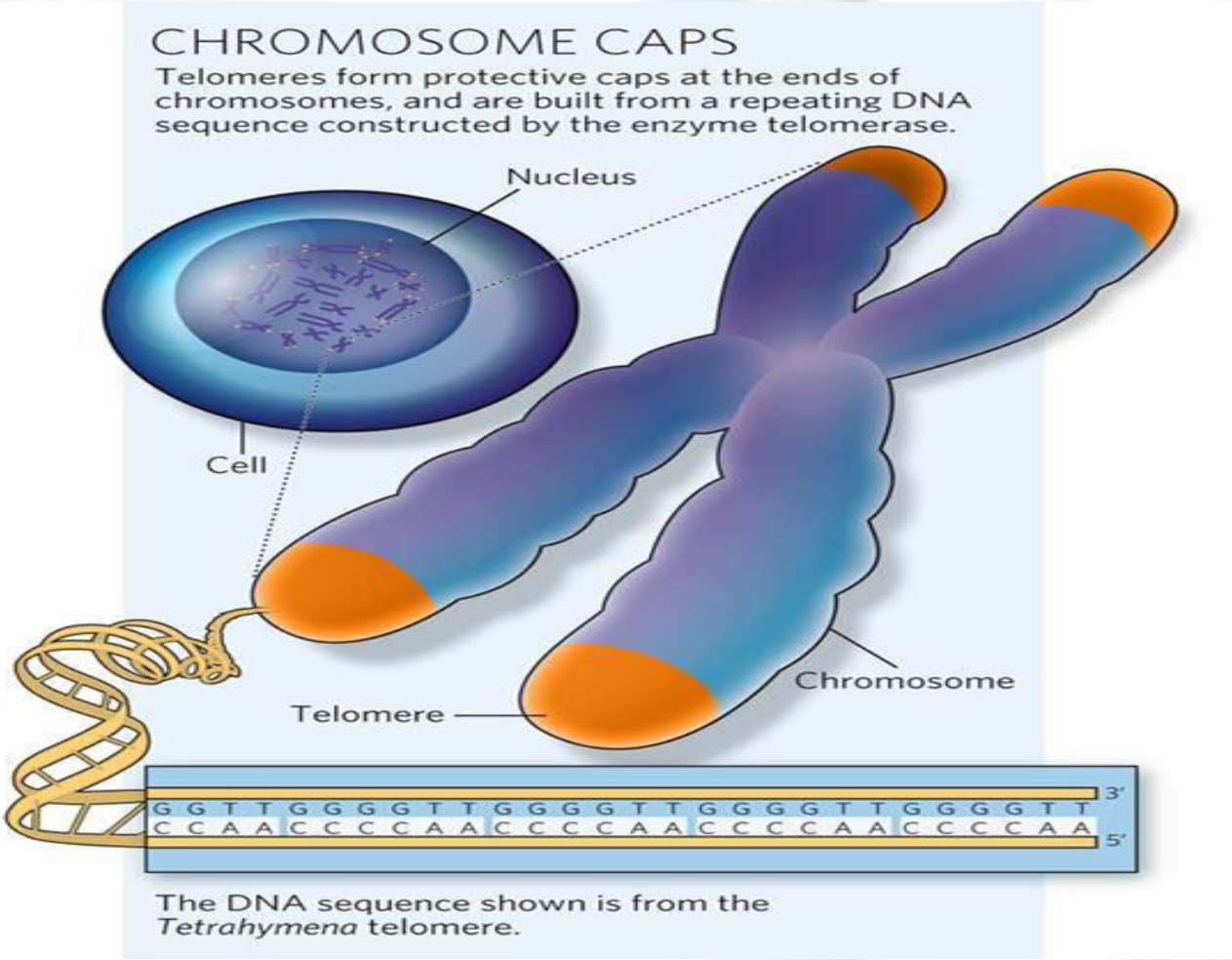
- The two ends of a chromosome are known as telomeres.
- Required for the replication and stability of the chromosomes.

Telomeres - Function

TELOMERES

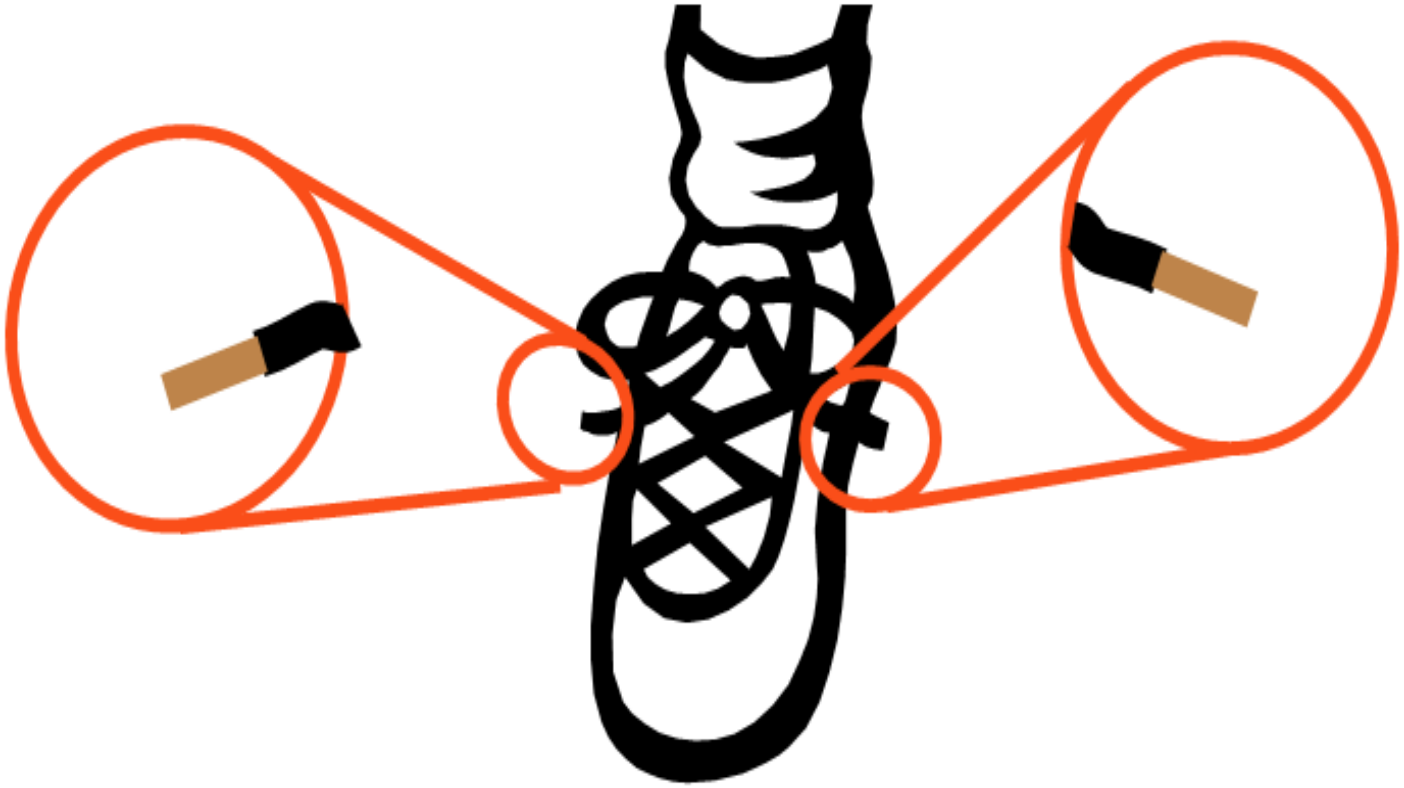
- When telomeres are damaged or removed due to chromosome breakage, the damaged chromosome ends can readily fuse or unite with broken ends of other chromosome.

Telomeres - Function



Telomeres - Function

TELOMERES ARE LIKE THE ENDS OF SHOES TRINGS



Telomeres - Function

TELOMERES

- Ends of broken chromosomes are sticky, whereas the normal end is not sticky, suggesting the ends of chromosomes have unique features.

Telomeres - Function

TELOMERE REPEAT SEQUENCES

- Telomeres have been isolated and characterized from several species.

Species	Repeat Sequence
Arabidopsis	**TTTAGGG**
Human	**TTAGGG**
Oxytricha	**TTTTGGGG**
Slime Mold	**TAGGG**
Tetrahymena	**TTGGGG**
Trypanosome	**TAGGG**

Telomeres - Function

HOW TELOMERE IS REPLICATED

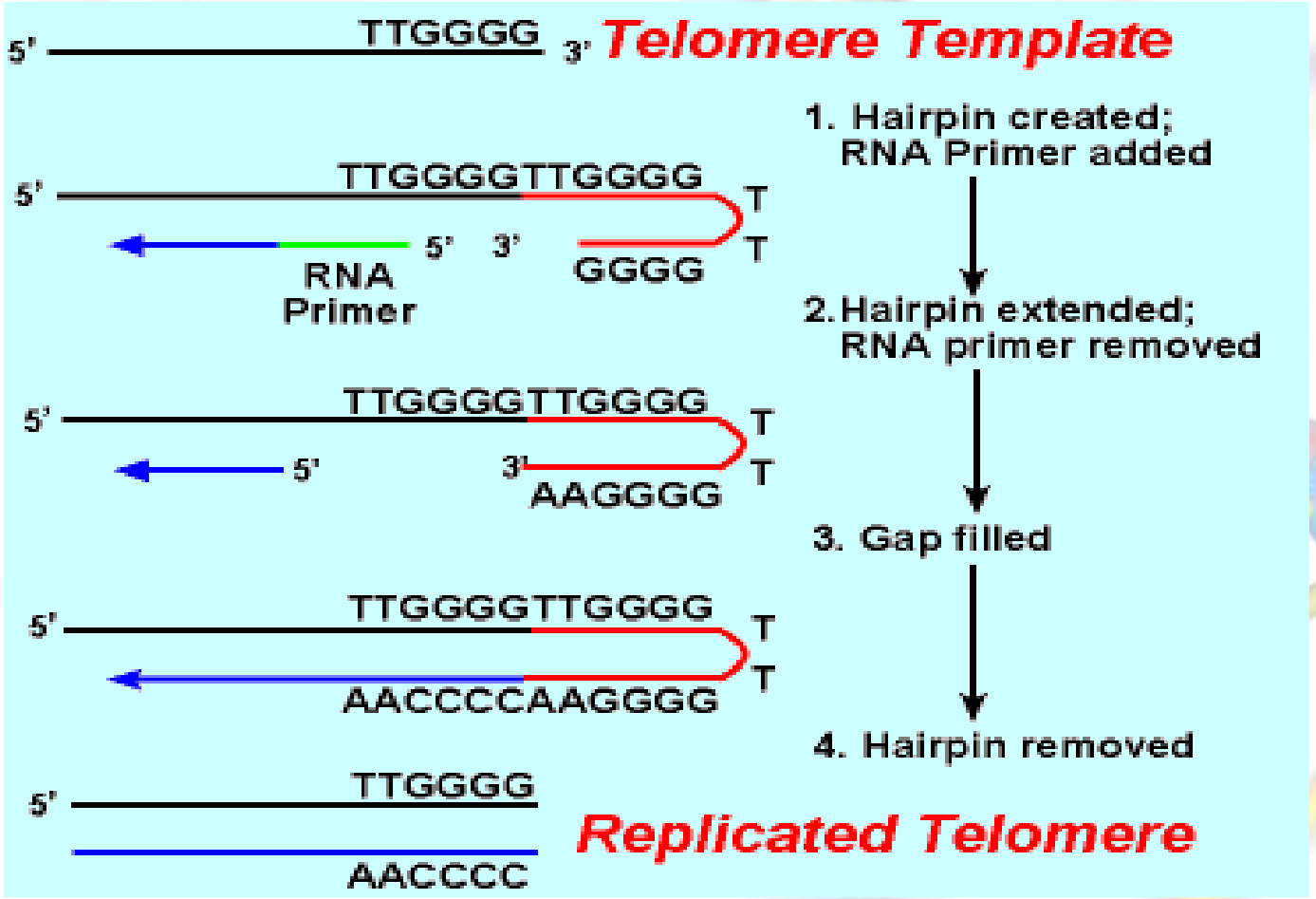
- The telomeres of this organism end in the sequence 5'-TTGGGG-3'.
- The telomerase adds a series of 5'-TTGGGG-3' repeats to the ends of the lagging strand.

Telomeres - Function

HOW TELOMERE IS REPLICATED

- Finally, the hairpin is removed at the 5'-TTGGGG-3' repeat.
- Thus the end of the chromosome is faithfully replicated.

Telomeres - Function

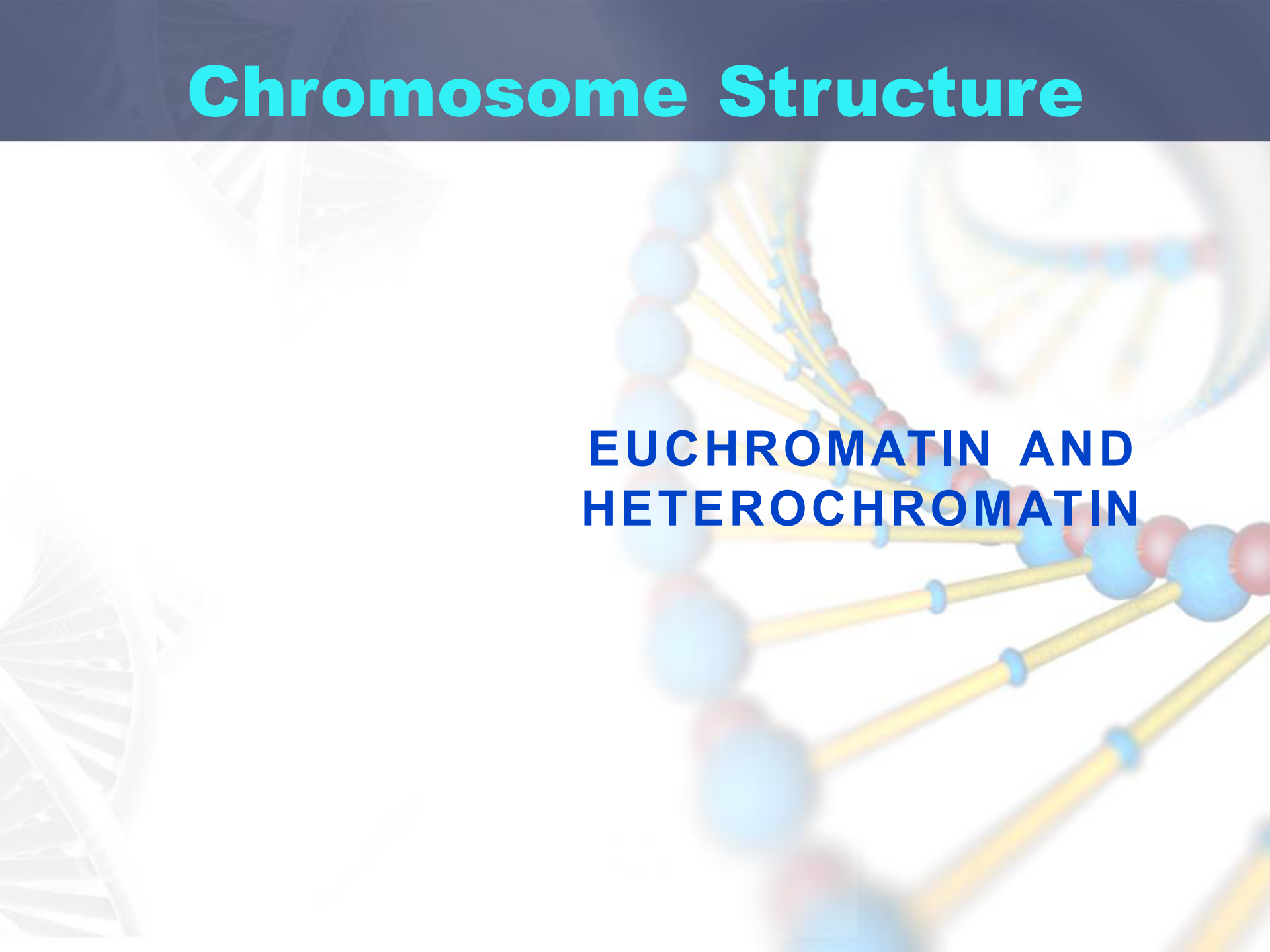


Telomeres - Function

HOW TELOMERE IS REPLICATED

- RNA Primer - Short stretches of ribonucleotides (RNA substrates) found on the lagging strand during DNA replication. Helps initiate lagging strand replication .

Chromosome Structure



EUCHROMATIN AND HETEROCHROMATIN

Euchromatin and Heterochromatin

INTRODUCTION

- Chromosomes may be identified by regions that stain in a particular manner when treated with various chemicals.
- Several different chemical techniques are used to identify certain chromosomal regions by staining.

Euchromatin and Heterochromatin

INTRODUCTION

- Darker bands are generally found near the centromere or on the ends (telomeres) of the chromosome.
- The position of the dark-staining are heterochromatic region or heterochromatin.

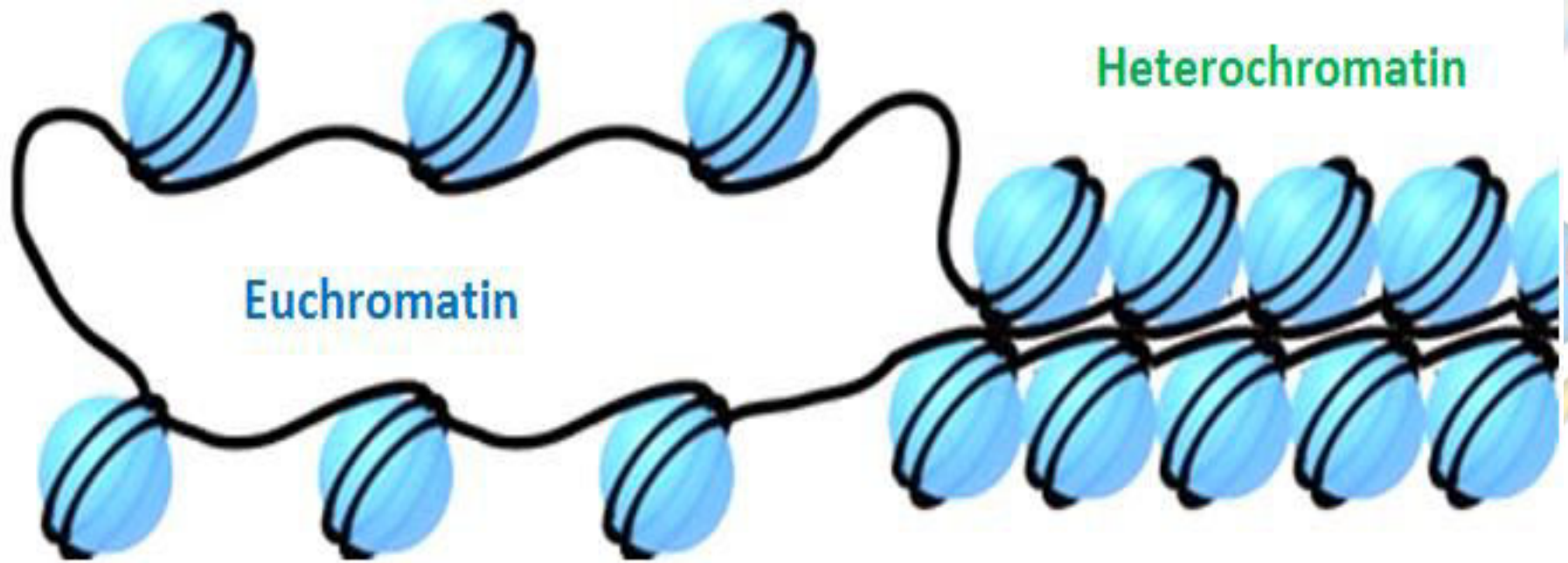
Euchromatin and Heterochromatin

INTRODUCTION

- Light staining regions are Euchromatin region or Euchromatin.

Euchromatin and Heterochromatin

HETERO AND EUCHROMATIN



Euchromatin and Heterochromatin

HETEROCHROMATIN

- Heterochromatin is classified into two groups:
  - Constitutive
  - Facultative

Euchromatin and Heterochromatin

HETEROCHROMATIN

- Constitutive heterochromatin remains permanently in the heterochromatic stage.

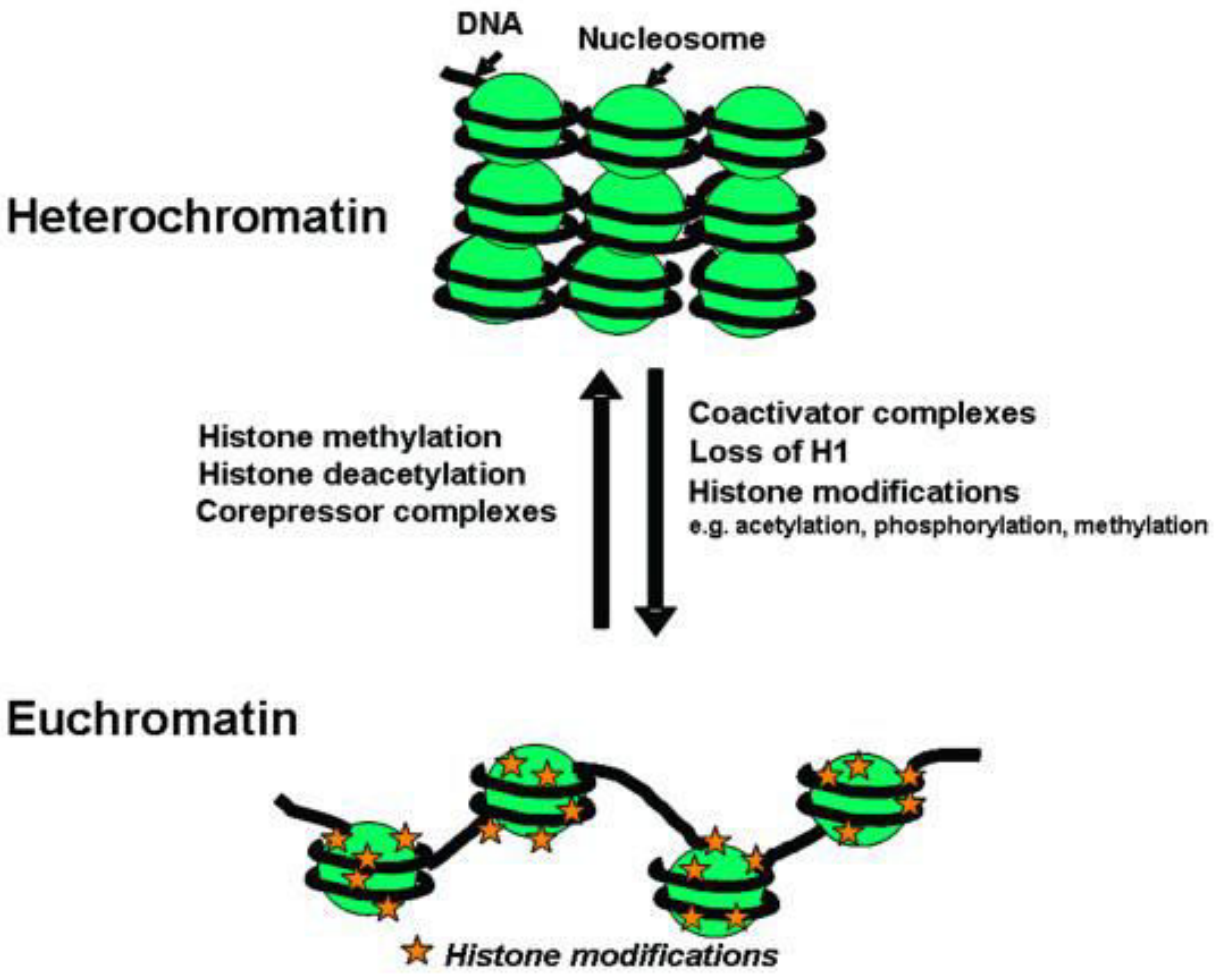
- It does not revert to the Euchromatin stage.

Euchromatin and Heterochromatin

FACULTATIVE

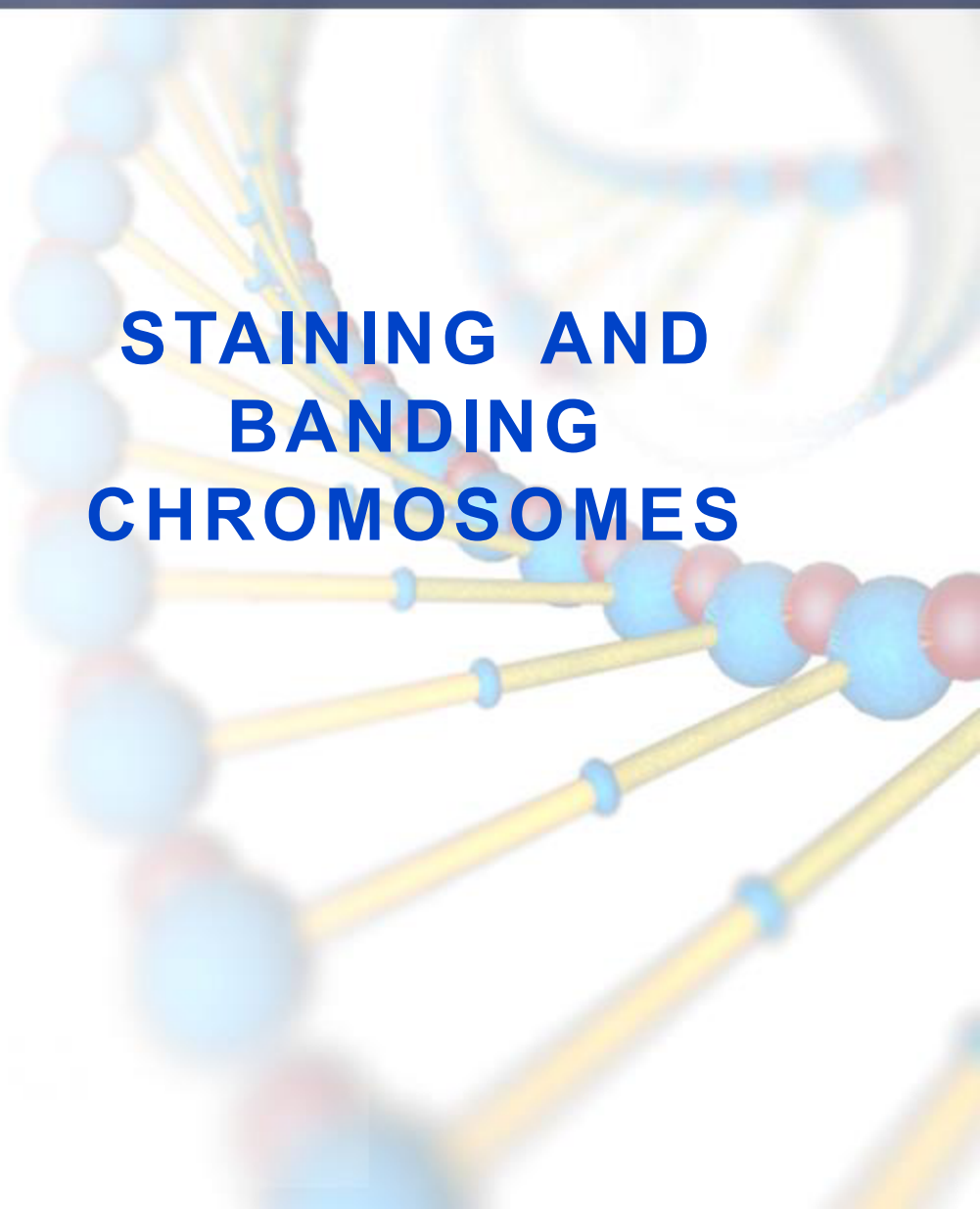
- Facultative heterochromatin consists of Euchromatin that takes on the staining and compactness characteristics of heterochromatin during some phase of development.

Euchromatin and Heterochromatin



Chromosome Structure

STAINING AND BANDING CHROMOSOMES

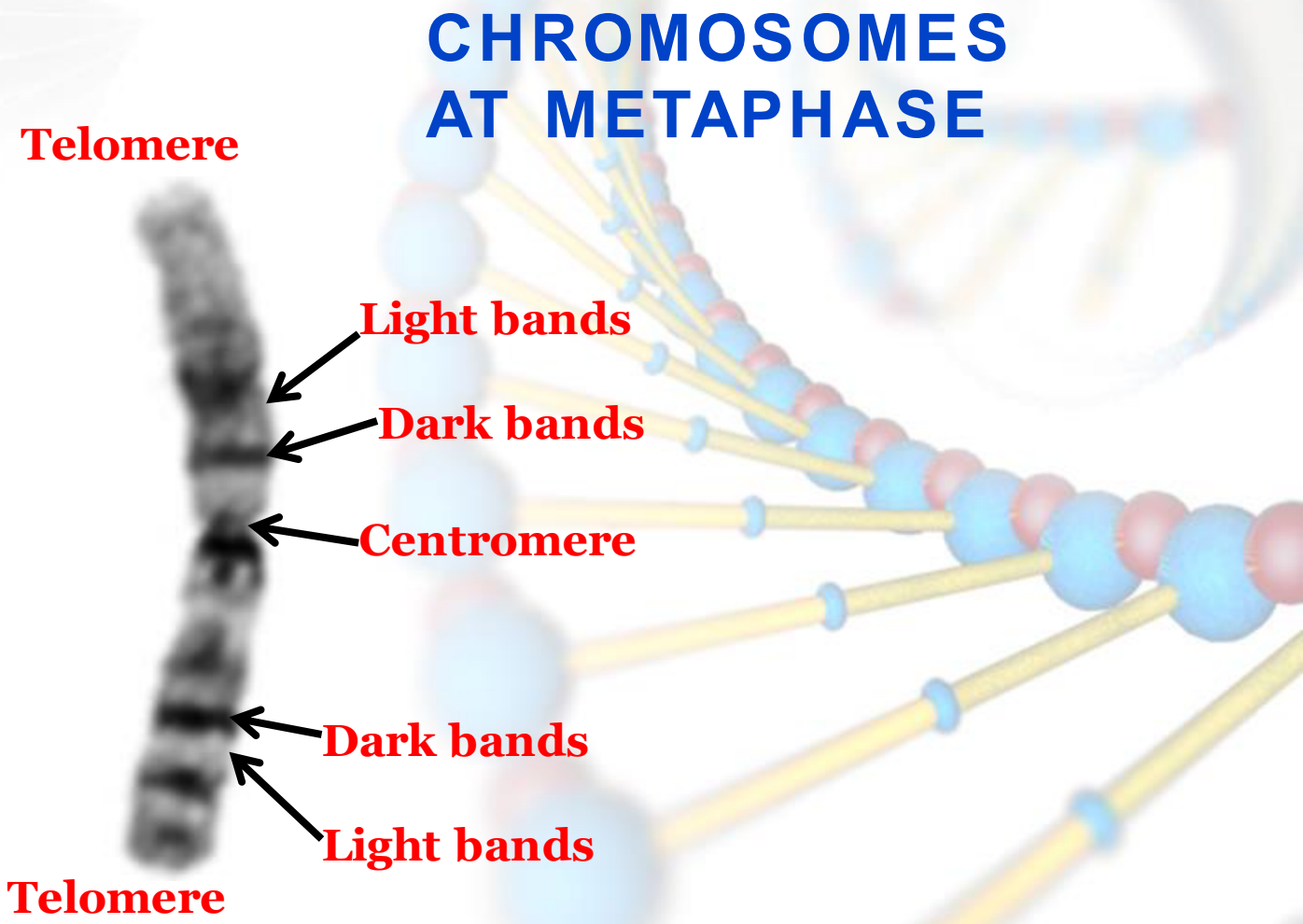


Staining and Banding Chromosomes

STAINING PROCEDURES

- Staining procedures have been developed in the past two decades and these techniques help to study the karyotype in plants and animals.

Staining and Banding Chromosomes



Staining and Banding Chromosomes

FEULGEN STAINING

- Cells are subjected to a mild hydrolysis in 1N HCl at 60 °C for 10 minutes.
- This treatment produces a free aldehyde group in deoxyribose molecules.

Staining and Banding Chromosomes

Q BANDING

- Q bands are the fluorescent bands observed after quinacrine mustard staining with UV light.

- The distal ends not stained by this technique.

Staining and Banding Chromosomes

R BANDING

- The R bands (reverse) are those located in the zones that do not fluoresce with the quinacrine mustard, that is they are between the Q bands and can be visualized as green.

Staining and Banding Chromosomes

C BANDING

- The C bands correspond to constitutive heterochromatin.

- The heterochromatin regions in a chromosome distinctly differ in their stainability from euchromatic region.

Staining and Banding Chromosomes

C BANDING

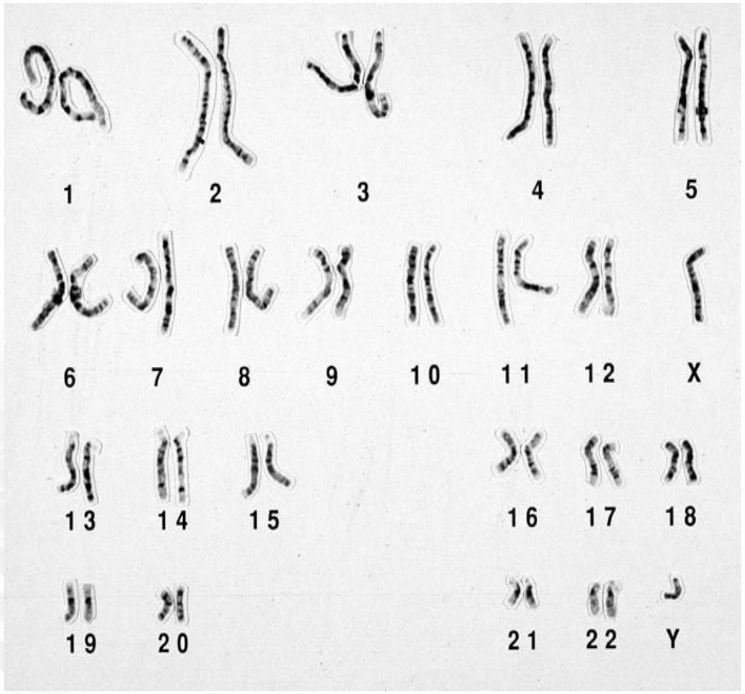


Staining and Banding Chromosomes

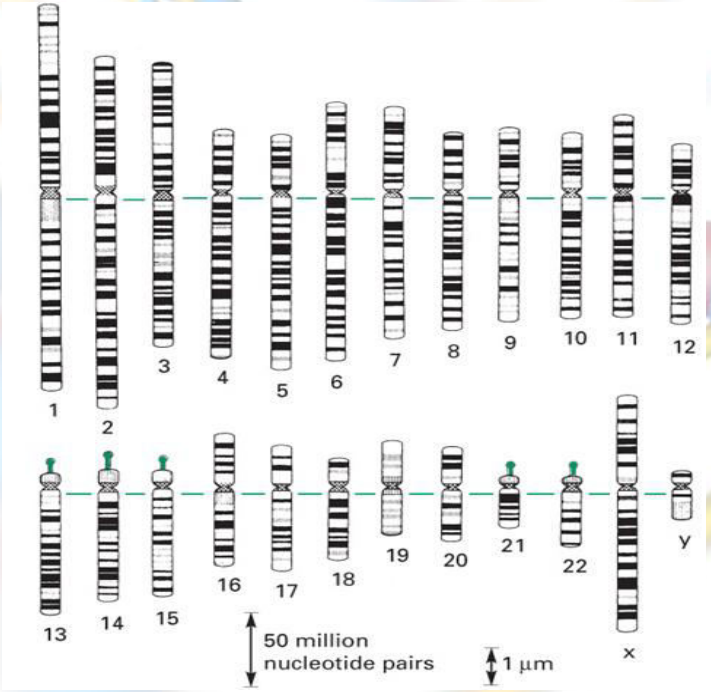
HUMAN CHROMOSOME BANDING PATTERNS

- Human body cells contain 46 chromosomes in 23 pairs – one of each pair inherited from each parent.

Staining and Banding Chromosomes

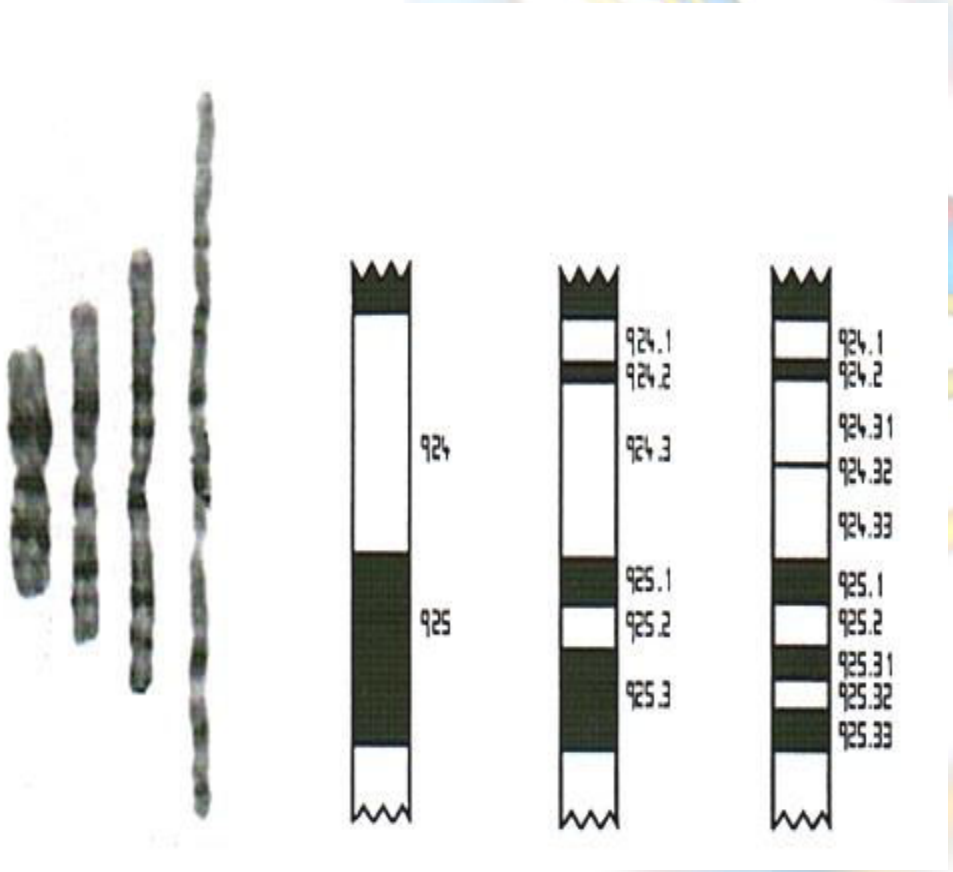


HUMAN CHROMOSOME BANDING



Staining and Banding Chromosomes

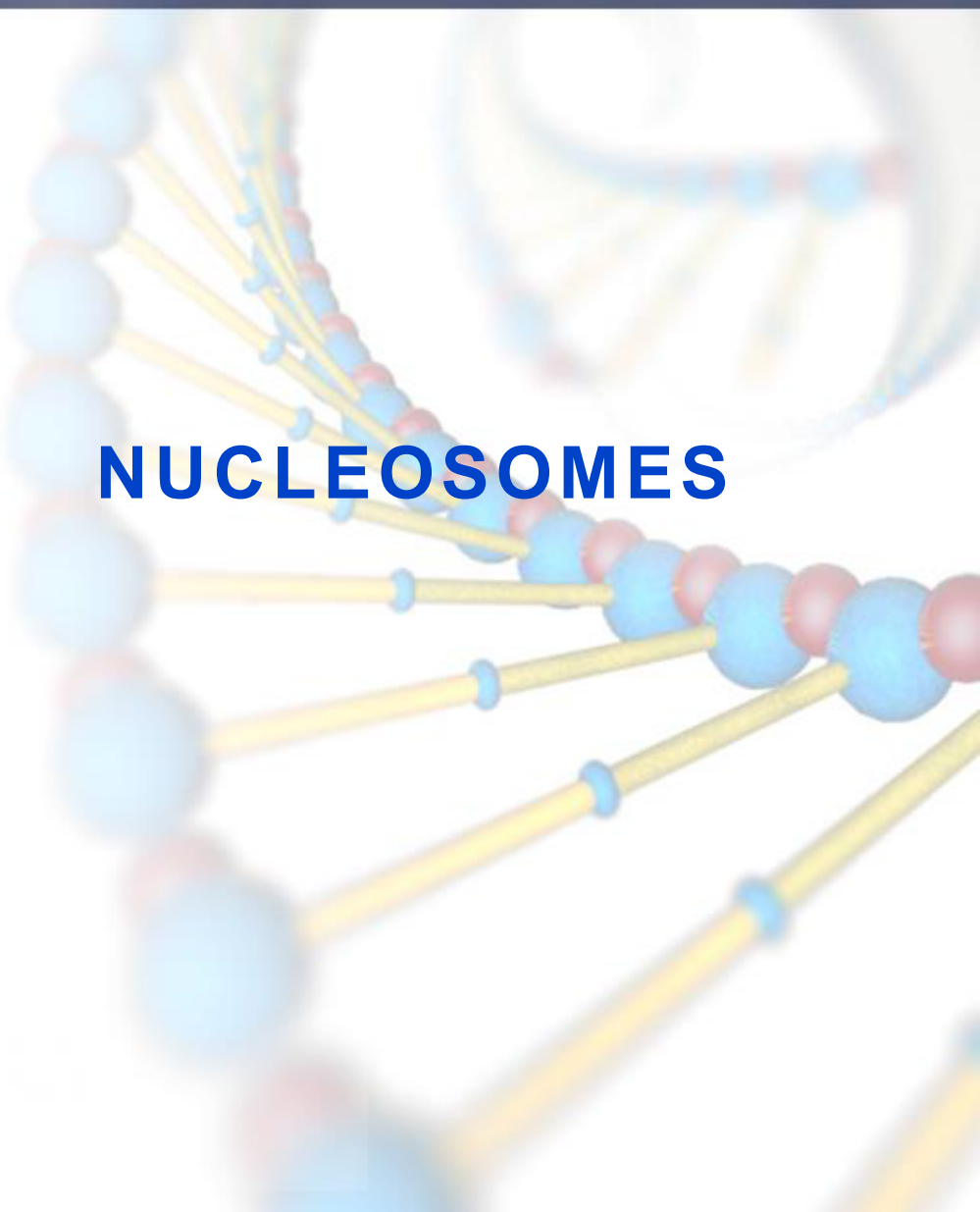
HUMAN CHROMOSOME BANDING



Staining and Banding Chromosomes

Q-banding	Quinacrine	Chromosome arms; mostly repetitive AT-rich DNA	Under UV light, distinct fluorescent banded pattern for each chromosome.
G-banding	Giemsa	Chromosome arms; mostly repetitive AT-rich DNA	Distinct banded pattern for each chromosome; same as Q-banding pattern except single additional band near centromere of chromosomes 1 and 16
R-banding	Variety of techniques	Chromosome arms; mostly unique GC-rich DNA	Reverse banding pattern of that observed with Q- or G-banding
C-banding	Variety of techniques	Centromere region of each chromosome and distal portion of Y chromosome; highly repetitive, mostly AT-rich DNA	Largest bands usually on chromosomes 1, 9, 16, and Y; chromosomes 7, 10, and 15 have medium-sized bands; size of C-bands highly variable from person to person

Chromosome Structure



NUCLEOSOMES

Nucleosomes

NUCLEOSOMES

- The nucleosome is composed of a core of eight histone proteins and DNA (147 bp) wrapped around them.
- The DNA between nucleosome ….linker DNA.
- Linker DNA 20-60 bp.

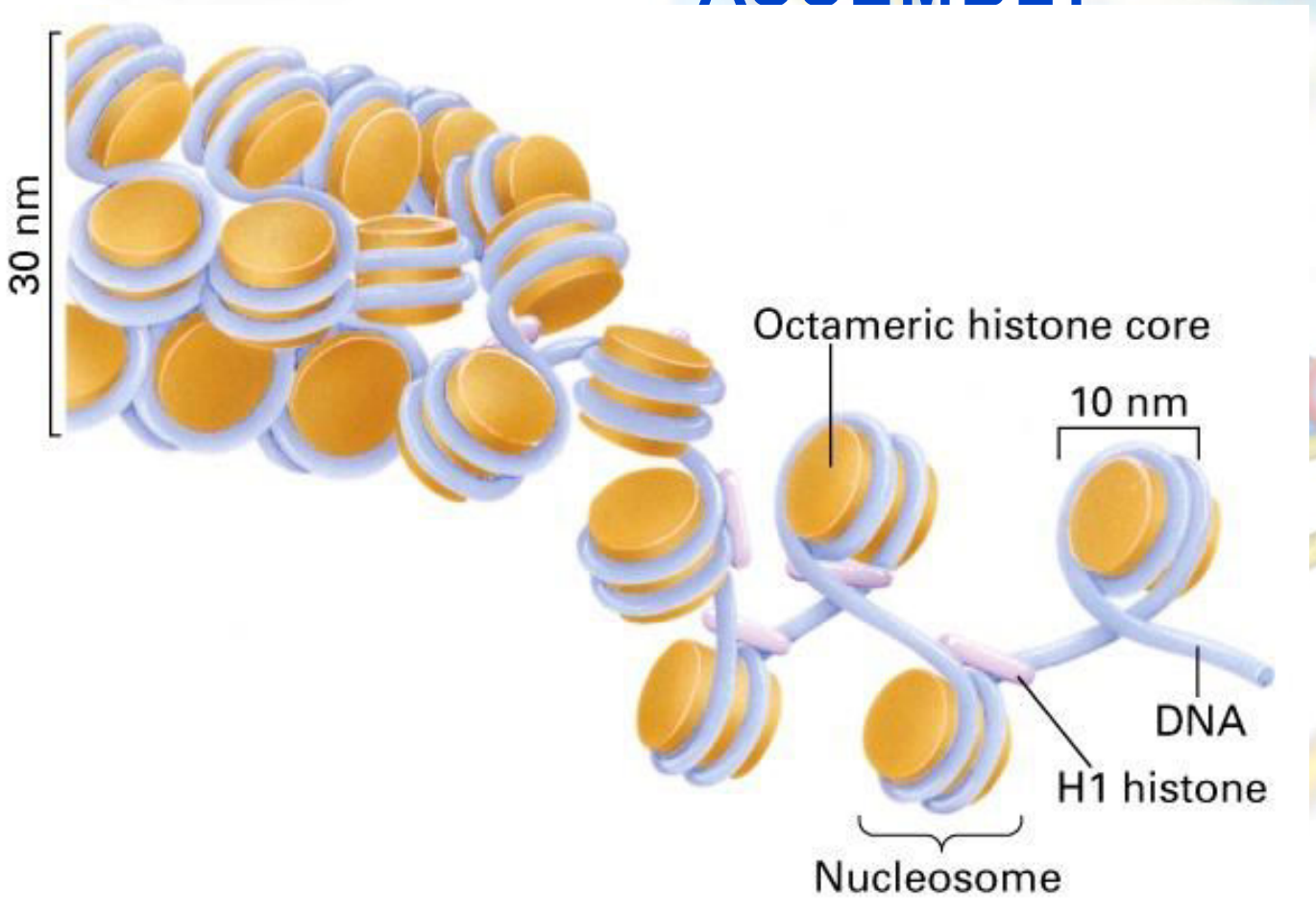
Nucleosomes

NUCLEOSOME ASSEMBLY

- Nucleosomes are assembled immediately after DNA replication.
- Two molecules each of four types of histones are packed by DNA to form a Nucleosomes.

Nucleosomes

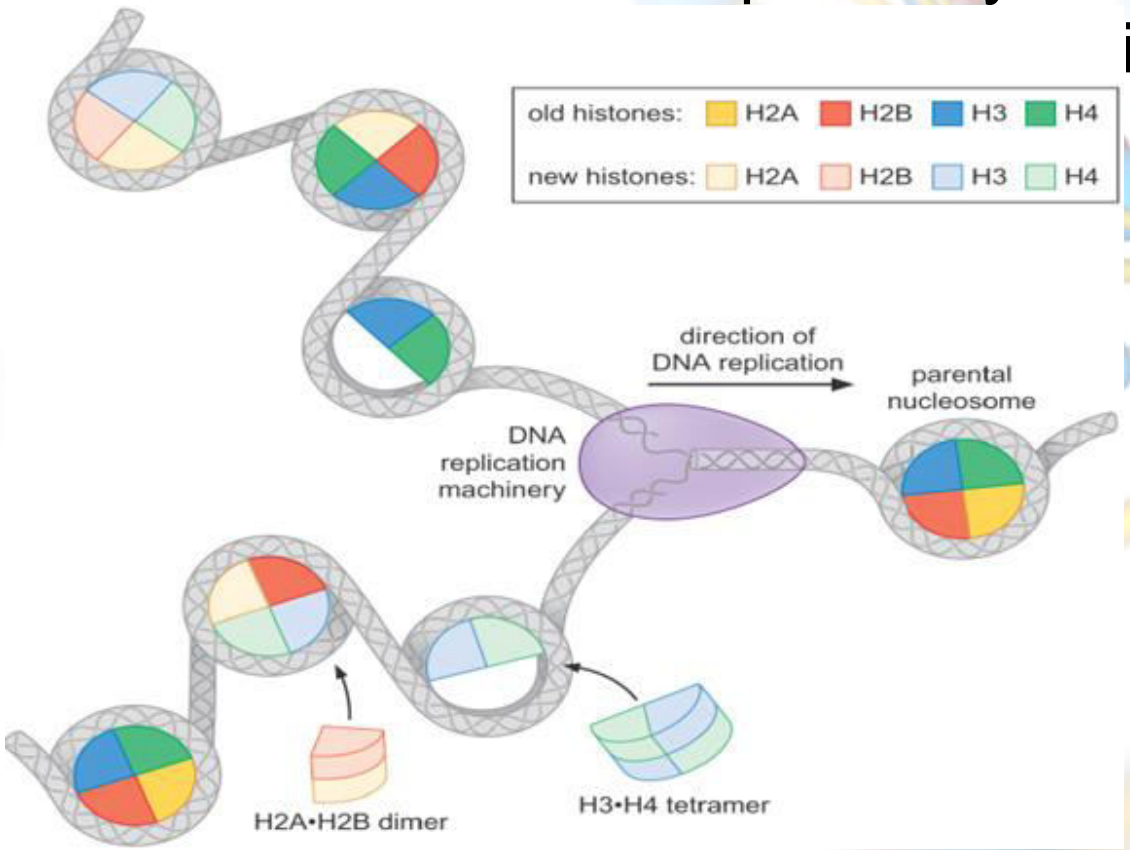
NUCLEOSOME ASSEMBLY



Nucleosomes

HISTONES

- Histones are small, positively charged
ins.



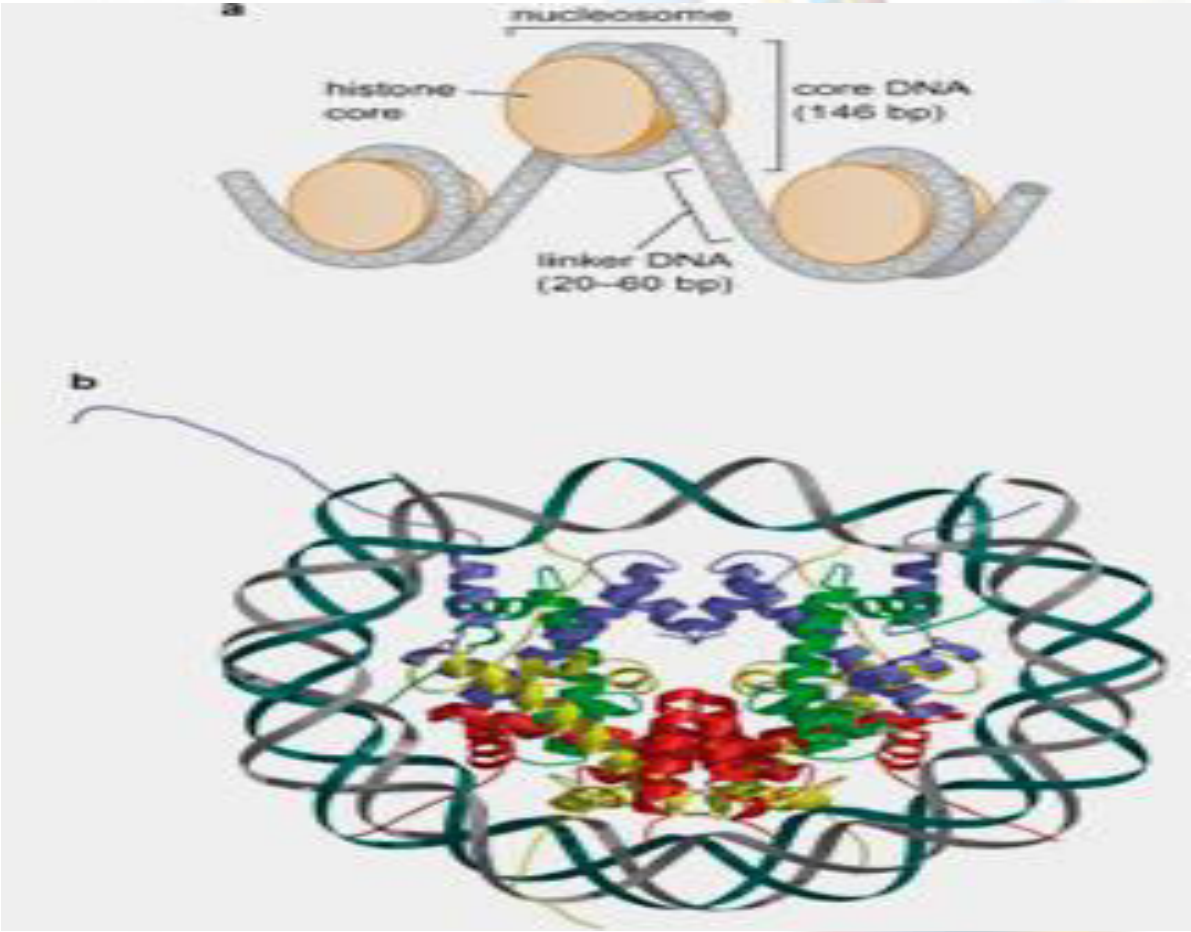
Nucleosomes

NUCLEOSOMES-FIVE HISTONES

- Five abundant histones are H1, H2A, H2B, H3 and H4.
- The core histones share a common structural fold, called histone-fold domain.

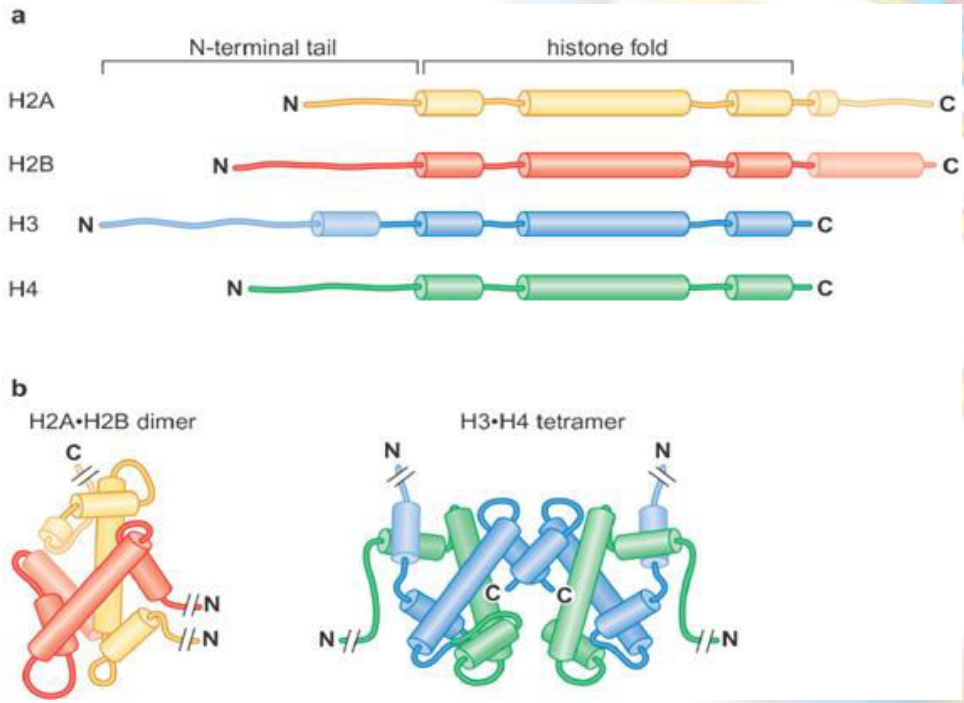
Nucleosomes

HISTONES AND LINKER DNA

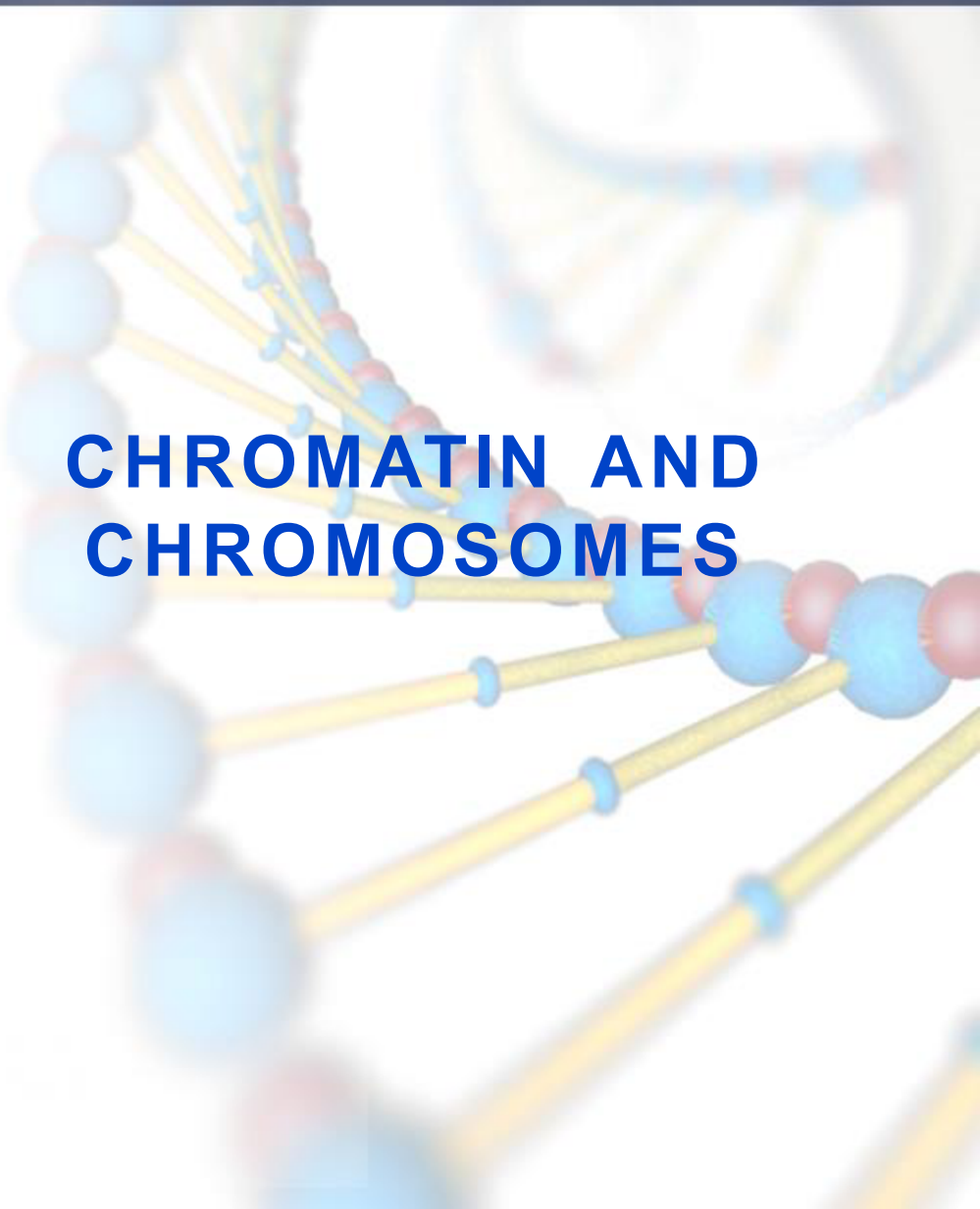


Nucleosomes

CORE HISTONES SHARE A COMMON STRUCTURAL FOLDS



Chromosome Structure



CHROMATIN AND CHROMOSOMES

Chromatin and Chromosomes

CHROMATIN

- The complexes between eukaryotic DNA and proteins are called chromatin.
- The major proteins of chromatin are the histones – small proteins containing a high proportion of basic amino acids arginine and lysine.

Chromatin and Chromosomes

CHROMATIN

- There are 5 major types of histones: H1, H2A, H2B, H3, and H4 – which are very similar among different sp of eukaryotes.
- The histones are extremely abundant proteins in eukaryotic cells.

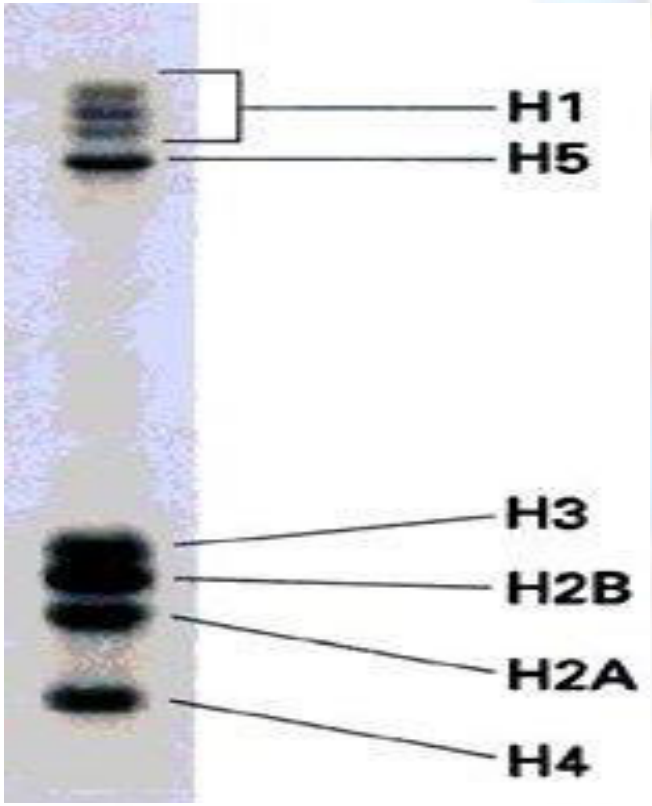
Chromatin and Chromosomes

MOLECULAR WEIGHT AND NUMBER OF AMINO ACIDS

Histone	Mol. Wt	No. of Amino acid	Percentage Lys + Arg
H1	22,500	244	30.8
H2A	13,960	129	20.2
H2B	13,774	125	22.4
H3	15,273	135	22.9
H4	11,236	102	24.5

Chromatin and Chromosomes

HISTONES - SIZE AND MOLECULAR WEIGHT



Chromatin and Chromosomes

NON-HISTONES

- Chromatin contains an approximately equal mass of a variety of nonhistone proteins.
- More than a thousand different types of proteins, which are involved in a DNA replication and gene expression.

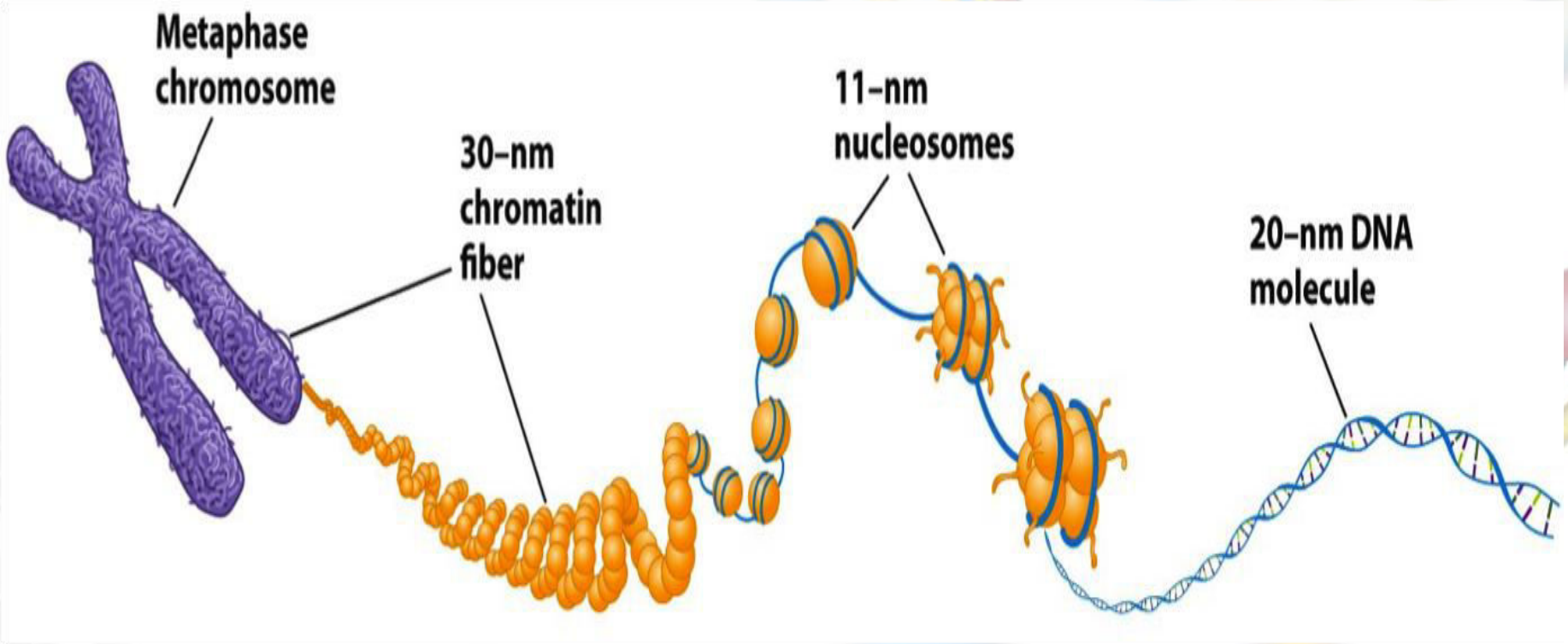
Chromatin and Chromosomes

NON-HISTONES

- The DNA of prokaryotes is similarly associated with proteins, some of which presumably function as histones do, packing DNA within the bacterial cell.

Chromatin and Chromosomes

COMPARISON

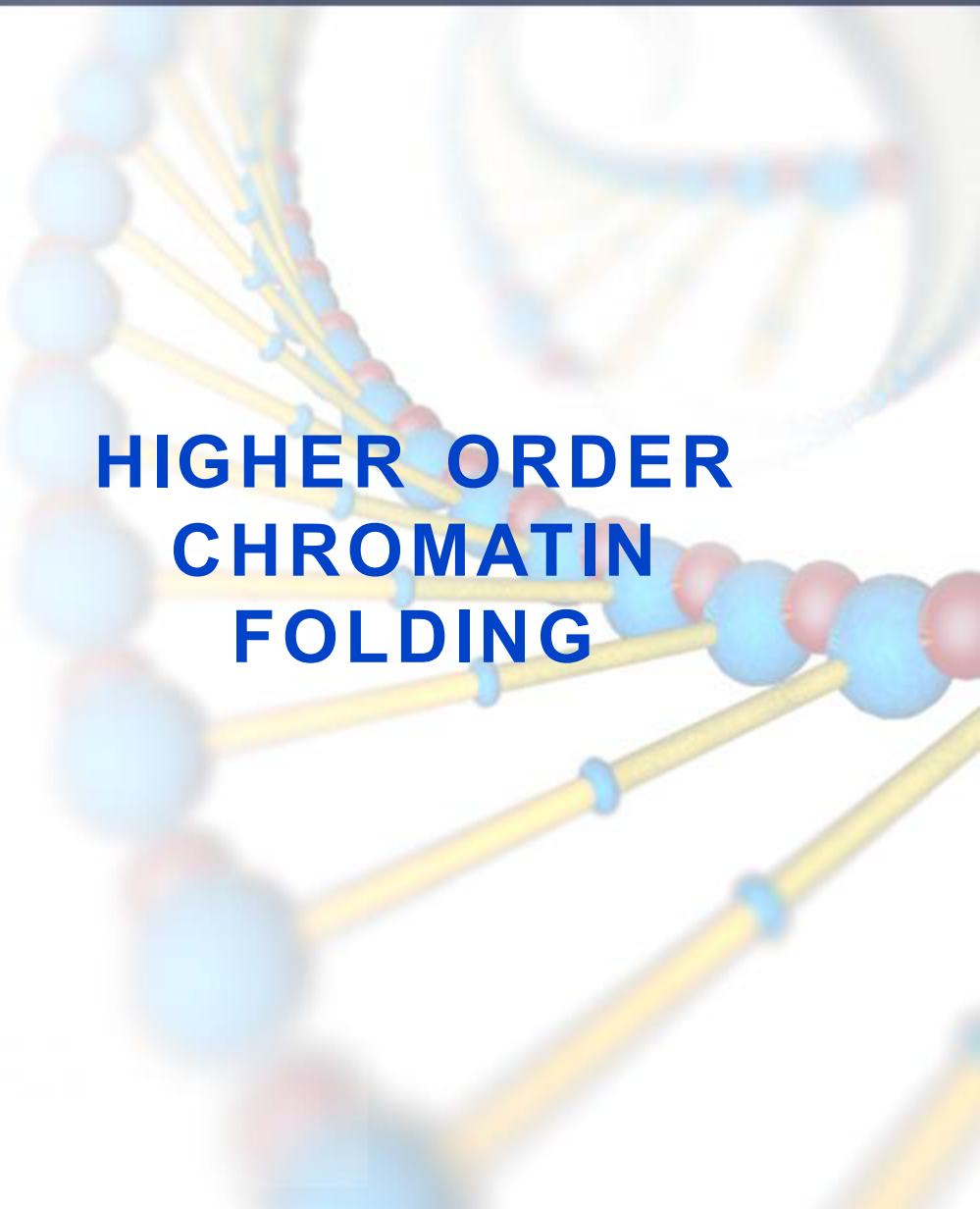


Chromatin and Chromosomes

COMPARISON

Chromosomes	Chromatin
• Tightly packaged DNA	• Unwound DNA
• Found only during cell division	• Found throughout Interphase
• DNA is not being used for macromolecule synthesis	• DNA *is* being used for macromolecule synthesis

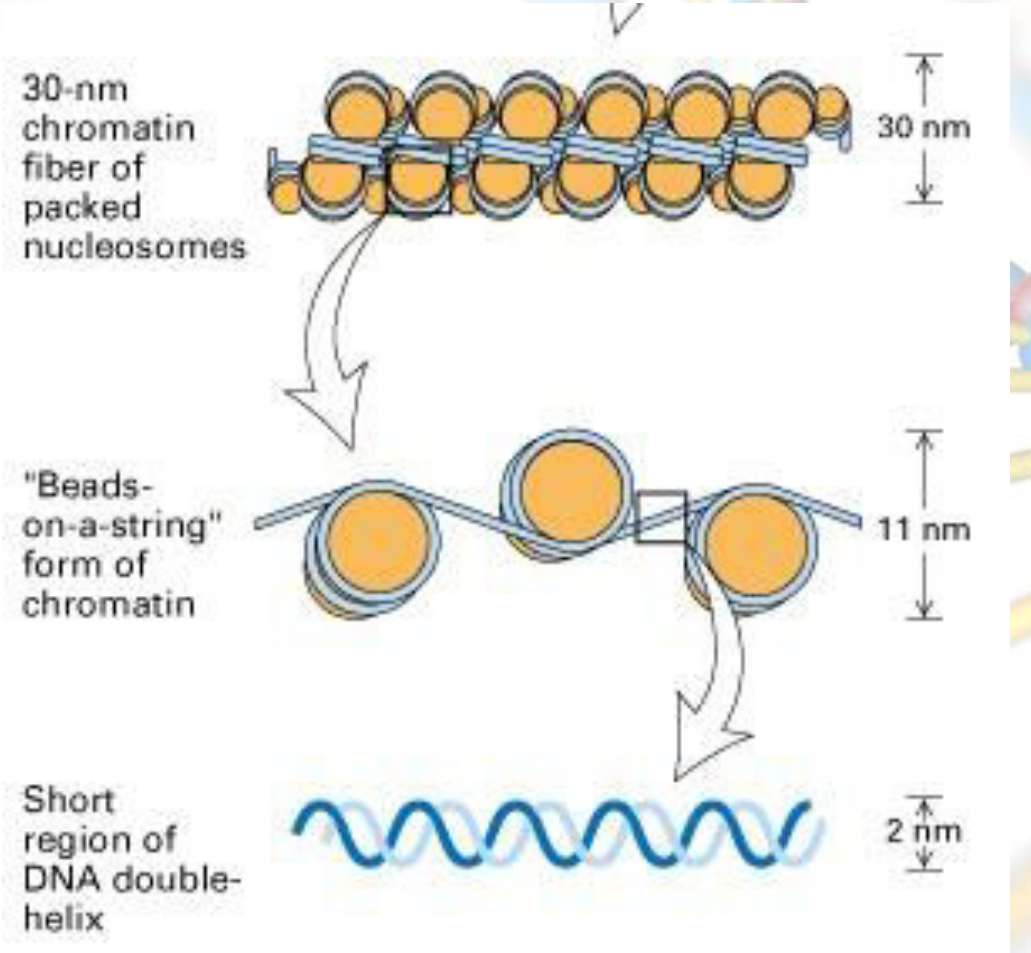
Chromosome Structure



HIGHER ORDER CHROMATIN FOLDING

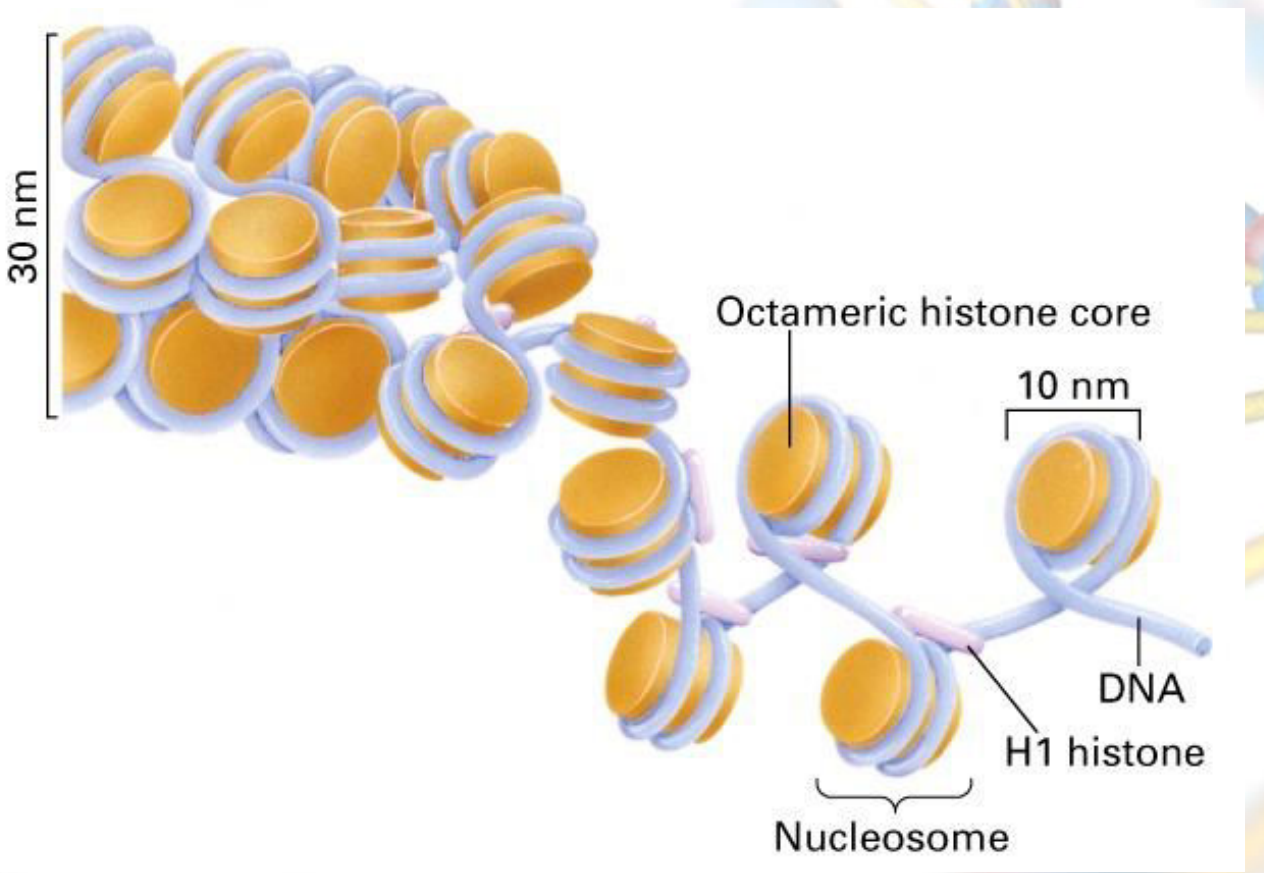
Higher Order Chromatin Folding

HIGHER ORDER FOLDING



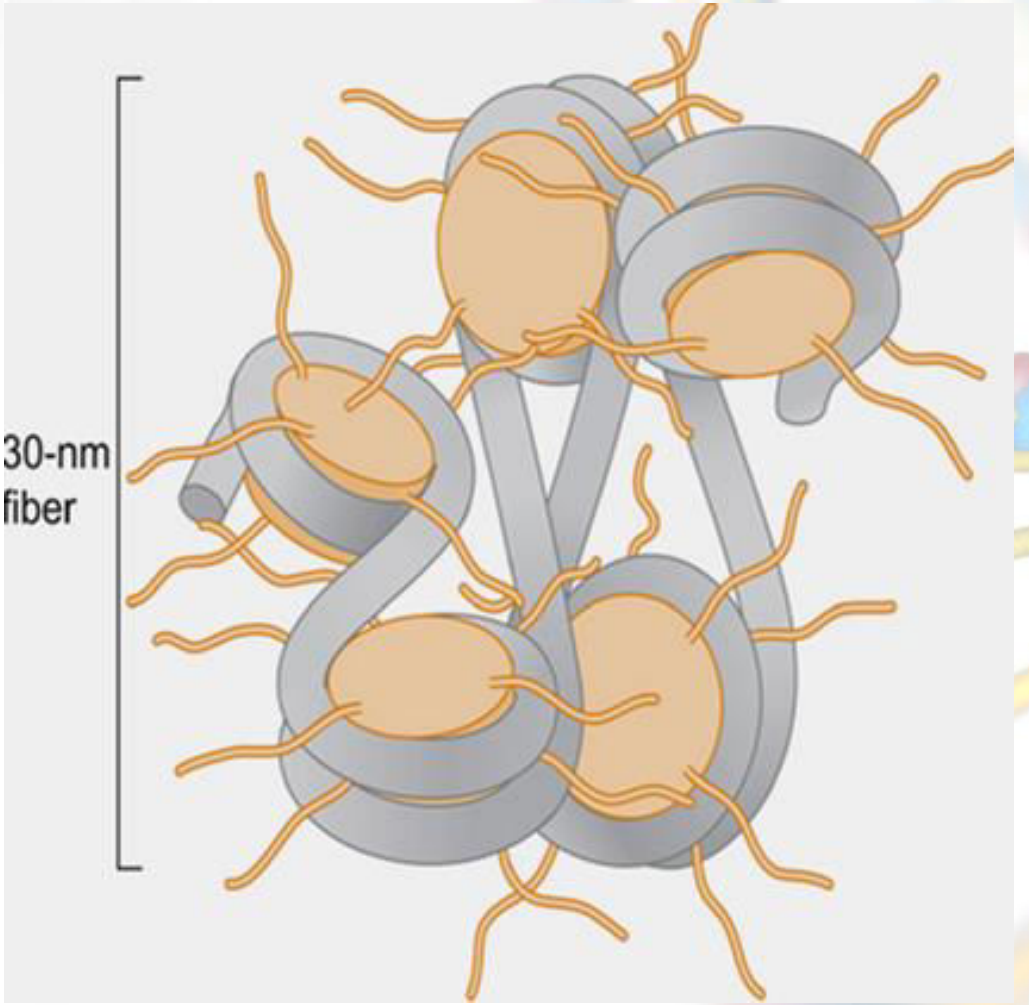
Higher Order Chromatin Folding

NUCLEOSOMES AND HIGHER ORDER



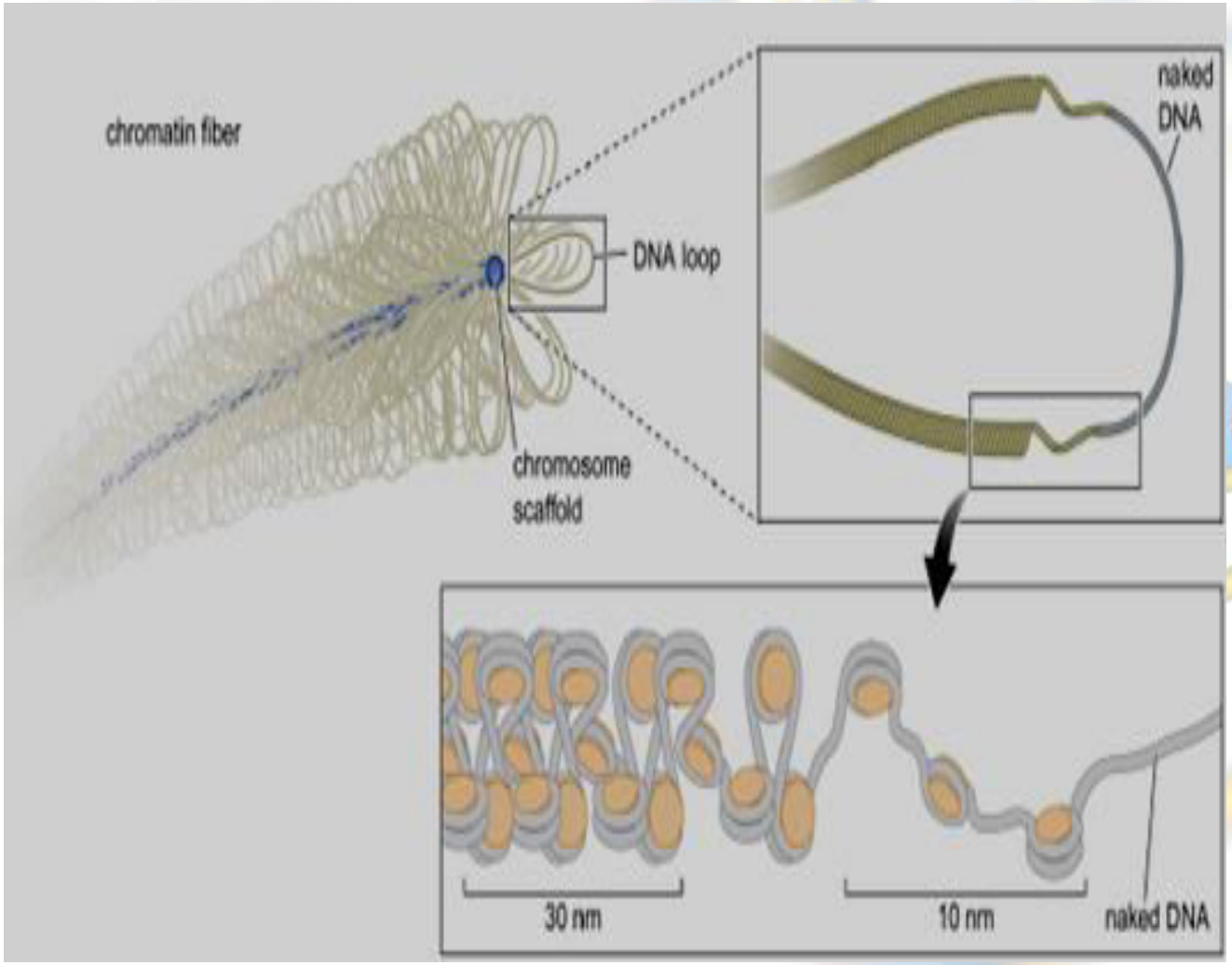
Higher Order Chromatin Folding

30 NM FIBRE



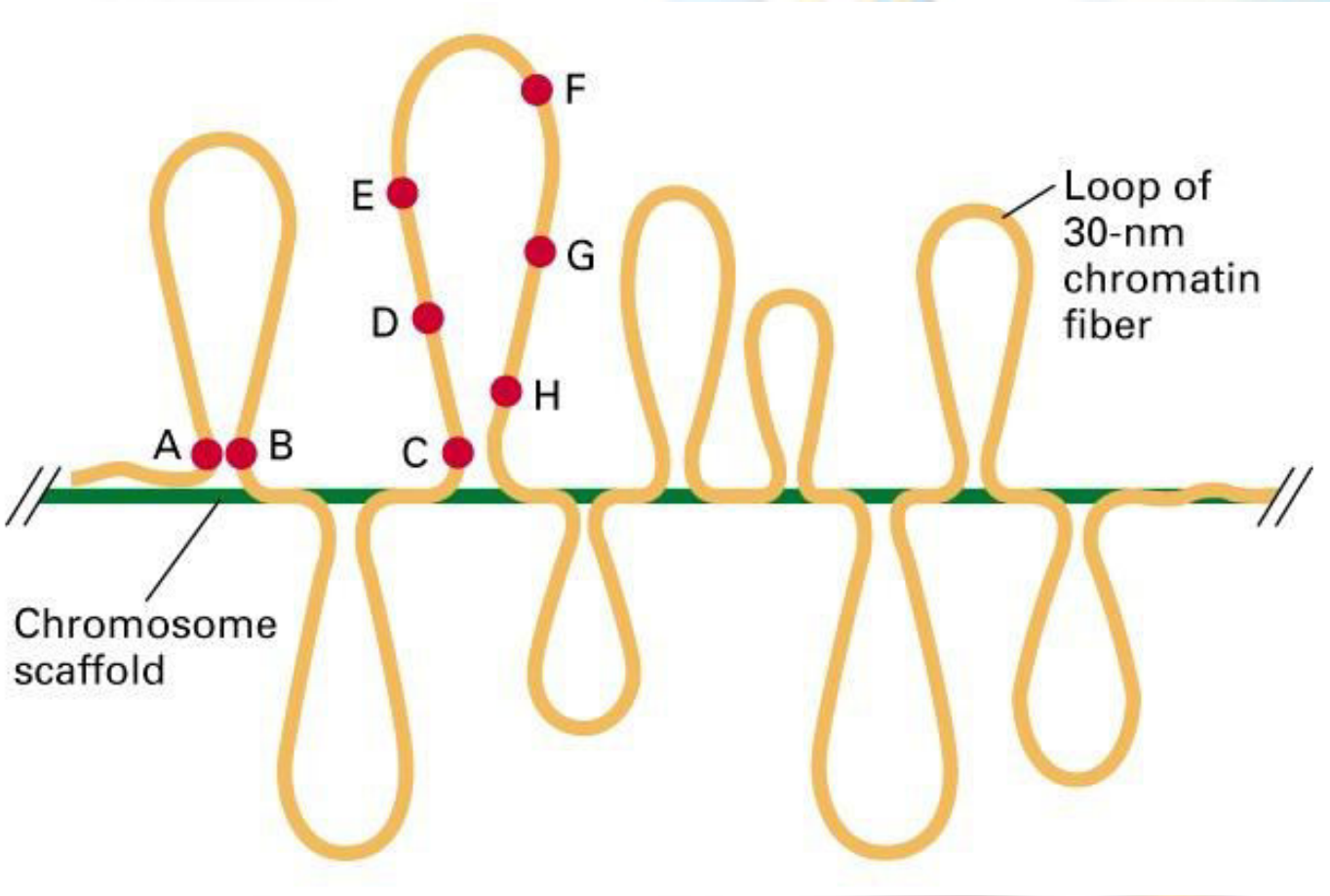
Higher Order Chromatin Folding

300 NM FIBRE

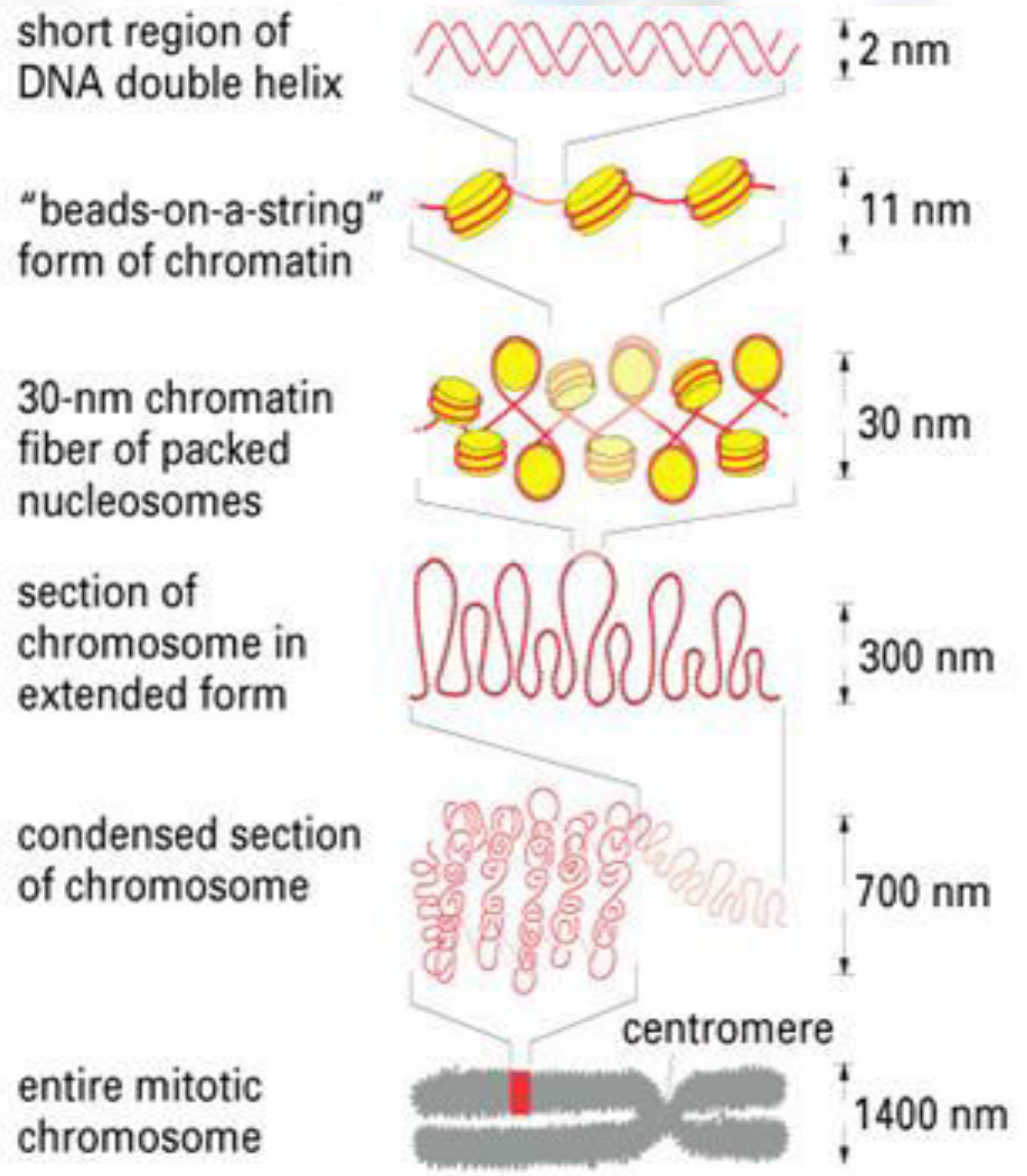


Higher Order Chromatin Folding

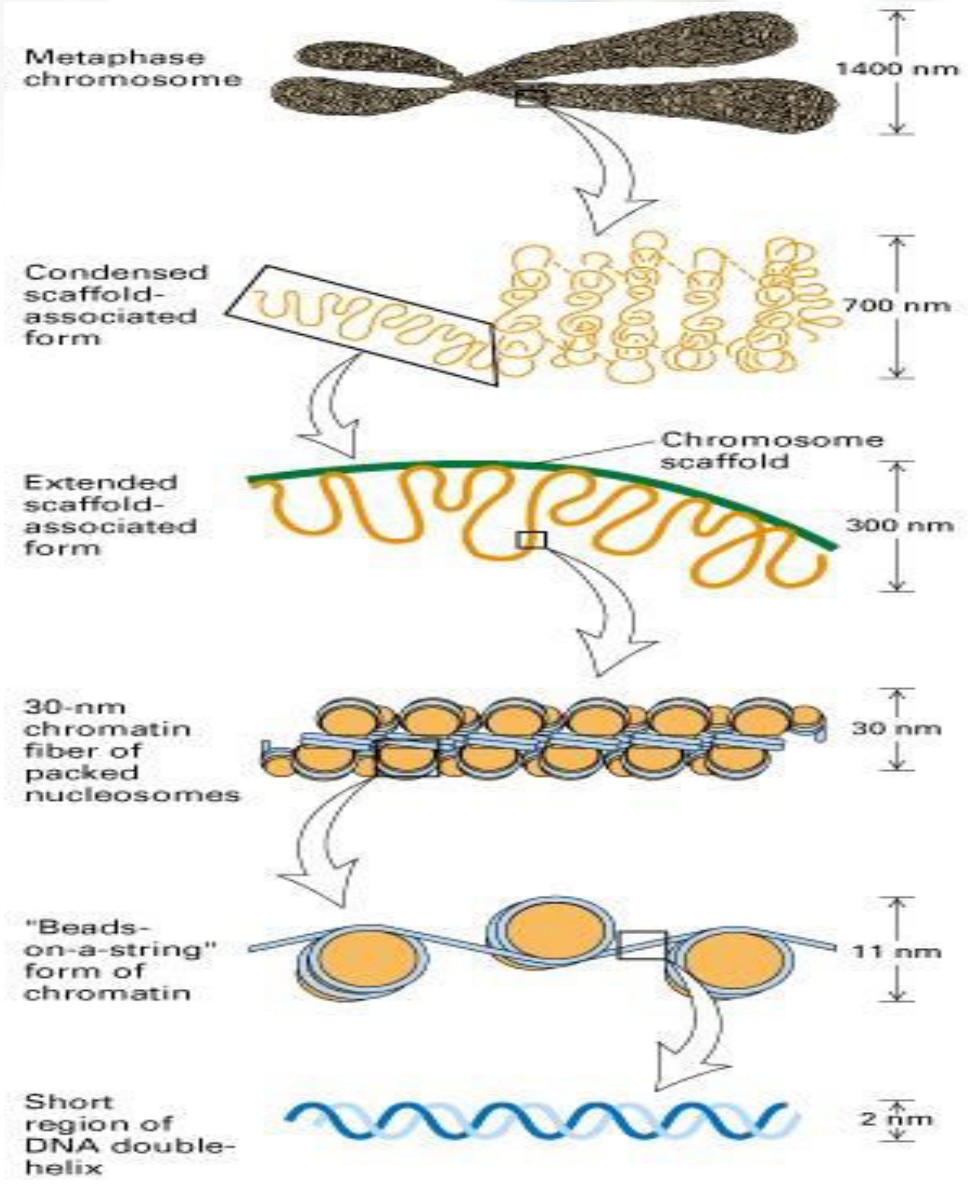
300 NM FIBRE



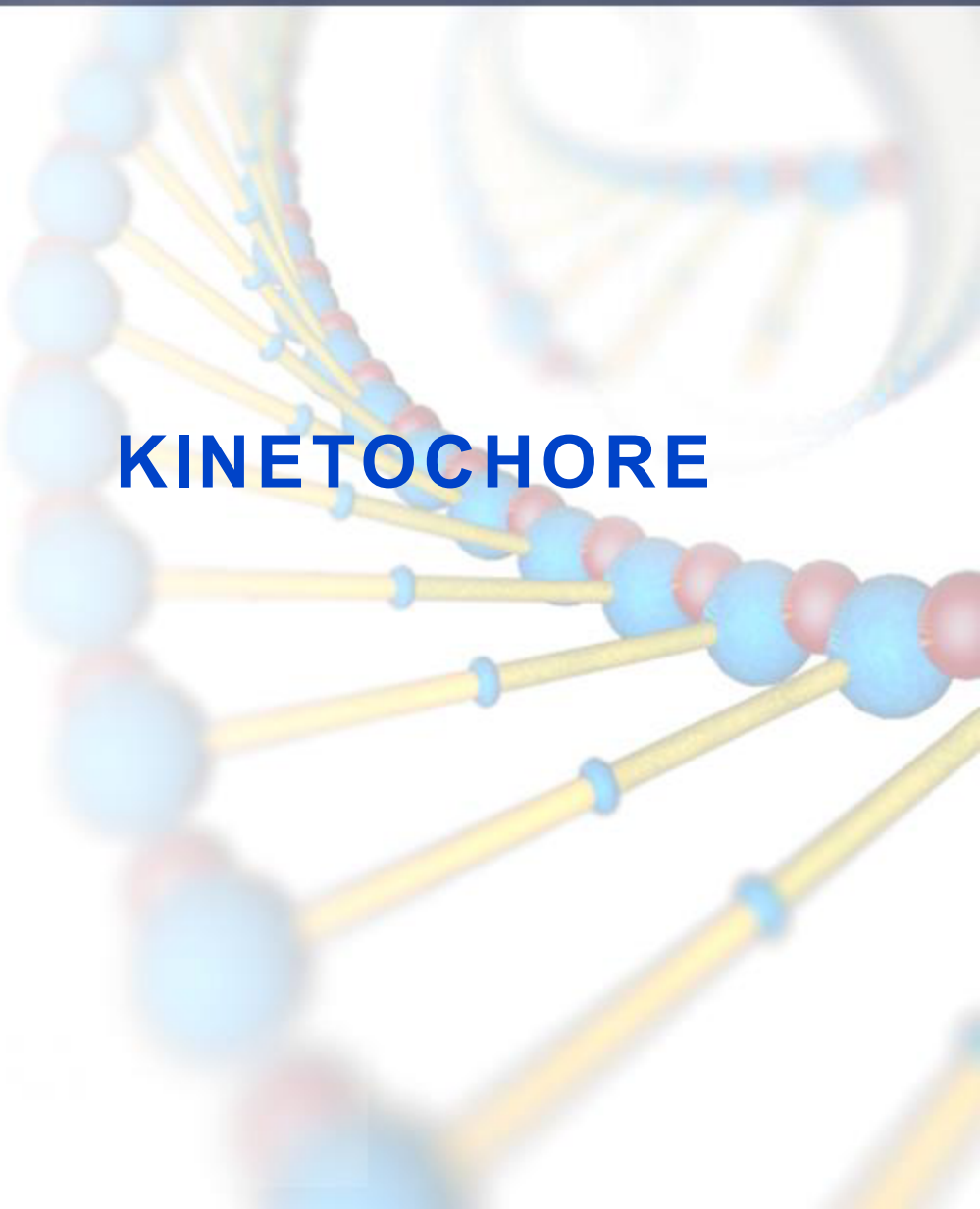
Higher Order Chromatin Folding



Higher Order Chromatin Folding



Chromosome Structure



Kinetochore

KINETOCHORE

- Within the centromere region, …locations where spindle fibers attach which consist of DNA as well as protein.
- The actual location where the attachment occurs is called the kinetochore and is composed of both DNA and protein.

Kinetochore

KINETOCHORE

- The DNA sequence within these regions is called CEN DNA ..120 base pairs long and consists of sub-domains, CDE-I, CDE-II and CDE-III.
- Mutations in the first two sub-domains have no effect upon segregation.

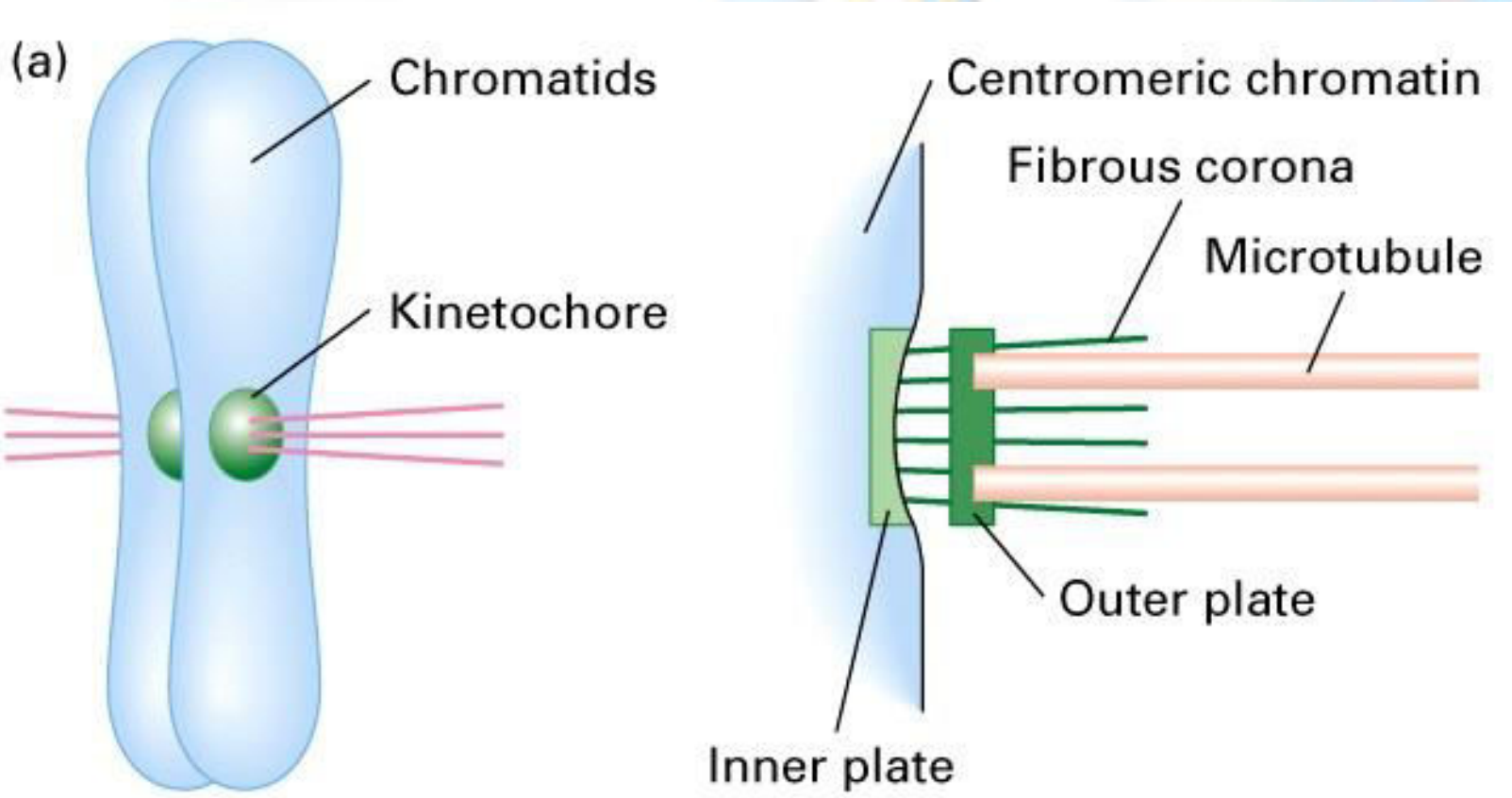
Kinetochore

KINETOCHORE

- But a point mutation in the CDE-III sub-domain completely eliminates the ability of the centromere to function during chromosome segregation.

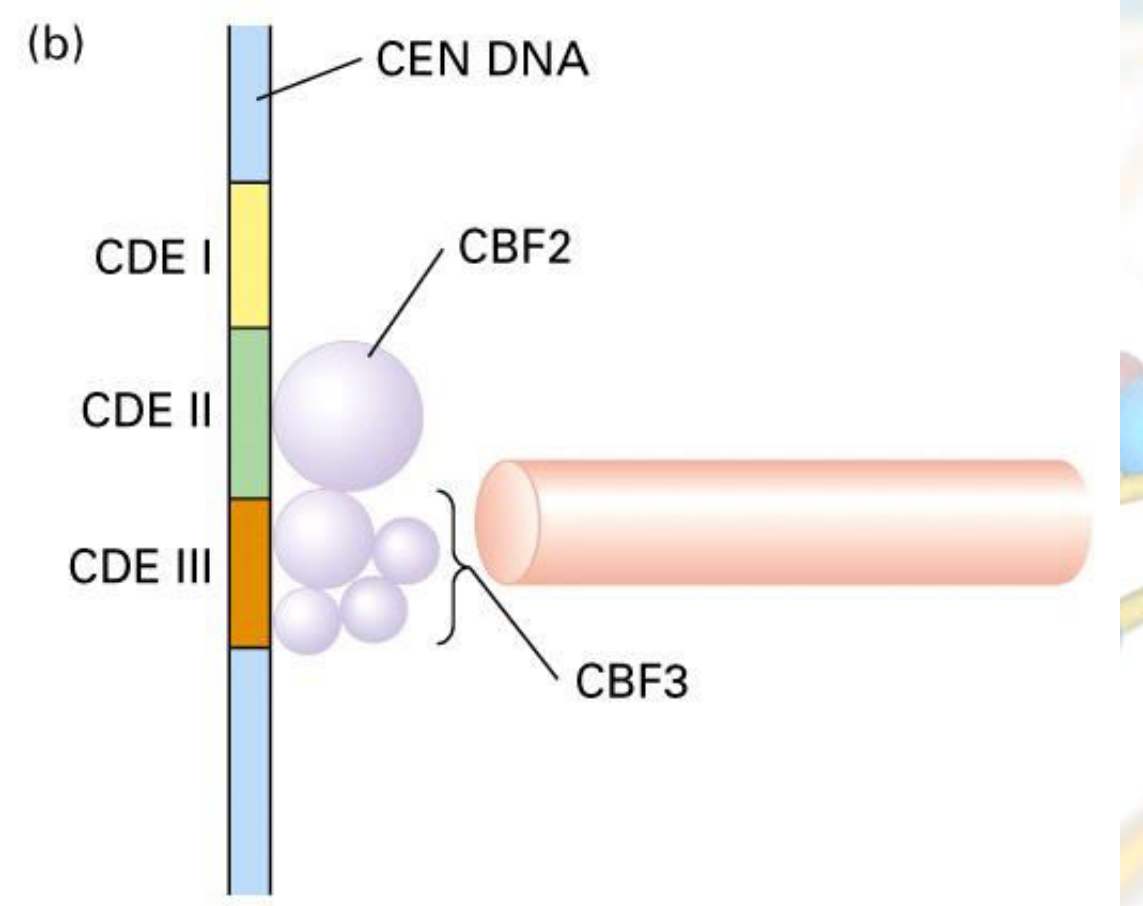
Kinetochore

KINETOCHORE



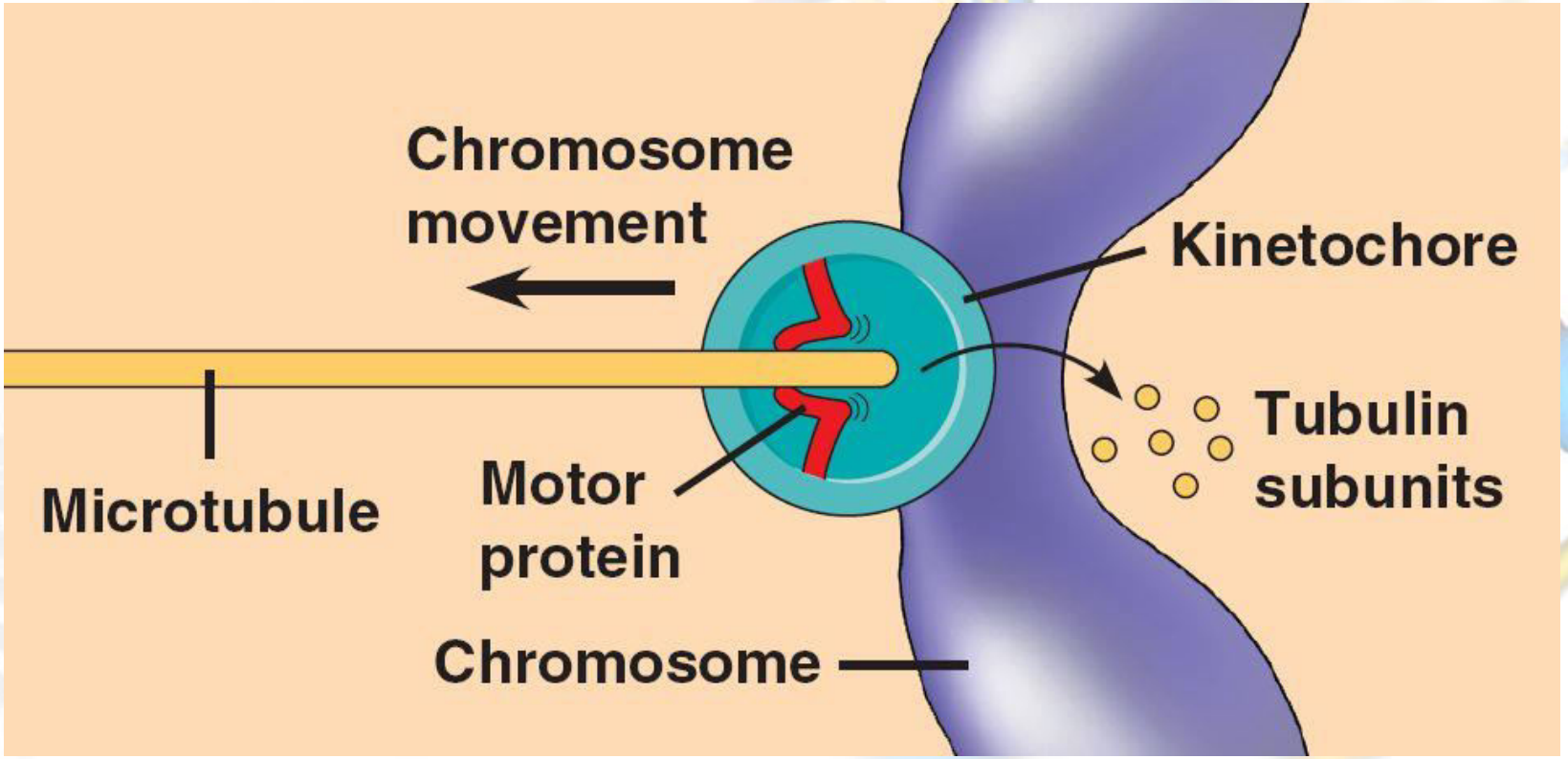
Kinetochore

KINETOCHORE

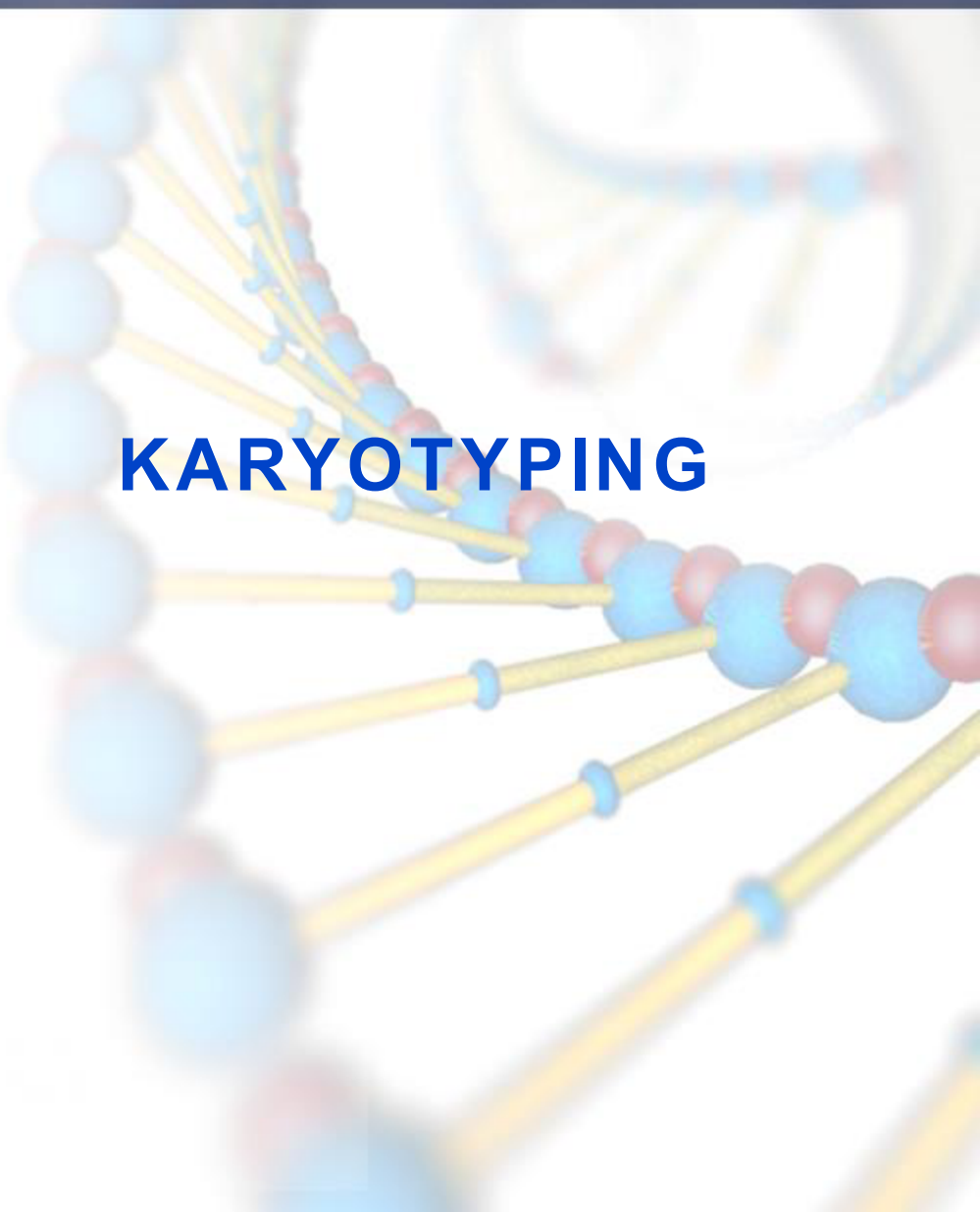


Kinetochore

KINETOCHORE



Chromosome Structure



KARYOTYPING

Karyotyping

IDENTIFYING CHROMSOMES

- Chromosomes identified by:
- Their size
- Their shape (the position of the centromere)
- Banding patterns produced by specific stains (Giemsa)

Karyotyping

KARYOTING

- Chromosomes are analyzed by organizing them into a karyotype.

Karyotyping

KARYOTYPE

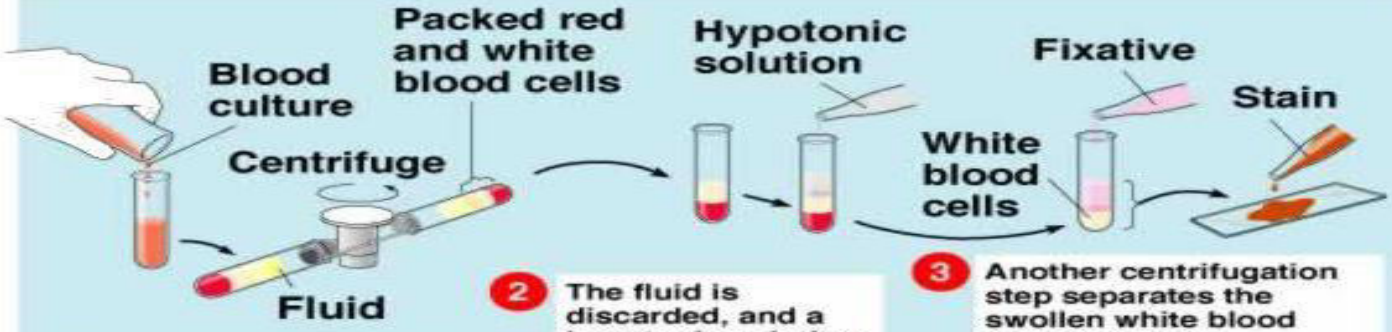
- Karyotype is the general morphology of the somatic chromosome.
- Generally, karyotypes represent by arranging in the descending order of size keeping their centromere in a straight line.

Karyotyping

IDIOTYPE

- Idiotype: the karyotype of a species may be represented diagrammatically, showing all the morphological features of the chromosome; such a diagram is known as Idiotype.

Karyotyping



1. The blood culture is centrifuged to separate the blood cells from the culture fluid.

2. The fluid is discarded, and a hypotonic solution is mixed with the cells. This makes the red blood cells swell and burst. The white blood cells swell but do not burst, and their chromosomes spread out.

3. Another centrifugation step separates the swollen white blood cells. The fluid containing the remnants of the red blood cells is poured off. A fixative (preservative) is mixed with the white blood cells. A drop of the cell suspension is spread on a microscope slide, dried, and stained.

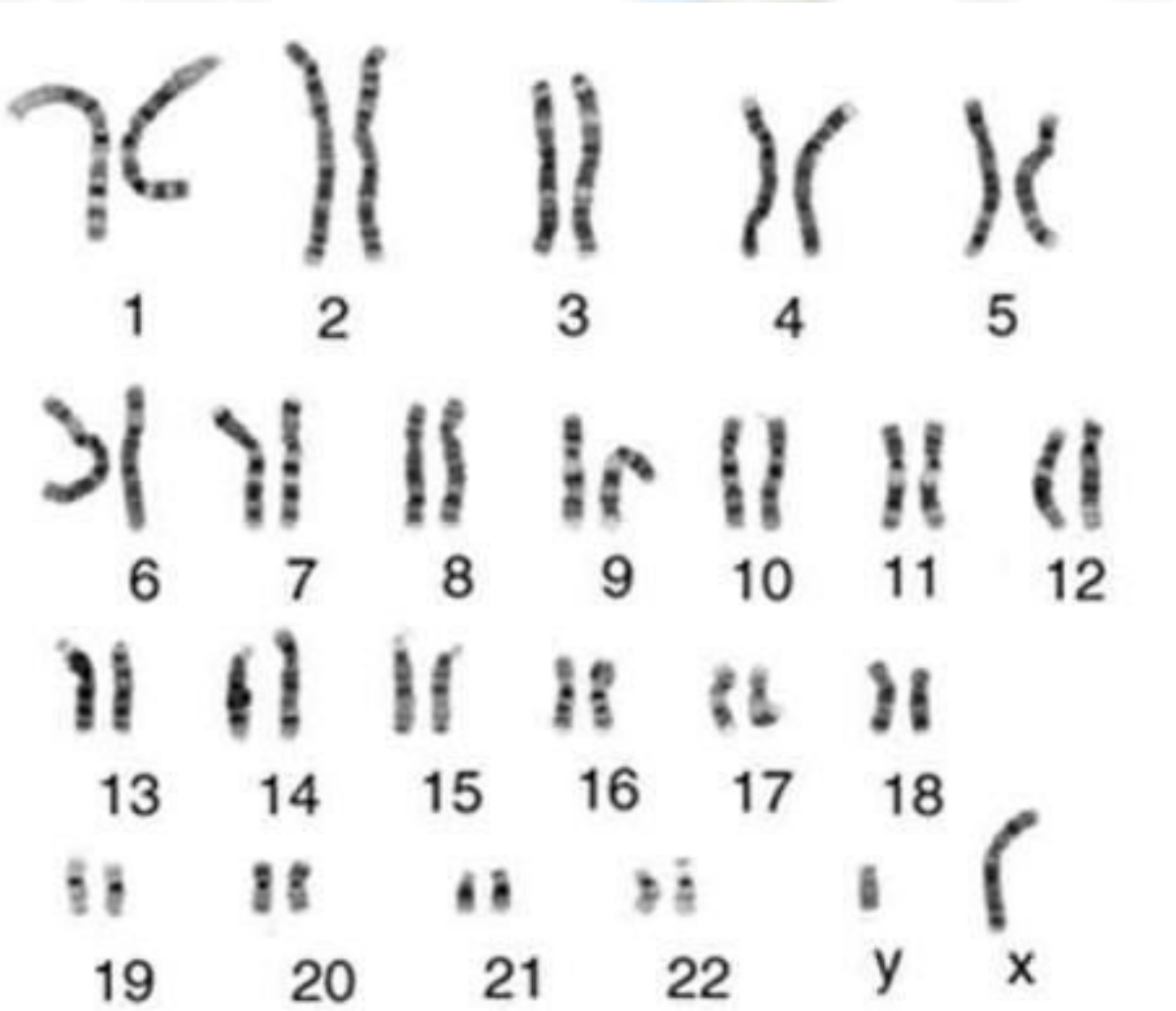




4. The slide is viewed with a microscope, and images of the metaphase chromosomes are sorted by size and shape on a computer.

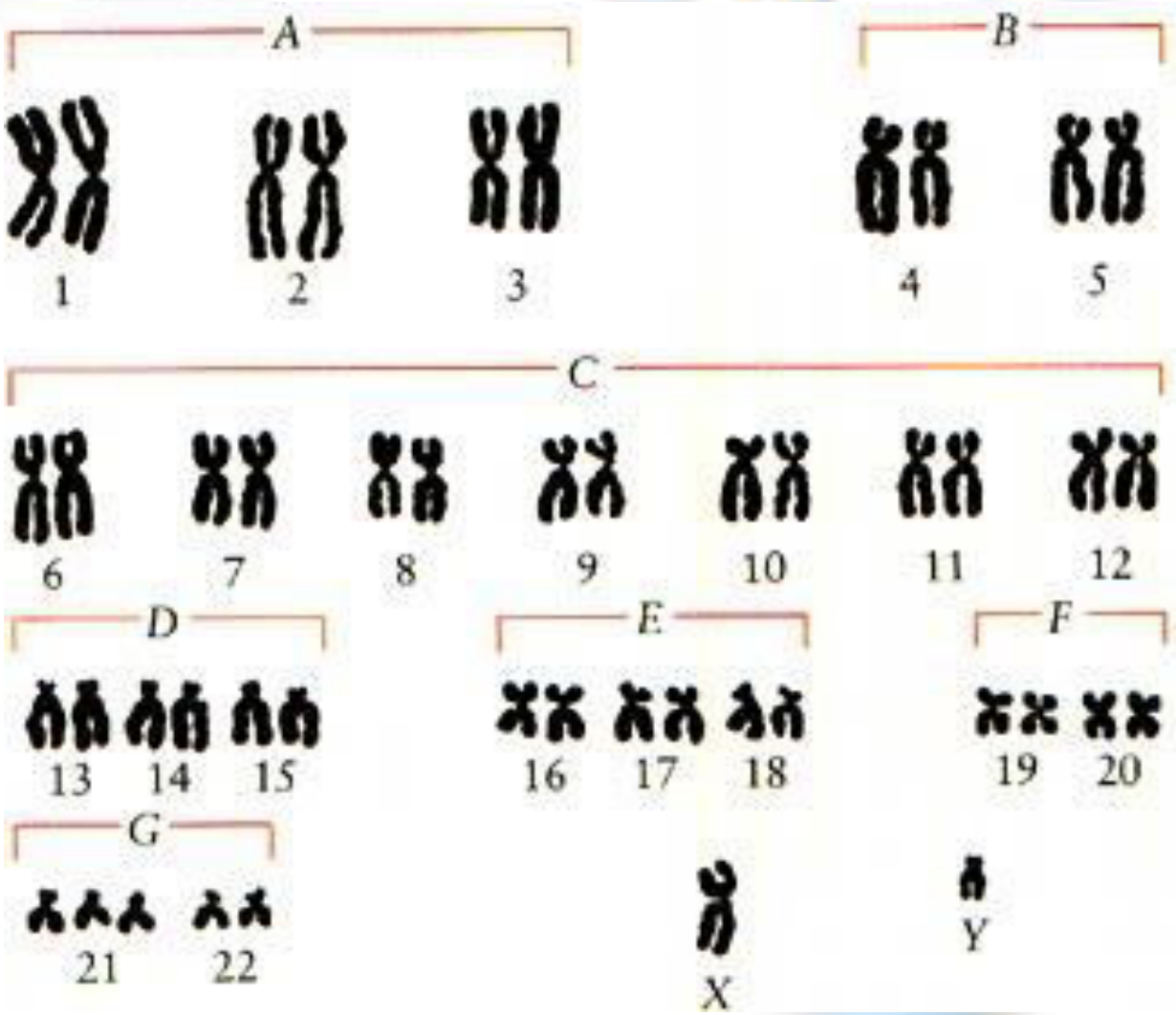
5. The resulting display is the karyotype. The 46 chromosomes here include 22 pairs of autosomes and 2 sex chromosomes, *X* and *Y*. Each of the chromosomes consists of two sister chromatids lying close together. (see Figure 8.12)

Karyotyping

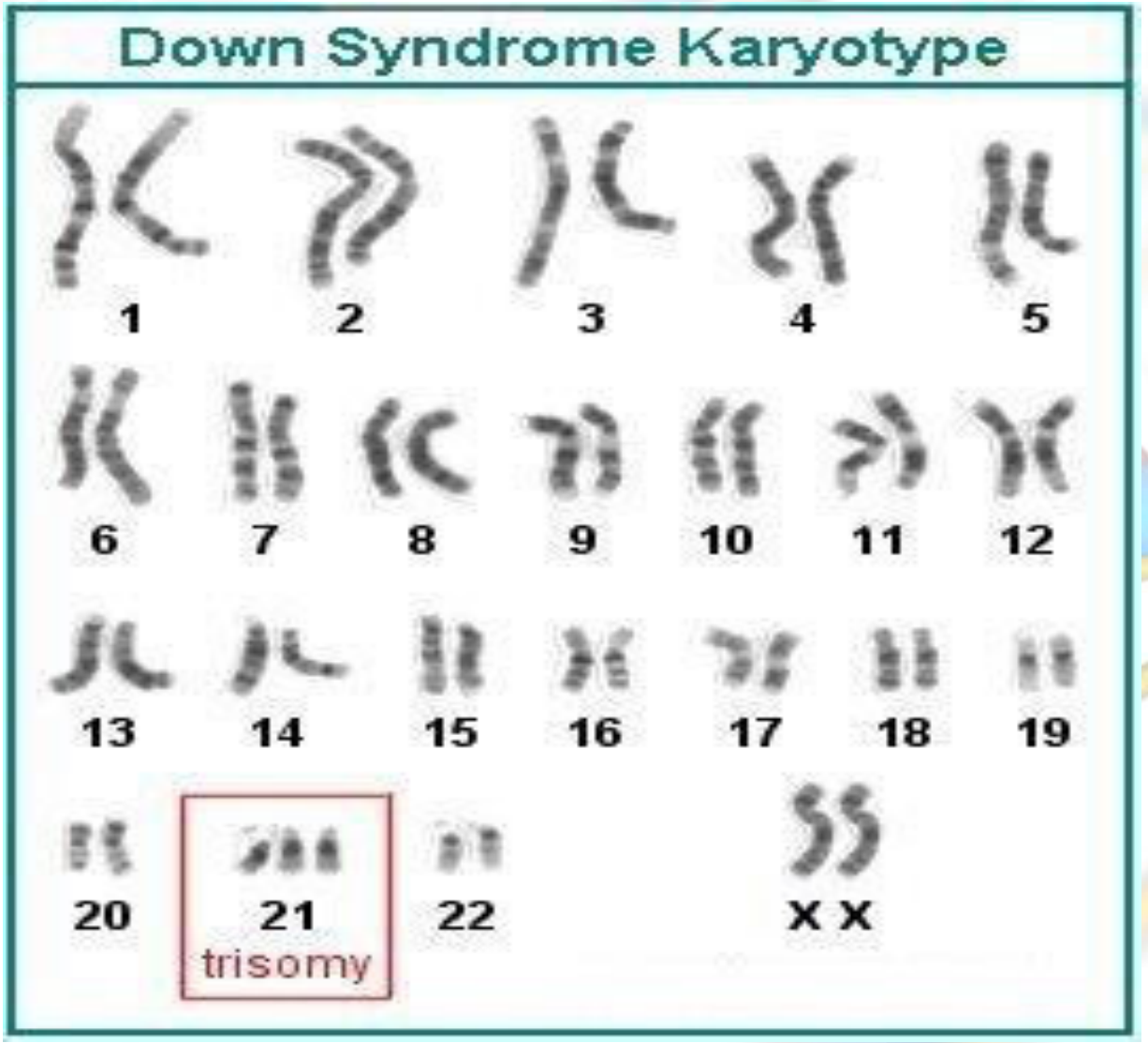


Karyotyping

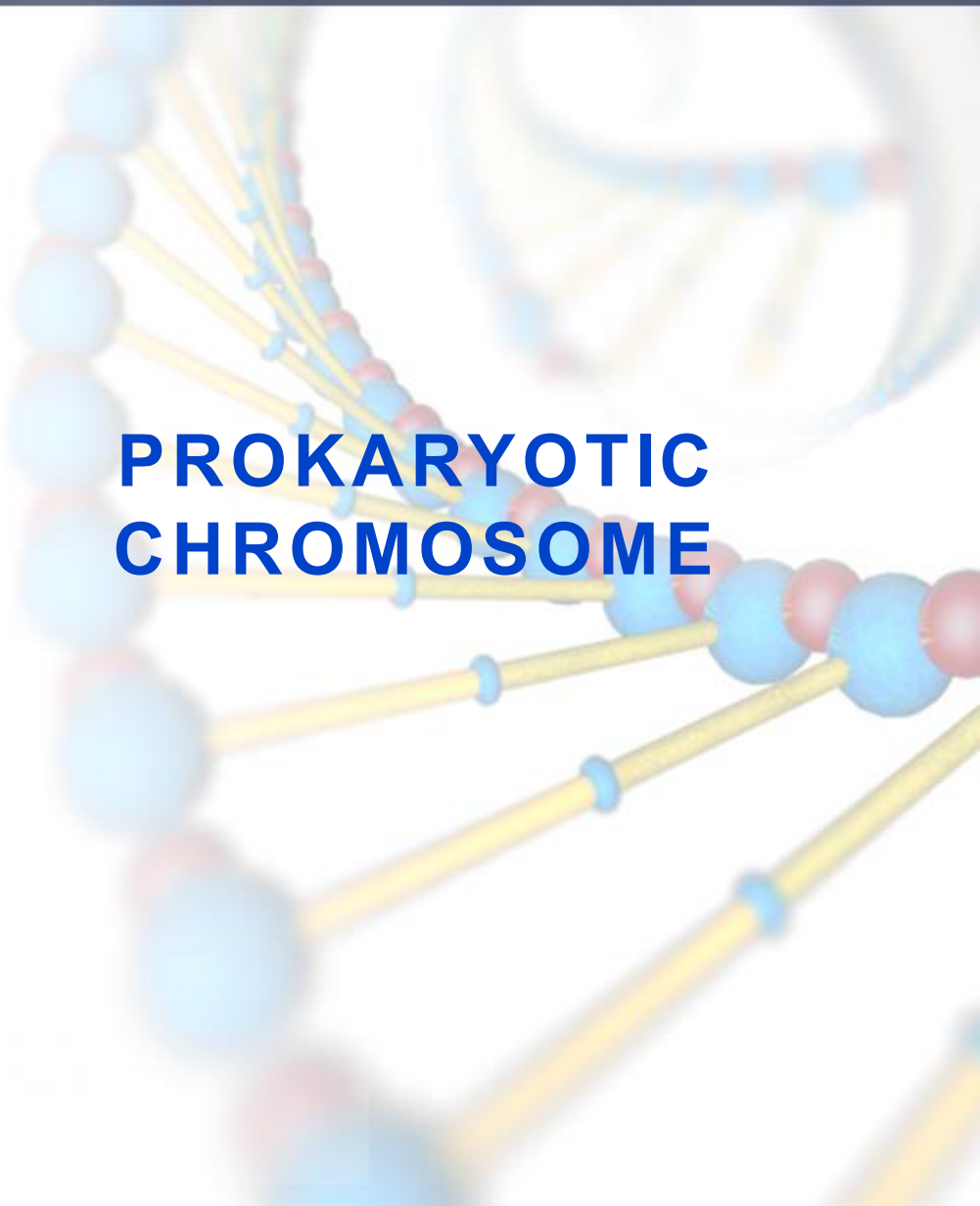
KARYOTYPE



Karyotyping



Chromosome Structure



PROKARYOTIC CHROMOSOME

Prokaryotic Chromosome

PROKARYOTIC CHROMOSOMES

- The prokaryotes usually have only one chromosome, and it bears little morphological resemblance to eukaryotic chromosomes.

Prokaryotic Chromosome

PROKARYOTIC CHROMOSOMES

- The genome length is smallest in RNA viruses.
- The number of gene may be as high as 150 in some larger bacteriophage genome.

Prokaryotic Chromosome

PROKARYOTIC CHROMOSOMES

- In E.coli, about 3000 to 4000 genes are organized into its one circular chromosome.
- The chromosome exists as a highly folded and coiled structure dispersed throughout the cell.

Prokaryotic Chromosome

PROKARYOTIC CHROMOSOMES

- There are about 50 loops in the chromosome of E. coli.
- These loops are highly twisted or supercoiled structure with about four million nucleotide pairs.

Prokaryotic Chromosome

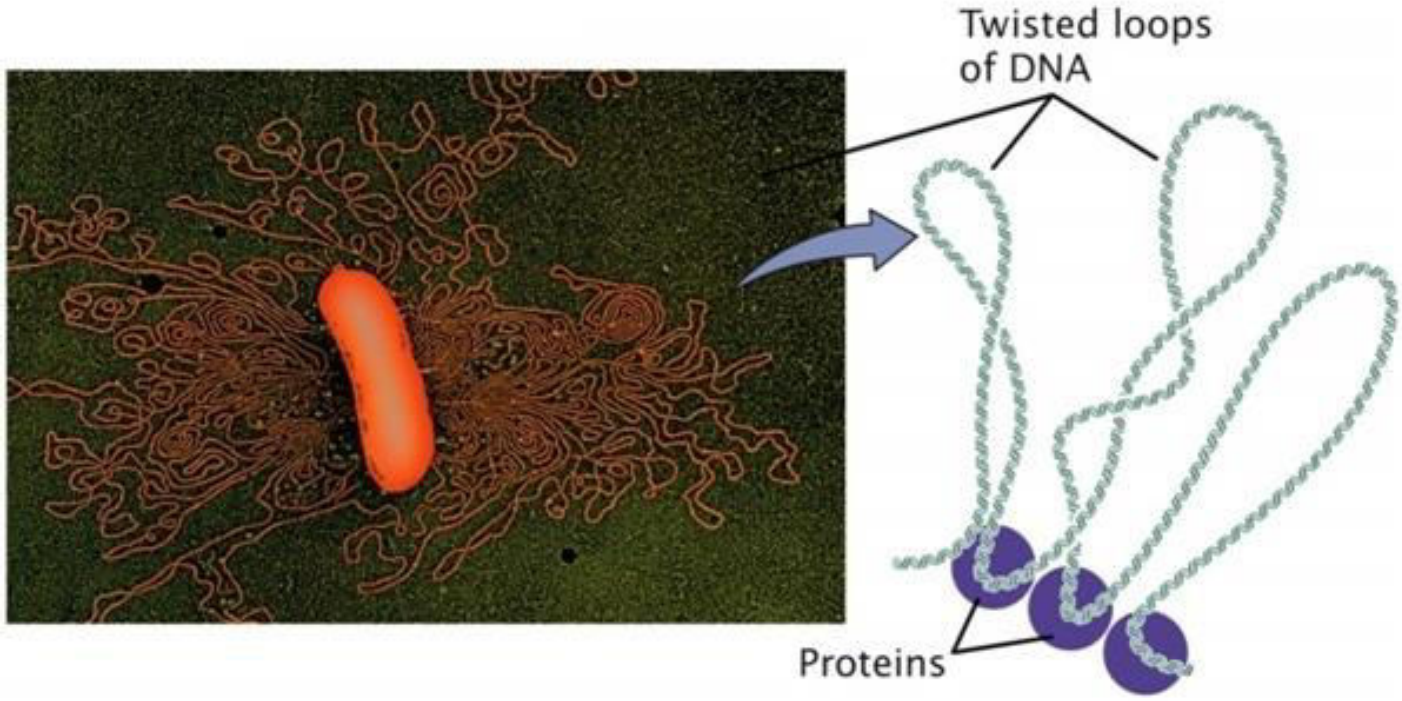
PROKARYOTIC CHROMOSOMES

- During replication of DNA, the coiling must be relaxed.
- DNA gyrase is necessary for the unwinding the coils.

Prokaryotic Chromosome

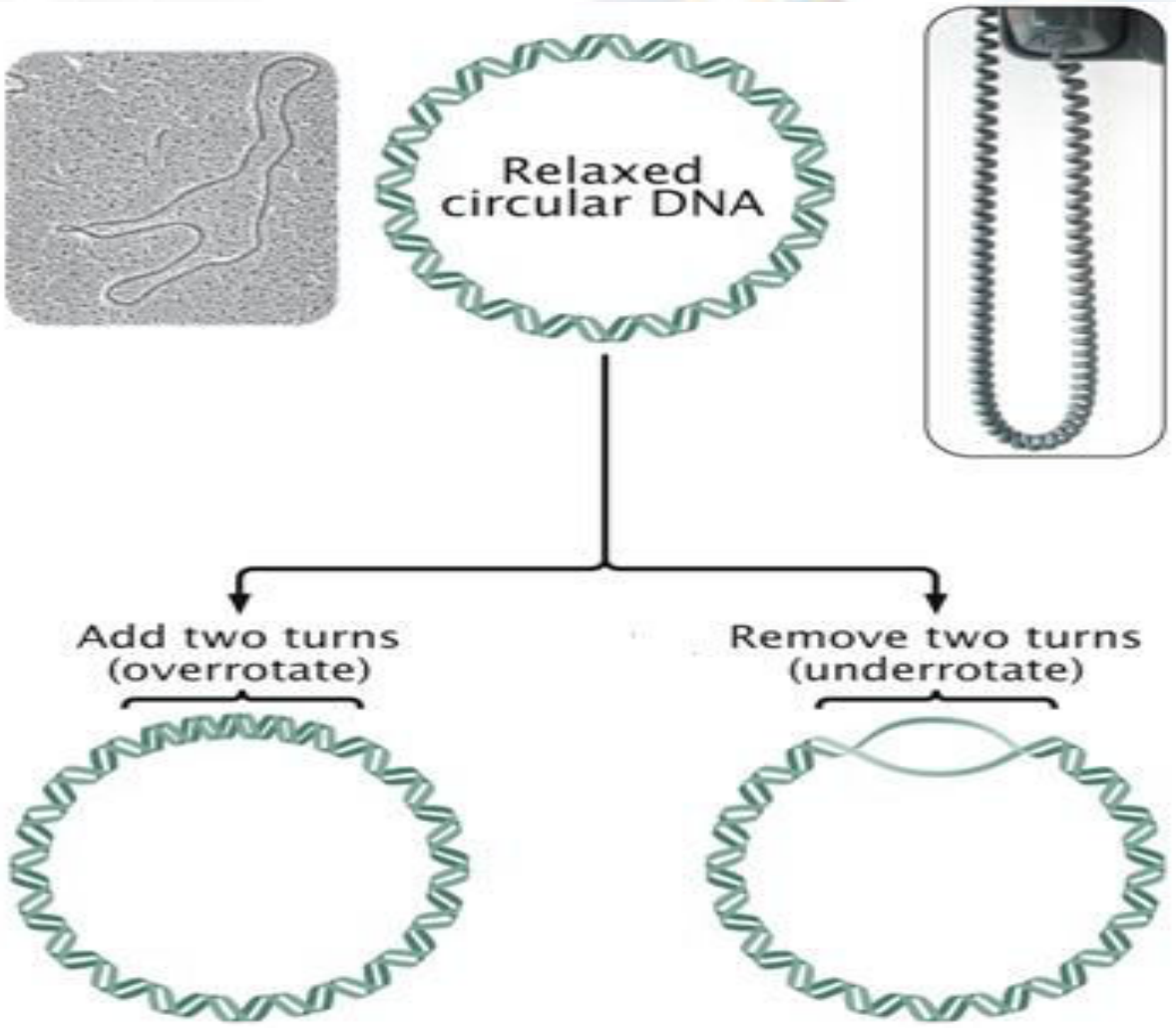
BACTERIAL CHROMOSOME

- Single, circular DNA molecule located in the nucleoid region of



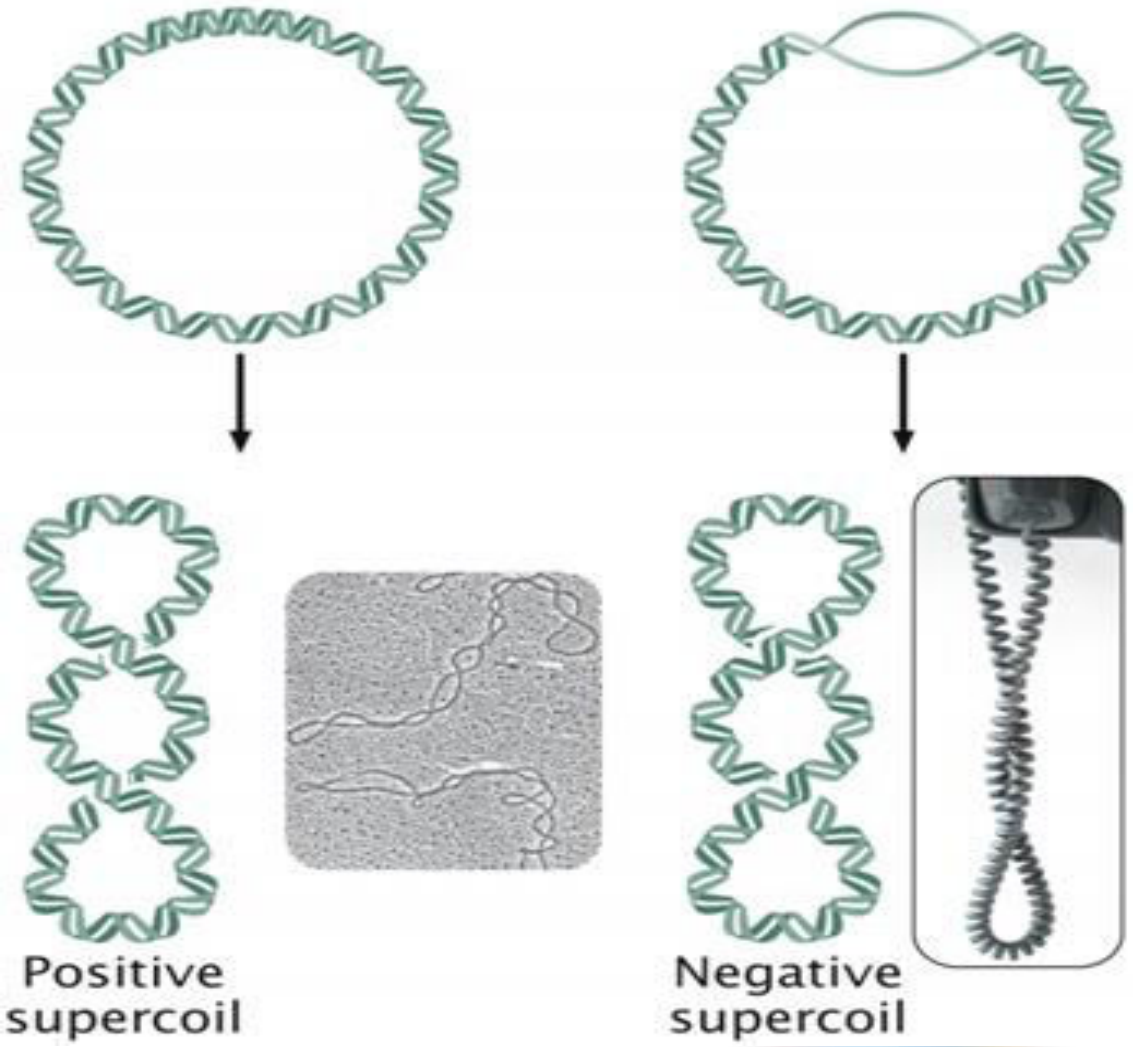
Prokaryotic Chromosome

SUPER COILING



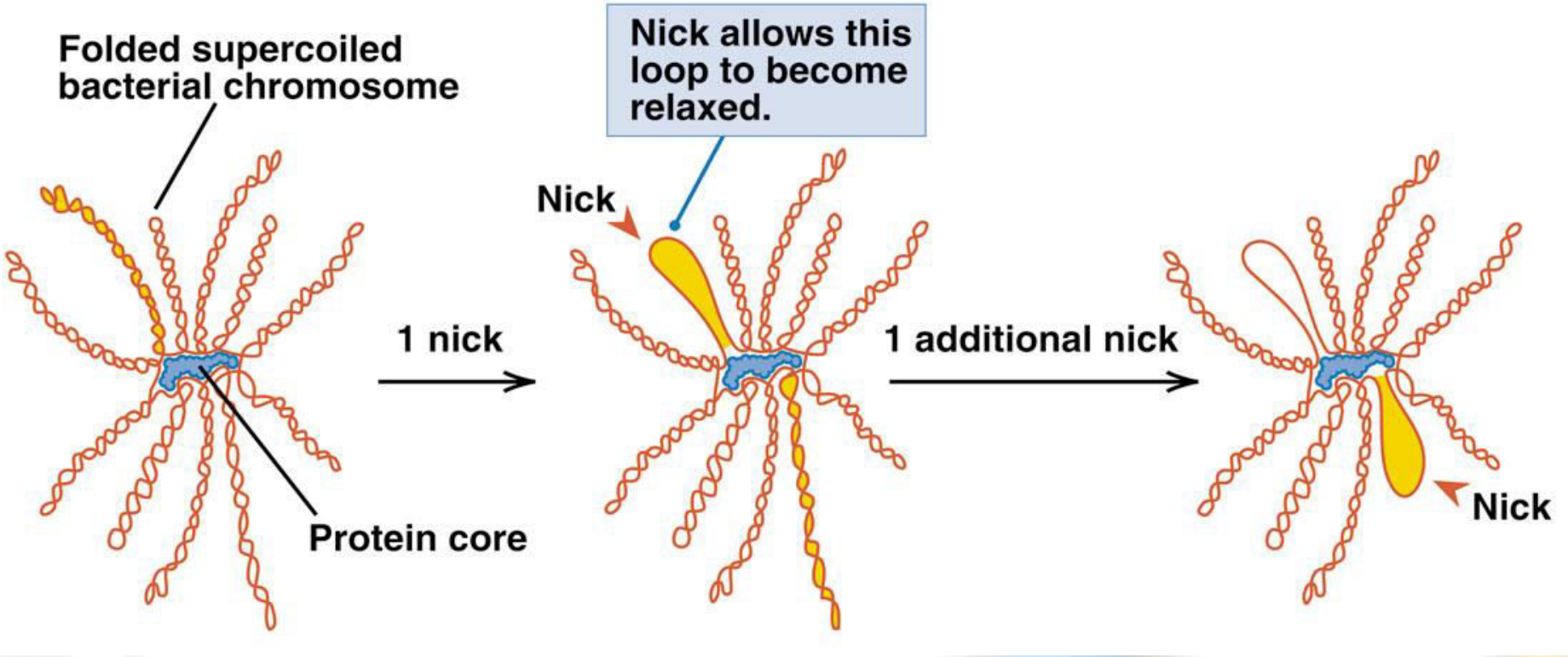
Prokaryotic Chromosome

SUPER COILING

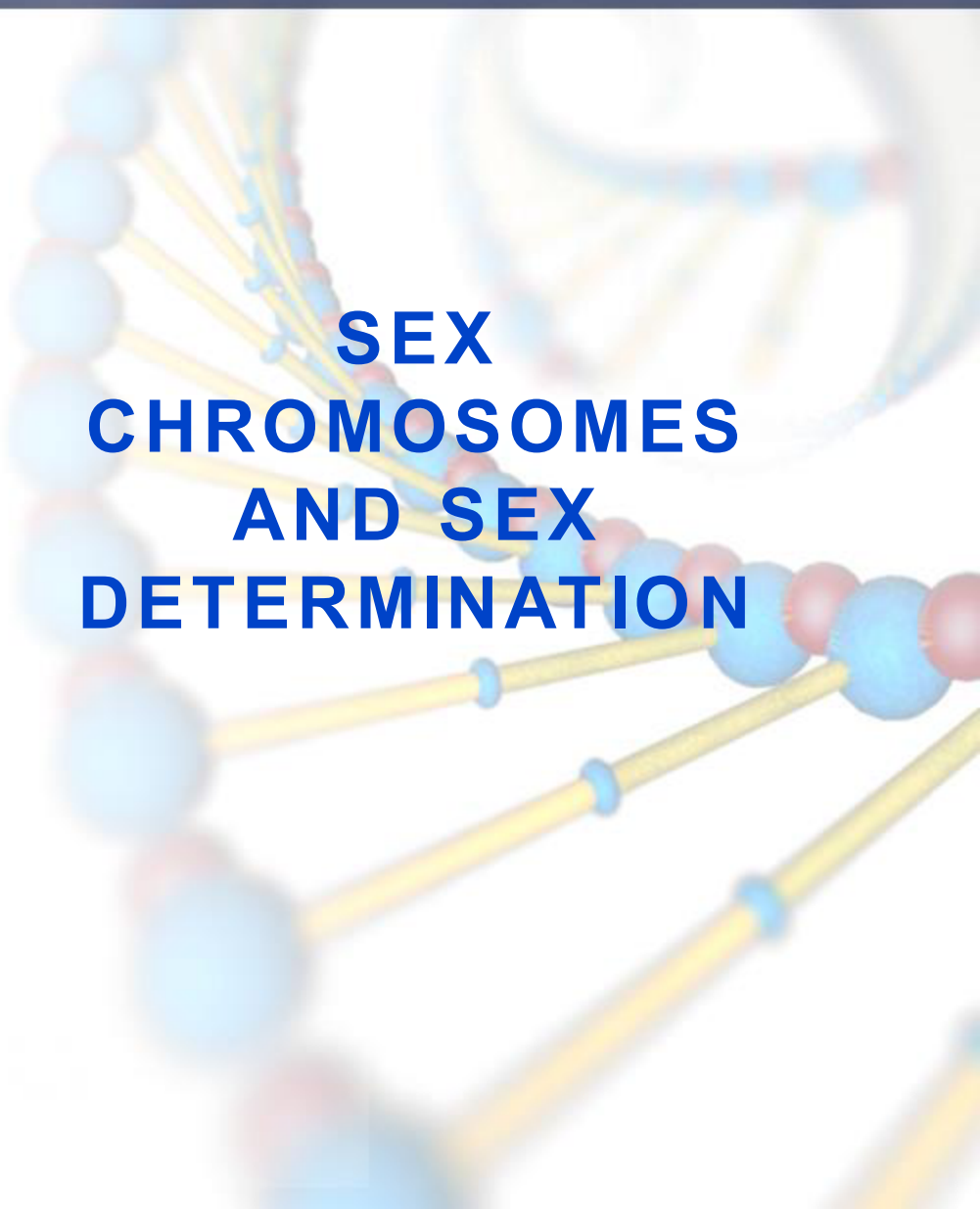


Prokaryotic Chromosome

MECHANISM OF FOLDING OF BACTERIAL CHROMOSOME



Chromosome Structure



SEX CHROMOSOMES AND SEX DETERMINATION

Sex Chromosomes and Sex Determination

SEX CHROMOSOMES

- Sex of many animals is determined by genes on chromosomes called sex chromosomes.
- Sex can be homogametic
- Sex can be heterogametic

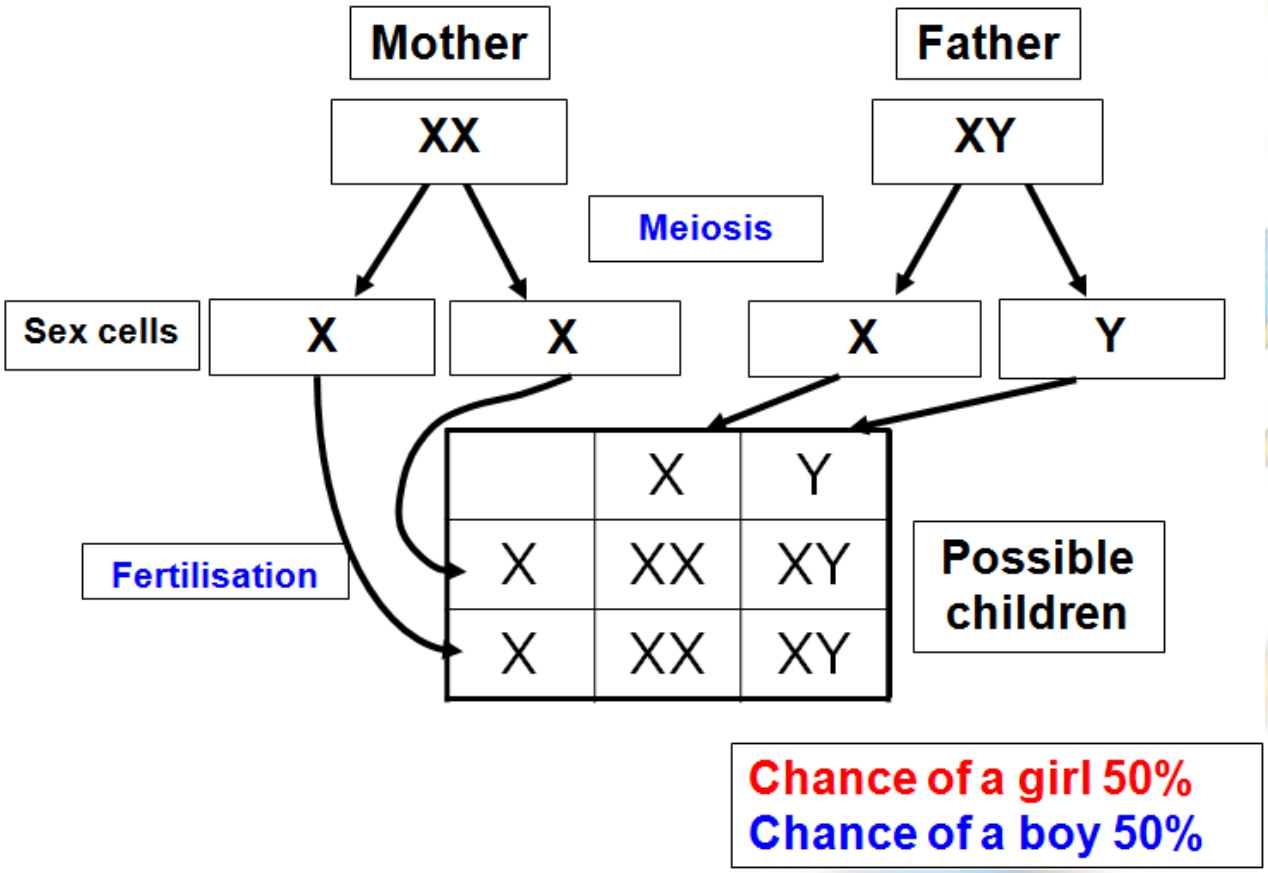
Sex Chromosomes and Sex Determination

SEX DETERMINATION IN DIFFERENT ANIMALS

HOMOGAMETIC SEX	HETEROGAMETIC SEX	SEX DETERMINATION
Female **XX**	Male **XY**	Presence of Y-chromosome = maleness (mammals and fish) Presence of second X-chromosome = femaleness (Drosophila, the fruit fly)
Male **ZZ**	Female **ZW**	Birds, amphibians, reptiles, butterflies, moths.
Female **XX**	Male **Xo**	Grasshoppers

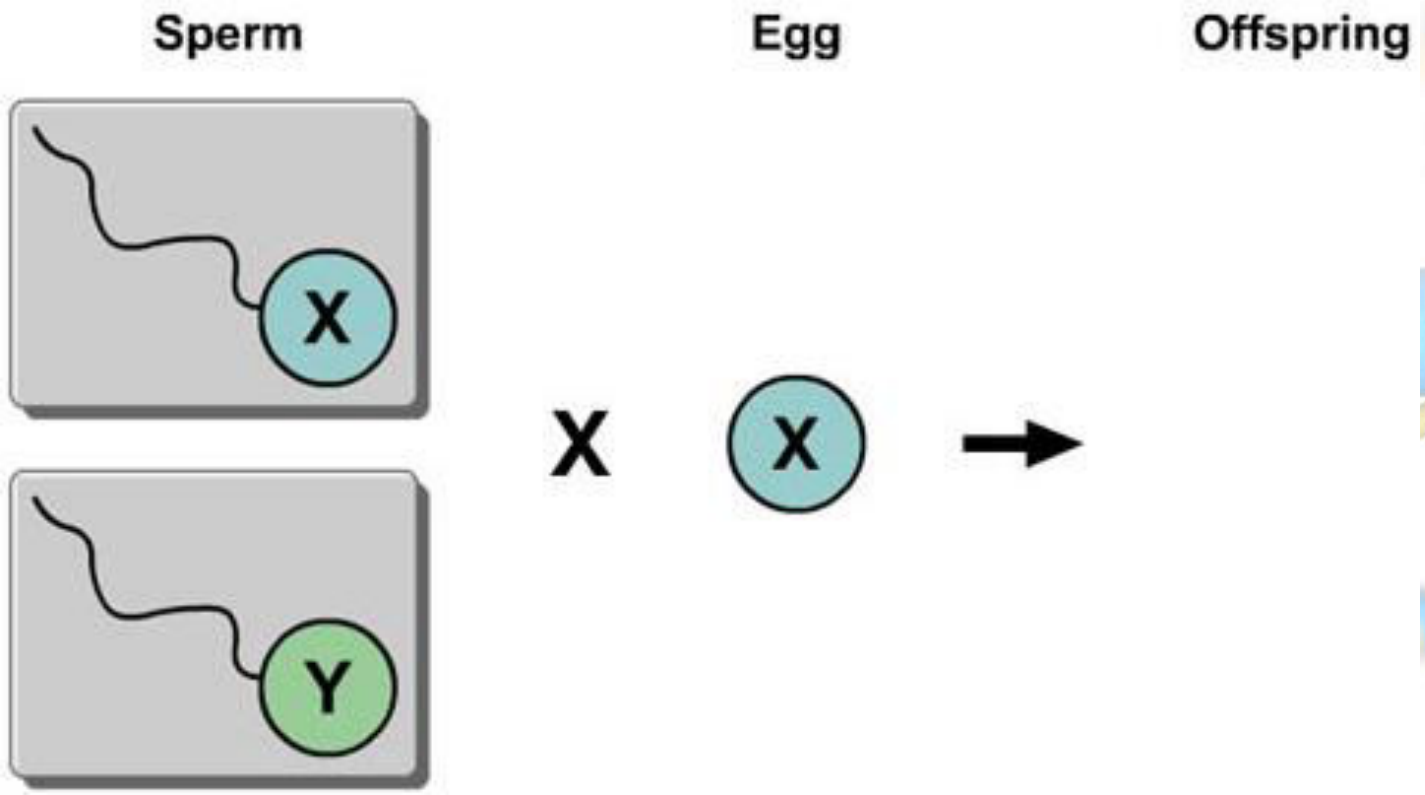
Sex Chromosomes and Sex Determination

INHERITANCE OF GENDER IN HUMANS



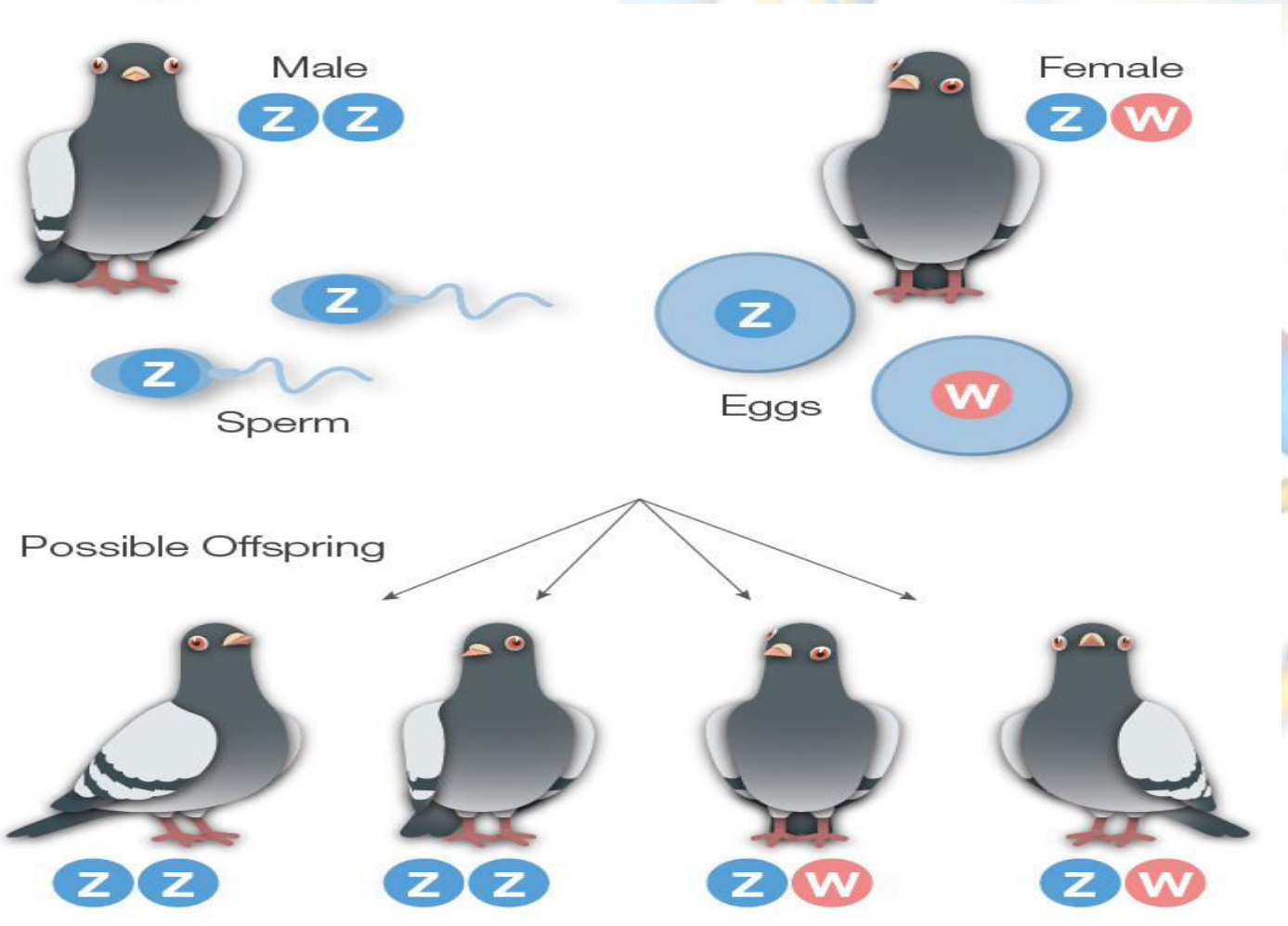
Sex Chromosomes and Sex Determination

HUMAN SEX DETERMINATION



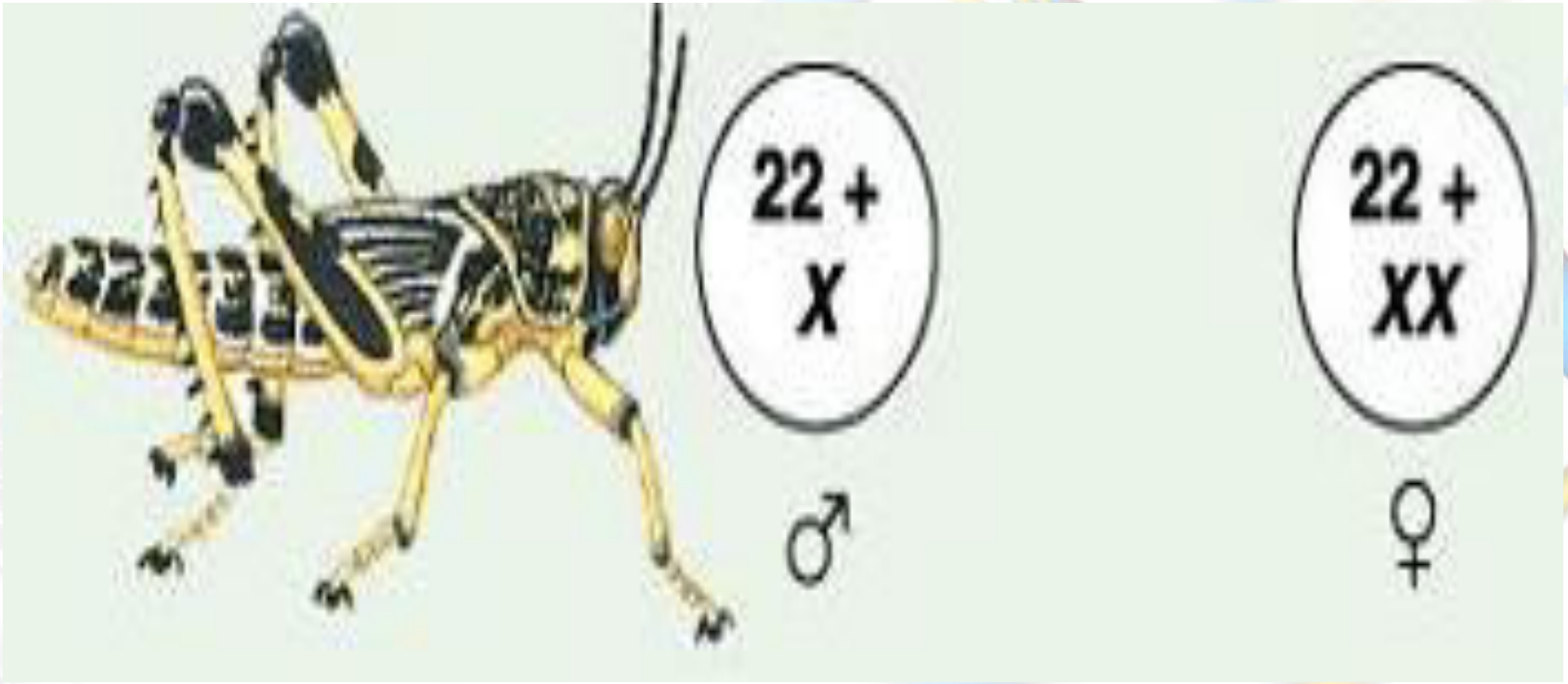
Sex Chromosomes and Sex Determination

SEX SYSTEM IN BIRDS

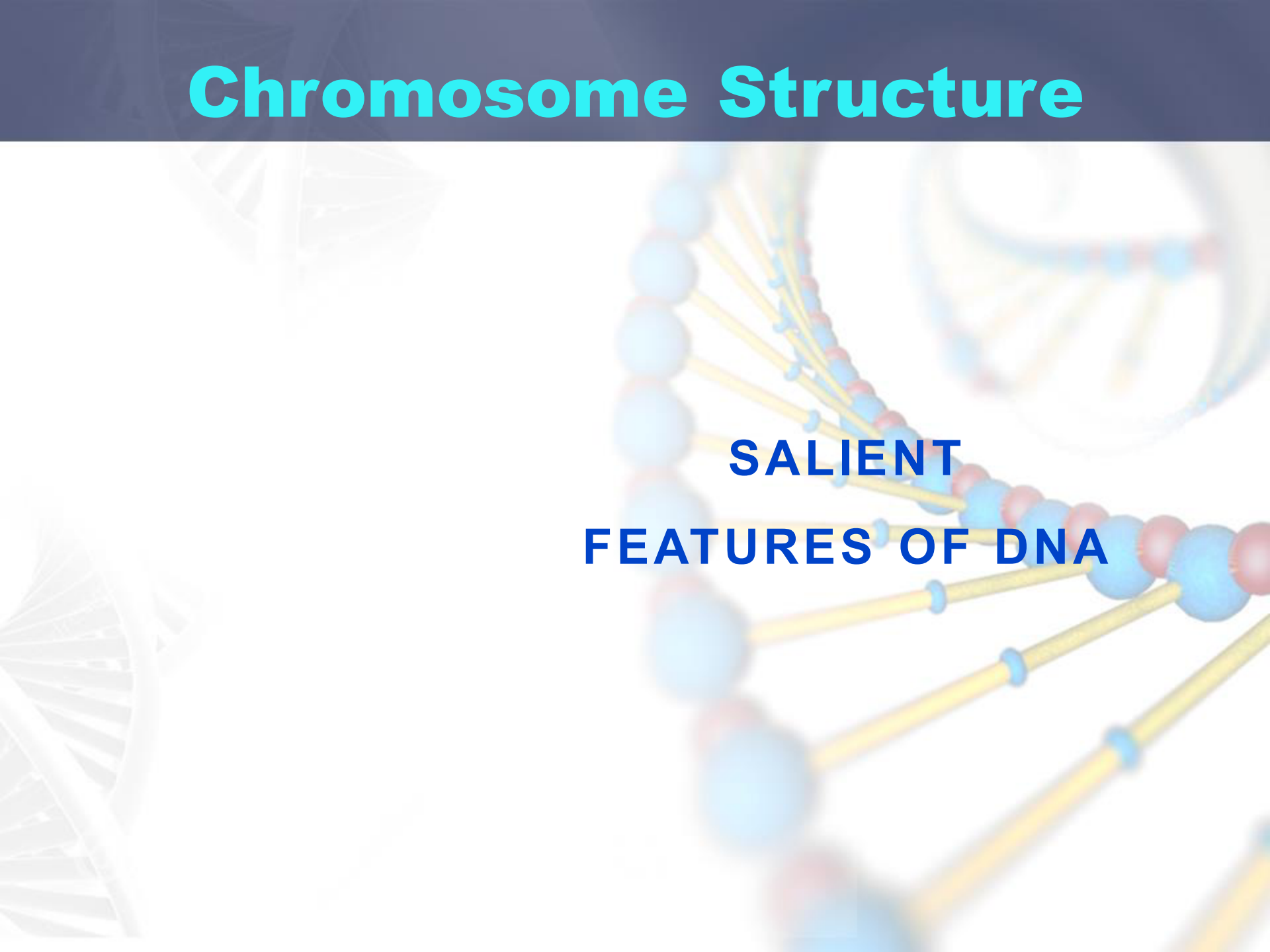


Sex Chromosomes and Sex Determination

XO SYSTEM IN GRASSHOPER



Chromosome Structure



SALIENT FEATURES OF DNA

Salient Features of DNA

DNA

- Watson: X-rays diffraction pattern projected by DNA.
- Erwin Chargaff demonstrated ratio of A:T are 1:1, and G:C are 1:1
- Chemical structure of nucleotide identified.

Salient Features of DNA

DNA

- Composed of 2 polymers of nucleotides.
- Run antiparallel.
- Molecule structure resembles a spiral staircase.
- Complementary base pairing:

  A-T, C-G

Salient Features of DNA

DNA

- Genetic information in the linear sequence of nucleotides.
- Genetic information is to synthesize proteins.
- DNA forms double helix with two complimentary strands holding - hydrogen bonds.

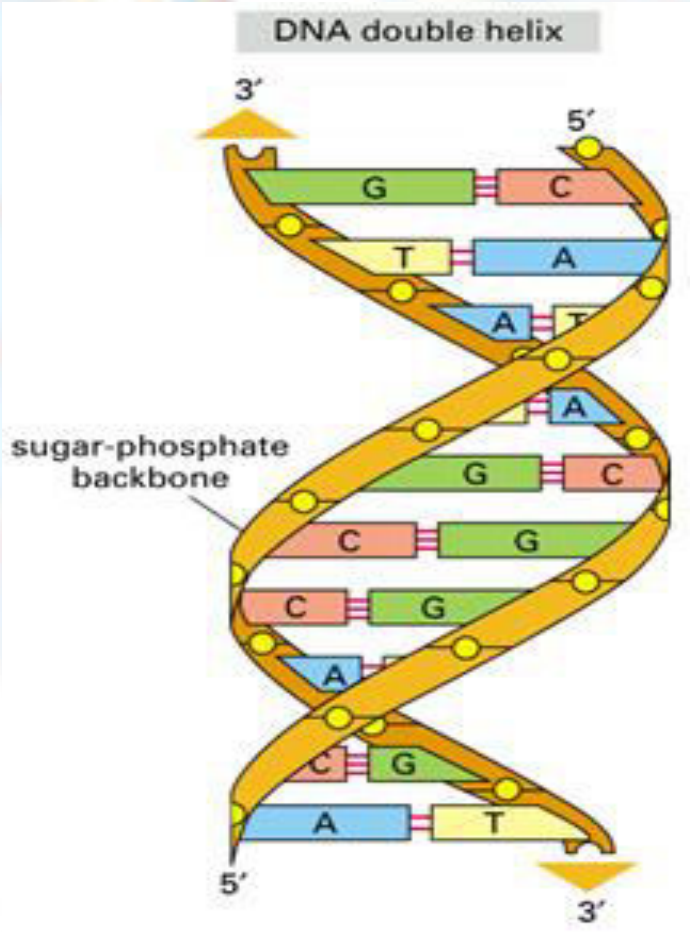
Salient Features of DNA

DNA

- DNA duplication occurs using one strand of parental DNA as template to form complimentary pairs with a new DNA strand.

Salient Features of DNA

DNA



Salient Features of DNA

DNA

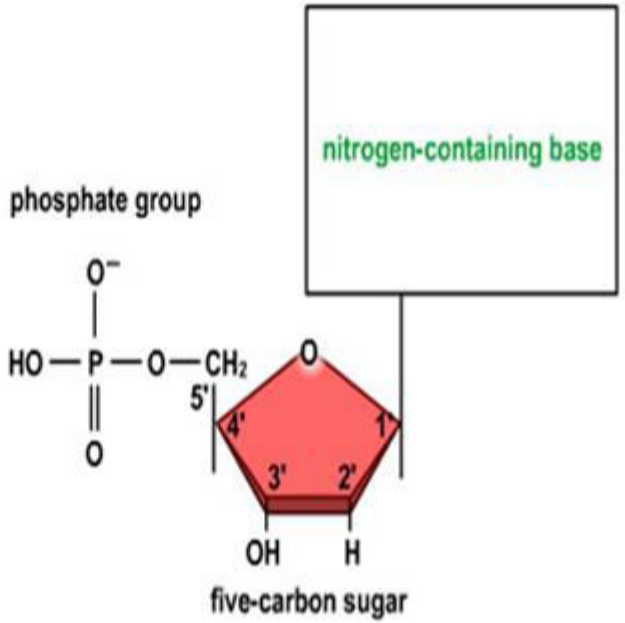
- The structure of Nucleotides includes: A phosphate sugar backbone, with one :he 5 nitrogenous ses (ATCG,U).

adenine (A)

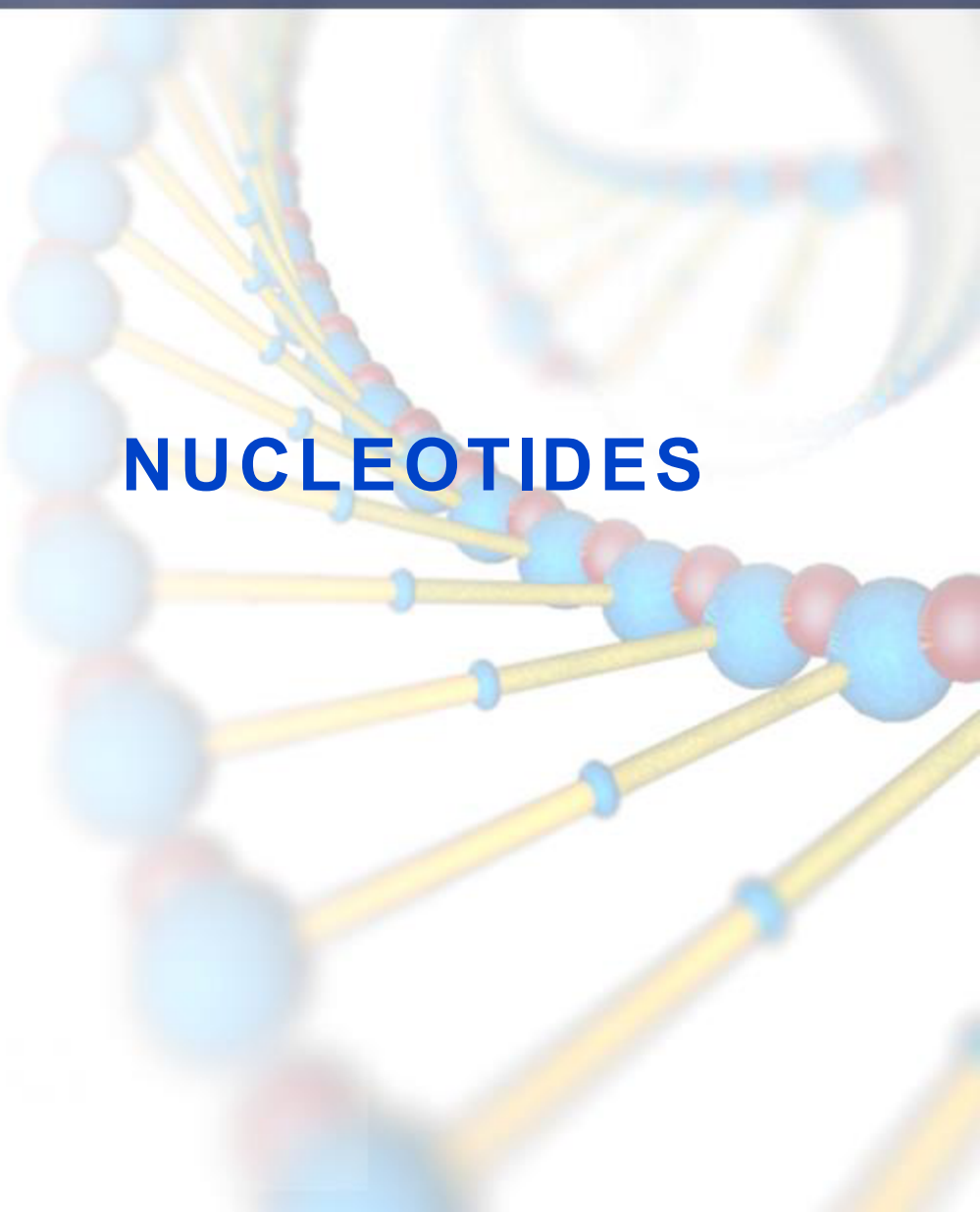
guanine (G)

thymine (T)

cytosine (C)



Chromosome Structure



NUCLEOTIDES

Nucleotides

NUCLEOTIDE

- Made of three components
  - 5 carbon sugar (pentose)
  - nitrogenous base
  - phosphate group

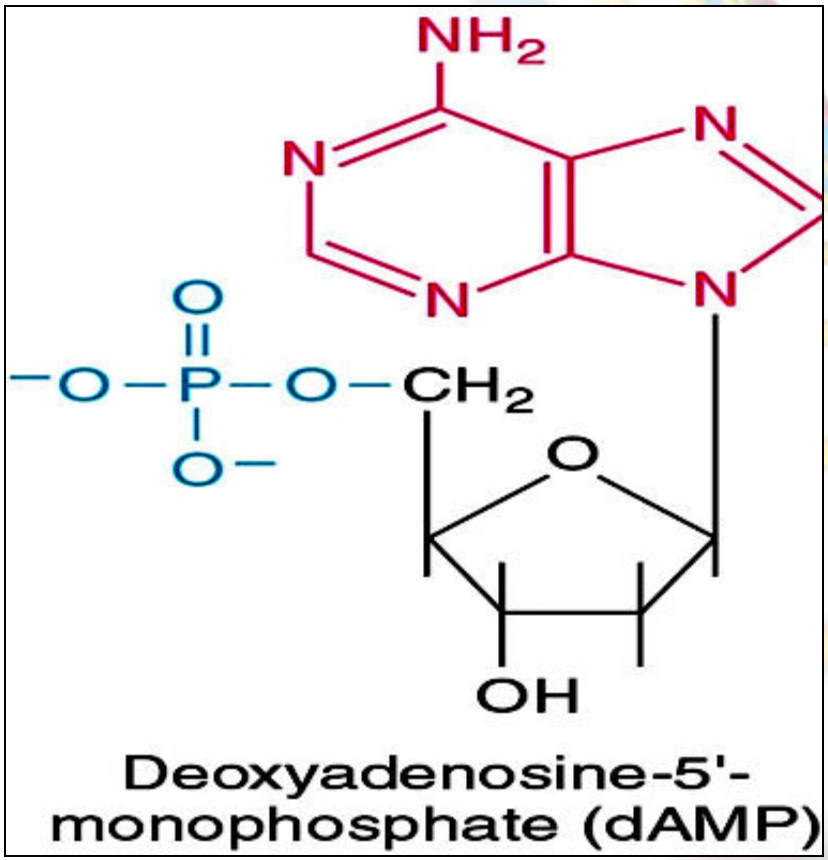
Nucleotides

NUCLEOTIDE

- Oligonucleotide – short polymer (<10)
- Polynucleotide – long polymer (>10)
- Nucleoside = monomer of sugar + base

Nucleotides

NUCLEOTIDE MONOMER



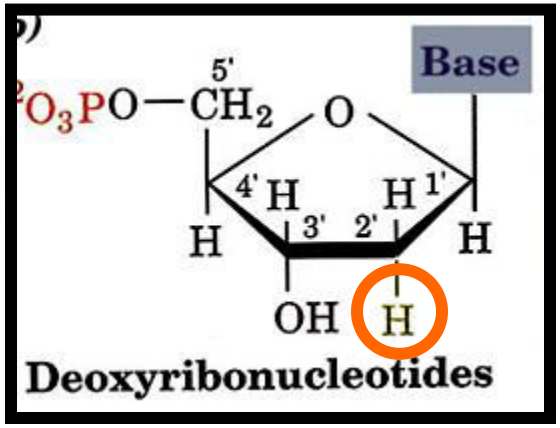
Deoxyadenosine-5'-monophosphate (dAMP)

Nucleotides

SUGAR MOLECULES

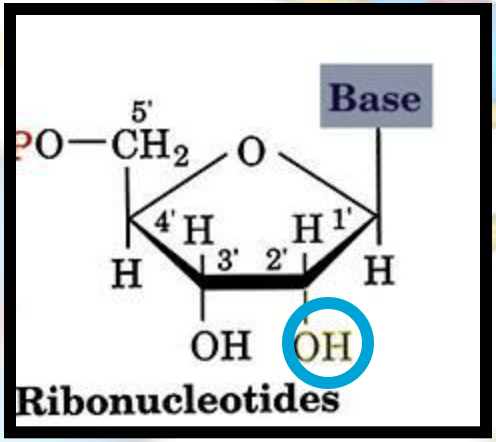
- Two types of sugar molecules

DNA



2'-deoxy-D-ribose

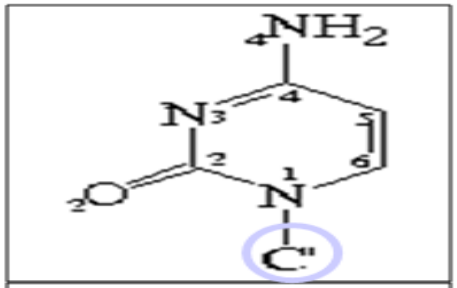
RNA



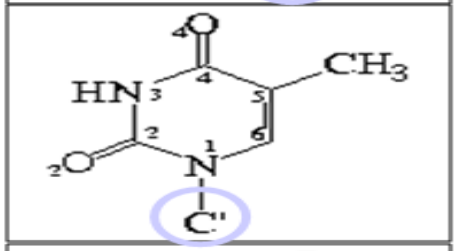
2'-D-ribose

Nucleotides

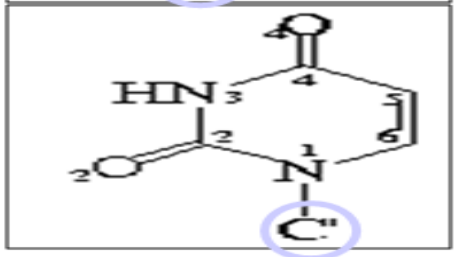
NITROGENOUS BASES
PYRIMIDINES



cytosine



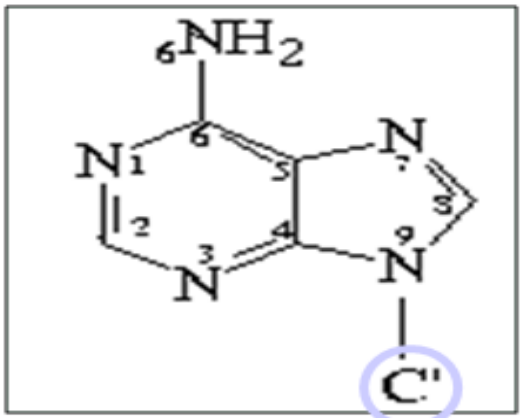
thymine



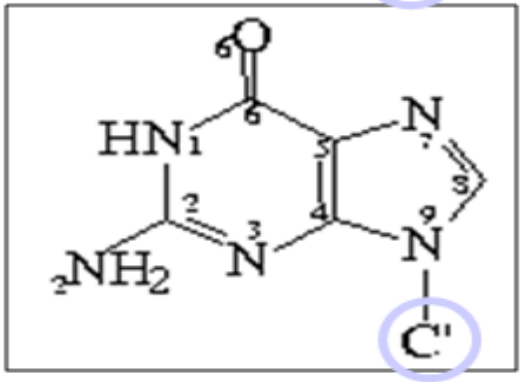
uracil

Nucleotides

NITROGENOUS BASES PURINES

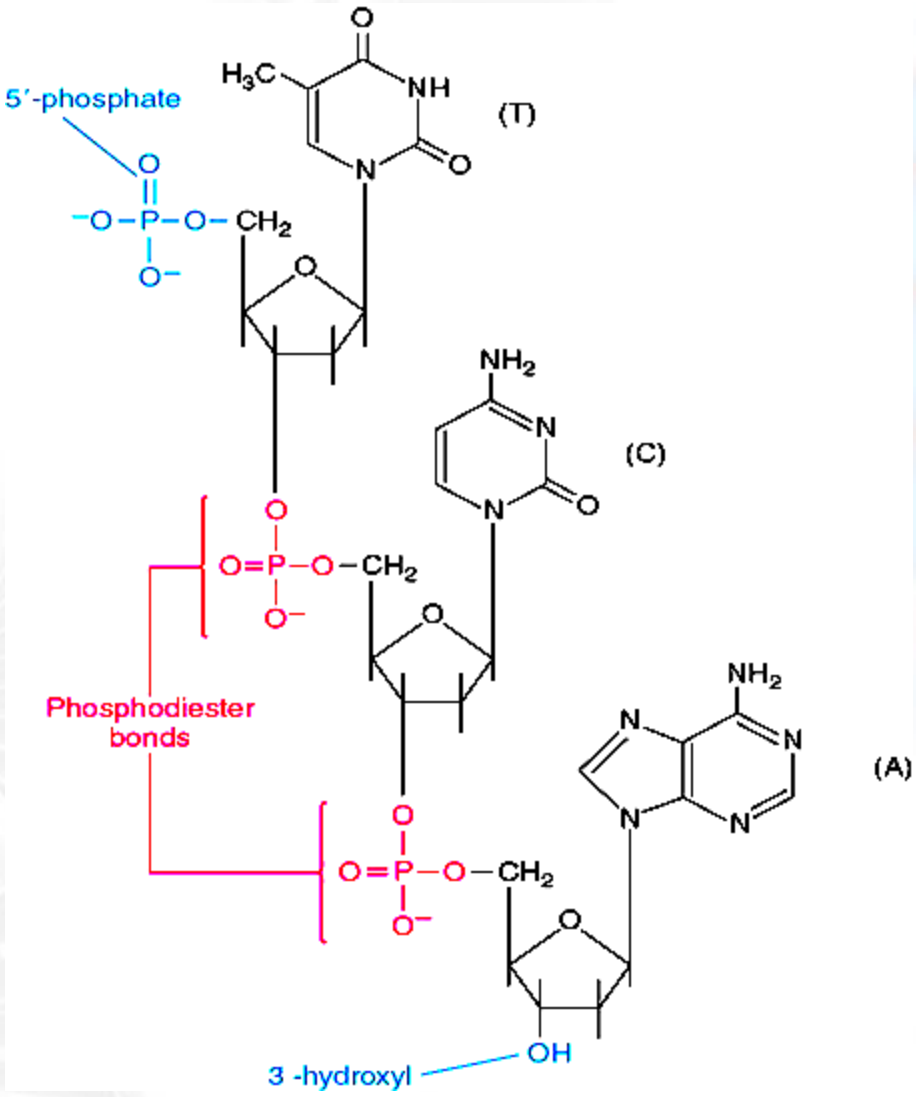


adenine



guanine

Nucleotides

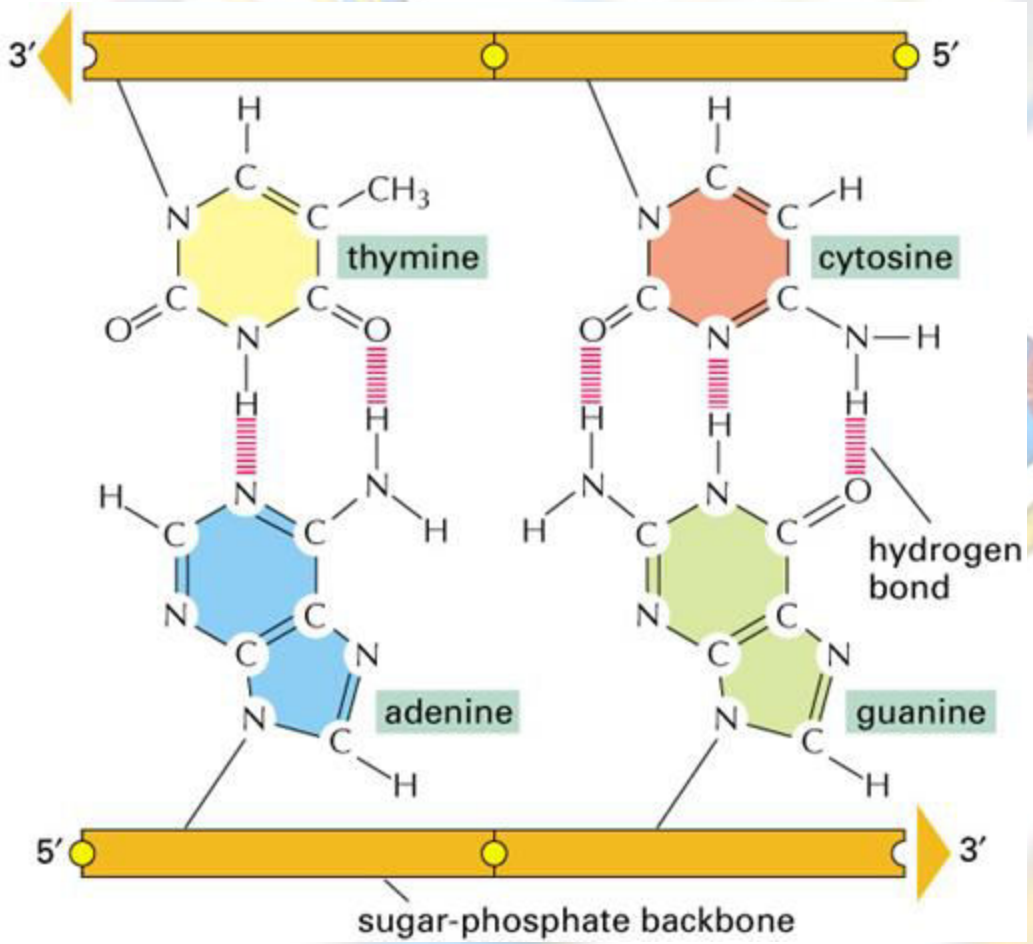


POLYNUCLEOTIDE LINKAGE

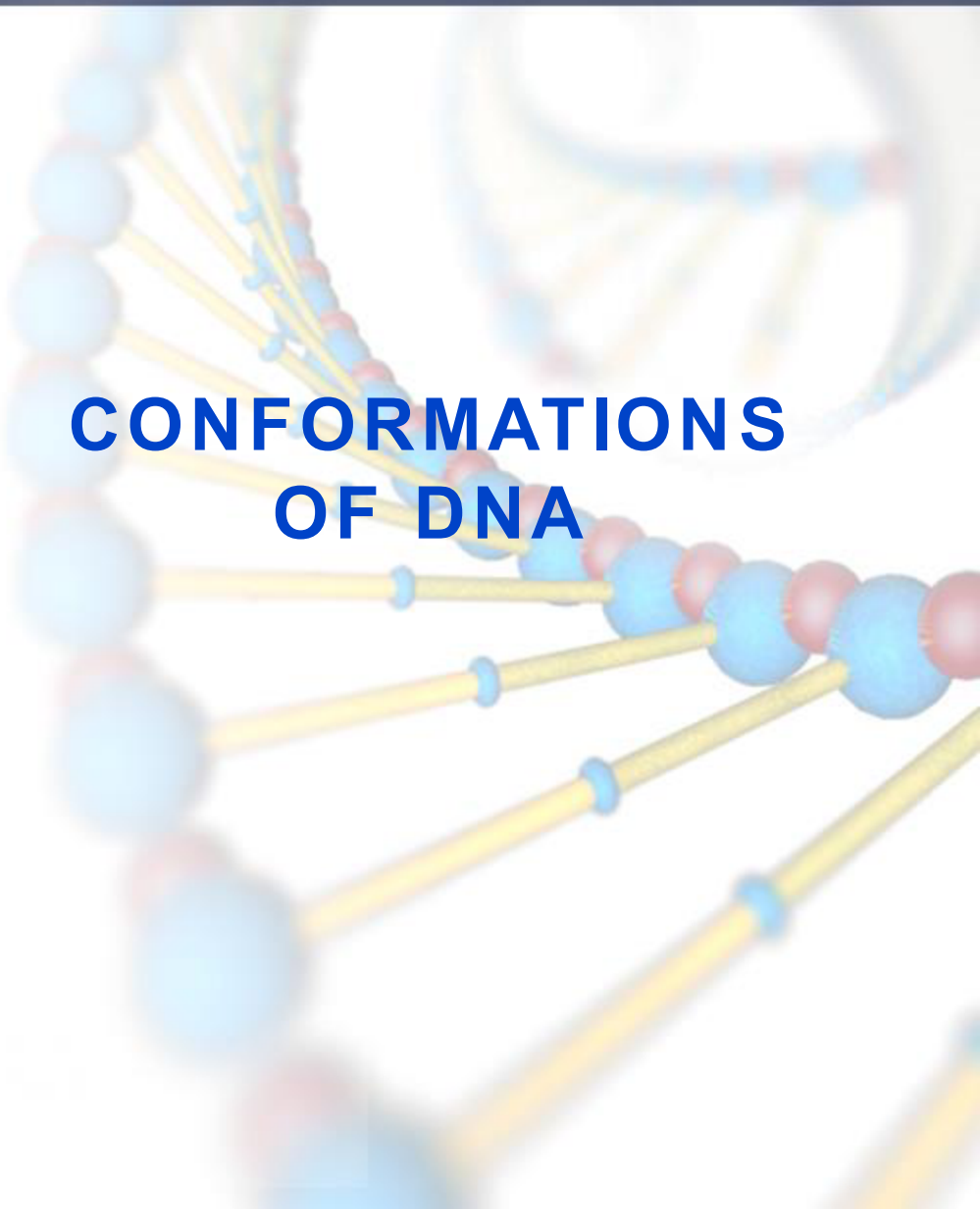
Nucleotides

PAIRING OF

A – T , C – G



Chromosome Structure



CONFORMATIONS OF DNA

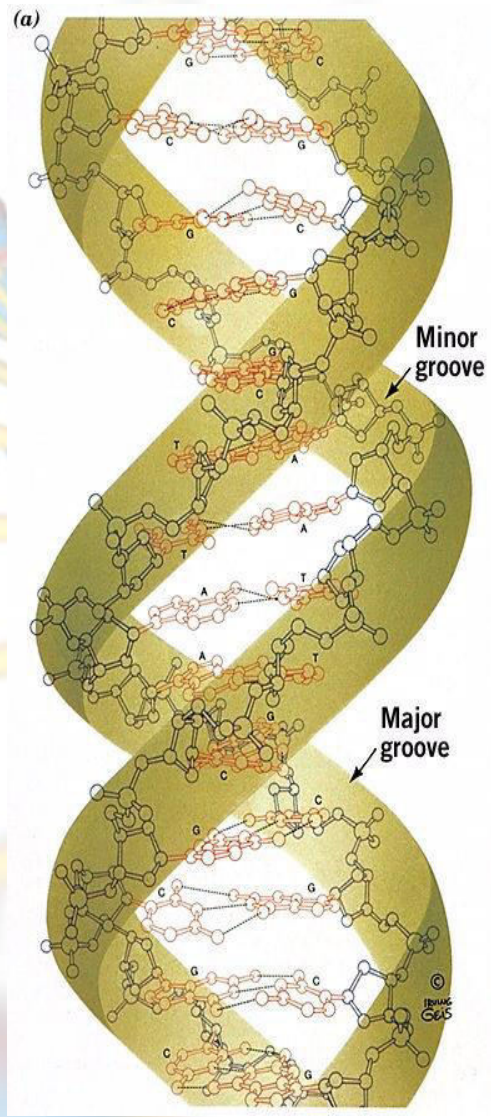
Conformations of DNA

B-DNA

- Right-handed helix.
- Intermediate.
- Planes of the base pairs nearly perpendicular to the helix axis
- Tiny central axis.
- Wide + deep major groove.
- Narrow + deep minor groove.

Conformations of DNA

B-DNA



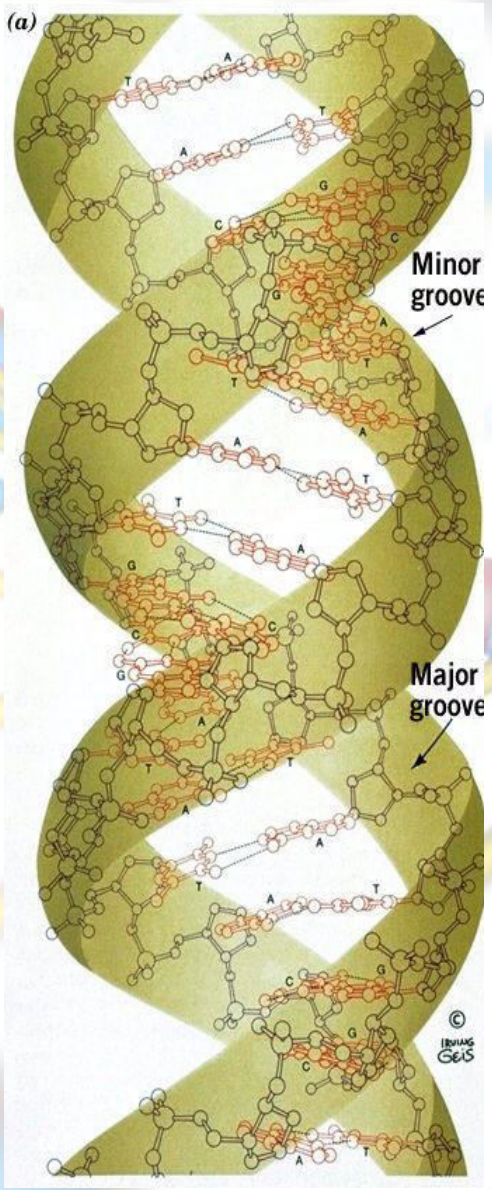
Conformations of DNA

A-DNA

- Right-handed helix.
- Widest .
- Planes of the base pairs inclined to the helix axis.
- Narrow + deep major groove.
- Wide + shallow minor groove.

Conformations of DNA

A-DNA



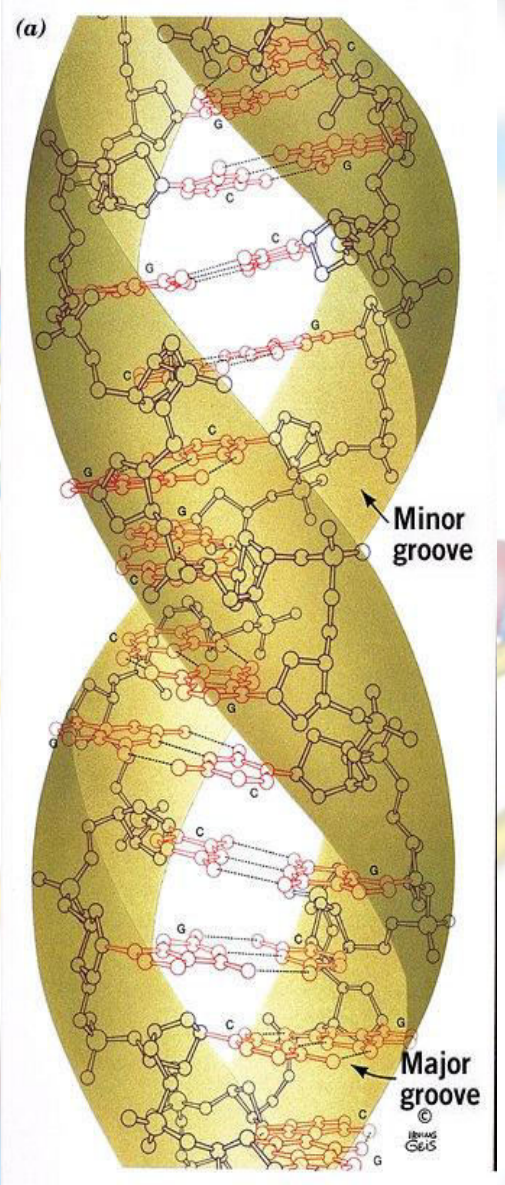
Conformations of DNA

Z-DNA

- Left-handed helix.
- Narrowest .
- Planes of the base pairs nearly perpendicular to the helix axis.
- No internal spaces.
- No major groove.
- Narrow + deep minor groove.

Conformations of DNA

Z-DNA



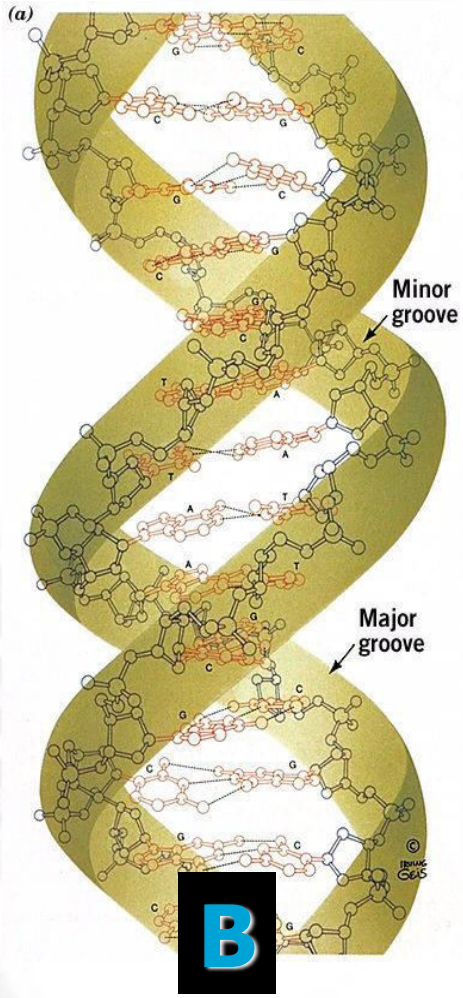
Conformations of DNA

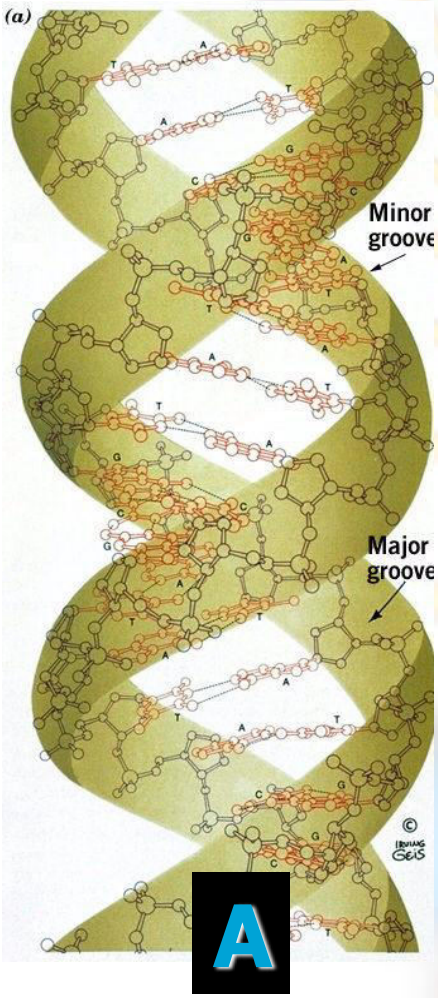
COMPARISON

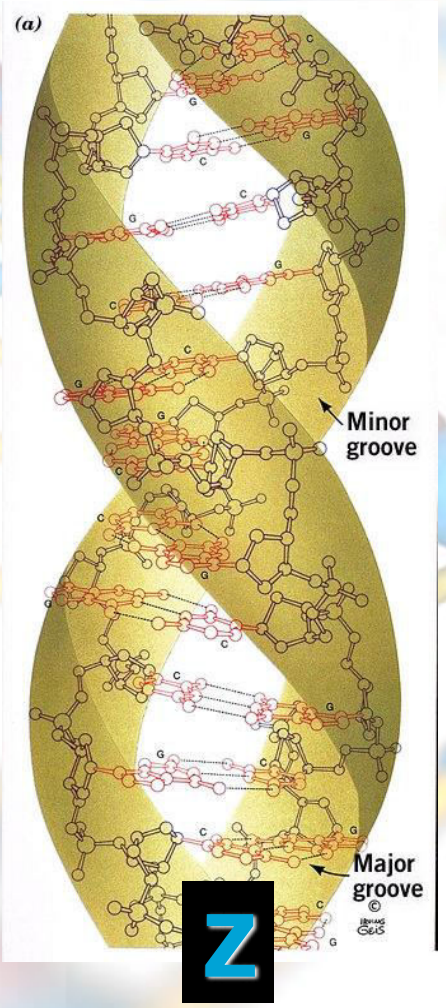
	A- DNA	B-DNA	Z-DNA
Helix	Right-handed	Right-handed	Left-handed
Width	Widest	Intermediate	Narrowest
Planes of bases	planes of the base pairs inclined to the helix axis	planes of the base pairs nearly perpendicular to the helix axis	planes of the base pairs nearly perpendicular to the helix axis
Central axis	6A hole along helix axis	tiny central axis	no internal spaces
Major groove	Narrow and deep	Wide and deep	No major groove
Minor groove	Wide and shallow	Narrow and deep	Narrow and deep

Conformations of DNA

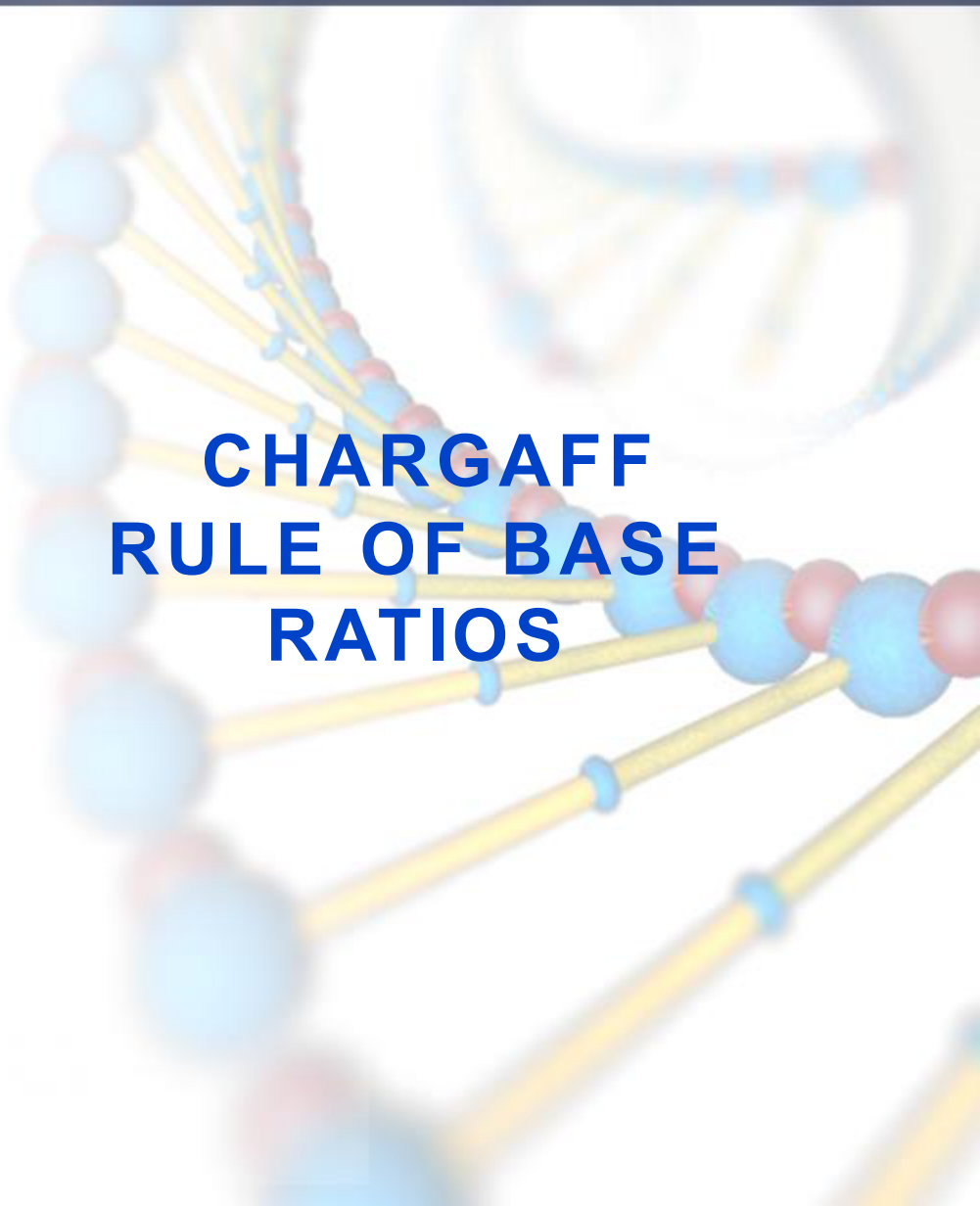
CONFORMATIONS







Chromosome Structure



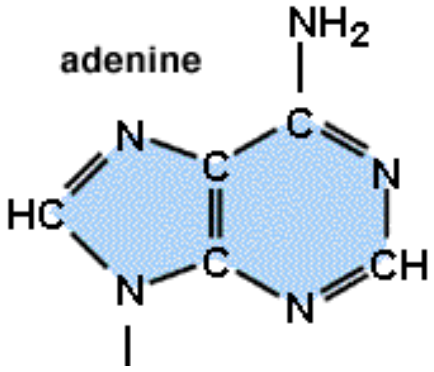
CHARGAFF
RULE OF BASE
RATIOS

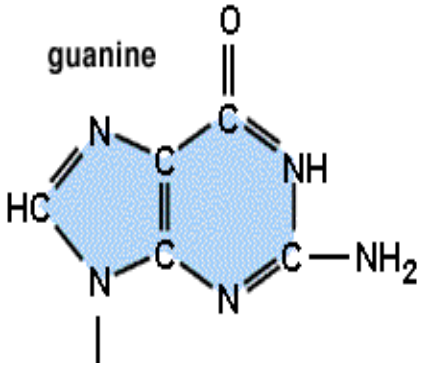
Chargaff Rule of Base Ratios

CHARGAFF

- 1944 - Study of DNA and its four chemical bases - adenine, cytosine, guanine and thymine.
- Discover amounts of adenine and thymine were equal, as were the amounts of cytosine and guanine.

Chargaff Rule of Base Ratios

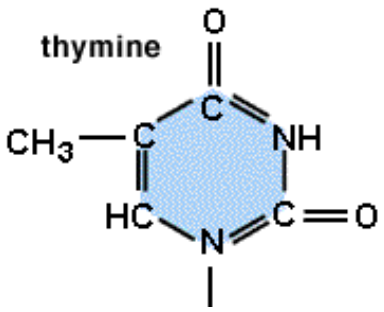


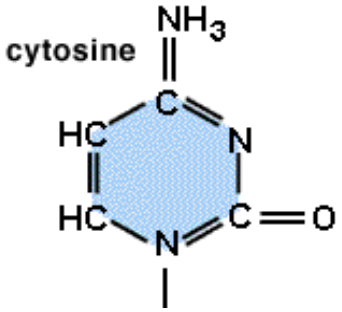


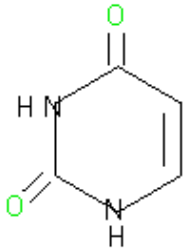
PURINES

- Consist of a six-membered and a five-membered nitrogen-containing ring, fused together.
- Adenine = 6-amino purine
  Guanine = 2-amino-6-oxy purine .

Chargaff Rule of Base Ratios







Uracil

PYRIMIDINES

- Have only a six-membered nitrogen-containing ring.
- Uracil = 2,4-dioxy pyrimidine
- Thymine = 2,4-dioxy-5-methyl pyrimidine
- Cytosine = 2-oxy-4-amino pyrimidine

Chargaff Rule of Base Ratios

CHARGAFF'S FIRST PARITY RULE

- Stated that purines pair with pyrimidines, specifically A with T and C with G

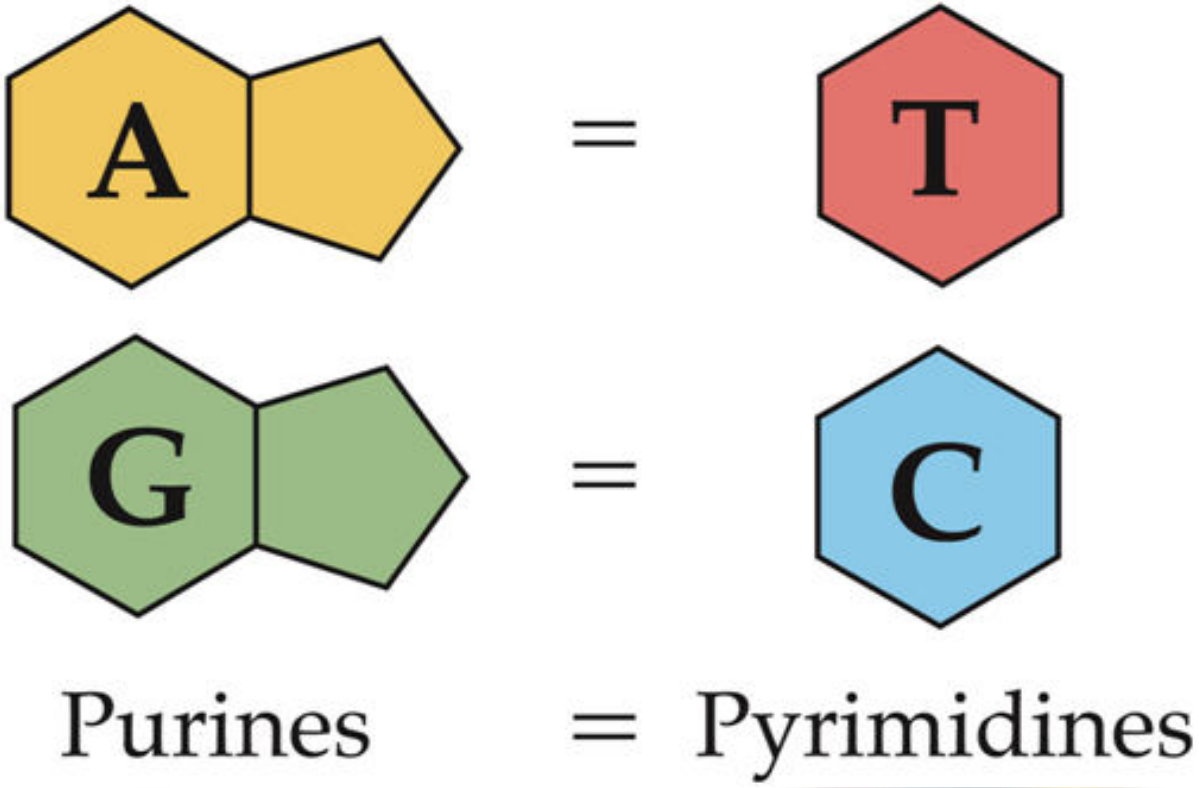
Chargaff Rule of Base Ratios

CHARGAFF'S SECOND PARITY RULE

- % of A approximately equals % of T
- % of G approximately equals % of C

Chargaff Rule of Base Ratios

CHARGAFF'S SECOND PARITY RULE



Chargaff Rule of Base Ratios

G-C RULE

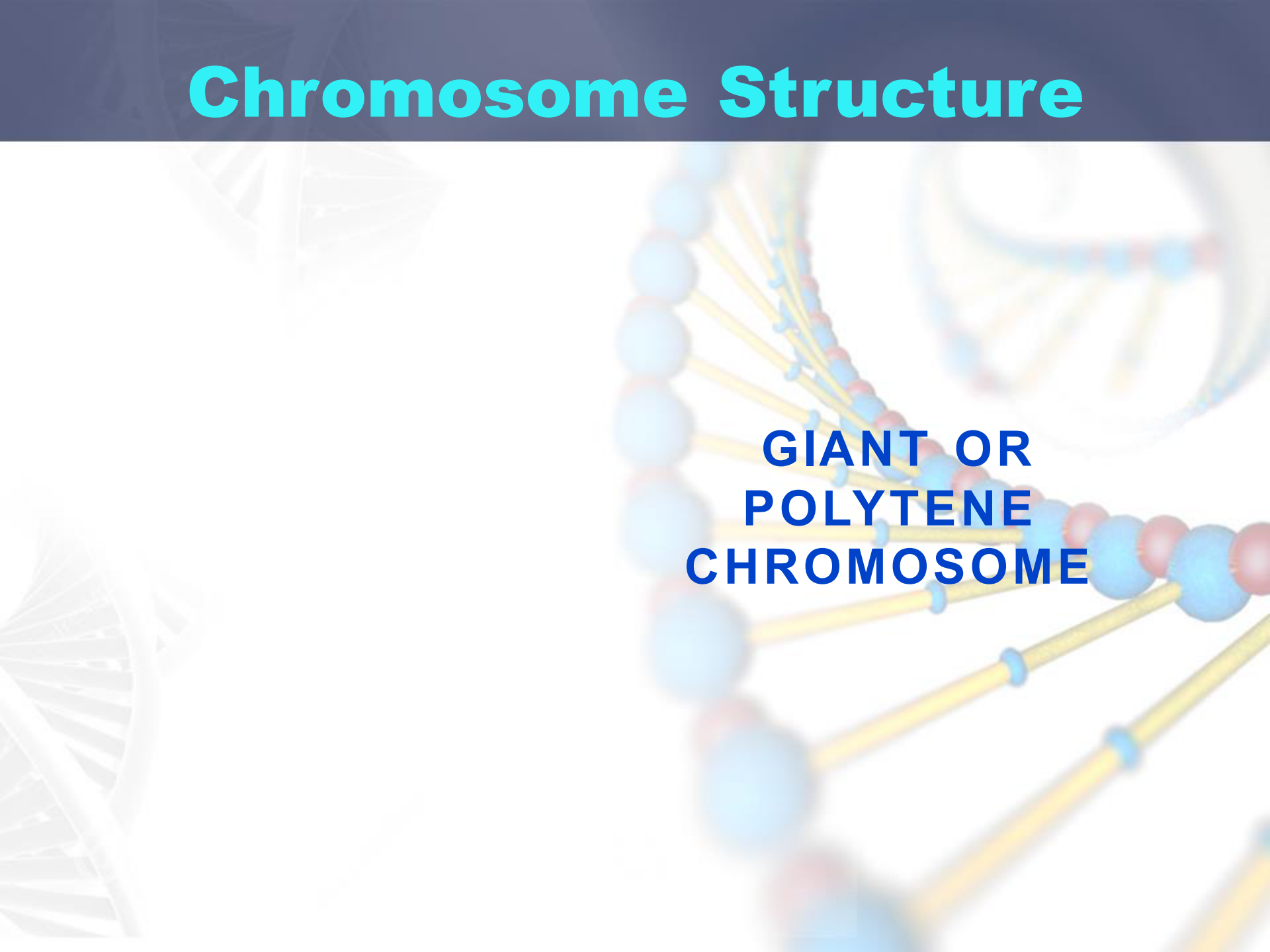
- Stated that (G+C) % is constant within a species, but often differs between species.

Chargaff Rule of Base Ratios

CLUSTER RULE

- Stated that pyrimidines often occur in clusters, and hence on the complementary strand purines do likewise.

Chromosome Structure



GIANT OR POLYTENE CHROMOSOME

Giant or Polytene Chromosome

GIANT CHROMOSOME

- Found in salivary glands of some Diptera.
- These chromosomes are very long and thick.
- Hence they are known as Giant chromosomes.

Giant or Polytene Chromosome

GIANT CHROMOSOME

- They are first discovered by Balbiani in 1881 in dipteran salivary glands and thus also known as salivary gland chromosomes.

Giant or Polytene Chromosome

GIANT CHROMOSOME

- Giant chromosomes have also been discovered in suspensors of young embryos of many plants, but these do not show the bands so typical of salivary gland chromosomes.

Giant or Polytene Chromosome

GIANT CHROMOSOME

- The total length of D. melanogater giant chromosomes is about 2,000μ.

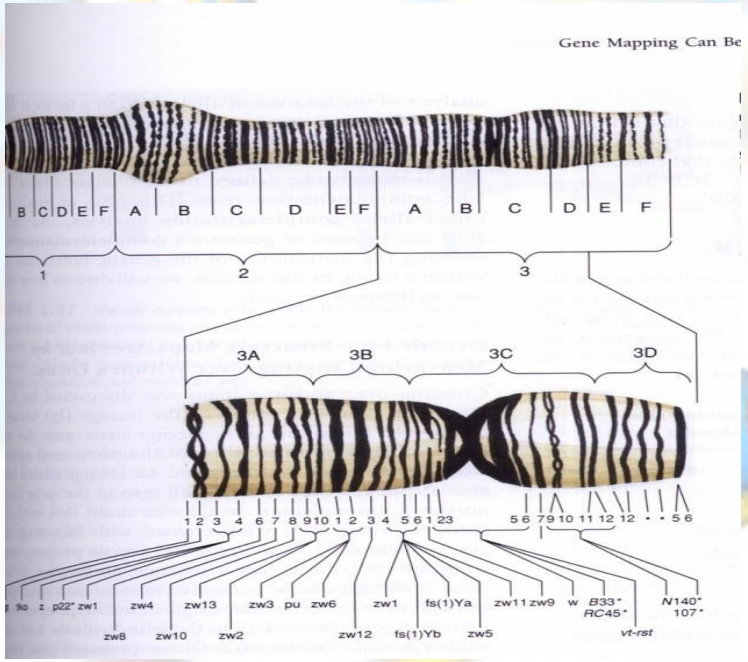
Giant or Polytene Chromosome

GIANT CHROMOSOME

- Giant chromosomes are made dark staining regions called “bands”.
- Separated by relatively light or non-staining “interband” regions. In Drosophila about 5000 bands can be recognized.

Giant or Polytene Chromosome

GIANT CHROMOSOME



Giant or Polytene Chromosome

GIANT CHROMOSOME

- Some of these bands are as thick as 0.5μ, while some may be only 0.05μ thick.
- About 25,000 base-pairs are now estimated for each band.

Giant or Polytene Chromosome

GIANT CHROMOSOME

- These chromosomes are also known as “Polytene chromosome”, and the condition is referred to as “Polytene”

Giant or Polytene Chromosome

GIANT CHROMOSOME

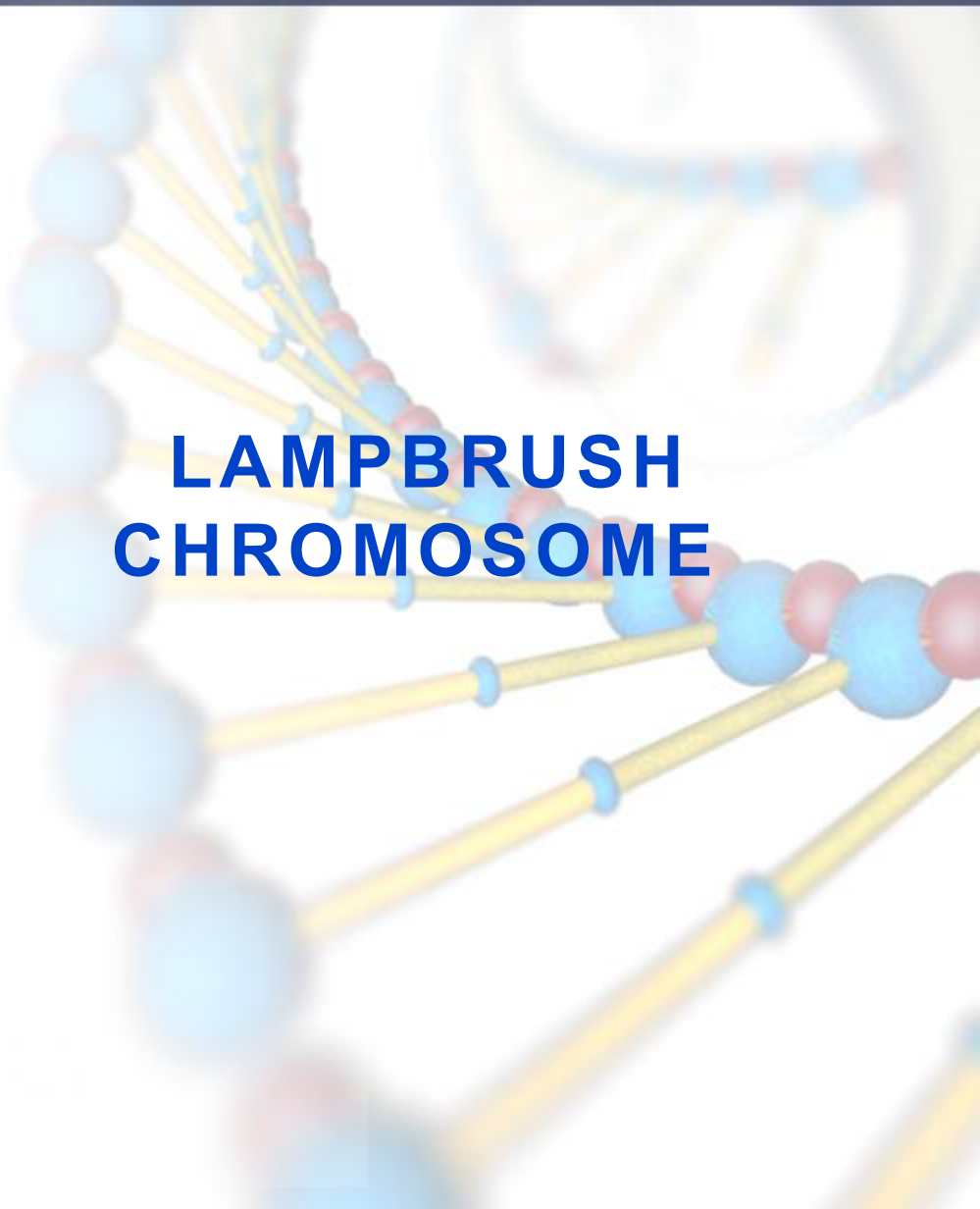
- The numerous strands of these chromosomes are produced due to repeated replication of the paired chromosomes without any nuclear or cell division.

Giant or Polytene Chromosome

GIANT CHROMOSOME

- It is estimated that giant chromosomes of Drosophila have about 1,024 strands.
- In the case of Chironomous may have about 4,096 strands.

Chromosome Structure



Lampbrush Chromosome

LAMPBRUSH CHROMOSOME

- It was given this name because it is similar in appearance to the brushes used to clean lamp chimneys.
- First observed by Flemming in 1882.

Lampbrush Chromosome

LAMPBRUSH CHROMOSOME

- These are found in oocytic nuclei of vertebrates (sharks, amphibians, reptiles and birds) as well as in some invertebrates.
- Also found in some plants.

Lampbrush Chromosome

LAMPBRUSH CHROMOSOME

- One loop represent one chromatid, i.e., one DNA molecule. The size of the loop may be ranging the average of 9.5 µm to about 200 µm

Lampbrush Chromosome

LAMPBRUSH CHROMOSOME

- Each lampbrush chromosome contains a central axial region, where the two chromatids are highly condensed.
- Each chromosome has several chromomeres distributed over its length

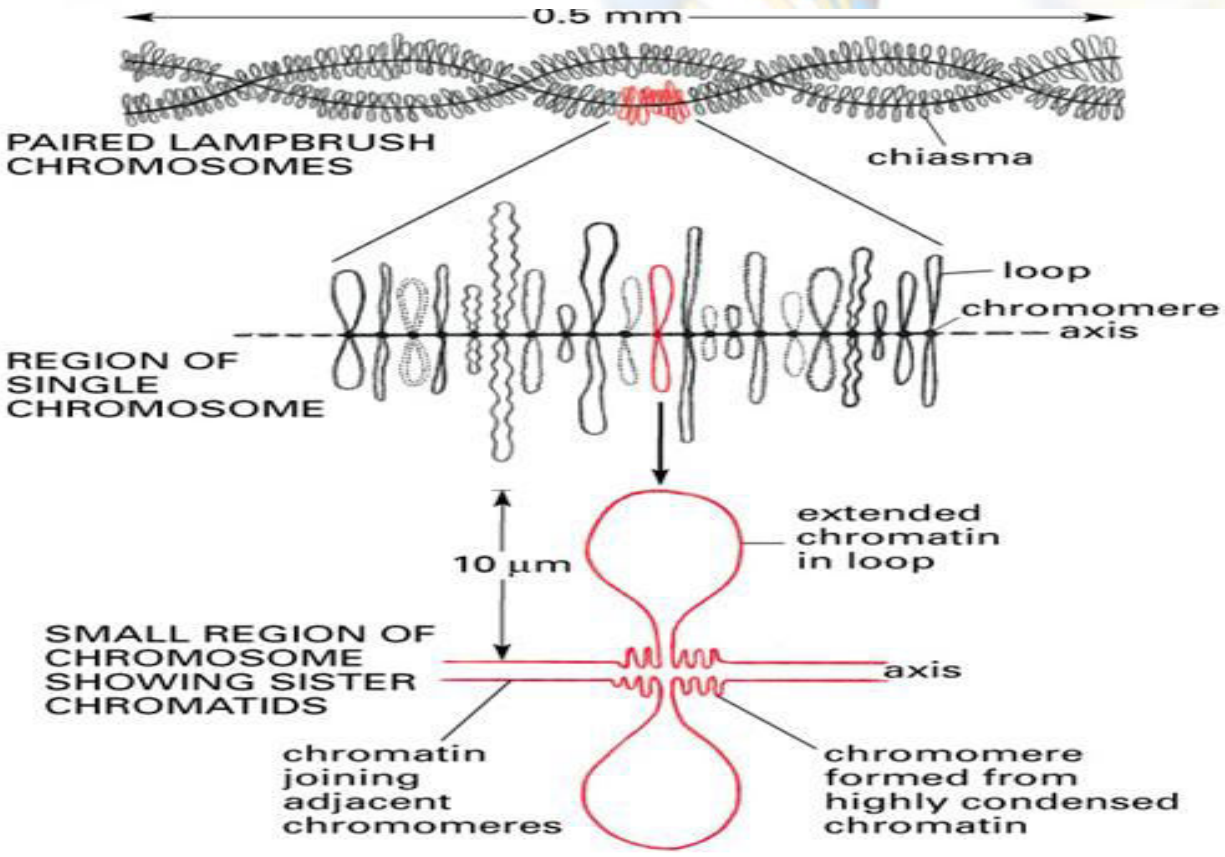
Lampbrush Chromosome

LAMPBRUSH CHROMOSOME

- One loop represent one chromatid, i.e., one DNA molecule. The size of the loop may be ranging the average of 9.5 µm to about 200 µm

Lampbrush Chromosome

LAMPBRUSH CHROMOSOME



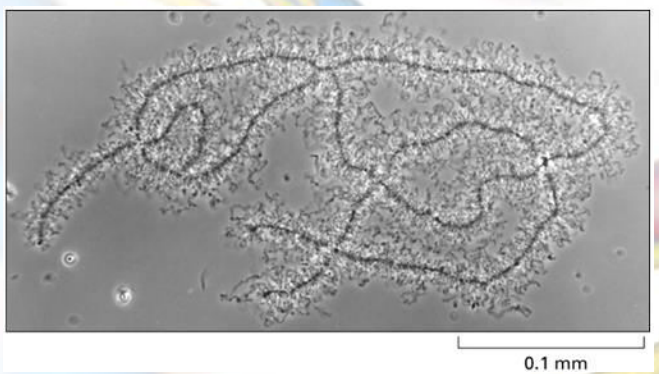
Lampbrush Chromosome

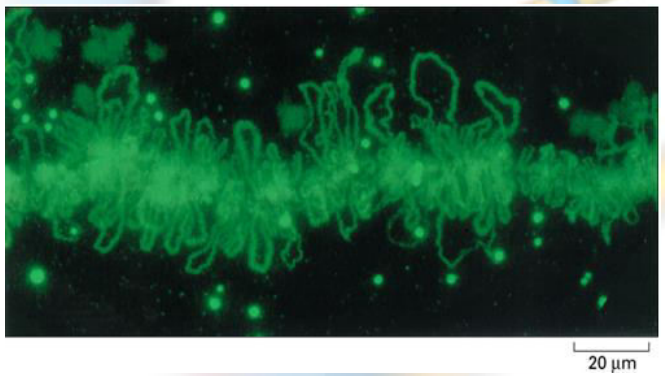
LAMPBRUSH CHROMOSOME

- One end of each loop is thinner (thin end) than the other end (thick end).

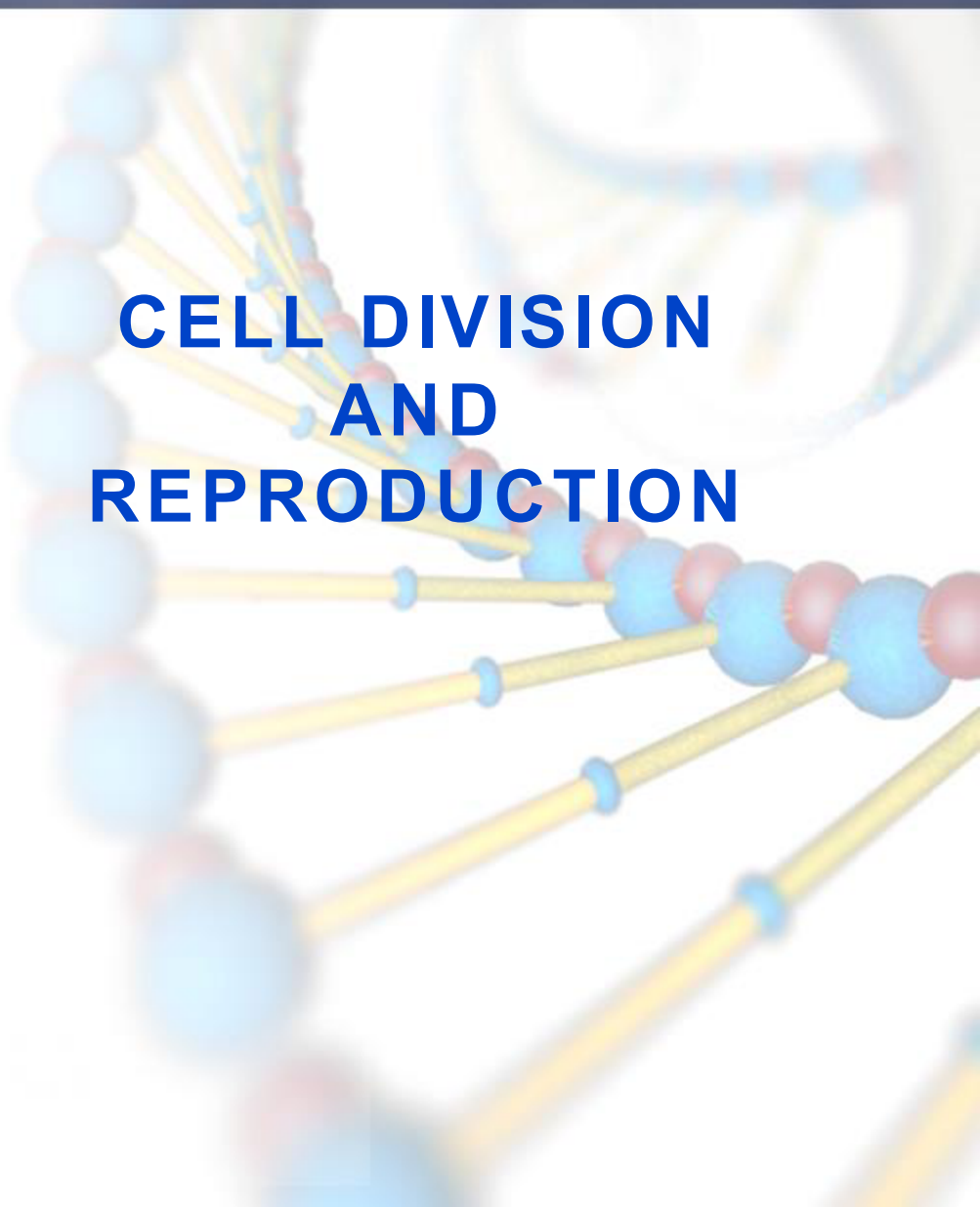
Lampbrush Chromosome

PHASE-CONTRAST AND FLUORESCENT MICROGRAPHS





Essentials of Genetics



CELL DIVISION AND REPRODUCTION

Cell Division & Reproduction

CELL DIVISION - FUNDAMENTAL PROCESS

- Cell division - a fundamental process.
- It is required for growth in unicellular and multicellular organisms.

Cell Division & Reproduction

CELL DIVISION - FUNDAMENTAL PROCESS

- Cell division – a process that divides the cell into two daughter cells.
- It is necessary for reproduction in unicellular and multicellular organisms.

Cell Division & Reproduction

REPLACEMENT OF WORN OUT CELLS

- Cell division also helps to replace worn out cells in multicellular organisms.

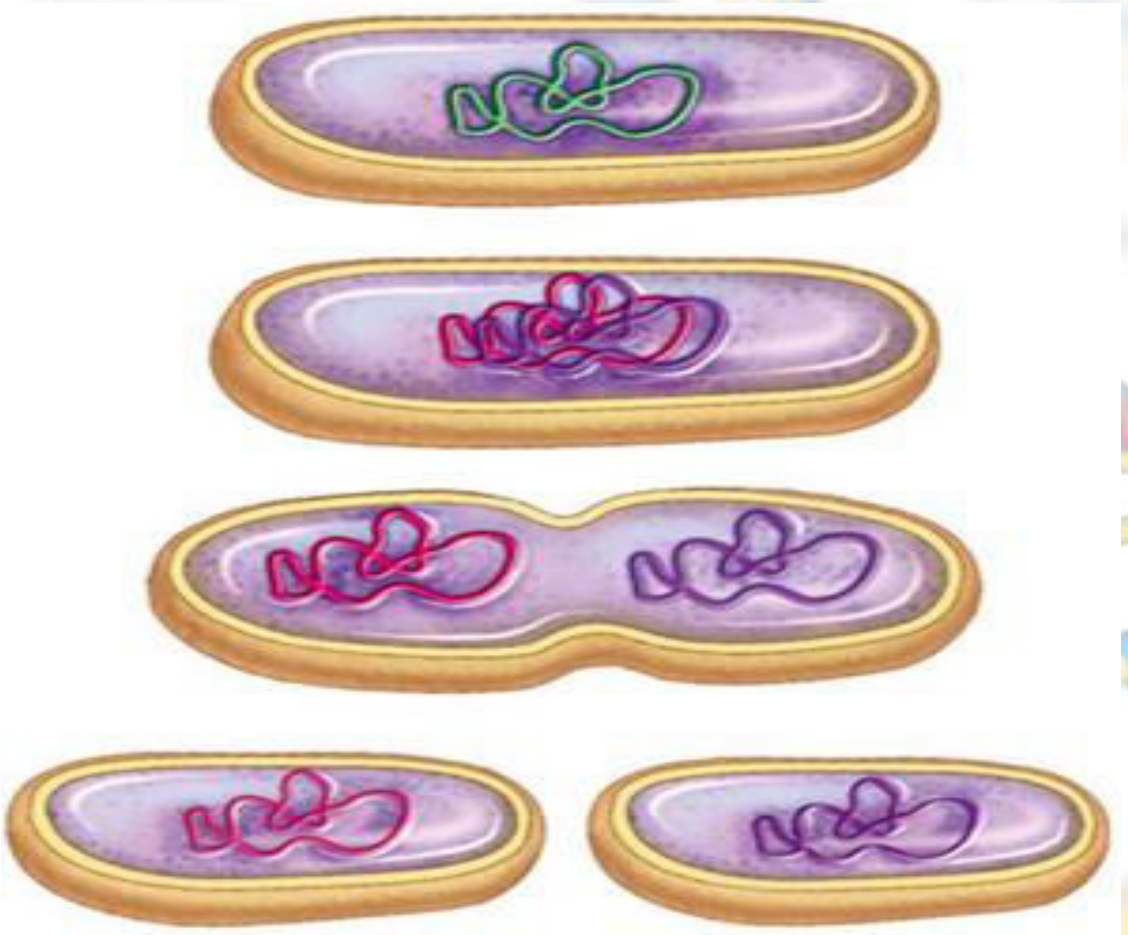
Cell Division & Reproduction

REPRODUCTION IN PROKARYOTES - BINARY FISSION

- Prokaryotes simply divide their cells into two daughter cells by binary fission.

Cell Division & Reproduction

REPRODUCTION IN PROKARYOTES



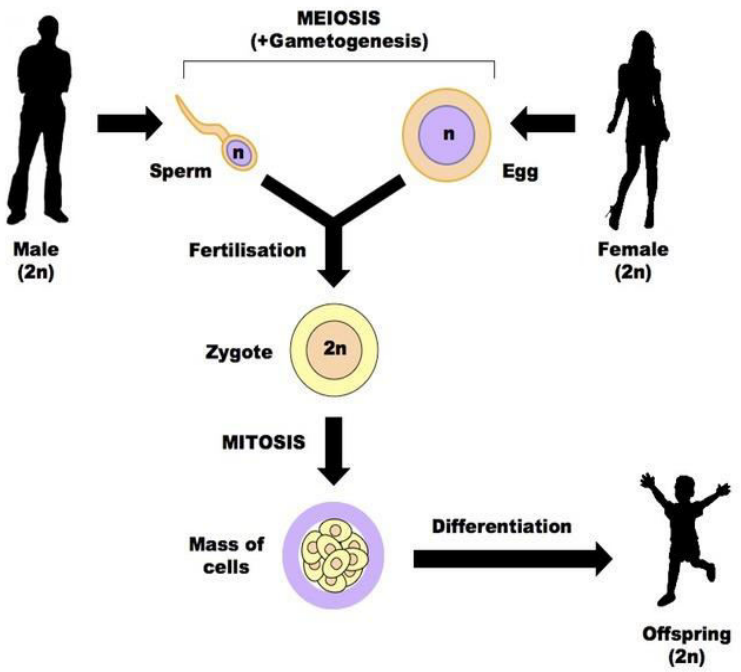
Cell Division & Reproduction

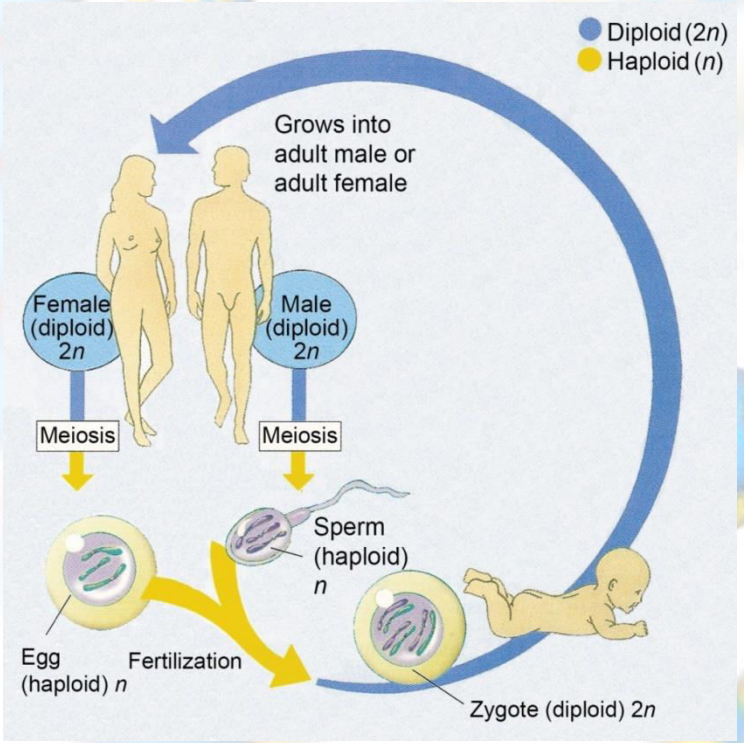
REPRODUCTION IN EUKARYOTES

- Eukaryotes must divide their nucleus and other organelles before cell division.
- Before the nucleus divides, the genetic material replicates.

Cell Division & Reproduction

REPRODUCTION IN EUKARYOTES



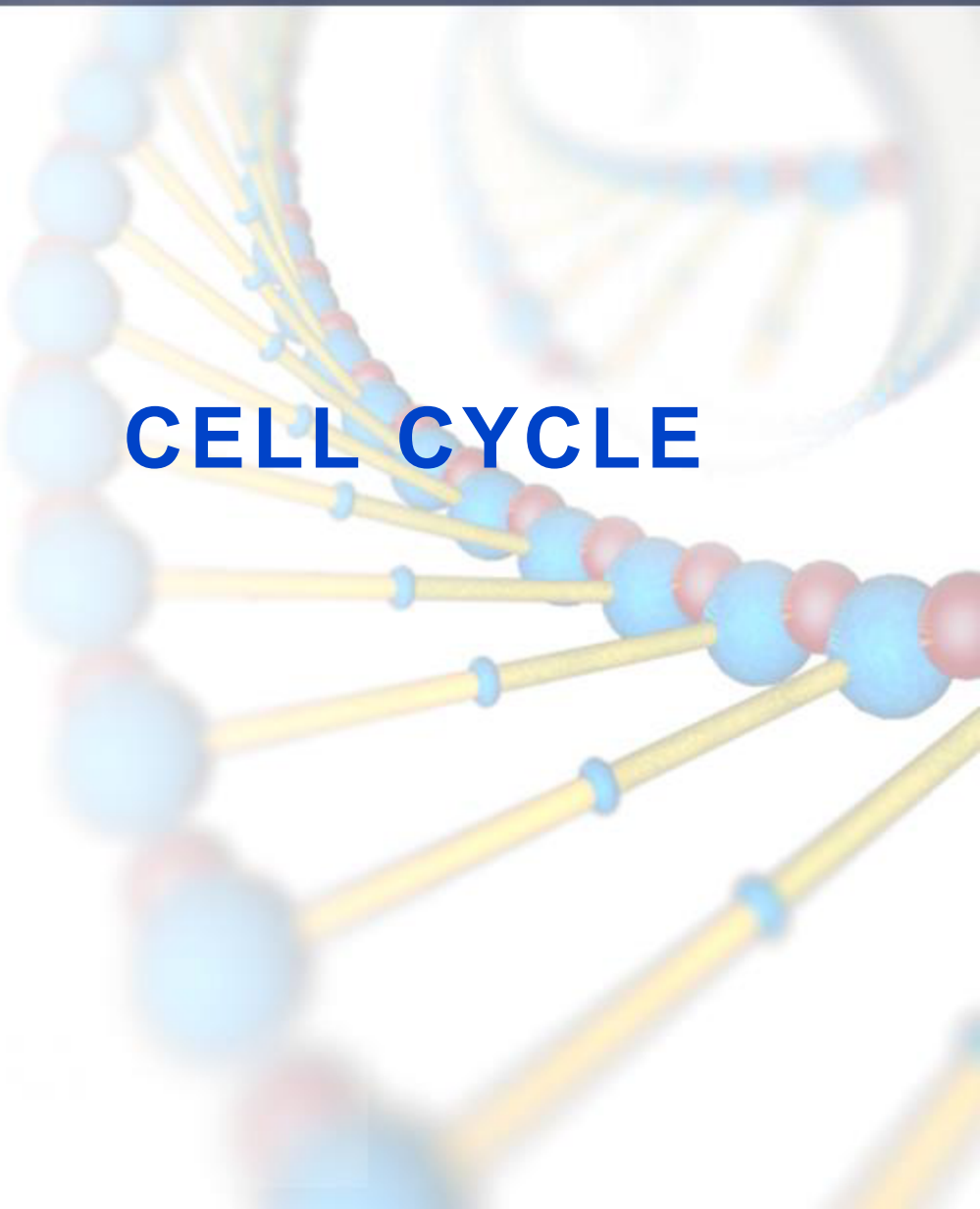


Cell Division & Reproduction

CONCLUSION

- Cell division and reproduction are two fundamental processes in prokaryotes and eukaryotes.

Essentials of Genetics



CELL CYCLE

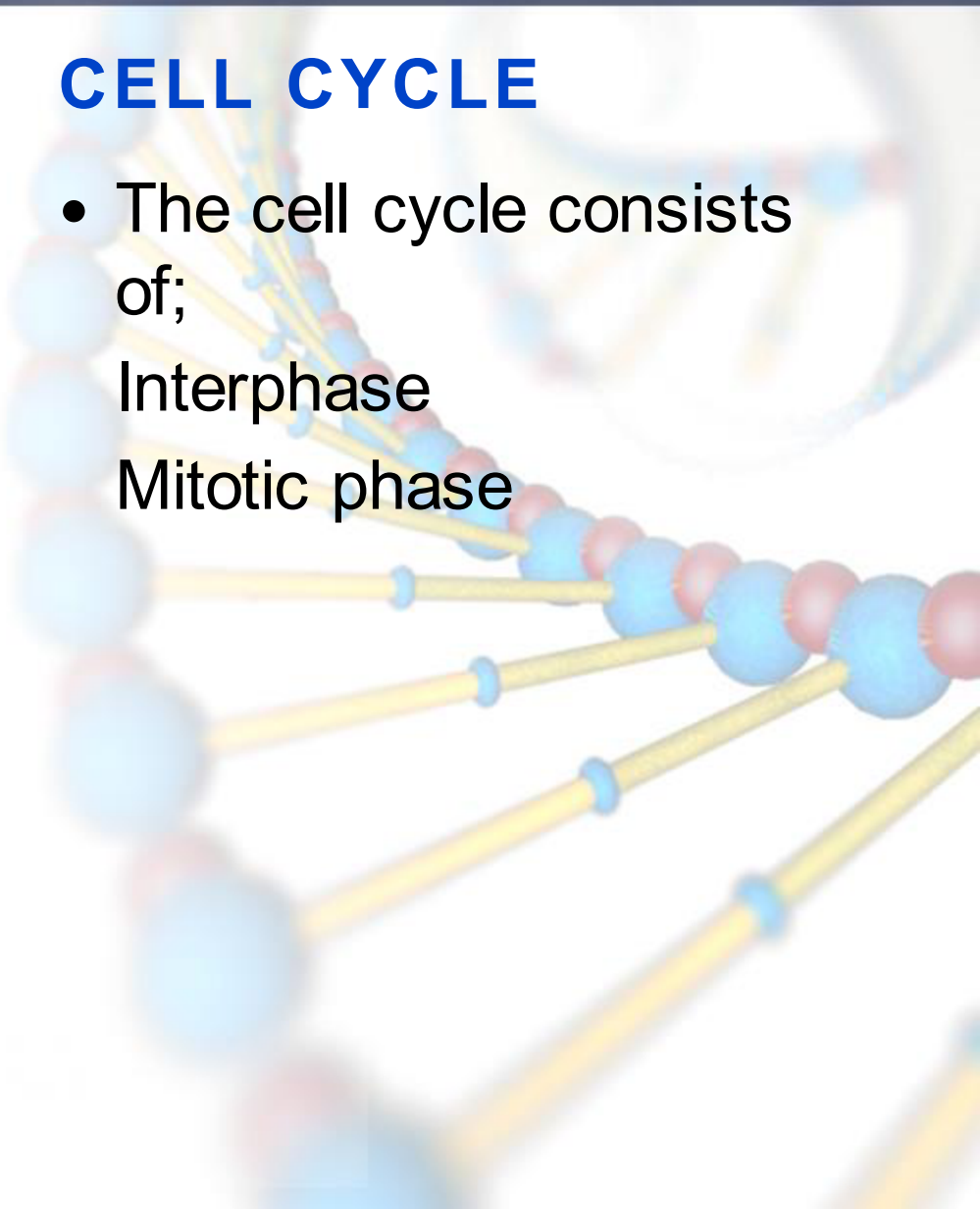
Cell Cycle

CELL CYCLE

- The cell cycle consists of;

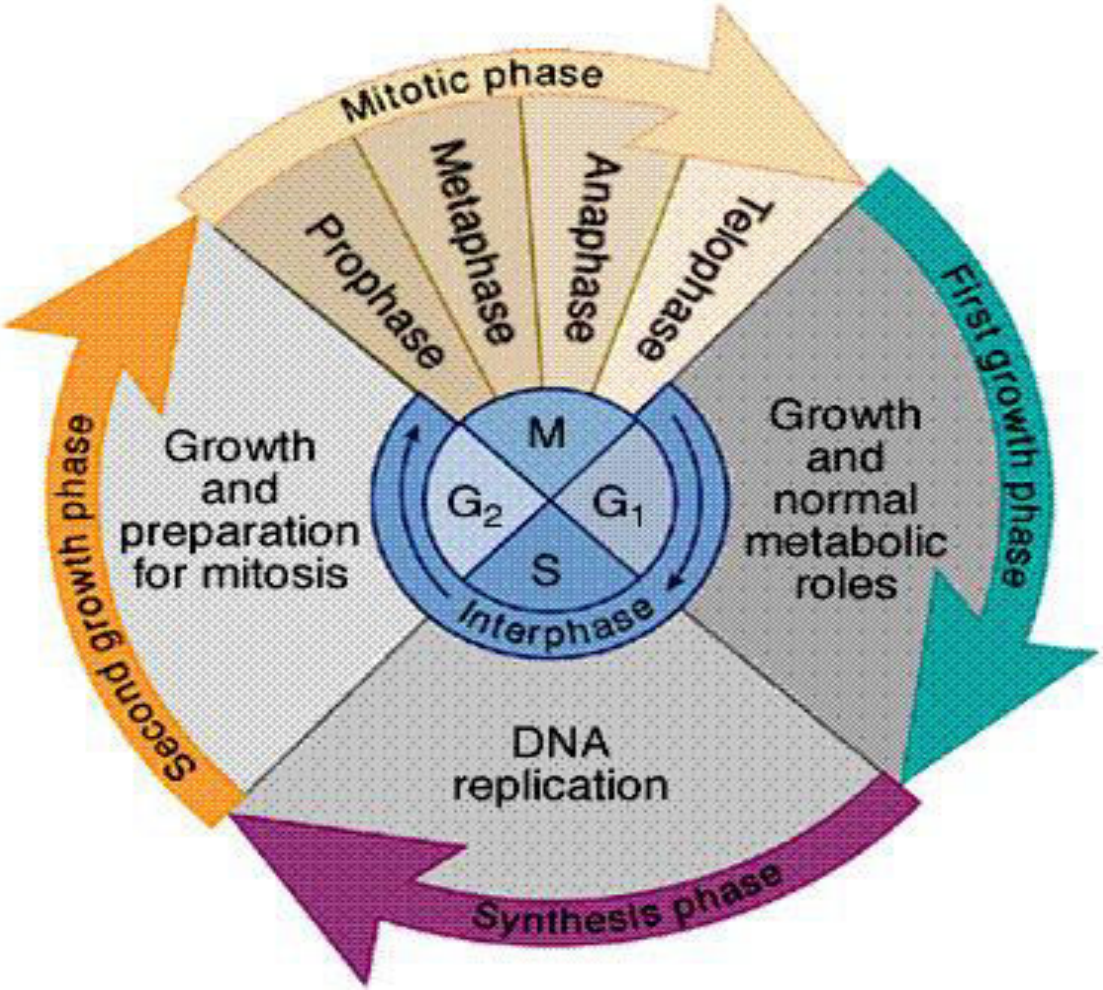
  Interphase

  Mitotic phase



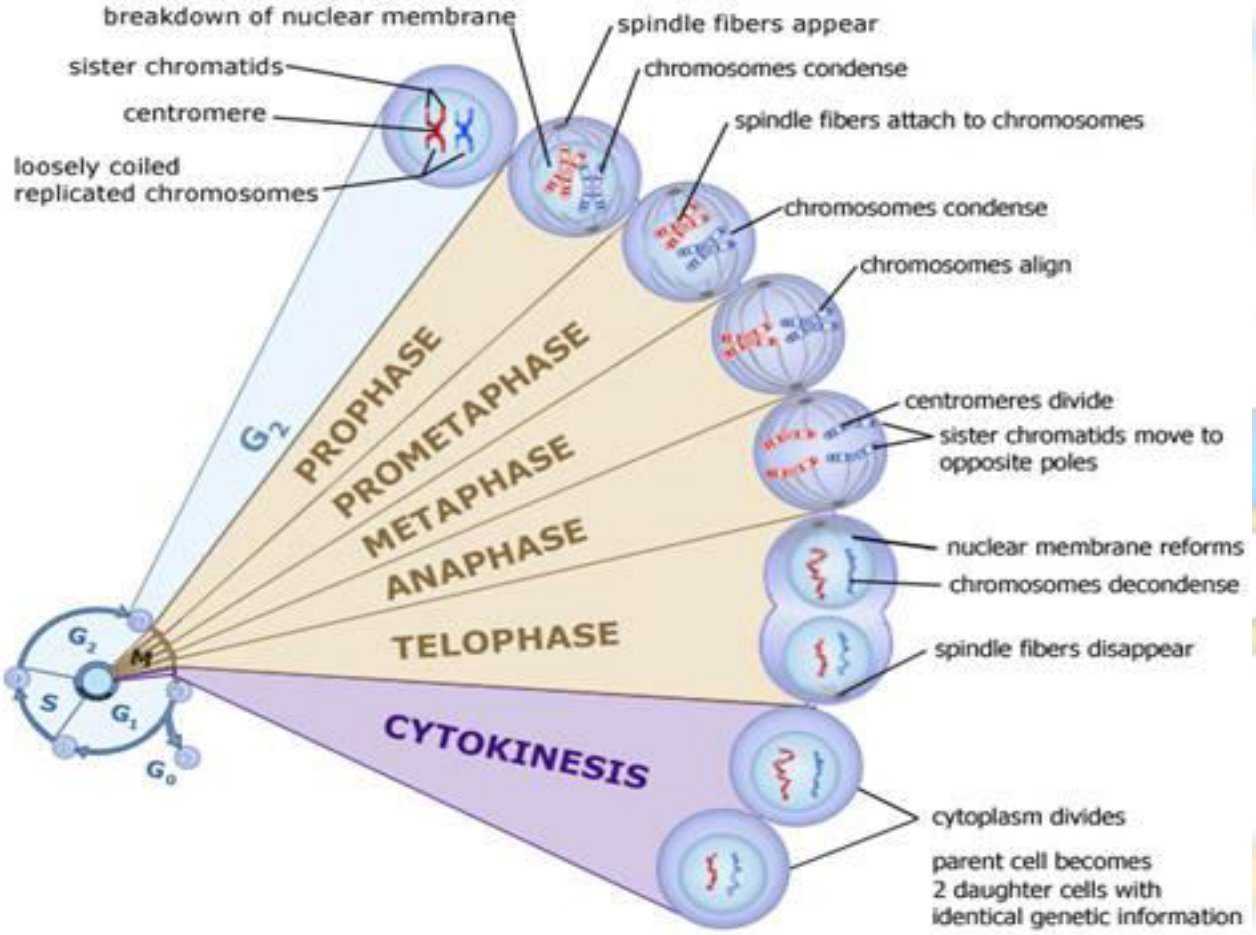
Cell Cycle

PHASES OF CELL CYCLE



Cell Cycle

MITOTIC PHASE



Cell Cycle

G0 PHASE

- G0 phase also called as post-mitotic phase.
- Eukaryotes cells generally enter into G0 state from G1.

Cell Cycle

INTERPHASE

- Before a cell enter into cell division, it needs to take nutrients.
- Interphase is a series of changes that takes place in a newly formed cells.

Cell Cycle

INTERPHASE

- Interphase also called as resting stage - no activity related to cell division.
- Interphase lasts for 90% of the total time required for the cell cycle.

Cell Cycle

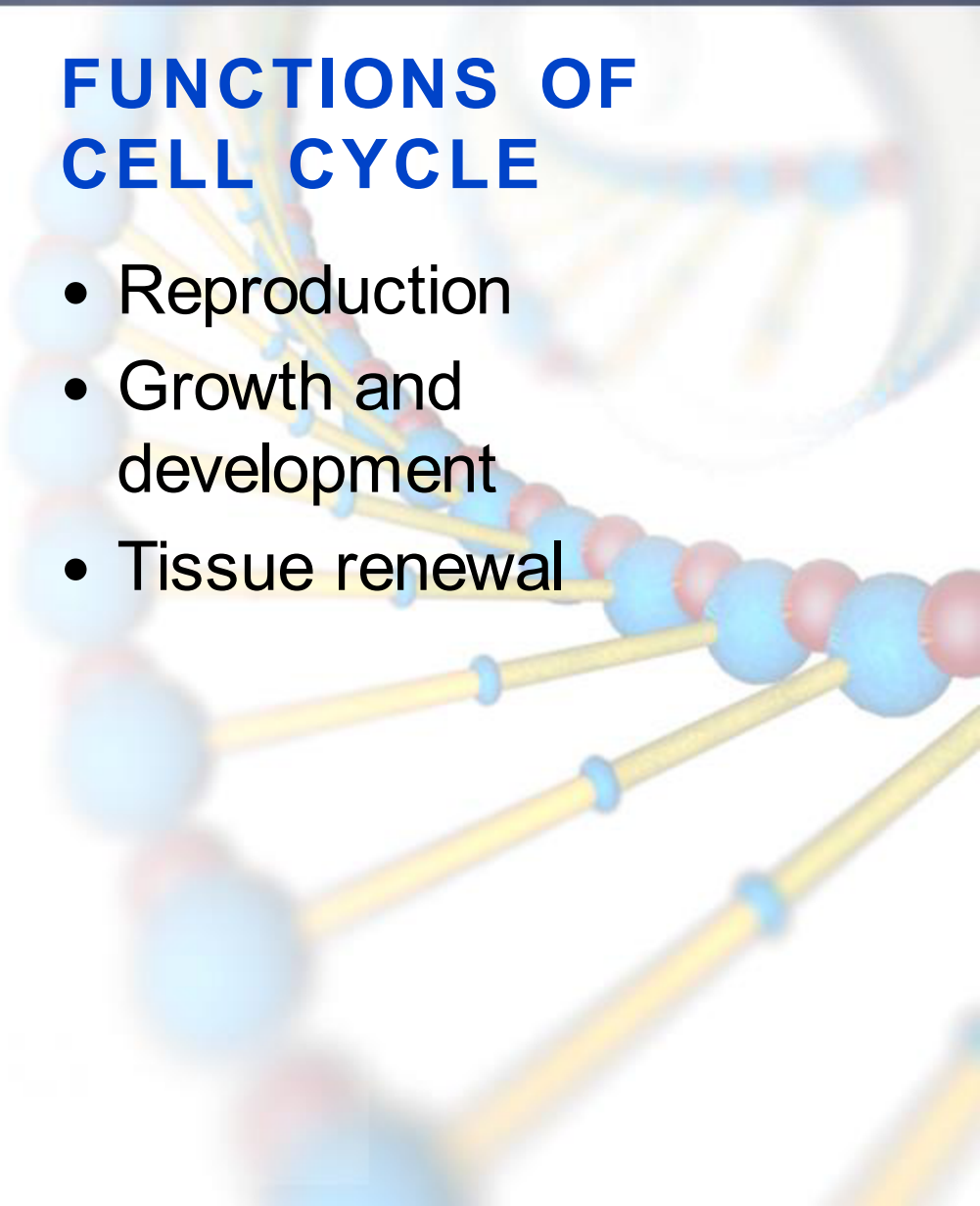
INTERPHASE

- Interphase proceeds in three stages;
  - G1 stage
  - S stage
  - G2 stage

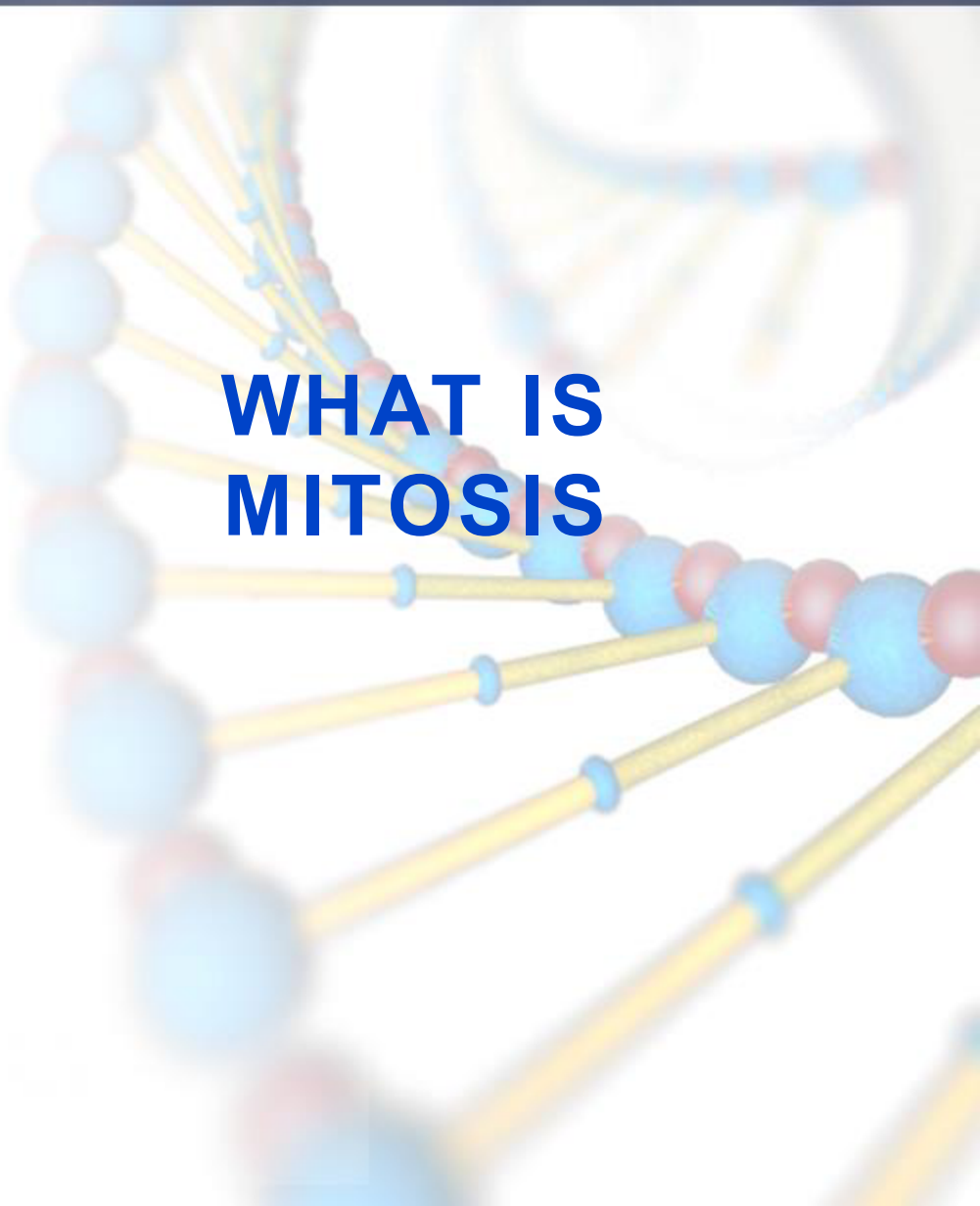
Cell Cycle

FUNCTIONS OF CELL CYCLE

- Reproduction
- Growth and development
- Tissue renewal



Essentials of Genetics



WHAT IS MITOSIS

What is Mitosis

MITOSIS

- A single cell divides into two identical daughter cells.
- Some haploid & diploid cells divide by mitosis.

What is Mitosis

MITOSIS

- Produces two new cells from a single cell.
- Both new cells are genetically identical to the original cell.

What is Mitosis

MITOSIS

- Each cell has to have its own cytoplasm and DNA.



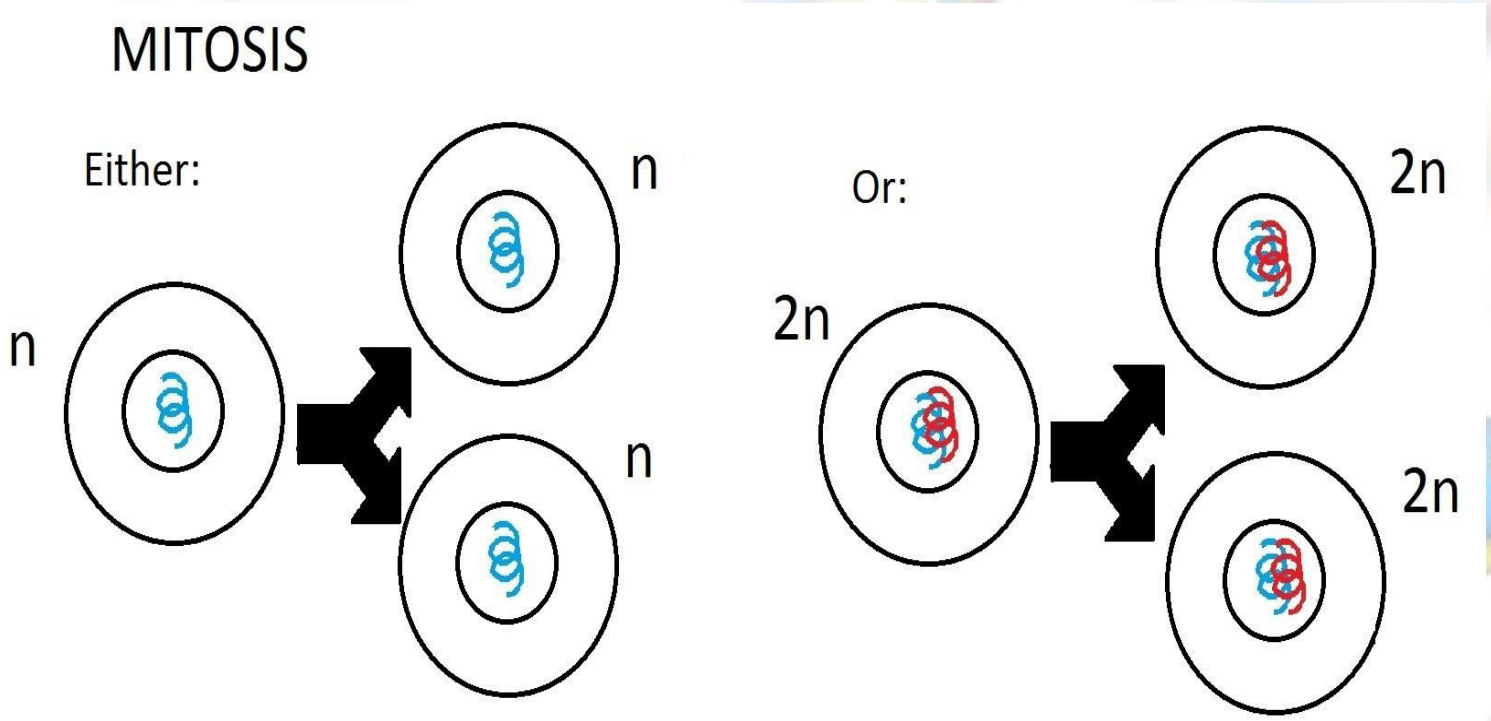
What is Mitosis

CHROMOSOMES REPLICATE BEFORE MITOSIS

- Before mitosis, DNA replicate to produce identical copies.
- Two sister chromatids have to break apart.

What is Mitosis

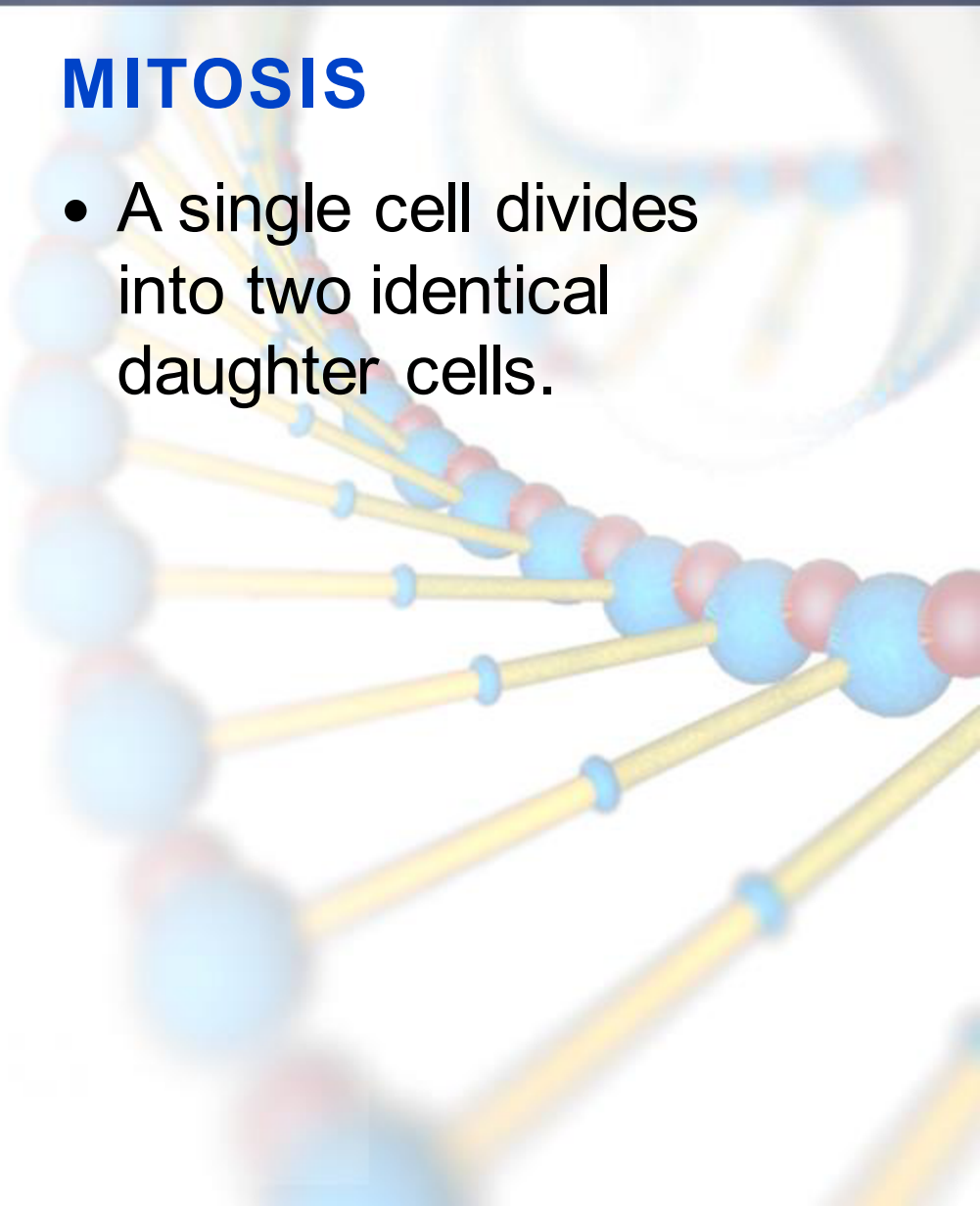
MITOSIS: HAPLOID/DIPLOID CELLS



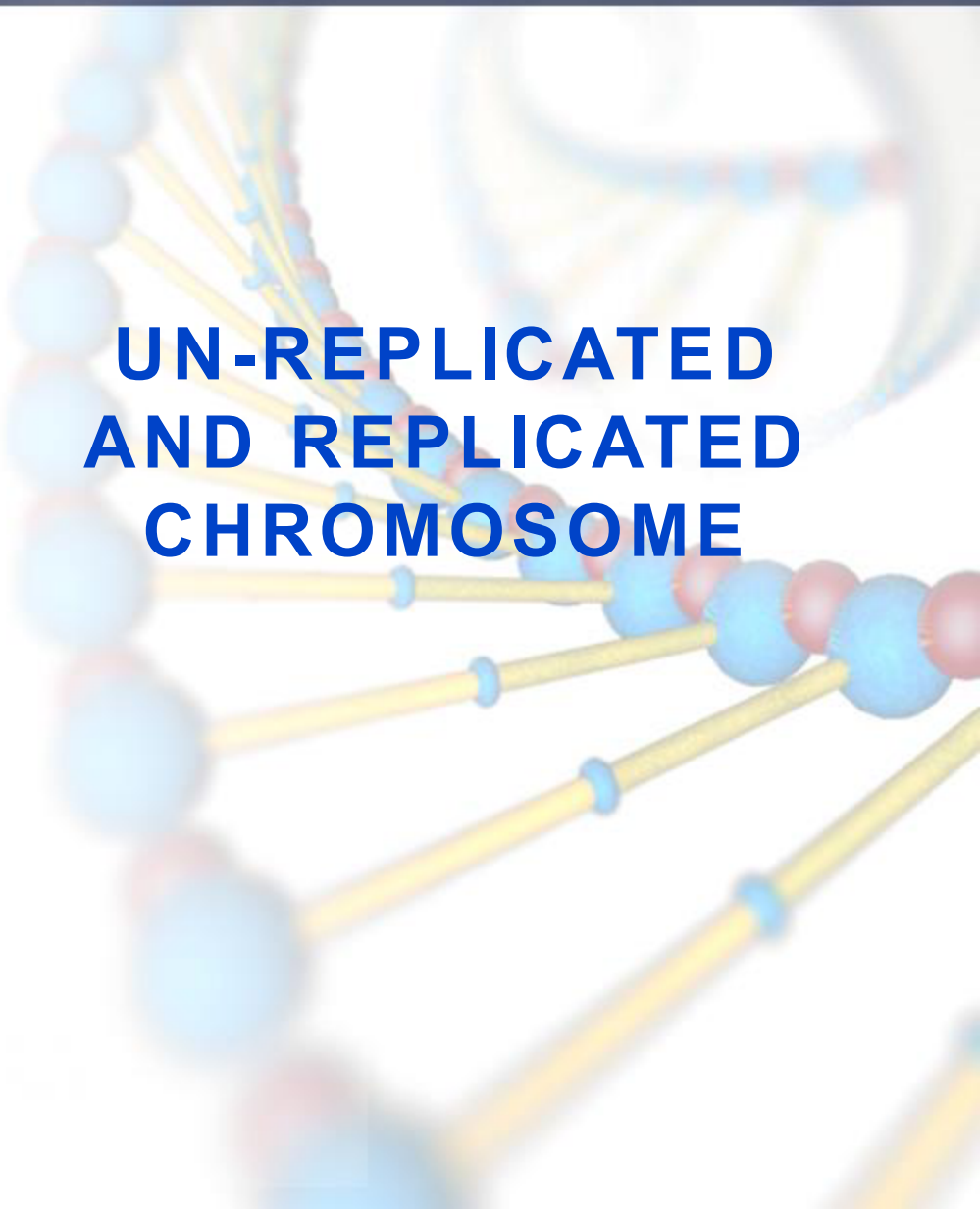
What is Mitosis

MITOSIS

- A single cell divides into two identical daughter cells.



Essentials of Genetics



UN-REPLICATED AND REPLICATED CHROMOSOME

Un-replicated/Replicated Chromosome

CHROMOSOMES ARE REPLICATED

- Before cell division;
- Chromosomes (DNA) are replicated (duplicated).

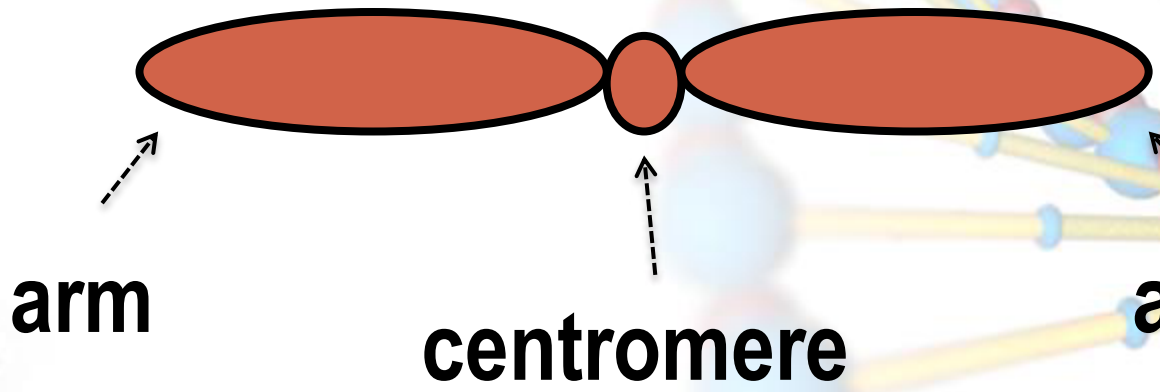
Un-replicated/Replicated Chromosome

CHROMOSOMES ARE REPLICATED

- Duplicated chromosome – attached at their centromere.
- As long as attached, known as sister chromatids.

Un-replicated/Replicated Chromosome

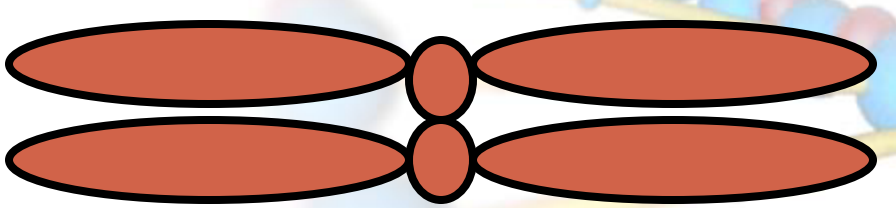
UN-REPLICATED



Un-replicated/Replicated Chromosome

DUPLICATED

Duplicated chromosome



Un-replicated/Replicated Chromosome

DUPLICATED CHROMOSOMES ARE DUE TO DNA REPLICATION

- Entire DNA of each of the chromosome replicates.
- Replication is semiconservative.

Un-replicated/Replicated Chromosome

DNA REPLICATION

- Double helical structure is opened by enzyme gryrases and topoisomerase.
- These enzymes remove helix and produces a nick to uncoiled DNA.

Un-replicated/Replicated Chromosome

DNA REPLICATION

- One strand of DNA is called as leading strand while other strand is called as lagging strand.

Un-replicated/Replicated Chromosome

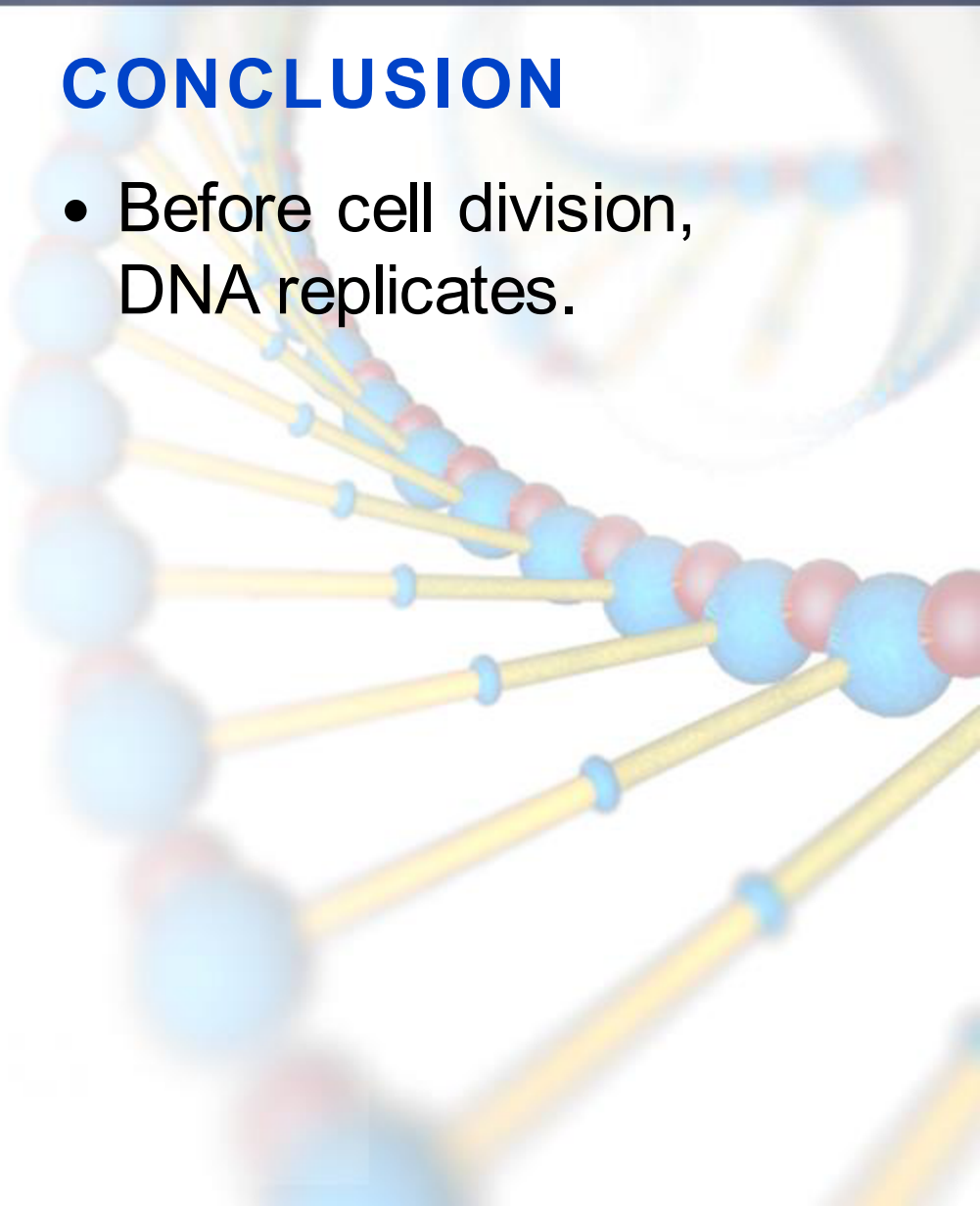
DNA REPLICATION

- Continuous replication in leading strand.
- Discontinuous replication in lagging strands, i.e in forms of small fragments which are called as Okazaki fragments.

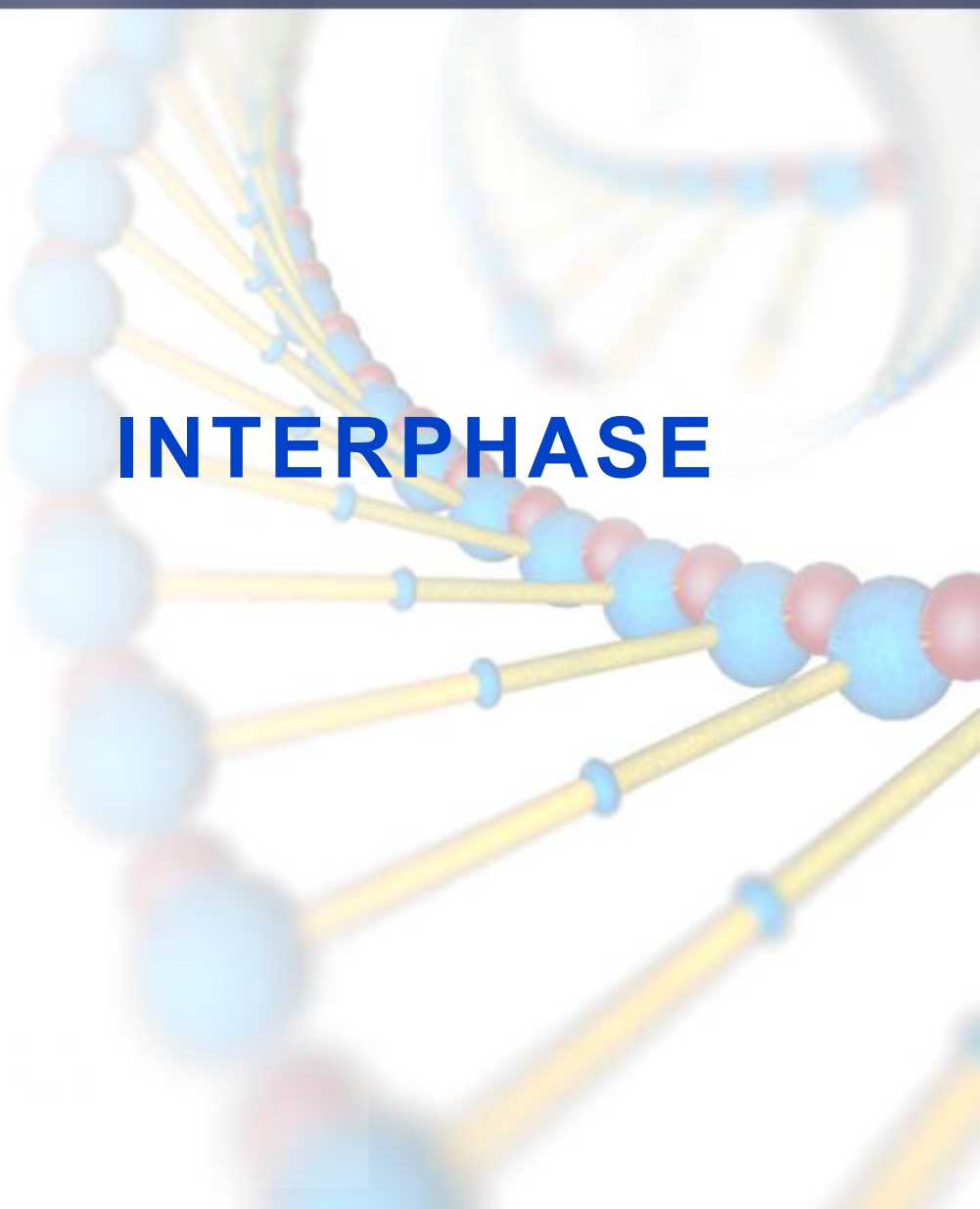
Un-replicated/Replicated Chromosome

CONCLUSION

- Before cell division, DNA replicates.



Essentials of Genetics

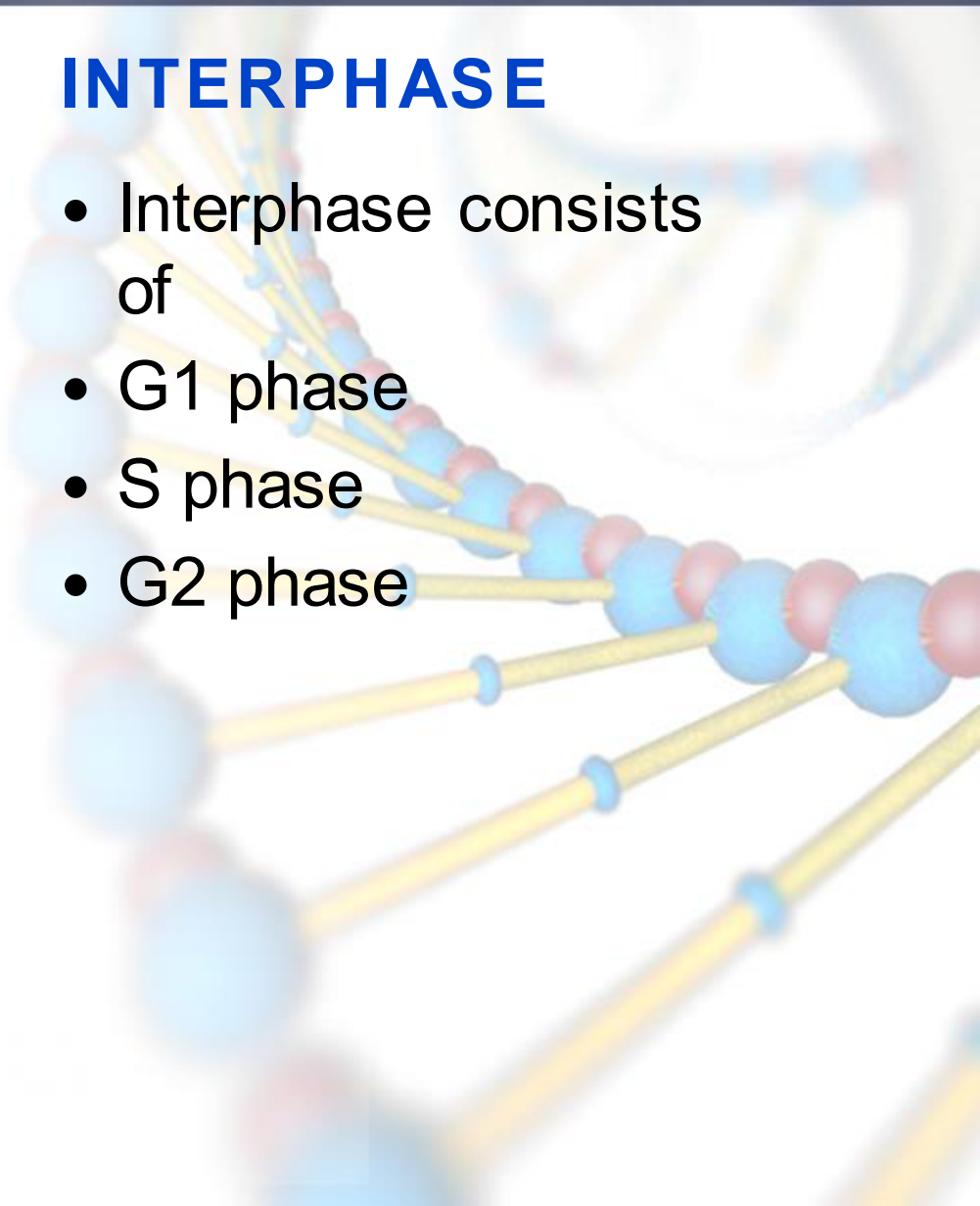


INTERPHASE

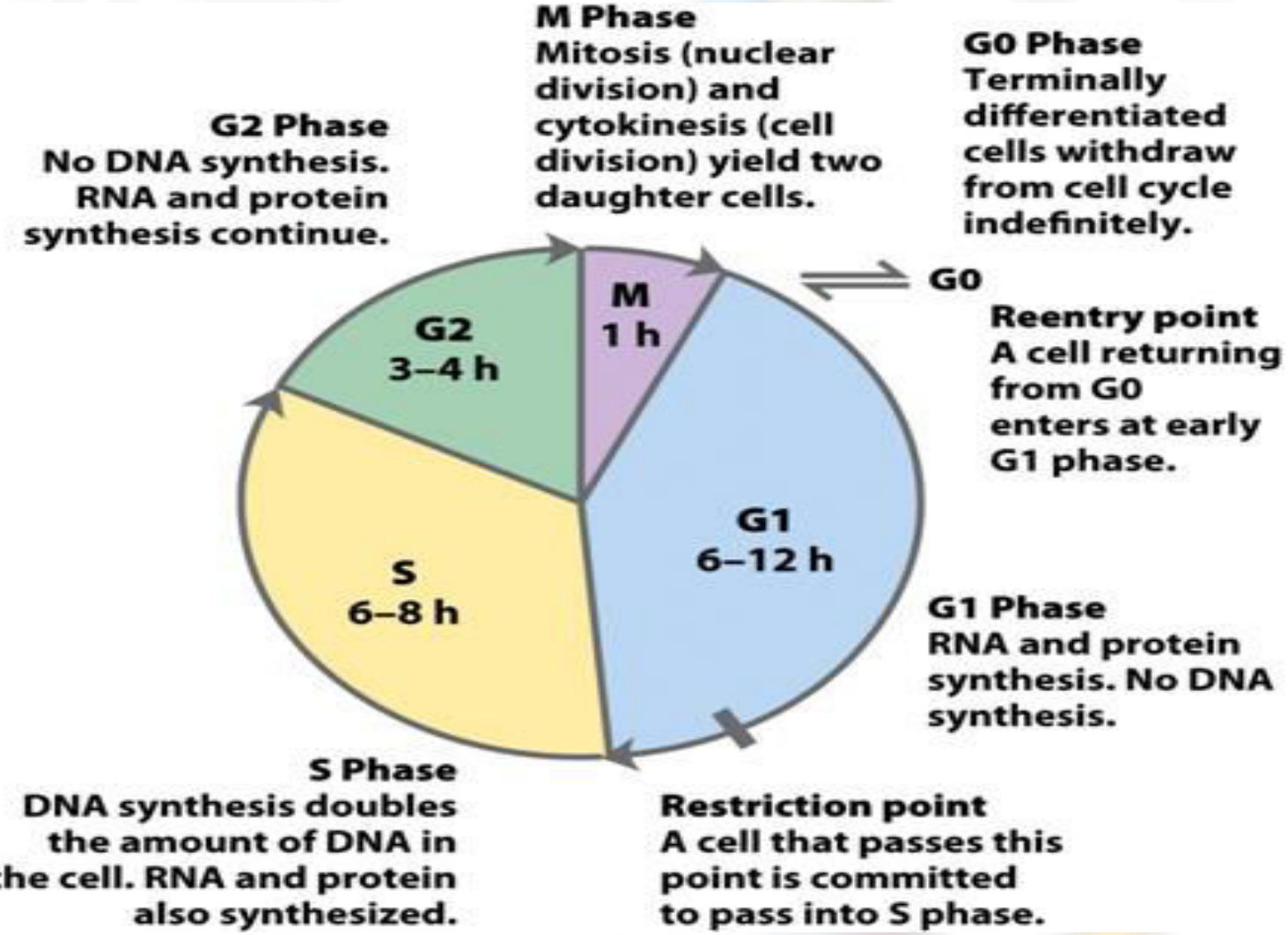
Interphase

INTERPHASE

- Interphase consists of
- G1 phase
- S phase
- G2 phase



Interphase


M Phase
Mitosis (nuclear division) and cytokinesis (cell division) yield two daughter cells.
G0 Phase
Terminally differentiated cells withdraw from cell cycle indefinitely.
G2 Phase
No DNA synthesis. RNA and protein synthesis continue.
G0
Reentry point
A cell returning from G0 enters at early G1 phase.
M
1 h
G2
3–4 h
G1
6–12 h
S
6–8 h
G1 Phase
RNA and protein synthesis. No DNA synthesis.
S Phase
DNA synthesis doubles the amount of DNA in the cell. RNA and protein also synthesized.
Restriction point
A cell that passes this point is committed to pass into S phase.

Interphase

INTERPHASE SUMMARY

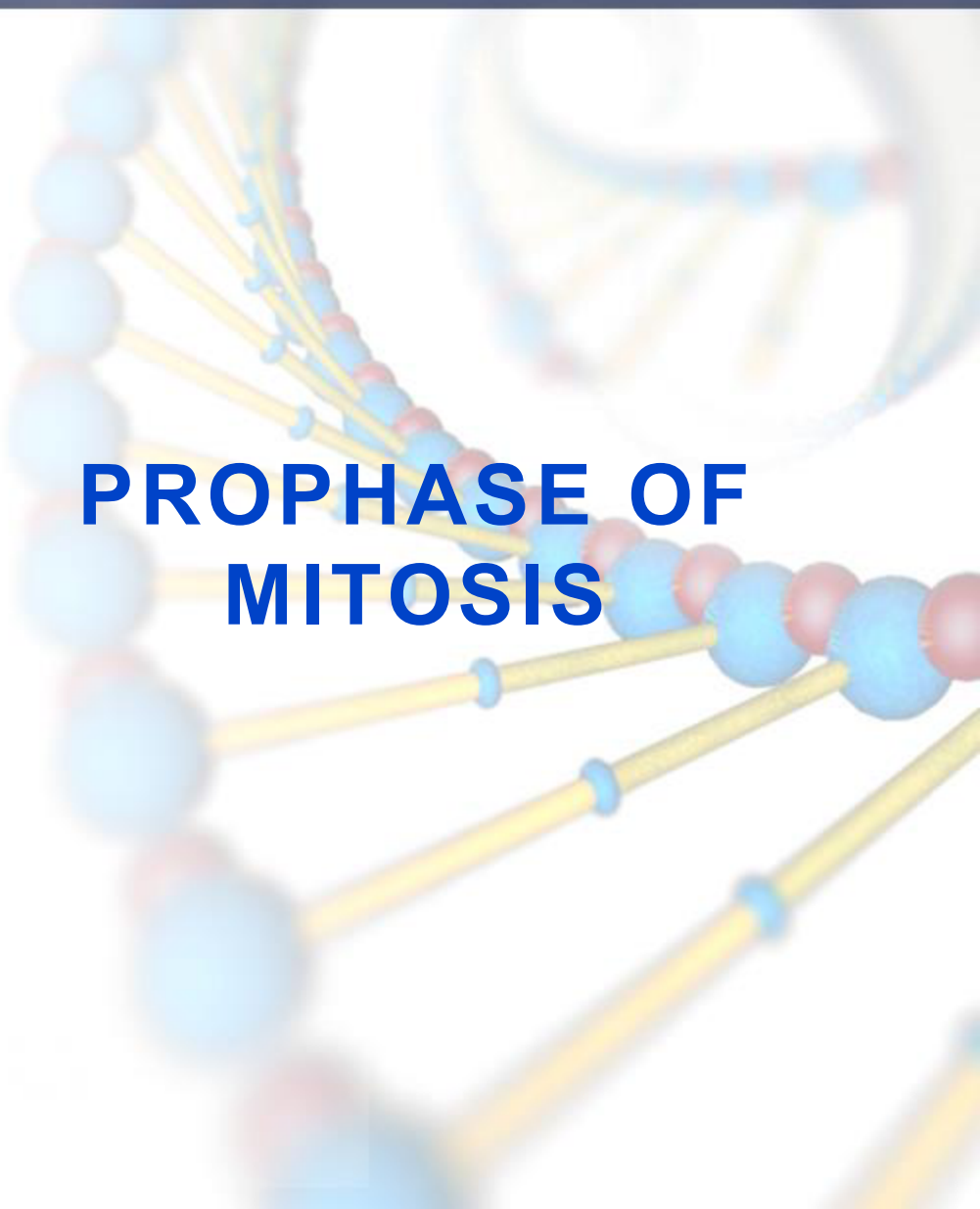
- Interphase ~ 90% of the time.
- G1: new cell absorbs nutrients and grows larger.

Interphase

INTERPHASE SUMMARY

- S phase: Synthesis of new DNA for daughter cells.
- G2: Cell continues to grow, gets too large, needs to divide.

Essentials of Genetics



PROPHASE OF MITOSIS

Prophase of Mitosis

PROPHASE

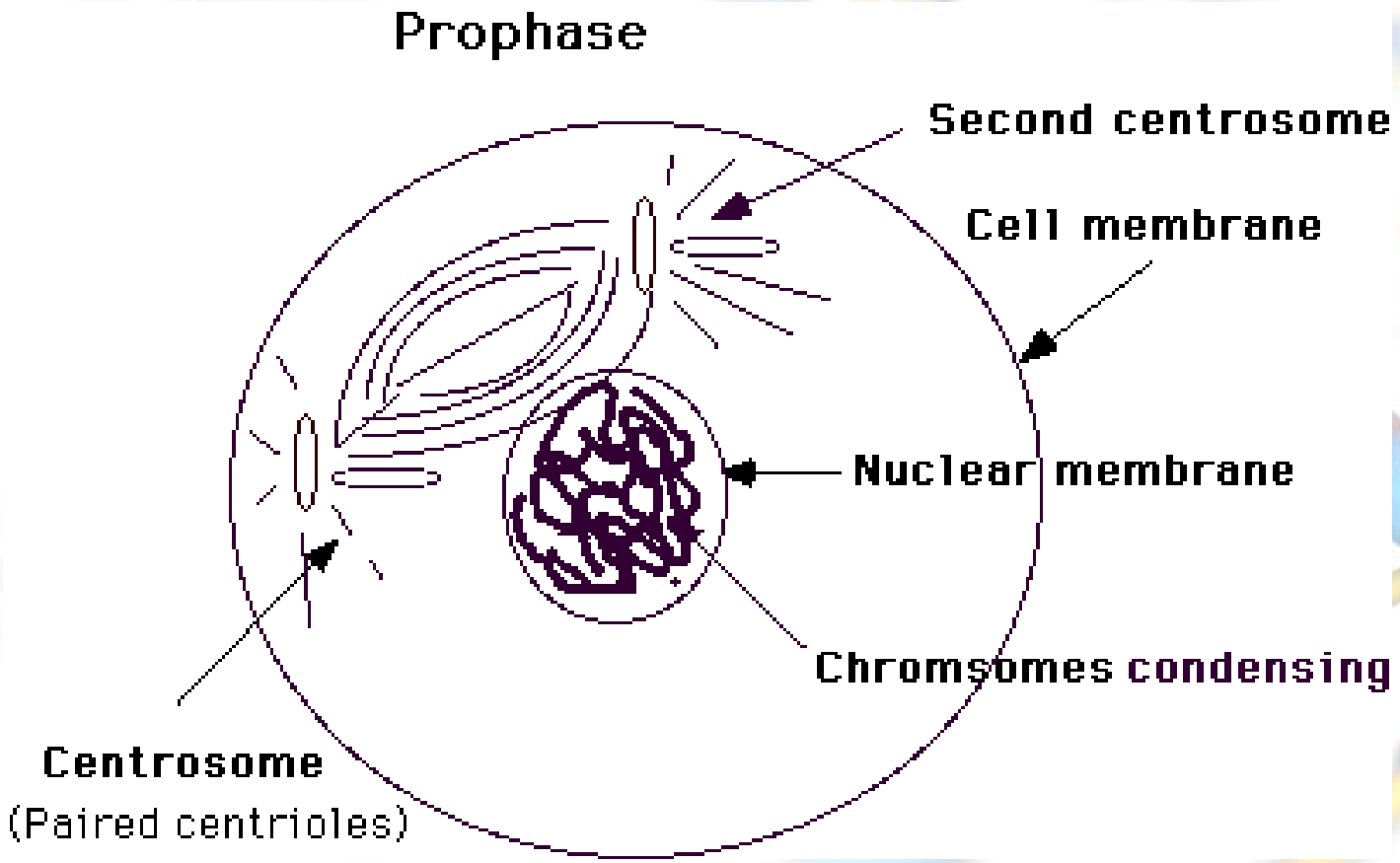
- Prophase is characterized by four events:
- Chromosomes condensed - more visible.
- Nuclear membrane disappears.

Prophase of Mitosis

PROPHASE

- Centrioles are separated and take positions on the opposite poles of the cell.
- Spindle fibers form and radiate toward the center of the cell.

Prophase of Mitosis



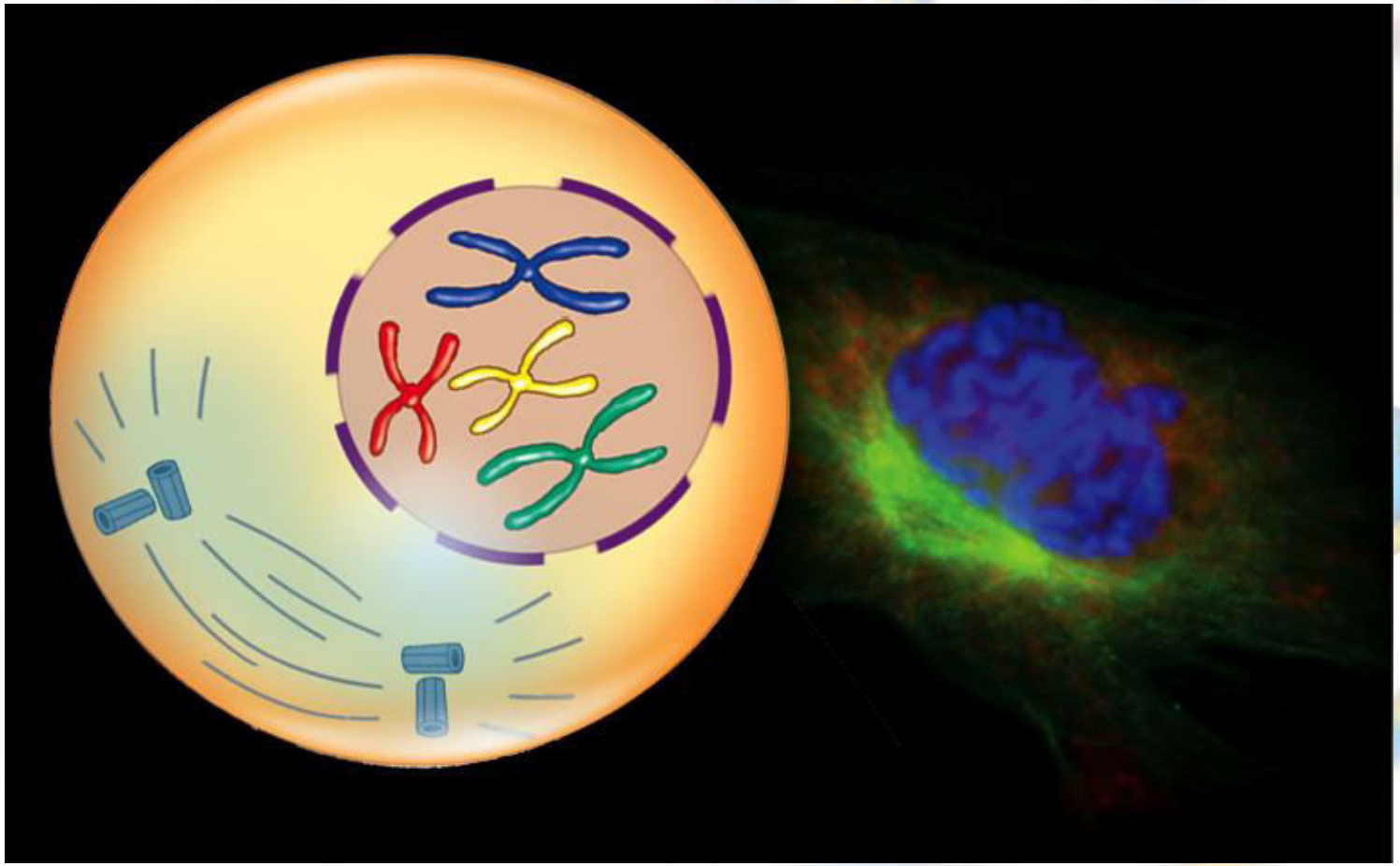
Prophase of Mitosis

PROPHASE – CHROMOSOMES ARE CONDENSED

- Chromosomes condense at the start of mitosis.
- DNA wraps around histones that condense it.

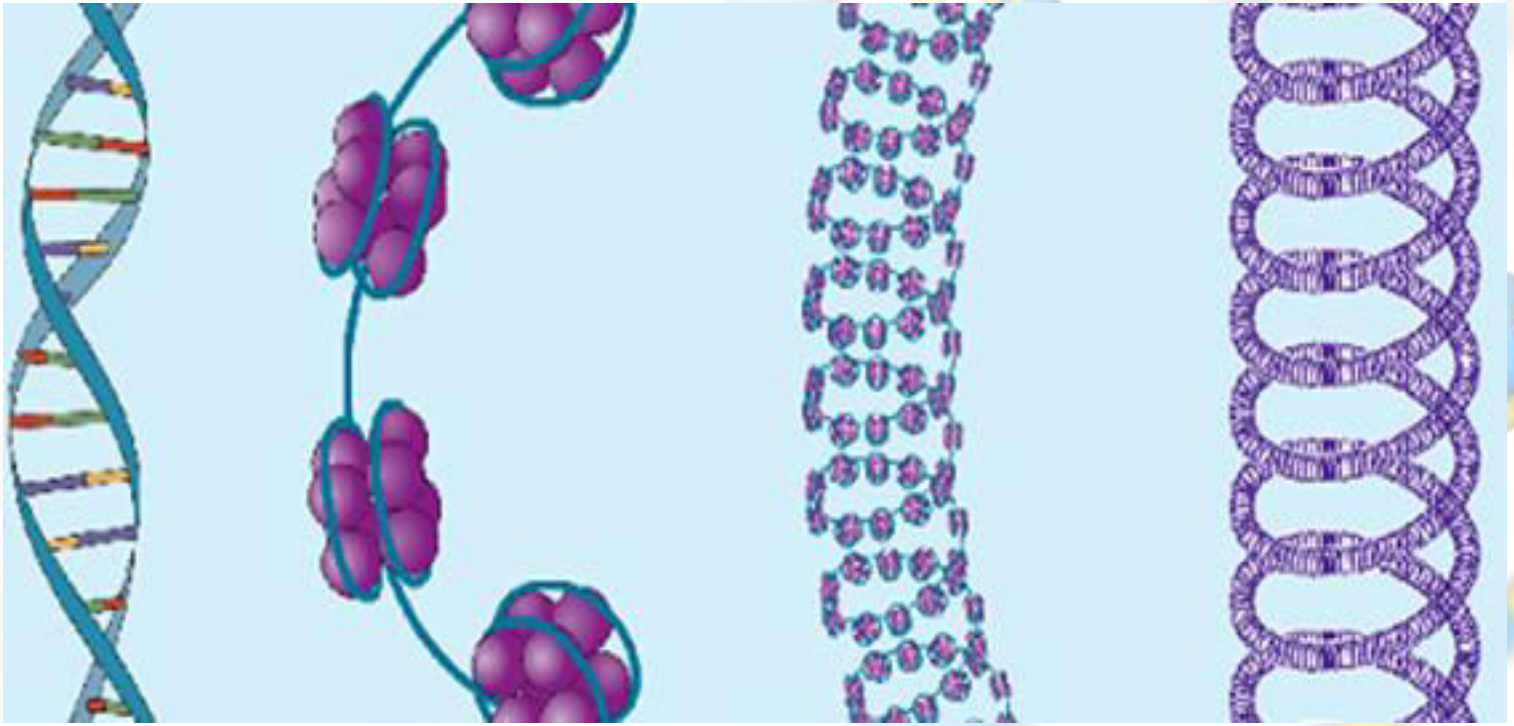
Prophase of Mitosis

PROPHASE –CHROMOSOMES ARE CONDENSED



Prophase of Mitosis

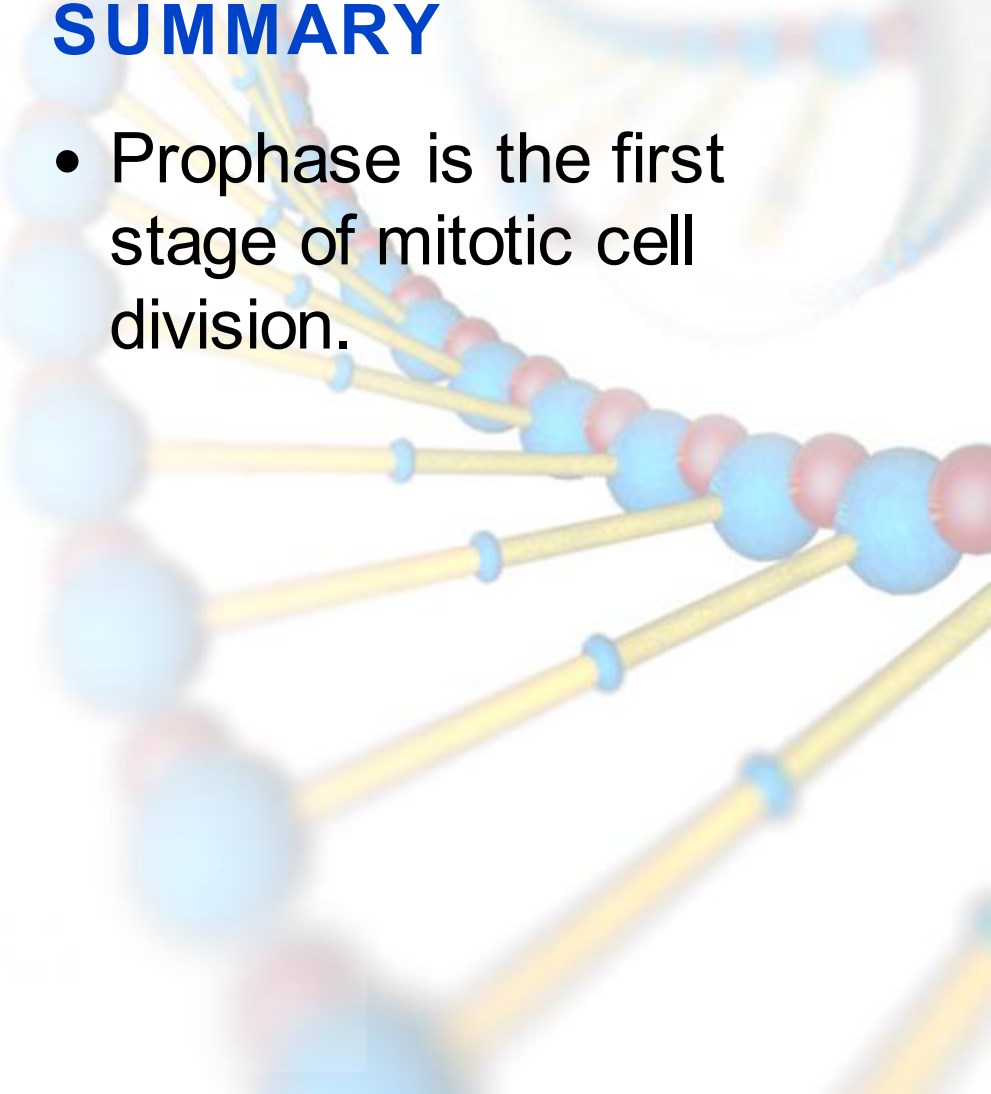
PROPHASE- CHROMOSOMES ARE CONDENSED- Different levels of condensation/folding



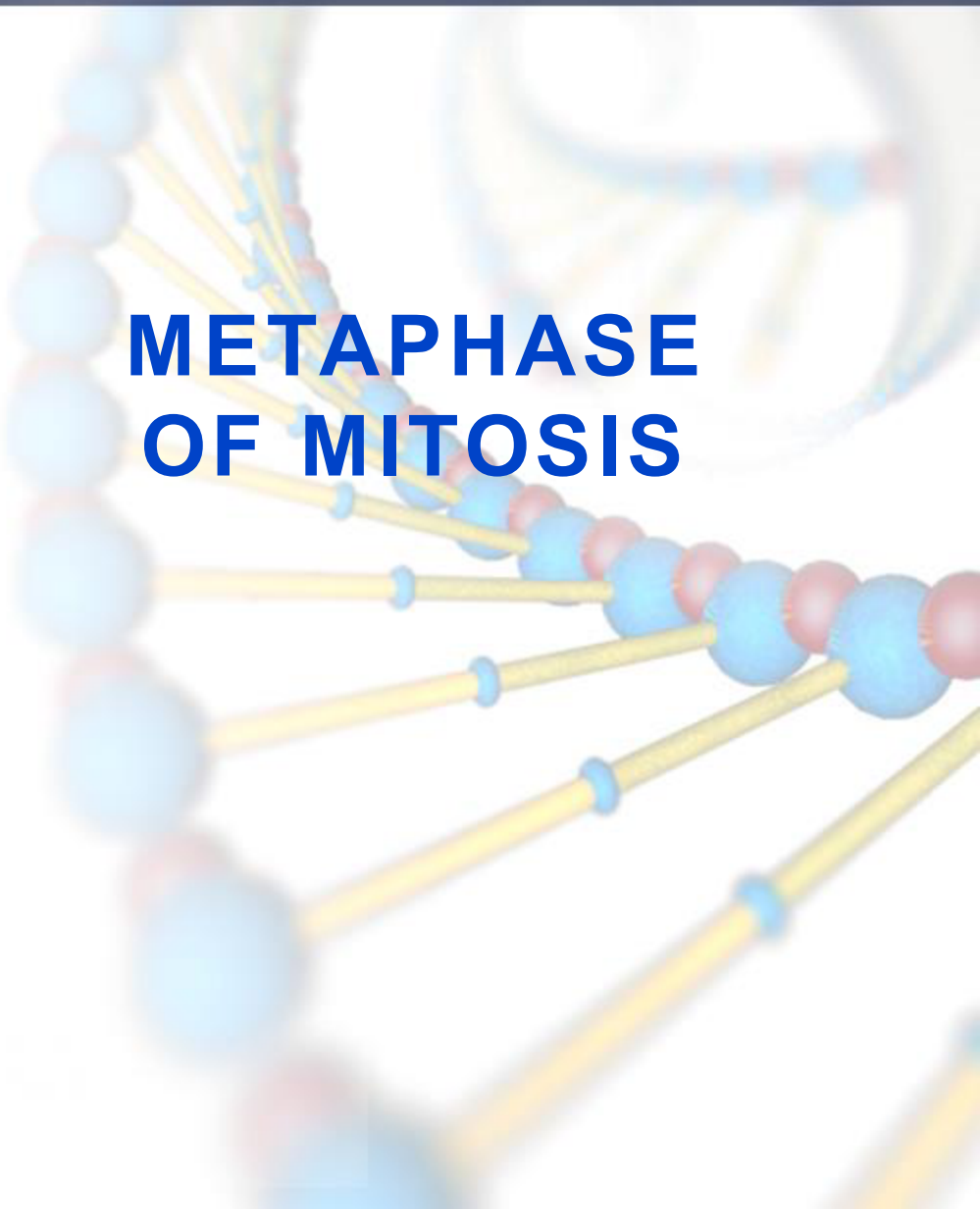
Prophase of Mitosis

PROPHASE SUMMARY

- Prophase is the first stage of mitotic cell division.



Essentials of Genetics



METAPHASE OF MITOSIS

Metaphase of Mitosis

METAPHASE

- During metaphase, two events occurs;
- Chromosomes line up across the middle of the cell.

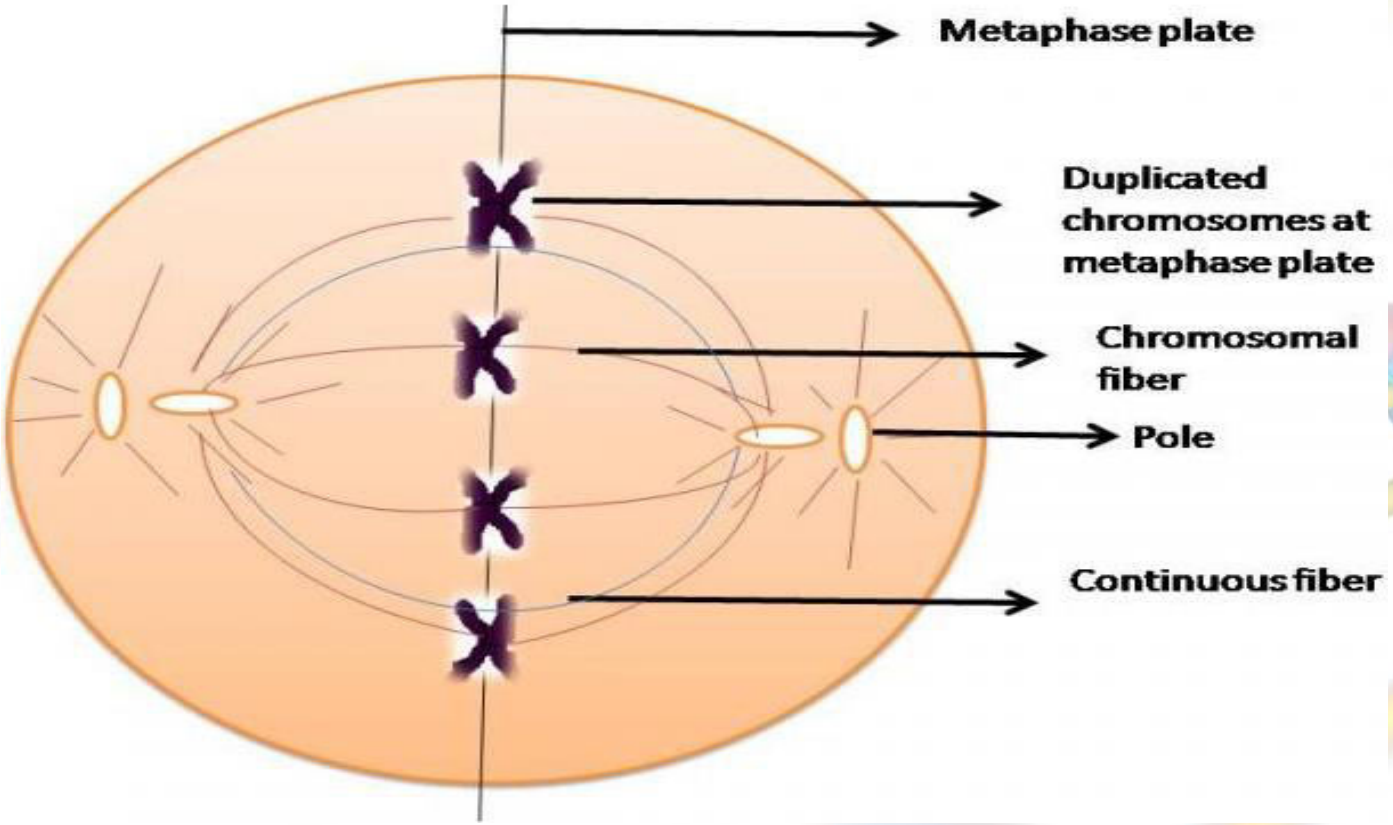
Metaphase of Mitosis

METAPHASE

- Spindle fibers connect the centromere of each sister chromatid to the poles of the cell, where centrioles are present.

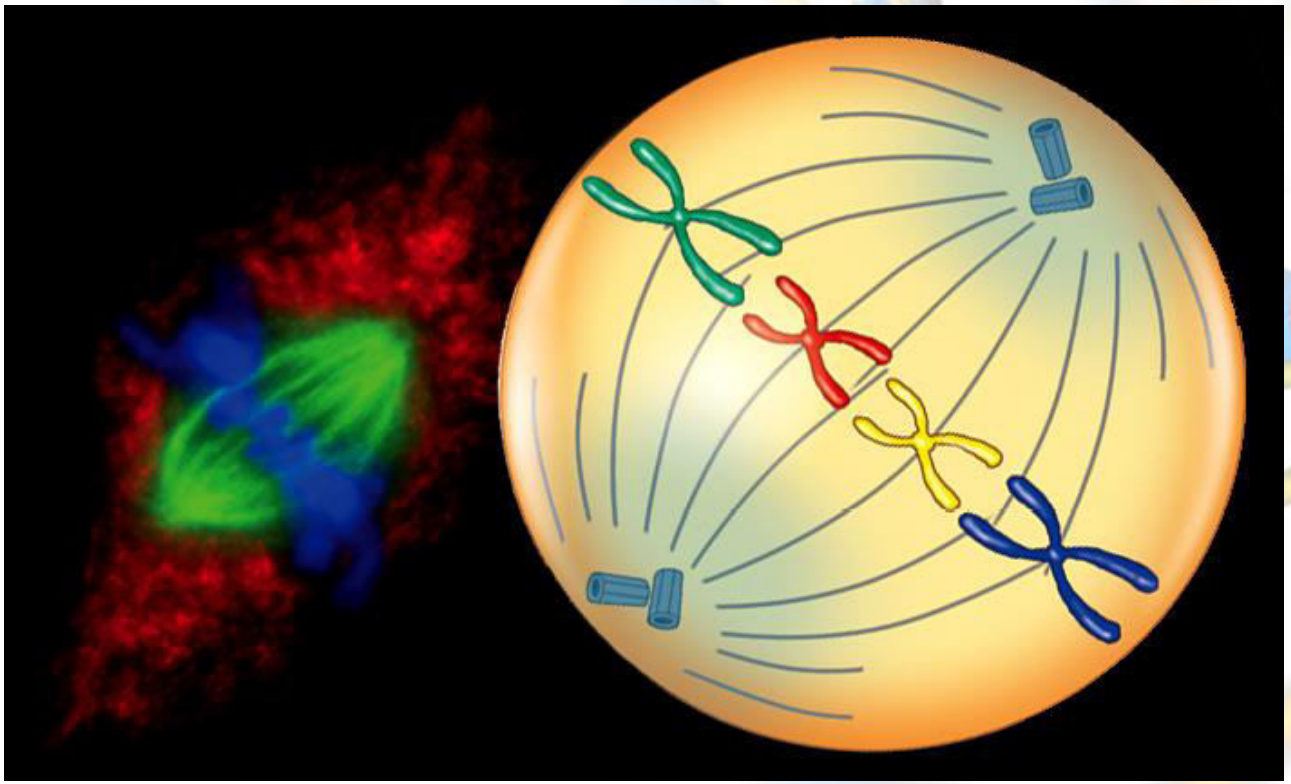
Metaphase of Mitosis

METAPHASE – CHROMOSOMES ARE ALIGNED IN THE MIDDLE



Metaphase of Mitosis

METAPHASE - Chromosomes are aligned in the middle, - Spindle fibre are attached to centromere



Metaphase of Mitosis

METAPHASE-SUMMARY

- During metaphase, two events occurs;
- Chromosomes lined up in the middle line.
- Spindle fibers connect to centromere.

Essentials of Genetics

ANAPHASE OF MITOSIS

Anaphase of Mitosis

ANAPHASE

- During anaphase, three events occurs;
- Centromere split – as a result sister chromatids are separated.
- Sister chromatids becomes individual chromosomes.

Anaphase of Mitosis

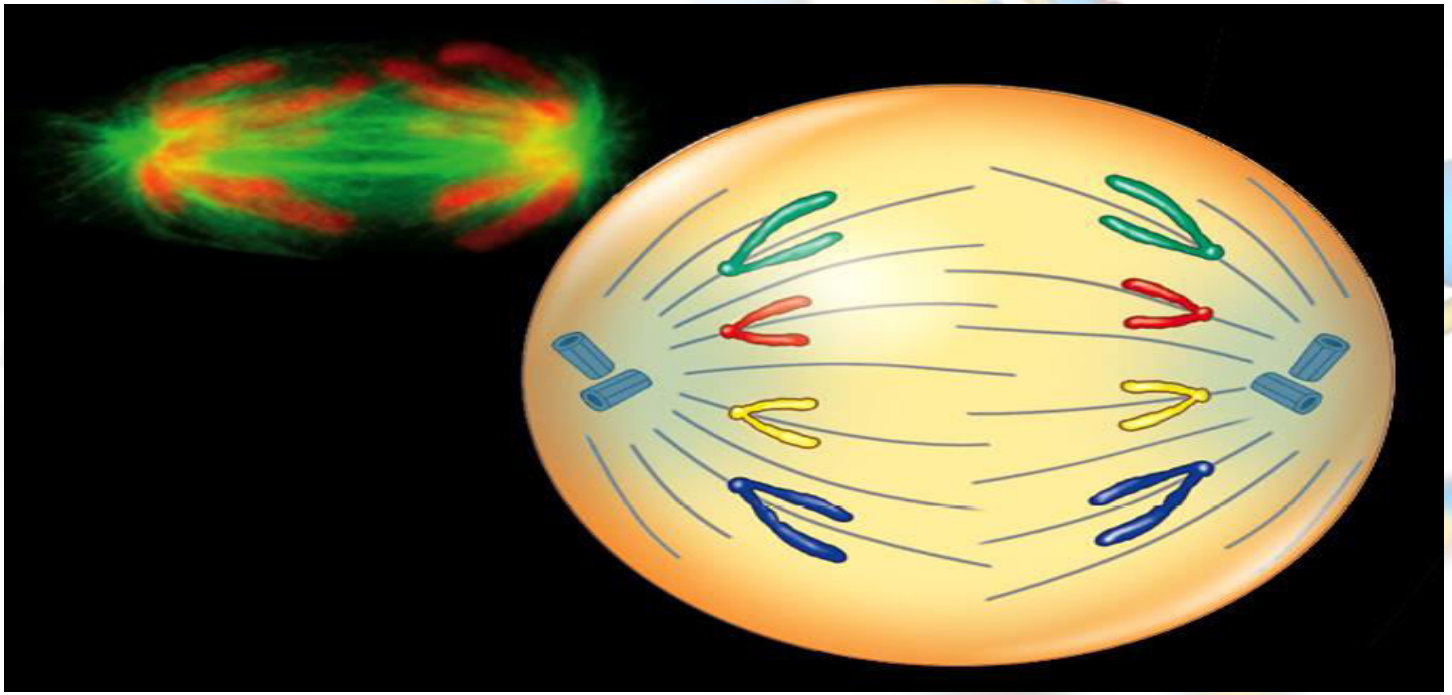
ANAPHASE

- Separated chromatids move to opposite poles of the cell.

Anaphase of Mitosis

ANAPHASE

Securin is a protein which inhibits a protease known as separase. Destruction of securin results in split of centromeres and the new daughter chromosomes are pulled toward the poles.



Anaphase of Mitosis

CELL IN AN OVAL SHAPE

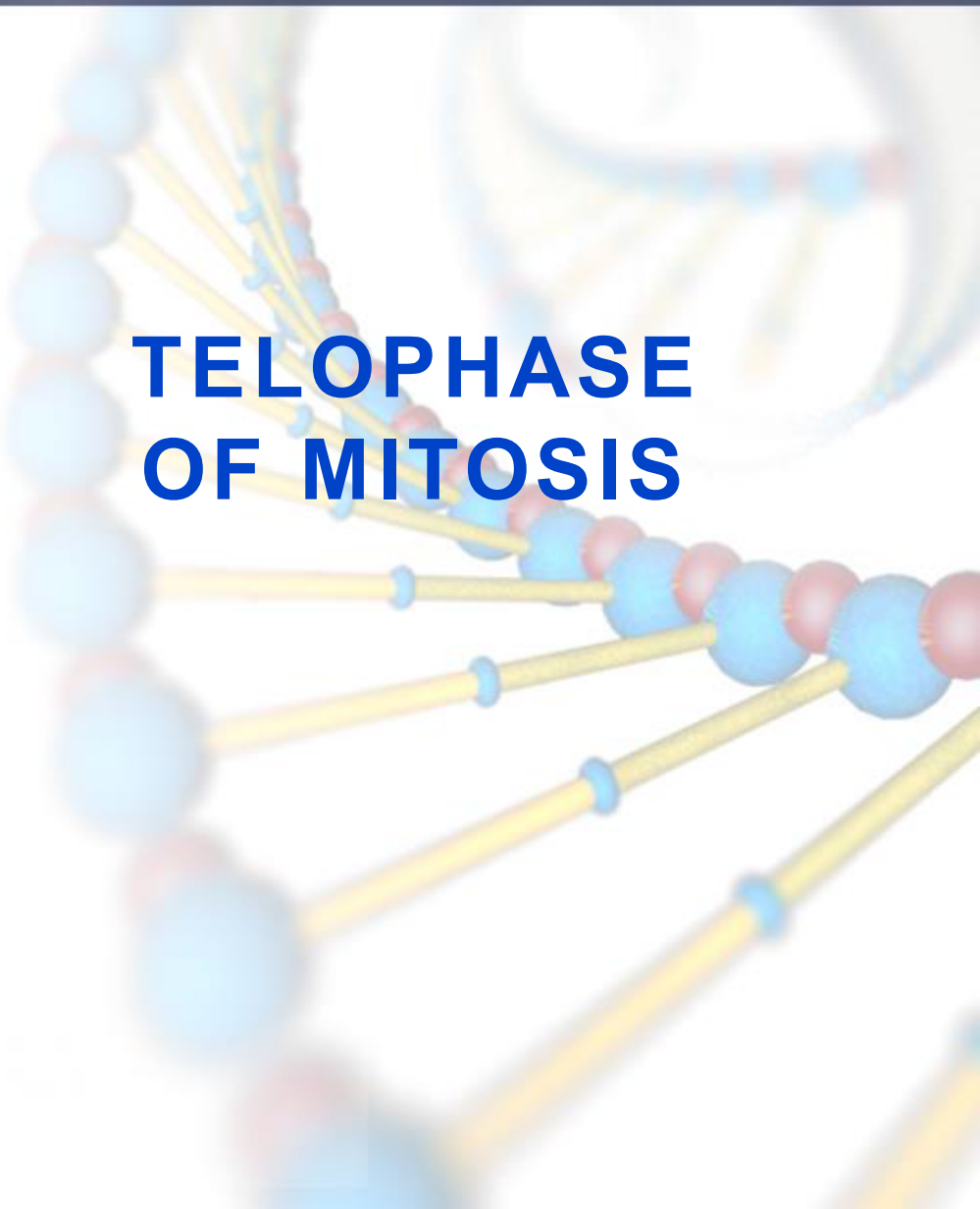
- Chromosomes are drawn to each side of the cell.
- Non-kinetochore spindle fibers push against each other, that stretches the cell into an oval shape.

Anaphase of Mitosis

ANAPHASE SUMMARY

- Centromere split and chromosomes are separated.
- Chromosomes arrived at poles.
- Cell stretches into an oval shape.

Essentials of Genetics



TELOPHASE OF MITOSIS

Telophase of Mitosis

TELOPHASE

- Telophase consists of four events:
- Chromosomes start to uncoil.
- Nuclear envelope forms around the chromosomes at each pole of the cell.

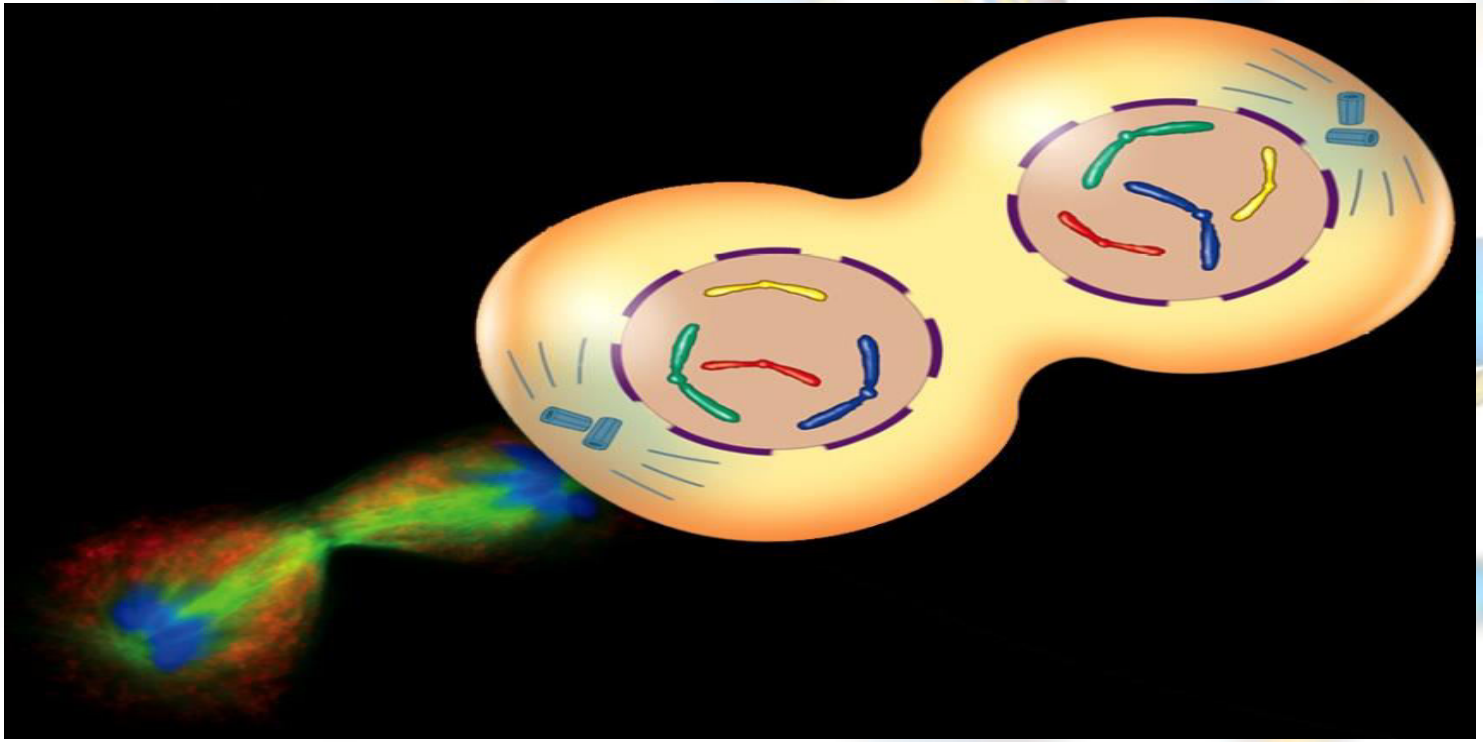
Telophase of Mitosis

TELOPHASE

- Spindle fibers break down and dissolve.
- Daughter nucleus is formed in each of the daughter cell.

Telophase of Mitosis

TELOPHASE Nuclear membranes are formed around each set of chromatids. Nucleoli also reappear. Chromosomes unwind back into chromatin material.



Telophase of Mitosis

TELOPHASE – EFFECTS OF PROPHASE ARE REVERSED

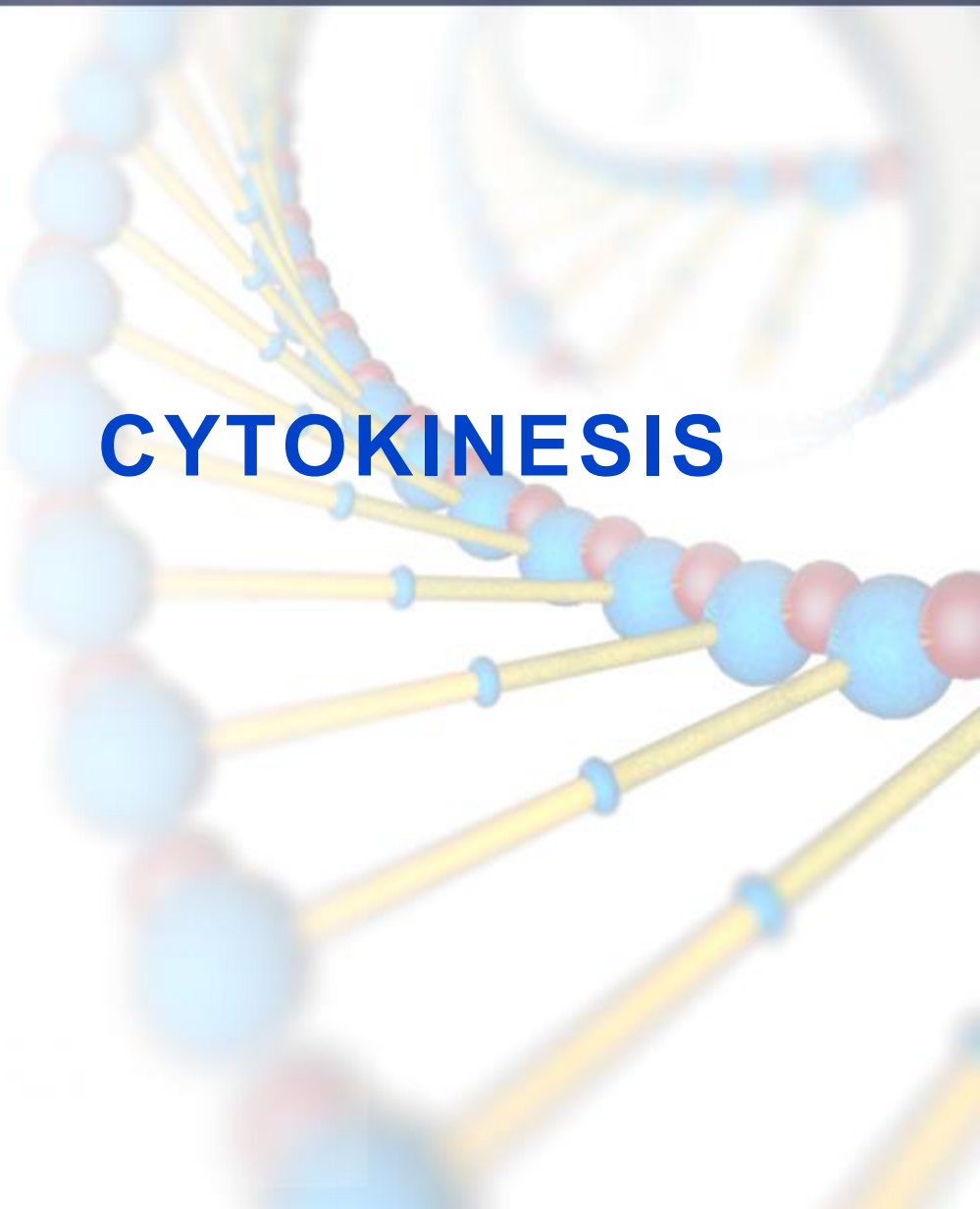
- During telophase, the effects of prophase and pro-metaphase are reversed.

Telophase of Mitosis

TELOPHASE SUMMARY

- Spindle fibers break down and nuclear members develops around each daughter nucleus.

Essentials of Genetics



CYTOKINESIS

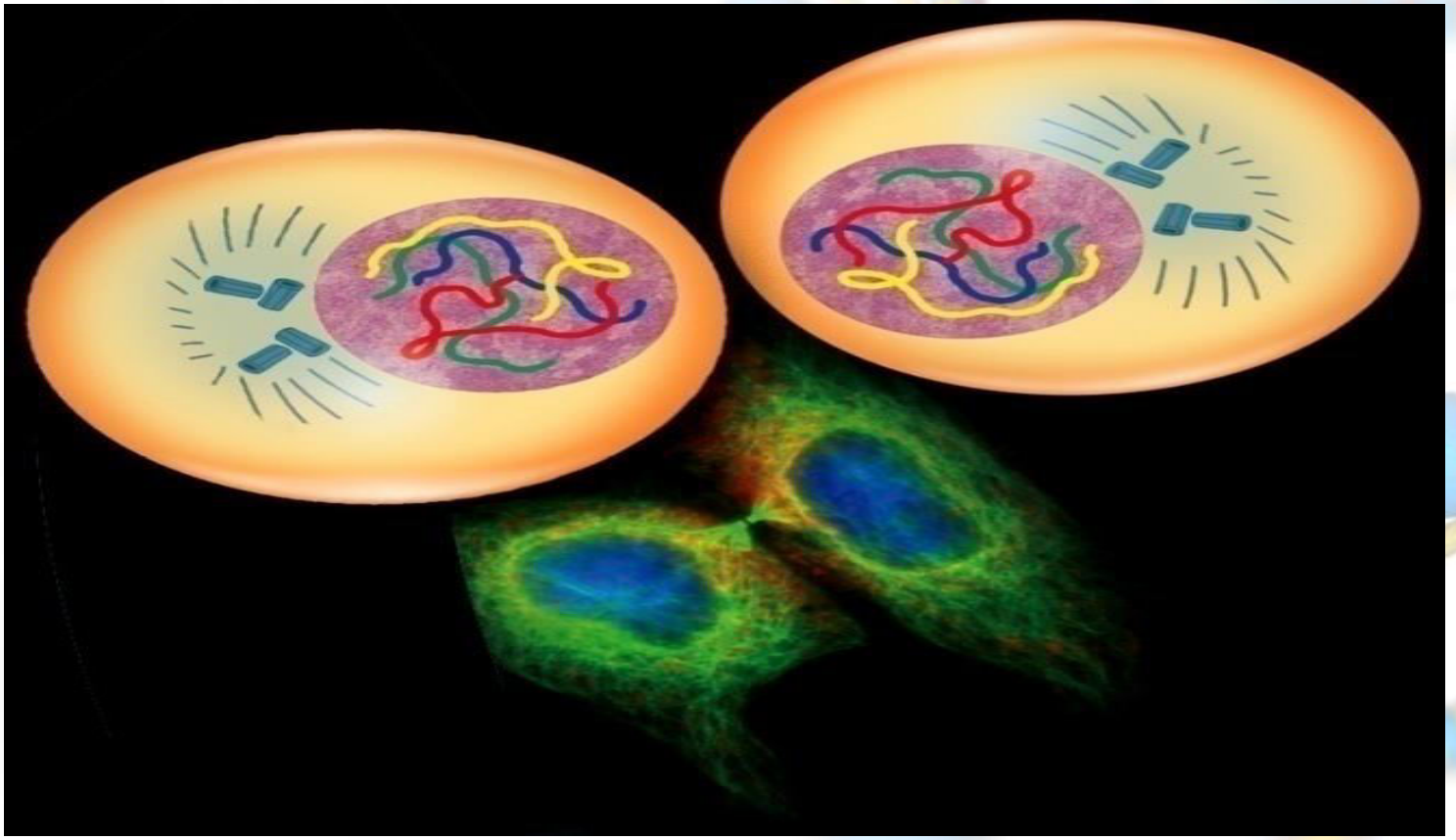
Cytokinesis

CYTOKINESIS

- Cytokinesis is the division of the cytoplasm into two individual cells.
- Process differs in plants and animal cells.

Cytokinesis

CYTOKINESIS



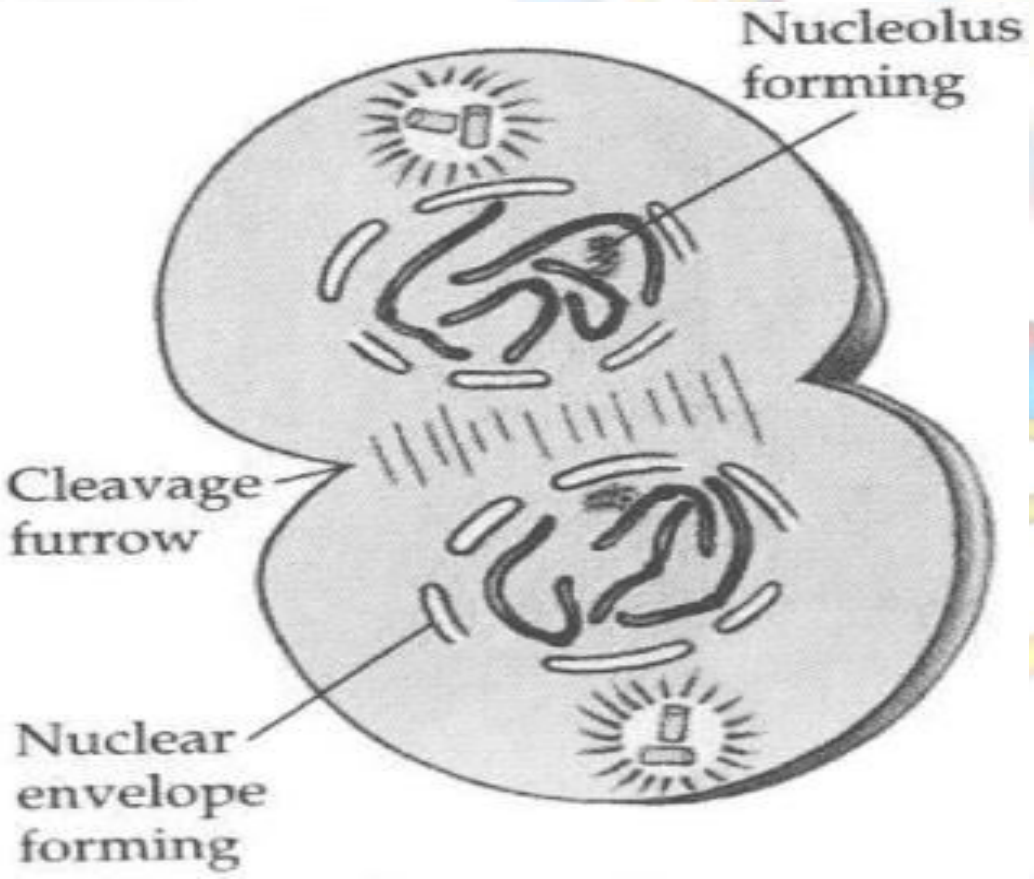
Cytokinesis

CYTOKINESIS IN ANIMAL CELLS

- Cell membrane forms a cleavage furrow that eventually pinches the cell into two equal parts.
- Each part containing its own nucleus and cytoplasmic organelles.

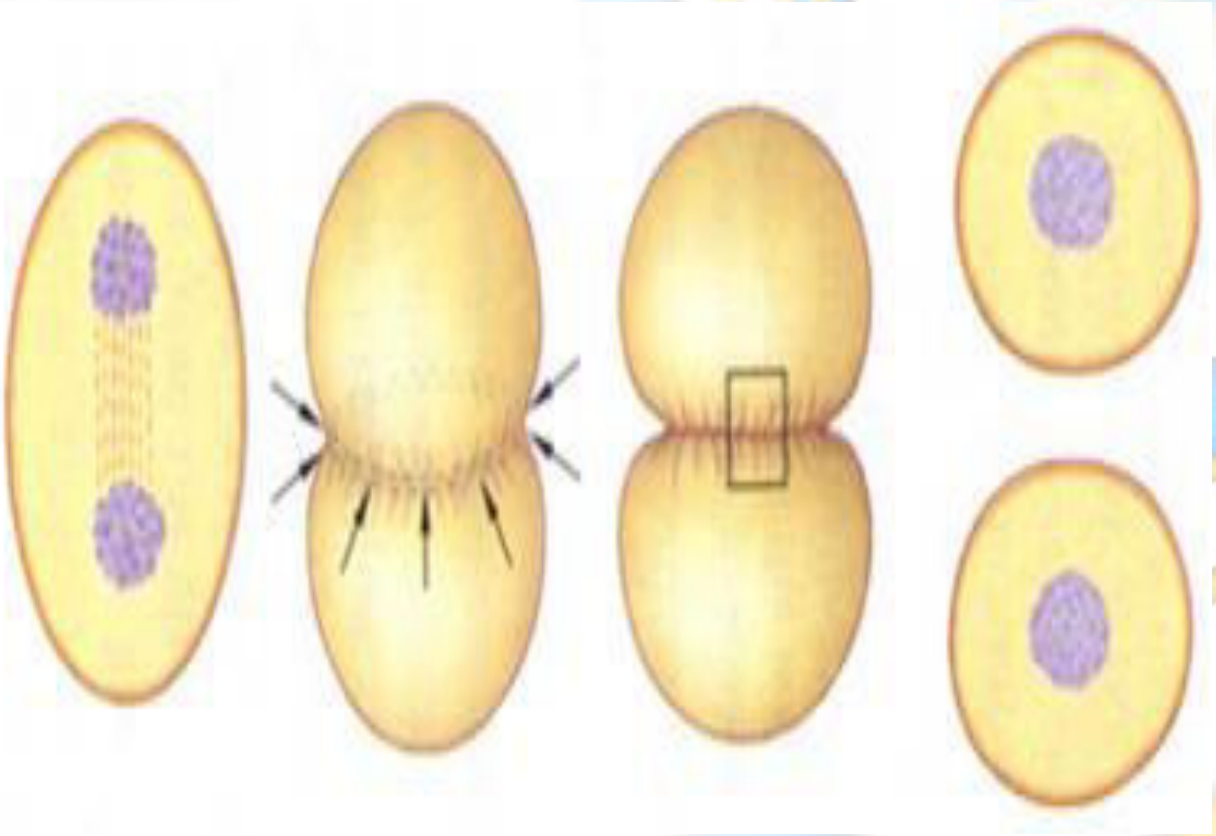
Cytokinesis

CYTOKINESIS IN ANIMAL CELL



Cytokinesis

CYTOKINESIS IN ANIMAL CELL



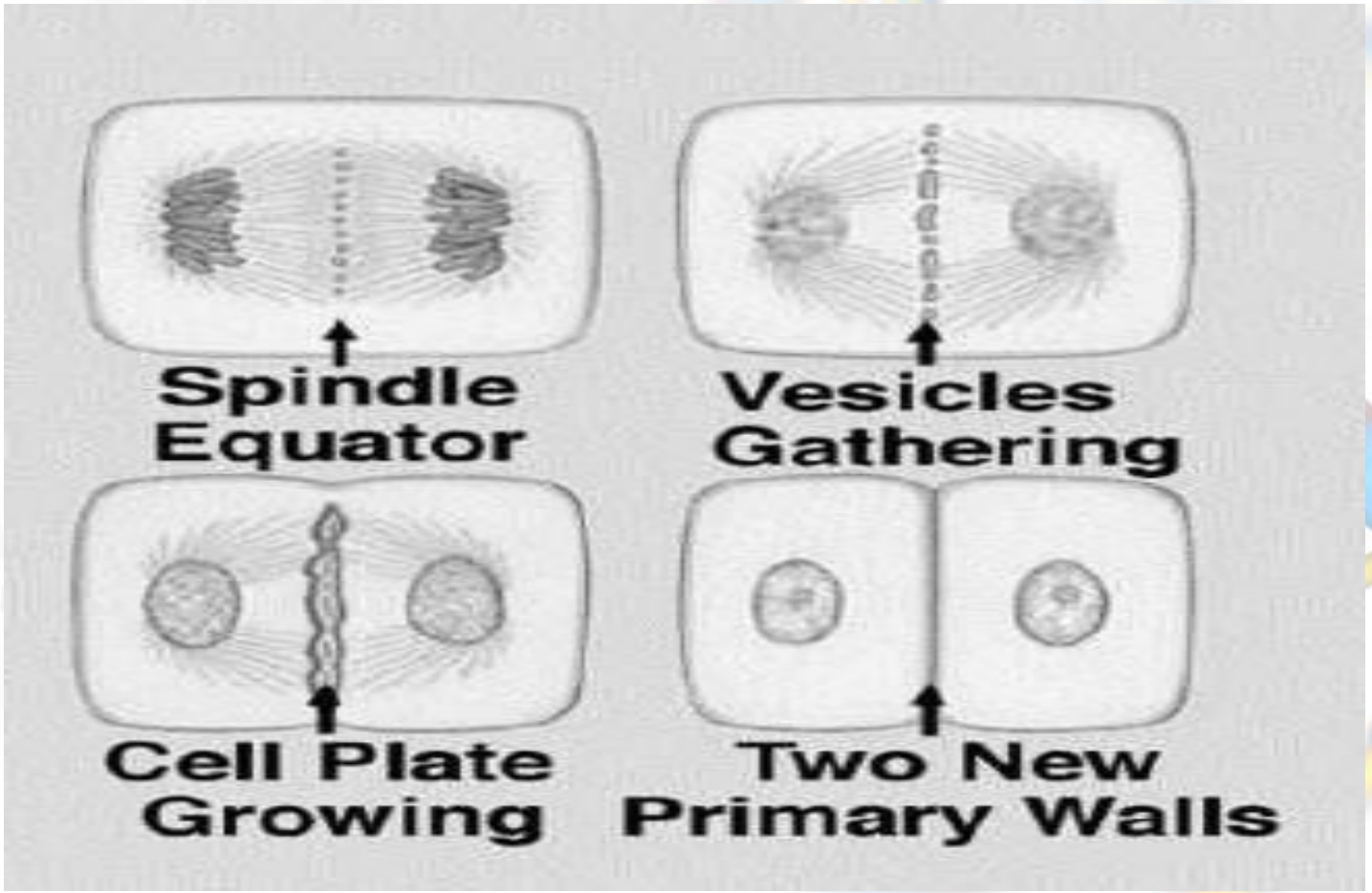
Cytokinesis

CYTOKINESIS IN PLANTS CELLS

- A cell plate is formed in midway that divides the nuclei.
- Cell plate gradually develops into a separating membrane.

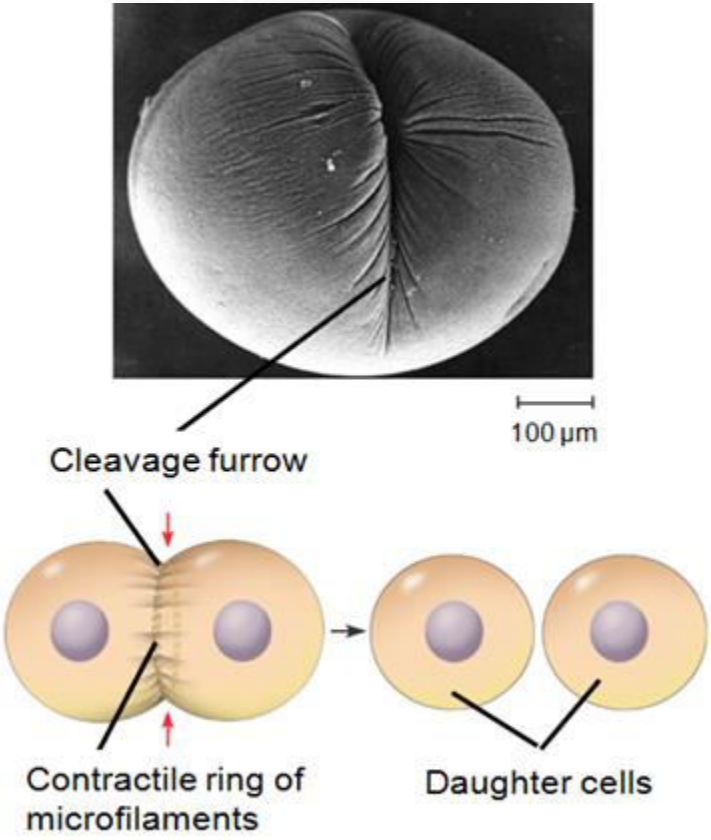
Cytokinesis

PLANTS CELL- CYTOKINESIS

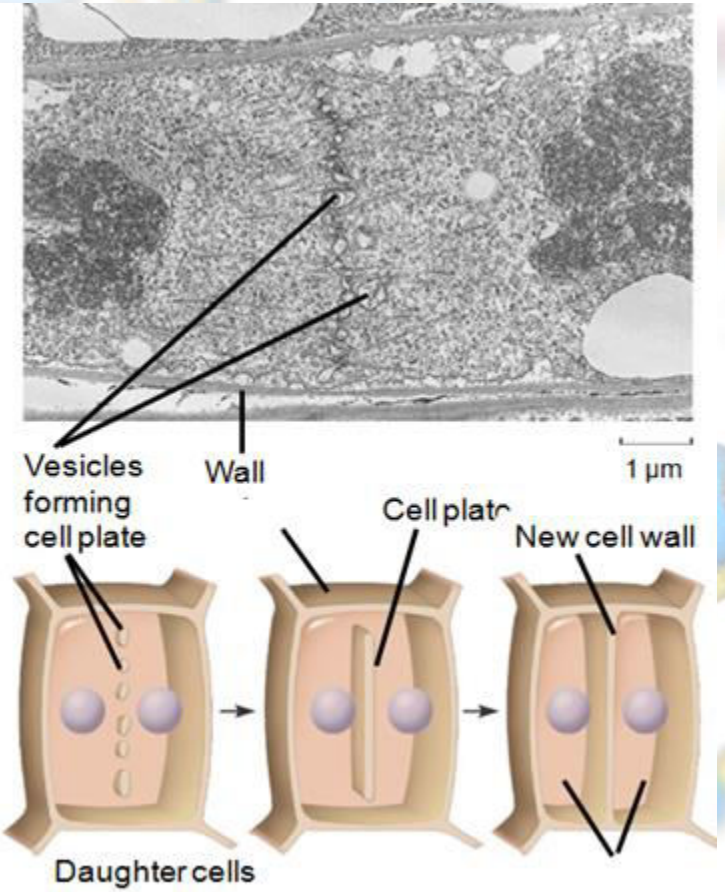


Cytokinesis

CYTOKINESIS - COMPARISON



(a) Cleavage of an animal cell (SEM)



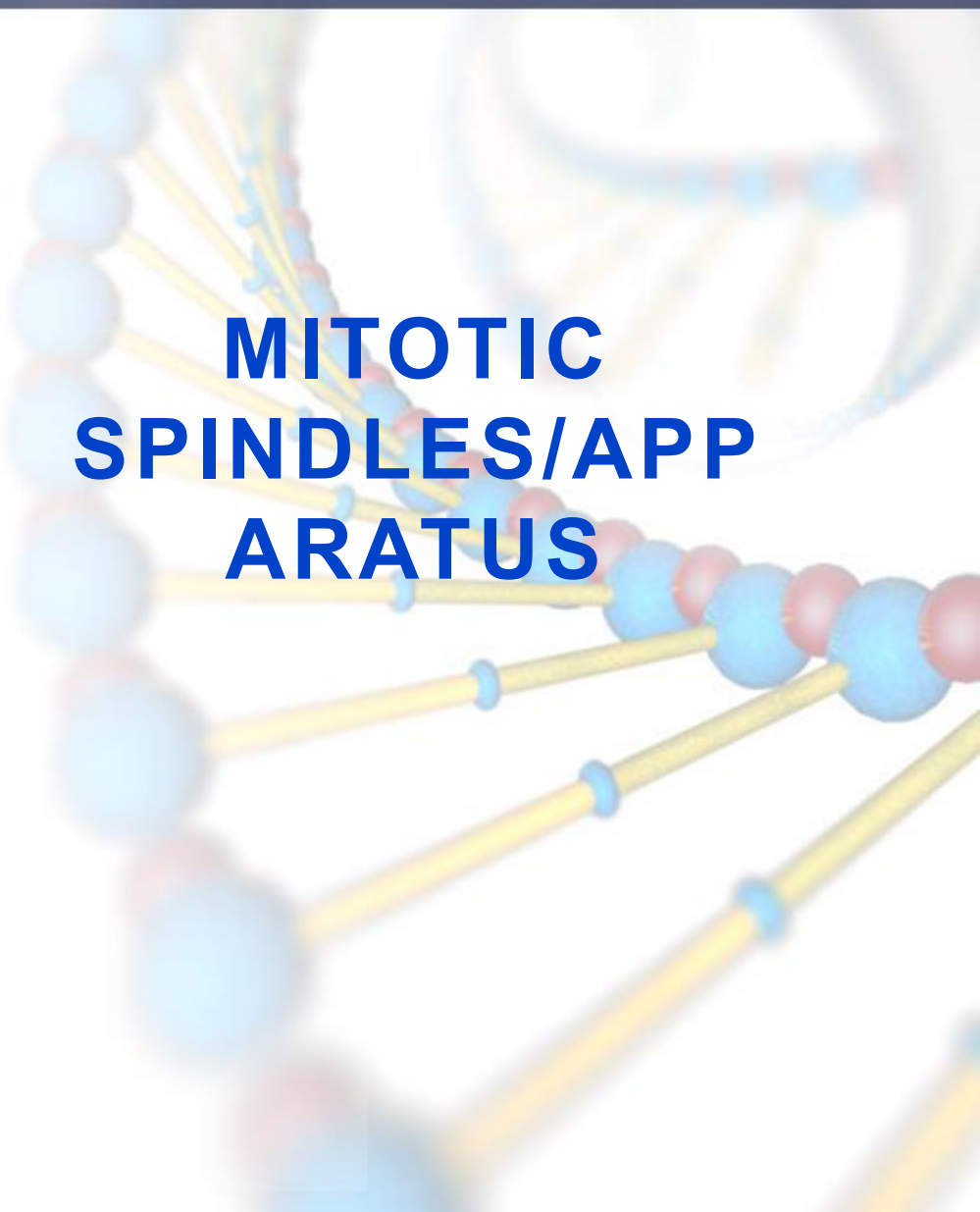
(b) Cell plate formation in a plant cell (SEM)

Cytokinesis

CYTOKINESIS SUMMARY

- Cytokinesis differs in animal and plant cells.
- In animal cells, the membrane pinches and divides the cell.
- In plant cells, a cell plate forms that divides the cell.

Essentials of Genetics



MITOTIC SPINDLES/APP ARATUS

Mitotic Spindles

MITOTIC APPARATUS/ SPINDLES

- The mitotic apparatus/spindles include
  - Centrosomes
  - Spindle microtubules
  - Asters

Mitotic Spindles

MITOTIC APPARATUS/ SPINDLES

- The apparatus of microtubules controls chromosome movement during mitosis.

Mitotic Spindles

CENTROSOME REPLICATES

- Centrosome replicates, forming two centrosomes that migrate to opposite poles.
- Assembly of spindle microtubules begins in the centrosome.

Mitotic Spindles

MITOTIC SPINDLES – ASTERS

- Asters are the radial array of short microtubules extends from each centrosome.

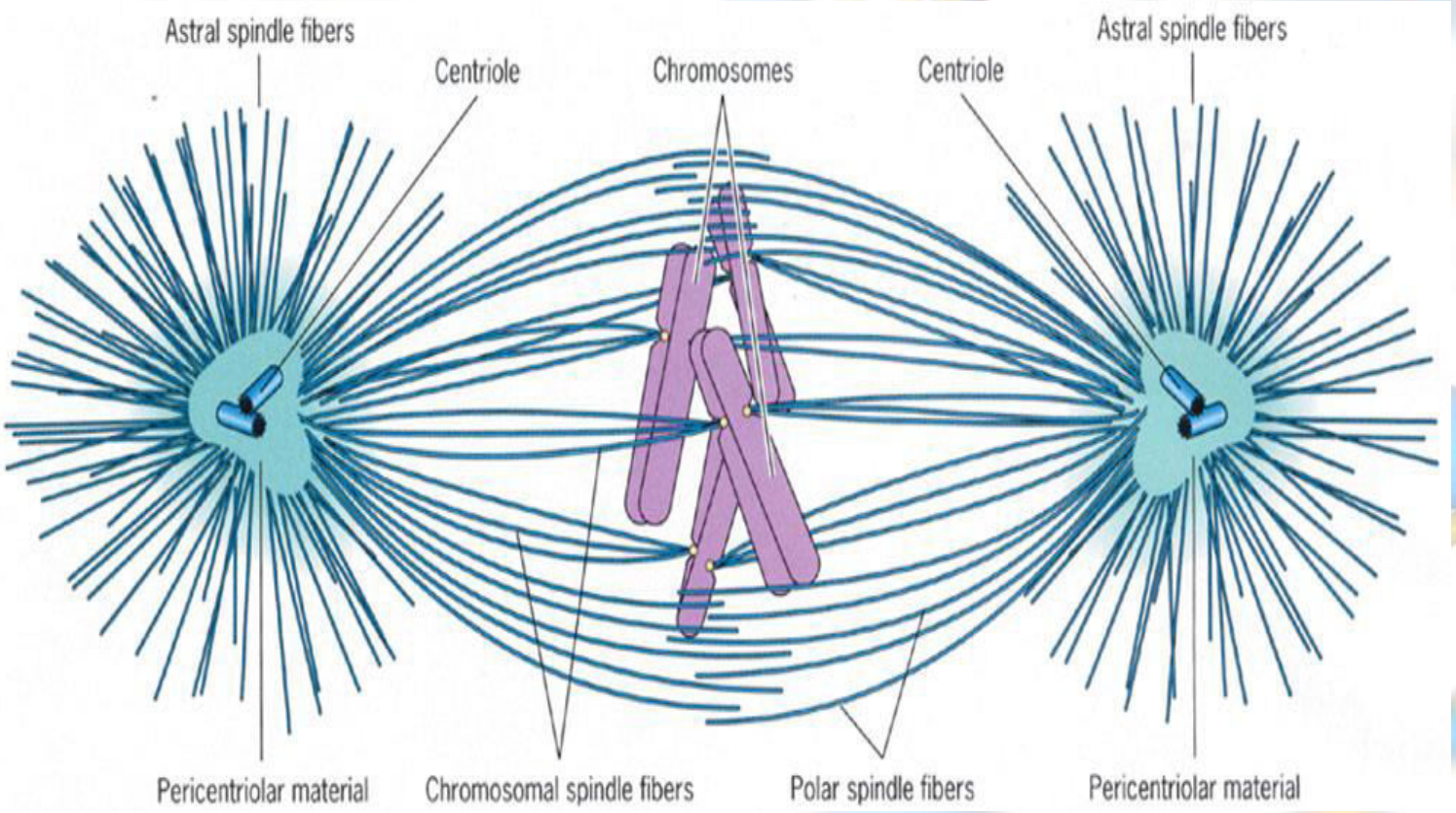
Mitotic Spindles

SPINDLES MICROTUBULES ATTACH TO KINETOCHORES

- Spindle microtubules attach to the kinetochores of chromosomes and move the chromosomes to the metaphase plate.

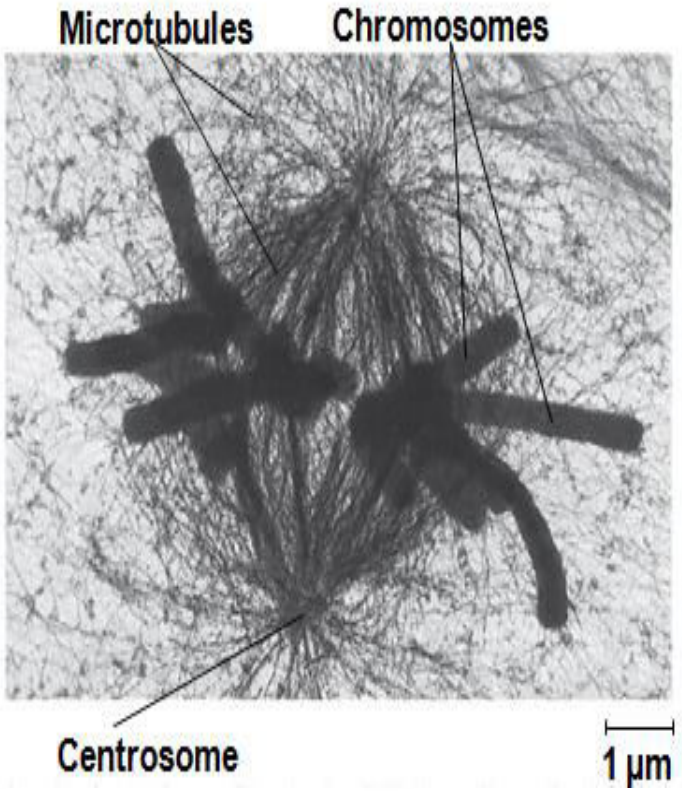
Mitotic Spindles

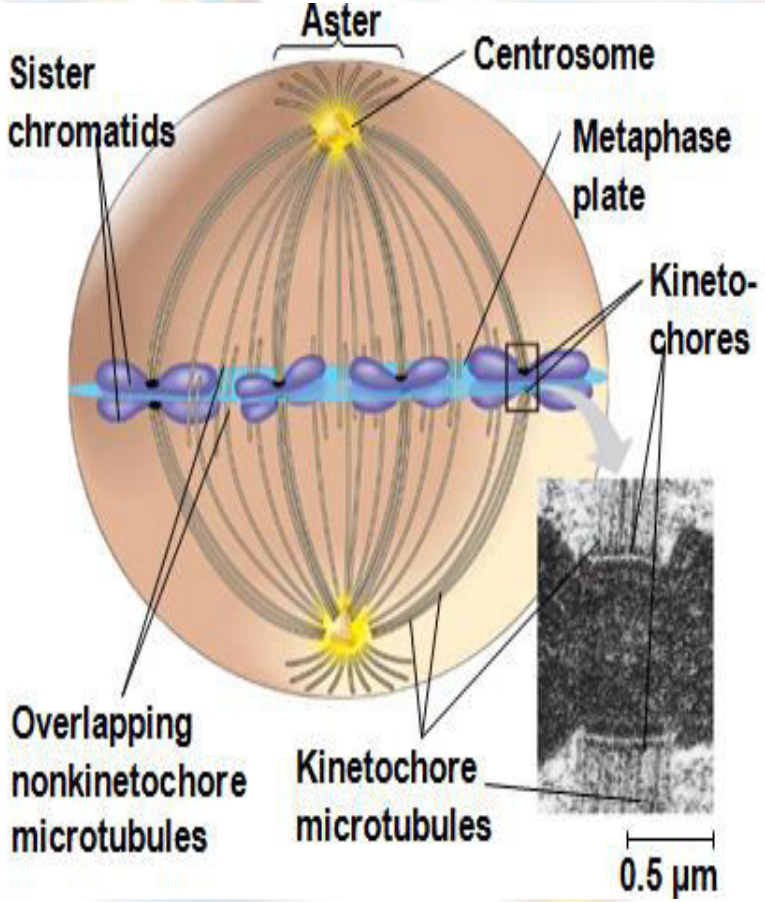
MITOTIC SPINDLES



Mitotic Spindles

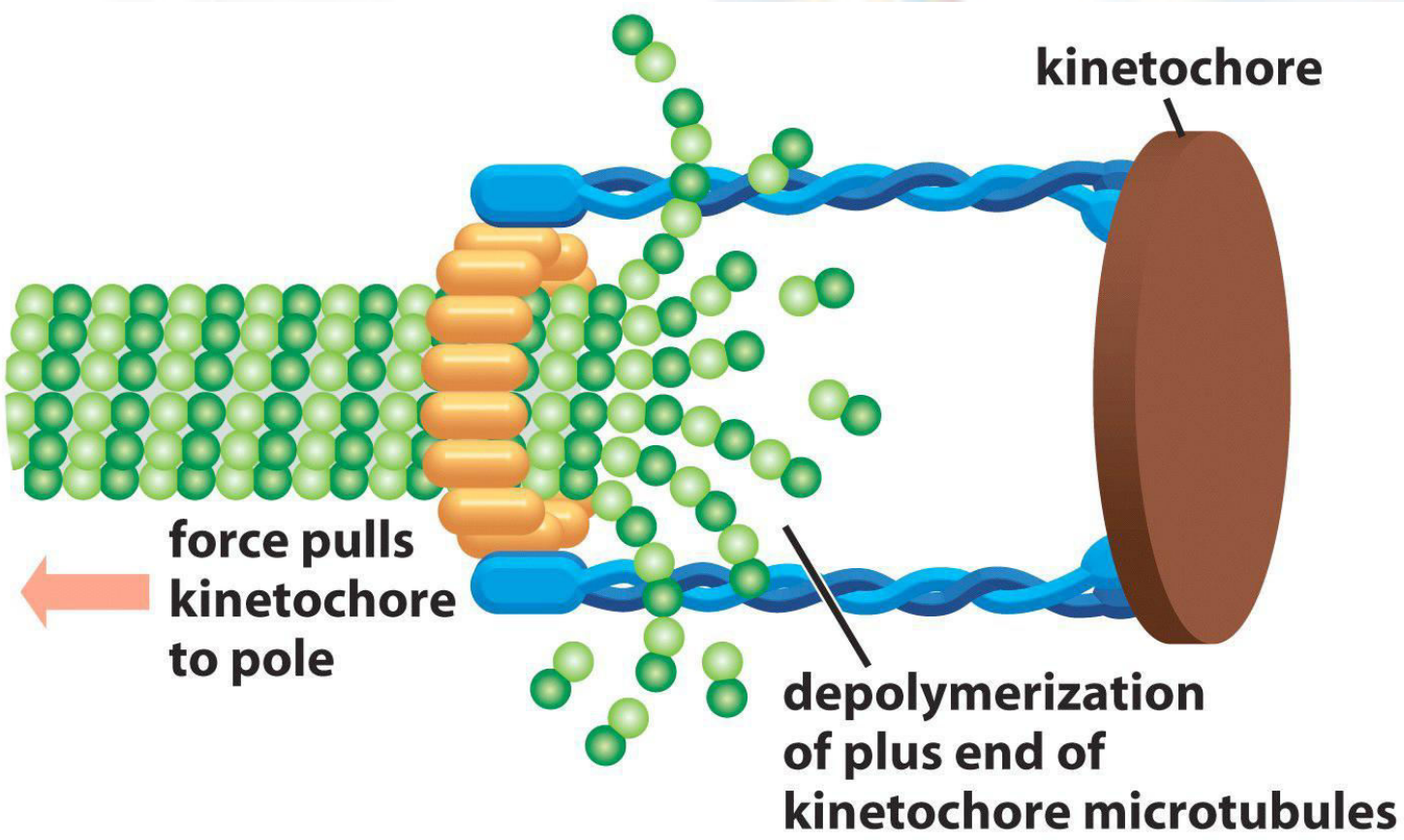
MITOTIC SPINDLES





Mitotic Spindles

MITOTIC SPINDLES



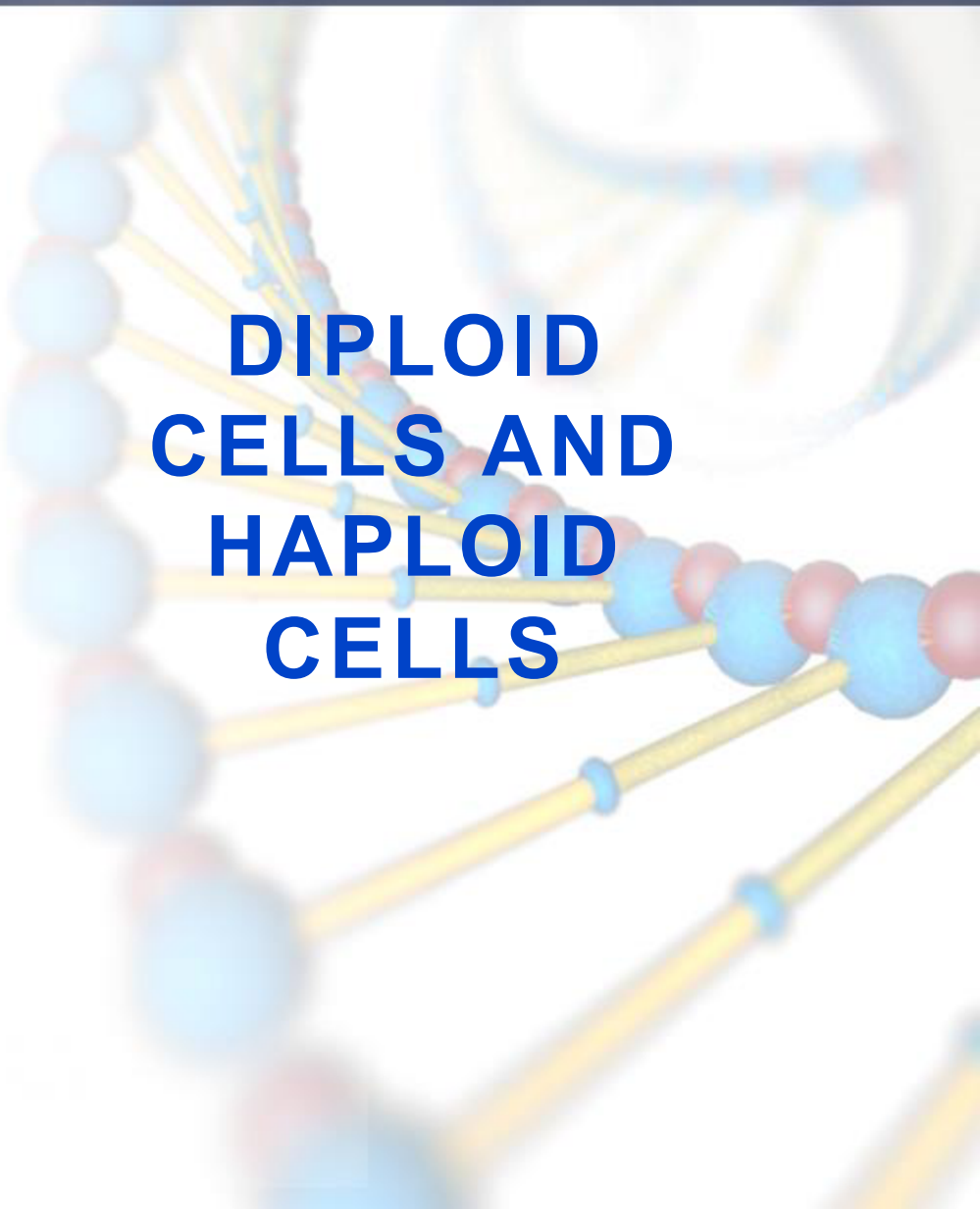
Mitotic Spindles

MITOTIC SPINDLES

- In anaphase, sister chromatids separate and move along the kinetochore microtubules toward opposite ends of the cell.

Essentials of Genetics

DIPLOID CELLS AND HAPLOID CELLS



Diploid Cells & Haploid Cells

DIPLOID CELLS

- Diploid cells are body cells.
- They are called as (2n) cells.
- Examples: skin cells, brain cells, bone cells, blood cells.
- In humans, ~ 220 types of somatic cells.

Diploid Cells & Haploid Cells

HAPLOID CELLS

- Sex cells produce gametes.
- Gametes are haploid (n) cells.
- Sperms and ova are gametes which are produced in males and females respectively.

Diploid Cells & Haploid Cells

MALE GAMETES AND FEMALE GAMETES

- The male gametes are the sperms and are produced in the male gonad - testes.
- The female gametes are the ova (ovum) and are produced in the female gonad - ovaries.

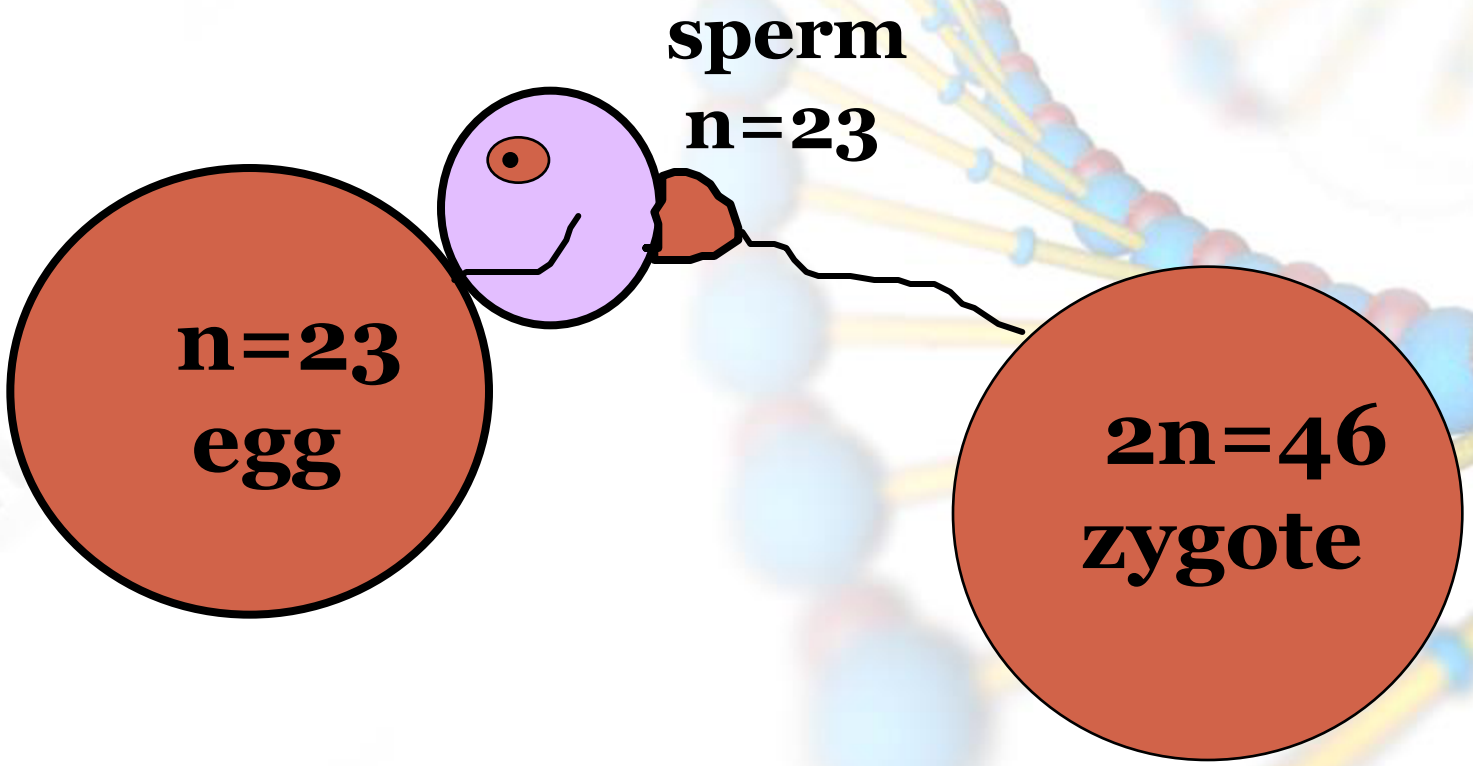
Diploid Cells & Haploid Cells

FERTILIZATION

- Fertilization is a process of fusion of spermatozoa and ova to produce a cell called zygote.

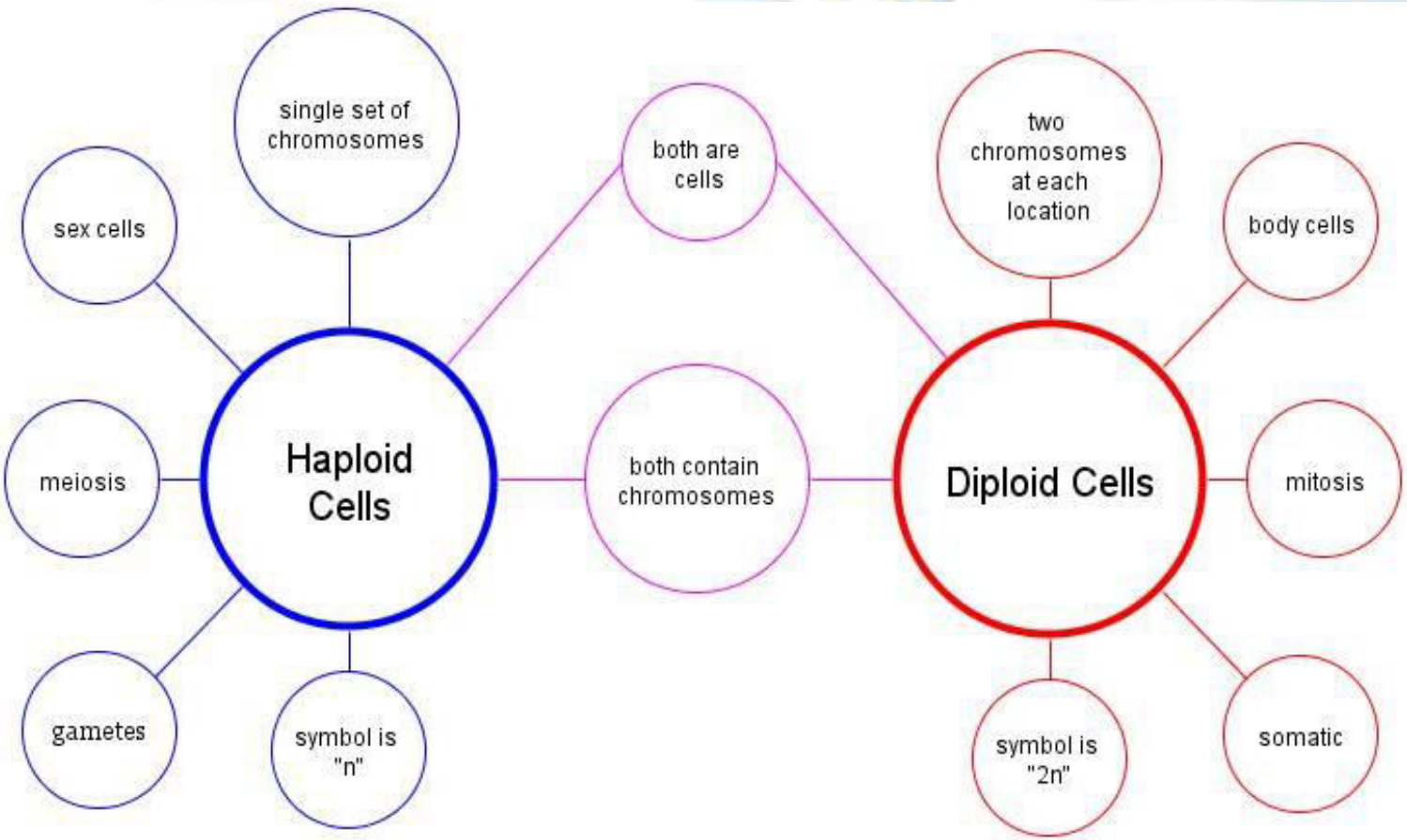
Diploid Cells & Haploid Cells

FERTILIZATION



Diploid Cells & Haploid Cells

DIPLOID - SOMATIC CELLS
HAPLOID - SEX CELLS



Diploid Cells & Haploid Cells

DIPLOID NUMBER IS REDUCED TO HAPLOID BY MEIOSIS

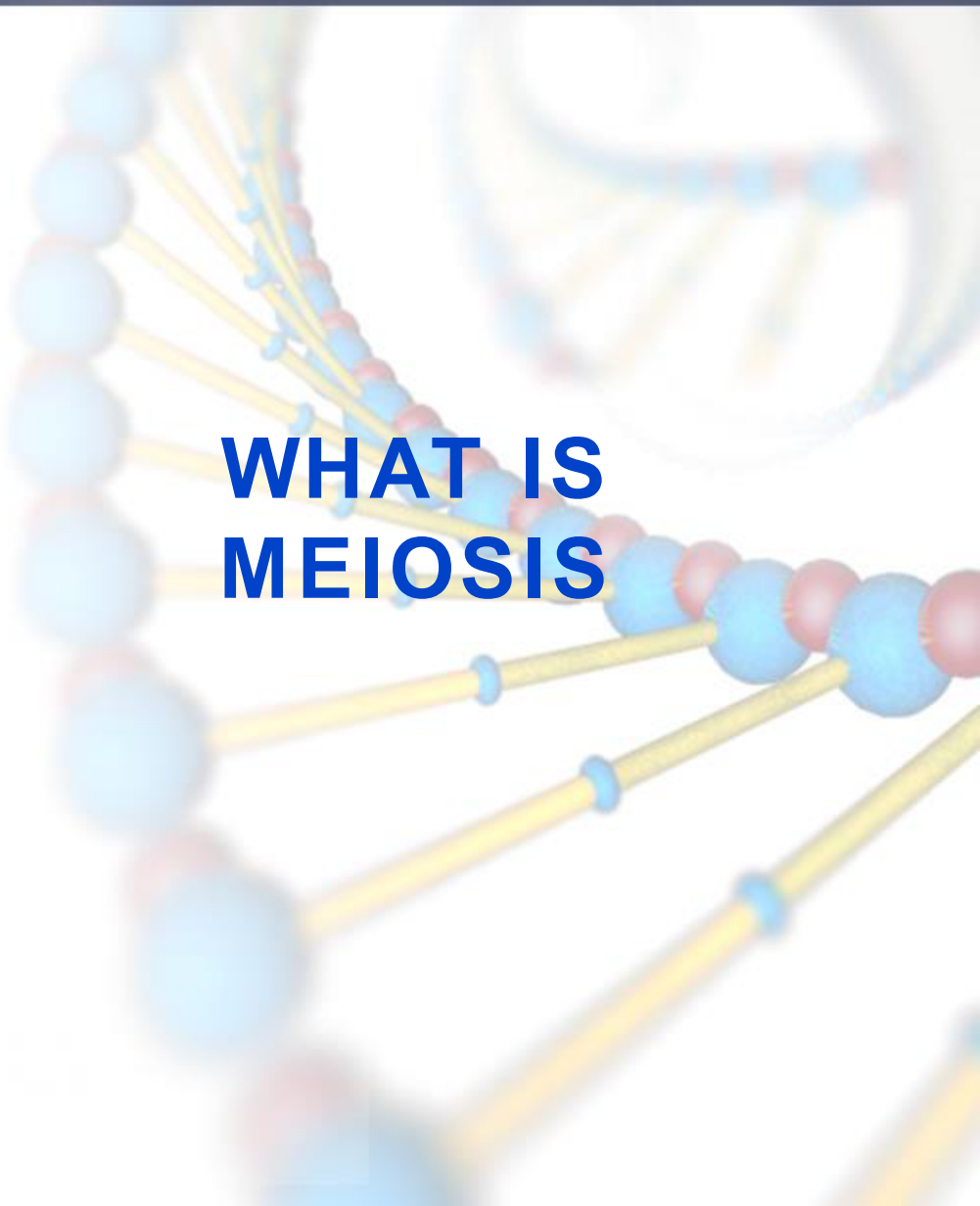
- Meiosis is a process that reduces diploid number into haploid.

Diploid Cells & Haploid Cells

DIPLOID NUMBER IS REDUCED TO HAPLOID BY MEIOSIS

- Meiosis is a process that reduces diploid number into haploid.

Essentials of Genetics



WHAT IS MEIOSIS

What is Meiosis

MEIOSIS

- Meiosis - diploid cells are reduced to haploid cells.
- Process by which gametes (sex cells) are produced.

What is Meiosis

CHROMOSOME NUMBERS WOULD BE DOUBLED WITHOUT MEIOSIS

- If meiosis did not occur, the chromosomes number in each new generation would be doubled.
- The offsprings would die.

What is Meiosis

MEIOSIS

- Meiosis is a two cell divisions process.

  Meiosis I

  Meiosis II

- There is only one duplication of chromosomes during meiosis I.

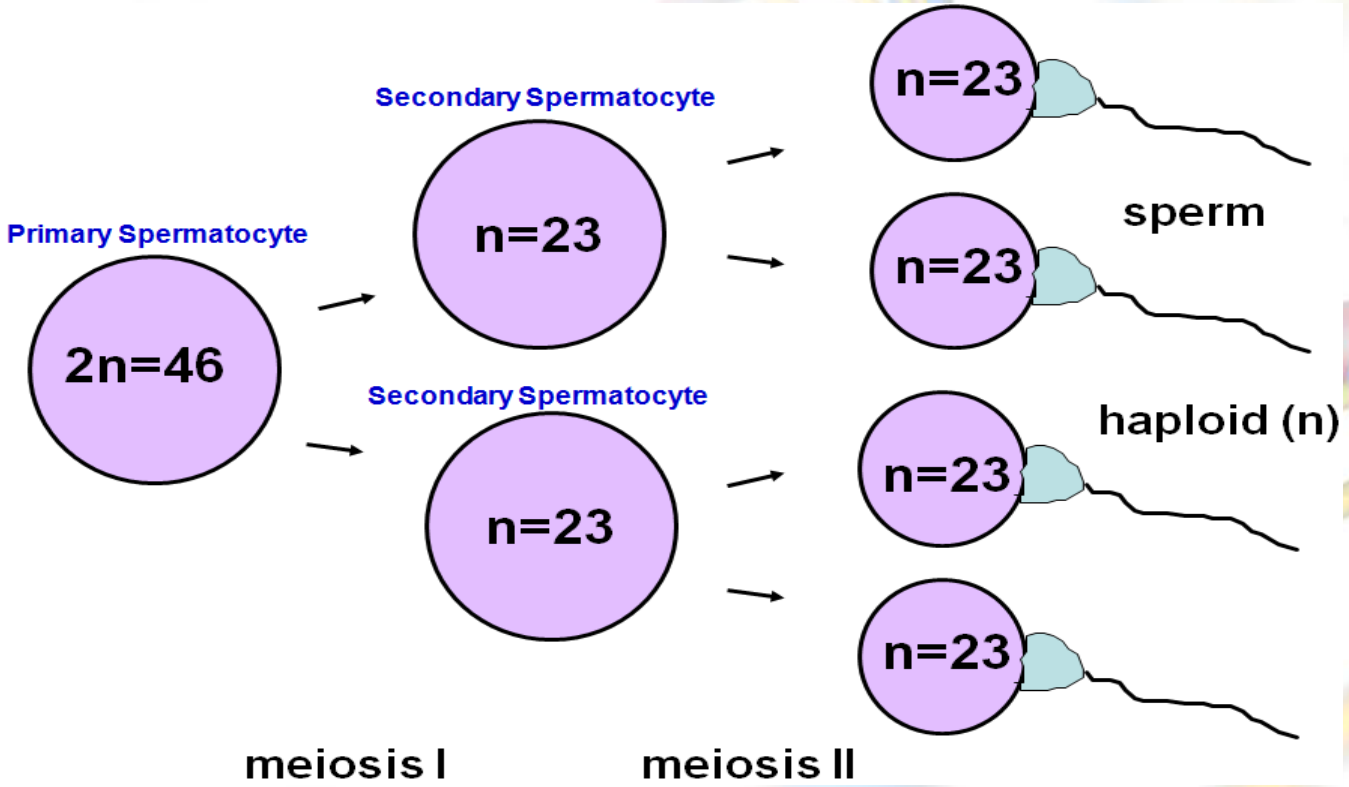
What is Meiosis

SPERMATOGENESIS AND OOGENESIS

- Meiosis in males is called spermatogenesis and produces sperms.
- Meiosis in females is called oogenesis and produces ova.

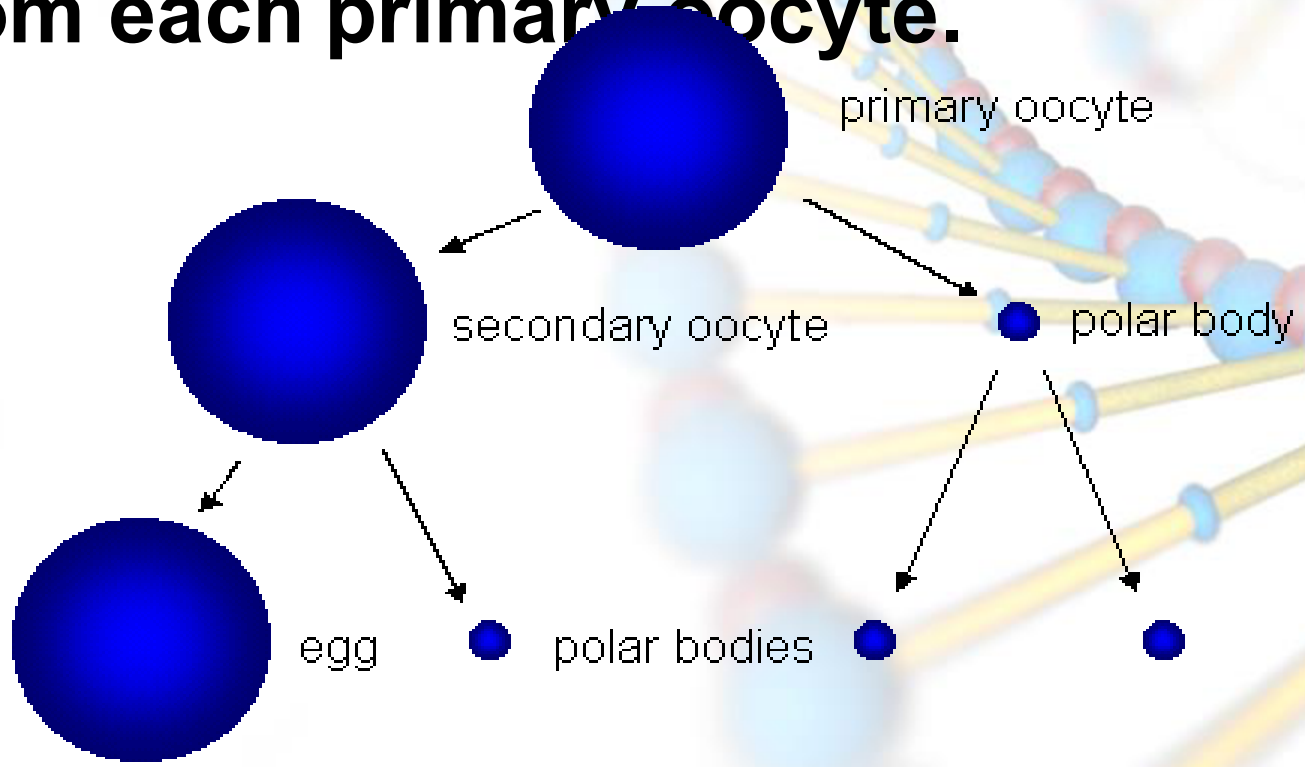
What is Meiosis

SPERMATOGENESIS: Four sperm cells are produced from primary sp



What is Meiosis

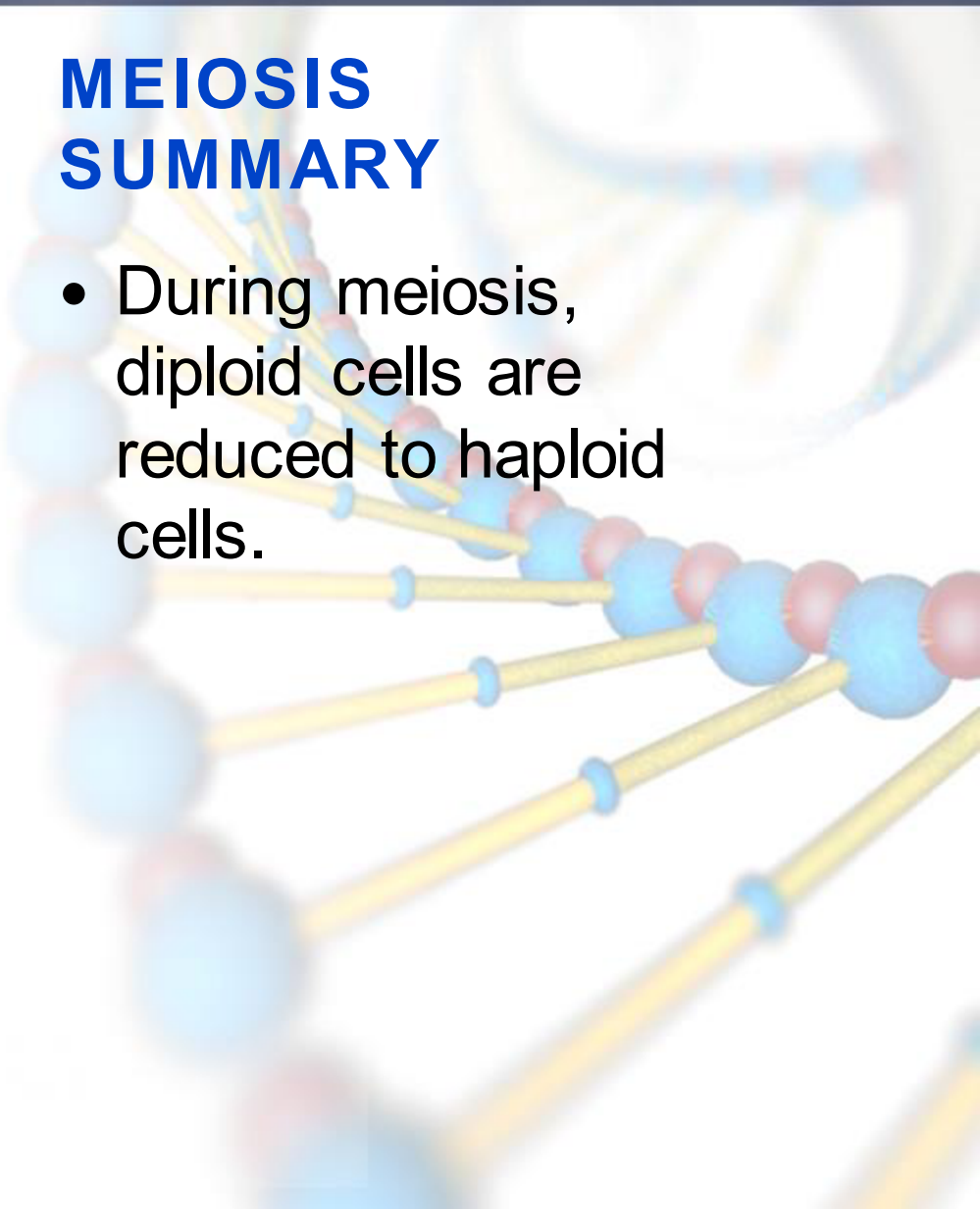
OOGENESIS: The polar bodies die. Only one ovum (egg) is produced from each primary oocyte.



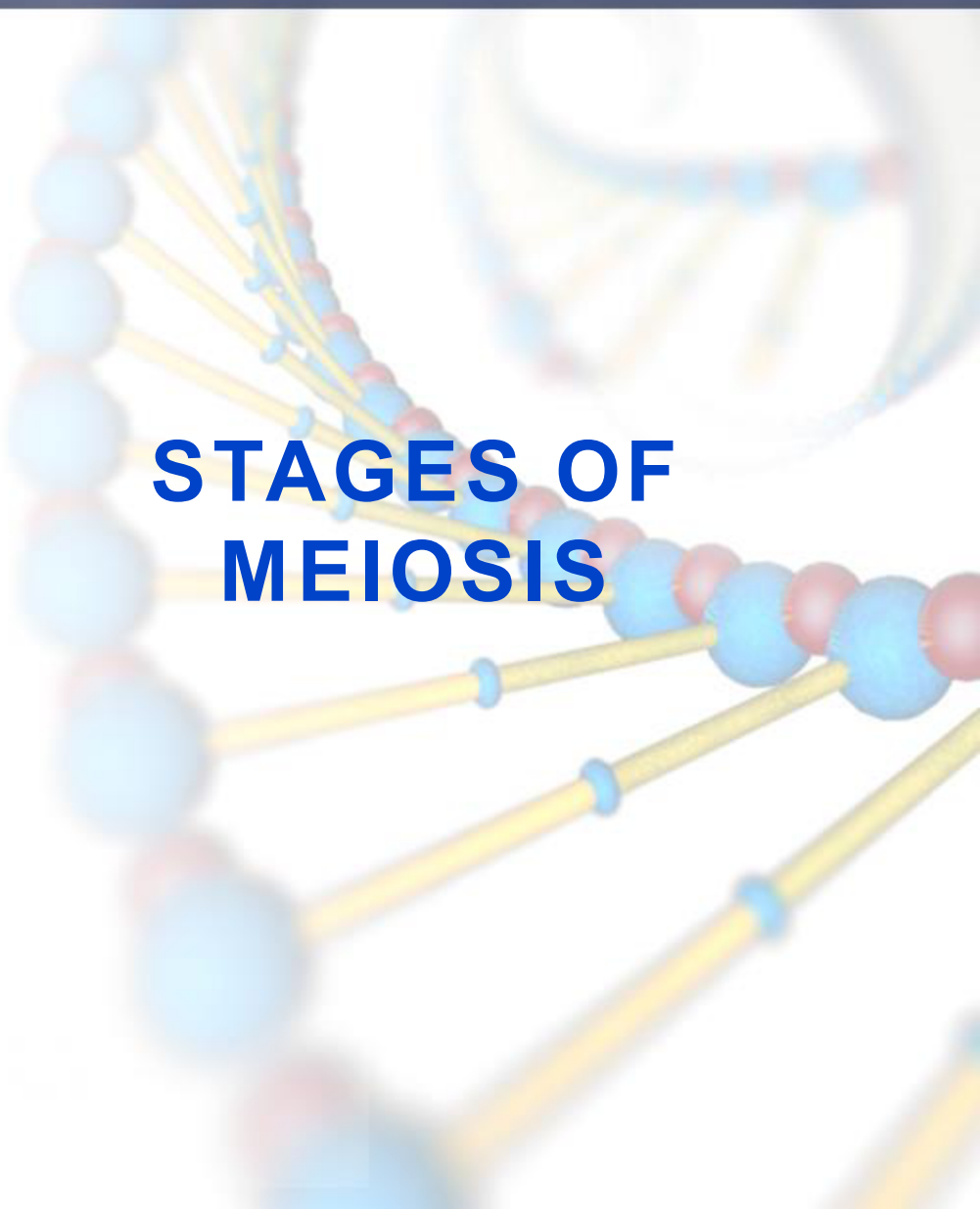
What is Meiosis

MEIOSIS SUMMARY

- During meiosis, diploid cells are reduced to haploid cells.



Essentials of Genetics



STAGES OF MEIOSIS

Stages of Meiosis

MEIOSIS

- Two stages of meiosis
  - Meiosis I
  - Meiosis II
- During meiosis I, homologous chromosomes separate.
- During meiosis II, sister chromatids separate.

Stages of Meiosis

MEIOSIS I – REDUCTIONAL DIVISION

- Meiosis I results in two haploid daughter cells.
- It is called the reductional division.
- Homologous chromosomes separate.

Stages of Meiosis

MEIOSIS II

- Second cell division where sister chromatids separate.
- As a result of meiosis II, two haploid cells becomes four haploid cells.
- Meiosis II behaves like mitosis, where no reduction in chromosomes number.

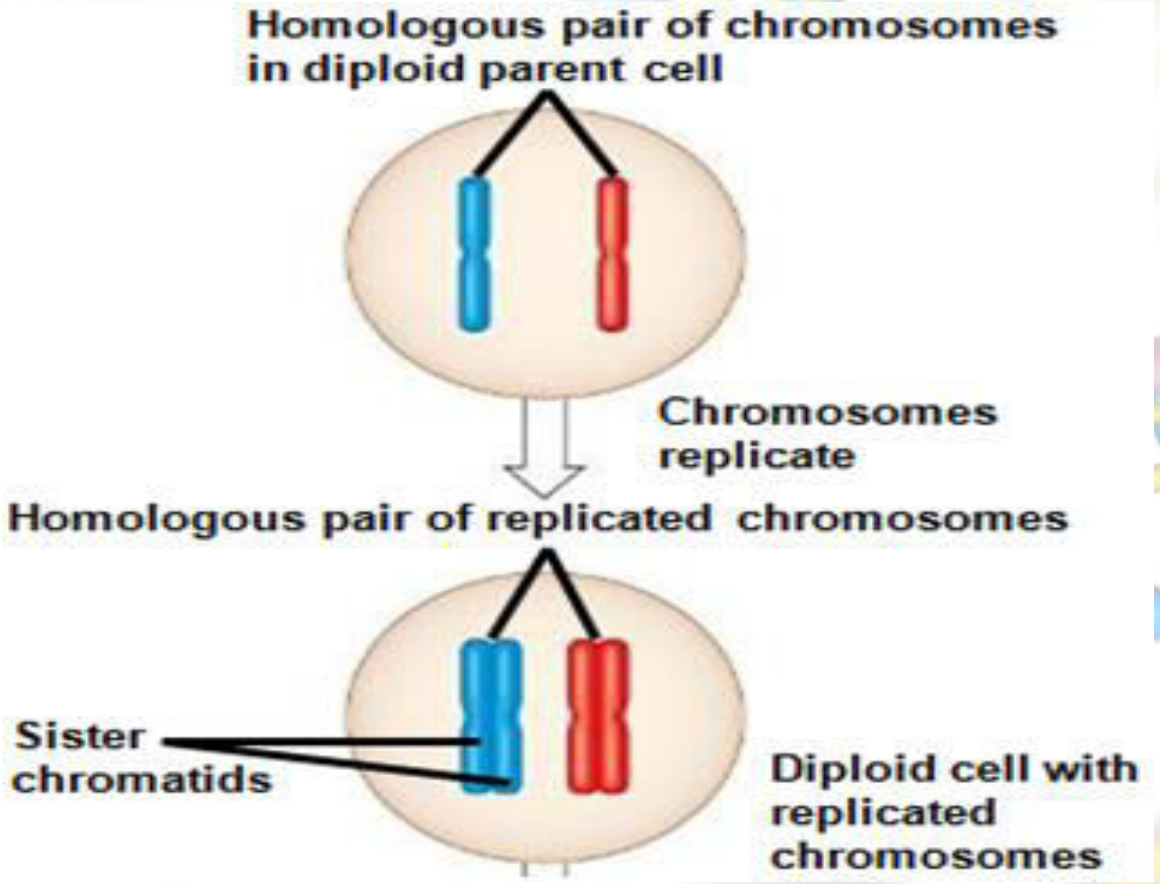
Stages of Meiosis

MEIOSIS II – EQUATIONAL DIVISION

- Meiosis II results in four haploid daughter cells with unreplicated chromosomes; it is called equational division.

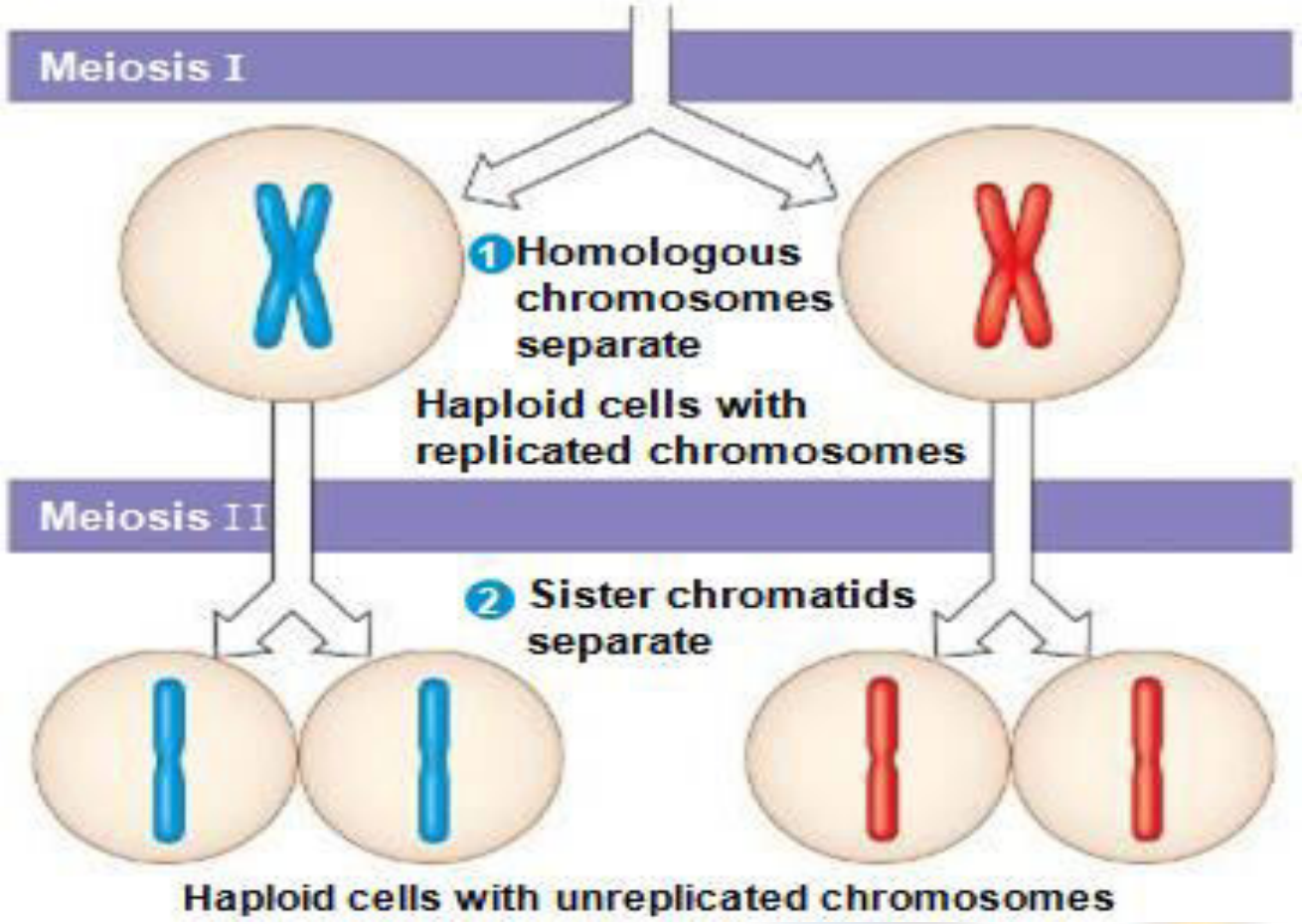
Stages of Meiosis

INTERPHASE CHROMOSOMES REPLICATES



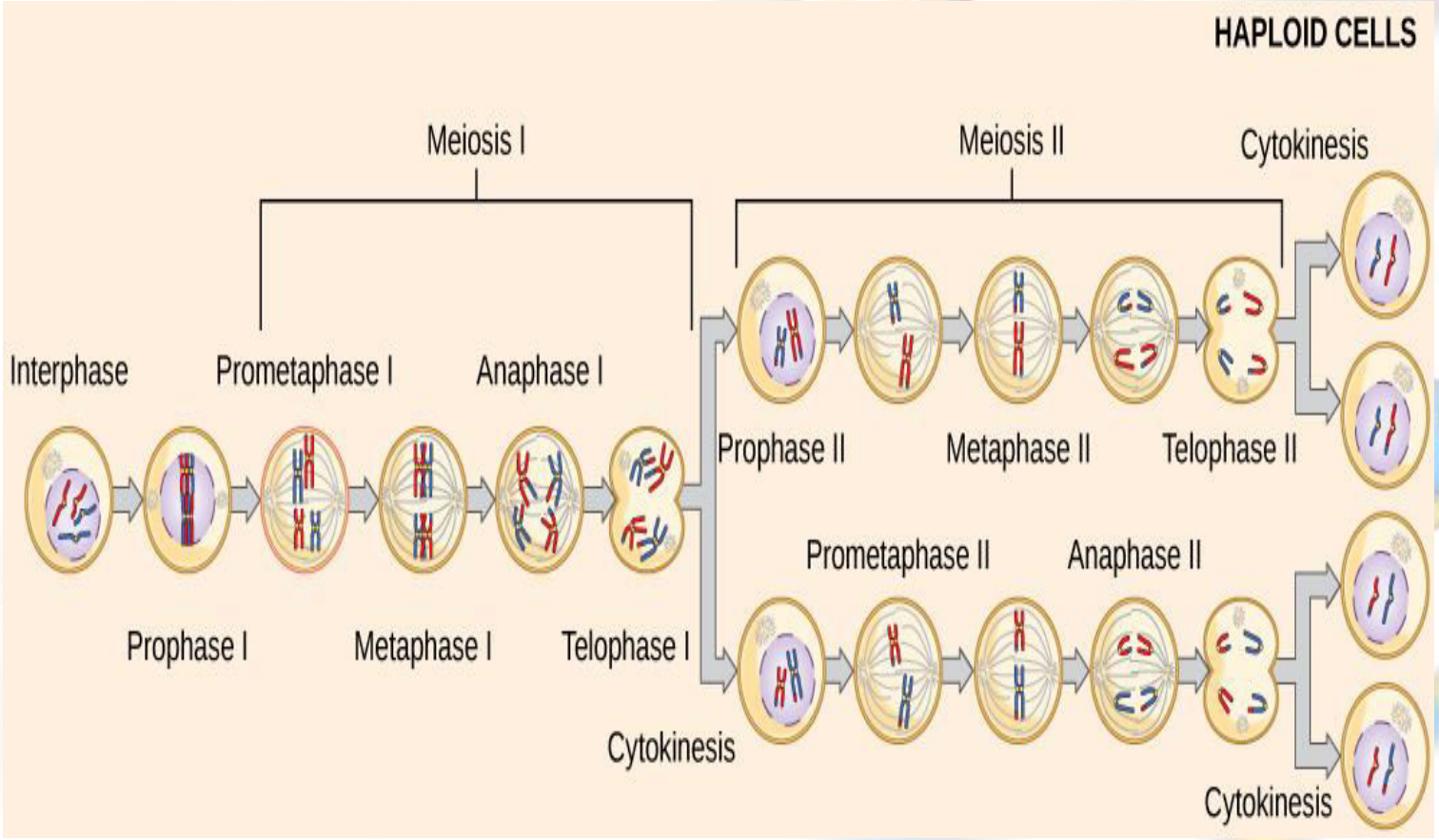
Stages of Meiosis

MEIOSIS I, MEIOSIS II



Stages of Meiosis

MEIOSIS - FOUR HAPLOID CELLS

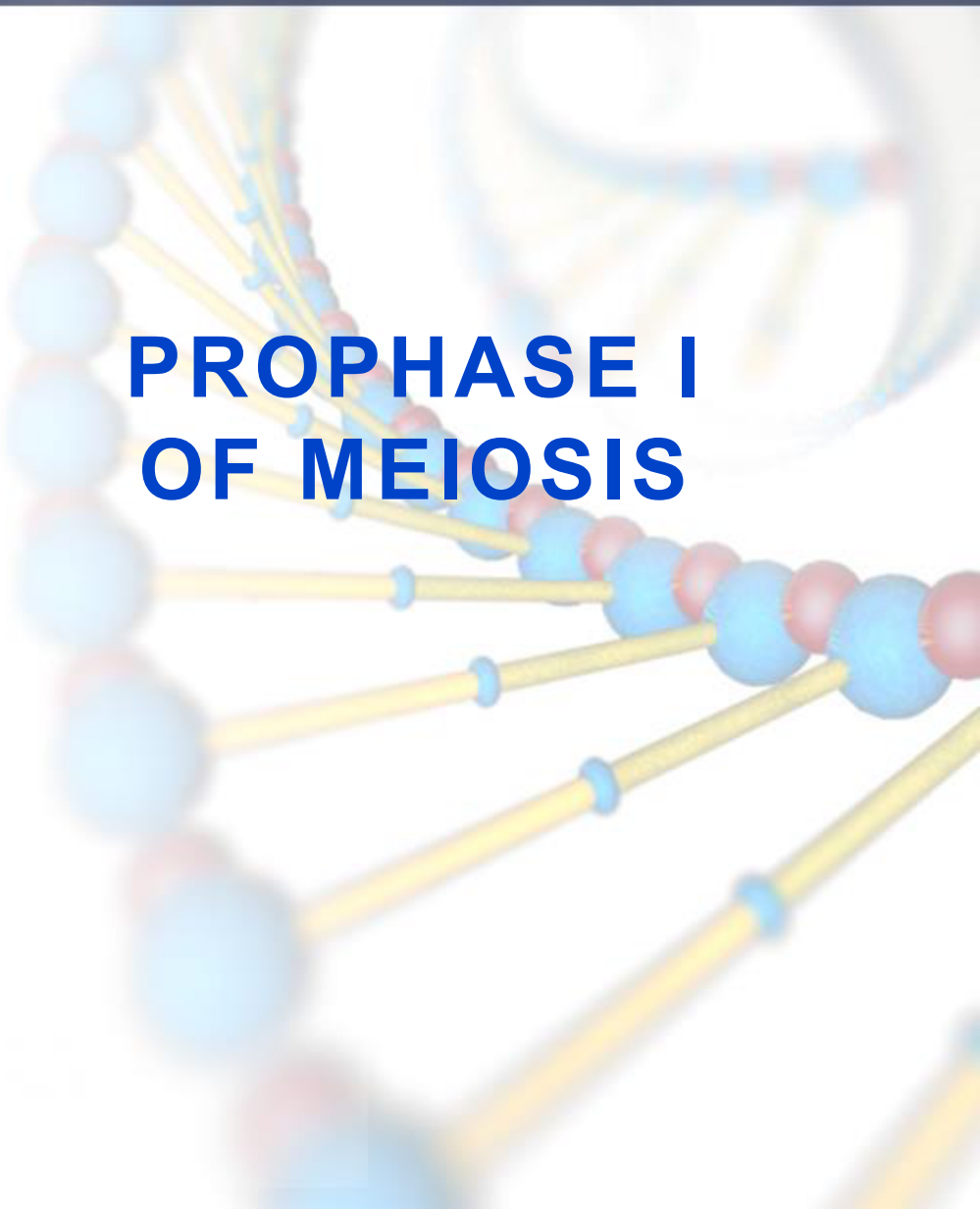


Stages of Meiosis

MEIOSIS RESULTS IN FOUR HAPLOID CELLS

- Meiosis results in four haploid cells.

Essentials of Genetics



PROPHASE I OF MEIOSIS

Prophase I of Meiosis

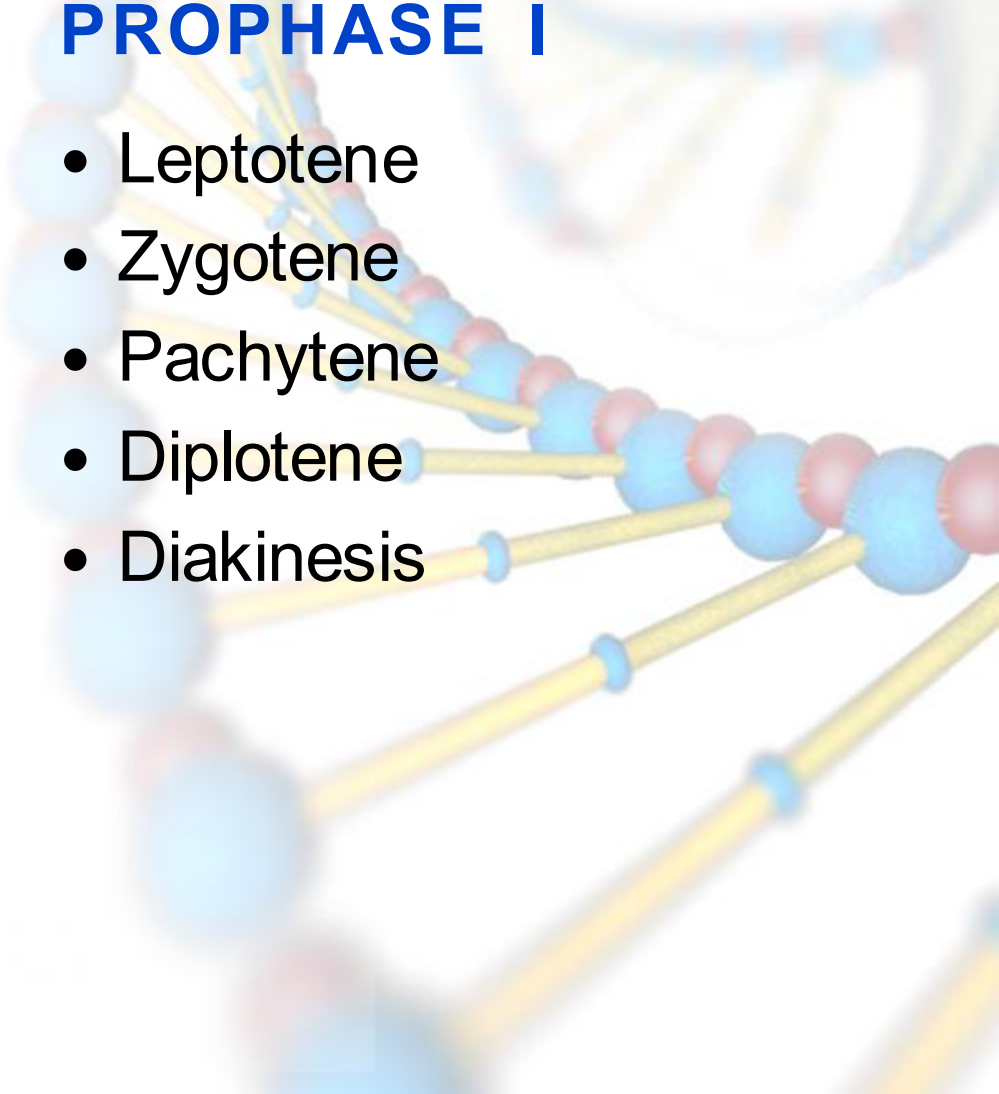
PROPHASE I

- Prophase I, occupies more than 90% of the time required for meiosis.
- Five sub stages of prophase I

Prophase I of Meiosis

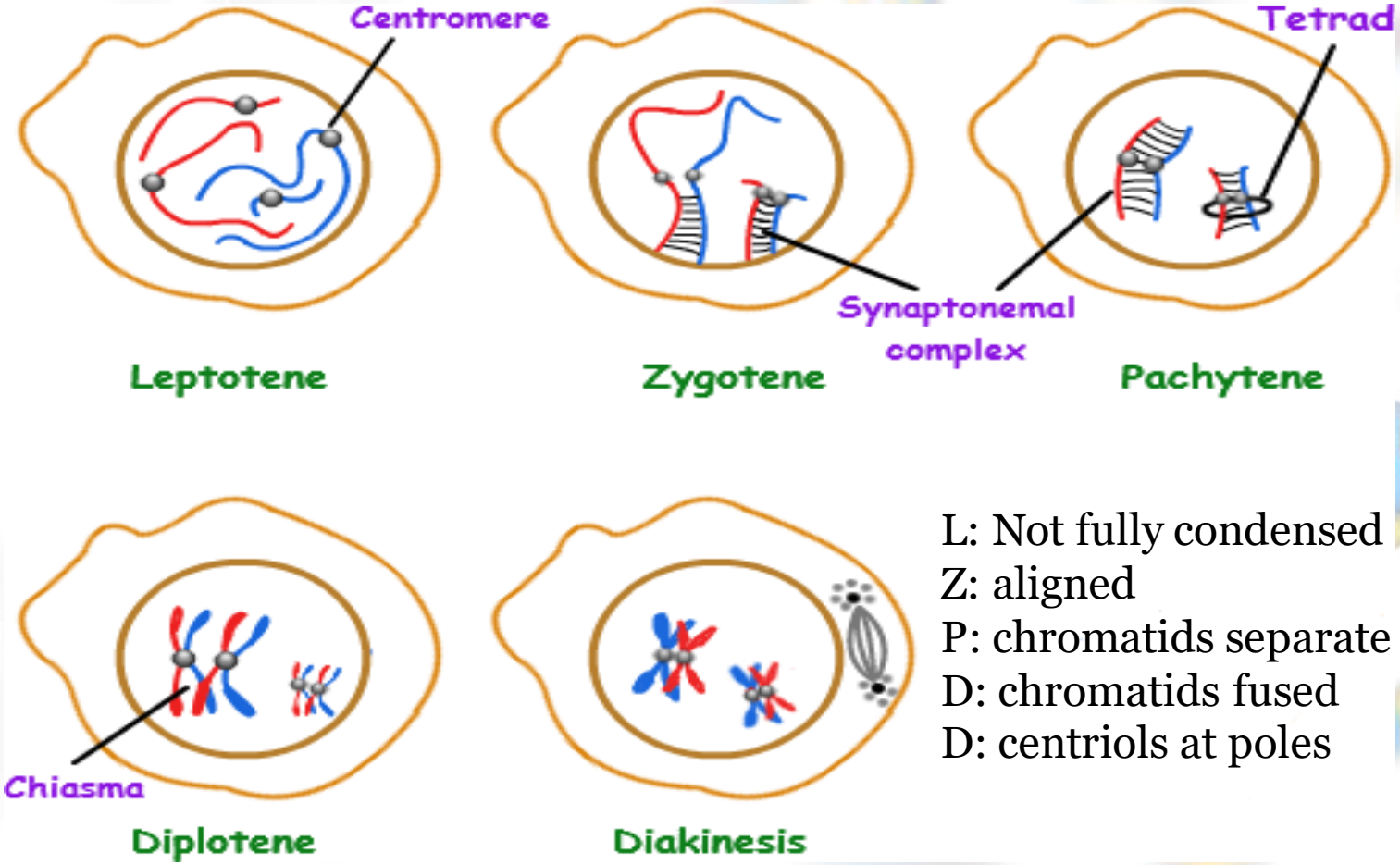
SUB STAGES OF PROPHASE I

- Leptotene
- Zygotene
- Pachytene
- Diplotene
- Diakinesis



Prophase I of Meiosis

SUB STAGES OF PROPHASE I



L: Not fully condensed
Z: aligned
P: chromatids separate
D: chromatids fused
D: centriols at poles

Prophase I of Meiosis

EVENTS OF PROPHASE I

- The duplicated homologous chromosomes pairs.
- The nucleolus disappears.
- Centrioles migrate to opposite poles of the cell.
- Meiotic spindle forms between the two pairs of centrioles.

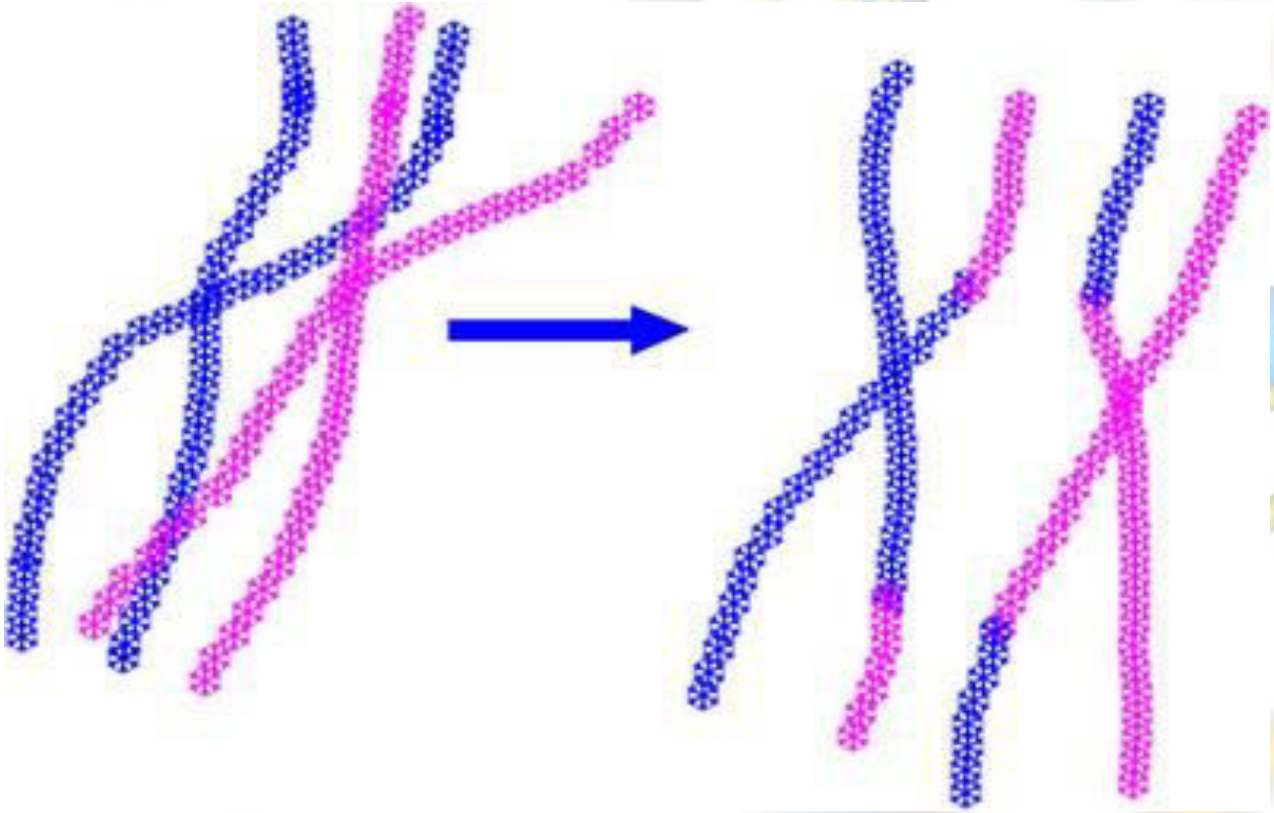
Prophase I of Meiosis

CHIASMATA FORMATION DURING PROPHASE I

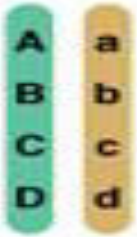
- During pairing up of chromosomes, nonsister chromatids exchange DNA segments.
- Each pair of chromosomes forms a tetrad, a group of four chromatids.

Prophase I of Meiosis

PAIRING AND EXCHANGE OF CHROMOSOME FRAGMENTS



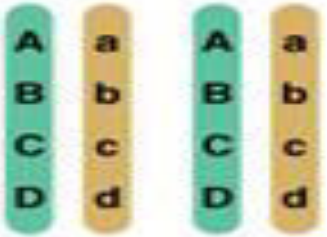
Prophase I of Meiosis



Homologous Chromosomes



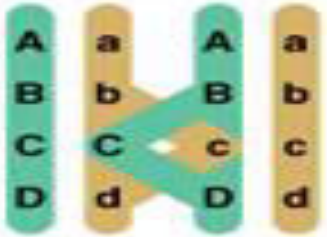
Chromatids



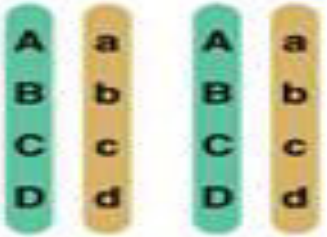
No cross-over



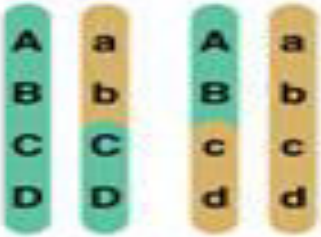
Cross-over



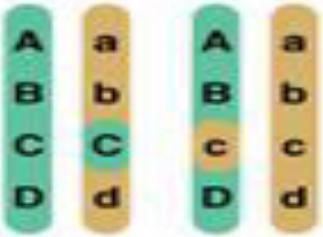
Double cross-over



No recombination in gametes



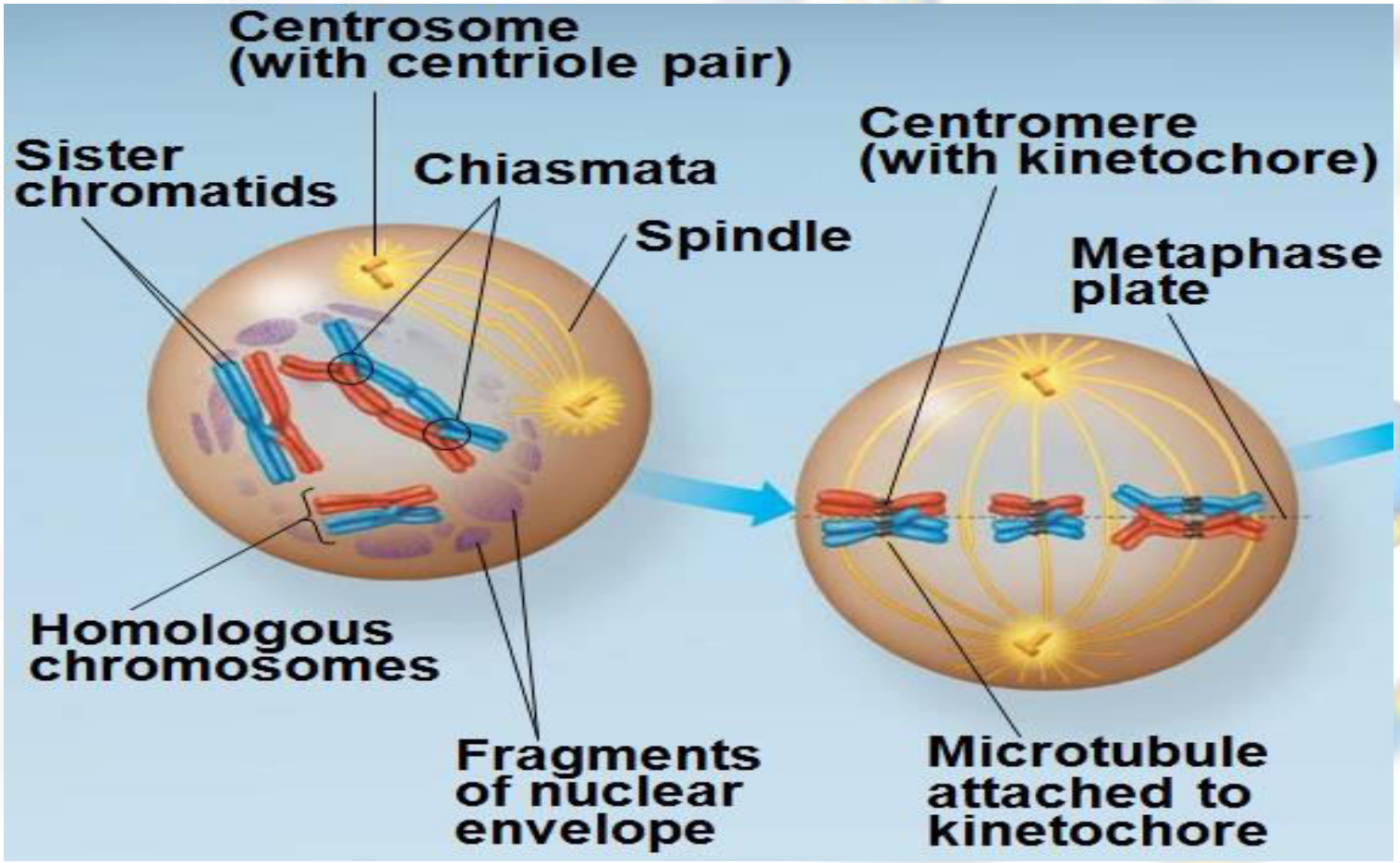
Recombination in gametes



Recombination in gametes

Prophase I of Meiosis

PROPHASE I – CHIASMATA



Prophase I of Meiosis

PROPHASE I SUMMARY

- Homologous chromosomes pairs.
- Nucleolus disappears.
- Meiotic spindle forms.
- Nuclear membrane disappears.

Essentials of Genetics

METAPHASE I OF MEIOSIS

Metaphase I of Meiosis

METAPHASE I

- In metaphase I, tetrads line up at the metaphase plate, with one chromosome facing each pole.

Metaphase I of Meiosis

MICROTUBULES ARE ATTACHED TO KINETOCHORE OF ONE CHROMOSOME

- Microtubules from one pole are attached to kinetochore of one chromosome of each tetrad.

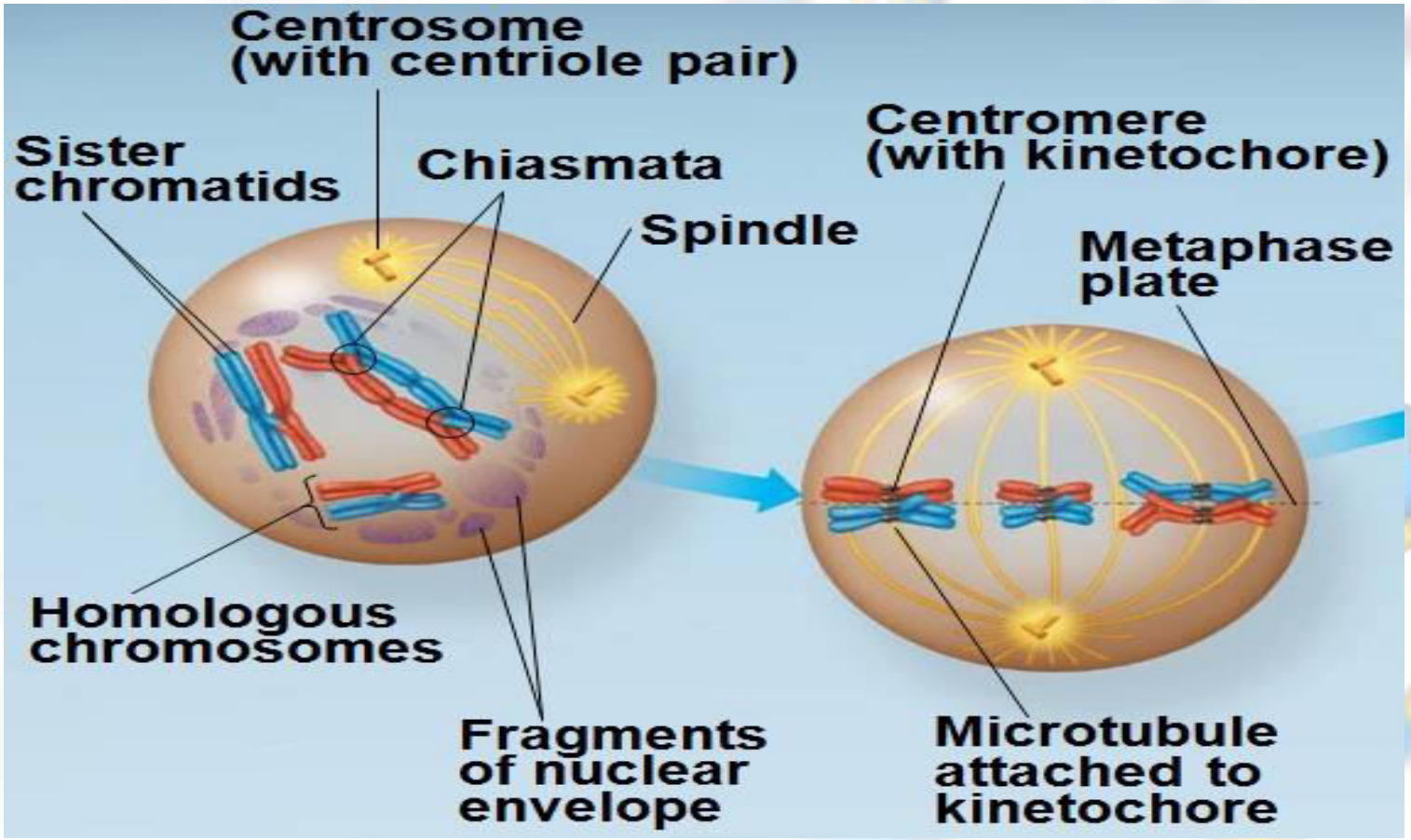
Metaphase I of Meiosis

MICROTUBULES ARE ATTACHED TO KINETOCHORE OF OTHER CHROMOSOME

- Microtubules from the other pole are attached to the kinetochore of the other chromosome.

Metaphase I of Meiosis

METAPHASE I



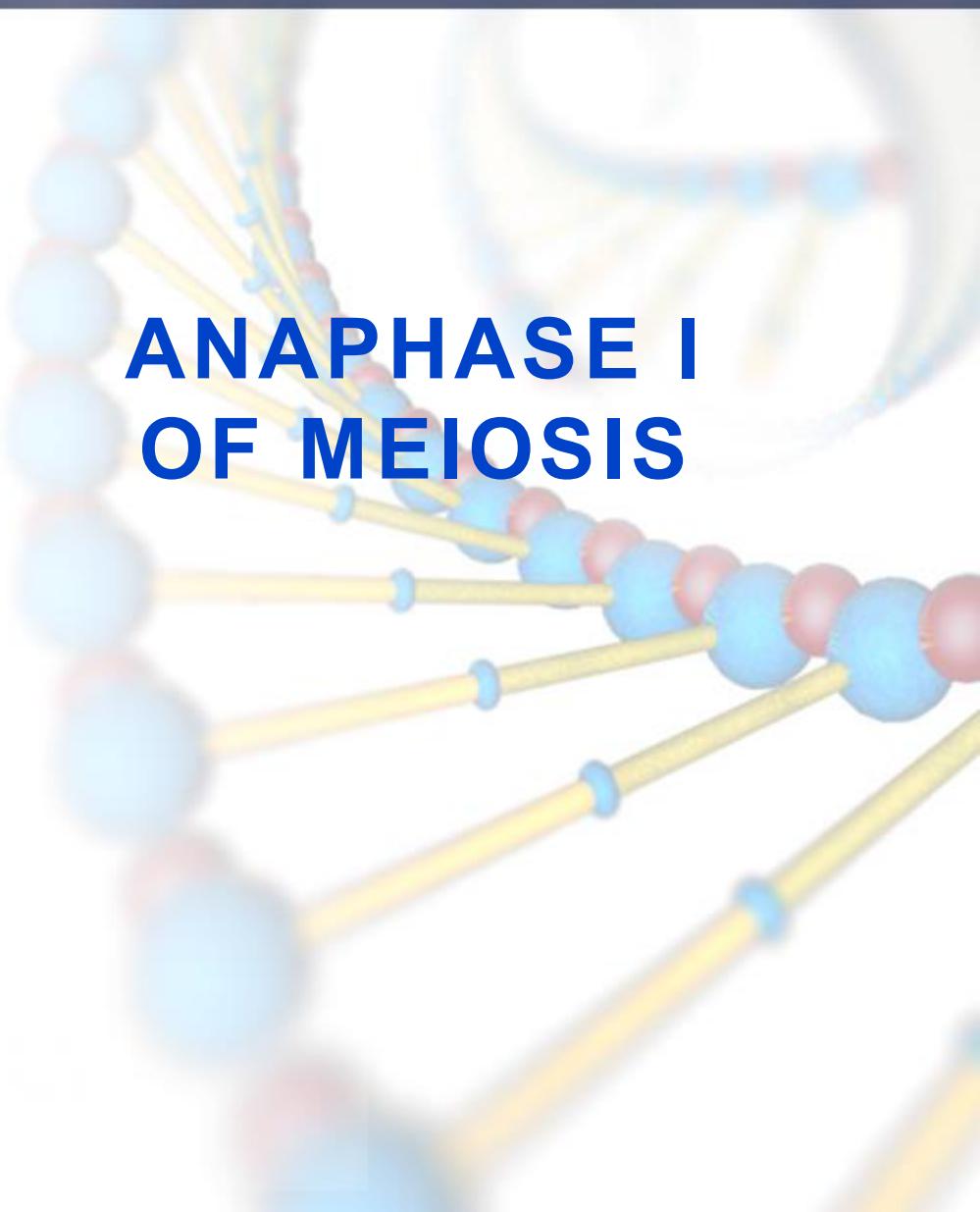
Metaphase I of Meiosis

METAPHASE I

- Chromosomes in the middle forming metaphase plate and spindle fibers are attached.

Essentials of Genetics

ANAPHASE I OF MEIOSIS



Anaphase I of Meiosis

ANAPHASE I

- Pairs of homologous chromosomes separate.
- One chromosome moves toward each pole, guided by the spindle apparatus.

Anaphase I of Meiosis

ANAPHASE I – SISTER CHROMATIDS ATTACHED AT CENTROMERE

- Sister chromatids remain attached at the centromere of each of the homologue chromosome and move as one unit toward the pole.

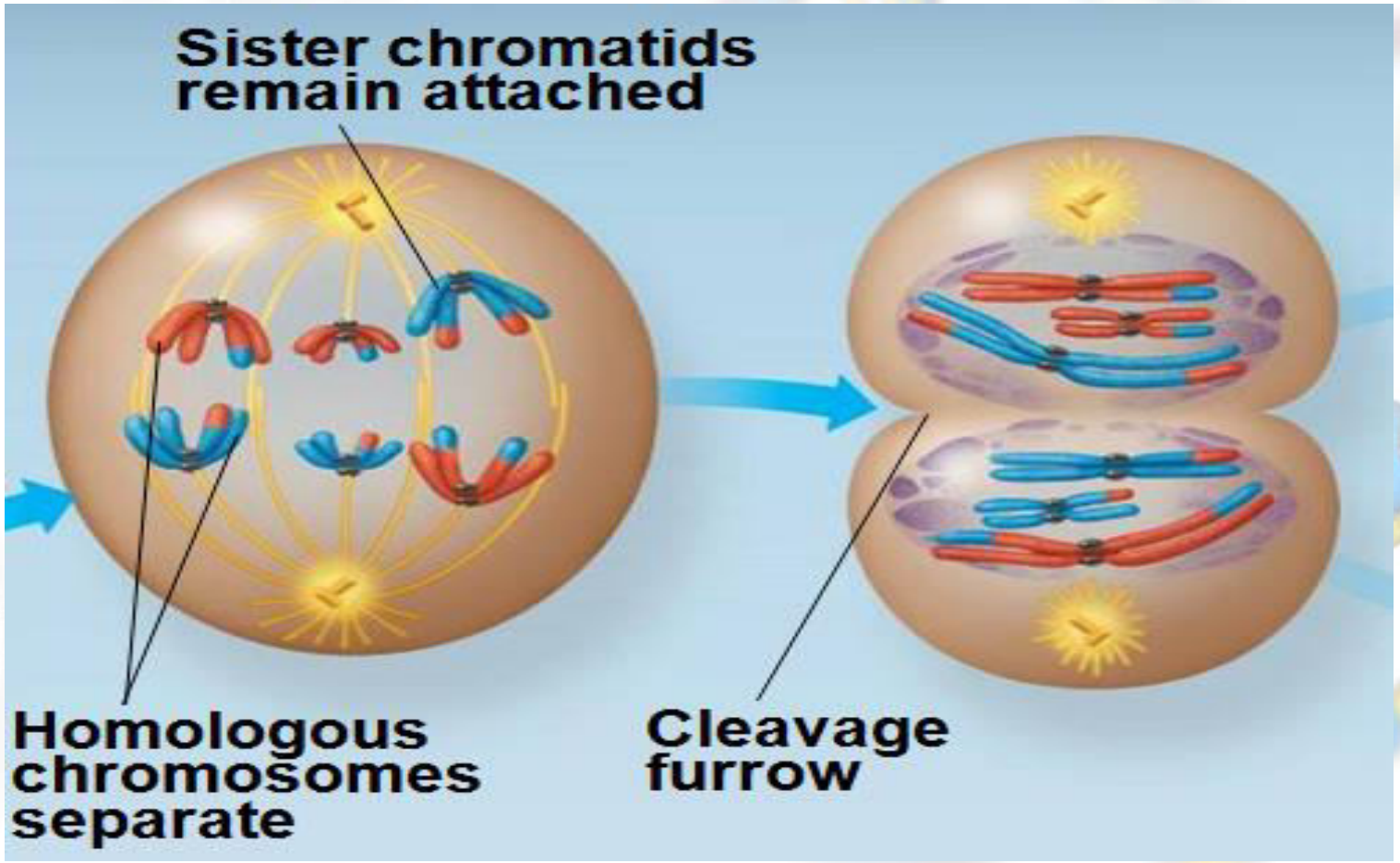
Anaphase I of Meiosis

DIFFERENCE BETWEEN ANAPHASE OF MITOSIS AND MEIOSIS I

- Sister chromatids remained joined in meiosis I, whereas in mitosis they separate.
- In mitosis, centromere splits into two parts while in meiosis it do not

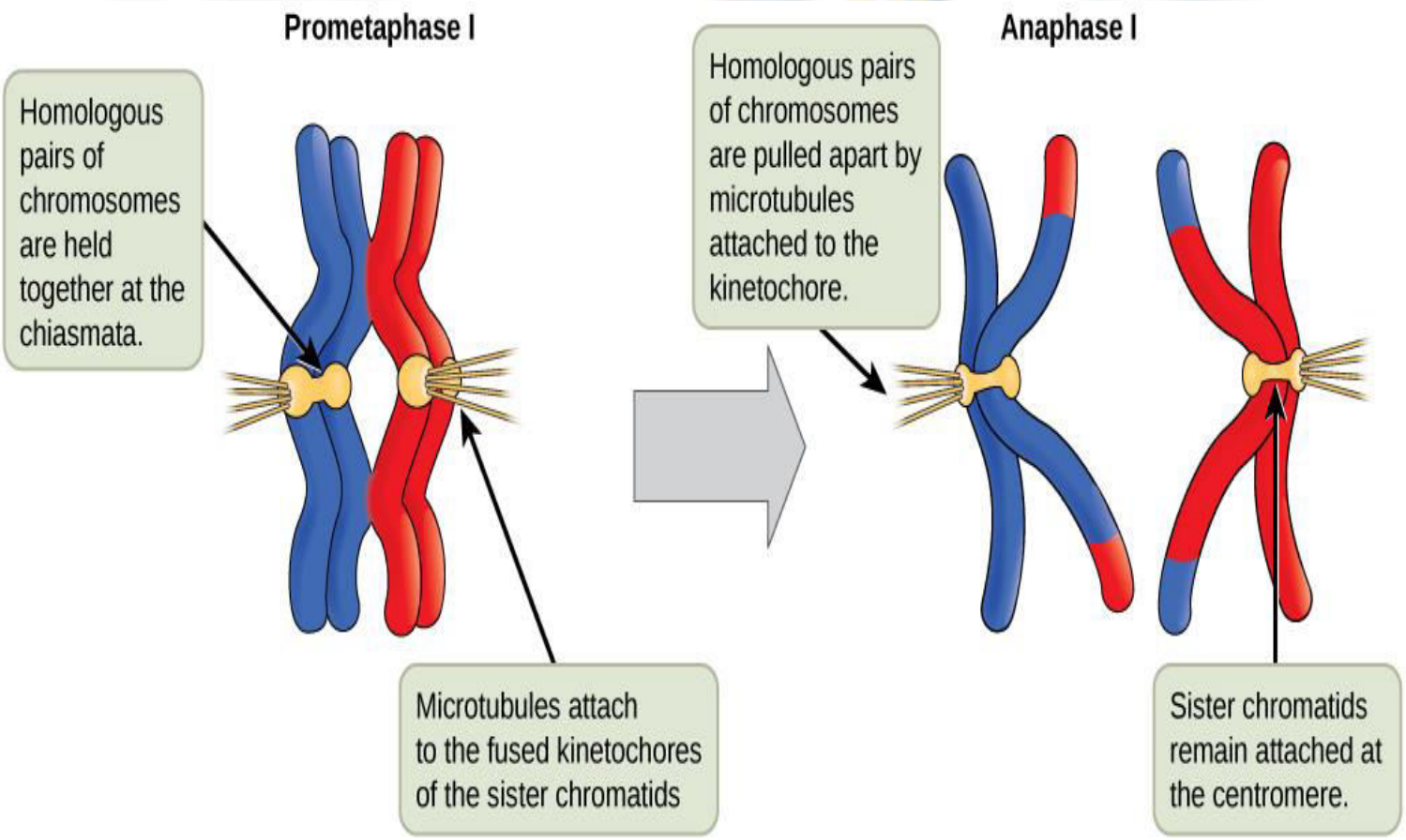
Anaphase I of Meiosis

ANAPHASE I



Anaphase I of Meiosis

ANAPHASE I



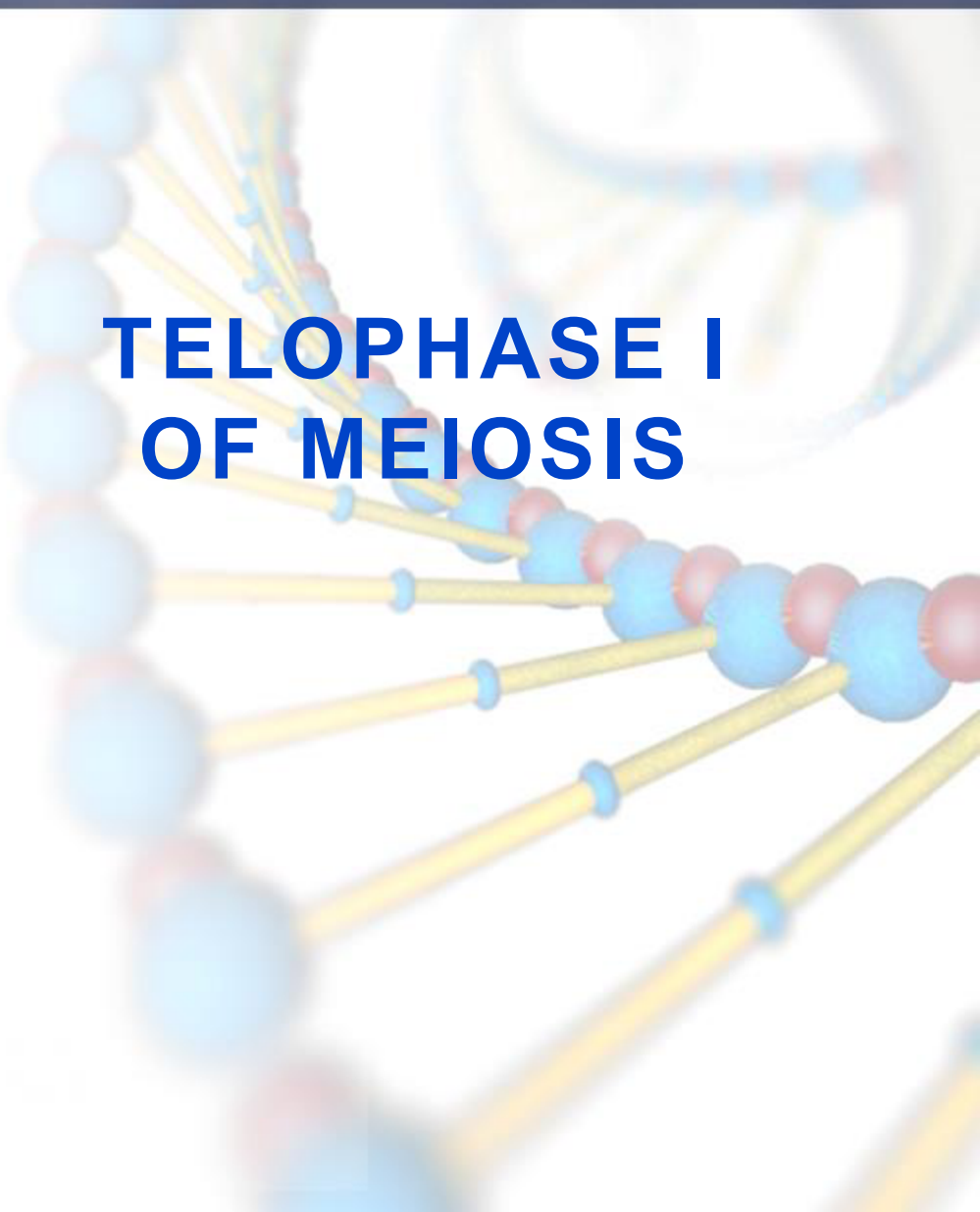
Anaphase I of Meiosis

ANAPHASE I SUMMARY

- Pairs of homologous chromosomes separate.
- One chromosome moves toward one pole while other chromosome moves to other pole of the cell.

Essentials of Genetics

TELOPHASE I OF MEIOSIS



Telophase I of Meiosis

TELOPHASE I

- Haploid set of chromosomes arrived at each of the pole.
- Each chromosome still consists of two sister chromatids.

Telophase I of Meiosis

CYTOKINESIS- TWO HAPLOID CELLS

- Cytokinesis usually occurs simultaneously, forming two haploid daughter cells.

Telophase I of Meiosis

CYTOKINESIS – ANIMAL CELL AND PLANTS CELL

- In animal cells, a cleavage furrow is formed.
- In plant cells, a cell plate formed.

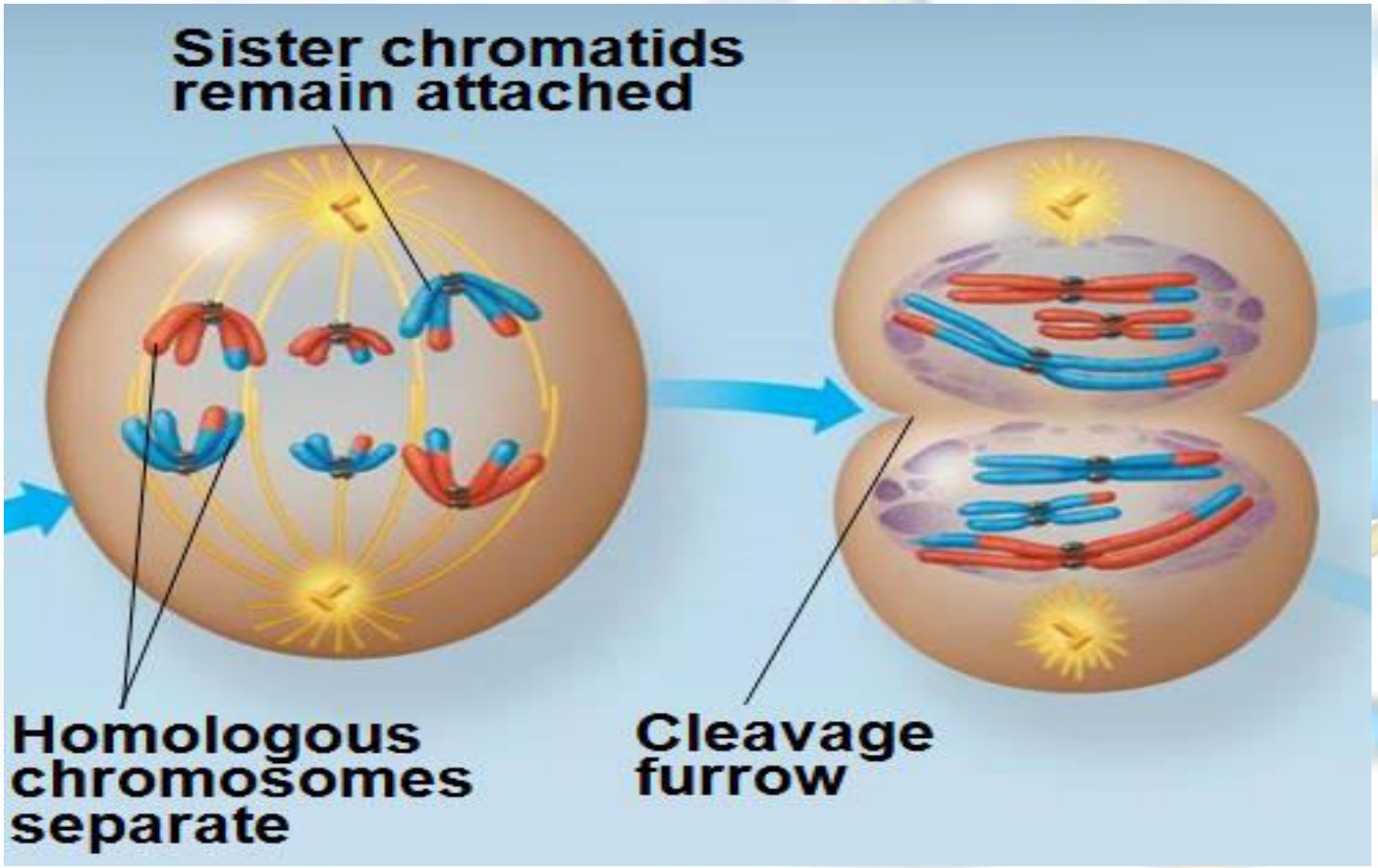
Telophase I of Meiosis

NO REPLICATION BETWEEN MEIOSIS I AND MEIOSIS II

- There is no chromosome replication between the end of meiosis I and beginning of meiosis II.
- Chromosomes are already replicated.

Telophase I of Meiosis

TELOPHASE I

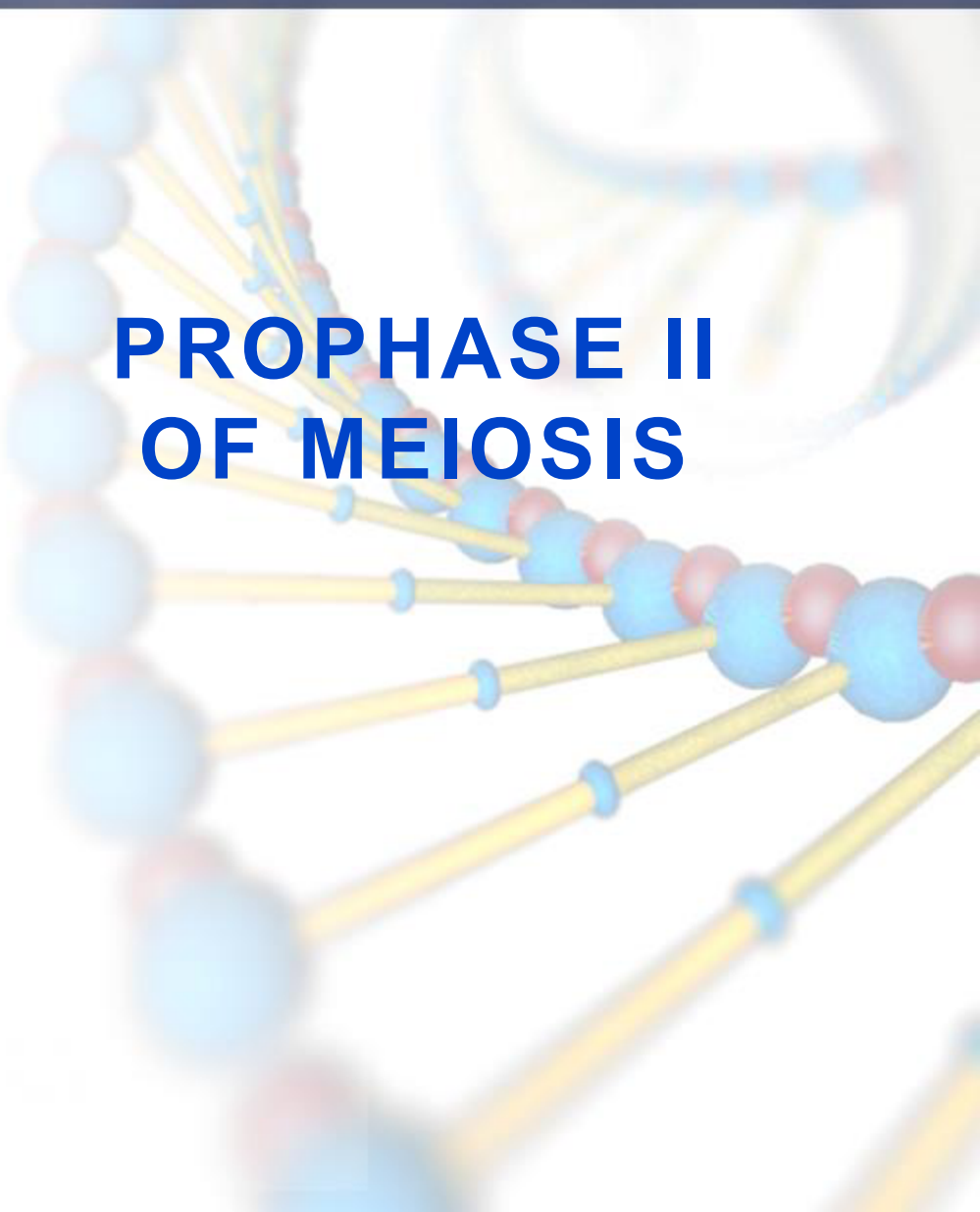


Telophase I of Meiosis

TELOPHASE I

- Haploid set of chromosomes arrived at each of the pole.
- Meiosis I results, a diploid cell produced two haploid cells.

Essentials of Genetics



PROPHASE II OF MEIOSIS

Prophase II of Meiosis

MEIOSIS II SIMILAR TO MITOSIS

- Meiosis II also occurs in four phases:
- Prophase II
- Metaphase II
- Anaphase II
- Telophase II and cytokinesis

Prophase II of Meiosis

PROPHASE II

- No replication of Chromosomes as they are already replicated.
- Centrioles are separated and take positions on the opposite poles of the cell.

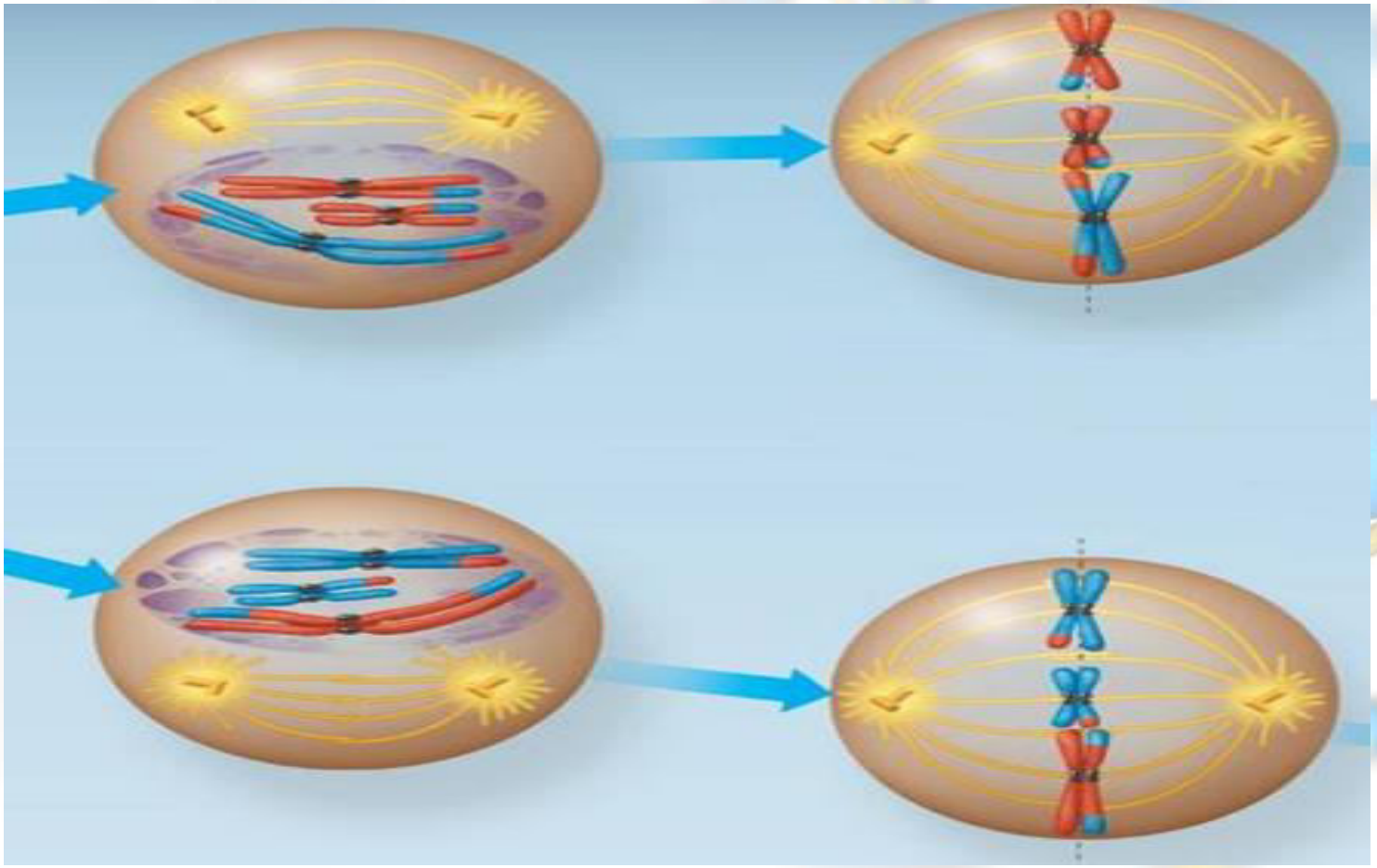
Prophase II of Meiosis

PROPHASE II

- Nuclear membrane disappears.
- Spindle fibers form and radiate toward the center of the cell.

Prophase II of Meiosis

PROPHASE II

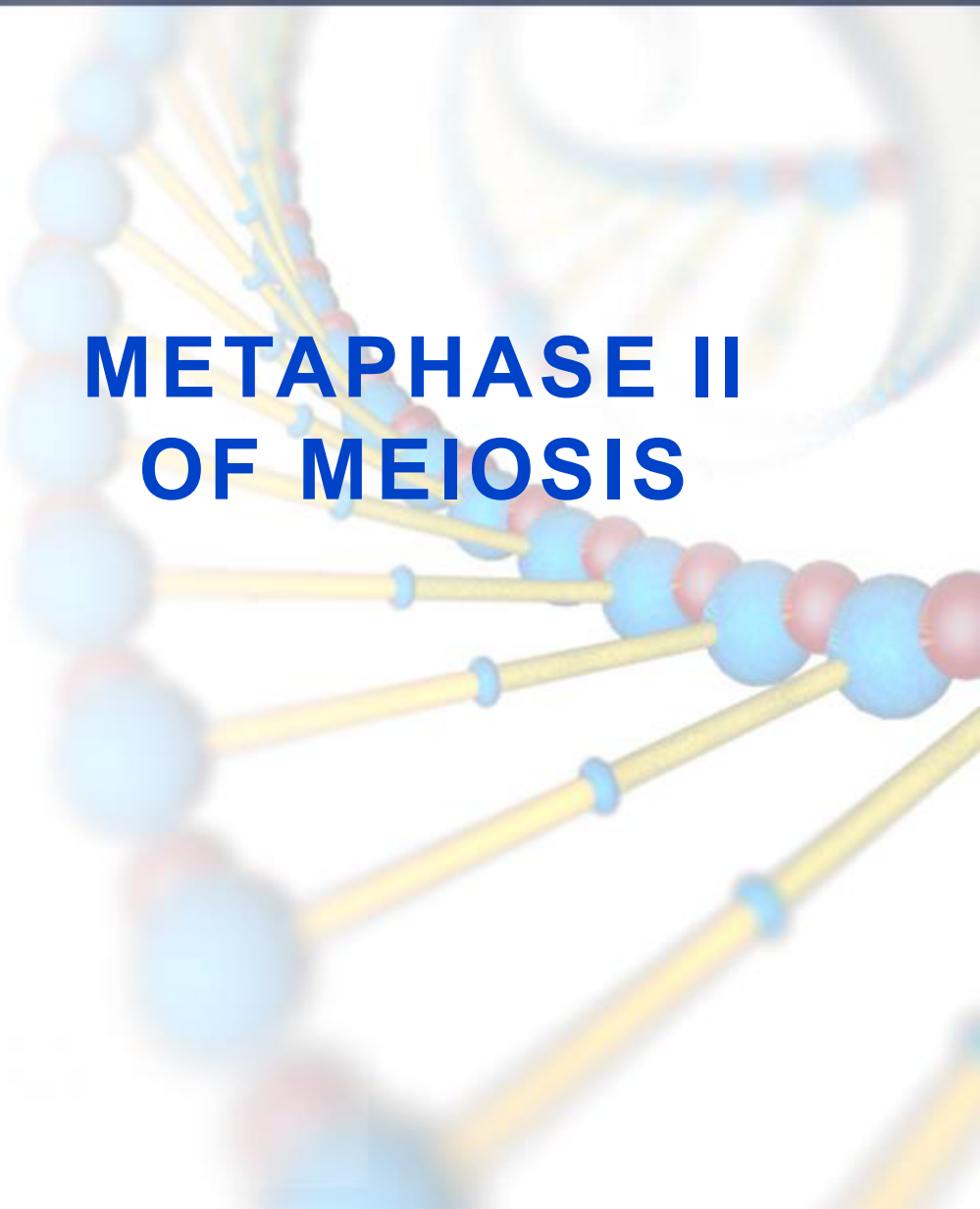


Prophase II of Meiosis

PROPHASE II SUMMARY

- Nuclear membrane disappears.
- Spindle fibers form and radiate toward the center of the cell.
- Centrioles take the position on poles.

Essentials of Genetics



METAPHASE II OF MEIOSIS

Metaphase II of Meiosis

METAPHASE II

- In metaphase II, sister chromatids are arranged at the metaphase plate.

Metaphase II of Meiosis

METAPHASE II

- The kinetochores of sister chromatids attach to microtubules extending from opposite poles.
- Chromosomes lined up across the middle of the cell.

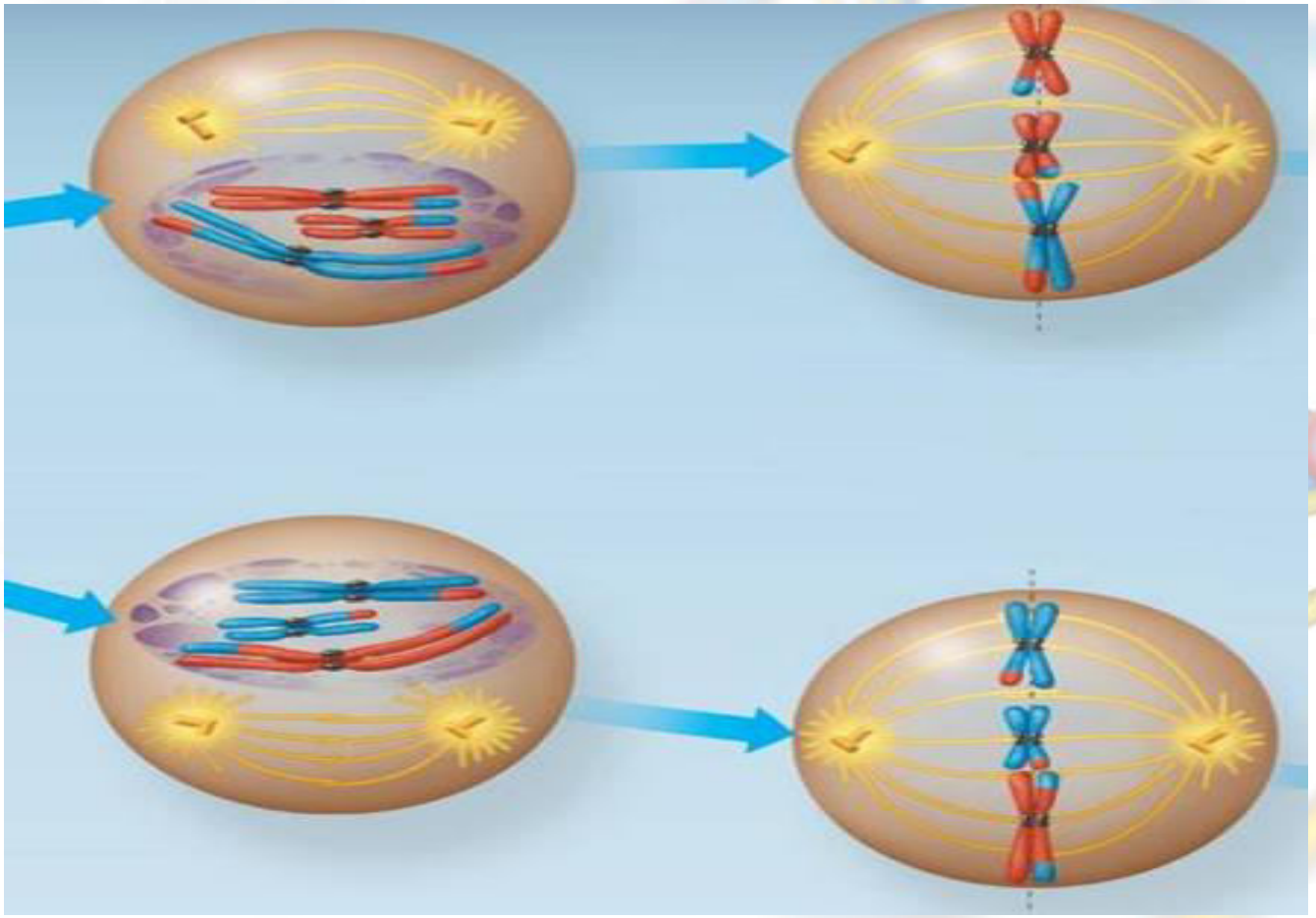
Metaphase II of Meiosis

SISTER CHROMATIDS OF A CHROMOSOME ARE NOT IDENTICAL

- Due to crossing over during meiosis I, the two sister chromatids of each chromosome are no longer genetically identical.

Metaphase II of Meiosis

METAPHASE II



Metaphase II of Meiosis

METAPHASE II SUMMARY

- The kinetochores of sister chromatids attach to microtubules extending from opposite poles.
- Chromosomes lined up across the middle of the cell.

Essentials of Genetics

ANAPHASE II OF MEIOSIS

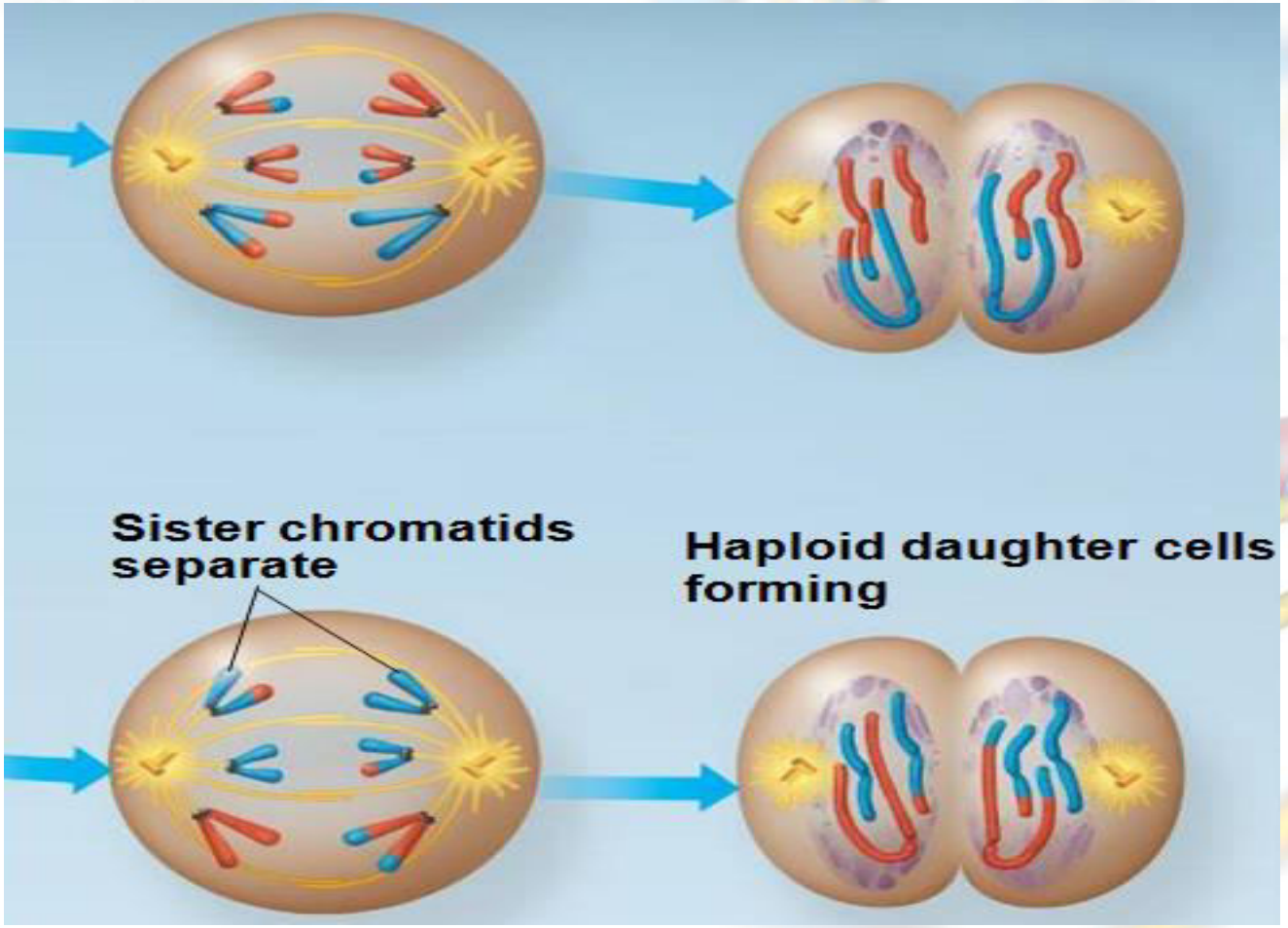
Anaphase II of Meiosis

ANAPHASE II

- Centromere split – as a result sister chromatids are separated.
- Sister chromatids becomes individual chromosomes.
- Separated chromatids move to opposite poles of the cell

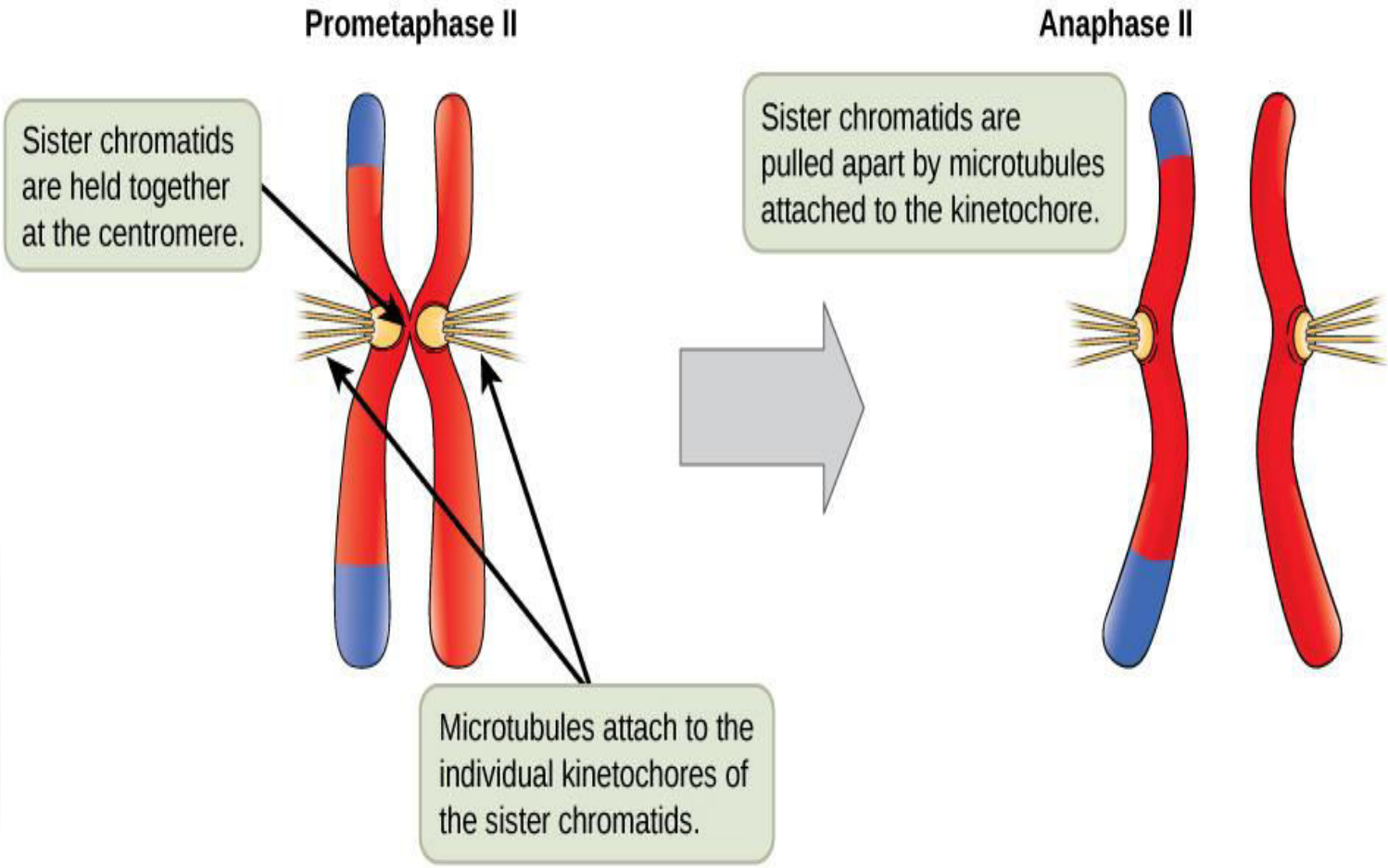
Anaphase II of Meiosis

ANAPHASE II



Anaphase II of Meiosis

ANAPHASE II

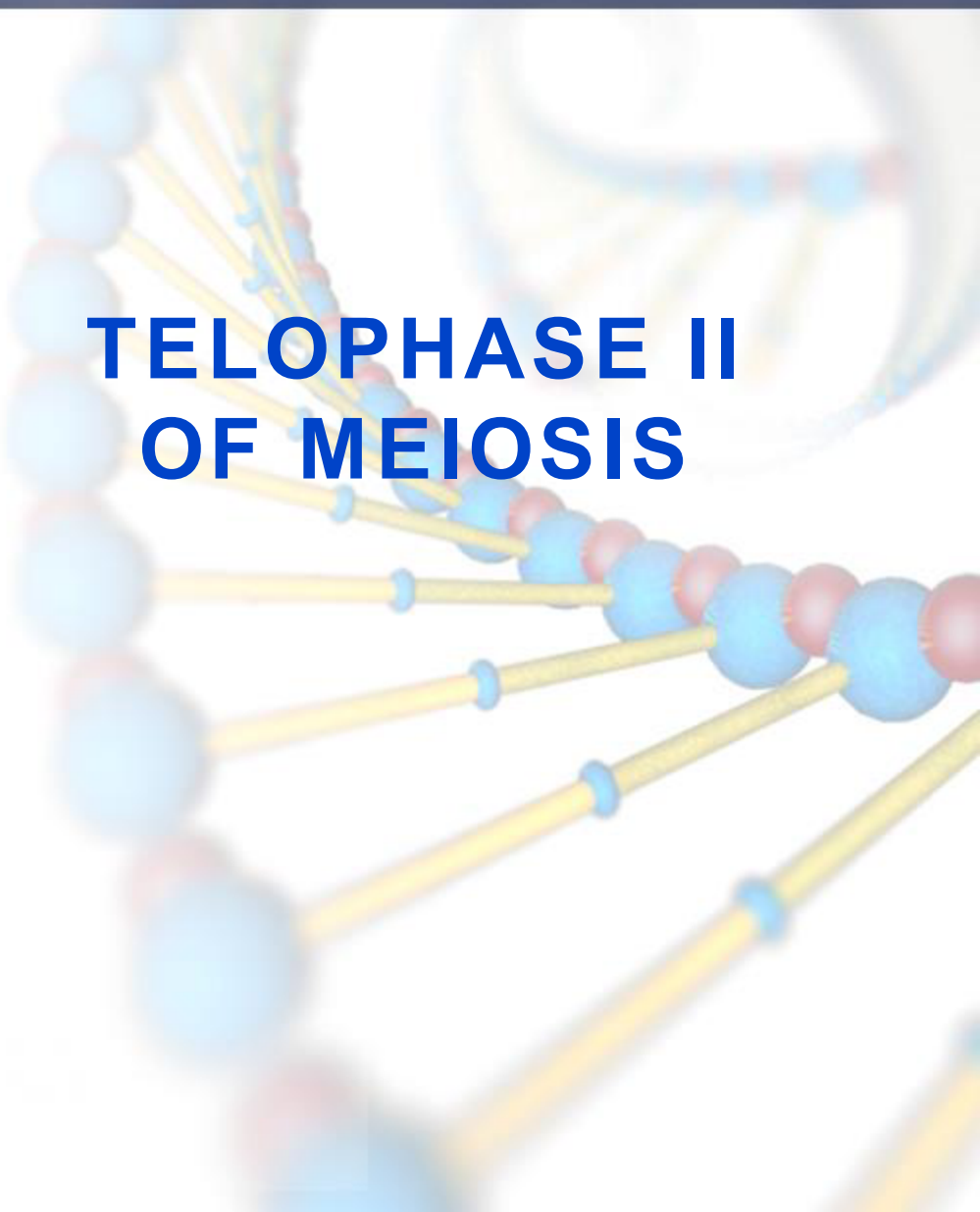


Anaphase II of Meiosis

ANAPHASE II SUMMARY

- Sister chromatids becomes individual chromosomes.
- Separated chromatids move to opposite poles of the cell.

Essentials of Genetics



TELOPHASE II OF MEIOSIS

Telophase II of Meiosis

TELOPHASE II

- In Telophase II, the chromosomes arrive at opposite poles.
- Nuclear membrane forms, and the chromosomes begin de-condensing.

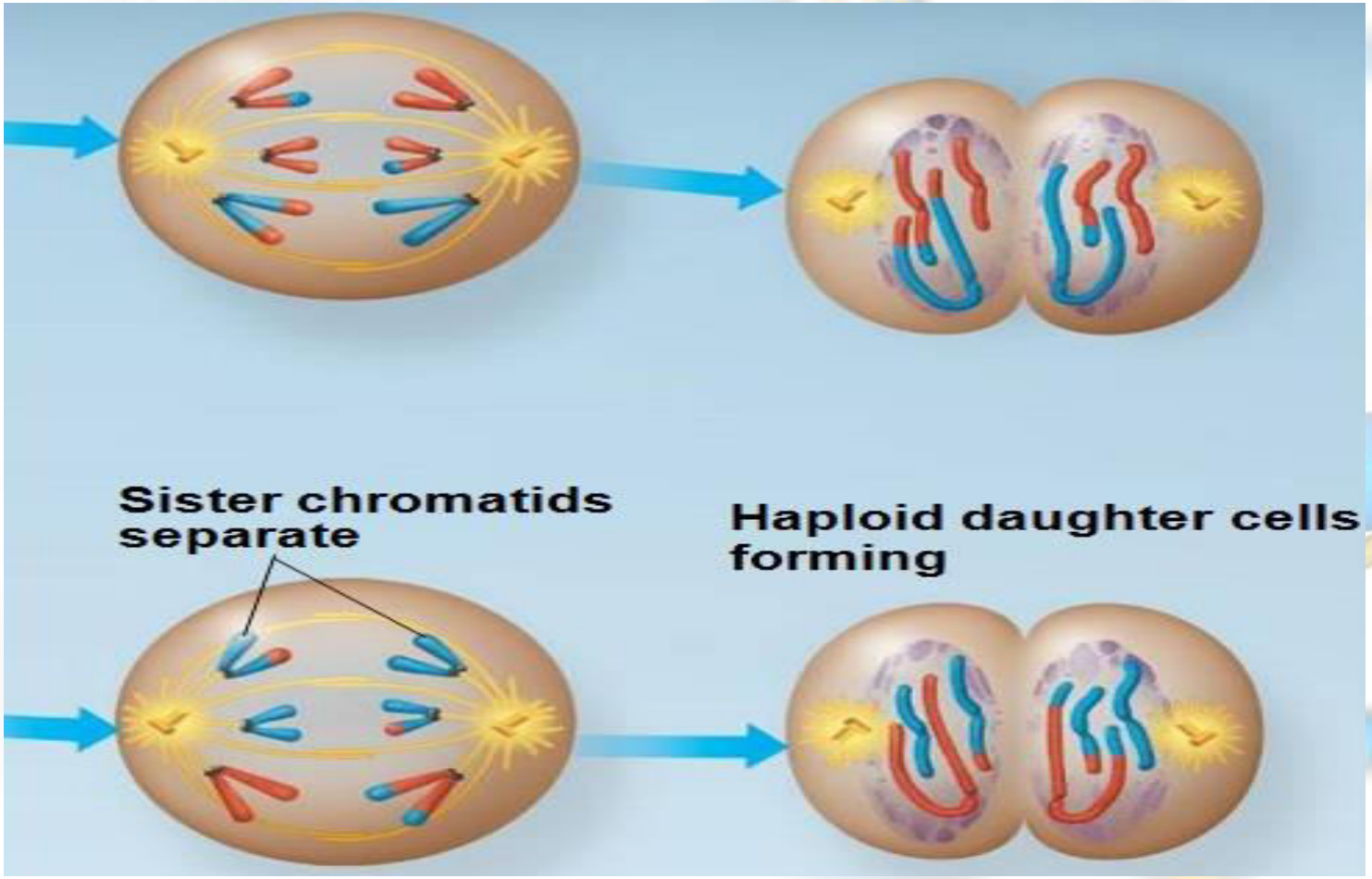
Telophase II of Meiosis

TELOPHASE II

- Cytokinesis occurs.
- Now, there are four daughter cells, each with a haploid set of chromosomes.
- Each daughter cell is genetically distinct from the others and from the parent cell.

Telophase II of Meiosis

TELOPHASE II



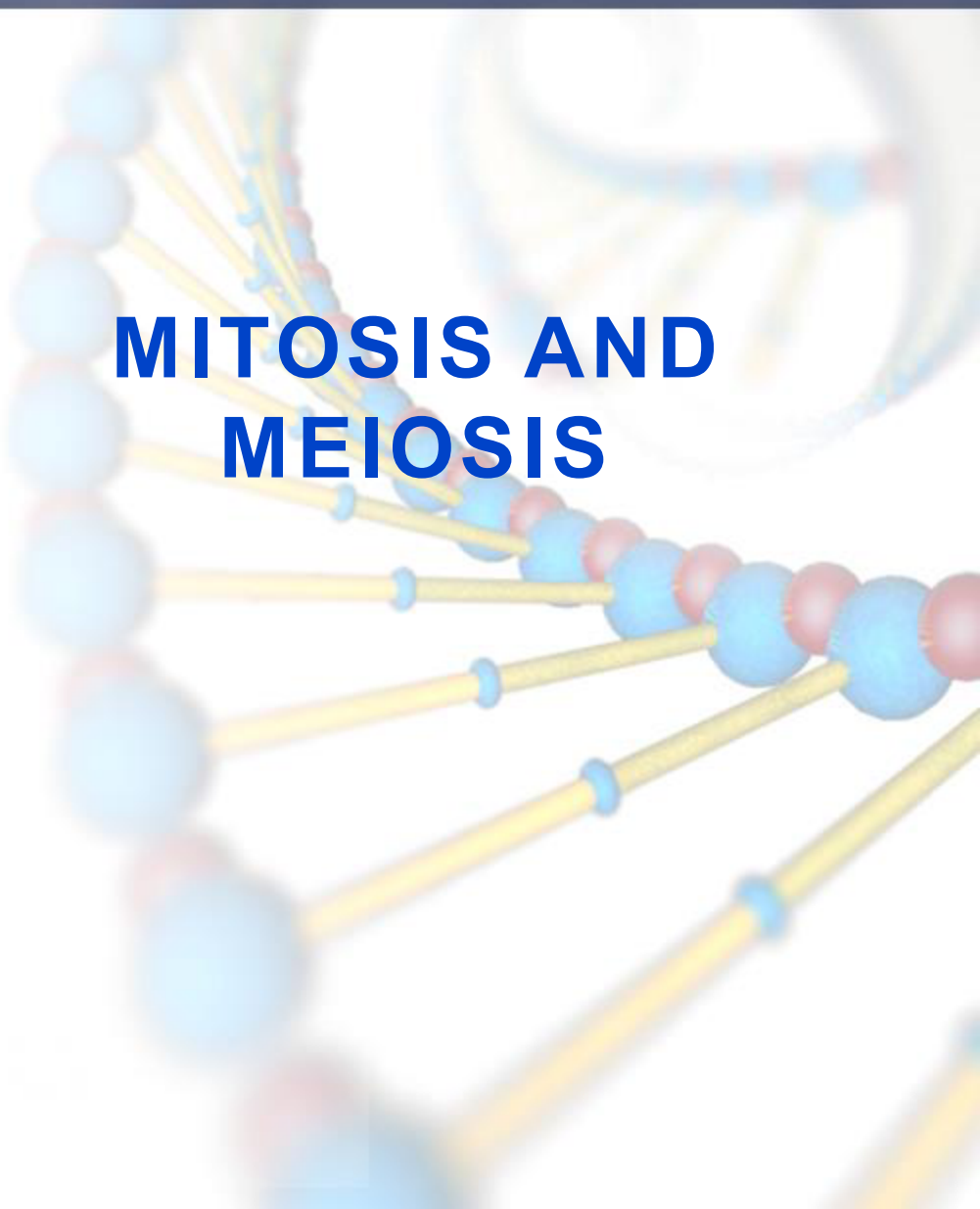
Telophase II of Meiosis

TELOPHASE II SUMMARY

- Four daughter cells are produced each with a haploid set of chromosomes.
- Each daughter cell is genetically distinct from the others and from the parent cell.

Essentials of Genetics

MITOSIS AND MEIOSIS



Mitosis and Meiosis

MITOSIS

- Mitosis conserves the number of chromosome sets, producing cells that are genetically identical to the parent cell.

Mitosis and Meiosis

MEIOSIS

- Meiosis reduces the number of chromosomes sets from two (diploid) to one (haploid), producing cells that differ genetically from each other and from the parent cell.

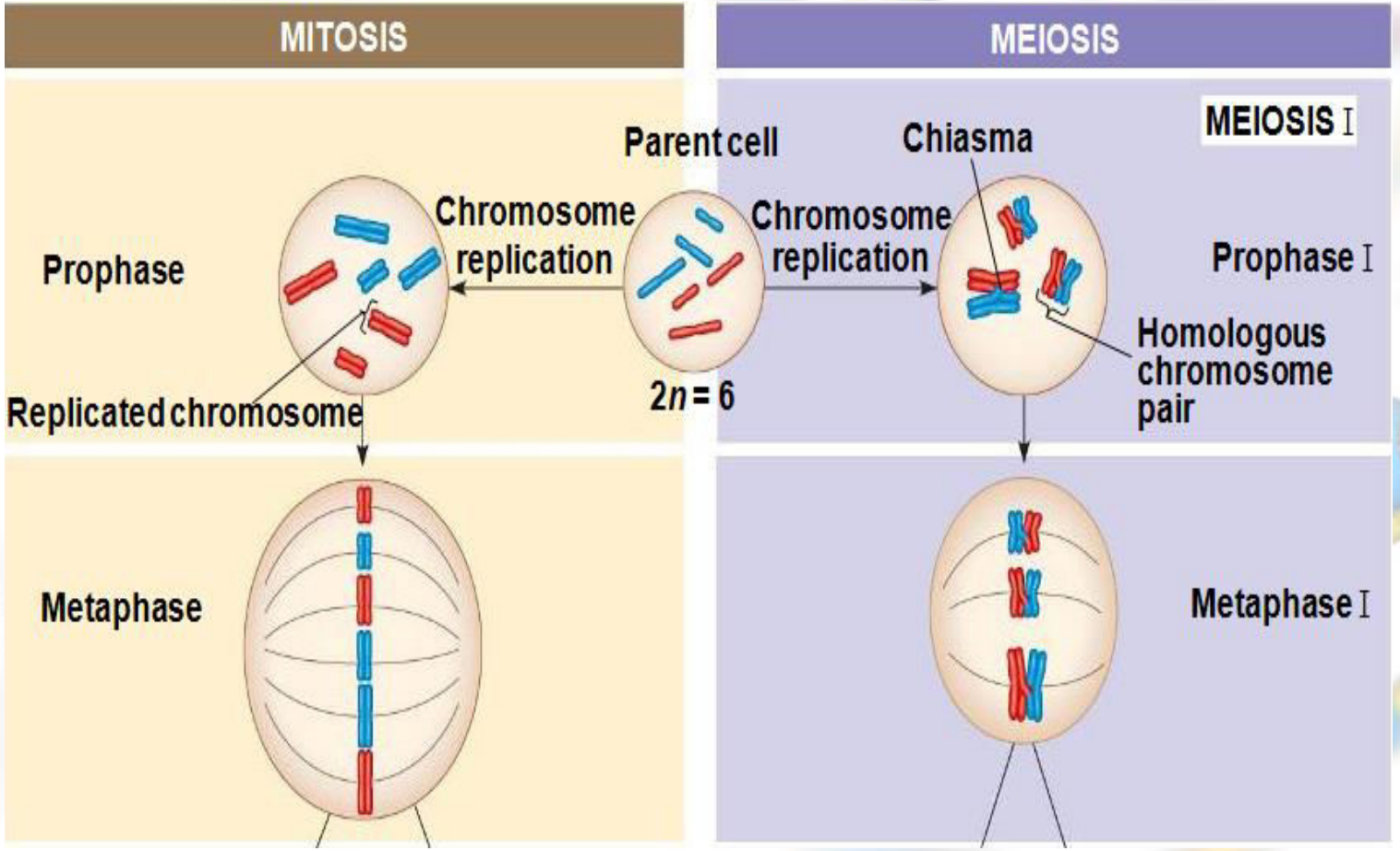
Mitosis and Meiosis

COMPARISON

- The mechanism for separating sister chromatids is virtually identical in meiosis II and mitosis.

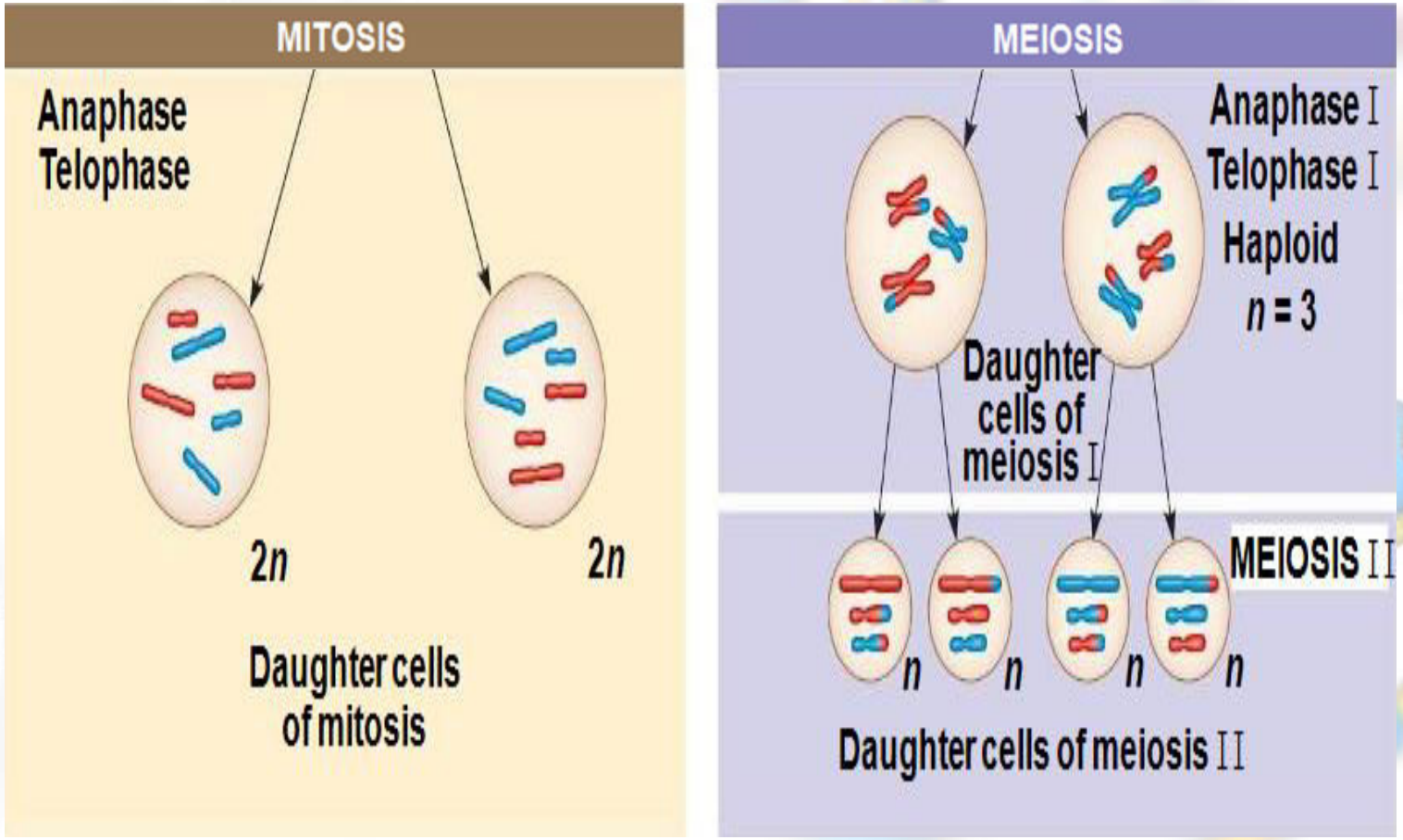
Mitosis and Meiosis

COMPARISON



Mitosis and Meiosis

COMPARISON

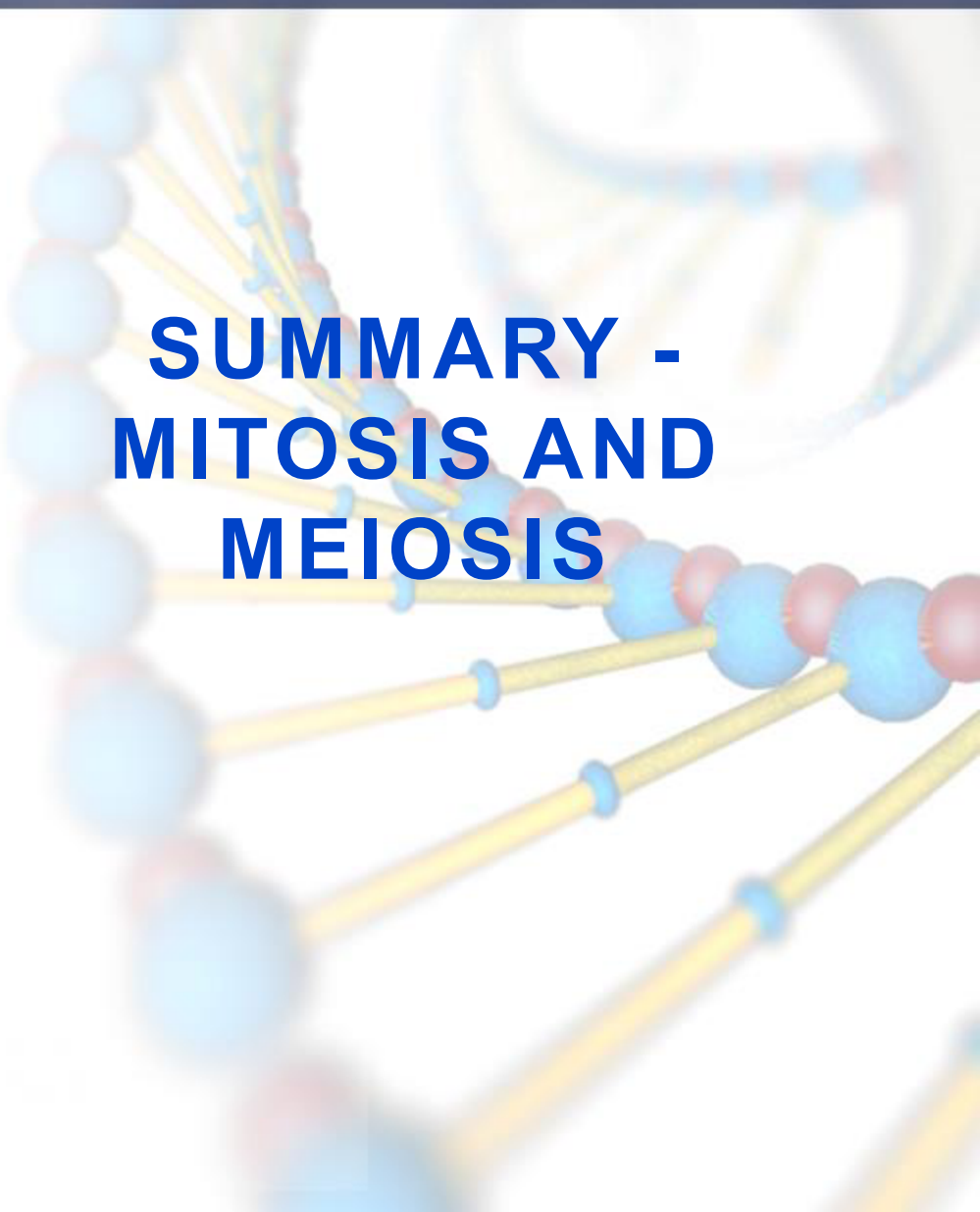


Mitosis and Meiosis

COMPARISON

- Meiosis reduces the number of chromosomes sets from diploid to one haploid and produced four daughter cells which not identical to each other and parent cell.
- Mitosis – produces two daughter cells which are identical to parent cell.

Essentials of Genetics

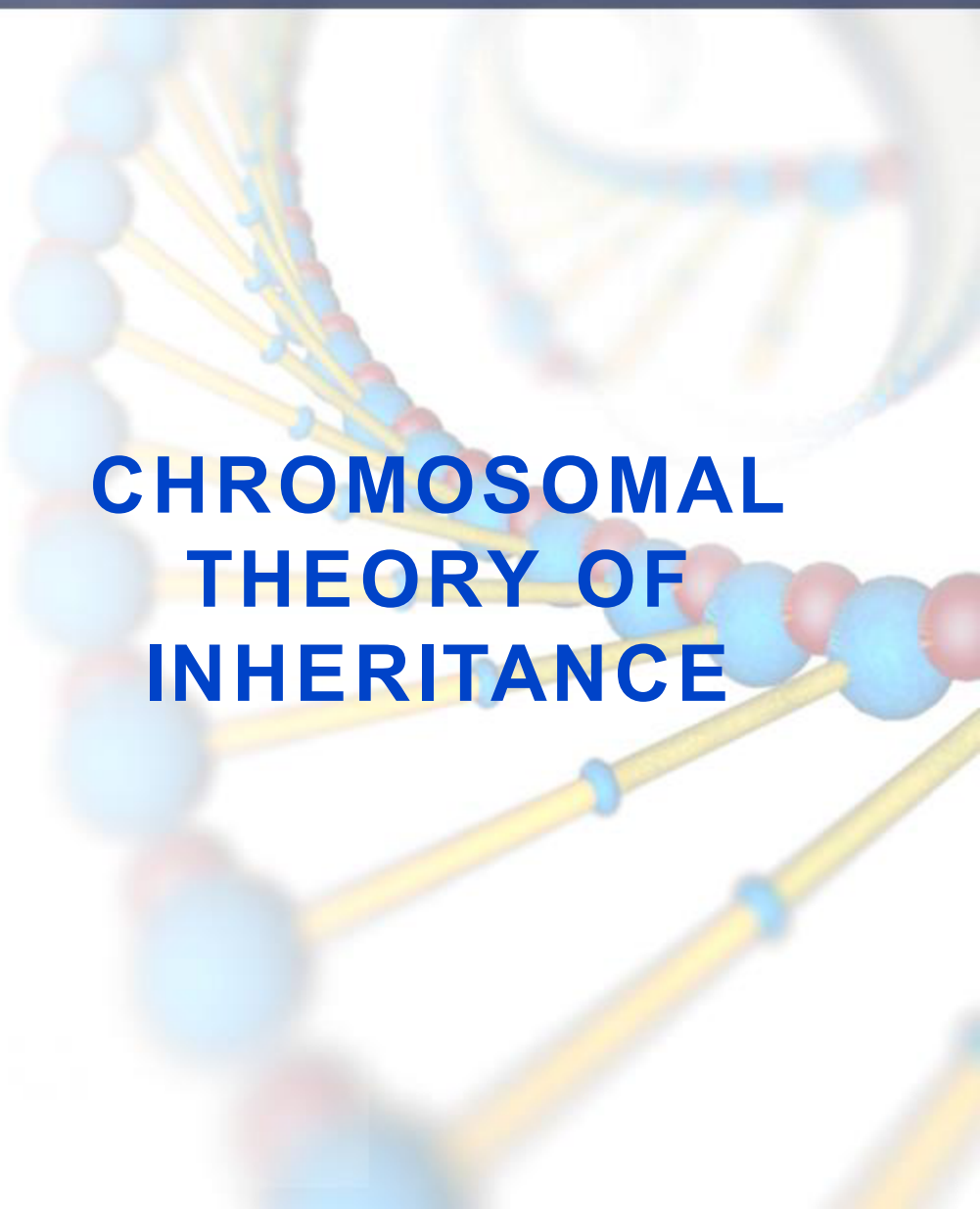


SUMMARY - MITOSIS AND MEIOSIS

Summary - Mitosis & Meiosis

SUMMARY		
Property	**Mitosis**	**Meiosis**
DNA replication	Occurs during interphase before mitosis begins	Occurs during interphase before meiosis I begins
Number of divisions	One, including prophase, metaphase, anaphase, and telophase	Two, each including prophase, metaphase, anaphase, and telophase
Synapsis of homologous chromosomes	Does not occur	Occurs during prophase I along with crossing over between nonsister chromatids; resulting chiasmata hold pairs together due to sister chromatid cohesion
Number of daughter cells and genetic composition	Two, each diploid (*2n*) and genetically identical to the parent cell	Four, each haploid (*n*), containing half as many chromosomes as the parent cell; genetically different from the parent cell and from each other
Role in the animal body	Enables multicellular adult to arise from zygote; produces cells for growth, repair, and, in some species, asexual reproduction	Produces gametes; reduces number of chromosomes by half and introduces genetic variability among the gametes

Essentials of Genetics



CHROMOSOMAL THEORY OF INHERITANCE

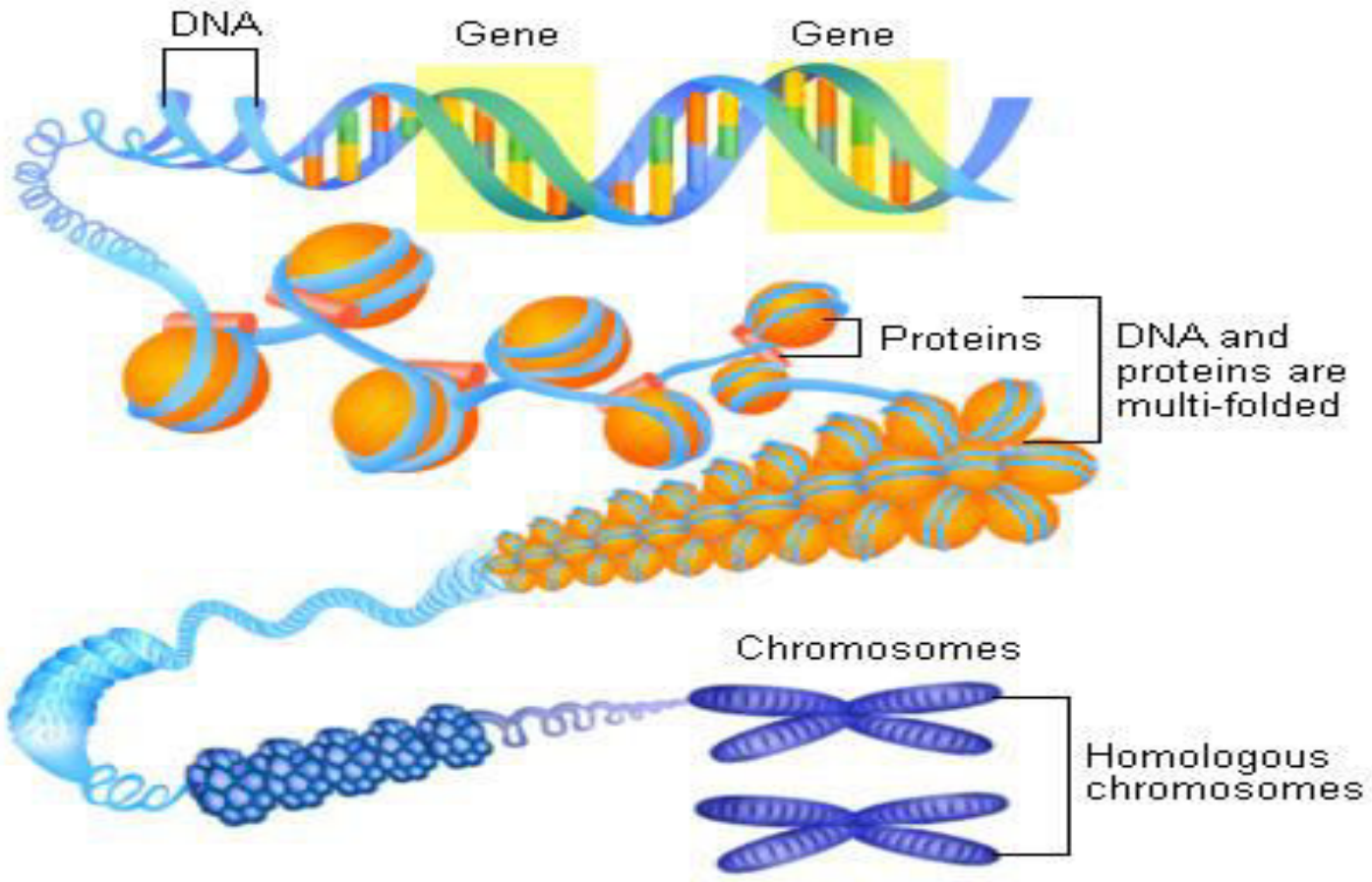
Chromosomal Theory of Inheritance

CHROMOSOMAL THEORY OF INHERITANCE

- The chromosomal theory of inheritance states that genes have specific loci on chromosomes.

Chromosomal Theory of Inheritance

GENES ON CHROMOSOMES



Chromosomal Theory of Inheritance

BEHAVIOR OF CHROMOSOMES – ACCORDING TO MENDEL S' LAWS

- Theory states that behavior of chromosomes is according to Mendel's laws of segregation and independent assortment.

Chromosomal Theory of Inheritance

MENDEL S' FACTORS AND CHROMOSOMES BEHAVIOR

- Around 1900, cytologists and geneticists began to see parallels between the behavior of chromosomes and the behavior of Mendel's factors.

Chromosomal Theory of Inheritance

ALLELES ARE SEPARATED - PRESENT ON HOMOLOGOUS CHROMOSOMES

- Homologous chromosomes separate and alleles segregate during meiosis.

Chromosomal Theory of Inheritance

FERTILIZATION RESTORES PAIRED CONDITION

- Fertilization restores the paired condition for both chromosomes and genes.

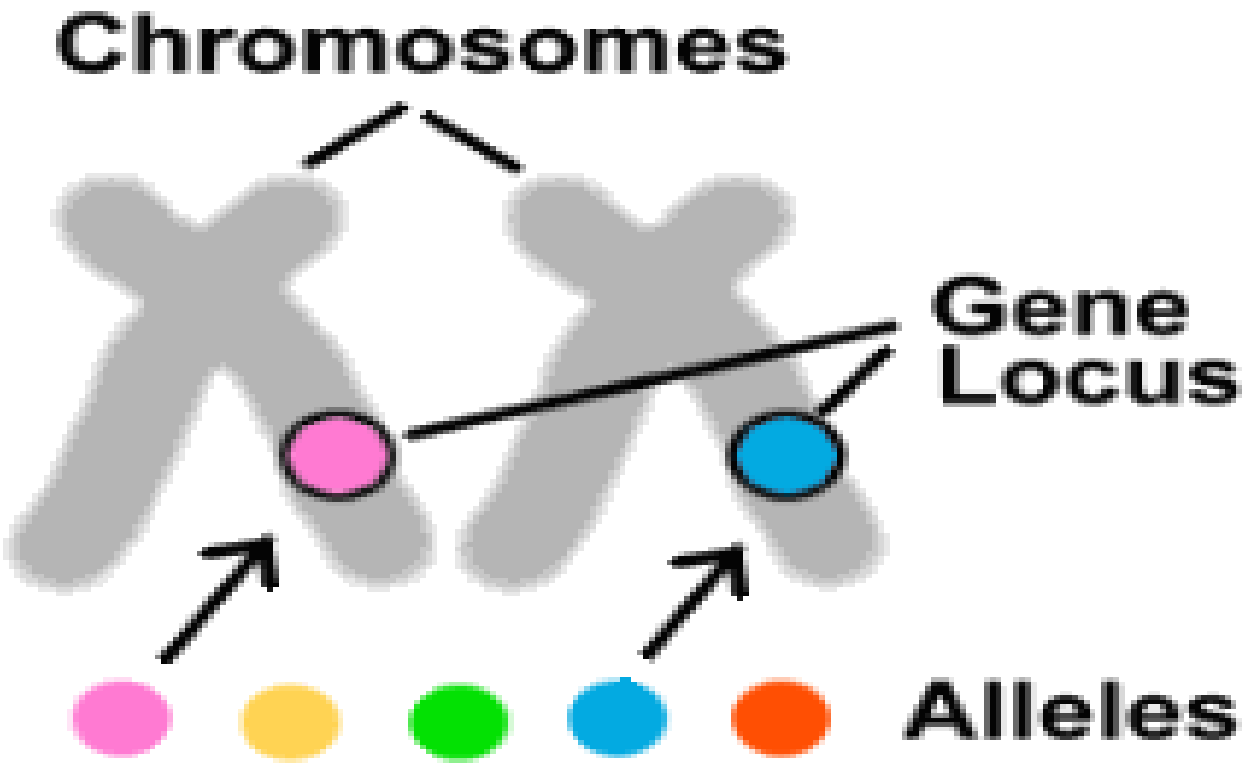
Chromosomal Theory of Inheritance

EVIDENCE THAT CHROMOSOMES ARE LOCATION OF GENES

- T. H. Morgan was the first to associate a specific gene/trait with a specific chromosome.
- Convincing evidence that chromosomes are the location of Mendel's heritable factors.

Chromosomal Theory of Inheritance

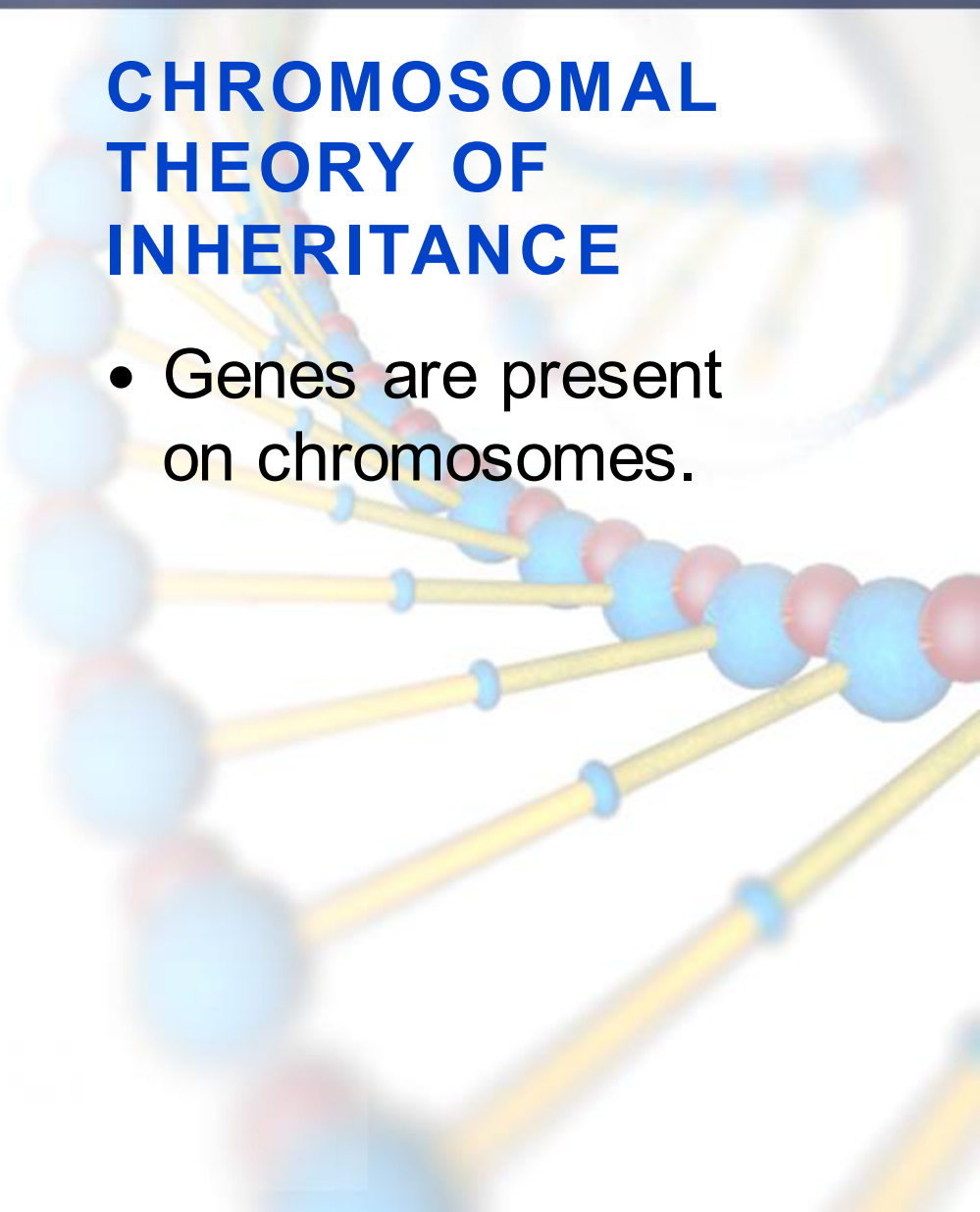
GENES ON CHROMOSOMES



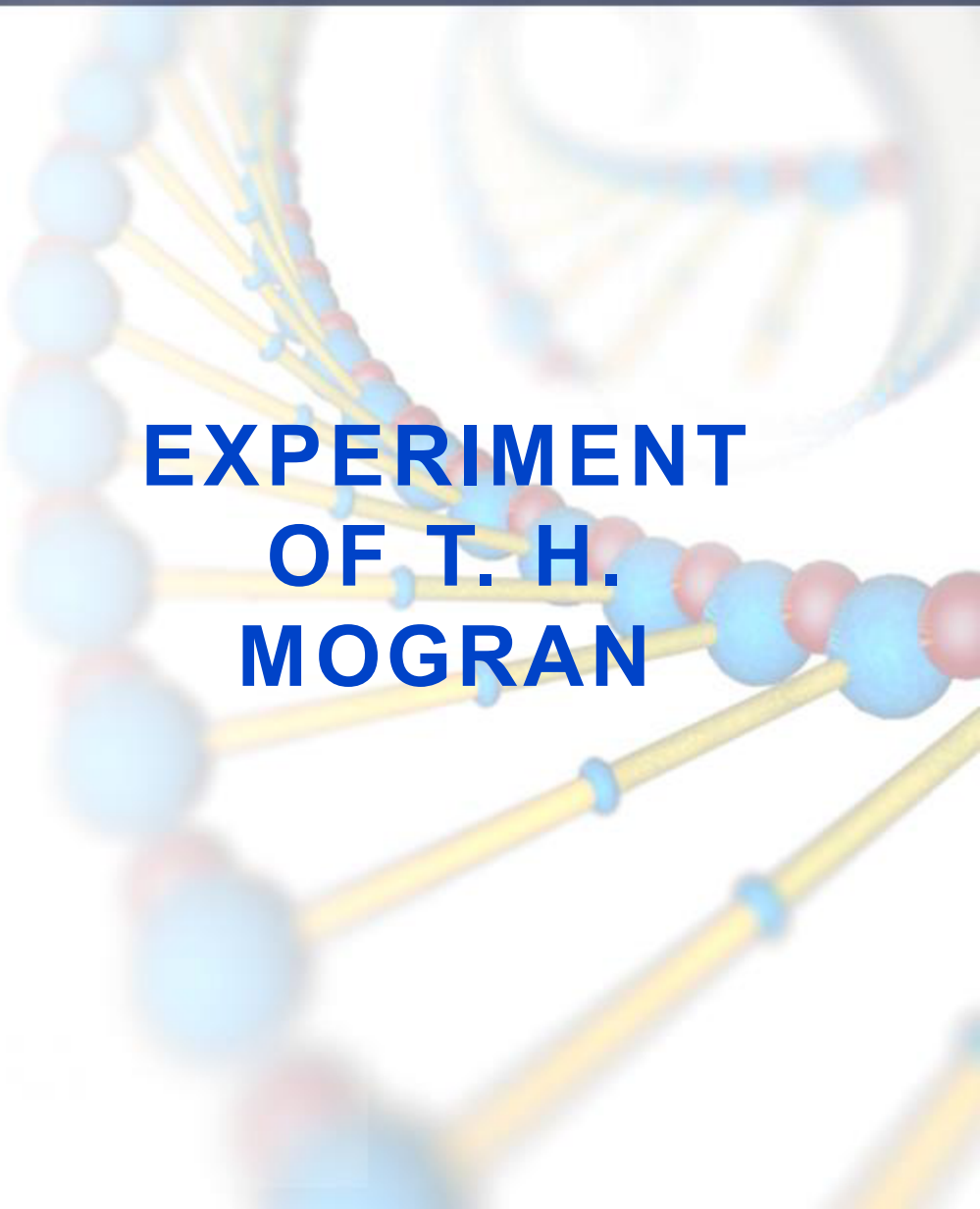
Chromosomal Theory of Inheritance

CHROMOSOMAL THEORY OF INHERITANCE

- Genes are present on chromosomes.



Essentials of Genetics



EXPERIMENT OF T. H. MOGRAN

Experiment of T. H. Morgan

EYE COLOR TRAIT IS LINKED WITH SEX OF DROSOPHILA

- Morgan with an experiment proved that eye color trait is linked with sex of the drosophila.

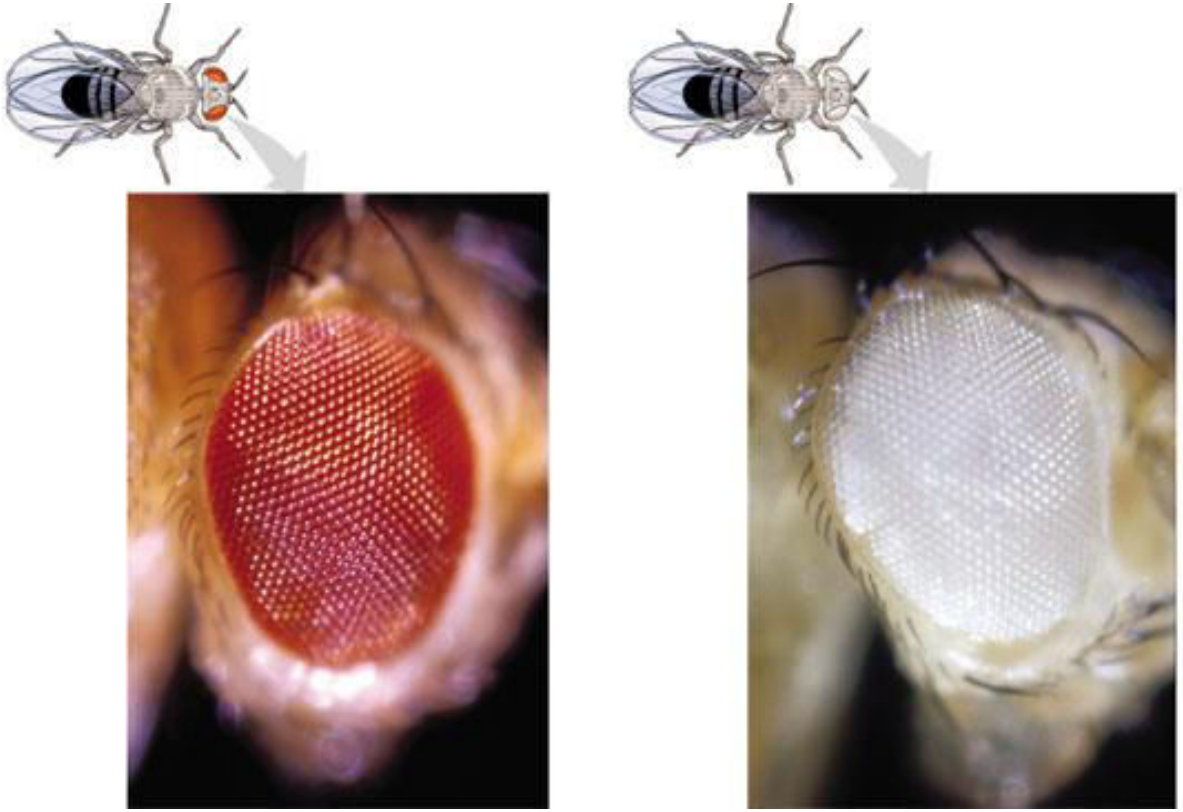
Experiment of T. H. Morgan

EXPERIMENT

- Male flies were with white eyes (mutant).
- Female flies were with red eyes (wild type).

Experiment of T. H. Morgan

MALE FLIES WITH WHITE EYES
FEMALE FLIES WITH RED EYES



Experiment of T. H. Morgan

PROGENY F1 WAS WITH RED EYES

- All the flies produced during F1 generation were with red eyes.

Experiment of T. H. Morgan

PROGENY F2 SHOWED 3:1 RATIO

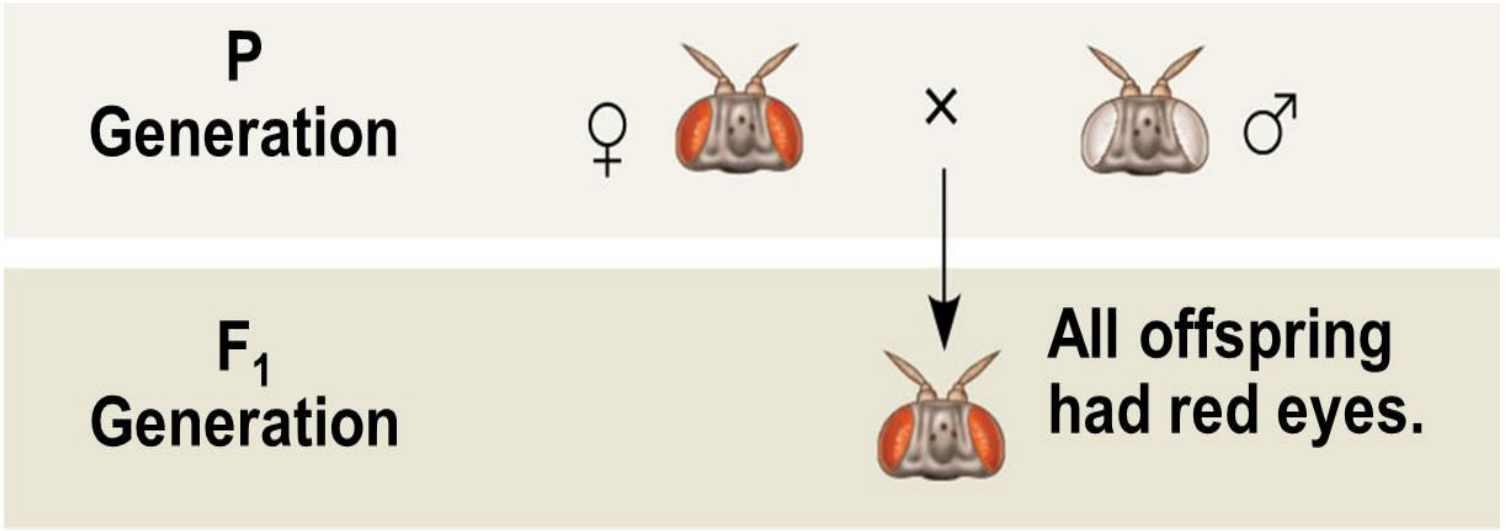
- F2 generation showed 3:1 (three red eye: one white eye).
- White-eyed trait appeared only in males.
- All females and half males with red eye.

Experiment of T. H. Morgan

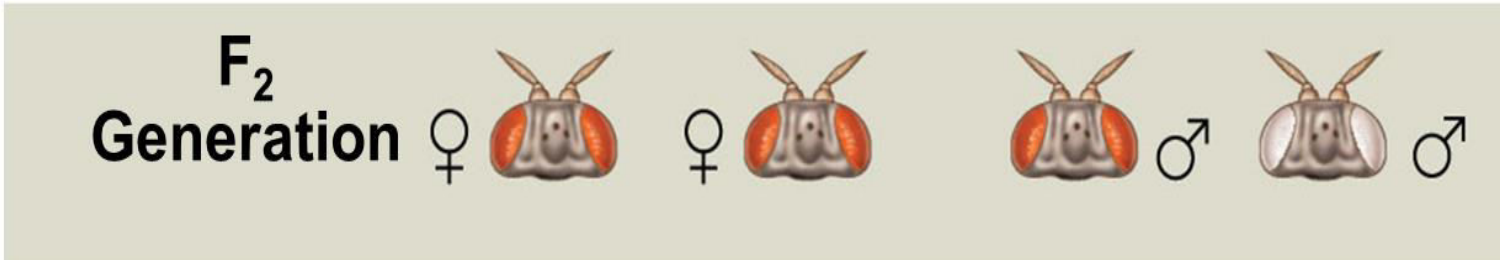
WHITE EYED TRAIT APPEARED ONLY IN MALES

- White-eyed trait appeared only in male fruit flies (Drosophila melanogaster).

Experiment of T. H. Morgan

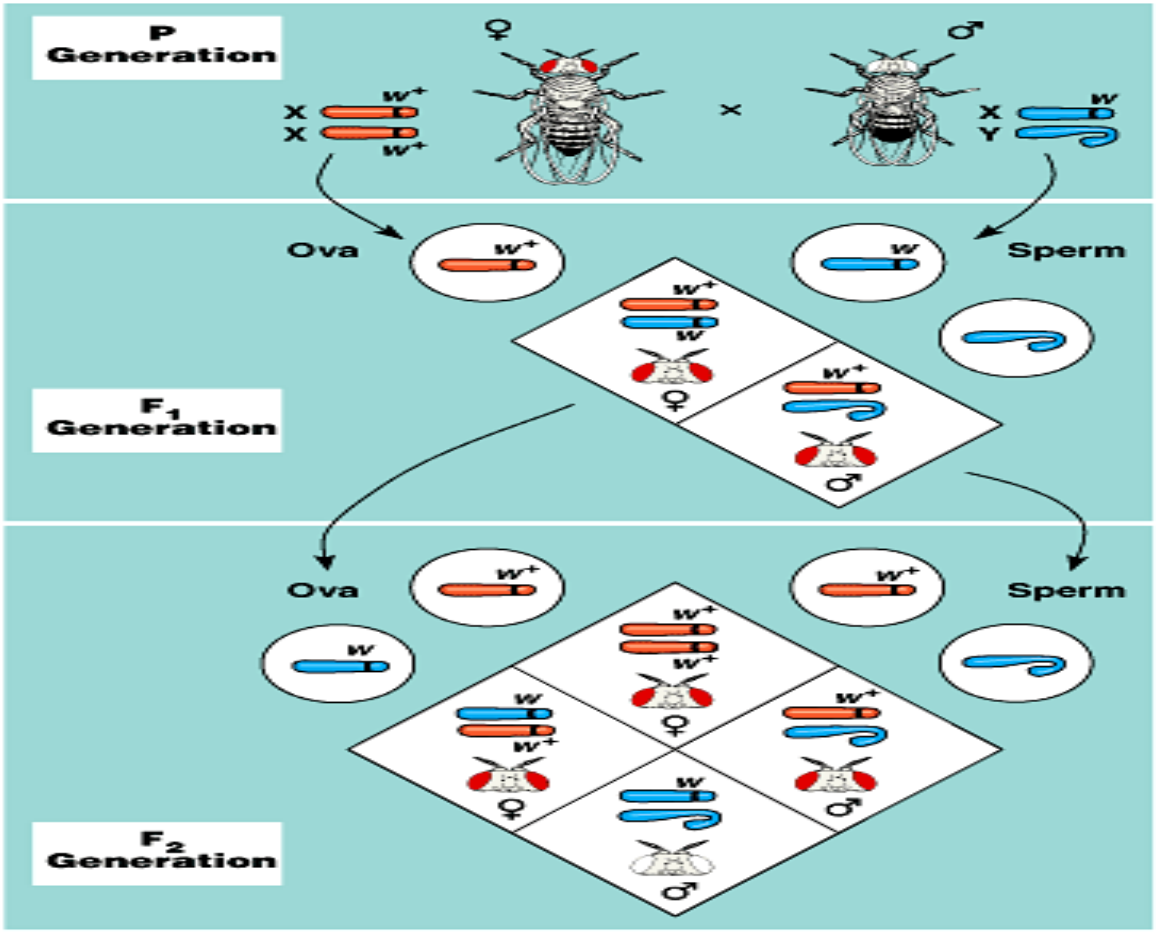


RESULTS

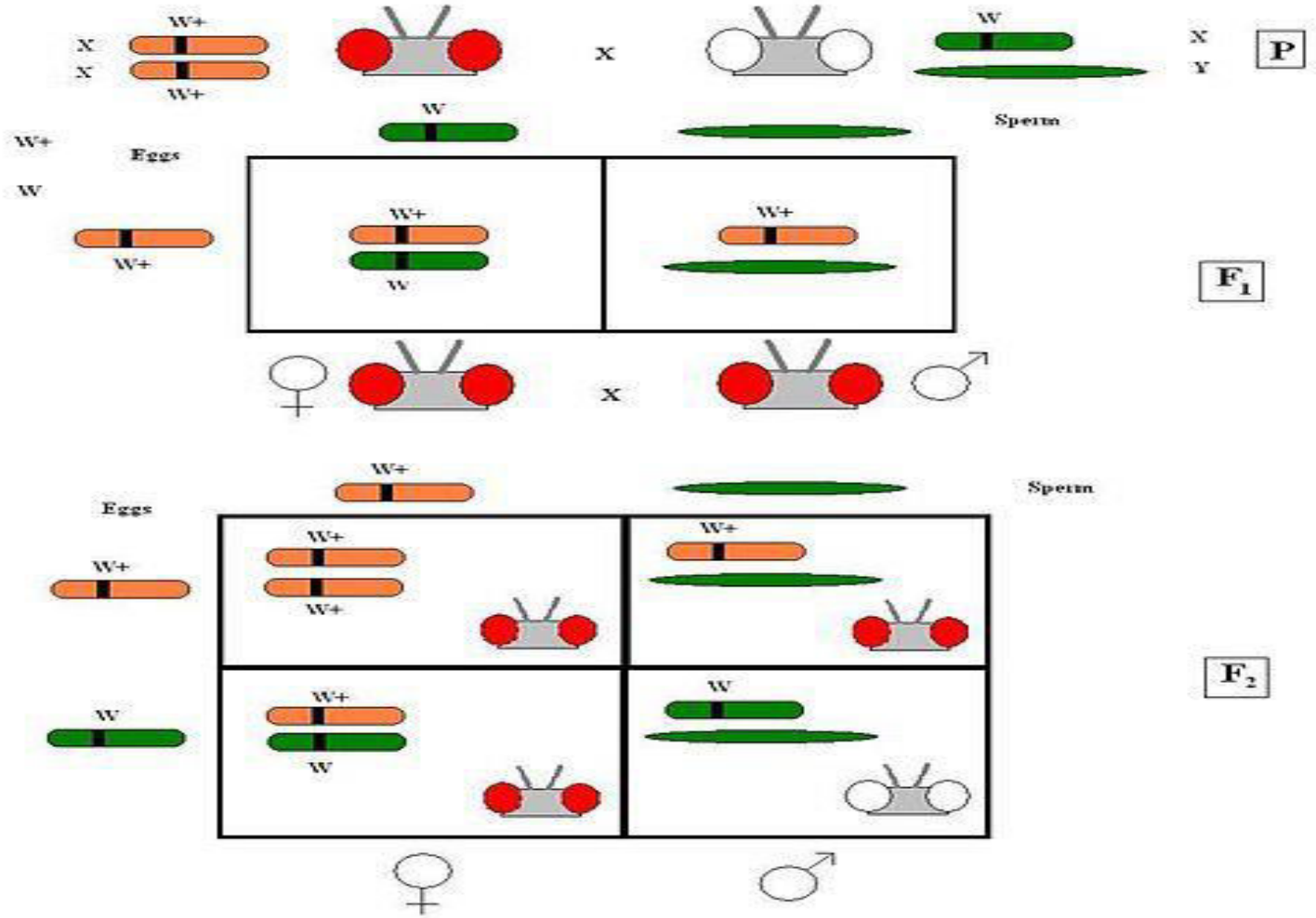


Experiment of T. H. Morgan

MORGAN EXPERIMENT



Experiment of T. H. Morgan

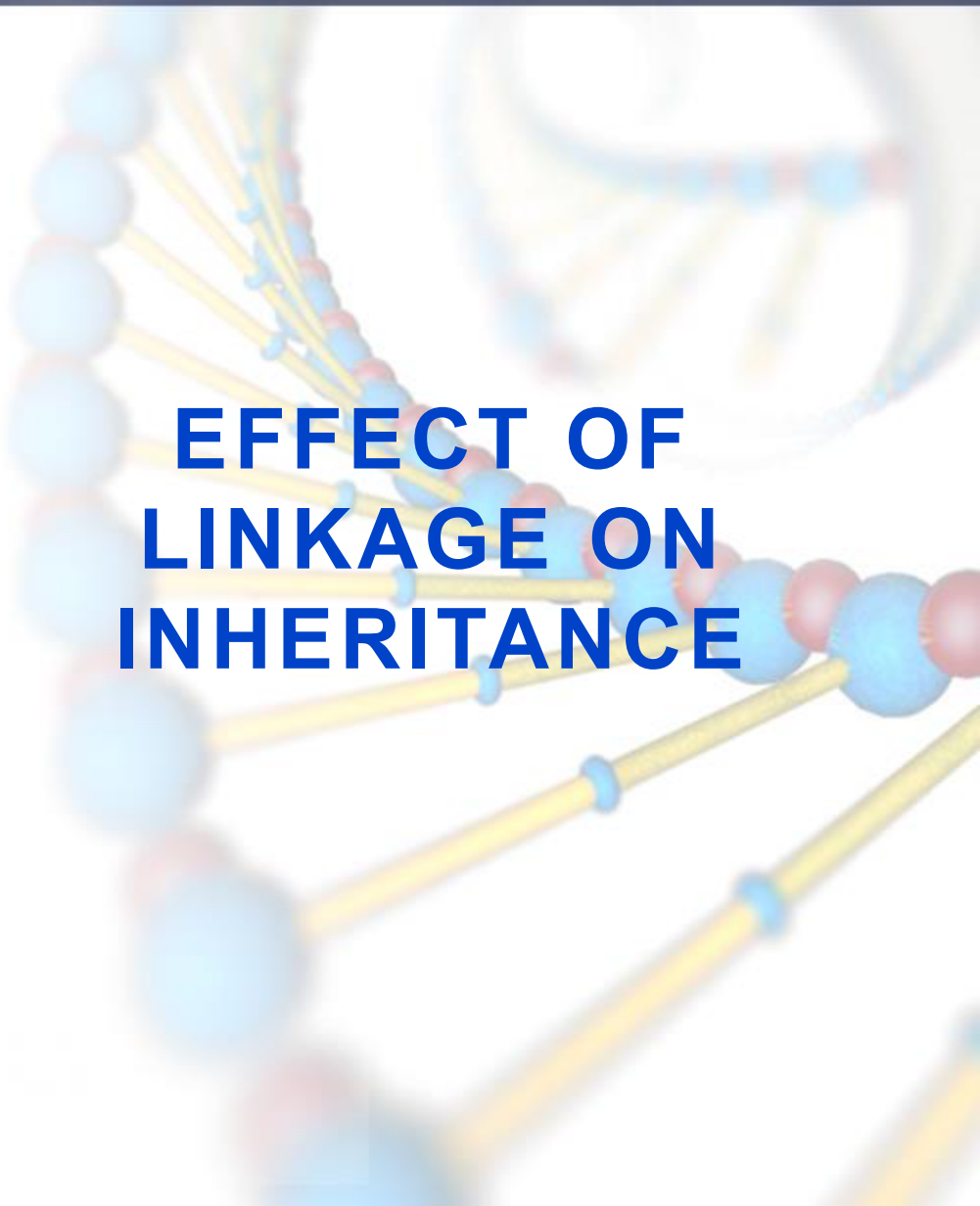


Experiment of T. H. Morgan

MORGAN'S DISCOVERY

- Transmission of X chromosome correlates with inheritance of eye color trait.
- First solid evidence indicating that a gene is associated with a specific chromosome.

Essentials of Genetics



EFFECT OF LINKAGE ON INHERITANCE

Effect of linkage on Inheritance

LINKED GENES INHERIT TOGETHER

- Genes located on the same chromosome, if are linked, will inherit together.

Effect of linkage on Inheritance

EFFECT OF LINKAGE ON INHERITANCE

- Morgan did another experiment with fruit flies to see how linkage affects the inheritance of two different characters.

Effect of linkage on Inheritance

BODY COLOR – WILD TYPE GRAY, MUTANT BLACK

WING SIZE – WILD TYPE NORMAL, MUTANT VESTIGIAL



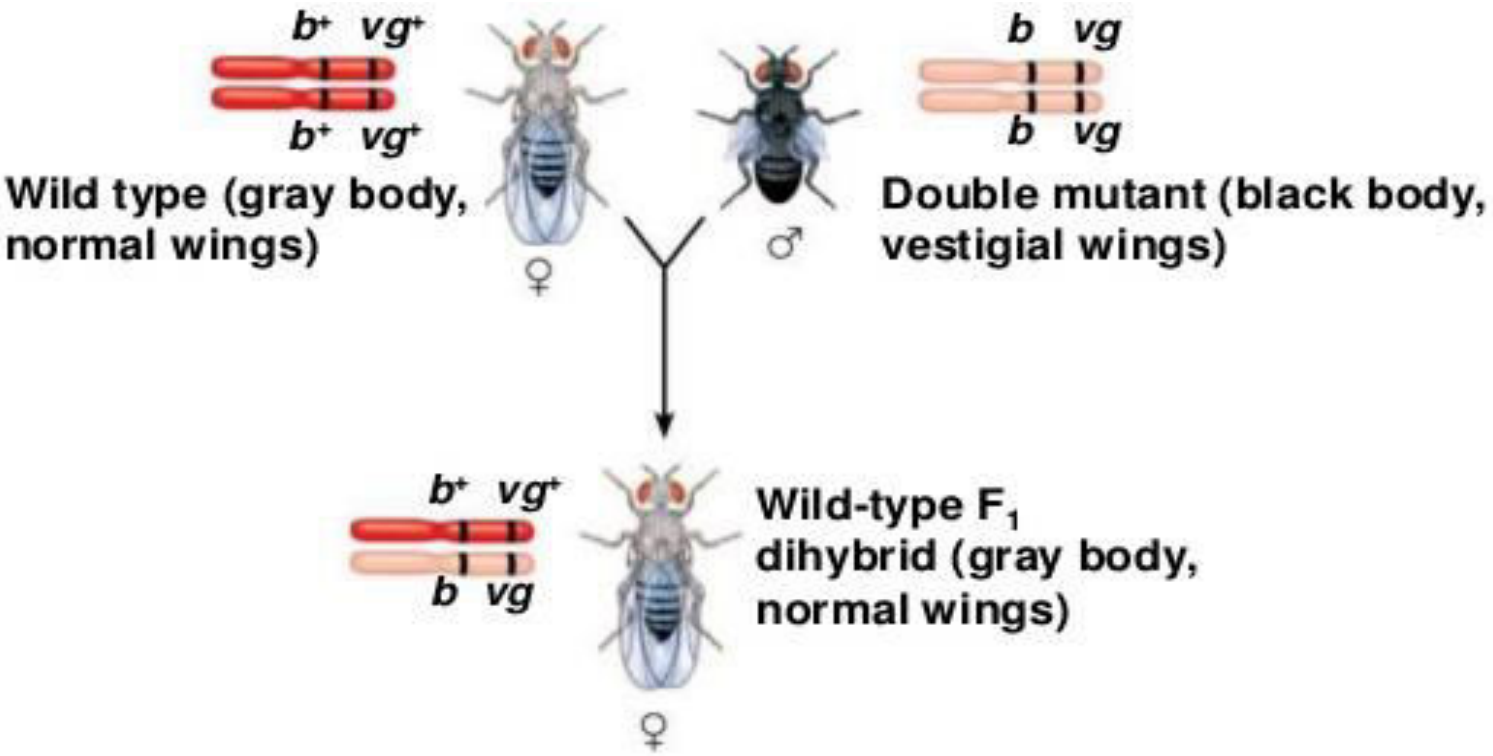
Wild type (gray body, normal wings)



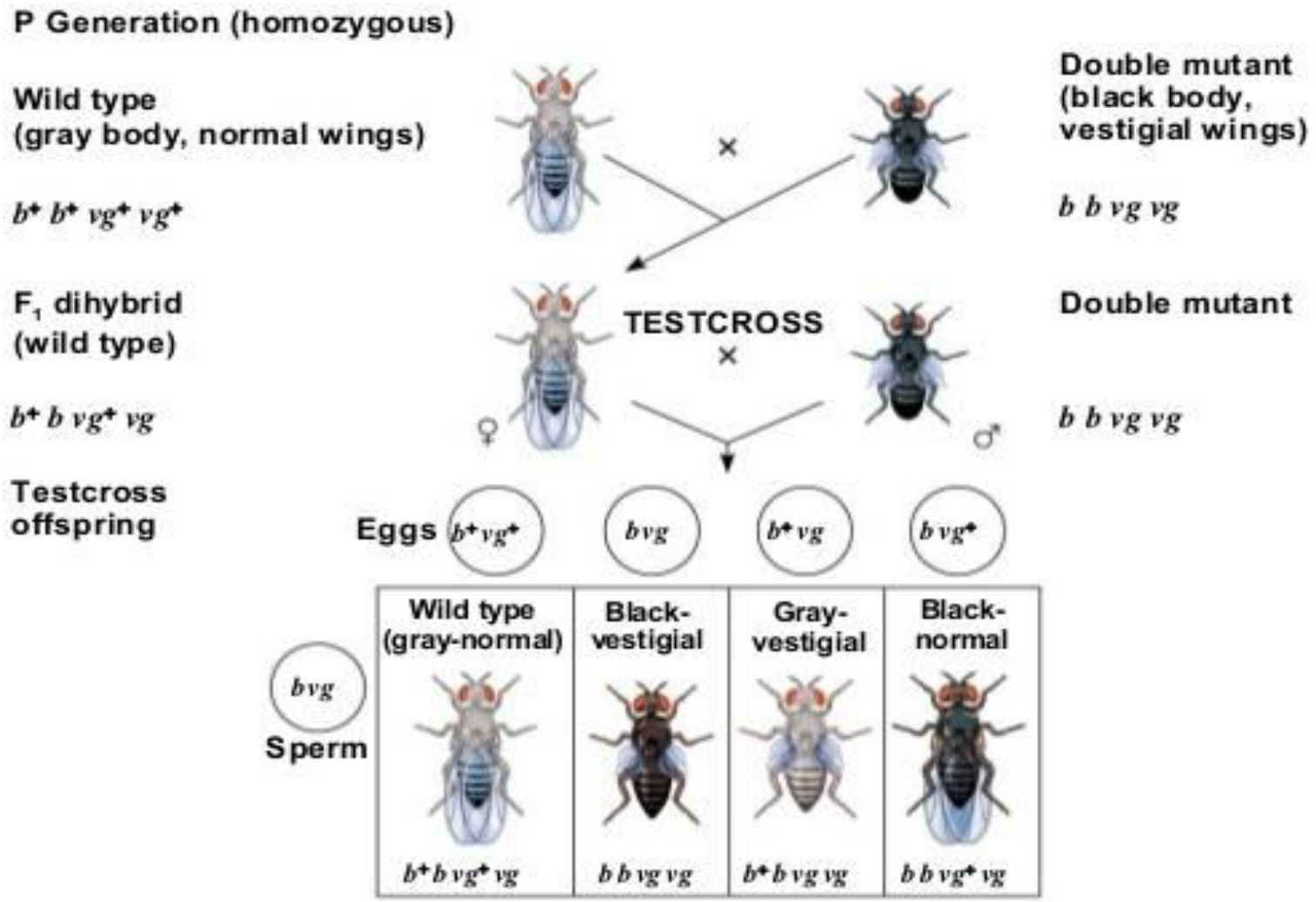
Double mutant (black body, vestigial wings)

Effect of linkage on Inheritance

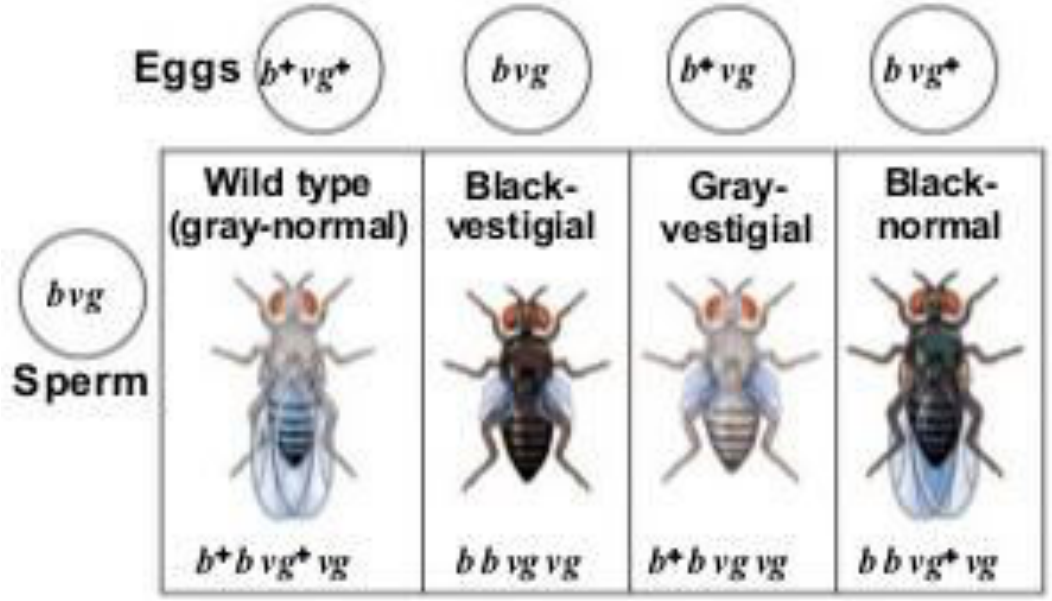
CROSS



Effect of linkage on Inheritance

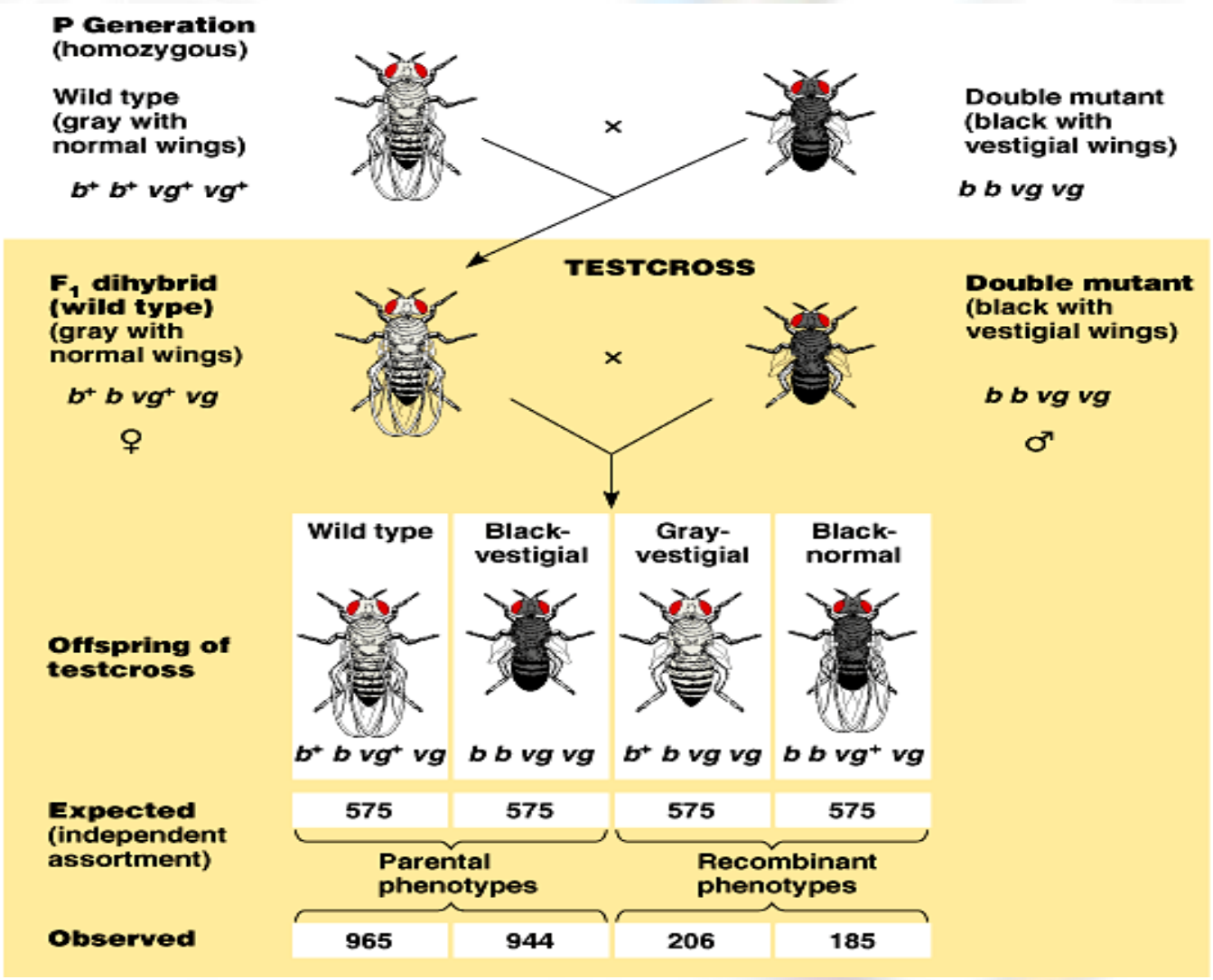


Effect of linkage on Inheritance



If gene are located on different chromosomes	1 (575)	1 (575)	1 (575)	1 (575)
If genes are located on same chromosomes	1 (1150)	1 (1150)	0	0
Morgan Results	965	944	206	185

Effect of linkage on Inheritance



Effect of linkage on Inheritance

RESULTS - LINKAGE AND CROSS OVER

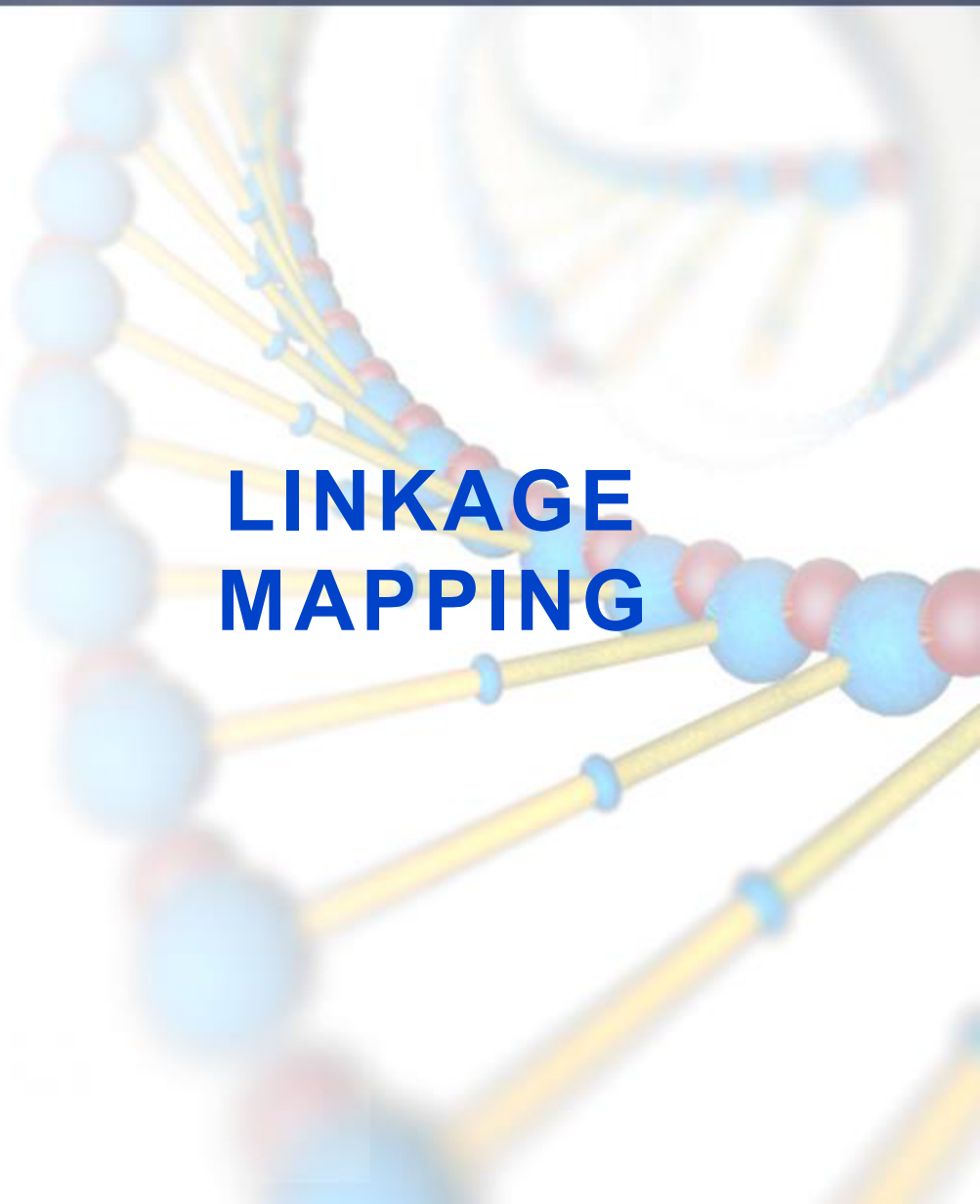
- According to independent assortment, four phenotypes in 1:1:1:1 ratio.
- Observed large number of wild-type and double-mutant flies.

Effect of linkage on Inheritance

MORGAN REASONED

- Body color and wing shape are inherited together because their genes are on the same chromosome.
- New phenotypic variations --- crossing over and linkage that break the genes physical connection.

Essentials of Genetics



LINKAGE MAPPING

Linkage Mapping

LINKAGE MAP

- Linkage map is an ordered list of the genetic loci along a particular chromosome.
- A map of the genes on a chromosome based on linkage analysis.

Linkage Mapping

LINKAGE MAP DOES NOT SHOW PHYSICAL DISTANCE

- A linkage map does not show the physical distances between genes but rather their relative positions, as determined by how often two gene loci are inherited together.

Linkage Mapping

POSITION OF GENES IN TERMS OF RECOMBINATION FREQUENCY

- A linkage map is a genetic map of a species that shows the position of its known genes or genetic markers relative to each other in terms of recombination frequency.

Linkage Mapping

LINKAGE MAPPING USED FOR IDENTIFYING LOCATION OF DISEASED GENES

- Linkage mapping is critical for identifying the location of genes that cause genetic diseases.

Linkage Mapping

GENETIC MAP IS BASED ON RECOMBINATION FREQUENCIES

- A genetic map is a map based on the frequencies of recombination between markers during crossover of homologous chromosomes.

Linkage Mapping

GREATER FREQUENCY MORE DISTANCE

- The greater the frequency of recombination (segregation) between two genetic markers, the further apart they are assumed to be.

Linkage Mapping

MARKERS USED

- Microsatellites
- Restriction fragment length polymorphisms (RFLPs).

Essentials of Genetics

LINKAGE MAP OF DROSOPHILA CHROMOSOME

Linkage map of Drosophila Chromosome

LINKAGE MAP

- A linkage map is a genetic map of a species that shows the position of its known genes or genetic markers relative to each other in terms of recombination frequency.

Linkage map of Drosophila Chromosome

FREQUENCY OF RECOMBINANT OFFSPRING REFLECT THE DISTANCE

- Frequency of recombinant offspring reflected distances between genes on chromosomes.

Linkage map of Drosophila Chromosome

GREATER THE DISTANCE, MORE CHANCES OF CROSS OVER

- The greater the distance between two genes, the more points between them where crossing over can occur.

Linkage map of Drosophila Chromosome

STURTEVANT USED RECOMBINATION FREQUENCIES TO MAP POSITION

- He used recombination frequencies from fruit fly crosses to map the position of genes along chromosomes.

Linkage map of Drosophila Chromosome

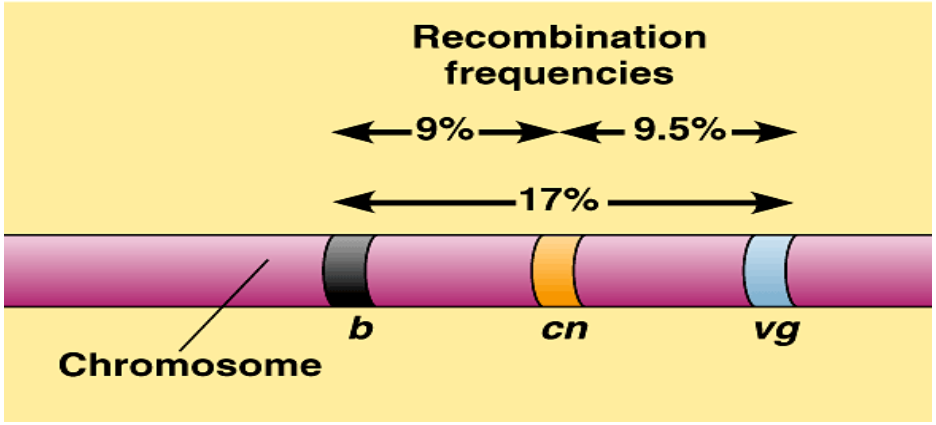
TEST CROSS DESIGNED TO MAP THREE GENES

- The test cross design to map the relative position of three fruit fly genes, body color (b), wing size (vg), and eye color (cn).

Linkage map of Drosophila Chromosome

LINKAGE MAP

- Recombination frequency between cn and b is 9%
- Between cn and vg is 9.5%
- Recombination frequency between b and vg is 17%



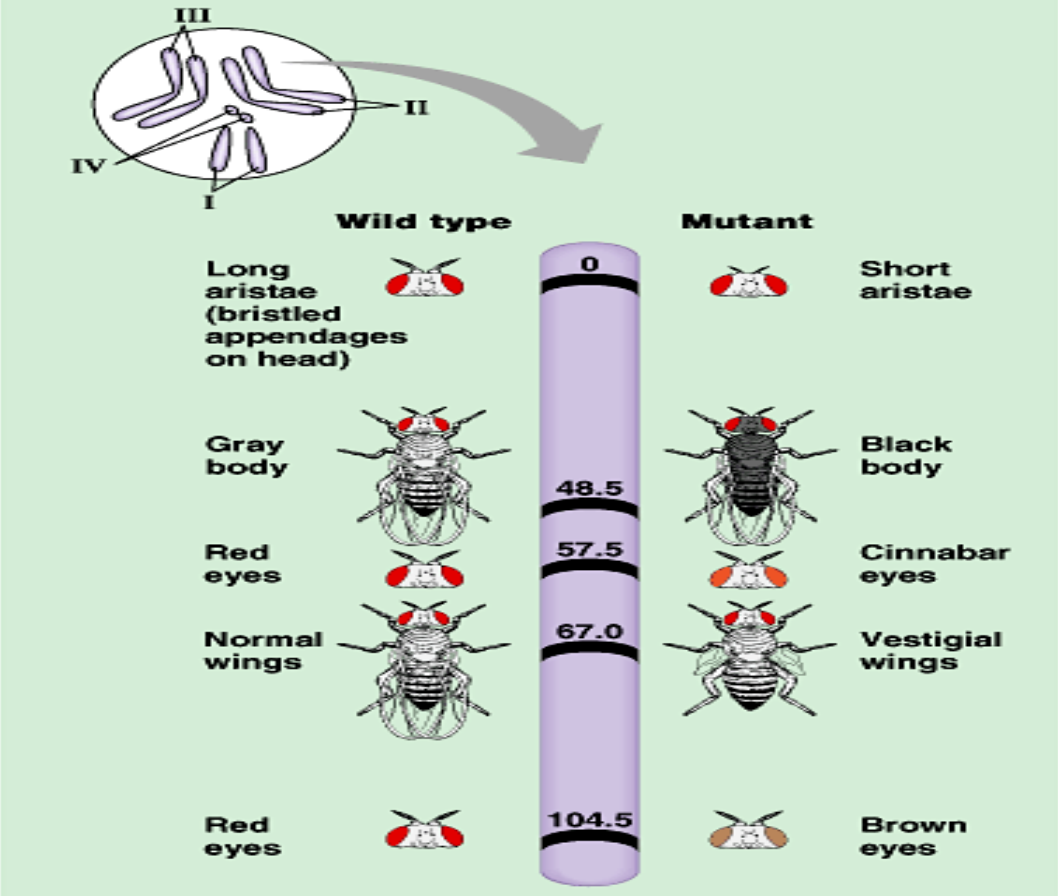
Linkage map of Drosophila Chromosome

EYE COLOR GENE BETWEEN THE TWO OTHERS

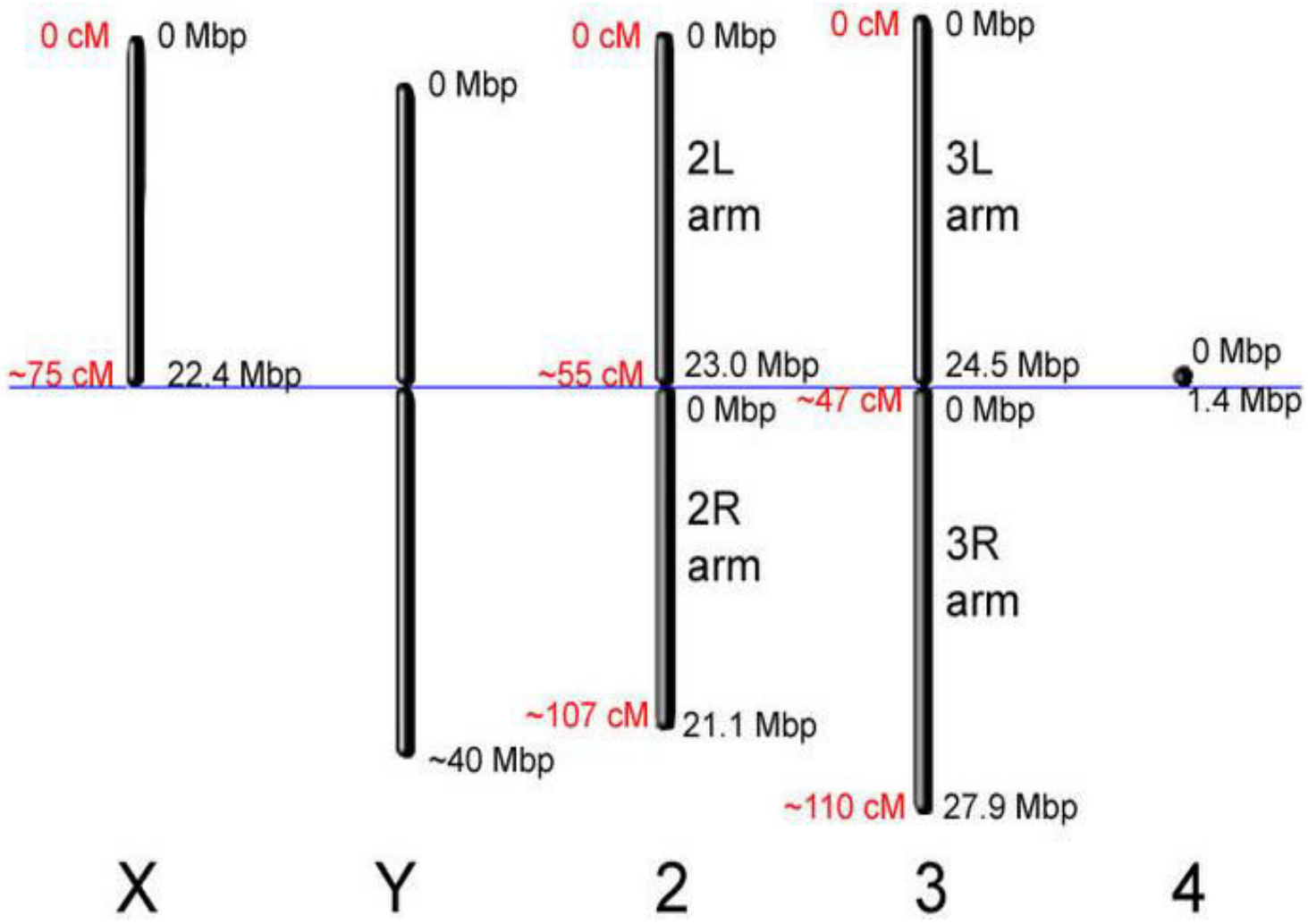
- The only possible arrangement of these three genes places the eye color gene between the other two.

Linkage map of Drosophila Chromosome

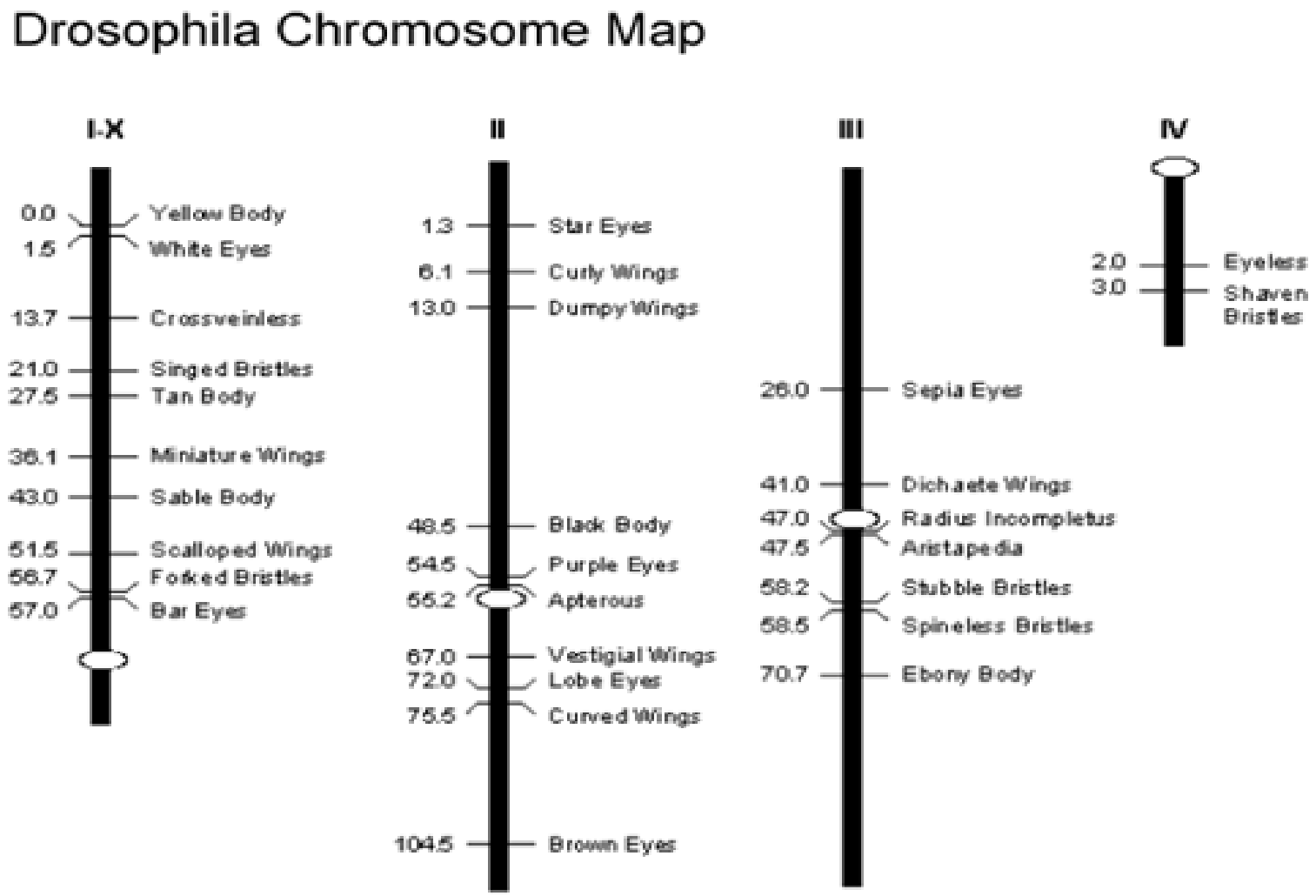
LINKAGE MAP OF DROSOPHILA CHROMOSOME 2



Linkage map of Drosophila Chromosome



Linkage map of Drosophila Chromosome

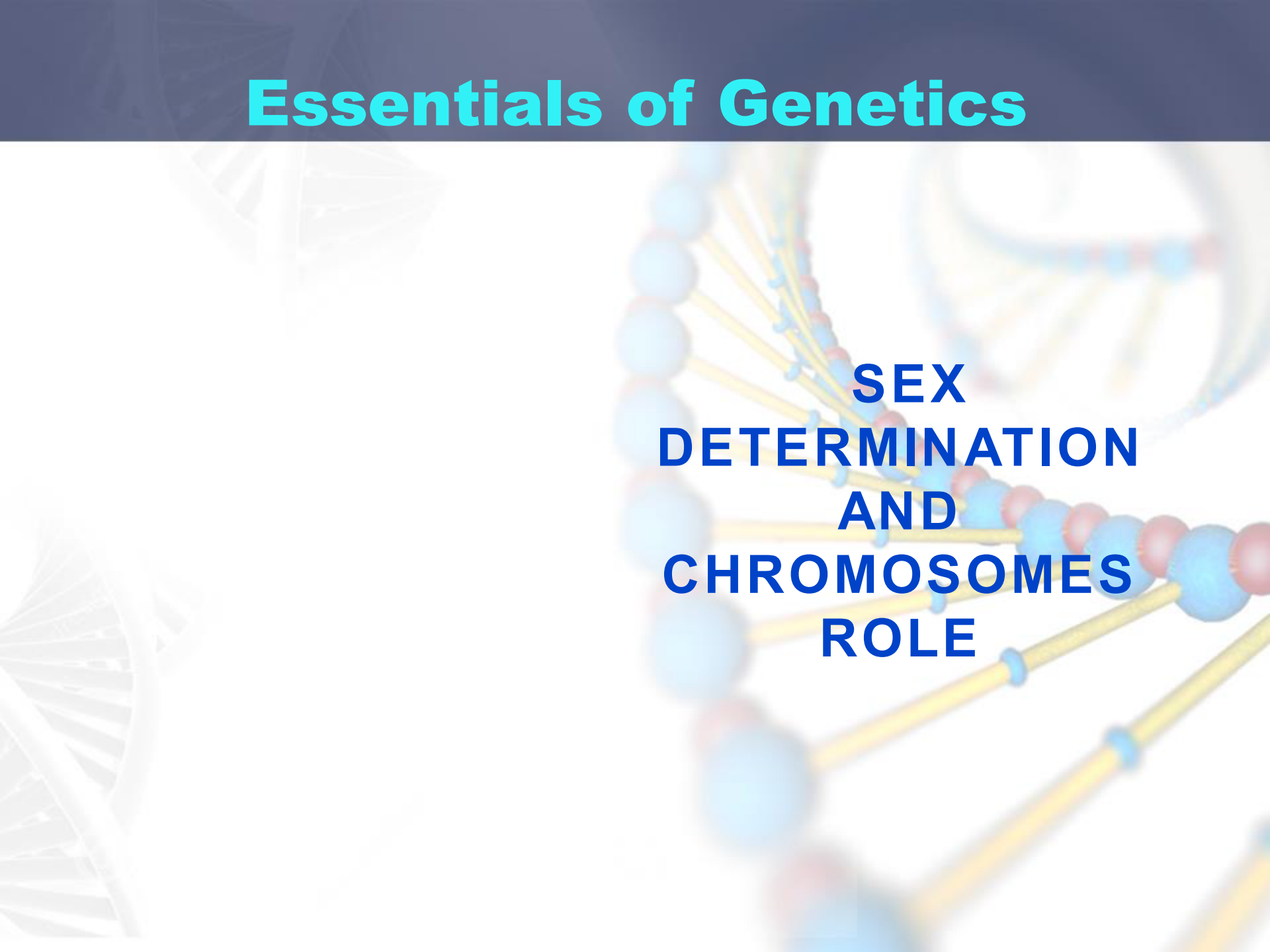


Linkage map of Drosophila Chromosome

MAP UNITS - MENTIMORGAN

- Sturtevant expressed distance between genes - as map units.
- One map unit (1 cM) is equivalent to a 1% recombination frequency.

Essentials of Genetics



SEX DETERMINATION AND CHROMOSOMES ROLE

Sex Determination-Chromosomes Role

GENETIC BASIS OF SEX – DISCOVERY OF SEX CHROMOSOMES

- Understanding of genetic basis of sex was with the discovery of sex chromosomes in the early 1900s.

Sex Determination-Chromosomes Role

INSECTS CHROMOSOMES

- Analyses of male and female insect chromosomes.
- Most chromosomes were present in equal numbers in both males and females.
- One /two additional chromosome.

Sex Determination-Chromosomes Role

CHROMOSOMAL DIFFERENCE ARE RESPONSIBLE FOR SEX DETERMINATION

- Analyses of other species revealed that chromosomal differences are primarily responsible for sex determination.

Sex Determination-Chromosomes Role

SEX DETERMINATION

- Chromosomal basis for determining sex
- The X-Y System
- The X-O System
- The Z-W System
- The haploid-diploid System

Sex Determination-Chromosomes Role

SEX DETERMINATION IN HUMANS

- Female homogametic XX
- Male heterogametic XY
- Sex of offspring depends on the sperm has an X or Y

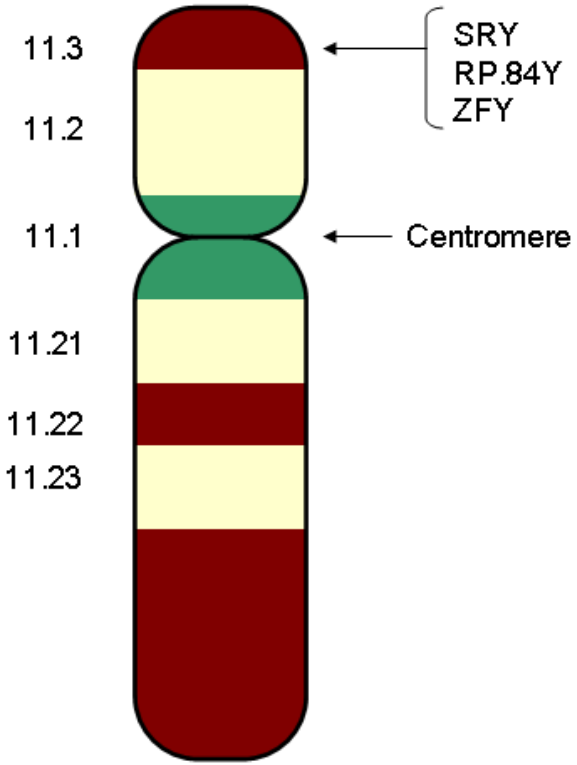
Sex Determination-Chromosomes Role

SRY GENE - Y CHROMOSOME

- Researchers have found the SRY gene (sex determining region of the Y chromosome
- In individuals with the SRY gene, the generic embryonic gonads are modified into testes.

Sex Determination-Chromosomes Role

SEX DETERMINING REGION ON Y CHROMOSOME

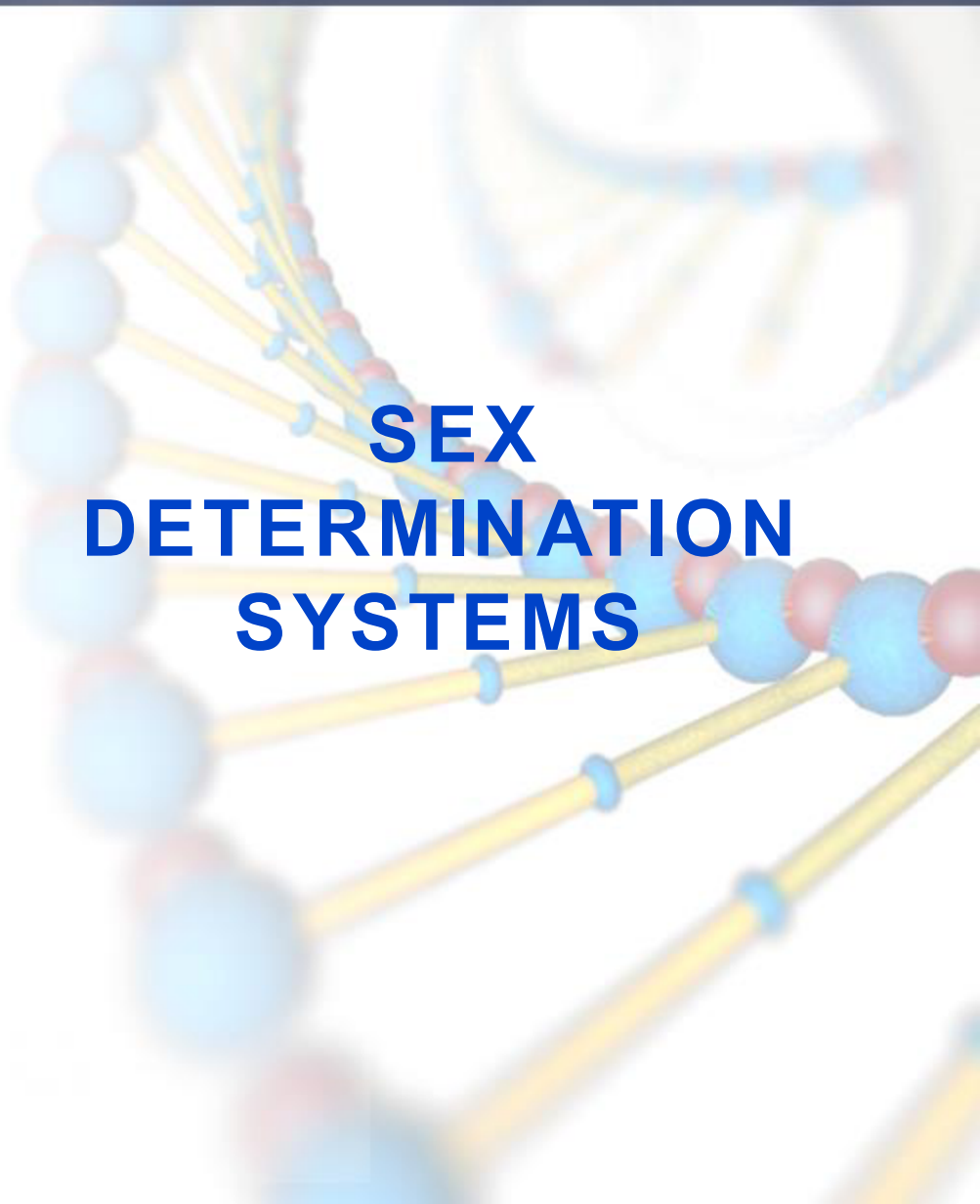


Sex Determination-Chromosomes Role

OTHER GENES ON Y CHROMOSOME

- In addition, other genes on the Y chromosome are necessary for the production of functional sperms.
- In individuals lacking the SRY gene, the generic embryonic gonads develop into ovaries.

Essentials of Genetics



SEX DETERMINATION SYSTEMS

Sex Determination Systems

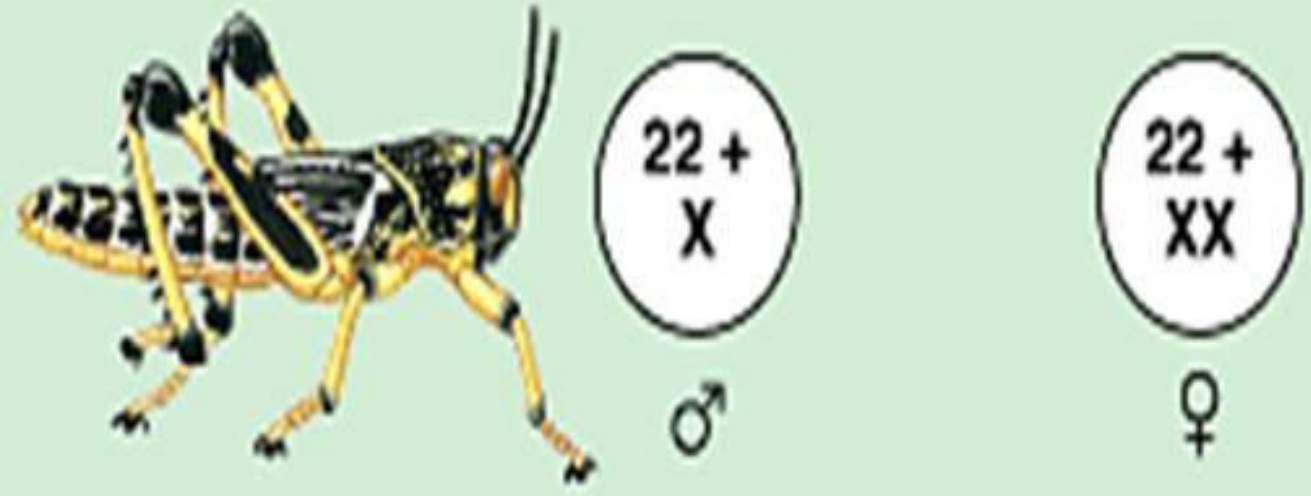
X-O SYSTEM

- In insects, only one type of chromosome.
- Females XX, Males XO
- Sex of offspring, sperm X or O

Sex Determination Systems

X-O SYSTEM

- In insects, only one type of chromosome.
- Females XX, Males XO
- Sex of offspring, sperm X or O



Sex Determination Systems

Z-W SYSTEM

- In birds, some fishes and some insects.
- Females ZW, Males ZZ.

Sex Determination Systems

Z-W SYSTEM

- In birds, some fishes and some insects.
- Females ZW, Males ZZ.



Sex Determination Systems

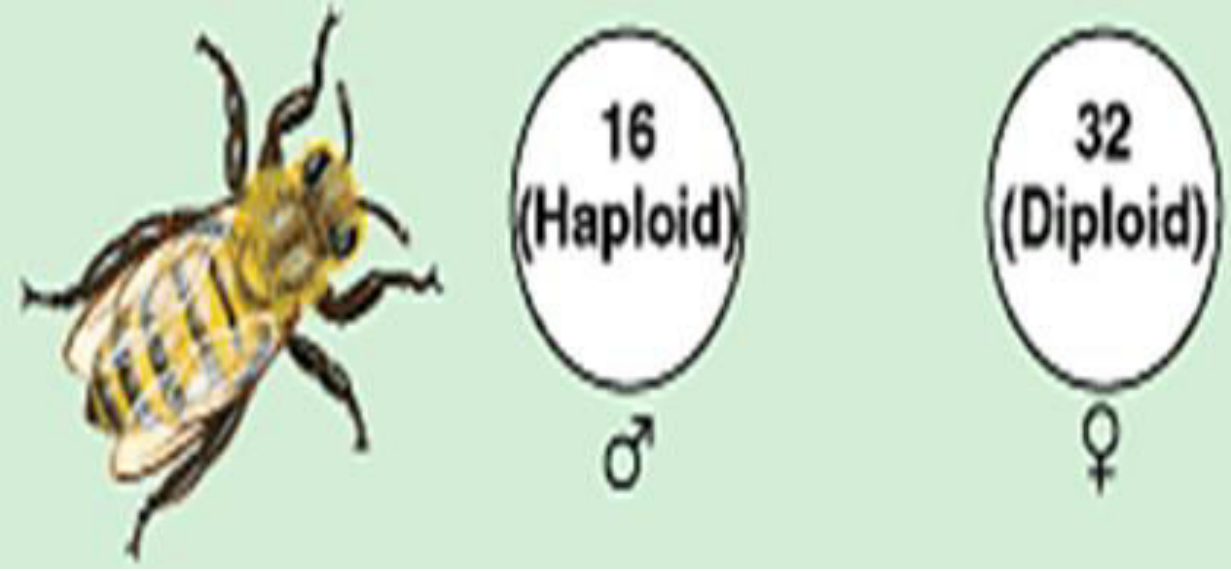
HAPLOID-DIPLOID

- No sex chromosomes in most species of ants and bees.
- Females fertilized ova diploid
- Males unfertilized ova haploid

Sex Determination Systems

HAPLOID-DIPLOID

- No sex chromosomes in most species of ants and bees.
- Females fertilized ova diploid
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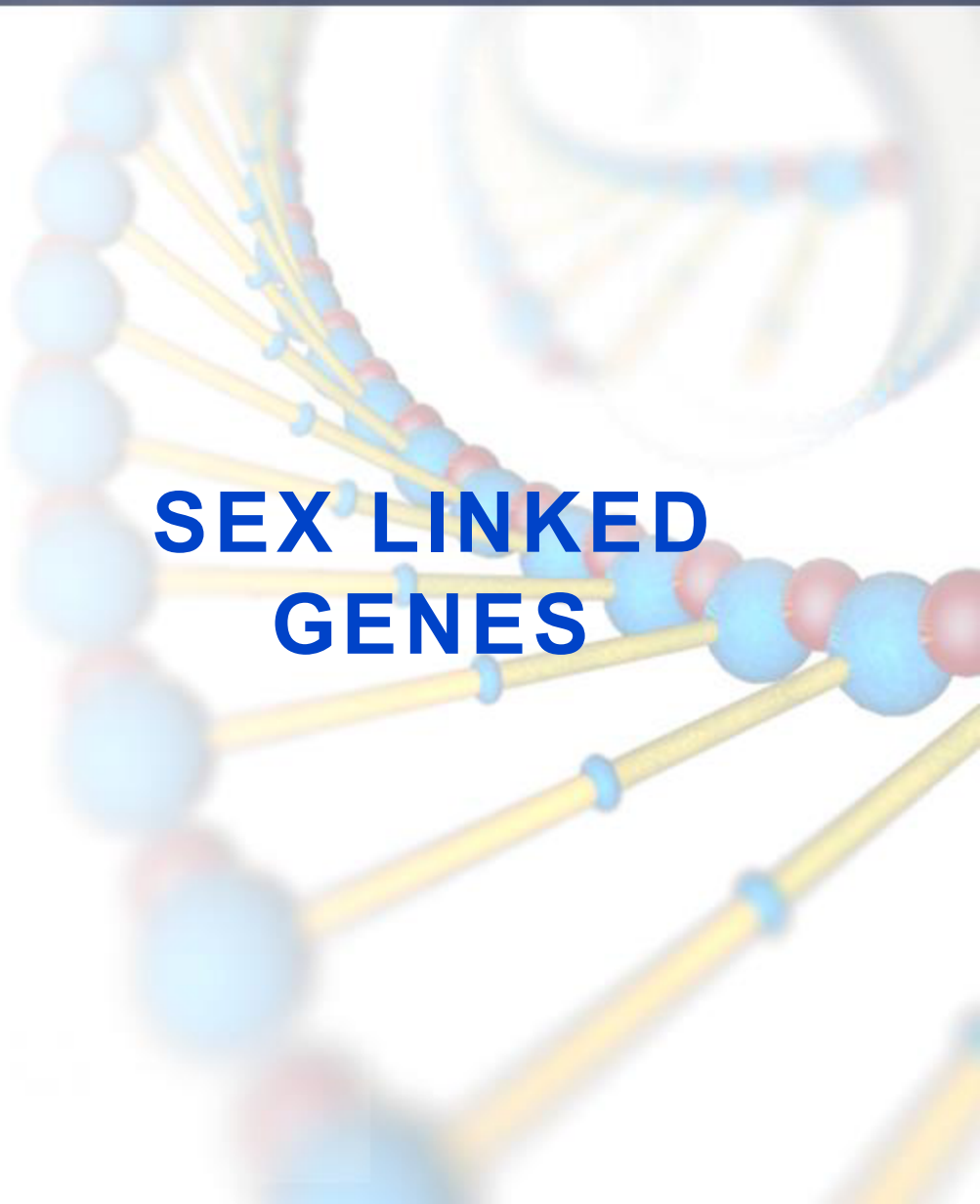


Sex Determination Systems

SEX DETERMINATION CONCLUSION

- Different organism have different systems of sex determination.

Essentials of Genetics



SEX LINKED GENES

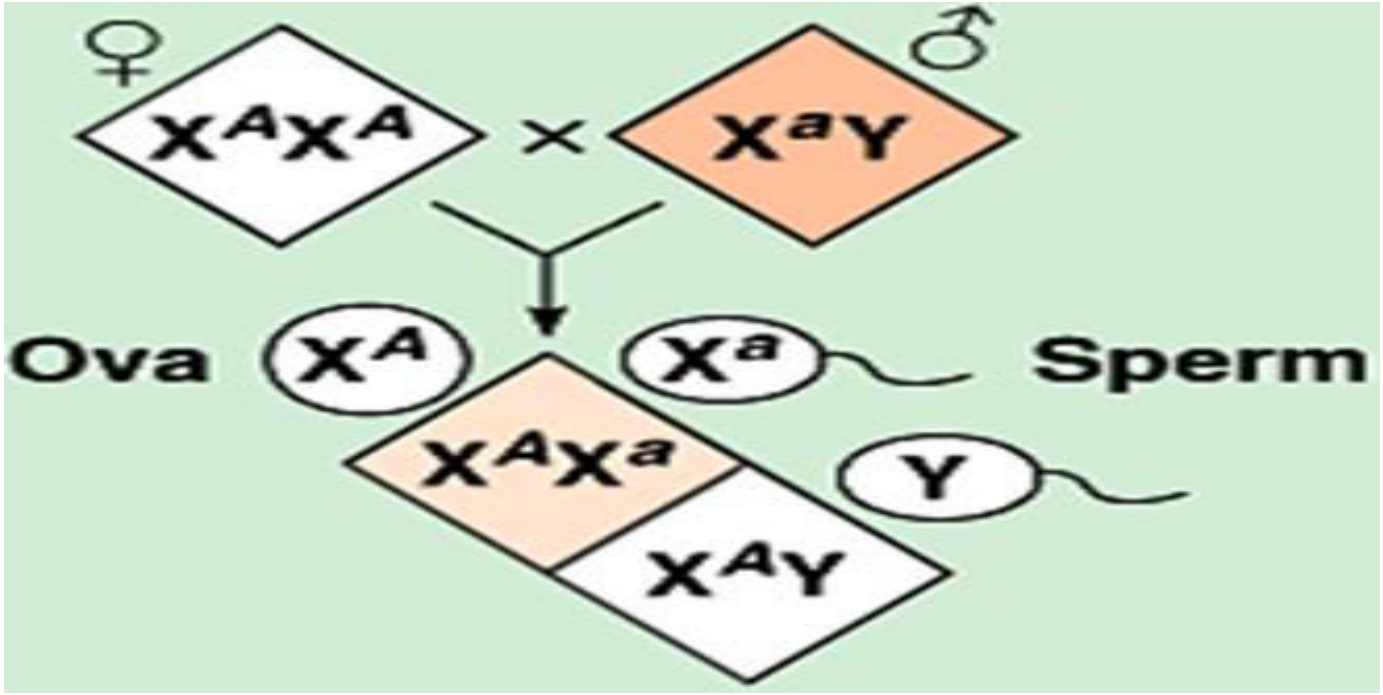
Sex Linked Genes

INHERITANCE OF SEX LINKED GENES

- A gene located on either sex chromosome a sex-linked.
- The sex chromosomes have genes for many characters unrelated to sex.

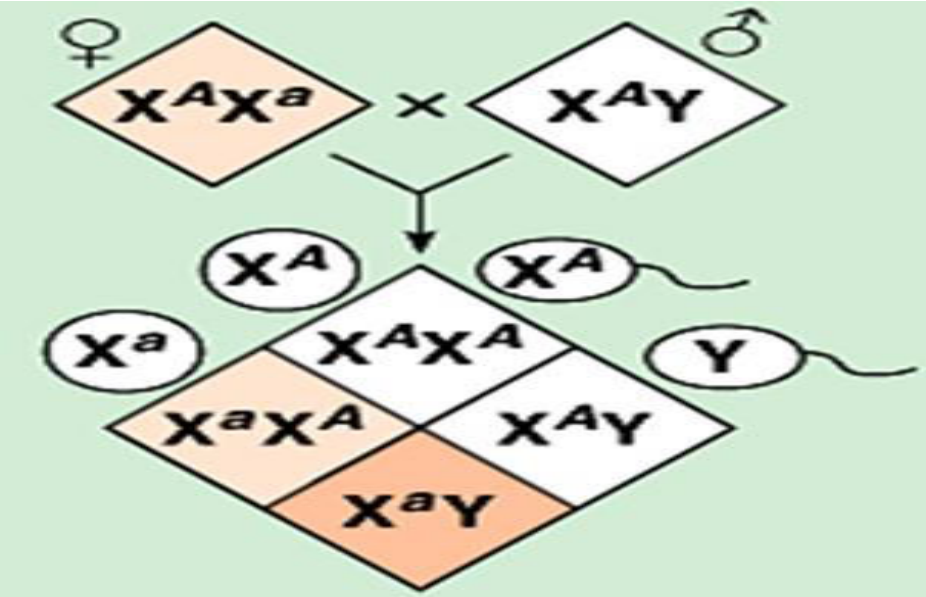
Sex Linked Genes

FATHER WITH DISORDER WILL TRANSMIT THE MUTANT ALLELE TO ALL DAUGHTERS BUT TO NO SONS



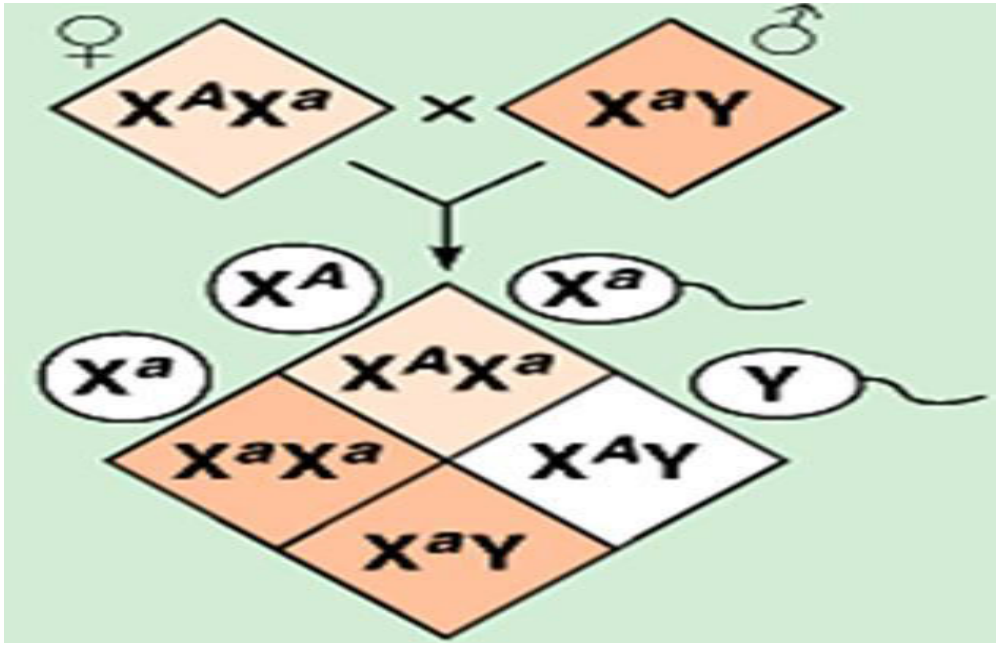
Sex Linked Genes

CARRIER FEMALE MATES NORMAL MALE, 50% CHANCE THAT EACH DAUGHTER WILL BE A CARRIER; 50% CHANCE THAT EACH SON WILL HAVE THE DISORDER.



Sex Linked Genes

IF A CARRIER FEMALE MATES A MALE WITH DISORDER, 50% CHANCE THAT EACH CHILD BORN WITH DISORDER, REGARDLESS OF SEX.



Sex Linked Genes

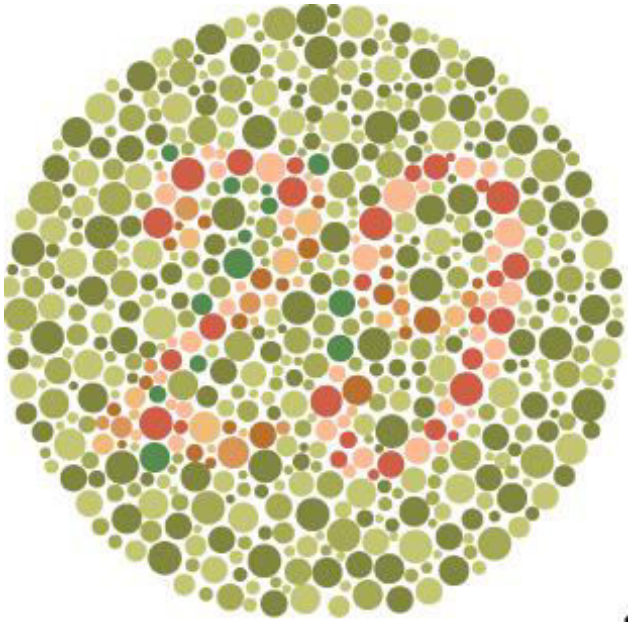
HUMAN SEX LINKED DISORDERS

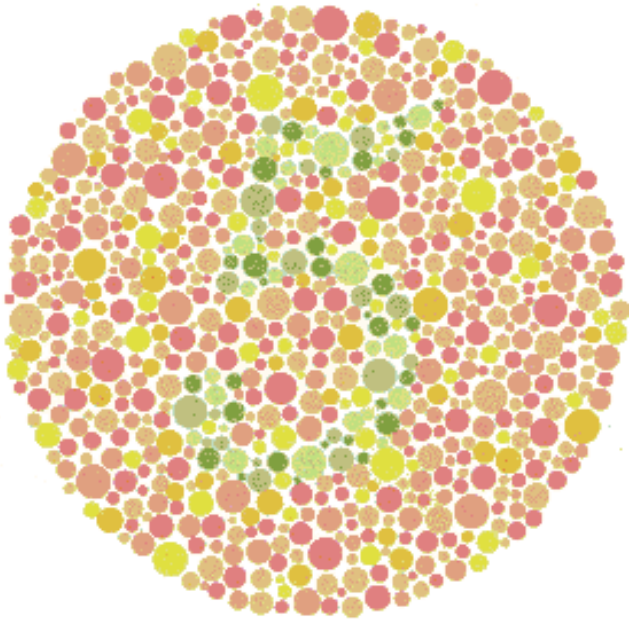
- Color blindness
- Duchenne muscular dystrophy
- Hemophilia

Sex Linked Genes

COLOR BLINDNESS -

- Normal Female XX, Normal male XY
- Color Blind XcXc, Carrier XcX, Color Blind Male XcY





Sex Linked Genes

NORMAL FATHER
COLOR BLIND MOTHER

- All daughters are carriers, sons are color blind.

	X	Y
X^c	$X X^c$	$X X^c$
X^c	$X X^c$	$X^c Y$

Sex Linked Genes

NORMAL MOTHER
COLOR BLIND FATHER

- All daughters are carriers, sons are normal

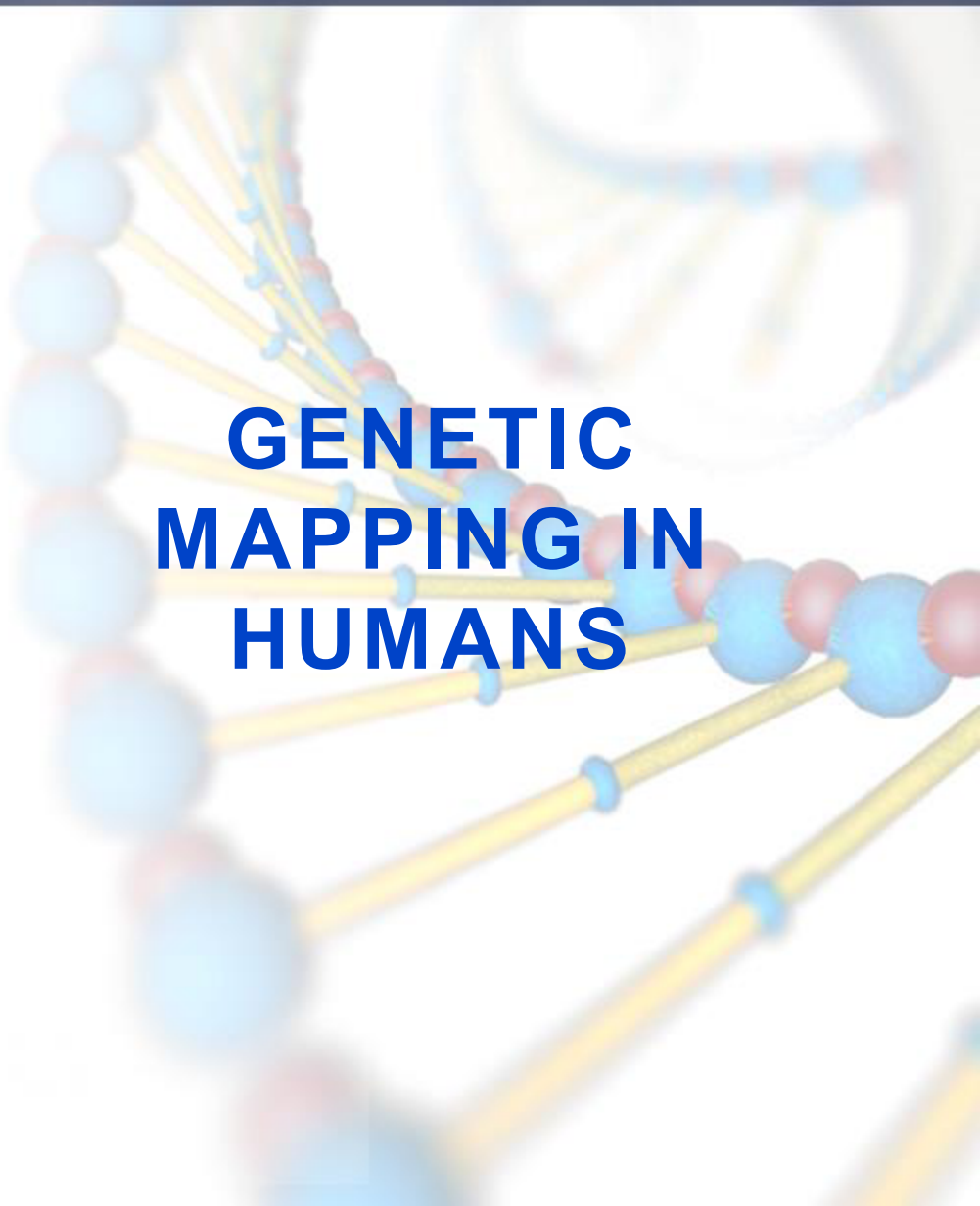
	X^c	Y
X	$X X^c$	X Y
X	$X X^c$	X Y

Sex Linked Genes

SEX LINKED GENES/TRAITS

- A gene located on either of sex chromosome is called sex-linked and traits are called as sex-linked traits.

Essentials of Genetics



GENETIC MAPPING IN HUMANS

Genetic Mapping in Humans

GENETIC MAPPING

- Genetic mapping - also called linkage mapping provides clues about which chromosome contains the gene and precisely where it lies on that chromosome.

Genetic Mapping in Humans

SINGLE GENE DISORDERS IDENTIFIED

- Genetic maps have been used successfully to find the single gene responsible inherited disorders, like cystic fibrosis and muscular dystrophy.

Genetic Mapping in Humans

HOW GENETIC MAPPING PERFORMED

- To produce a genetic map, researchers collect blood or tissue samples from family members where a certain disease or trait is prevalent.

Genetic Mapping in Humans

HOW GENETIC MAPPING PERFORMED

- Scientists isolate DNA from samples and examine it for the unique patterns of bases.
- Molecular techniques are referred to as genotyping polymorphisms or markers.

Genetic Mapping in Humans

DNA MARKERS HELP IN LOCATION OF GENES

- Before researchers identify the gene responsible for the disease or trait, DNA markers can tell them roughly where the gene is on chromosome.

Genetic Mapping in Humans

MORE MARKERS INCREASES CHANCE OF LINKAGE

- The more DNA markers there are on a genetic map, the more likely it is that one will be closely linked to a disease gene.

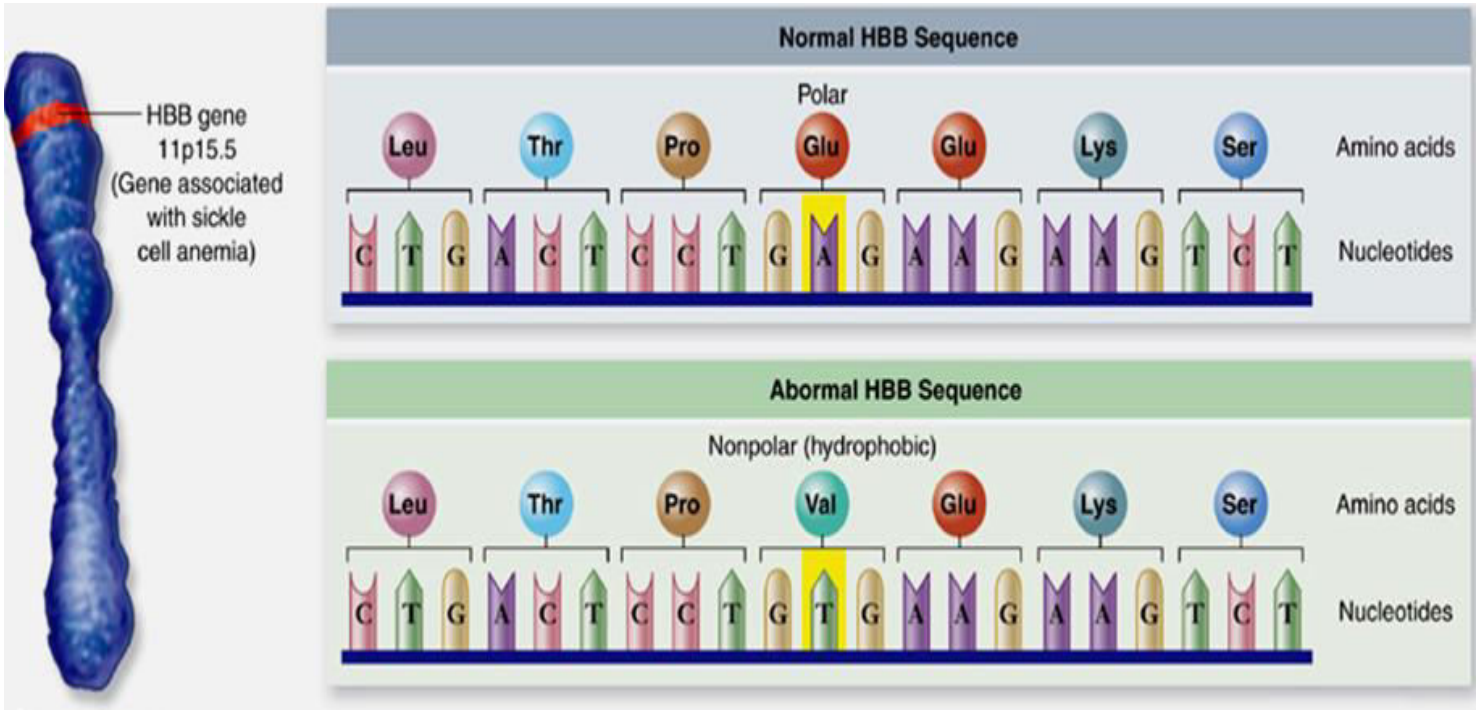
Genetic Mapping in Humans

TYPES OF GENETIC MARKERS

- Several different types of genetic markers, the type most used on genetic maps is known as a microsatellite and SNPs.

Genetic Mapping in Humans

GENETIC MAPPING IN HUMANS



Genetic Mapping in Humans

GENETIC MAPPING PROVIDE CLUES- LOCATION OF GENES

- Genetic mapping provides clues about the location of genes on chromosomes.